

《三命无常》工程真源定稿

完整玩法、数值、地图、内容、交互与工程协议的唯一开发蓝图

项目	定案值	项目	定案值
产品 ID	SANMING_WUCHANG	文档状态	RELEASE_BLUEPRINT_V4R2 / 108—179 三轮审计闭合并通过 CCK_V4_2 复核 / 唯一工程真源，允许据此开发
模式	30 人单人乱斗	生命	每人 3 条
平台	Android / iOS / Windows	同服	三端同服同局
地图	百眼迷城 · 等效跨度 840 米	目标局长	18–20 分钟；之后雷暴加时
内容量	38 英雄 / 57 主动 / 44 被动	装备与怪物	48 装备 / 22 机械原型 / 38 视觉形象
定稿日期	2026-07-21	适用范围	LAUNCH_SCOPE：产品、客户端、服务端、数据与内容生产； POST_LAUNCH_NOT_AUTHORIZED 须另行立项
规则集版本	3.1.0	规则哈希	ea742316cbb722fe...

唯一性声明 | 运行时唯一值只存在于附录 E 中完整命名空间化的工程键；正文、玩家文案和汇总表均为同一工程键生成的只读投影。构建器发现投影与工程键不一致时必须失败，禁止人工二选一。

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0 真源权限与工程约定

执行原则 | 客户端、服务端、数据生成器、BOT、回放和测试只读取附录 E 导出的规则快照；正文解释设计意图，资产表指导内容生产，二者不得直接驱动运行时。空值必须显式写为 NONE、[]、NOT_APPLICABLE 或 INHERIT_DEFAULT。

约定	唯一规则
坐标	世界原点为地图几何中心；X 向东、Y 向北、Z 向上；文档用米，运行时位置量化为毫米整数。
时间	服务端固定 20Hz；时间窗均采用[开始时刻,结束时刻)半开区间；结束 tick 先关旧状态再验证输入。
数值	伤害内部使用有符号 64 位 Q4；生命提交四舍五入且远离 0；金币向下取整；位置取最近毫米且中点远离 0。
随机	每局一个根种子，英雄、出生、商店、怪物、掉落、五行、天劫等使用相互隔离的随机流；客户端不获知未来流状态。
ID	英雄 H001-H038；专属主动 H001-H038；地图主动 D1-D22 中的 19 个有效 ID；被动 B01-B44；装备 W/B/P/G；地图实体使用命名空间+ID。
未知字段	闭合枚举遇到未知值直接拒绝加载；缺少必填字段禁止启动对局；不得用默认猜测替代。
冲突	本文的规则表、内容表和工程字段同时由同一份定稿数据生成；如展示文字与工程字段冲突，停止构建并修正文档，不允许开发者自行选一边。
投影生成	TRUTH_TO_PROSE_GENERATOR_V1 为唯一投影通道：正文机制列、玩家文案与所有汇总表由附录 E 工程键单向生成，产物携带 source_hash；手工编辑投影即构建失败。
投影校验	构建器逐字段比对投影与工程键；漂移即 G_PROJECTION_DRIFT 失败，禁止人工二选一（与冲突条款

约定	唯一规则
	同源) 。

0.1 稳定规则域 ID 与 AI 开发入口

规则域 ID	唯一责任	运行时权威
RULE-CORE-001	版本、ID、单位、空值、投影与冲突政策	E.1/E.23
RULE-LIFE-001	三命、致死、掉落、复活、奖励	E.1/E.2
RULE-INPUT-001	三端输入与 HUD	E.6/E.20/E.24
RULE-COMBAT-001	普攻、伤害、控制、五行、视野	E.3—E.7
RULE-ACTIVE-001	57 主动与类型化效果	§ 6/E.23
RULE-PASSIVE-001	44 被动与替换	§ 7/E.15
RULE-EQUIP-001	48 装备、格位和效果	§ 8/E.15
RULE-MAP-001	区域、地表、路网、墙体、POI	§ 9/附录 D/E.8—E.9
RULE-STORM-001	天劫与灭世雷暴	§ 10/E.11
RULE-PVE-001	怪物 AI、战斗、重生和视觉	§ 11/E.12—E.13
RULE-ECON-001	掉落、地面物、商店、黑山、空投	§ 12/E.14—E.19
RULE-MATCH-001	匹配、BOT、断线、MMR	§ 13/E.21
RULE-META-001	局外、商业化与设置	§ 14/E.24
RULE-NET-001	网络、后端、事务、随机、回放	§ 15/E.21—E.22
RULE-ASSET-001	美术、音频、本地化、资产	§ 16
RULE-RELEASE-001	AI 开发顺序、性能和发布门槛	§ 17/E.25

唯一开发顺序：规则内核→共享类型与闭合枚举→地图物理层→战斗→PVE→经济→UI→网络与后端→局外→自动测试与发布门槛。任何后续阶段不得反向发明前序规则。

0.2 封闭术语表

术语	工程键	唯一含义
R 点	revive_point	系统在当前阶段合法候选池中确定的复活目的地
宏区	macro_zone	用于分散出生或商店抽取的一组互斥微点
微点	micro_point	具备精确 XYZ、朝向和物理合法性的单个锚点

术语	工程键	唯一含义
实际伤害	actual_hp_or_shield_damage	经过无敌、闪避、减伤与护盾流程后提交的正生命或护盾损失
破隐	stealth_break	隐身状态结束；不等同于获得隔墙锁定
现形	revealed	被指定真视来源持续追踪；锁定仍需墙体与射程合法
真视	true_sight	可把范围内隐身目标置为 REVEALED 的信息效果
未击	final_damage_owner	把目标推进合法死亡/怪物死亡的最后伤害归属者
真死	true_death	所有致死保护失败后消耗一条命并提交 的死亡

1 产品定位与设计宪法

支柱	定案
极简操作	移动、普攻、唯一主动、情境交互四类核心动作；有手就会，深度来自路线、时机、资源和构筑。
自由优先	除非破坏公平、产生严重干扰或造成技术不可实现，玩家可刷野、追击、苟活、伏击、拾荒、换英雄、拉怪、治人或背叛。
三条命	死亡不是立刻退场；前两次死亡随机复活，第三次才淘汰。每条命都可能更换路线、英雄与构筑。
玩家驱动生态	怪物容量固定、死后迁移读秒。玩家杀得多，PVE 经济周转就高；没人刷野，PVE 产出就自然接近 0；不做隐藏橡皮筋。
公共故事点	龙王、核心 BOSS、商店、空投与地面装备公开竞争；金币经验归未击，BOSS 掉落公开，制造打王、截胡、伏击和反杀。
战争迷雾迷宫	玩家学习区块和路线规律，但每局出生、资源微点、商店、怪物迁移、猪窝、龙宫、伏石和决赛庭发生变化。
同服公平	手机、平板和 PC 同规则同数值同视野；只允许输入辅助和 UI 差异，不允许伤害、移动、掉率或信息补偿。
服务端权威	客户端只提交意图；伤害、碰撞、视野、随机、物品、金币、CD、掉落和胜负均由服务端确认。

2 单局流程、胜负与三条命

项目	规则
参赛	30 人，SOLO，每人 3 条命；最终仅一名参与者未淘汰即获胜。
并列	同一服务端 tick 完成全部已提交致死链；若剩余者同时淘汰，则并列第 1，局外冠军奖励各自完整发放，局内经济不复制。

项目	规则
保险	若对局运行到 30 分钟仍无合法胜者，服务端以 VOID_ABORT 结束：不计算胜负与 MMR，发放固定参与修为补偿，保留诊断录像；客户端显示统一异常结算原因并提供重新匹配。
英雄选择	开局 500 金；15 秒内从 38 英雄等权随机的 3 个不重复选项中选 1；250 金可整组三项重随，余额足够即可继续；超时取当前最左项。
起点	从 40 个出生宏区中选 30 个不同宏区，每区再随机选 1 个微位；每宏区最多 1 人。
目标局长	18-20 分钟；20:00 安全半径归零后用确定性灭世雷暴继续到胜负产生。

2.1 等级与经验

等级	升到下一级经验	累计经验
1	50	50
2	70	120
3	90	210
4	110	320
5	130	450
6	155	605
7	180	785
8	210	995
9	240	1235
10	275	1510
11	315	1825
12	450	2275
13	550	2825
14	700	3525
15	封顶；后续经验丢弃	3525

保底经验：每 10 秒获得 10 经验；ALIVE、TRUE_DEATH_RESOLVING、SOUL_FLIGHT、REVIVE_PROTECTION 均计时。经验溢出可连续升级。

2.2 致死链、掉落与复活

顺位	处理
1	DAMAGE_IMMUNITY_OR_DODGE
2	SHIELD_ABSORPTION

顺位	处理
3	HP_DAMAGE_COMMIT
4	B19_FEIGN_DEATH_ROLL
5	B20_PASSIVE_REVIVE_IF_CHARGE_AVAILABLE
6	G1_NINE_TURN_PILL_IF_OWNED
7	CONSUME_ONE_LIFE_AND_COMMIT_TRUE_DEATH
8	ELIMINATE_IF_NO_LIFE_REMAINS

- 只有所有致死保护都失败才消耗一条命；假死、九转金丹和被动复活均不触发真死掉落与全局重置。
- 第 1、2 次真死：掉落全部手牌装备，身穿 3 件保留；第 3 次真死：手牌和身穿 3 件全部公开落地；金币永不掉落。
- 第 1、2 次真死使当前主动变为 READY；被动内置 CD 继续计时且不因死亡清零。临时增减益、DoT、控制、召唤物、陷阱、冰墙和装备临时层清除；等级、经验、金币、被动等级、装备及明确声明的永久层保留。第 3 次真死淘汰后不再创建技能运行时状态。
- 真死结算后成为无碰撞无敌灵魂，以 18m/s 飞往分阶段复活候选池选出的 R 点。基础飞行时长 clamp(3,直线距离/18,10)秒；到点重验，不合法则按同一确定性评分转向次选，额外最多 3 秒，随后使用本章规定的安全回退点。
- 复活为满生命并获得 3 秒保护：可移动、可进入伏石，但不能攻击、施法、拾取、购物、开箱或赌博；任何进攻/交互意图先解除保护再重验同一请求。
- 怪物最后一击玩家永远记怪物击杀，玩家不获奖励；所有者死亡后的已释放弹道继续结算并保留原所有者归属。

2.3 玩家击杀奖励

项目	公式
金币	$500 + 100 * \text{VICTIM_LEVEL} + \text{MIN}(\text{FLOOR}(0.10 * \text{VICTIM_WALLET_SNAPSHOT}), 800)$
受害者钱包	不扣除；钱包奖励由系统新发、上限 800 且不受 G5 倍率
第三条命终结	额外 500 金
经验	$\text{MIN}(180, 60 + 8 * \text{VICTIM_LEVEL})$
助攻	0 金币； $\text{MIN}(60, 20 + 3 * \text{VICTIM_LEVEL})$ 经验。8 秒内对受害者造成实际伤害或施加硬控且非末击者获得一次
怪物/环境补刀	0 玩家击杀奖励

3 三端输入、镜头、HUD 与信息公平

动作	手机 / 平板	Windows PC
移动	左摇杆	WASD

动作	手机 / 平板	Windows PC
普攻	普攻键；按住连续普攻；拖向目标可显式锁定 2 秒	鼠标左键；按住连续普攻
唯一主动	轻点智能施法；按住拖拽精瞄，松手施放	鼠标右键；短按智能施法；按住 ≥ 0.22 秒且指针移动 $\geq 4\text{px}$ 显示预览，松手施放
交互	情境交互键；短按执行高优先级目标/再次短按轮换；长按打开遗宝堆列表	E 短按执行高优先级目标，再按轮换重叠目标；E 长按 ≥ 0.35 秒打开附近遗宝堆列表
取消	拖回取消区/专用取消手势	Esc
支持范围	Android、iOS	Windows；不支持手柄

- 逻辑战斗视口固定 16:9；宽屏只扩展装饰，不增加敌情、锁定距离或小地图信息。
- 基础视野 30 米，单位锁定必须同时满足直接视线和合法射程；长距瞄准镜头前移最多 20 米，只给预览，不给额外敌情。
- 固定斜俯视镜头，不允许自由旋转或拖屏侦察。UI 事件必须吞掉世界点击，避免点按钮误放技能。
- 索敌扇区全角 24°；并列顺序：最近有效伤害过自己的目标→屏幕夹角最小→Nav 距离最短→实体 ID。
- 任何重要信息不得只靠颜色或声音；五行至少使用符号+文字+配色+阵纹，CD 使用数字+扇形遮罩。

3.1 1280×720 逻辑 HUD

工程字段	唯一值 / 执行语义
logical_canvas.width	1280
logical_canvas.height	720
logical_canvas.aspect	16:9
logical_canvas.min_safe_inset	24
fair_viewport.interactive_aspect	16:9
fair_viewport.ultrawide_extra_gameplay_information	否
common.player_block.anchor	TOP_LEFT
common.player_block.x	32
common.player_block.y	28
common.player_block.portrait	72
common.player_block.hp_w	240
common.player_block.hp_h	28
common.storm_block.anchor	TOP_CENTER
common.storm_block.x	490
common.storm_block.y	24
common.storm_block.w	300
common.storm_block.h	42
common.minimap.anchor	TOP_RIGHT
common.minimap.x	32
common.minimap.y	28

工程字段	唯一值 / 执行语义
common.minimap.w	188
common.minimap.h	188
common.minimap.expanded	480
common.passive_slot	52
common.equipment_slot	48
common.status_icon	32
mobile.joystick.x	150
mobile.joystick.y	585
mobile.joystick.outer_radius	92
mobile.joystick.drag_radius	76
mobile.joystick.deadzone	12
mobile.joystick.floating_zone_w	280
mobile.joystick.floating_zone_h	250
mobile.attack.x	1120
mobile.attack.y	590
mobile.attack.radius	58
mobile.active.x	995
mobile.active.y	620
mobile.active.radius	50
mobile.interact.x	1170
mobile.interact.y	465
mobile.interact.radius	46
mobile.cancel.x	885
mobile.cancel.y	470
mobile.cancel.radius	64
mobile.min_touch_target	72
mobile.critical_touch_target	96
mobile.min_gap	12
mobile.drawer_width	420
mobile.drawer_max_height	560
pc.move	W / A / S / D
pc.basic_attack	MOUSE_LEFT
pc.active_skill	MOUSE_RIGHT
pc.precision_aim_hold_s	0.22
pc.interact	E
pc.cancel	ESC
pc.map	M

工程字段	唯一值 / 执行语义
pc.inventory_panel_hold	TAB
pc.camera_center	SPACE
pc.drawer_width	360
controller_supported	否
responsive.safe_area	OS_SAFE_INSETS_PLUS_24_LOGICAL_PX
responsive.aspect_policy	16_9_GAMEPLAY_VIEW; EXTRA_WIDTH_DECORATIVE_ONLY; NO_EXTRA_WORLD_INFORMATION
responsive.ui_scale_user_percent	80 / 90 / 100 / 110 / 120 / 130
responsive.text_scale_user_percent	90 / 100 / 110 / 120 / 130
responsive.tablet	ANCHOR_RELAYOUT_WITH_SAME_GAMEPLAY_INFORMATION_DENSITY
responsive.pc_ultrawide	LETTERBOX_OR_DECORATIVE_SIDEBARS; NO_GAMEPLAY_CAMERA_GAIN
accessibility.subtitles	ON_BY_DEFAULT; SIZE_FOLLOWS_TEXT_SCALE
accessibility.windows_key_rebind	ALL_GAMEPLAY_KEYS_EXCEPT_SYSTEM_ESCAPE; CONFLICT_VALIDATION_REQUIRED

4 英雄、等级、属性与五行

主动成长铁律 | 主动技能没有技能等级。所有固定伤害、固定治疗、固定护盾、固定召唤属性、百分比、CD、距离、范围与时长终局不变；只有技能明确声明的属性系数读取释放时当前面板属性。

属性处理	规则
面板顺序	基础等级行→所有固定加成求和→百分比加成相乘→临时倍率
最大生命变化	保持当前生命百分比，向下取整且至少 1；卸装不能直接杀死
移动速度	1 游戏移速单位=0.01m/s；无总上限，鞋类不同名可叠加，同名身穿唯一
攻击速度	无总上限；当前已提交攻击不回算，下一次攻击使用新攻速
普攻暴击	基础暴击率 0%，基础暴击伤害×2.0；怪物不暴击
五行	金克木、木克土、土克水、水克火、火克金；攻击方克制目标时×1.5，否则×1.0；没有防守方反向减伤。
五行适用	玩家 PVP 的普通普攻/技能/DoT，以及五行猪和五行龙王的普攻与技能。真伤、百分比、天劫、反伤不走五行。
怪物五行	只有肥猪与龙王有五行；普通近战、远程、飞行、首领和核心 BOSS 均无五行。

5 战斗结算、控制、视野与隐身

5.1 普攻

项目	规则
合法目标	其他存活玩家、敌对怪物、可受伤的敌对召唤物
前摇期间	可全速移动；施放主动、硬控或真死取消；普通移动输入不取消
释放复验	释放 tick 再次验证距离与直接视线
攻击间隔	1/当前每秒攻击次数，从释放 tick 开始计时
远程弹道	锁定提交目标中心；释放后失去视野仍继续；被实体墙、风墙、冻河阻挡；每 tick 连续扫掠，半径 0.12m
所有者死亡	已释放弹道继续，保持原归属；目标在命中前淘汰则弹道销毁

5.2 伤害结算顺序

顺序	步骤
1	VALIDATE_TYPED_SOURCE_TARGET_LIFE_AND_CAST_OR_IMPACT_REQUIREMENTS
2	RESOLVE_PROJECTILE_OR_SHAPE_COLLISION
3	RESOLVE_FULL_IMMUNITY_DODGE_AND_BLOCK_IF_ELIGIBLE
4	CALCULATE_FIXED_PLUS_CURRENT_ATTRIBUTE_COEFFICIENT_Q4
5	APPLY_CRITICAL_IF_ELIGIBLE
6	APPLY_SOURCE_CONDITIONAL_AMPLIFIERS
7	APPLY_BIDIRECTIONAL_ELEMENT_IF_RULE_REQUIRES
8	APPLY_TARGET_PERCENT_REDUCTION_UNLESS_DAMAGE_FORM_BYPASSES
9	APPLY_TARGET_FIXED_REDUCTION_UNLESS_DAMAGE_FORM_BYPASSES
10	CLAMP_POSITIVE_NONIMMUNE_DAMAGE_TO_MINIMUM_ONE
11	ABSORB_SHIELDS_NEWEST_FIRST
12	COMMIT_INTEGER_HP_DAMAGE
13	RUN_POST_DAMAGE_AND_LETHAL_TRANSACTION
14	APPEND_LEDGER_REPLAY_ANALYTICS

5.3 伤害类型

类型	暴击	五行	比例减伤	固定减伤	护盾	无敌
BASIC	是	是	是	是	是	是
SKILL	否	是	是	是	是	是
DOT	否	是	是	否	是	是
PERCENT	否	否	是	否	是	是

类型	暴击	五行	比例减伤	固定减伤	护盾	无敌
REFLECT	否	否	是	否	是	是
TRUE	否	否	否	否	是	是
STORM	否	否	否	否	是	是

5.4 控制与强制位移

项目	规则
hard_control_same_kind	KEEP_LATER_END_TIME; DO_NOT_ADD_DURATIONS
different_hard_controls	COEXIST; EFFECTIVE_RESTRICTIONS_ARE_UNION
slow	ONLY_STRONGEST_PERCENT_APPLIES; DURATION_TRACKED_PER_SOURCE
damage_reduction	MULTIPLICATIVE_REMAINING_UNLESS_EXPLICIT_FIXED
silence	BLOCK_ACTIVE_CAST_START; DOES_NOT_CANCEL_ALREADY_COMMITTED_INSTANT_CAST
root	BLOCK_TRANSLATION_AND_DASH; BASIC_AND_ACTIVE_CASTS_STILL_ALLOWED_UNLESS ALSO_SILENCED
invulnerable	DAMAGE_ZERO; TARGETABILITY_PRESERVED_UNLESS SKILL SAYS OTHERWISE
forced_movement_into_solid	STOP_AT_LAST_LEGAL_CAPSULE_POSITION
hard_control_taxonomy	hard_reducible=STUN / FREEZE / PETRIFY / ROOT / FEAR / TRANSFORM / KNOCKUP; soft_not_reduced_by_P18=SLOW / SILENCE; immunity_and_cleanse_are_not_duration_resistance=是
target_duration_resistance	SAME_CATEGORY_TAKE_STRONGEST_LOWEST_MULTIPLIER; NEVER_MULTIPLY
duration_resolution_order	IF_IMMUNE_OR_CLEANSSED_THEN_ZERO_ELSE_MAX(EFFECT_MINIMUM,(BASE_PLUS_SOURCE_FLAT_EXTENSION)*TARGET_STRONGEST_MULTIPLIER)
forced_movement_resistance	SAME_CATEGORY_TAKE_STRONGEST_LOWEST_MULTIPLIER; B4_AND_P18_NEVER_MULTIPLY
control_application_snapshot	SOURCE_EXTENSION_AND_TARGET_RESISTANCE_SNAPSHOT_AT_APPLICATION; UNEQUIP_DOES_NOT_RETROACTIVELY_CHANGE_RUNNING_CONTROL

5.5 战争迷雾、伏石与揭露

信息态	可锁定	表现与边界
UNKNOWN	否	无世界或小地图敌人位置
VISIBLE	是	直接视线+视野范围；世界渲染并可合法锁定
DIRECTION	否	只给模糊区域/方向
POSITION_SNAPSHOT	否	记录快照 tick 的小地图点，不跟随
REVEALED	是	持续精确位置；仍需满足技能射程与墙体规则

- 玩家对玩家造成实际生命或护盾伤害时，攻击者向该受击者暴露精确小地图位置 3 秒；只提供小地图位置，不提供隔墙锁定，持续时间可刷新但复制频率最多 1Hz。

- 伏石为石体周围 2 米非方向环带：圈外看不见圈内，同一环带互见；普攻释放、敌对主动提交或承受实际生命/护盾伤害会破隐；拾取不破隐。
- 伏石、主动隐身、装备隐身都属于 STEALTH；普通战争迷雾与烟雾遮挡不同隐身。隐身不能被直接锁定，但实体碰撞和盲放范围技能仍有效。
- 照妖镜 20 米/3 秒、照妖珠 20 米常驻、核心 BOSS 真视眼 10 米可揭露；墙体仍阻挡揭露与锁定。
- 核心 BOSS 真视眼共享给 10 米内所有玩家；BOSS 死亡后在死亡点保留 10 秒。
- 灭世雷暴不会强制显形，也不会禁用伏石、隐身、治疗或护盾。

6 主动技能（57 项）

- 每名英雄出生自带 1 个专属主动；地图另有 19 个通用主动。任意时刻仅持有 1 个主动。
- 通用主动落地后公开争夺；替换需确认，新主动从完整 CD 开始；太白换英雄得到新专属时 READY；前两次真死复活后当前主动 READY。
- 所有 57 个主动的 level_scaling=NONE；运行时只读取工程参数，绝不解析展示文案。
- 施法非法时沿同方向缩短至最近合法点；仍非法则不施放、不进入 CD。动态墙体及所有通行合法性只由真实碰撞几何和持续时间决定。

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H001	芭蕉风墙	专属	H001	18	target=DIRECTION; cast=INSTANT; range=15m; shape=LINE_WALL; radius=0m; width=12m; duration=3s	type=WIND_WALL; length_m=12; duration_s=3; knockback_m=5; solid=否; blocks_projectile_tag=BLOCKABLE_BY_WIND_WALL

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H002	三昧火墙	专属	H002	24	target=DIRECTION; cast=INSTANT; range=15m; shape=LINE_WALL; radius=0m; width=15m; duration=5s	type=FIRE_WALL; length_m=15; thickness_m=1; duration_s=5; cross_damage=fixed=100; attack_coef=0.4; burn_damage_per_s=30; burn_duration_s=3; burn_refresh=是; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; owner_credit=CASTER; cross_damage_form=SKILL; cross_hit_kind=ABILITY; burn_damage_form=DOT; burn_hit_kind=PERIODIC; cross_trigger=FULL_OUTSIDE_TO_INSIDE_OR_INSIDE_TO_OUTSIDE; cross_reentry_lock_per_target_s=0.5; standing_overlap_does_not_retrigger=是
H003	盘丝阵	专属	H003	36	target=GROUND_POINT; cast=INSTANT; range=30m; shape=CIRCLE; radius=8m; width=0m; duration=4s	type=DAMAGE_SLOW_ZONE; radius_m=8; duration_s=4; tick_s=1; damage=fixed=40; attack_coef=0.2; slow_percent=25; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=PERIODIC; owner_credit=CASTER

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H004	倒马毒	专属	H004	36	target=VISIBLE_ENEMY; cast=INSTANT; range=20m; shape=NONE; radius=0m; width=0m; duration=4s	type=CONDITIONAL_POISON_DETONATE; damage_per_stack_per_s=5; duration_s=4; damage_target_max_hp_percent_per_stack=3; max_count=10; stack_scope=CASTER_OWNED_DOT_AND_TOXIN_ONLY; when_no_owned_dot_add_stacks=3; when_owned_dot_consume_all_owned_dot_and_toxin_stacks=是; other_owner_stacks_unchanged=是; stack_dot_damage_form=DOT; stack_dot_hit_kind=PERIODIC; detonation_damage_form=PERCENT; detonation_hit_kind=ABILITY
H005	金光	专属	H005	30	target=VISIBLE_ENEMY; cast=INSTANT; range=20m; shape=NONE; radius=0m; width=0m; duration=1.5s	type=PETRIFY_TARGET; duration_s=1.5; target_invulnerable=是; target_cannot_act=是

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H006	毒雾	专属	H006	55	target=GROUND_POINT; cast=INSTANT; range=30m; shape=CIRCLE; radius=8m; width=0m; duration=8s	type=CONTAGIOUS_DAMAGE_ZONE; radius_m=8; duration_s=8; tick_s=1; damage=fixed=40; attack_coef=0.2; outside_residual_s=3; infection_radius_m=2; check_interval_s=0.5; max_generation=2; stack=否; refresh=是; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=PERIODIC; owner_credit=CASTER; infection_instance_scope=PER_CASTER_PER_ROOT_EFFECT; generation_origin=ROOT_EFFECT_INSTANCE_ID; same_caster_reinfection=REFRESH_DURATION_NO_STACK; different_caster_infection=INDEPENDENT
H007	三昧神风	专属	H007	50	target=GROUND_POINT; cast=INSTANT; range=30m; shape=CIRCLE; radius=10m; width=0m; duration=4s	type=DAMAGE_SLOW_ZONE; radius_m=10; duration_s=4; tick_s=1; damage=fixed=60; attack_coef=0.3; slow_percent=40; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=PERIODIC; owner_credit=CASTER

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H008	八卦炉火	专属	H008	62	target=GROUND_POINT; cast=INSTANT; range=30m; shape=CIRCLE; radius=12m; width=0m; duration=0s	type=DELAYED_AREA_DAMAGE; radius_m=12; telegraph_s=1.5; damage=fixed=400; attack_coef=1.2; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER
H009	大闹天宫	专属	H009	40	target=SELF; cast=INSTANT; range=0m; shape=NONE; radius=0m; width=0m; duration=5s	type=SELF_COMBAT_BUFF; duration_s=5; attack_speed_add_percent=80; passive_effect_magnitude_multiplier=2; passive_proc_probability_multiplier=1; incoming_damage_multiplier=1.2; passive_magnitude_authority=H009_EXACT_FIELD_WHITELIST; excluded_field_classes=PROC_CHANCE / DURATION / COOLDOWN / COUNT / ECONOMY / PERMANENT_GROWTH / LETHAL_PROTECTION / STORM_PROBABILITY; transform_additive=VALUE*2; transform_delta_from_one=1+(VALUE-1)*2; transform_remaining_from_one=MAX(0,1-(1-VALUE)*2)

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H010	天眼	专属	H010	25	target=SELF; cast=INSTANT; range=0m; shape=NONE; radius=0m; width=0m; duration=0s	type=ARM_NEXT_BASIC; consume_on=VALID_BASIC_HIT_RESOLUTION; guaranteed_crit=是; extra_target_missing_hp_percent=15; kill_resets_active_cd=是; consume_if_damage_zero_due_to_dodge_or_immunity=是; do_not_consume_if_no_hit_entity=是; kill_reset_timing=AFTER_LETHAL_COMMIT; ; arm_persistence=UNTIL_NEXT_LEGAL_BASIC_NO_TIMEOUT; cleared_on=TRUE_DEATH / ACTIVE_REPLACED / HERO_SWAP
H011	风火轮	专属	H011	18	target=DIRECTION; cast=INSTANT; range=25m; shape=LINE; radius=0m; width=3m; duration=0s	type=DASH_STOP_FIRST; distance_m=25; width_m=3; damage=fixed=100; attack_coef=0.5; stun_s=0.8; interruptible_by_hard_control=是; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H012	六耳幻身	专属	H012	42	target=SELF; cast=INSTANT; range=0m; shape=NONE; radius=0m; width=0m; duration=8s	type=SPAWN_DECOYS; count=2; duration_s=8; hp_inherit_percent=30; move_speed_inherit_percent=30; attack=0; unit_collision=否; copy_owner_visual_and_hp_percent=是; decoy_max_hp=FLOOR(OWNER_CURRENT_MAX_HP*0.3); decoy_current_hp=FLOOR(DECOY_MAX_HP*OWNER_CURRENT_HP/OWNER_CURRENT_MAX_HP); decoy_death_credit=NONE; decoy_gold_xp_drop=NONE
H013	金翅俯击	专属	H013	20	target=VISIBLE_ENEMY; cast=INSTANT; range=40m; shape=NONE; radius=0m; width=0m; duration=0s	type=TARGET_BEHIND_BLINK_AND_ARM_CRIT; range_m=40; behind_offset_m=1.5; arm_next_basic_crit=是; candidate_directions=REAR_THEN_CLOCKWISE_45_DEG_EIGHT_POINTS; candidate_radius_m=1.5; candidate_query=PLAYER_CAPSULE_SWEEP_AND_DESTINATION_LEGALITY; no_candidate=FAIL_NO_COOLDOWN
H014	借尸还魂	专属	H014	35	target=DIRECTION; cast=INSTANT; range=15m; shape=POINT; radius=0m; width=0m; duration=1s	type=BLINK_LEAVE_DELAYED_DECOY; distance_m=15; delay_s=1; damage=fixed=120; attack_coef=0.5; blink_uses_capsule_sweep=是; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H015	九齿钉耙	专属	H015	38	target=DIRECTION; cast=INSTANT; range=8m; shape=CONE; radius=8m; width=0m; duration=2s	type=CONE_DAMAGE_SLOW; range_m=8; angle_deg=60; damage=fixed=160; attack_coef=0.7; slow_percent=30; slow_s=2; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER
H016	龙马奔腾	专属	H016	18	target=DIRECTION; cast=INSTANT; range=25m; shape=LINE; radius=0m; width=3m; duration=0s	type=PIERCING_DASH; distance_m=25; width_m=3; damage=fixed=100; attack_coef=0.5; knockback_m=2; hit_each_target_once=是; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER
H017	狮吼	专属	H017	60	target=SELF; cast=INSTANT; range=0m; shape=CIRCLE; radius=12m; width=0m; duration=0s	type=RADIAL_DAMAGE_KNOCK_TO_EDGE; ; radius_m=12; damage=fixed=150; attack_coef=0.7; stun_s=1; blocked_knock_still_stuns=是; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER; knock_direction=CASTER_TO_TARGET_RADIAL; knock_destination=SKILL_EDGE_OR_LAST_LEGAL_SWEEP_POINT; actor_collision=STOP_LAST_LEGAL_POINT

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H018	旋风	专属	H018	38	target=SELF; cast=INSTANT; range=0m; shape=CIRCLE; radius=8m; width=0m; duration=3s	type=MOBILE_CHANNEL_AOE; duration_s=3; tick_s=1; radius_m=8; damage_per_tick=fixed=100; attack_coef=0.5; self_move_multiplier=0.5; hard_control_interrupts=是; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=PERIODIC; owner_credit=CASTER
H019	金刚琢	专属	H019	45	target=GROUND_POINT; cast=INSTANT; range=25m; shape=CIRCLE; radius=10m; width=0m; duration=3s	type=DELAYED_SILENCE_ZONE; range_m=25; radius_m=10; telegraph_s=0.4; silence_s=3
H020	石罗汉	专属	H020	54	target=SELF_FORWARD; cast=INSTANT; range=3m; shape=POINT; radius=0m; width=0m; duration=20s	type=SPAWN_SUMMON; archetype=STONE_ARHAT; count=1; duration_s=20; hp=500; attack=30

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H021	紫金葫芦	专属	H021	56	target=SELF; cast=CHANNEL; range=0m; shape=CIRCLE; radius=15m; width=0m; duration=0s	type=CHARGED_RADIAL_PULL; charge_s=1; radius_m=15; max_targets=8; damage=fixed=120; attack_coef=0.6; landing_ring_m=2; wall_blocks=是; immune_tiers=LEADER/ DRAGON / CORE_BOSS; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARG ET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER; max_target_selection_order=THREAT_DES C / DISTANCE_MM_ASC / ENTITY_ID_ASC; landing_slot_layout=EIGHT_DISTINCT_RIN G_SLOTS_AROUND_CASTER; landing_slot_order=CLOCKWISE_FROM_CA STER_FORWARD; occupied_slot=NEXT_LEGAL_SLOT_ELSE_L AST_LEGAL_SWEEP_POINT
H022	砸钱	专属	H022	24	target=VISIBLE_ENEMY; cast=INSTANT; range=20m; shape=NONE; radius=0m; width=0m; duration=0s	type=SPEND_GOLD_TRUE_DAMAGE; minimum_balance_to_cast=500; cost=FLOOR(CLAMP(CURRENT_GOLD*0.20, 500,5000)); damage=FLOOR(COST*0.10); range_m=20; shieldable=是; atomic_deduct_and_cast=是; damage_form=TRUE; source_faction=PLAYER_OWNED; element_rule=NONE; hit_kind=ABILITY; owner_credit=CASTER

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H023	缠身	专属	H023	45	target=VISIBLE_ENEMY; cast=INSTANT; range=30m; shape=NONE; radius=0m; width=0m; duration=8s	type=TARGET_DOT_REVEAL; duration_s=8; tick_s=2; damage_per_tick=40; reveal=EXACT_WORLD_AND_MINIMAP; clear_on_caster_true_death=是; damage_form=DOT; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=PERIODIC; owner_credit=CASTER
H024	虎箭	专属	H024	35	target=DIRECTION; cast=INSTANT; range=1040m; shape=LINE; radius=0m; width=1.5m; duration=0s	type=GLOBAL_LINE_PROJECTILE; projectile_speed_mps=45; max_lifetime_s=24; damage=fixed=150; attack_coef=1; distance_bonus_percent_per_10m=5; distance_bonus_cap_percent=50; blocked_by=STATIC_SOLID / WIND_WALL / FROZEN_RIVER; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER; distance_bonus_origin=PROJECTILE_RELEASE_POSITION; distance_bonus_measure=ACCUMULATED_PROJECTILE_TRAVEL_DISTANCE_TO_HIT_POINT

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H025	鹿血	专属	H025	45	target=SELF; cast=INSTANT; range=0m; shape=CIRCLE; radius=5m; width=0m; duration=6s	type=BASIC_LIFESTEAL_AND_DAMAGE_AURA; duration_s=6; basic_lifesteal_add_percent=50; radius_m=5; tick_s=1; damage_per_tick=30; aura_heal_equals_actual_hp_damage=是; shield_damage_does_not_heal=是; damage_form=DOT; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=PERIODIC; owner_credit=CASTER
H026	定慧诀	专属	H026	30	target=VISIBLE_ENEMY; cast=INSTANT; range=20m; shape=NONE; radius=0m; width=0m; duration=2s	type=TARGET_DAMAGE_HARD_CONTROL; range_m=20; damage=fixed=60; attack_coef=0.3; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER; control_type=STUN; control_duration_s=2; control_uses_normal_tenacity=是
H027	行愿	专属	H027	42	target=VISIBLE_ANY_HERO_OR_SELF; cast=INSTANT; range=50m; shape=NONE; radius=0m; width=0m; duration=0s	type=VISIBLE_ANY_HERO_HEAL; range_m=50; heal=220; faction_filter=ANY_HERO_INCLUDING_ENEMY_OR_SELF; quick_cast_no_target=SELF

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H028	袖里乾坤	专属	H028	62	target=VISIBLE_ENEMY; cast=INSTANT; range=30m; shape=NONE; radius=0m; width=0m; duration=8s	type=TARGET_DAMAGE_AMP_MARK; range_m=30; duration_s=8; caster_damage_multiplier=1.3; on_credited_true_death_transfer=NEAREST_VISIBLE_ENEMY_WITHIN_20M; transfer_refresh_s=8; no_legal_transfer=END; transfer_distance_metric=DIRECT_WORLD_DISTANCE_MM; equal_distance_tie_break=ENTITY_ID_ASC; caster_true_death=CLEAR_MARK
H029	五行山	专属	H029	70	target=GROUND_POINT; cast=INSTANT; range=30m; shape=CIRCLE; radius=8m; width=0m; duration=0s	type=DELAYED_AREA_DAMAGE_SLOW; range_m=30; radius_m=8; telegraph_s=1.2; damage=fixed=300; attack_coef=1; slow_percent=40; slow_s=2; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER
H030	甘露	专属	H030	50	target=GROUND_POINT_OR_SELF; cast=INSTANT; range=30m; shape=CIRCLE; radius=8m; width=0m; duration=4s	type=ANY_HERO_HEAL_ZONE; range_m=30; radius_m=8; duration_s=4; tick_s=1; heal_per_tick=40; faction_filter=ANY_HERO
H031	宝塔	专属	H031	100	target=SELF; cast=INSTANT; range=0m; shape=RING_WALL; radius=10m; width=0m; duration=5s	type=PAGODA_RING_SOLID; radius_m=10; duration_s=5; telegraph_s=0.6; wall_thickness_m=1.6; height_m=4; segments=32; destructible=否; instance_limit=1
H032	袈裟	专属	H032	95	target=SELF; cast=INSTANT; range=0m; shape=NONE; radius=0m; width=0m; duration=3s	type=MOBILE_INVULNERABILITY; duration_s=3; can_move=是; can_basic=是; can_cast=是; world_interactions_during_invulnerability=FORBID_PICKUP_SHOP_CHEST_GAMBLE_TAIBAI; combat_actions_remain_as_DECLARED=是

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H033	流沙	专属	H033	50	target=GROUND_POINT; cast=INSTANT; range=30m; shape=CIRCLE; radius=8m; width=0m; duration=4s	type=DAMAGE_SLOW_ANTIMOBILITY_ZONE; range_m=30; radius_m=8; duration_s=4; tick_s=1; damage=fixed=50; attack_coef=0.25; slow_percent=60; blocks_mobility_cast_start=是; does_not_cancel_committed_mobility=是; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=PERIODIC; owner_credit=CASTER; mobility_tag_authority=MobilityTag_CLOSED_ENUM_AND_ACTIVE_REGISTRY; zone_checks=CAST_START_POSITION_INSIDE_VOLUME
H034	黑风罩	专属	H034	58	target=SELF; cast=INSTANT; range=0m; shape=NONE; radius=0m; width=0m; duration=4s	type=SELF_REDUCTION_SPEED_BUFF; duration_s=4; incoming_non_true_remaining_multiplier=0.2; move_speed_add_percent=15
H035	象鼻缚	专属	H035	28	target=VISIBLE_ENEMY; cast=INSTANT; range=60m; shape=PROJECTILE; radius=0m; width=1m; duration=2s	type=PROJECTILE_ROOT; range_m=60; projectile_speed_mps=30; root_s=2; root_allows_basic_and_active=是; blocked_by=STATIC_SOLID / WIND_WALL / FROZEN_RIVER; projectile_radius_m=0.5; homing=否; collision_query=3D_CAPSULE_SWEEP
H036	冻河	专属	H036	34	target=DIRECTION; cast=INSTANT; range=15m; shape=LINE_WALL; radius=0m; width=10m; duration=40s	type=DESTRUCTIBLE_ICE_WALL; range_m=15; length_m=10; thickness_m=1; height_m=2.5; hp=300; duration_s=40; instance_limit=1; new_replaces_old=是

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
H037	羊灵之光	专属	H037	35	target=VISIBLE_ENEMY_OR_SELF; cast=INSTANT; range=20m; shape=NONE; radius=0m; width=0m; duration=1.5s	type=DUAL_PETRIFY; range_m=20; duration_s=1.5; enemy_invulnerable=是; enemy_cannot_act=是; self_invulnerable=是; self_cannot_act=是; cooldowns_and_buffs_continue=是; dot_ticks_attempt_and_resolve_zero_due_to_invulnerability=是; timers_do_not_pause=是
H038	迷烟	专属	H038	28	target=GROUND_POINT; cast=INSTANT; range=30m; shape=CIRCLE; radius=8m; width=0m; duration=4s	type=SMOKE_VISION_ZONE; range_m=30; radius_m=8; duration_s=4; enemy_inside_direct_vision_cap_m=2; blocks_vision_across_boundary=是; caster_vision_unchanged=是; true_sight_overrides=是; area_skills_can_blind_cast=是
D1	钩锁	通用	地图通用	28	target=DIRECTION; cast=INSTANT; range=80m; shape=PROJECTILE; radius=0m; width=1m; duration=0s	type=PROJECTILE_PULL; range_m=80; projectile_speed_mps=60; damage=fixed=80; attack_coef=0.4; pull_to_caster_front_m=1.5; blocked_by=STATIC_SOLID / WIND_WALL / FROZEN_RIVER; immune_tiers=DRAGON / CORE_BOSS; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER; pull_destination=NEAREST_LEGAL_POINT_IN_FRONT_OF_CASTER; no_legal_pull_point=DAMAGE_COMMITS_PULL_FAILS

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
D3	雷击	通用	地图通用	24	target=VISIBLE_ENEMY; cast=INSTANT; range=70m; shape=NONE; radius=0m; width=0m; duration=0s	type=LOCKED_DAMAGE_STUN; range_m=70; damage=fixed=140; attack_coef=0.8; stun_s=1; miss=ONLY_IF_TARGET_INVULNERABLE_A T_IMPACT; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARG ET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER; impact_delay_s=0.2; after_lock_revalidation=INVULNERABILITY_ ONLY; range_los_not_rechecked=是
D4	变羊	通用	地图通用	32	target=VISIBLE_ENEMY; cast=INSTANT; range=45m; shape=PROJECTILE; radius=0m; width=1m; duration=2s	type=PROJECTILE_POLYMORPH; range_m=45; projectile_speed_mps=30; duration_s=2; disable_basic=是; disable_active=是; move_speed_multiplier=1.3; preserve_movement_class=是; on_polymorph=INTERRUPT_ACTIVE_WIND UP_AND_CHANNEL; existing_owned_entities=PERSIST_NORMA LLY
D6	闪现	通用	地图通用	15	target=DIRECTION; cast=INSTANT; range=15m; shape=POINT; radius=0m; width=0m; duration=0s	type=CAPSULE_SWEEP_BLINK; distance_m=15; max_continuous_solid_chord_m=1.5; illegal_endpoint=LAST_LEGAL_ALONG_DIRECTION; post_cast_lock_s=0
D7	潜行	通用	地图通用	32	target=SELF; cast=INSTANT; range=0m; shape=NONE; radius=0m; width=0m; duration=5s	type=SELF_STEALTH_SPEED; duration_s=5; move_speed_multiplier=1.3; break_on=BASIC_RELEASE/ ACTIVE_CAST_COMMIT; pickup_does_not_break=是; break_attack_bonus=0

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
D9	箭雨	通用	地图通用	24	target=GROUND_POINT; cast=INSTANT; range=35m; shape=CIRCLE; radius=8m; width=0m; duration=0s	type=AERIAL_DELAYED_AREA_DAMAGE; range_m=35; radius_m=8; telegraph_s=0.8; damage=fixed=130; attack_coef=0.6; wind_wall_blocked=否; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER
D10	电链	通用	地图通用	85	target=VISIBLE_ENEMY; cast=INSTANT; range=30m; shape=NONE; radius=0m; width=0m; duration=0s	type=CHAIN_DAMAGE; initial_range_m=30; chain_range_m=15; max_targets=6; damage=fixed=300; attack_coef=1; each_jump_multiplier=0.85; target_repeat=否; next_target=NEAREST_LEGAL_THEN_ENTITY_ID; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER
D11	轮盘	通用	地图通用	50	target=VISIBLE_ENEMY; cast=INSTANT; range=40m; shape=NONE; radius=0m; width=0m; duration=0s	type=UNIFORM_RANDOM_DAMAGE; range_m=40; integer_inclusive=1/999; rng_stream=ACTIVE_SKILL; crit=否; rejected_cast_consumes_rng=否; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER; can_crit=否; shieldable=是

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
D12	标记	通用	地图通用	70	target=VISIBLE_ENEMY; cast=INSTANT; range=60m; shape=NONE; radius=0m; width=0m; duration=90s	type=TARGET_REVEAL_BOUNTY_MARK; range_m=60; duration_s=90; caster_kill_bonus_created_gold=800; clear_on=DURATION_END / CASTER_TRUE_DEATH / TARGET_TRUE_DEATH / RECAST_REPLACE; exact_reveal_recipient=CASTER_ONLY; other_players_receive_no_information= 是; B41_independent=是; reveal_does_not_grant_through_wall_lock =是; wall_blocked_target=POSITION_INTEL_ON LY
D13	换位	通用	地图通用	62	target=TWO_VISIBLE_UNITS; cast=INSTANT; range=50m; shape=NONE; radius=0m; width=0m; duration=0s	type=ATOMIC_SWAP_TWO_VISIBLE_UNITS ; range_m=50; legal_types=PLAYER / ORDINARY_MONSTER / FLYING_MONSTER / PIG / DAMAGEABLE_SUMMON; immune_types=LEADER / DRAGON / CORE_BOSS; either_destination_illegal=FAIL_NO_CD; selection_state=WAIT_FIRST_TARGET / WAIT_SECOND_TARGET / COMMIT_OR_CANCEL; second_target_timeout_s=3; cancel_or_timeout=NO_COOLDOWN; second_invalid=KEEP_WAITING_UNTIL_TI MEOUT; second_target_same_as_first=INVALID_KE EP_WAITING

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
D14	回溯	通用	地图通用	95	target=SELF; cast=INSTANT; range=0m; shape=NONE; radius=0m; width=0m; duration=0s	type=SELF_REWIND; lookback_s=5; hp_result=MAX(CURRENT_HP,HISTORICAL_HP_CLAMPED_TO_CURRENT_MAX_HP); position_result=HISTORICAL_IF_LEGAL_ELSE_CURRENT; position_illegal_does_not_cancel_hp_restore=是; cooldowns_rewind=否; gold_inventory_status_rewind=否; history_sample_hz=20; history_buffer_s=6; sample_choice=LATEST_VALID_ALIVE_SAMPLE_AT_OR_BEFORE_NOW_MINUS_5S
D15	妙手空空	通用	地图通用	38	target=VISIBLE_MONSTER_OR_ENEMY; cast=INSTANT; range=30m; shape=NONE; radius=0m; width=0m; duration=0s	type=MONSTER_EXECUTE_OR_PLAYER_GOLD_STEAL; range_m=30; execute_types=ORDINARY_MONSTER / FLYING_MONSTER; xp_and_drops=NORMAL_ONCE; immune_types=LEADER / DRAGON / CORE_BOSS / PLAYER_OWNED_SUMMON; player_transfer_gold=500; insufficient=TRANSFER_ALL_REMAINING; never_creates_gold_on_player=是; pig_execute_current_hp_lte_percent=30; execute_base_gold_multiplier=3; gold_multiplier_scope=BASE_MONSTER_GOLD_ONLY; xp_and_drop_multiplier=1; kill_credit=CASTER_NORMAL_MONSTER_KILL; on_kill_triggers_once=EQUIP_B8 / B32 / B33 / B40 / G9; does_not_trigger_basic_hit_chain=是

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
D16	聚财	通用	地图通用	100	target=SELF; cast=INSTANT; range=0m; shape=NONE; radius=0m; width=0m; duration=0s	type=CREATE_GOLD; amount=450; uses_per_match=UNLIMITED; ledger=GOLD_CREATE_ACTIVE_D16; true_death_ready_rule=ACTIVE_READY_BUT_MATCH_GOLD_LEDGER_REMAINS; economy_sim_source=GOLD_CREATE_ACTIVE_D16; uses_per_life_not_capped=是
D17	连赏	通用	地图通用	55	target=SELF; cast=INSTANT; range=0m; shape=NONE; radius=0m; width=0m; duration=0s	type=TOGGLE_PLAYER_BOUNTY; toggle_cd_s=55; base_bounty=500; on_each_credited_player_true_death=BOUNTY=MIN(500*2^COUNT,32000); cap=32000; player_killer_gets_created_gold_percent=50; monster_final_blow=DESTROY_WITHOUT_PAYOUT; manual_off=RESET_AND_OFF; true_death=PAY_IF_PLAYER_KILLER_THEN_RESET_AND_OFF; numeric_storage=SIGNED_INT64_SATURATING; head_marker_shows_capped_value=是
D18	淘金狂热	通用	地图通用	16	target=SELF; cast=INSTANT; range=0m; shape=CIRCLE; radius=80m; width=0m; duration=20s	type=SCAN_OLD_EQUIPMENT; radius_m=80; minimum_ground_age_s=10; mark_duration_s=20; snapshot_at_cast=是; newly_aging_items_not_added=是; scan_ignores_fog_and_walls=是; recipient=CASTER_ONLY; marker_reveals_owner_or_source=否
D19	豆兵	通用	地图通用	50	target=SELF_FORWARD; cast=INSTANT; range=3m; shape=POINT; radius=0m; width=0m; duration=15s	type=SPAWN_SUMMON; archetype=BEAN_SOLDIER; count=3; duration_s=15; hp=150; attack=15; attack_attribute_inherit=0; triggers_owner_passives=否

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
D20	陷阱	通用	地图通用	16	target=GROUND_POINT_OR_OWN_T RAP; cast=INSTANT; range=3m; shape=CIRCLE; radius=6m; width=0m; duration=90s	type=VISIBLE_TRAP_OR_REMOTE_DETONA TE; placement_range_m=3; duration_s=90; instance_limit=3; fourth_removes_oldest=是; arming_s=0.5; body_blocks=否; trigger_radius_m=0.75; trigger_damage=fixed=180; trigger_attack_coef=0.8; root_s=1.5; reveal_s=8; remote_range=GLOBAL; remote_damage=fixed=100; remote_attack_coef=0.6; remote_radius_m=6; damage_form=SKILL; source_faction=PLAYER_OWNED; element_rule=OWNER_ELEMENT_IF_TARG ET_HAS_ELEMENT; hit_kind=ABILITY; owner_credit=CASTER; all_capsule_trajectory_cross_triggers=是; exceptions=INVULNERABLE / PHYSICALLY_ABOVE_TRIGGER_VOLUME; remote_selection_tap=NEAREST_TRAP_TO _AIM_VECTOR; remote_selection_hold=OPEN_OWN_TRAP _SELECTOR; remote_selection_tie=TRAP_INSTANCE_ID _ASC; trap_hp=250; destructible_by_enemy=是; destroy_reward=NONE; destroy_explosion=否; blocks_movement=否; no_own_trap_press=PLACEMENT_MODE
D21	冰棺	通用	地图通用	75	target=SELF; cast=INSTANT; range=0m; shape=NONE; radius=0m; width=0m; duration=4s	type=SELF_LOCK_INVULNERABILITY; duration_s=4; can_move=否; can_basic=否; can_cast=否

ID	名称	类型	归属	CD(s)	瞄准与几何	完整机制
D22	铁布衫	通用	地图通用	45	target=SELF; cast=INSTANT; range=0m; shape=NONE; radius=0m; width=0m; duration=5s	type=SELF_SHIELD; amount=600; duration_s=5; absorbs=BASIC / SKILL / DOT / PERCENT / REFLECT / TRUE / STORM; shield_absorption_order=EARLIEST_EXPIRY _TICK_FIRST / CREATION_SEQUENCE_ASC; same_source_new_cast=INDEPENDENT_IN STANCE

7 被动技能（44 项）

- 每名玩家最多 4 个被动槽；技能书空槽直接学习 1 级，同名未满 5 级则消耗书升 1 级。
- 同名 5 级书不可拾取；四槽满时拾取不同名书必须明确选择被替换槽，取消则书留在地面公开。
- 技能宝石进入 HUD 宝石仓，不占手牌且不能主动丢地；可自由分配给未满级被动；全部被动 5 级时拒绝拾取或购买。
- 真死保留被动与等级，但重置内置 CD 并清临时状态；替换被动时清其实体、未来计时器和永久层，已独立施加的 DoT 按合同继续。

ID	名称	类别	触发	玩家说明	工程数值(L1-L5)
B01	冰霜	控制类	BASIC_HIT_PROC	普攻有 10%/13%/16%/20%/25%概率使 目标减速 30%/40%/50%/60%/80%，持续 2/2/2.5/2.5/3 秒。5 级触发时还 使目标周围 5 米敌人减速 35%。	chance=10 / 13 / 16 / 20 / 25; slow_percent=30 / 40 / 50 / 60 / 80; duration_s=2 / 2 / 2.5 / 2.5 / 3; l5_aoe_radius_m=5; l5_aoe_slow_percent=35
B02	麻痹	控制类	BASIC_HIT_PROC	普攻有 8%/10%/12%/15%/18% 概率使目标沉默 0.8/1/1.2/1.4/1.5 秒。5 级沉默 结束时，目标主动技能剩余 CD 增加 3 秒。	chance=8 / 10 / 12 / 15 / 18; silence_s=0.8 / 1 / 1.2 / 1.4 / 1.5; l5_on_end_active_cd_add_s=3

ID	名称	类别	触发	玩家说明	工程数值(L1-L5)
B03	击退	控制类	BASIC_HIT_PROC	普攻有 20%/23%/26%/30%/35%概率击退目标 3/3.5/4/4.5/5 米。5 级暴击触发时还击退目标周围 5 米敌人 3 米。	chance=20 / 23 / 26 / 30 / 35; knockback_m=3 / 3.5 / 4 / 4.5 / 5; l5_on_critical_aoe_radius_m=5; l5_aoe_knockback_m=3
B04	致盲	控制类	BASIC_HIT_PROC	普攻有 10%/12%/14%/17%/20%概率使目标致盲，持续 2/2/2.5/2.5/3 秒；致盲期间普攻有 50%/60%/70%/80%/80%概率落空。5 级同时禁止暴击。	chance=10 / 12 / 14 / 17 / 20; basic_miss_percent=50 / 60 / 70 / 80 / 80; duration_s=2 / 2 / 2.5 / 2.5 / 3; l5_disable_critical=是
B05	眩晕	控制类	BASIC_HIT_PROC	普攻有 5%/6%/7%/8%/10%概率眩晕目标 0.4/0.5/0.6/0.7/0.8 秒；对同一目标内置 CD 2 秒。5 级还眩晕周围 3 米敌人 0.3 秒。	chance=5 / 6 / 7 / 8 / 10; stun_s=0.4 / 0.5 / 0.6 / 0.7 / 0.8; internal_cd_s=2; internal_cd_scope=PER_TARGET; l5_aoe_radius_m=3; l5_aoe_stun_s=0.3
B06	暴击	伤害类	BASIC_HIT_PROC	普攻有 8%/10%/13%/16%/20%概率暴击，暴击伤害为 180%/190%/200%/210%/230%。5 级暴击伤害的 30%绕过护盾直接扣生命。	chance=8 / 10 / 13 / 16 / 20; critical_damage_percent=180 / 190 / 200 / 210 / 230; l5_ignore_shield_percent=30; damage_event_model=ONE_PARENT_WITH_SHIELD_CHILD_AND_HP_CHILD; proc_and_lifesteal_count=ONCE_PER_PARENT; l5_split_rounding=TOTAL_CRIT_FIRST_HP_CHILD_FLOOR_REMAINDER_TO_SHIELD_CHILD

ID	名称	类别	触发	玩家说明	工程数值(L1-L5)
B07	溅射	伤害类	BASIC_HIT_PROC	普攻有 10%/13%/16%/20%/25%概率对 目标周围 3 米造成主伤害 40%/45%/50%/55%/60%的溅 射。5 级半径×1.5 且主目标也 承受一次溅射；溅射不能再次触 发溅射。	chance=10 / 13 / 16 / 20 / 25; splash_damage_percent=40 / 45 / 50 / 55 / 60; base_radius_m=3; l5_radius_multiplier=1.5; l5_main_target_receives_splash=是; splash_cannot_splash=是
B08	燃灰	伤害类	BASIC_HIT_STACK	普攻给目标叠加燃灰；达到 5/5/5/4/4 层时造成其已损生命 8%/10%/12%/12%/15%的伤害 并清层。5 级触发后给周围 5 米敌人各施加 2 层。	stacks_required=5 / 5 / 5 / 4 / 4; lost_hp_damage_percent=8 / 10 / 12 / 12 / 15; damage_form=PERCENT; l5_spread_stacks=2; spread_radius_m=5
B09	毒素	伤害类	BASIC_HIT_DOT	普攻给目标叠毒，每层每秒 3/4/5/6/8 伤害，最多 3/4/4/5/5 层，持续 3/3/4/4/4 秒并刷新。5 级满层时毒伤 ×1.5。	damage_per_stack_s=3 / 4 / 5 / 6 / 8; max_stacks=3 / 4 / 4 / 5 / 5; duration_s=3 / 3 / 4 / 4 / 4; refresh_duration=是; l5_full_stack_damage_multiplier=1.5
B10	斩杀	伤害类	DIRECT_DAMAGE_CON DITIONAL	1 级 生命低于 20% 伤害+25% 2 级 22% +30% 3 级 25% +35% 4 级 28% +40% 5 级 30% +50%。质变：斩杀击杀时回复自身 10%最大生命	hp_threshold_percent=20 / 22 / 25 / 28 / 30; damage_multiplier=1.25 / 1.3 / 1.35 / 1.4 / 1.5; l5_kill_heal_max_hp_percent=10

ID	名称	类别	触发	玩家说明	工程数值(L1-L5)
B11	连击	伤害类	BASIC_HIT_PROC	1 级 8% 追加 1 次普攻 2 级 10% 追加 1 次 3 级 12% 最多 2 次 4 级 15% 最多 2 次 5 级 20% 最多 3 次。质变：连击最后一刀 50%概率触发一个其他被动（从已装备其他 3 个被动中随机选一个，无视该被动原概率，无论只触发一次）。额外普攻不能再次触发连击；没有其他已拥有被动时，5 级附加触发不发生	chance=8 / 10 / 12 / 15 / 20; max_extra_basics=1 / 1 / 2 / 2 / 3; extra_basic_may_trigger_other_passives=是; cannot_trigger_self=是; l5_last_hit_other_passive_chance=50; l5_other_passive_selection=UNIFORM_FROM_OTHER_THREE; TRIGGER_ONCE_IGNORE_PROC_CHANCE; extra_basic_cannot_trigger_B11=是; l5_candidate_pool=B11_CLOSED_POOL_INTERSECT_OWNED_OTHERS; no_candidate=NO_EXTRA_PASSIVE_TRIGGER
B12	狼灵	伤害类	BASIC_HIT_PROC	1 级 6% 召唤狼灵 15 秒 150 血 20 攻 2 级 8% 200/25 3 级 10% 250/30 4 级 12% 300/35 5 级 15% 350/40。质变：最多同时 2 只	chance=6 / 8 / 10 / 12 / 15; summon_hp=150 / 200 / 250 / 300 / 350; summon_attack=20 / 25 / 30 / 35 / 40; duration_s=15; max_instances=1 / 1 / 1 / 1 / 2
B13	火灵	伤害类	BASIC_HIT_PROC	1 级 6% 召唤火灵环绕 8 秒触碰 60 伤+点燃 3 秒（每秒 6 点） 2 级 8% 80 伤 3 级 10% 100 伤 4 级 12% 120 伤 5 级 15% 150 伤。质变：最多 2 个火灵	chance=6 / 8 / 10 / 12 / 15; contact_damage=60 / 80 / 100 / 120 / 150; duration_s=8; contact_internal_cd_s=1; contact_internal_cd_scope=PER_TARGET; burn_duration_s=3; max_instances=1 / 1 / 1 / 1 / 2; burn_damage_per_s=6 / 8 / 10 / 12 / 15

ID	名称	类别	触发	玩家说明	工程数值(L1-L5)
B14	冷箭	伤害类	BASIC_HIT_PROC	1 级 10% 自动射一箭 60 伤 2 级 12% 80 伤 3 级 14% 100 伤 4 级 16% 120 伤 5 级 20% 150 伤。质变：冷箭可暴击享受暴击加成	chance=10/12/14/16/20; damage=60/80/100/120/150; projectile_speed_mps=30; l5_can_crit=是
B15	闪避	生存类	INCOMING_BASIC_PRE_HIT_PROC	1 级 10% 躲避普攻 2 级 13% 3 级 16% 4 级 20% 5 级 25%。质变：成功闪避后 2 秒内移速+30%	chance=10/13/16/20/25; result=DODGE_ZERO_DAMAGE; l5_move_speed_percent=30; l5_duration_s=2
B16	铁皮	生存类	INCOMING_BASIC_FIXED_REDUCTION	1 级 减 6 点普攻伤害 2 级 减 10 3 级 减 14 4 级 减 18 5 级 减 25。质变：10%概率完全格挡一次普攻	reduction=6/10/14/18/25; minimum_final_damage=1; l5_full_block_chance=10; l5_block_order=AFTER_B15_BEFORE_B17
B17	护盾	生存类	INCOMING_BASIC_PROCC_PRE_DAMAGE	1 级 8% 护盾吸 60 伤 2 级 10% 80 伤 3 级 13% 100 伤 4 级 16% 130 伤 5 级 20% 160 伤。质变：护盾被打破时对周围造成 100 点 AOE 伤害	chance=8/10/13/16/20; shield=60/80/100/130/160; duration_s=10; absorbs_triggering_hit=是; l5_break_aoe_damage=100; l5_break_aoe_radius_m=3
B18	嗜血	生存类	MISSING_HP_SCALING	1 级 每损 10%生命攻击+3% 2 级 +4% 3 级 +5% 4 级 +6% 5 级 +8%。质变：血量低于 30%时额外获得 10%吸血	attack_percent_per_missing_10pct_hp=3/4/5/6/8; step_rounding=FLOOR_MISSING_PERCENT/10; l5_below_hp_percent=30; l5_lifesteal_percent=10
B19	假死	生存类	LETHAL_PROC	1 级 5% 致死免死回 10%血 2 级 6% 12% 3 级 7% 15% 4 级 8% 18% 5 级 10% 回 20%血。质变：触发后自动向随机方向闪现	chance=5/6/7/8/10; heal_max_hp_percent=10/12/15/18/20; l5_random_blink=是; random_blink_distance_m=8; blink_rng_stream=COMBAT_PROC; blink_direction_draw=SINGLE_DRAW_EIGHT_DIRECTIONS; blink_destination=FARTHEST_LEGAL_POINT_ALONG_DIRECTION; all_points_illegal=NO_BLINK_NO_REDRAW; post_success_retrigger_lock_s=1

ID	名称	类别	触发	玩家说明	工程数值(L1-L5)
B20	复活	生存类	LETHAL_PROTECTION_ONCE_MATCH	1 级 死亡原地复活回 25%血 (每局 1 次) 2 级 30% 3 级 35% 4 级 40% 5 级 50%。质变: 复活后 5 秒内伤害+30%且免控。这次充能属于整局, 顶掉、重新获得或换英雄都不会恢复	heal_max_hp_percent=25 / 30 / 35 / 40 / 50; charge=1; consume_life=否; l5_buff_s=5; l5_damage_multiplier=1.3; l5_control_immune=是; charge_ledger_key=MATCH_ID / PLAYER_ID / B20; removed_or_reacquired_does_not_restore=是; hero_swap_does_not_restore=是
B21	回复	生存类	OUT_OF_ANY_COMBAT_REGEN	1 级 脱战 5 秒后每秒回 4 血 2 级 6 点 3 级 8 点脱战缩为 4 秒 4 级 10 点 5 级 15 点脱战缩为 3 秒。质变: 脱战后第一次普攻伤害+30%	delay_s=5 / 5 / 4 / 4 / 3; hp_per_s=4 / 6 / 8 / 10 / 15; l5_first_basic_damage_multiplier=1.3; first_basic_consumes_bonus=是
B22	反伤	反击类	POST_INCOMING_BASIC_REFLECT	1 级 反弹所受普攻 10% 2 级 13% 3 级 16% 4 级 20% 5 级 25%。质变: 反弹伤害可暴击	actual_hp_plus_shield_damage_percent=10 / 13 / 16 / 20 / 25; cannot_trigger_reflect=是; l5_can_crit=是; l5_crit_uses_owner_crit_snapshot_without_triggering_proc=是
B23	反击	反击类	INCOMING_ACTUAL_DAMAGE_PROC	1 级 6% 受击自动反击 80%伤害 2 级 8% 85% 3 级 10% 90% 4 级 13% 95% 5 级 16% 100%。质变: 反击必定暴击。只对直接伤害反击, 同一来源 0.5 秒内最多触发一次	chance=6 / 8 / 10 / 13 / 16; counter_damage_attack_coefficient=0.8 / 0.85 / 0.9 / 0.95 / 1; damage_form=REFLECT; cannot_trigger_basic_passives=是; l5_forced_critical=是; eligible_damage=DIRECT_ACTUAL_HP_OR_SHIELD_DAMAGE_ONLY; dot_and_periodic_ineligible=是; internal_cd_s=0.5; internal_cd_scope=PER_SOURCE_ENTITY

ID	名称	类别	触发	玩家说明	工程数值(L1-L5)
B24	化伤	反击类	POST_SKILL_DAMAGE_IF_SURVIVES	1 级 受技能伤害后将其中 10% 转为回血 2 级 13% 3 级 16% 4 级 20% 5 级 25%。质变：转化时将转化量的 50%反弹给施法者。化伤是受到伤害后的回血，不能防止致死。	heal_percent_actual_hp_plus_shield_damage=10 / 13 / 16 / 20 / 25; cannot_prevent_lethal=是; overheal_discarded=是; l5_reflect_pre_overheal_conversion_percent=50; cannot_trigger_reflect=是; l5_reflect_amount=FLOOR(PRE_OVERHEAL_AMOUNT*0.5); l5_reflect_target=ATTACKER_OF_TRIGGERING_DAMAGE; l5_reflect_damage_form=REFLECT; l5_reflect_cannot_recurse=是
B25	暴怒	反击类	WHEN_CRITICALLY_HIT	1 级 被暴击时下次普攻+20% 2 级 28% 3 级 36% 4 级 44% 5 级 55%。质变：触发后 2 秒内攻速+40%	next_basic_damage_multiplier=1.2 / 1.28 / 1.36 / 1.44 / 1.55; one_charge=是; l5_attack_speed_percent=40; l5_duration_s=2; monster_cannot_crit=是
B26	背刺	战术类	DIRECT_DAMAGE_FROM_BACK_ARC	1 级 从背后攻击+35% 2 级 42% 3 级 50% 4 级 58% 5 级 70%。质变：背后攻击必定暴击	back_arc_deg=120; damage_multiplier=1.35 / 1.42 / 1.5 / 1.58 / 1.7; eligible_damage_forms=BASIC; l5_forced_critical=是
B27	疾跑	战术类	INCOMING_ACTUAL_DAMAGE_PROC	1 级 10% 受击移速+20% 2 秒 2 级 13% +25% 3 级 16% +30% 2.5 秒 4 级 20% +35% 5 级 25% +40% 3 秒。质变：加速期间免疫减速	chance=10 / 13 / 16 / 20 / 25; move_speed_percent=20 / 25 / 30 / 35 / 40; duration_s=2 / 2 / 2.5 / 2.5 / 3; l5_slow_immune=是

ID	名称	类别	触发	玩家说明	工程数值(L1-L5)
B28	猎杀	战术类	LOW_HP_HUNT	1 级 感知 30 米内残血敌人（低于 30%）对其移速+15% 2 级 40 米 +18% 3 级 50 米 +20% 4 级 60 米 +22% 5 级 80 米 +25%。质变：对残血敌人伤害 +20%。只追踪当前可见或已被显形的残血敌人	detect_range_m=30 / 40 / 50 / 60 / 80; target_hp_below_percent=30; move_speed_toward_target_percent=15 / 18 / 20 / 22 / 25; does_not_grant_vision=是; l5_damage_multiplier=1.2; eligible_target_information_state=VISIBLE_OR_REVEALED; hidden_unrevealed_target_grants_no_speed_or_direction_leak=是; multi_candidate_selection=NAV_PATH_DISTANCE_MM_ASC / ENTITY_ID_ASC; reselect_interval_s=0.25
B29	伏击	战术类	FIRST_DIRECT_HIT_AFTER_GAP	1 级 对 5 秒内未受过你伤害的目标首击 +25% 2 级 35% 3 级 45%时间缩为 4 秒 4 级 55% 5 级 70%时间缩为 3 秒。质变：触发伏击时额外暴露目标视野 3 秒	qualification_gap_s=OWNER_OUTPUTGOING_DIRECT_DAMAGE_GAP_S=5 / 5 / 4 / 4 / 3; damage_multiplier=1.25 / 1.35 / 1.45 / 1.55 / 1.7; eligible=BASIC / SKILL; dot_does_not_trigger=是; l5_reveal_target_s=3; trigger_requires_positive_hp_or_shield_damage=是; zero_damage_hit_does_not_consume_or_reset=是

ID	名称	类别	触发	玩家说明	工程数值(L1-L5)
B30	残影	战术类	MOVING_AFTERIMAGE	1 级 移动每 2 秒留一个残影 1.5 秒触碰减速 20% 2 级 减速 25%持续 2 秒 3 级 每 1.5 秒留一个减速 30% 4 级 每 1.5 秒留一个持续 2.5 秒 5 级 每秒留一个残影减速 35%持续 3 秒。质变：残影被触碰时爆炸 80 点 AOE	spawn_interval_s=2 / 2 / 1.5 / 1.5 / 1; afterimage_lifetime_s=1.5; slow_percent=20 / 25 / 30 / 30 / 35; slow_duration_s=1.5 / 2 / 2 / 2.5 / 3; l5_explosion_damage=80; l5_radius_m=3; each_enemy_triggers_each_afterimage_once=是; afterimage_visible_to=ALL; afterimage_solid=否; spawn_at=OWNER_HISTORICAL_PATH_POSITION; trigger_radius_m=0.8
B31	偷钱	经济类	BASIC_HIT_PROC_TRANSFER_GOLD	1 级 6% 普攻偷 30 金 2 级 8% 40 金 3 级 10% 50 金 4 级 13% 60 金 5 级 16% 80 金。质变：偷金币时回复偷取额 10% 的生命	chance=6 / 8 / 10 / 13 / 16; gold=30 / 40 / 50 / 60 / 80; insufficient=TRANSFER_REMAINING; not_create=是; l5_heal_percent_transferred_gold=10; eligible_basic=MAIN_BASIC_ONLY; extra_basic_ineligible=是; internal_cd_s=0.4; internal_cd_scope=PER_TARGET
B32	贪婪	经济类	ON_KILL_GOLD_CREATE	1 级 击杀野怪额外+40 金 2 级 +60 3 级 +80 4 级 +100 5 级 +150。质变：击杀英雄也触发+250 金	monster_gold=40 / 60 / 80 / 100 / 150; eligible_monsters=ORDINARY_MONSTER / FLYING_MONSTER / PIG / LEADER / DRAGON / CORE_BOSS; l5_enemy_player_true_death_gold=250; summon_kill=否

ID	名称	类别	触发	玩家说明	工程数值(L1-L5)
B33	寻宝	经济类	ON_MONSTER_KILL_NESTED_CHEST	1 级 8%击杀野怪掉宝箱（100 金） 2 级 10%（120 金） 3 级 13%（120 金，10%含 1 宝石） 4 级 16%（120 金，15%含 1 宝石） 5 级 20%（120 金，20%含 1 宝石，5%含 G1-G9 随机金装）	chest_chance=8 / 10 / 13 / 16 / 20; chest_gold=100 / 120 / 120 / 120 / 120; gem_chance=0 / 0 / 10 / 15 / 20; gold_equipment_chance=0 / 0 / 0 / 0 / 5; gold_pool=G1 / G2 / G3 / G4 / G5 / G6 / G7 / G8 / G9; summon_kill=否; nested_contents_roll_at=CHES T_SPAWN_COMMIT; contents_snapshot_immutable =是
B34	利息	经济类	SERVER_MINUTE_INTE REST	1 级 每分钟获得持有金币 5%利息（上限 300 金） 2 级 6%上限 400 3 级 7%上限 500 4 级 8%上限 600 5 级 10%上限 800。质变：利息上限翻倍（1600 金）	percent=5 / 6 / 7 / 8 / 10; cap_gold=300 / 400 / 500 / 600 / 1600; snapshot=COMMITTED_WALLET _AT_EXACT_60S_BOUNDARY_B EFORE_INTEREST; compound= 是; eligible_life_states=ALIVE / TRUE_DEATH_RESOLVING / SOUL_FLIGHT / REVIVE_PROTECTION; eliminated_ineligible=是
B35	拾荒	经济类	SELL_AND_PICKUP_E CONOMY	1 级 卖装备额外+10%金 2 级 13% 3 级 16% 4 级 20% 5 级 25%。质变：捡起地上装备时立刻获得该装备卖价的 20%金币	sell_bonus_percent=10 / 13 / 16 / 20 / 25; l5_pickup_base_sell_percent=20; l5_reward_once_per_item_instance_global ly=是; owner_dropped_never_rewards_same_ow ner=是

ID	名称	类别	触发	玩家说明	工程数值(L1-L5)
B36	奔腾	改规则类	CONTINUOUS_MOVEMENT_STACK	1 级 移动每秒叠 1 层每层+1%移速最多 5 层停下清零 2 级 每层 1.2%最多 6 层 3 级 每层 1.5%最多 6 层 4 级 每层 1.8%最多 7 层 5 级 每层 2%最多 8 层。质变：叠满 8 层时普攻附带移速值 20%的额外伤害	stack_interval_s=1; move_speed_percent_per_stack=1 / 1.2 / 1.5 / 1.8 / 2; max_stacks=5 / 6 / 6 / 7 / 8; clear_on=MOVE_INPUT_ZERO_OR_ACTUAL_SPEED_LT_0_1_FLOOR_0_2S; l5_full_stack_basic_extra_damage_move_speed_value_percent=20; extra_can_crit=是; move_speed_value_unit=GAME_MOVE_SPEED_INTEGER; l5_extra_damage_formula=FLOOR(CURRENT_GAME_MOVE_SPEED*0.20)
B37	共鸣	改规则类	OWNED_SUMMON_DEATH	1 级 召唤物死亡回 10 血 2 级 15 血 3 级 20 血+30 点 AOE 4 级 25 血+40 点 AOE 5 级 30 血+50 点 AOE。质变：召唤物死亡效果提升 50%（45 血+75 点 AOE）。世界伤害（雷暴/陷阱等非敌对单位来源）、自然到期、替换或主人死亡清理的召唤物不触发共鸣	heal=10 / 15 / 20 / 25 / 45; aoe_damage=0 / 0 / 30 / 40 / 75; aoe_radius_m=4; l5_values_already_include_50pct_bonus=是; eligible_death_causes=HOSTILE_PLAYER_DAMAGE / HOSTILE_MONSTER_DAMAGE; ineligible_removal_causes=WORLD_DAMAGE / NATURAL_EXPIRY / REPLACEMENT / OWNER_TRUE_DEATH_CLEANUP / OWNER_ELIMINATION_CLEANUP

ID	名称	类别	触发	玩家说明	工程数值(L1-L5)
B38	逆境	改规则类	WHILE_HARD_CONTROLLED	1 级 被控制时受伤-15% 2 级 20% 3 级 25% 4 级 30% 5 级 35%。质变：被控制时每秒回 2%最大生命	incoming_damage_remaining_multiplier=0.85 / 0.8 / 0.75 / 0.7 / 0.65; l5_heal_max_hp_percent_per_s=2; soft_control_not_eligible=是; continuous_control_timer_instances=ONE_PER_OWNER; first_heal_after_s=1; overlapping_hard_control=EXTEND_CONTINUITY_NO_EXTRA_TIMER
B39	石像	改规则类	OUT_OF_ANY_COMBAT_SUMMON	1 级 脱战 7 秒后自动召唤石像跟随 250 血 0 攻 2 级 300 血 3 级 350 血脱战缩为 6 秒 4 级 400 血 5 级 500 血脱战缩为 5 秒。质变：石像被摧毁时对周围造成 120 点 AOE 伤害	delay_s=7 / 7 / 6 / 6 / 5; statue_hp=250 / 300 / 350 / 400 / 500; statue_attack=0; follow_owner=是; one_instance=是; l5_destroy_aoe_damage=120; l5_radius_m=4; statue_move_speed_mps=3; statue_collision_radius_m=0.6 ; statue_collision_height_m=1.8 ; statue_aggro=NONE; statue_targetable=是; statue_destroy_credit=ATTACKER_ONLY_NO_REWARD; forbidden_block_sole_corridor=是; owner_block_yield_after_s=0.5 ; yield_fail=TEMPORARY_NO_COLLISION

ID	名称	类别	触发	玩家说明	工程数值(L1-L5)
B40	强韧	养成类	ON_ELIGIBLE_KILL_PERMANENT_MAX_HP	1 级 每击杀+2 生命无上限 2 级 +4 3 级 +6 4 级 +7 5 级 +10。质变：每击杀 50 个单位额外+160 生命（可多次触发）	hp_per_kill=2 / 4 / 6 / 7 / 10; eligible=ORDINARY_MONSTER / FLYING_MONSTER / PIG / LEADER / DRAGON / CORE_BOSS / ENEMY_PLAYER_TRUE_DEATH; summon_kill=否; l5_every_50_kills_bonus_hp=160; no_cap=是
B41	赏金	赏金体系	ON_OWNER_TRUE_DEATH_MARK_KILLER	1 级 你被击杀后击杀者头出现赏金标记 90 秒，任何人击杀该目标+400 金 2 级 +600 金 3 级 +800 金标记持续 120 秒 4 级 +1000 金 5 级 +1500 金。质变：标记期间所有玩家小地图可见赏金目标大致方位	payout_gold=400 / 600 / 800 / 1000 / 1500; duration_s=90 / 90 / 120 / 120 / 120; merge=MAX_PAYOUT_MAX_DURATION; monster_killer=NO_MARK; monster_final_blow_on_mark=DESTROY_NO_PAYOUT; l5_public_minimap_direction=visible_to=ALL_NON_ELIMINATED_PLAYERS; sector_count=8; sector_zero_center_deg=0; sector_formula=FLOOR(MOD(ATAN2_DEG(target_y-viewer_y,target_x-viewer_x)+22.5+360,360)/45); update_interval_s=3; marker=MINIMAP_EDGE_AT_SECTOR_CENTER; shows_distance=否; ends_with_mark=是
B42	悬赏	赏金体系	ON_ENEMY_BOUNTY_TARGET_TRUE_DEATH	1 级 击杀带赏金标记敌人时主动技能 CD-10 秒 2 级 -15 秒 3 级 -20 秒 4 级 -25 秒 5 级 -30 秒。质变：击杀赏金目标后 5 秒内移速+40%	active_cd_reduce_s=10 / 15 / 20 / 25 / 30; minimum_remaining_s=0; l5_move_speed_percent=40; l5_duration_s=5
B43	避雷	天劫体系	NORMAL_STORM_ZONE_MODIFIER	1 级 天劫区内被劈概率-20% (50%→40%) 2 级 -30% (→35%) 3 级 -40% (→30%) 天劫区内移速+10% 4 级 -50% (→25%) 5 级 -60% (→20%) 移速+15%。质变：天劫区内普攻伤害+10%	normal_hit_probability=0.4 / 0.35 / 0.3 / 0.25 / 0.2; move_speed_percent=0 / 0 / 10 / 10 / 15; l5_basic_damage_multiplier=1.1; does_not_change_apocalypse_probability_or_damage=是
B44	雷暴	天劫体系	BASIC_HIT_PROC	1 级 8% 普攻触发落雷 80 点 AOE (半径 3 米) 2 级 10% 90 点 3 级 13% 100 点 AOE 范围 4 米 4 级 16% 110 点 5 级 20% 130 点 AOE 范围 5 米。质变：天劫区内触发概率 1.5 倍 (30%)	chance=8 / 10 / 13 / 16 / 20; damage=80 / 90 / 100 / 110 / 130; radius_m=3 / 3 / 4 / 4 / 5; damage_form=SKILL; inherits_element=是; l5_normal_storm_zone_chance_multiplier=1.5; l5_zone_chance=30

8 装备、手牌、技能书与宝石

容量	规则
身穿	3 格；同名装备身穿唯一；不同名鞋数量不设限制，正常叠加
手牌	基础 1 格；穿戴 P8 布袋额外+1。移除布袋导致超容时，最近获得的非 P8 手牌自动公开落脚下。

容量	规则
满容购买/拾取	必须在同一原子请求内明确丢弃或替换旧装备；商品或地面物被抢走时不得扣金、丢旧装或毁旧装。
换装	脱离 PVP 战斗 3 秒；引导 0.5 秒；可移动；任何实际生命/护盾伤害打断。
死亡	第 1、2 次真死掉全部手牌；第 3 次真死手牌+身穿全掉；装备地面公开。
地面寿命	白/蓝/紫/金装备、技能书、通用主动和 BOSS 掉落在安全区内不自然消失；连续处于天劫外 60 秒才销毁。

ID	名称	品质	属性	效果	买价	卖价	商店
W1	精铁棍	白	ATTACK_FLAT=15	type=NONE	600	240	是
W2	粗布战衣	白	MAX_HP_FLAT=220	type=NONE	600	240	是
W3	草鞋	白	MOVE_SPEED_FLAT=18	type=SHOE_ONLY	600	240	是
W4	铜护腕	白	ATTACK_FLAT=10; MAX_HP_FLAT=100	type=NONE	600	240	是
W5	轻甲背心	白	MAX_HP_FLAT=110; MOVE_SPEED_FLAT=10	type=NONE	600	240	是
W6	行者束带	白	ATTACK_FLAT=8; MOVE_SPEED_FLAT=10	type=NONE	600	240	是
B1	狼牙棒	蓝	ATTACK_FLAT=20	type=DAMAGE_MULTIPLIER; target_classes=ORDINARY_MONSTER / FLYING_MONSTER / PIG / LEADER / DRAGON / CORE_BOSS; multiplier=1.3	2000	800	是
B2	虎皮裙	蓝	MAX_HP_FLAT=280	type=FIXED_DAMAGE_REDUCTION; eligible_damage_forms=BASIC; amount=8; minimum_final_damage=1	2000	800	是
B3	避水珠	蓝	MAX_HP_FLAT=200	type=PERCENT_DAMAGE_REDUCTION; eligible_damage_forms=SKILL / DOT; remaining_multiplier=0.85	2000	800	是
B4	定风珠	蓝	MAX_HP_FLAT=200	type=FORCED_MOVEMENT_DISTANCE_MULTIPLIER; eligible=KNOCKBACK / PULL; multiplier=0.5	2000	800	是
B5	药葫芦	蓝	MAX_HP_FLAT=150	type=REGEN; out_of_any_combat_s=5; hp_per_s=8; tick_s=1	2000	800	是
B6	磨刀石	蓝	ATTACK_FLAT=12	type=CRIT_DAMAGE_BONUS; add_percentage_points=30	2000	800	是
B7	铜镜	蓝	MAX_HP_FLAT=200	type=CONDITIONAL_BONUS_DAMAGE_REDUCTION; bonus_sources=BACKSTAB_B26 / AMBUSH_B29; bonus_portion_multiplier=0.5	2000	800	是
B8	聚宝袋	蓝	MAX_HP_FLAT=120	type=ON_MONSTER_KILL_GOLD_CREATE; amount=30; eligible=ORDINARY_MONSTER / FLYING_MONSTER / PIG / LEADER / DRAGON / CORE_BOSS; summon_kill=否	2000	800	是
B9	夜巡灯	蓝	ATTACK_FLAT=10	type=DIRECT_VISION_RADIUS_MULTIPLIER; multiplier=1.2; does_not_ignore_walls_or_stealth=是	2000	800	是
B10	驭兽鞭	蓝	ATTACK_FLAT=12	type=SUMMON_ATTACK_MULTIPLIER; multiplier=1.3	2000	800	是

ID	名称	品质	属性	效果	买价	卖价	商店
B11	避雷木牌	蓝	MAX_HP_FLAT=200	type=STORM_DAMAGE_REDUCTION; remaining_multiplier=0.7; applies=NORMAL_STORM / APOCALYPSE_STORM	2000	800	是
B12	千里靴	蓝	MOVE_SPEED_FLAT=26	type=SHOE_ONLY	2000	800	是
B13	登云履	蓝	MOVE_SPEED_FLAT=20	type=SLOW_MAGNITUDE_MULTIPLIER; multiplier=0.5; is_shoe=是	2000	800	是
B14	蛰伏靴	蓝	MOVE_SPEED_FLAT=20	type=OFFENSIVE_STEALTH_BREAK_MOVE_SPEED; is_shoe=是; eligible_transition=STEALTH_TO_VISIBLE; eligible_causes=OWNER_BASIC_RELEASE_AGAINST_H OSTILE / OWNER_HOSTILE_ACTIVE_CAST_COMMIT; excluded_causes=REVEALED_BY_ENEMY / ACTUAL_DAMAGE_TAKEN / VOLUNTARY_CANCEL / STEALTH_TIMEOUT; final_move_speed_multiplier=1.3; duration_s=2; internal_cooldown_s=8; once_per_stealth_episode= 是; stacking=REFRESH_DURATION_NO_MAGNITUDE_STAC K; eligible_hidden_sources=FUSHI_HIDDEN / ACTIVE_STEALTH / EQUIPMENT_STEALTH; excluded_hidden_sources=FOG_OF_WAR_ONLY / SMOKE_OCCLUSION_ONLY; hostile_active_definition=ACTIVE_COMMIT_THAT_TARG ETS_HOSTILE_OR_CREATES_PLAYER_OWNED_DAMAG E_CONTROL_TRAP_OR_HOSTILE_SUMMON; clear_speed_buff_on=UNEQUIP_B14 / OWNER_TRUE_DEATH / ITEM_INSTANCE_DESTROYED; internal_cooldown_on_unequip_or_true_death=CLEAR _WITH_ITEM_EFFECT_STATE	2000	800	是
P1	七星剑	紫	ATTACK_FLAT=35	type=CONDITIONAL_DAMAGE_MULTIPLIER; condition=TARGET_CURRENT_HP_PERCENT_LT_30; multiplier=1.25	6000	2400	是
P2	火尖枪	紫	ATTACK_FLAT=28	type=BASIC_ON_HIT_DOT; damage_per_s=20; duration_s=2; tick_s=1; refresh_not_stack=是; damage_form=DOT; inherits_element=是	6000	2400	是
P3	幌金绳	紫	ATTACK_FLAT=18	type=OWNER_HARD_CONTROL_DURATION_ADD; seconds=0.5; does_not_apply_to_soft_control=是	6000	2400	是
P4	人参果	紫	MAX_HP_FLAT=400	type=ON_ENEMY_PLAYER_TRUE_DEATH_HEAL; max_hp_percent=30; overheal_discarded=是	6000	2400	是
P5	万年灵芝	紫	MAX_HP_FLAT=500	type=REGEN; out_of_combat_required=否; hp_per_s=8; tick_s=1	6000	2400	是
P6	风袋	紫	MAX_HP_FLAT=200	type=ACTIVE_COOLDOWN_RATE; remaining_time_multiplier_on_equip=0.8; cooldown_duration_multiplier=0.8; preserve_remaining_percent=是	6000	2400	是

ID	名称	品质	属性	效果	买价	卖价	商店
P7	招魂幡	紫	ATTACK_FLAT=18	type=SUMMON_HP_AND_ATTACK_MULTIPLIER; multiplier=1.6	6000	2400	是
P8	布袋	紫	NONE	type=EXTRA_HAND_SLOT; amount=1	6000	2400	是
P9	照妖珠	紫	MAX_HP_FLAT=250	type=TRUE_SIGHT; radius_m=20; walls_block=是; shared_with_others=否	6000	2400	是
P10	赌坊金牌	紫	MAX_HP_FLAT=150	type=BLACK_MOUNTAIN_OUTCOME_OVERRIDE; probability_authority=§ 12.5 及 § E.18 黑山规则; rule=BIG_WIN_DOUBLES_FROM_BASE; FLAT_UNCHANGED; INCREASE_DEDUCTED_FROM_LOSS; service_active_condition=HEISHAN_SERVICE_OPEN_AT _EFFECT_EVALUATION_TICK; after_service_close=EXISTING_ITEM_RETAINS_MAX_HP _150_AND_CAN_BE_WORN_HELD_DROPPED_OR_SOL D; SPECIAL_EFFECT_DORMANT; new_random_generation_eligibility=HEISHAN_SERVICE _OPEN_AT_GENERATION_COMMIT_TICK; time_authority=§ 12.3 及 § E.17 商店规则 =HEISHAN].windows_s_half_open	6000	2400	是
P11	判官笔	紫	ATTACK_FLAT=25	type=ON_ENEMY_PLAYER_TRUE_DEATH _ACTIVE_READY; ready=是; kill_during_committed_cast=CAST_CON TINUES; post_cast_cooldown=READY; kill_during_cooldown=READY	6000	2400	是
P12	荆棘甲	紫	MAX_HP_FLAT=400	type=REFLECT; eligible_damage_forms=BASIC; actual_post_mitigation_damage_percent=20; cannot_trigger_reflect=是	6000	2400	是
P13	夜行斗篷	紫	MOVE_SPEED_FLAT=25	type=STATIONARY_STEALTH; stationary_required_s=2; breaks_on=MOVE_OVER_0_1M / ATTACK / CAST / ACTUAL_DAMAGE_TAKEN; walls_normal=是; forced_displacement_over_m=0.1_BREA KS; camera_or_facing_rotation=DOES_NOT _BREAK; position_noise_under_or_equal_m=0.1_ DOES_NOT_BREAK	6000	2400	是

ID	名称	品质	属性	效果	买价	卖价	商店
P14	血玉葫芦	紫	ATTACK_FLAT=25	type=BASIC_LIFESTEAL; actual_hp_damage_percent=10; does_not_heal_from_shield_damage=是	6000	2400	是
P15	疾风靴	紫	MOVE_SPEED_FLAT=30	type=SLOW_IMMUNITY; is_shoe=是	6000	2400	是
P16	连珠履	紫	MOVE_SPEED_FLAT=24	type=BASIC_SAME_TARGET_ATTACK_SPEED_STACK; is_shoe=是; eligible_hit=OWNER_BASIC_POST_IMMUNITY_HP_OR_SHIELD_DAMAGE_GT_ZERO; attack_speed_percent_per_stack=4; maximum_stacks=6; maximum_attack_speed_percent=24; duration_s=3; refresh_on_eligible_hit=是; hit_different_target=CLEAR_OLD_STACKS_THEN_ADD_ONE_STACK_TO_NEW_TARGET; target_true_death_or_invalid=CLEAR_IMMEDIATELY; summon_dot_reflect_and_equipment_procs=INELIGIBLE; B11_extra_basic=ELIGIBLE_IF_POST_IMMUNITY_HP_OR_SHIELD_DAMAGE_GT_ZERO; proc_generated_non_basic_damage=INELIGIBLE; each_damage_event_maximum_stack_gain=1; clear_on=UNEQUIP_P16 / OWNER_TRUE_DEATH / ITEM_INSTANCE_DESTROYED / TARGET_TRUE_DEATH_OR_INVALID / DURATION_TIMEOUT	6000	2400	是
P17	摘星靴	紫	MOVE_SPEED_FLAT=22	type=BASIC_ATTACK_RANGE_FLAT; is_shoe=是; meters=0.7; owner_basic_only=是; active_and_summon_unchanged=是; direct_visibility_and_release_revalidation_still_required=是; applies_at=BASIC_TARGET_ACQUIRE / BASIC_RELEASE_RANGE_REVALIDATION; does_not_extend_projectile_lifetime_or_homing_distance=是	6000	2400	是

ID	名称	品质	属性	效果	买价	卖价	商店
P18	磐岩履	紫	MOVE_SPEED_FLAT=24	type=HARD_CONTROL_AND_FORCED_MOVEMENT_RESIST; is_shoe=是; forced_movement_distance_multiplier=0.3; eligible_forced_movement=KNOCKBACK / PULL / KNOCKUP / THROW; hard_control_duration_multiplier=0.3; eligible_hard_control=STUN / FREEZE / PETRIFY / ROOT / FEAR / TRANSFORM / KNOCKUP; excluded_soft_control=SLOW / SILENCE; hard_control_minimum_duration_s=0.3; same_category_resistance_stacking=TAKE_STRONGEST_T_LOWEST_MULTIPLIER; duration_order=MAX_0_30S((BASE_DURATION+SOURCE_FLAT_EXTENSION) * TARGET_STRONGEST_MULTIPLIER); minimum_applies_only_if_pre_resistance_duration_gt_zero=是; immunity_or_cleanses_zero_stays_zero=是; forced_movement_minimum_distance_m=0; airborne_control_uses=KNOCKUP_CONTROL_DURATION_AND_KNOCKUP_FORCED_MOVEMENT_DISTANCE_BOTH_RESOLVE_ONCE; clear_state_on_unequip=NO_PERSISTENT_STATE; REEVALUATE_ONLY_FUTURE_CONTROL_APPLICATIONS	6000	2400	是
G1	九转金丹	金	NONE	type=LETHAL_PROTECTION_CONSUMABLE; priority_after=B20; hp_percent=100; invulnerable_s=2; consume_item=是; consume_life=否	不售	4000	否
G2	混元珠	金	ATTACK_FLAT=25	type=ELEMENT_COUNTER_MULTIPLIER_OVERRIDE; normal=1.5; owner_counters_target=2; incoming_unchanged=是	不售	4000	否
G3	渡劫铃	金	MAX_HP_FLAT=350	type=STORM_DAMAGE_REDUCTION; remaining_multiplier=0.3; applies=NORMAL_STORM / APOCALYPSE_STORM	不售	4000	否

ID	名称	品质	属性	效果	买价	卖价	商店
G4	筋斗云	金	MOVE_SPEED_FLAT=40	type=FLIGHT_OUT_OF_COMBAT; out_of_any_combat_s=5; capsule_bottom_height_m=3; land_on_any_actual_damage_dealt_or_received=是; is_shoe=是; world_interaction=AUTO_LAND_BEFORE_PICKUP_SHOP_CHEST_GAMBLE_TAIBAL; combat_active_cast_does_not_auto_land_unless_damage_dealt=是; no_legal_landing_within_6m=INTERACTION_FAILS_KEEP_FLYING_NO_COST_NO_UI	不售	4000	否
G5	聚宝盆	金	NONE	type=ALL_GOLD_INCOME_MULTIPLIER; multiplier=1.5; eligible_ledger_classes=G5_EXACT_LEDGER_WHITELIST; bonus_ledger_class=GOLD_CREATE_G5_BONUS; does_not_reapply_to_own_bonus=是; excludes=BLACK_MOUNTAIN_GROSS_RETURN / PLAYER_KILL_WALLET_BONUS / REFUND / ROLLBACK; black_mountain_only_positive_net_profit_eligible_once=是	不售	4000	否
G6	降魔杵	金	ATTACK_FLAT=60	type=CONDITIONAL_DAMAGE_MULTIPLIER; condition=TARGET_CURRENT_HP_GREATER_THAN_OWNER_CURRENT_HP_AT_DAMAGE_SNAPSHOT; multiplier=1.25	不售	4000	否

ID	名称	品质	属性	效果	买价	卖价	商店
G7	顺风耳	金	MAX_HP_FLAT=300	type=GLOBAL_ACTIVE_CAST_PING; enemy_active_cast_only=是; minimap_exact_cast_position_snapshot_s=1; does_not_grant_vision_or_lock=是; coalesce_window_s=0.25; coalesce_spatial_cell_m=10; max_simultaneous_markers=8; overflow_policy=KEEP_MOST_RECENT_THEN_CAST_EVENT_ID_ASC	不售	4000	否
G8	雷公凿	金	ATTACK_FLAT=20; ATTACK_SPEED_PERCENT=40	type=STAT_ONLY	不售	4000	否
G9	噬魂戒	金	NONE	type=PERMANENT_ATTACK_ON_KILL; ordinary_amount=2; enemy_player_true_death_amount=10; eligible=ORDINARY_MONSTER / FLYING_MONSTER / PIG / LEADER / DRAGON / CORE_BOSS / ENEMY_PLAYER_TRUE_DEATH; summon_kill=否; no_cap=是	不售	4000	否
G10	金箍棒	金	ATTACK_FLAT=80	type=BASIC_ATTACK_RANGE_FLAT; meters=3	不售	4000	否

9 百眼迷城地图总则与精确几何

项目	定案
世界	百眼迷城；非对称复合区块网络；世界原点=几何中心
尺度	等效跨度 840 米；可玩包围盒约 839×685 米；初始安全半径 520 米覆盖最远边界 512.2 米
路网	192 节点、405 条可走边；主路 12m、连路 7m、风险路 4.5m、庭/斗场 8m、猪窝 5m、商店/出生支路 6m；连续无决策长廊≤60m
墙体	47 个实体多边形；道路和墙从同一几何生成，Nav 边不得穿墙。所有飞行、闪现与投射物只读真实碰撞体和高度。
高台	3 处可走顶面；实体侧壁；每处恰有 2 条线性坡道；提供信息优势但不放永久高价值资源
区域	断金坊、蛛丝峡、龙脊洲、百足城、烬水市、迷魂田；中央万劫三庭=惊蛰庭、玄冥庭、梵天庭
出生	40 宏区×2 微位=80；每局 30 个不同宏区，每区选 1 微位；每宏区最多 1 人
资源	108 普通/飞行怪槽；12 猪窝选 8；5 龙宫选 2；4 首领场；3 核心庭同屏 1 核心；167 宝箱点
商店	16 宏区×3 微点=48 锚点；营业实例按时间窗迁移
伏石	30 个备选点（24 世界+6 三庭），每局启用 19；先锁 8 个活跃猪窝关系点、4 关键岔路、每庭二选一，再随机补足

项目	定案
六区工程边界	六个沿道路、墙体和外边界制作的互斥闭合多边形；共享边界只存一份路径；所有坐标实体导入时物化唯一 region_id
三维地表	地面、坡道、高台顶面均为可走三角面；Z 由表面重心插值；所有墙、动态墙和边界使用真实 3D 挤出体统一判定移动、视线、弹道、飞行与闪现

9.1 六区职责

区块	主体验与路线职责
断金坊	低风险断墙巷战；多段折角；新手可读的觅食与撤退
蛛丝峡	折线峡谷、窄关和截击；长廊以侧门、掩体龛和可穿墙洞拆分
龙脊渊	高台与双龙宫；垂直信息优势；远程暴露面明确
百足城	密路游走与多向切入；适合追击、反绕和短线换区
烬水市	商店—龙宫—关口三角；消费路线和争龙路线互相截断
迷魂田	猪窝袋路、伏击点与拾荒回路；追猪收益与被堵风险绑定
万劫三庭	三种核心 BOSS/决赛舞台；每庭 3 条独立接近路线、环廊、伏石、反侦察面、6 复活位与 2 终局商位

9.1a 六区可计算边界

region_id	中文名	共享边界路径	闭合多边形顶点（米）
REGION_BURNING_WATER	烬水市	DISTRICT_PATH_0 / DISTRICT_PATH_5	0.0/0.0; 80.0/25.0; 170.0/55.0; 270.0/100.0; 416.8/111.9; 384.8/10.7; 419.5/-106.6; 368.8/-245.2; 250.0/-120.0; 150.0/-70.0; 75.0/-30.0; 0.0/0.0
REGION_DRAGON_RIDGE	龙脊渊	DISTRICT_PATH_1 / DISTRICT_PATH_0	0.0/0.0; 20.0/80.0; 10.0/170.0; 38.0/260.0; 38.4/322.5; 161.0/341.1; 302.2/317.1; 400.8/245.2; 416.8/111.9; 270.0/100.0; 170.0/55.0; 80.0/25.0; 0.0/0.0

region_id	中文名	共享边界路径	闭合多边形顶点（米）
REGION_BROKEN_GOLD	断金坊	DISTRICT_PATH_2 / DISTRICT_PATH_1	0.0/0.0; -70.0/45.0; - 135.0/100.0; -220.0/170.0; - 340.1/333.1; -201.5/325.1; - 110.9/343.8; 38.4/322.5; 38.0/260.0; 10.0/170.0; 20.0/80.0; 0.0/0.0
REGION_SPIDER_GORGE	蛛丝峡	DISTRICT_PATH_3 / DISTRICT_PATH_2	0.0/0.0; -70.0/-30.0; - 140.0/-80.0; -240.0/-120.0; - 401.3/-165.2; -374.7/-50.6; - 412.0/58.6; -420.0/293.2; - 340.1/333.1; -220.0/170.0; - 135.0/100.0; -70.0/45.0; 0.0/0.0
REGION_LOST_SOUL_FIELD	迷魂田	DISTRICT_PATH_4 / DISTRICT_PATH_3	0.0/0.0; 15.0/-80.0; -5.0/- 170.0; 11.0/-260.0; 11.7/- 341.1; -116.2/-317.1; - 233.5/-341.1; -353.4/- 271.8; -401.3/-165.2; - 240.0/-120.0; -140.0/-80.0; - 70.0/-30.0; 0.0/0.0
REGION_CENTIPEDE_CITY	百足城	DISTRICT_PATH_5 / DISTRICT_PATH_4	0.0/0.0; 75.0/-30.0; 150.0/- 70.0; 250.0/-120.0; 368.8/- 245.2; 256.9/-322.5; 129.0/- 306.5; 11.7/-341.1; 11.0/- 260.0; -5.0/-170.0; 15.0/- 80.0; 0.0/0.0

- 共享边界路径为唯一权威几何；六区多边形仅由外边界段与下表路径拼接生成。构建器必须验证：每条路径顶点在两侧多边形中按序完整出现、六区并集覆盖全图无缝隙、任意两区交集面积为 0。

path_id	相邻区对	顶点数	有序顶点（米，自圆心向外）
DISTRICT_PATH_0	REGION_BURNING_WATER ↔ REGION_DRAGON_RIDGE	5	0/0; 80/25; 170/55; 270/100; 416.8/111.9
DISTRICT_PATH_1	REGION_DRAGON_RIDGE ↔	5	0/0; 20/80; 10/170;

	REGION_BROKEN_GOLD		38/260; 38.4/322.5
DISTRICT_PATH_2	REGION_BROKEN_GOLD ↔ REGION_SPIDER_GORGE	5	0/0; -70/45; -135/100; - 220/170; -340.1/333.1
DISTRICT_PATH_3	REGION_SPIDER_GORGE ↔ REGION_LOST_SOUL_FIELD	5	0/0; -70/-30; -140/-80; - 240/-120; -401.3/-165.2
DISTRICT_PATH_4	REGION_LOST_SOUL_FIELD ↔ REGION_CENTIPEDE_CITY	5	0/0; 15/-80; -5/-170; 11/- 260; 11.7/-341.1
DISTRICT_PATH_5	REGION_CENTIPEDE_CITY ↔ REGION_BURNING_WATER	5	0/0; 75/-30; 150/-70; 250/- 120; 368.8/-245.2

9.1b 三维地表与物体行为矩阵

对象类	移动	视线	弹道	飞行	闪现	可破坏	唯一工程语义
STATIC_BOUND	阻挡	阻挡	阻挡	体积相交时 阻挡	胶囊扫掠	否	读取墙多边形、Z0 和高度
STATIC_VAULT	阻挡	阻挡	阻挡	体积相交时 阻挡	连续实体弦长≤1.5m才可跨越，否则停在最后合法点	否	没有语义禁飞/禁闪标签
H031_PAGODA	阻挡	阻挡	按实体环墙 阻挡	体积相交时 阻挡	墙厚 1.6m 自然阻挡	否	5 秒或施法者真死清除
H036_FROZEN_RIVER	阻挡	阻挡	阻挡	体积相交时 阻挡	墙厚 1m， 按 1.5m 弦长规则可跨	300HP	40 秒；新墙替换旧墙
FUSHI_STONE	阻挡实体穿入石体	石体阻挡； 2m 环带另走隐身规则	石体阻挡	体积相交时 阻挡	胶囊扫掠	否	石体与隐身环带是两个不同体积
SMOKE_ZONE	不阻挡	按烟雾边界遮挡	不阻挡	不阻挡	不阻挡	否	视觉区，不是实体墙

对象类	移动	视线	弹道	飞行	闪现	可破坏	唯一工程语义
RAMP_WALKABLE	可走	不额外阻挡	不额外阻挡	可越过	合法落点	否	线性 Z 插值；坡度由坐标生成

9.2 地图总图与分层图



百眼迷城 840 米灰盒总图：道路、墙体、区块、资源和三庭的统一空间关系。





资源层：普通怪槽、猪窝、龙宫、首领场、核心庭与宝箱池。



商店层：16 宏区与 48 微锚点。





伏石层：世界与三庭备选伏石关系。

10 天劫、雷眼、决赛庭与灭世雷暴

时间	阶段	安全半径	圆心	事件
0:00-5:00	保持	r520	原点	全图发育
5:00-7:30	收缩	r520→320	原点	第一段
7:30-9:40	保持	r320	原点	中段发育
9:40-12:00	收缩	r320→220	原点	第二段
12:00-13:00	保持	r220	原点	12:00 公布三庭之一为决赛庭
13:00-15:45	收缩+移心	r220→140	向决赛庭移动 1/3	第三段
15:45-18:00	收缩+移心	r140→90	到达决赛庭中心	第四段
18:00-19:00	收缩	r90→60	决赛庭	庭内聚合
19:00-20:00	收缩	r60→0	决赛庭	终局例外段
20:00 以后	灭世雷暴	无几何安全点	全图	至胜负产生

类型	定案规则
普通天劫	20:00 前；每 3 秒对圈外存活玩家独立判定 50%；命中造成 20%最大生命天劫伤害并眩晕 0.5 秒；护盾可吸、无敌可挡；不走五行，只允许 B11 和 G3 按剩余倍率减免。
避雷被动	B43 按等级把命中概率改为 40%/35%/30%/25%/20%；不影响 20:00 后的确定性雷暴。
灭世雷暴	20:00 进入；首个伤害事件 20:01；每秒确定性伤害；事件 t 秒的伤害百分比= $2+\text{floor}((t-1200)/5)$ ，无眩晕、无概率、无上限。
雷暴目标	所有未淘汰玩家实体和存活核心 BOSS；核心 BOSS 每次承受 $\text{FLOOR}(\text{该事件雷暴百分比} \times \text{其最大生命快照} \times 50\%)$ 的天劫伤害；其他怪物与召唤物不受灭世雷暴。
玩法自由	雷暴不显形、不禁伏石、不禁隐身、不禁治疗与护盾；墙体仍阻挡视线和锁定。
地面资源	安全半径归零时不存在中心安全点，全部地面物继续/开始 60 秒天劫销毁倒计时。



11 PVE 生态、怪物、首领、龙王与核心 BOSS

PVE 调性 | 怪物是“有牙的资源”：少量小怪可稳定处理，拉十几二十只会因近战围位、远程和错帧攻击迅速变危险；AOE 构筑用更高瞬时风险换更快清群，但 PVE 不压过玩家之间的主戏。

机械层级	同时存量	定位
近战普通	58	三圈层基础资源；主动仇恨；6 个近战围位上限
远程普通	38	提供站位压力；被墙阻挡弹道
飞行	12	无视低矮地形路线但仍受锁定和范围伤害
五行肥猪	8	高价值诱饵；冲锋后逃向袋底，再转身战斗
首领	4	4 个固定首领场；肉盾/输出两种机械模板
五行龙王	2	5 座龙宫中启用 2 座；14,000HP 五行小 BOSS
核心 BOSS	1	三庭同屏 1 只；22,000HP；公共技能+6 种签名机制
总计	123	固定容量；不按在线人数动态加怪

11.1 AI、仇恨与迁移

- 除猪的特殊逃跑循环外，所有怪物均主动发现并攻击进入范围的合法玩家；猪也会主动预警、冲锋并在袋底战斗。
- 状态：栖息/巡逻→发现→预警→战斗追击→回巢→死亡/迁移；怪物不是被动木桩。
- 同巢声响可联动；同一组首次攻击错开 0-0.5 秒，同组命中至少间隔 0.15 秒；跨组不提供保护。
- 回巢时免疫硬控、获得 60%减伤，1.5 秒后每秒回复 8%，但不是无敌；不允许无限风筝白嫖。
- 普通怪死后在本区其他迁移位读秒重生，优先距死亡点 30-55 米，避开玩家 12 米和直接视野；无合法位每 5 秒重试。
- 普通/飞行怪攻击不随时间增长；新刷生命按 0/5/10/15 分钟快照×1.00/1.05/1.10/1.15。首领、龙王、核心按各自 HP/攻击/CD 曲线成长。
- 天劫吞没刷新节点即停止后续刷新；圈外存活怪连续覆盖 60 秒后消散，不迁入内圈堆积。

11.2 机械原型完整表

ID	名称	层级	五行	HP	攻	攻速	移速	射程	发现	重生	前摇	后摇	弹速
MEL_OUTER	OUTER 近战小妖	ORDINARY_MELEE	NONE	110	18	0.8	2.5	1.5	10	40	0.25	0.15	0
MEL_MIDDLE	MIDDLE 近战小妖	ORDINARY_MELEE	NONE	300	35	0.8	2.5	1.5	10	40	0.25	0.15	0
MEL_INNER	INNER 近战小妖	ORDINARY_MELEE	NONE	600	60	0.5	2.5	1.5	10	40	0.4	0.15	0
RNG_OUTER	OUTER 远程小妖	ORDINARY_RANGED	NONE	88	22	0.8	2.5	12	12	40	0.35	0.2	22
RNG_MIDDLE	MIDDLE 远程小妖	ORDINARY_RANGED	NONE	240	42	0.8	2.5	12	12	40	0.35	0.2	22
RNG_INNER	INNER 远程小妖	ORDINARY_RANGED	NONE	480	72	0.5	2.5	12	12	40	0.35	0.2	22
FLY_OUTER	OUTER 飞行小妖	FLYING	NONE	80	15	0.8	3	2	15	45	0.3	0.15	0
FLY_MIDDLE	MIDDLE 飞行小妖	FLYING	NONE	200	30	0.8	3	2	15	45	0.3	0.15	0
FLY_INNER	INNER 飞行小妖	FLYING	NONE	400	50	0.8	3	2	15	45	0.3	0.15	0

ID	名称	层级	五行	HP	攻	攻速	移速	射程	发现	重生	前摇	后摇	弹速
PIG_METAL	METAL 肥猪	PIG	METAL	1200	0	0	2.6	1	8	120	NON E	NON E	0
PIG_WOOD	WOOD 肥猪	PIG	WOOD	1200	0	0	2.6	1	8	120	NON E	NON E	0
PIG_WATER	WATER 肥猪	PIG	WATER	1200	0	0	2.6	1	8	120	NON E	NON E	0
PIG_FIRE	FIRE 肥猪	PIG	FIRE	1200	0	0	2.6	1	8	120	NON E	NON E	0
PIG_EARTH	EARTH 肥猪	PIG	EARTH	1200	0	0	2.6	1	8	120	NON E	NON E	0
LEADER_TANK	肉盾首领	LEADER	NONE	7000	120	0.6	2.2	2	25	180	0.5	0.3	0
LEADER_DPS	输出首领	LEADER	NONE	4000	220	1	2.4	8	25	180	0.4	0.2	26
DRAGON_METAL	METAL 龙王	DRAGON_MINIBOSS	METAL	14000	200	1	2.8	4	30	240	0.45	0.25	0
DRAGON_WOOD	WOOD 龙王	DRAGON_MINIBOSS	WOOD	14000	200	1	2.8	4	30	240	0.45	0.25	0
DRAGON_WATER	WATER 龙王	DRAGON_MINIBOSS	WATER	14000	200	1	2.8	4	30	240	0.45	0.25	0
DRAGON_FIRE	FIRE 龙王	DRAGON_MINIBOSS	FIRE	14000	200	1	2.8	4	30	240	0.45	0.25	0
DRAGON_EARTH	EARTH 龙王	DRAGON_MINIBOSS	EARTH	14000	200	1	2.8	4	30	240	0.45	0.25	0
CORE_COMMON	核心 BOSS 公共骨架	CORE_BOSS	NONE	22000	300	1	2	5	90	300	0.60	0.35	0

ID	碰撞 r	碰撞 h	转向°/s	加速 s	脱链边界	HP 阶段倍率	攻击阶段倍率	伤害形态	五行规则	命中类	归属
MEL_OUTER	0.5	1.8	360	0.3	type=RADIUS; m=25	1/1.05/1.1/1.15	1/1/1/1	BASIC	NONE	DIRECT	MONSTER
MEL_MIDDLE	0.5	1.8	360	0.3	type=RADIUS; m=25	1/1.05/1.1/1.15	1/1/1/1	BASIC	NONE	DIRECT	MONSTER
MEL_INNER	0.5	1.8	360	0.3	type=RADIUS; m=25	1/1.05/1.1/1.15	1/1/1/1	BASIC	NONE	DIRECT	MONSTER
RNG_OUTER	0.5	1.8	360	0.3	type=RADIUS; m=25	1/1.05/1.1/1.15	1/1/1/1	BASIC	NONE	DIRECT	MONSTER
RNG_MIDDLE	0.5	1.8	360	0.3	type=RADIUS; m=25	1/1.05/1.1/1.15	1/1/1/1	BASIC	NONE	DIRECT	MONSTER
RNG_INNER	0.5	1.8	360	0.3	type=RADIUS; m=25	1/1.05/1.1/1.15	1/1/1/1	BASIC	NONE	DIRECT	MONSTER
FLY_OUTER	0.4	1.4	360	0.3	type=RADIUS; m=25	1/1.05/1.1/1.15	1/1/1/1	BASIC	NONE	DIRECT	MONSTER
FLY_MIDDLE	0.4	1.4	360	0.3	type=RADIUS; m=25	1/1.05/1.1/1.15	1/1/1/1	BASIC	NONE	DIRECT	MONSTER

ID	碰撞 r	碰撞 h	转向°/s	加速 s	脱链边界	HP 阶段倍率	攻击阶段倍率	伤害形态	五行规则	命中类	归属
FLY_INNER	0.4	1.4	360	0.3	type=RADIUS; m=25	1/1.05/1.1/1.15	1/1/1/1	BASIC	NONE	DIRECT	MONSTER
PIG_METAL	0.8	1.6	300	0.4	type=PIG_NEST_BAG	1/1/1/1	1/1/1/1	NONE	BIDIRECTIONAL_FIVE_ELEMENT	DIRECT	MONSTER
PIG_WOOD	0.8	1.6	300	0.4	type=PIG_NEST_BAG	1/1/1/1	1/1/1/1	NONE	BIDIRECTIONAL_FIVE_ELEMENT	DIRECT	MONSTER
PIG_WATER	0.8	1.6	300	0.4	type=PIG_NEST_BAG	1/1/1/1	1/1/1/1	NONE	BIDIRECTIONAL_FIVE_ELEMENT	DIRECT	MONSTER
PIG_FIRE	0.8	1.6	300	0.4	type=PIG_NEST_BAG	1/1/1/1	1/1/1/1	NONE	BIDIRECTIONAL_FIVE_ELEMENT	DIRECT	MONSTER
PIG_EARTH	0.8	1.6	300	0.4	type=PIG_NEST_BAG	1/1/1/1	1/1/1/1	NONE	BIDIRECTIONAL_FIVE_ELEMENT	DIRECT	MONSTER
LEADER_TANK	1.2	2.4	240	0.5	type=RESOURCE_FIELD_PLUS_ONE_CORRIDOR	1/1/1.1/1.2	1/1/1/1	BASIC	NONE	DIRECT	MONSTER
LEADER_DPS	1.2	2.4	240	0.5	type=RESOURCE_FIELD_PLUS_ONE_CORRIDOR	1/1/1.1/1.2	1/1/1/1	BASIC	NONE	DIRECT	MONSTER
DRAGON_METAL	1.5	2.8	200	0.5	type=DRAGON_ARENA	1/1/1.1/1.1	1/1/1.05/1.05	BASIC	BIDIRECTIONAL_FIVE_ELEMENT	DIRECT	MONSTER
DRAGON_WOOD	1.5	2.8	200	0.5	type=DRAGON_ARENA	1/1/1.1/1.1	1/1/1.05/1.05	BASIC	BIDIRECTIONAL_FIVE_ELEMENT	DIRECT	MONSTER
DRAGON_WATER	1.5	2.8	200	0.5	type=DRAGON_ARENA	1/1/1.1/1.1	1/1/1.05/1.05	BASIC	BIDIRECTIONAL_FIVE_ELEMENT	DIRECT	MONSTER
DRAGON_FIRE	1.5	2.8	200	0.5	type=DRAGON_ARENA	1/1/1.1/1.1	1/1/1.05/1.05	BASIC	BIDIRECTIONAL_FIVE_ELEMENT	DIRECT	MONSTER
DRAGON_EARTH	1.5	2.8	200	0.5	type=DRAGON_ARENA	1/1/1.1/1.1	1/1/1.05/1.05	BASIC	BIDIRECTIONAL_FIVE_ELEMENT	DIRECT	MONSTER
CORE_COMMON	2	3.6	160	1	type=WHOLE_COURT	1/1/1.1/1.2	1/1/1.05/1.1	BASIC	NONE	DIRECT	MONSTER

11.2.1 怪物技能逐条工程表

技能 ID	使用者	CD	前摇	形状	伤害	附加规则	形态	五行	命中	归属
PIG_CHARGE	五行肥猪	6	0.6	type=LINE; length_m=3; width_m=1.6	80	knockback_m=2	SKILL	BIDIRECTIONAL_FIVE_ELEMENT	ABILITY	MONSTER
LEADER_SMASH	肉盾首领	8	1	type=CIRCLE; radius_m=5	150	slow_percent=30; slow_s=2	SKILL	NONE	ABILITY	MONSTER
LEADER_CONE	输出首领	6	1	type=CONE; range_m=8; angle_deg=60	200	无	SKILL	NONE	ABILITY	MONSTER
DRAGON_BREATH	五行龙王	7	1.2	type=CONE; range_m=10; angle_deg=70	280	inherits_element=是	SKILL	BIDIRECTIONAL_FIVE_ELEMENT	ABILITY	MONSTER
DRAGON_TAIL	五行龙王	10	0.6	type=CIRCLE; radius_m=5	180	knockback_m=3; inherits_element=是	SKILL	BIDIRECTIONAL_FIVE_ELEMENT	ABILITY	MONSTER

技能 ID	使用者	CD	前摇	形状	伤害	附加规则	形态	五行	命中	归属
DRAGON_DASH	五行龙王	12	1	type=LINE_CAPSULE; length_m=8; width_m=2.4; ; height_m=3; telegraph_s=1	120	condition=TOP_THREAT_AT_LEAST_12M_FOR_CONTINUOUS_2S_AND_REACHABLE; max_distance_m=8; slow_percent=25; slow_s=1.5; inherits_element=是	SKILL	BIDIRECTIONAL_FIVE_ELEMENT	ABILITY	MONSTER
CORE_RING	核心 BOSS 公共骨架	10	1.5	type=CIRCLE; radius_m=10	400	first_cast_delay_s=5	SKILL	NONE	ABILITY	MONSTER
CORE_METEOR	核心 BOSS 公共骨架	20	2	type=CIRCLE; radius_m=6	500	first_cast_delay_s=10; target=RANDOM_LIVING_THREAT_TARGET_CAST_POSITION	SKILL	NONE	ABILITY	MONSTER

11.3 核心 BOSS 六种签名机制

ID	名称	完整签名机制
CORE_V1_EARTHBREAK	地裂镇压	cooldown_s=16; windup_s=1.2; shape=type=THREE_PARALLEL_LINES; length_m=20; width_m=2; gap_m=3; damage=350; knockup_s=0.6; damage_spec=DamageForm=SKILL; SourceFaction=MONSTER; ElementRule=NONE; HitKind=ABILITY; OwnerCredit=MONSTER
CORE_V2_FIRELANE	烈焰回廊	cooldown_s=18; windup_s=1.2; shape=type=THREE_RADIAL_LANES_WITH_ONE_GAP; length_m=18; width_m=3; duration_s=6; damage_per_s=100; damage_spec=DamageForm=DOT; SourceFaction=MONSTER; ElementRule=NONE; HitKind=PERIODIC; OwnerCredit=MONSTER
CORE_V3_POISONPOOL	毒沼追猎	cooldown_s=20; windup_s=1.5; target_count=3; shape=type=CIRCLE; radius_m=5; duration_s=8; damage_per_s=70; damage_spec=DamageForm=DOT; SourceFaction=MONSTER; ElementRule=NONE; HitKind=PERIODIC; OwnerCredit=MONSTER
CORE_V4_WINDCHARGE	狂风冲阵	cooldown_s=16; windup_s=1.2; shape=type=LINE; length_m=18; width_m=3; damage=350; knockback_m=5; stops_at_solid=是; damage_spec=DamageForm=SKILL; SourceFaction=MONSTER; ElementRule=NONE; HitKind=ABILITY; OwnerCredit=MONSTER
CORE_V5_THUNDERCHAIN	天雷点名	cooldown_s=18; windup_s=1.5; target=RANDOM_LIVING_THREAT_TARGET; damage=300; jumps=4; jump_range_m=6; decay_percent_per_jump=20; damage_spec=DamageForm=SKILL; SourceFaction=MONSTER; ElementRule=NONE; HitKind=ABILITY; OwnerCredit=MONSTER

ID	名称	完整签名机制
CORE_V6_MIRRORSHADOW	镜影惑阵	cooldown_s=24; windup_s=1.5; summon_count=2; summon_hp=2000; summon_attack=60; summon_attack_speed_per_s=0.8; duration_s=10; decoy_cannot_drop_or_credit=是; summon_move_speed_mps=2.6; summon_attack_range_m=1.8; summon_detection_range_m=12; summon_collision_radius_m=0.65; summon_collision_height_m=2; summon_targeting=HIGHEST_CORE_BOSS_THREAT_VISIBLE_AND_REACHABLE_ELSE_NEAREST_VISIBLE_PLAYER; summon_leash=WHOLE_CURRENT_COURT; summon_movement_collision=是; mirror_basic_damage_spec=E.34_CORE_MIRROR 载荷

11.4 龙王与核心重生窗口

- 龙王仅当死亡时刻<11:00 才启动 240 秒重生；到期严格早于 15:00。15:00 仍存活者可继续击杀，龙宫被天劫完全覆盖后才进入 5 秒无敌退场且无掉落。
- 核心 BOSS 仅当死亡时刻<14:00 才启动 300 秒重生；下一只换庭。12:00 雷眼后，非决赛庭存活 BOSS 可继续被击杀，直至该庭战斗区被圈边吞没才 5 秒撤庭并按生命比例迁至决赛庭。
- 核心 BOSS 金币经验归未击；装备、技能书、宝石和地图主动落地后立即公开。雷暴击杀核心时无人获得金币经验，但公开掉落照常生成。
- 核心 BOSS 拥有 10 米真视并共享给区域内玩家，死亡后 10 秒仍生效，防止隐身者无风险抢装，但不禁止其隐身或拾取。

12 掉落、经济、商店、空投与黑山

12.1 怪物掉落

来源	金币	经验	书	宝石	装备	通用主动	G10
ORDINARY_OUTER	90	28	10	30	无	0	0
ORDINARY_MIDDLE	140	40	10	30	无	0	0
ORDINARY_INNER	180	56	10	30	无	0	0
FLYING_OUTER	140	40	20	30	无	0	0
FLYING_MIDDLE	200	56	20	30	无	0	0
FLYING_INNER	260	78	20	30	无	0	0
PIG	700	80	55	60	chance_percent=25; quality_percent=WHITE=60; BLUE=30; PURPLE=10	0	0
LEADER	1400	200	1	2	chance_percent=75; quality_percent=WHITE=35; BLUE=40; PURPLE=25	10	0
DRAGON	2000	300	0	3	count=1; quality_percent=BLUE=55; PURPLE=35; GOLD=10; gold_pool=G1_G9	0	0

来源	金币	经验	书	宝石	装备	通用主动	G10
CORE_BOSS	3000	400	2	3	count=1; quality_percent=PURPLE=75; GOLD=25; gold_pool=G1_G9	1	5

- 怪物与核心 BOSS 金币经验只归最后伤害所有者；地面物立即公开，不设尾刀保护。世界击杀怪物时金币经验为 0，但公开物品掉落仍滚动。
- 龙王未击者五行克制龙王时，蓝装升紫、紫装升金、金装保持；升级后按对应五行主题权重重抽。猪只受双向五行伤害，不触发品质升档。
- 定向掉落是隐藏规律：技能书有 50%概率进入怪物倾向池，装备有 35%概率进入倾向池；只改变同品质合法 ID 权重，不改变掉率和品质率；无合法项回退全局合法池。
- 金箍棒 G10 只由核心 BOSS 独立 5%额外掉落；龙王、普通金装池、黑山和商店均不生成 G10。

12.2 地面物、拾取与遗宝堆

类别	安全区寿命	天劫规则
SKILL_BOOK	永久	连续圈外 60 秒销毁；回圈重置倒计时
WHITE_EQUIPMENT	永久	连续圈外 60 秒销毁；回圈重置倒计时
BLUE_EQUIPMENT	永久	连续圈外 60 秒销毁；回圈重置倒计时
PURPLE_EQUIPMENT	永久	连续圈外 60 秒销毁；回圈重置倒计时
GOLD_EQUIPMENT	永久	连续圈外 60 秒销毁；回圈重置倒计时
GENERIC_ACTIVE	永久	连续圈外 60 秒销毁；回圈重置倒计时
BOSS_DROP	永久	连续圈外 60 秒销毁；回圈重置倒计时
GEM	60 秒	连续圈外 60 秒销毁；回圈重置倒计时
NORMAL_CHEST	60 秒	连续圈外 60 秒销毁；回圈重置倒计时
AIR_DROP_CHEST	120 秒	连续圈外 60 秒销毁；回圈重置倒计时

- 拾取距离 2 米且需直接视线；同一地面实例并发拾取时服务端接收序最小者成功。
- 同一 4 米网格≥2 件物品时复制为一个遗宝堆网络实体，但每件 instance_id、时间、品质、销毁计时和账本保持独立；客户端每页 12 项。
- 地面记录无硬删除上限；8192 条报警、256 复制簇软上限只触发分页和 LOD，绝不因性能压力删装备或技能书。
- 宝石进入 HUD 仓，不占格、不落地；技能书被学习即消耗；装备只能在商店卖出并销毁，不回到货架。

12.3 六商店数量、时间窗与库存

实例	类型	营业半开窗口	每窗开启事件
LAND_A	LAND	[00:30,03:30) / [03:30,06:30) / [06:30,09:30) / [09:30,12:30) / [12:30,15:00) / [15:00,20:00)	RELOCATE_AND_ROLL_NEW_LISTIN G_AT_OPEN

实例	类型	营业半开窗口	每窗开启事件
SHOE_A	SHOE	[00:45,03:45) / [03:45,06:45) / [06:45,09:45) / [09:45,13:15)	RELOCATE_AND_ROLL_NEW_LISTING_AT_OPEN; FINAL_SHORT_REMAINDER_MERGED_NO_RESTOCK
TAIBAI	TAIBAI	[01:00,04:00) / [04:00,07:00) / [07:00,10:00) / [10:00,13:00) / [13:00,15:00) / [15:00,20:00)	RELOCATE_SERVICE_AT_OPEN
HEISHAN	HEISHAN	[01:15,04:15) / [04:15,07:15) / [07:15,10:15) / [10:15,12:30)	RELOCATE_SERVICE_AT_OPEN
LAND_B	LAND	[01:30,04:30) / [04:30,07:30) / [07:30,10:30) / [10:30,14:45)	RELOCATE_AND_ROLL_NEW_LISTING_AT_OPEN; FINAL_SHORT_REMAINDER_MERGED_NO_RESTOCK
SHOE_B	SHOE	[01:45,04:45) / [04:45,07:45) / [07:45,10:45) / [10:45,14:00)	RELOCATE_AND_ROLL_NEW_LISTING_AT_OPEN; FINAL_SHORT_REMAINDER_MERGED_NO_RESTOCK

店型	单实例单轮容量	价格与规则
土地公	白色非鞋 5 格（五件各 1）；蓝色非鞋 3 格不放回；紫色非鞋 70%出 1 格；技能宝石 3；千里眼符 1；照妖镜 1	白 600/卖 240；蓝 2000/卖 800；紫 6000/卖 2400；宝石 250；千里眼 300；照妖镜 500
鞋匠	草鞋 1；蓝鞋 3 选 2 不放回；紫鞋 40%出 1；千里眼符 1；照妖镜 1	装备货架只卖鞋；回收任意装备；不回收宝石
太白	服务，无货架	换英雄 1500 金；个人 CD120 秒；脱 PVP 5 秒后引导 3 秒；同英雄可换但保留当前专属剩余 CD；不同英雄的新专属 READY
黑山	服务，无货架；每玩家每局 3 次	12:30 关闭；四类赌局；持 P10 使用专属概率表；区域禁赌时替换为固定炼化配方

整局理论上架容量：白装 58、蓝装 46、紫装期望 10.2（范围 0-18）、技能宝石 30、千里眼符 18、照妖镜 18。这是上架量，不保证售出。

- 常态最多同时 6 家：土地公×2、鞋匠×2、太白×1、黑山×1。商店从 16 宏区×3 微点中迁移，同类尽量异宏区且间距≥100 米，不落天劫预告区、唯一通路或实体墙。
- 打开界面不预留库存；购买提交时原子验证窗口、实例版本、商品、金币、生命、距离 2.5 米、直视、脱 PVP 3 秒和目标槽；售罄保持空格至下一轮。
- 12:30 黑山关闭；13:15 鞋 A 关闭；14:00 鞋 B 关闭；14:45 鞋 B 关闭；15:00 鞋 A 和太白迁入决赛庭；安全半径归零时两家同时关闭。

- 15:00 土地公 A 按正常概率补最后完整一轮，之后不再补货。鞋店关闭后仍可去土地公出售鞋。技能宝石只可在土地公按 100 金出售。
- 商店不提供安全区；PVP 战斗中不能开，受到玩家来源的实际生命/护盾伤害立即关闭；怪物或天劫伤害不设置 PVP 时间戳。

12.4 太白换英雄

- 引导开始时锁定选择，允许在 3 米商店系绳内移动；离开、玩家伤害、死亡或店关会取消，取消不扣金、不进入 CD。
- 提交顺序：重验→扣 1500→启动 120 秒 CD→清旧主动所有者绑定实体→替换英雄身体与五行→保持当前生命百分比至少 1→保留等级/经验/金币/被动/宝石/装备/永久层和运行中的状态计时→换新专属并设 READY→记账。
- 换英雄不免费治疗、不净化负面状态。同英雄换回合法，但保留当前专属主动剩余 CD；只有换成不同英雄时，新英雄专属主动设为 READY。

12.5 黑山老妖

赌局	普通大赢/平/亏	P10大赢/平/亏	输入与结果
被动	15% / 50% / 35%	30% / 50% / 20%	输入 1 个非空被动；大赢换未持有被动且等级+1；平换未持有被动同等级；亏损失输入
装备	15% / 40% / 45%	30% / 40% / 30%	白→蓝、蓝→紫、紫→G1-G9；平换同品质不同 ID；亏毁掉；G10 禁入
主动	10% / 50% / 40%	20% / 50% / 30%	大赢=37 个其他专属之一；平=19 地图主动中不同项；亏=当前英雄默认专属；结果均从完整 CD 开始
金币	10% / 40% / 50%	20% / 40% / 40%	倍金下注 500-5000、步进 100，回报 $\times 2/\times 1/\times 0.5$ ；普通长期期望 85%，P10 为 100%；紫装模式固定下注 2000，结果=紫装/返 2000/返 1000

12.6 空投与宝箱

工程字段	唯一值 / 执行语义
times_s	360 / 720 / 1080
warning_s	15
reservation.radius_m	3
reservation.blocks_dynamic_solid_placement_query	是
reservation.blocks_player_movement	否
player_overlap.nav_search_m	4
player_overlap.if_no_point	MOVE_AIRDROP_TO_NEXT_LEGAL_CANDIDATE
player_overlap.never_push_through_wall_or_storm	是
fallback_priority	NO_SOLID_OVERLAP / OPENABLE_SPACE / OTHER_CHEST_DISTANCE / SHOP_DISTANCE / MONSTER_DISTANCE
final_chest.authored_backup_points	4
final_chest.boss_body_has_priority	CHEST_RELOCATES_TO_NEXT_BACKUP; BOSS_NEVER_MOVES

工程字段	唯一值 / 执行语义
final_chest.choose	FIRST_LEGAL_POINT_AFTER_BOSS_BODY_AND_SOLID_CHECK_IN_ARRAY_ORDER_ROTATED_BY_AIR_DROP_RNG
channel_s	3
interrupts	ACTUAL_NON_STORM_DAMAGE / FORCED_MOVEMENT / STUN / KNOCKUP / PULL / MOVE_OVER_0_25M / TRUE_DEATH
does_not_interrupt_alone	SLOW / SILENCE / ROOT / NORMAL_STORM_DAMAGE_WITHOUT_STUN
drop_placement	SEARCH_8_DIRECTIONS_ON_RING_1_2_TO_2_0M_ELSE_LOOT_CLUSTER_AT_CHEST
reward.opener_gold	1200
reward.equipment_count	1
reward.equipment_quality	GOLD
reward.equipment_pool	G1 / G2 / G3 / G4 / G5 / G6 / G7 / G8 / G9
reward.selection	UNIFORM_AIR_DROP_RNG
reward.equipment_public_ground	是
reward.gold_to_first_successful_channeler	是
reward.single_commit	是
normal_B33_chest.eligible_final_blow_targets	ORDINARY_MONSTER / FLYING_MONSTER / PIG / LEADER / DRAGON / CORE_BOSS
normal_B33_chest.summon_final_blow_uses_owner_passive	否
normal_B33_chest.chest_trigger_percent_by_level	8 / 10 / 13 / 16 / 20
normal_B33_chest.chest_gold_by_level	100 / 120 / 120 / 120 / 120
normal_B33_chest.gem_chance_percent_by_level	0 / 0 / 10 / 15 / 20
normal_B33_chest.gold_equipment_chance_percent_by_level	0 / 0 / 0 / 0 / 5
normal_B33_chest.gold_equipment_pool	G1 / G2 / G3 / G4 / G5 / G6 / G7 / G8 / G9
normal_B33_chest.spawn_at	MONSTER_DEATH_POINT_ELSE_NEAREST_LEGAL_WITHIN_3M
normal_B33_chest.ground_expiry_s	60
normal_B33_chest.open_channel_s	0.35
normal_B33_chest.public_first_COMMIT_wins	是
normal_B33_chest.nested_roll_order	ROLL_CHEST / CREATE_CHEST_WITH_GOLD / ROLL_GEM_IF_LEVEL_AT_LEAST_3 / ROLL_GOLD_EQUIPMENT_IF_LEVEL_5
fallback_min_distances_m.OTHER_CHEST	12
fallback_min_distances_m.SHOP_INTERACTION_PAD	15
fallback_min_distances_m.LIVING_MONSTER_BODY	12
fallback_min_distances_m.LIVING_PLAYER_CAPSULE	3

13 匹配、BOT、新手与断线

系统	定案
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系统	定案
匹配池	三端同池、30 人 SOLO；不支持手柄；优先真人，等待 20 秒后用 BOT 补齐
隐藏 MMR	服务端计算，不向玩家显示单局变化；BOT 对局按同一结算进入 MMR；公式、搜索半径与新手保护见本节工程字段
新手	前六局逐步减少 BOT 占比，并逐步提高剩余 BOT 的行为强度，不给玩家数值加成，不弹强制教程；通过一次性气泡教会移动、普攻、主动、互动、三条命、迷雾与天劫
BOT 公平	D1-D5 难度出生时锁定；使用与真人相同的输入、经济、事务、视野、迷雾和三命接口；禁止读取迷雾外玩家位置
断线	3 秒后托管保命；120 秒内可无缝取回；超时 BOT 打到底；所有状态、随机流和事务保留
挂机	90 秒无有效输入进入保命托管；不额外扣分或剥夺资产

13.1 匹配工程字段

工程字段	唯一值 / 执行语义
player_state.initial_rating	1500
player_state.initial_uncertainty	350
player_state.min_rating	400
player_state.max_rating	2600
player_state.min_uncertainty	80
rating_formula.elo_divisor	400
rating_formula.placement_weight	0.8
rating_formula.combat_weight	0.2
rating_formula.combat_divisor	5
rating_formula.delta_min	-120
rating_formula.delta_max	120
rating_formula.full_weight_humans	10
rating_formula.uncertainty_decay	0.9
rating_formula.inactive_grace_days	14
rating_formula.inactive_uncertainty_per_day	3
search[0].from_s	0
search[0].to_s	5
search[0].rating_radius	100
search[1].from_s	5
search[1].to_s	10
search[1].rating_radius	200
search[2].from_s	10
search[2].to_s	15
search[2].rating_radius	350
search[3].from_s	15

工程字段	唯一值 / 执行语义
search[3].to_s	20
search[3].rating_radius	500
normal_bot_fill_at_s	20
low_population_exception_at_s	35
lobby_size	30
new_player_bot_caps.match_1	27
new_player_bot_caps.match_2_3	22
new_player_bot_caps.match_4_6	12
new_player_bot_caps.emergency	29
bot_rating_bands[0].rating_min	400
bot_rating_bands[0].rating_max	1099
bot_rating_bands[0].mix.D1	0.2
bot_rating_bands[0].mix.D2	0.6
bot_rating_bands[0].mix.D3	0.2
bot_rating_bands[1].rating_min	1100
bot_rating_bands[1].rating_max	1499
bot_rating_bands[1].mix.D2	0.35
bot_rating_bands[1].mix.D3	0.55
bot_rating_bands[1].mix.D4	0.1
bot_rating_bands[2].rating_min	1500
bot_rating_bands[2].rating_max	1849
bot_rating_bands[2].mix.D2	0.1
bot_rating_bands[2].mix.D3	0.55
bot_rating_bands[2].mix.D4	0.3
bot_rating_bands[2].mix.D5	0.05
bot_rating_bands[3].rating_min	1850
bot_rating_bands[3].rating_max	2149
bot_rating_bands[3].mix.D3	0.3
bot_rating_bands[3].mix.D4	0.55
bot_rating_bands[3].mix.D5	0.15
bot_rating_bands[4].rating_min	2150
bot_rating_bands[4].rating_max	2600
bot_rating_bands[4].mix.D4	0.55
bot_rating_bands[4].mix.D5	0.45
cross_input_pool	是
controller_supported	否
bot_events_affect_mmr	否

工程字段	唯一值 / 执行语义
fill_policy.normal_fill_s	20
fill_policy.new_player_match_1_to_6_fill_s	35
fill_policy.new_player_emergency_bot_cap	29
rating_formula_exact.expected_pair	$1/(1+10^{((R_j-R_i)/400)})$
rating_formula_exact.expected	MEAN_OF_PAIR_EXPECTATIONS_VS_HUMANS_ONLY
rating_formula_exact.placement	IF HUMAN_COUNT>=4 THEN 1-(HUMAN_ORDINAL_PLACEMENT-1)/(HUMAN_COUNT-1) ELSE UNRATED; TIES_USE_AVERAGE_HUMAN_ORDINAL
rating_formula_exact.combat	CLAMP(SUM_FIVE_HUMAN_ONLY_COMBAT_METRIC_Z/5,-1,1); EACH_ZERO_VARIANCE_METRIC_Z=0
rating_formula_exact.human_weight	$\text{MIN}(1, (\text{HUMAN_COUNT}-1)/9)$
rating_formula_exact.K	$32+88*(\text{UNCERTAINTY}-80)/270$
rating_formula_exact.delta	$\text{ROUND_HALF_AWAY_FROM_ZERO}(\text{CLAMP}(K*\text{HUMAN_WEIGHT}*(0.8*(\text{PLACEMENT}-\text{EXPECTED})+0.2*\text{COMBAT}), -120, 120))$
rating_formula_exact.rating	$\text{CLAMP}(\text{OLD_RATING}+\text{DELTA}, 400, 2600)$
rating_formula_exact.bots_ignored	是
rating_formula_exact.uncertainty_afterRatedMatch	IF HUMAN_COUNT>=4 THEN MAX(80,FLOOR(OLD*0.9)) ELSE OLD
rating_formula.minimum_humans_to_rate	4
rating_formula.ordinal_population	HUMANS_ONLY
rating_formula.single_or_zero_variance_metric	Z_SCORE=0
rating_formula.bots_do_not_shift_human_ordinal	是
rating_formula.unrated_result_when_humans_lt_4	MMR_DELTA=0; MATCH_STILL_COUNTS_FOR_NON_RATED_STATS

13.2 BOT 工程字段

工程字段	唯一值 / 执行语义
obeys_fog_and_same_input_interface	是
utility_actions	SURVIVE / STORM_ROUTE / FIGHT_PLAYER / FARM_MONSTER / FIND_BUILD_DROP / SHOP / SELECT_DRAGON_BY_ELEMENT / PUBLIC_EVENT / AMBUSH_OR_HIDE
profiles.FARMER.farm	1
profiles.FARMER.pvp	0.35
profiles.FARMER.shop	0.8

工程字段	唯一值 / 执行语义
profiles.FARMER.boss	0.55
profiles.HUNTER.farm	0.35
profiles.HUNTER.pvp	1
profiles.HUNTER.shop	0.5
profiles.HUNTER.boss	0.7
profiles.SCAVENGER.farm	0.5
profiles.SCAVENGER.pvp	0.45
profiles.SCAVENGER.shop	0.9
profiles.SCAVENGER.boss	0.4
revive	USE_SAME_SERVER_REVIVE_ASSIGNMENT_AS_HUMAN
final_storm	MAXIMIZE_SURVIVAL_AND_LEGAL_DAMAGE; STEALTH_REMAINS_LEGAL
build_replace	EXPECTED_VALUE_USING_VISIBLE_OWN_BUILD_ONLY
utility_formula	$PROFILE_WEIGHT * NEED_SCORE * SUCCESS_PROBABILITY * EXPECTED_VALUE / (1 + PATH_TIME_S / 30 + DEATH_RISK * 3)$
recompute_interval_s	0.25
action_contracts.FARM_MONSTER.need	XP_DEFICIT / BUILD_DROP_NEED / LOCAL_PACK_TTK / THIRD_PARTY_RISK
action_contracts.FARM_MONSTER.never_reads_hidden_drop_weights	是
action_contracts.SHOP.requires_known_shop	是
action_contracts.SHOP.expected_gain_uses_own_build_and_visible_listing	是
action_contracts.SHOP.travel_budget_s	90
action_contracts.SELECT_DRAGON_BY_ELEMENT.uses_public_dragon_element	是
action_contracts.SELECT_DRAGON_BY_ELEMENT.bidirectional_counter_in_risk	是
action_contracts.BUILD_REPLACE.uses_only_own_inventory_and_public_rules	是
action_contracts.BUILD_REPLACE.replacement_modal_obeyes_same_transaction_as_human	是
action_contracts.FINAL_STORM.may_hide_heal_fight_or_boss	是
action_contracts.FINAL_STORM.no_forced_reveal_knowledge	是
difficulty.locked_at_spawn	是
difficulty.D1.reaction_s	0.8
difficulty.D1.perception_refresh_s	0.5
difficulty.D1.aim_error_deg	14
difficulty.D1.dodge_probability_percent	5

工程字段	唯一值 / 执行语义
difficulty.D1.retreat_hp_percent	20
difficulty.D1.combo_depth	1
difficulty.D1.decision_noise_percent	30
difficulty.D2.reaction_s	0.6
difficulty.D2.perception_refresh_s	0.35
difficulty.D2.aim_error_deg	10
difficulty.D2.dodge_probability_percent	15
difficulty.D2.retreat_hp_percent	28
difficulty.D2.combo_depth	1
difficulty.D2.decision_noise_percent	22
difficulty.D3.reaction_s	0.42
difficulty.D3.perception_refresh_s	0.25
difficulty.D3.aim_error_deg	7
difficulty.D3.dodge_probability_percent	30
difficulty.D3.retreat_hp_percent	35
difficulty.D3.combo_depth	2
difficulty.D3.decision_noise_percent	15
difficulty.D4.reaction_s	0.28
difficulty.D4.perception_refresh_s	0.18
difficulty.D4.aim_error_deg	4
difficulty.D4.dodge_probability_percent	48
difficulty.D4.retreat_hp_percent	40
difficulty.D4.combo_depth	3
difficulty.D4.decision_noise_percent	8
difficulty.D5.reaction_s	0.2
difficulty.D5.perception_refresh_s	0.12
difficulty.D5.aim_error_deg	2
difficulty.D5.dodge_probability_percent	62
difficulty.D5.retreat_hp_percent	45
difficulty.D5.combo_depth	4
difficulty.D5.decision_noise_percent	4
forbidden	READ_FOG_HIDDEN_ENTITY / READ_FUTURE_RNG_OR_DROP / AIM_AFTER_PROJECTILE_COMMIT / DAMAGE_HP_SPEED_COOLDOWN_BONUS / DYNAMIC_DIFFICULTY_RUBBERBAND / TELEPORT_OR_IGNORE_NAV
same_input_and_transaction_interface_as_human	是

13.3 断线重连状态机

工程字段	唯一值 / 执行语义
connection.machine.states	CONNECTED / DEGRADED / DISCONNECTED_GRACE / BOT_TAKEOVER / RECONNECT_SYNC / RETURNED_TO_PLAYER / SESSION_ENDED
connection.machine.transitions[0].from_states	CONNECTED
connection.machine.transitions[0].to	DEGRADED
connection.machine.transitions[0].guard	PACKET_HEALTH_THRESHOLD
connection.machine.transitions[1].from_states	CONNECTED / DEGRADED / RETURNED_TO_PLAYER
connection.machine.transitions[1].to	DISCONNECTED_GRACE
connection.machine.transitions[1].guard	FIRST_TICK_AFTER_LAST_VALID_PACKET
connection.machine.transitions[2].from_states	DISCONNECTED_GRACE
connection.machine.transitions[2].to	CONNECTED
connection.machine.transitions[2].guard	VALID_PACKET_BEFORE_LAST_VALID_PACKET_PLUS_3S
connection.machine.transitions[3].from_states	DISCONNECTED_GRACE
connection.machine.transitions[3].to	BOT_TAKEOVER
connection.machine.transitions[3].guard	SERVER_TIME_REACHES_LAST_VALID_PACKET_PLUS_3S
connection.machine.transitions[4].from_states	BOT_TAKEOVER
connection.machine.transitions[4].to	RECONNECT_SYNC
connection.machine.transitions[4].guard	VALID_MATCH_TOKEN_BEFORE_OR_AT_BOT_TAKEOVER_TICK_PLUS_120S
connection.machine.transitions[5].from_states	RECONNECT_SYNC
connection.machine.transitions[5].to	RETURNED_TO_PLAYER
connection.machine.transitions[5].guard	FULL_SNAPSHOT_ACK_AND_INPUT_SEQUENCE_RESET
connection.machine.transitions[6].from_states	RETURNED_TO_PLAYER

工程字段	唯一值 / 执行语义
connection.machine.transitions[6].to	CONNECTED
connection.machine.transitions[6].guard	NEXT_SERVER_TICK
connection.machine.transitions[7].from_states	CONNECTED / DEGRADED / DISCONNECTED_GRACE / BOT_TAKEOVER / RECONNECT_SYNC / RETURNED_TO_PLAYER
connection.machine.transitions[7].to	SESSION_ENDED
connection.machine.transitions[7].guard	PLAYER_ELIMINATED_OR_MATCH_SETTLED_OR_ABORTED
bot_difficulty	D2
bot_has_no_extra_information	是
all_state_and_rng_preserved	是
new_device	REVOKE_OLD_SESSION_AFTER_NEW_DEVICE_AUTH_AND_SNAPSHOT_READY; NEVER_DUPLICATE_ENTITY
reconnect_input	DROP_OLD_CLIENT_INPUTS_AT_OR_BEFORE_SNAPSHOT_ACK_SEQUENCE
timers.bot_takeover_from_last_valid_packet_s	3
timers.player_reconnect_deadline_from_bot_takeover_s	120
timers.after_deadline	BOT_REMAINS_CONTROLLER_UNTIL_ENTITY_ELIMINATED_OR_MATCH_ENDS
timers.exact_deadline_is_inclusive	是
transition_encoding.from_states	NONEMPTY_ARRAY_OF_DECLARED_STATE_IDS
transition_encoding.to	ONE_DECLARED_STATE_ID
transition_encoding.guard	ONE_CLOSED_GUARD_ID_OR_SEMICOLON_AND_OF_GUARD_IDS
snapshot.chunked	是；快照分块编号 0..N-1，附整包 BLAKE3 哈希
snapshot.chunk_ack_required	逐块 ACK；缺块重发；整包哈希校验失败则整体重传
snapshot.bot_until_full_ack	确认完成前 BOT 继续接管
snapshot.handover	整包 ACK 后自下一 tick 交还控制，并按序补发快照 tick 之后的增量事件
version_mismatch.force_update	是；内容版本不符强制更新
version_mismatch.reconnect_if_seat_alive	更新完成且席位仍存在则重连入局；超时前 BOT 接管
version_mismatch.no_desertion_penalty	是；版本不符导致的离线不计逃跑处罚

14 局外成长、图鉴、战报与商业化

- 局外成长只形成身份、收藏和历史，不给予局内攻击、生命、移速、掉率、折扣、复活或额外候选。38 英雄全部免费。
- 修为轨道 50 级；经验来自名次、击杀、PVE 里程碑和玩法行为；奖励为头像、头像框、称号、名片和纯外观。
- 图鉴：38 英雄、101 技能（38 专属+19 通用+44 被动）、48 装备；状态为未发现/见过/用过/精通，只进不退。定向掉落权重不展示。
- 最近 20 局战报由服务端结算快照生成；最多 8 个故事节点，只引用真实事件；不展示隐藏 MMR、BOT 身份和反作弊信号。
- 首发只直接销售纯外观；无英雄抽卡、装备/技能抽卡、数值战令、广告复活、复活币、第四条命、付费候选位、微属性、付费掉率或真钱黑山。
- 皮肤不得改变碰撞、攻击帧、实际高度、射程、预警几何、脚步响度、五行符号、迷雾识别或可见性；低端机降级不得改变玩法信息。

14.1 局外系统工程字段

工程字段	唯一值 / 执行语义
xiuwei.only_increases	是
xiuwei.per_match_cap	200
xiuwei.active_participation.per_active_minute	2
xiuwei.active_participation.cap	20
xiuwei.active_participation.completed_match_bonus	20
xiuwei.active_participation.afk_or_bot_takeover_time_earns	否
xiuwei.placement.30_21	0
xiuwei.placement.20_11	10
xiuwei.placement.10_6	25
xiuwei.placement.5_4	40
xiuwei.placement.3	55
xiuwei.placement.2	70
xiuwei.placement.1	90
xiuwei.placement.co_champion_each	90
xiuwei.player_kill.each	5
xiuwei.player_kill.cap	50
xiuwei.dragon_last_hit.each	10
xiuwei.dragon_last_hit.cap	20
xiuwei.core_last_hit.each	25
xiuwei.core_last_hit.cap	50
xiuwei.bot_account_reward	0
xiuwei.invalid_match_reward	0
xiuwei.human_match_with_fill_bots_multiplier	1

工程字段	唯一值 / 执行语义
xiuwei.track[0].level	1
xiuwei.track[0].realm_zh	凡人 · 一层
xiuwei.track[0].required_from_previous	0
xiuwei.track[0].cumulative	0
xiuwei.track[0].reward_zh	初始形象(纯外观)
xiuwei.track[1].level	2
xiuwei.track[1].realm_zh	凡人 · 一层
xiuwei.track[1].required_from_previous	70
xiuwei.track[1].cumulative	70
xiuwei.track[1].reward_zh	头像框(纯外观)
xiuwei.track[2].level	3
xiuwei.track[2].realm_zh	凡人 · 二层
xiuwei.track[2].required_from_previous	80
xiuwei.track[2].cumulative	150
xiuwei.track[2].reward_zh	称号(纯外观)
xiuwei.track[3].level	4
xiuwei.track[3].realm_zh	凡人 · 二层
xiuwei.track[3].required_from_previous	80
xiuwei.track[3].cumulative	230
xiuwei.track[3].reward_zh	名片底纹(纯外观)
xiuwei.track[4].level	5
xiuwei.track[4].realm_zh	凡人 · 三层
xiuwei.track[4].required_from_previous	90
xiuwei.track[4].cumulative	320
xiuwei.track[4].reward_zh	边框特效(纯外观)
xiuwei.track[5].level	6
xiuwei.track[5].realm_zh	凡人 · 三层
xiuwei.track[5].required_from_previous	100
xiuwei.track[5].cumulative	420
xiuwei.track[5].reward_zh	击杀播报样式(纯外观)
xiuwei.track[6].level	7
xiuwei.track[6].realm_zh	炼气 · 一层
xiuwei.track[6].required_from_previous	110
xiuwei.track[6].cumulative	530
xiuwei.track[6].reward_zh	境界称号[炼气]+专属头像框(纯外观)
xiuwei.track[7].level	8
xiuwei.track[7].realm_zh	炼气 · 一层

工程字段	唯一值 / 执行语义
xiuwei.track[7].required_from_previous	110
xiuwei.track[7].cumulative	640
xiuwei.track[7].reward_zh	称号(纯外观)
xiuwei.track[8].level	9
xiuwei.track[8].realm_zh	炼气 · 二层
xiuwei.track[8].required_from_previous	120
xiuwei.track[8].cumulative	760
xiuwei.track[8].reward_zh	名片底纹(纯外观)
xiuwei.track[9].level	10
xiuwei.track[9].realm_zh	炼气 · 二层
xiuwei.track[9].required_from_previous	130
xiuwei.track[9].cumulative	890
xiuwei.track[9].reward_zh	边框特效(纯外观)
xiuwei.track[10].level	11
xiuwei.track[10].realm_zh	炼气 · 三层
xiuwei.track[10].required_from_previous	150
xiuwei.track[10].cumulative	1040
xiuwei.track[10].reward_zh	击杀播报样式(纯外观)
xiuwei.track[11].level	12
xiuwei.track[11].realm_zh	炼气 · 三层
xiuwei.track[11].required_from_previous	160
xiuwei.track[11].cumulative	1200
xiuwei.track[11].reward_zh	头像框(纯外观)
xiuwei.track[12].level	13
xiuwei.track[12].realm_zh	筑基 · 一层
xiuwei.track[12].required_from_previous	170
xiuwei.track[12].cumulative	1370
xiuwei.track[12].reward_zh	境界称号[筑基]+专属头像框(纯外观)
xiuwei.track[13].level	14
xiuwei.track[13].realm_zh	筑基 · 一层
xiuwei.track[13].required_from_previous	190
xiuwei.track[13].cumulative	1560
xiuwei.track[13].reward_zh	名片底纹(纯外观)
xiuwei.track[14].level	15
xiuwei.track[14].realm_zh	筑基 · 二层
xiuwei.track[14].required_from_previous	200
xiuwei.track[14].cumulative	1760

工程字段	唯一值 / 执行语义
xiuwei.track[14].reward_zh	边框特效(纯外观)
xiuwei.track[15].level	16
xiuwei.track[15].realm_zh	筑基 · 二层
xiuwei.track[15].required_from_previous	220
xiuwei.track[15].cumulative	1980
xiuwei.track[15].reward_zh	击杀播报样式(纯外观)
xiuwei.track[16].level	17
xiuwei.track[16].realm_zh	筑基 · 三层
xiuwei.track[16].required_from_previous	240
xiuwei.track[16].cumulative	2220
xiuwei.track[16].reward_zh	头像框(纯外观)
xiuwei.track[17].level	18
xiuwei.track[17].realm_zh	筑基 · 三层
xiuwei.track[17].required_from_previous	260
xiuwei.track[17].cumulative	2480
xiuwei.track[17].reward_zh	称号(纯外观)
xiuwei.track[18].level	19
xiuwei.track[18].realm_zh	金丹 · 一层
xiuwei.track[18].required_from_previous	280
xiuwei.track[18].cumulative	2760
xiuwei.track[18].reward_zh	境界称号[金丹]+专属头像框(纯外观)
xiuwei.track[19].level	20
xiuwei.track[19].realm_zh	金丹 · 一层
xiuwei.track[19].required_from_previous	300
xiuwei.track[19].cumulative	3060
xiuwei.track[19].reward_zh	边框特效(纯外观)
xiuwei.track[20].level	21
xiuwei.track[20].realm_zh	金丹 · 二层
xiuwei.track[20].required_from_previous	330
xiuwei.track[20].cumulative	3390
xiuwei.track[20].reward_zh	击杀播报样式(纯外观)
xiuwei.track[21].level	22
xiuwei.track[21].realm_zh	金丹 · 二层
xiuwei.track[21].required_from_previous	360
xiuwei.track[21].cumulative	3750
xiuwei.track[21].reward_zh	头像框(纯外观)
xiuwei.track[22].level	23

工程字段	唯一值 / 执行语义
xiuwei.track[22].realm_zh	金丹 · 三层
xiuwei.track[22].required_from_previous	390
xiuwei.track[22].cumulative	4140
xiuwei.track[22].reward_zh	称号(纯外观)
xiuwei.track[23].level	24
xiuwei.track[23].realm_zh	金丹 · 三层
xiuwei.track[23].required_from_previous	420
xiuwei.track[23].cumulative	4560
xiuwei.track[23].reward_zh	名片底纹(纯外观)
xiuwei.track[24].level	25
xiuwei.track[24].realm_zh	元婴 · 一层
xiuwei.track[24].required_from_previous	460
xiuwei.track[24].cumulative	5020
xiuwei.track[24].reward_zh	境界称号[元婴]+专属头像框(纯外观)
xiuwei.track[25].level	26
xiuwei.track[25].realm_zh	元婴 · 一层
xiuwei.track[25].required_from_previous	500
xiuwei.track[25].cumulative	5520
xiuwei.track[25].reward_zh	击杀播报样式(纯外观)
xiuwei.track[26].level	27
xiuwei.track[26].realm_zh	元婴 · 二层
xiuwei.track[26].required_from_previous	540
xiuwei.track[26].cumulative	6060
xiuwei.track[26].reward_zh	头像框(纯外观)
xiuwei.track[27].level	28
xiuwei.track[27].realm_zh	元婴 · 二层
xiuwei.track[27].required_from_previous	580
xiuwei.track[27].cumulative	6640
xiuwei.track[27].reward_zh	称号(纯外观)
xiuwei.track[28].level	29
xiuwei.track[28].realm_zh	元婴 · 三层
xiuwei.track[28].required_from_previous	630
xiuwei.track[28].cumulative	7270
xiuwei.track[28].reward_zh	名片底纹(纯外观)
xiuwei.track[29].level	30
xiuwei.track[29].realm_zh	元婴 · 三层
xiuwei.track[29].required_from_previous	690

工程字段	唯一值 / 执行语义
xiuwei.track[29].cumulative	7960
xiuwei.track[29].reward_zh	边框特效(纯外观)
xiuwei.track[30].level	31
xiuwei.track[30].realm_zh	化神 · 一层
xiuwei.track[30].required_from_previous	750
xiuwei.track[30].cumulative	8710
xiuwei.track[30].reward_zh	境界称号[化神]+专属头像框(纯外观)
xiuwei.track[31].level	32
xiuwei.track[31].realm_zh	化神 · 一层
xiuwei.track[31].required_from_previous	810
xiuwei.track[31].cumulative	9520
xiuwei.track[31].reward_zh	头像框(纯外观)
xiuwei.track[32].level	33
xiuwei.track[32].realm_zh	化神 · 二层
xiuwei.track[32].required_from_previous	880
xiuwei.track[32].cumulative	10400
xiuwei.track[32].reward_zh	称号(纯外观)
xiuwei.track[33].level	34
xiuwei.track[33].realm_zh	化神 · 二层
xiuwei.track[33].required_from_previous	950
xiuwei.track[33].cumulative	11350
xiuwei.track[33].reward_zh	名片底纹(纯外观)
xiuwei.track[34].level	35
xiuwei.track[34].realm_zh	化神 · 三层
xiuwei.track[34].required_from_previous	1030
xiuwei.track[34].cumulative	12380
xiuwei.track[34].reward_zh	边框特效(纯外观)
xiuwei.track[35].level	36
xiuwei.track[35].realm_zh	化神 · 三层
xiuwei.track[35].required_from_previous	1120
xiuwei.track[35].cumulative	13500
xiuwei.track[35].reward_zh	击杀播报样式(纯外观)
xiuwei.track[36].level	37
xiuwei.track[36].realm_zh	大罗 · 一层
xiuwei.track[36].required_from_previous	1220
xiuwei.track[36].cumulative	14720
xiuwei.track[36].reward_zh	境界称号[大罗]+专属头像框(纯外观)

工程字段	唯一值 / 执行语义
xiuwei.track[37].level	38
xiuwei.track[37].realm_zh	大罗 · 一层
xiuwei.track[37].required_from_previous	1320
xiuwei.track[37].cumulative	16040
xiuwei.track[37].reward_zh	称号(纯外观)
xiuwei.track[38].level	39
xiuwei.track[38].realm_zh	大罗 · 二层
xiuwei.track[38].required_from_previous	1430
xiuwei.track[38].cumulative	17470
xiuwei.track[38].reward_zh	名片底纹(纯外观)
xiuwei.track[39].level	40
xiuwei.track[39].realm_zh	大罗 · 二层
xiuwei.track[39].required_from_previous	1550
xiuwei.track[39].cumulative	19020
xiuwei.track[39].reward_zh	边框特效(纯外观)
xiuwei.track[40].level	41
xiuwei.track[40].realm_zh	大罗 · 三层
xiuwei.track[40].required_from_previous	1690
xiuwei.track[40].cumulative	20710
xiuwei.track[40].reward_zh	击杀播报样式(纯外观)
xiuwei.track[41].level	42
xiuwei.track[41].realm_zh	大罗 · 三层
xiuwei.track[41].required_from_previous	1830
xiuwei.track[41].cumulative	22540
xiuwei.track[41].reward_zh	头像框(纯外观)
xiuwei.track[42].level	43
xiuwei.track[42].realm_zh	齐天 · 一层
xiuwei.track[42].required_from_previous	1980
xiuwei.track[42].cumulative	24520
xiuwei.track[42].reward_zh	境界称号[齐天]+专属头像框(纯外观)
xiuwei.track[43].level	44
xiuwei.track[43].realm_zh	齐天 · 一层
xiuwei.track[43].required_from_previous	2150
xiuwei.track[43].cumulative	26670
xiuwei.track[43].reward_zh	名片底纹(纯外观)
xiuwei.track[44].level	45
xiuwei.track[44].realm_zh	齐天 · 二层

工程字段	唯一值 / 执行语义
xiuwei.track[44].required_from_previous	2340
xiuwei.track[44].cumulative	29010
xiuwei.track[44].reward_zh	边框特效(纯外观)
xiuwei.track[45].level	46
xiuwei.track[45].realm_zh	齐天 · 二层
xiuwei.track[45].required_from_previous	2540
xiuwei.track[45].cumulative	31550
xiuwei.track[45].reward_zh	击杀播报样式(纯外观)
xiuwei.track[46].level	47
xiuwei.track[46].realm_zh	齐天 · 三层
xiuwei.track[46].required_from_previous	2750
xiuwei.track[46].cumulative	34300
xiuwei.track[46].reward_zh	头像框(纯外观)
xiuwei.track[47].level	48
xiuwei.track[47].realm_zh	齐天 · 三层
xiuwei.track[47].required_from_previous	2980
xiuwei.track[47].cumulative	37280
xiuwei.track[47].reward_zh	称号(纯外观)
xiuwei.track[48].level	49
xiuwei.track[48].realm_zh	齐天 · 三层
xiuwei.track[48].required_from_previous	3240
xiuwei.track[48].cumulative	40520
xiuwei.track[48].reward_zh	名片底纹(纯外观)
xiuwei.track[49].level	50
xiuwei.track[49].realm_zh	齐天 · 三层
xiuwei.track[49].required_from_previous	3510
xiuwei.track[49].cumulative	44030
xiuwei.track[49].reward_zh	边框特效(纯外观)
collection.state_order	LOCKED / SEEN / USED / MASTERED
collection.monotonic	是
collection.rules.HERO.seen	OWN_OFFER_OR_DIRECT_VISIBLE_CUMULATIVE_1S
collection.rules.HERO.used	FORMAL_SPAWN_AND_ONE_VALID_INPUT
collection.rules.HERO.mastered	FINAL_PLACEMENT_LE_3
collection.rules.ACTIVE_EXCLUSIVE.seen	OWNED_OR_DIRECT_VISIBLE_SUCCESSFUL_CAST
collection.rules.ACTIVE_EXCLUSIVE.used	OWN_SUCCESSFUL_CAST
collection.rules.ACTIVE_EXCLUSIVE.mastered	OWN_SUCCESSFUL_CAST_AND_PLACEMENT_LE_3
collection.rules.ACTIVE_GENERIC.seen	DIRECT_VISIBLE_GROUND_OR_CAST_OR_SHOP_PREVIEW

工程字段	唯一值 / 执行语义
collection.rules.ACTIVE_GENERIC.used	EQUIP_AND_SUCCESSFUL_CAST
collection.rules.ACTIVE_GENERIC.mastered	SUCCESSFUL_CAST_AND_PLACEMENT_LE_3
collection.rules.PASSIVE.seen	DIRECT_VISIBLE_BOOK_OR_REPLACE_DRAWER
collection.rules.PASSIVE.used	LEARN_LEVEL_GTE_1
collection.rules.PASSIVE.mastered	REACH_LEVEL_5
collection.rules.EQUIPMENT.seen	DIRECT_VISIBLE_GROUND_SHOP_OR_LOOT_LIST
collection.rules.EQUIPMENT.used	SUCCESSFULLY_WORN_OR_UNIQUE_EFFECT_TRIGGERED
collection.rules.EQUIPMENT.mastered	PLACEMENT_LE_3_AND(WORN_CUMULATIVE_GTE_120S_OR_UNIQUE_EFFECT_TRIGGERED)
collection.fuzzy_or_fog_information_does_not_count_seen	是
collection.reward_threshold_used_percent	25 / 50 / 75 / 100
collection.rewards_gameplay_power	否
likes.max_targets_per_match	3
likes.self	否
likes.same_target_once	是
likes.window_after_settlement_s	300
likes.withdraw	否
likes.gameplay_reward	0
likes.bot_target	NPC_STAT_ONLY
battle_reports.retain_count	20
battle_reports.story_nodes_max	8
battle_reports.story_nodes_only_from_server_events	是

14.2 图鉴与点赞工程字段

工程字段	唯一值 / 执行语义
collection.machine.states	LOCKED / SEEN / USED / MASTERED
catalogs[0].id	HERO
catalogs[0].count	38
catalogs[0].seen	offered_or_direct_visible_1s
catalogs[0].used	spawned_with_valid_input
catalogs[0].mastered	placement_top3
catalogs[1].id	ACTIVE_EXCLUSIVE
catalogs[1].count	38
catalogs[1].seen	owned_or_direct_visible_cast
catalogs[1].used	successful_cast

工程字段	唯一值 / 执行语义
catalogs[1].mastered	successful_cast_and_top3
catalogs[2].id	ACTIVE_GENERIC
catalogs[2].count	19
catalogs[2].seen	direct_visible_drop_or_cast_or_shop_preview
catalogs[2].used	equipped_and_successful_cast
catalogs[2].mastered	successful_cast_and_top3
catalogs[3].id	PASSIVE
catalogs[3].count	44
catalogs[3].seen	direct_visible_book_or_replace_drawer
catalogs[3].used	learned_level_gte_1
catalogs[3].mastered	reached_level_5
catalogs[4].id	EQUIPMENT
catalogs[4].count	48
catalogs[4].seen	direct_visible_ground_or_shop_or_loot_pile
catalogs[4].used	equipped_or_unique_effect_triggered
catalogs[4].mastered	top3_and_equipped_120s_or_unique_effect_triggered
reward_thresholds_used_percent	25 / 50 / 75 / 100
rewards_gameplay_power	否
directed_drop_rules_visible	否
likes.per_match	3
likes.self_like	否
likes.per_target	1
likes.window_s	300
likes.gameplay_reward	0
likes.bot_target	NPC_STAT_ONLY
battle_report_limit	20
battle_report_story_nodes_max	8

14.3 商业化开关

工程字段	唯一值 / 执行语义
all_heroes_free	是
direct_cosmetic_sales_only	是
paid_random_boxes	否
ads_enabled	否
gameplay_paid_modifiers	否

工程字段	唯一值 / 执行语义
revive_coin	否
fourth_life	否
paid_flying_mount	否
paid_candidate_slots	否
skin_stats	否
paid_drop_rate	否
allowed_cosmetics	HERO_SKIN / AVATAR / AVATAR_FRAME / PROFILE_BACKGROUND / TITLE_STYLE / KILL_BROADCAST_STYLE
skin_may_change_collision	否
skin_may_change_telegraph_geometry	否
skin_may_reduce_visibility	否
real_money_black_mountain	否
launch_matrix.zh_CN.regions	CN=OPEN； 其余地区 CLOSED
launch_matrix.other_locales	全部 CLOSED； 本地化完成+当地法务签字后逐项开启， 每项开启记录签字证据 ID
launch_matrix.region_field_authority	服务端 region 字段； 客户端语言设置不改变地区闭合判定

15 网络、后端、事务、回放与反作弊

- 服务端 20Hz 权威模拟；近战斗 AI20Hz、远 AI5Hz、远召唤 10Hz 更新意图，但命中和伤害仍在权威 tick 结算。
- 客户端只提交移动、普攻、主动、取消、交互和 UI 事务意图；伤害、CD、随机、碰撞、视野、物品、金币、库存与胜负不由客户端决定。
- 所有购买、出售、拾取、技能书替换、主动替换、宝石分配、太白换人、黑山、真死与掉落使用幂等事务 ID 和服务端原子提交。
- 回放保存根种子、分离随机流消耗、输入、伤害、死亡、拾取、交易、商店、怪物、天劫事件和 5 秒快照；同规则版本重放哈希必须一致。
- 单局合作、救人、让资源、共同打 BOSS、背叛、拉怪害人和治疗陌生人均合法。预组队判定只能基于跨局重复模式及设备/IP/匹配证据，不可单局自动处罚。
- 第三次淘汰后观战延迟 5 秒，只显示目标当前直接可见画面与公开信息；不提供自由飞摄、背包、未发现商店或迷雾外实体。

15.1 网络合同

工程字段	唯一值 / 执行语义
server_tick_hz	20
maximum_accepted_past_input_ms	250
basic_attack_history_ms	250
movement_tolerance.window_s	1
movement_tolerance.extra_m	0.7

工程字段	唯一值 / 执行语义
far_ai.decision_may_lower_hz	是
far_ai.server_movement_collision_and_perception_hz	20
far_ai.client_snapshot_can_lower_hz	是
spectator.delay_s	5
spectator.all_information_from_same_delayed_snapshot	是
spectator.report_anchor_uses_original_match_tick	是
lag_comp.locked_cast_rewind_max_ms	250; 仅锁定类技能命中判定允许回看
lag_comp.projectile_no_victim_rewind	是; 投射物自服务端生成 tick 起只向前模拟
lag_comp.movement_walls_never_rewind	是; 位置与墙体状态永不倒带
prediction.local_movement_only	是; 仅本地移动做客户端预测
prediction.error_smooth_le_1_5m_ms	100; 误差 ≤ 1.5 米在 100ms 内平滑纠正
prediction.error_snap_gt_1_5m	立即硬校正到权威位置
clock.input_carries	单调 input_seq + 客户端估算 server_tick
clock.resync_interval_s	5
clock.authority	服务端只信自身 tick; 客户端估算仅用于插值与预测
input.duplicate_or_stale_seq	丢弃; 不产生任何回滚
input.movement_no_backdate	是; 移动输入按到达 tick 生效
input.basic_attack_history_ms	250; 仅用于命中验证, 不回滚受害者状态
input.transaction_at_arrival_tick	是; 交易/拾取/商店永远按服务端收到 tick 结算
interp.display_delay_ms	100; 客户端固定 100ms 表现插值
interp.authority_read_20hz_only	碰撞、距离与命中只读 20Hz 权威位置, 不读插值位置

15.2 后端与账号合同

工程字段	唯一值 / 执行语义
product_id	SANMING_WUCHANG
merge.preview_required	是
merge.commit_requires_preview_token	是
merge.never_silent_overwrite	是
merge.entitlements	SET_UNION_BY_ENTITLEMENT_ID; REVOKED_WINS_IF_SAME_PURCHASE_ID_REFUNDED

工程字段	唯一值 / 执行语义
merge.cosmetics	SET_UNION_BY_COSMETIC_ID
merge.xiuwei	TAKE_GREATER_CUMULATIVE_XIUWEI; DO_NOT_SUM
merge.collection	PER_CONTENT_REF_TAKE_MAX_STATE_ORDER_LOCKED_SEEN_USED_MASTERED
merge.mmr	TAKE_ACCOUNT_WITH_MORE_RATED_MATCHES; TIE_LATER_MMR_UPDATED_AT; TIE_LEXICOGRAPHIC_SMALLER_SOURCE_ACCOUNT_UUID
merge.lifetime_stats	RECOMPUTE_FROM_UNION_OF_UNIQUE_MATCH_IDS; NEVER_SUM_DUPLICATE_MATCH
merge.received_likes	UNION_UNIQUE_LIKE_TRANSACTION_IDS
merge.nickname	IF_DIFFERENT_REQUIRE_EXPLICIT_SOURCE_CHOICE_BEFORE_COMMIT
merge.cloud_settings	PER_FIELD_LATEST_SERVER_UPDATED_AT; TIE_TARGET_ACCOUNT
merge.device_local_settings	NOT_MERGED; KEEP_CURRENT_DEVICE_LOCAL
merge.battle_reports	UNION_BY_MATCH_ID_KEEP_NEWEST_20_BY_SETTLEMENT_TIME
api_groups	SESSION / ACCOUNT_BIND_MERGE_DELETE_EXPORT / MATCHMAKING / MATCH_RECONNECT_SETTLEMENT / PROFILE / COLLECTION / SETTINGS / BATTLE_REPORT / COSMETIC_CATALOG_OWNERSHIP / LIKE / REPORT / PAYMENT
commerce.entities	COSMETIC_CATALOG / ORDER / RECEIPT / ENTITLEMENT
commerce.refund_revokes_entitlement	是
commerce.gameplay_power_sale	否
age_gate_states	UNKNOWN / ADULT / MINOR / RESTRICTED
payment_requires_region_compliance	是
account_delete.cooling_days	7
account_delete.cancel_allowed_during_cooling	是
account_delete.export_available	是
account_delete.final_anonymize_gameplay	是
account_delete.legal_finance_and_anticheat_retention_by_region	是
retention.replay_normal_days	7
retention.replay_flagged_days	180
retention.battle_report_days	30
public_api.POST /v2/session/guest.auth	NONE
public_api.POST /v2/session/guest.idempotent	是
public_api.POST /v2/account/bind.auth	GUEST_OR_GLOBAL
public_api.POST /v2/account/bind.idempotent	是
public_api.POST /v2/account/merge/preview.auth	GLOBAL
public_api.POST /v2/account/merge/preview.idempotent	是
public_api.POST /v2/account/merge/commit.auth	GLOBAL
public_api.POST /v2/account/merge/commit.idempotent	是

工程字段	唯一值 / 执行语义
public_api.POST /v2/account/delete/request.auth	GLOBAL
public_api.POST /v2/account/delete/request.idempotent	是
public_api.POST /v2/account/delete/cancel.auth	GLOBAL
public_api.POST /v2/account/delete/cancel.idempotent	是
public_api.GET /v2/account/export.auth	GLOBAL
public_api.GET /v2/account/export.idempotent	是
public_api.GET /v2/profile/{id}.auth	ANY_ACCOUNT
public_api.GET /v2/profile/{id}.idempotent	是
public_api.PATCH /v2/profile.auth	GLOBAL
public_api.PATCH /v2/profile.idempotent	是
public_api.GET /v2/collection.auth	ANY_ACCOUNT
public_api.GET /v2/collection.idempotent	是
public_api.GET /v2/settings.auth	ANY_ACCOUNT
public_api.GET /v2/settings.idempotent	是
public_api.PUT /v2/settings.auth	ANY_ACCOUNT
public_api.PUT /v2/settings.idempotent	是
public_api.GET /v2/battle-reports.auth	ANY_ACCOUNT
public_api.GET /v2/battle-reports.idempotent	是
public_api.GET /v2/cosmetics/catalog.auth	ANY_ACCOUNT
public_api.GET /v2/cosmetics/catalog.idempotent	是
public_api.GET /v2/cosmetics/owned.auth	GLOBAL
public_api.GET /v2/cosmetics/owned.idempotent	是
public_api.POST /v2/like.auth	GLOBAL
public_api.POST /v2/like.idempotent	是
public_api.POST /v2/matchmaking/enqueue.auth	ANY_ACCOUNT
public_api.POST /v2/matchmaking/enqueue.idempotent	是
public_api.DELETE /v2/matchmaking/enqueue.auth	ANY_ACCOUNT
public_api.DELETE /v2/matchmaking/enqueue.idempotent	是
public_api.POST /v2/match/reconnect.auth	MATCH_SESSION
public_api.POST /v2/match/reconnect.idempotent	是
public_api.GET /v2/match/{id}/settlement.auth	PARTICIPANT
public_api.GET /v2/match/{id}/settlement.idempotent	是
public_api.POST /v2/report.auth	GLOBAL
public_api.POST /v2/report.idempotent	是
public_api.POST /v2/payment/create.auth	GLOBAL

工程字段	唯一值 / 执行语义
public_api.POST /v2/payment/create.idempotent	是
public_api.POST /v2/payment/verify.auth	GLOBAL
public_api.POST /v2/payment/verify.idempotent	是
public_api.POST /v2/payment/refund-webhook.auth	PROVIDER_SIGNATURE
public_api.POST /v2/payment/refund-webhook.idempotent	是
transaction_request_id_required_for_all_POST_PATCH_PUT	是
standard_errors	AUTH_REQUIRED / FORBIDDEN / NOT_FOUND / VERSION_CONFLICT / IDEMPOTENT_REPLAY / RATE_LIMITED / REGION_RESTRICTED / VALIDATION_FAILED
response_envelope.success.fields	request_id / server_time / data / resource_version
response_envelope.error.fields	request_id / server_time / error_code / message_key / retryable / details
concurrency.mutable_resources_require_expected_version	是
concurrency.mismatch_error	VERSION_CONFLICT
concurrency.idempotency_ttl_hours	48
concurrency.same_key_different_body	VALIDATION_FAILED
wire_types.uuid	LOWERCASE_RFC4122_STRING
wire_types.timestamp	UTC_RFC3339_MILLISECONDS
wire_types.money	NONNEGATIVE_INT64
wire_types.version	UINT64
wire_types.content_ref	本文对应章节
request_common.request_id	UUID_REQUIRED_FOR_POST_PUT_PATCH_DELETE
request_common.client_build	SEMPER
request_common.resource_version	UINT64_REQUIRED_FOR_MUTATION_OF_EXISTING_RESOURCE
response_common.success_fields	request_id / server_time / data / resource_version
response_common.error_fields	request_id / server_time / error.code / error.field_path / error.retryable / resource_version
endpoint_schemas.POST /v2/session/guest.request_required	request_id / device_install_id / platform / region
endpoint_schemas.POST /v2/session/guest.response_data	account_id / session_token / expires_at
endpoint_schemas.POST /v2/account/bind.request_required	request_id / provider / provider_token
endpoint_schemas.POST /v2/account/bind.response_data	account_id / bind_state
endpoint_schemas.POST /v2/account/merge/preview.request_required	request_id / source_account_id / target_account_id
endpoint_schemas.POST /v2/account/merge/preview.response_data	preview_token / expires_at / field_results / required_user_choices

工程字段	唯一值 / 执行语义
endpoint_schemas.POST /v2/account/merge/commit.request_required	request_id / preview_token / choices
endpoint_schemas.POST /v2/account/merge/commit.response_data	merged_account_id / merge_receipt_id
endpoint_schemas.POST /v2/matchmaking/enqueue.request_required	request_id / input_mode / region / party_size
endpoint_schemas.POST /v2/matchmaking/enqueue.constraints.party_size	1
endpoint_schemas.POST /v2/matchmaking/enqueue.response_data	ticket_id / estimated_wait_s
endpoint_schemas.DELETE /v2/matchmaking/enqueue.request_required	request_id / ticket_id
endpoint_schemas.DELETE /v2/matchmaking/enqueue.response_data	cancelled
endpoint_schemas.POST /v2/match/reconnect.request_required	request_id / match_id / last_received_server_tick / replay_hash
endpoint_schemas.POST /v2/match/reconnect.response_data	snapshot / events_after_tick / control_state
endpoint_schemas.GET /v2/match/{id}/settlement.response_data	match_id / placement / rewards / battle_report_id / settlement_hash
endpoint_schemas.GET /v2/match/{id}/settlement.path.id	UUID
endpoint_schemas.PUT /v2/settings.request_required	request_id / resource_version / changed_fields
endpoint_schemas.PUT /v2/settings.response_data	settings
endpoint_schemas.POST /v2/like.request_required	request_id / match_id / target_participant_slot
endpoint_schemas.POST /v2/like.response_data	accepted / remaining_match_likes
endpoint_schemas.POST /v2/report.request_required	request_id / match_id / target_participant_slot / category / description
endpoint_schemas.POST /v2/report.response_data	report_id / replay_retention_extended
endpoint_schemas.POST /v2/payment/create.request_required	request_id / catalog_sku / currency / quoted_price_minor
endpoint_schemas.POST /v2/payment/create.response_data	order_id / provider_payload
endpoint_schemas.POST /v2/payment/verify.request_required	request_id / order_id / provider_receipt
endpoint_schemas.POST /v2/payment/verify.response_data	order_state / entitlement_ids
endpoint_schemas.POST /v2/account/delete/request.request_required	request_id / confirm_phrase
endpoint_schemas.POST /v2/account/delete/request.response_data	delete_request_id / cooling_ends_at
endpoint_schemas.POST /v2/account/delete/cancel.request_required	request_id / delete_request_id

工程字段	唯一值 / 执行语义
endpoint_schemas.POST /v2/account/delete/cancel.response_data	cancelled
endpoint_schemas.GET /v2/account/export.request_required	NONE
endpoint_schemas.GET /v2/account/export.response_data	export_job_id / state
endpoint_schemas.GET /v2/account/export.response_optional	download_url_if_ready
endpoint_schemas.GET /v2/profile/{id}.request_required	NONE
endpoint_schemas.GET /v2/profile/{id}.response_data	public_profile
endpoint_schemas.GET /v2/profile/{id}.path.id	UUID
endpoint_schemas.PATCH /v2/profile.request_required	request_id / resource_version / changed_fields
endpoint_schemas.PATCH /v2/profile.response_data	profile
endpoint_schemas.GET /v2/collection.request_required	NONE
endpoint_schemas.GET /v2/collection.response_data	catalog_states / reward_claim_states
endpoint_schemas.GET /v2/settings.request_required	NONE
endpoint_schemas.GET /v2/settings.response_data	settings
endpoint_schemas.GET /v2/battle-reports.request_required	NONE
endpoint_schemas.GET /v2/battle-reports.response_data	reports
endpoint_schemas.GET /v2/battle-reports.response_optional	next_cursor
endpoint_schemas.GET /v2/battle-reports.query.cursor	OPTIONAL_OPAQUE_CURSOR_MAX256
endpoint_schemas.GET /v2/battle-reports.query.limit	UINT8_DEFAULT20_MIN1_MAX50
endpoint_schemas.GET /v2/cosmetics/catalog.request_required	NONE
endpoint_schemas.GET /v2/cosmetics/catalog.response_data	catalog / catalog_version
endpoint_schemas.GET /v2/cosmetics/owned.request_required	NONE
endpoint_schemas.GET /v2/cosmetics/owned.response_data	owned_cosmetic_ids
endpoint_schemas.POST /v2/payment/refund-webhook.request_required	provider_event_id / provider_signature / order_id / refund_state
endpoint_schemas.POST /v2/payment/refund-webhook.response_data	acknowledged
unknown_endpoint_field	REJECT_WITH_INVALID_ARGUMENT
all_public_endpoints_have_schema	是

工程字段	唯一值 / 执行语义
endpoint_field_wire_types.accepted	BOOLEAN
endpoint_field_wire_types.account_id	LOWERCASE_RFC4122_UUID
endpoint_field_wire_types.acknowledged	BOOLEAN
endpoint_field_wire_types.battle_report_id	LOWERCASE_RFC4122_UUID
endpoint_field_wire_types.bind_state	UTF8_NFC_1_TO_128
endpoint_field_wire_types.cancelled	BOOLEAN
endpoint_field_wire_types.catalog	ARRAY_COSMETIC_CATALOG_ENTRY
endpoint_field_wire_types.catalog_sku	UTF8_NFC_1_TO_128
endpoint_field_wire_types.catalog_states	MAP_TYPED_CONTENT_REF_TO_COLLECTION_STATE
endpoint_field_wire_types.catalog_version	INT64
endpoint_field_wire_types.category	UTF8_NFC_1_TO_128
endpoint_field_wire_types.changed_fields	MAP_SCHEMA_FIELD_ID_TO_TYPED_VALUE_MAX64
endpoint_field_wire_types.choices	MAP_MERGE_FIELD_ID_TO_SOURCE_ACCOUNT_ENUM_MAX32
endpoint_field_wire_types.confirm_phrase	UTF8_NFC_1_TO_128
endpoint_field_wire_types.control_state	ENUM_PLAYER_OR_BOT_TAKEOVER
endpoint_field_wire_types.cooling_ends_at	UTC_RFC3339_MILLISECONDS
endpoint_field_wire_types.currency	UTF8_NFC_1_TO_128
endpoint_field_wire_types.delete_request_id	LOWERCASE_RFC4122_UUID
endpoint_field_wire_types.description	UTF8_NFC_MAX500
endpoint_field_wire_types.device_install_id	LOWERCASE_RFC4122_UUID
endpoint_field_wire_types.download_url_if_ready	HTTPS_URL_MAX2048
endpoint_field_wire_types.entitlement_ids	ARRAY_ENTITLEMENT_UUID_MAX128
endpoint_field_wire_types.estimated_wait_s	INT64
endpoint_field_wire_types.events_after_tick	ARRAY_CANONICAL_REPLAY_EVENT_MAX4096
endpoint_field_wire_types.expires_at	UTC_RFC3339_MILLISECONDS
endpoint_field_wire_types.export_job_id	LOWERCASE_RFC4122_UUID
endpoint_field_wire_types.field_results	MAP_MERGE_FIELD_ID_TO_MERGE_RESULT_ENUM
endpoint_field_wire_types.input_mode	UTF8_NFC_1_TO_128
endpoint_field_wire_types.last_received_server_tick	INT64
endpoint_field_wire_types.match_id	LOWERCASE_RFC4122_UUID
endpoint_field_wire_types.merge_receipt_id	LOWERCASE_RFC4122_UUID
endpoint_field_wire_types.merged_account_id	LOWERCASE_RFC4122_UUID
endpoint_field_wire_types.next_cursor	UTF8_NFC_1_TO_128
endpoint_field_wire_types.order_id	LOWERCASE_RFC4122_UUID
endpoint_field_wire_types.order_state	UTF8_NFC_1_TO_128
endpoint_field_wire_types.owned_cosmetic_ids	ARRAY_COSMETIC_ID
endpoint_field_wire_types.party_size	INT64

工程字段	唯一值 / 执行语义
endpoint_field_wire_types.placement	INT64
endpoint_field_wire_types.platform	UTF8_NFC_1_TO_128
endpoint_field_wire_types.preview_token	OPAQUE_UTF8_NFC_MAX8192
endpoint_field_wire_types.profile	PRIVATE_PROFILE_OBJECT
endpoint_field_wire_types.provider	UTF8_NFC_1_TO_128
endpoint_field_wire_types.provider_event_id	LOWERCASE_RFC4122_UUID
endpoint_field_wire_types.provider_payload	PROVIDER_SPECIFIC_JSON_OBJECT_MAX16KB
endpoint_field_wire_types.provider_receipt	OPAQUE_UTF8_NFC_MAX8192
endpoint_field_wire_types.provider_signature	OPAQUE_UTF8_NFC_MAX8192
endpoint_field_wire_types.provider_token	OPAQUE_UTF8_NFC_MAX8192
endpoint_field_wire_types.public_profile	PUBLIC_PROFILE_OBJECT
endpoint_field_wire_types.quoted_price_minor	INT64
endpoint_field_wire_types.refund_state	UTF8_NFC_1_TO_128
endpoint_field_wire_types.region	UTF8_NFC_1_TO_128
endpoint_field_wire_types.remaining_match_likes	INT64
endpoint_field_wire_types.replay_hash	LOWERCASE_HEX_SHA256
endpoint_field_wire_types.replay_retention_extended	BOOLEAN
endpoint_field_wire_types.report_id	LOWERCASE_RFC4122_UUID
endpoint_field_wire_types.reports	ARRAY_BATTLE_REPORT_SUMMARY_MAX50
endpoint_field_wire_types.request_id	LOWERCASE_RFC4122_UUID
endpoint_field_wire_types.required_user_choices	ARRAY_MERGE_FIELD_ID_MAX32
endpoint_field_wire_types.resource_version	INT64
endpoint_field_wire_types.reward_claim_states	MAP_REWARD_ID_TO_CLAIM_STATE
endpoint_field_wire_types.rewards	MATCH_SETTLEMENT_REWARD_OBJECT
endpoint_field_wire_types.session_token	OPAQUE_UTF8_NFC_MAX8192
endpoint_field_wire_types.settings	SETTINGS_OBJECT_SETTINGS_SCHEMA_JSON
endpoint_field_wire_types.settlement_hash	LOWERCASE_HEX_SHA256
endpoint_field_wire_types.snapshot	AUTHORITATIVE_MATCH_SNAPSHOT_BINARY_BASE64_MAX4MB
endpoint_field_wire_types.source_account_id	LOWERCASE_RFC4122_UUID
endpoint_field_wire_types.state	UTF8_NFC_1_TO_128
endpoint_field_wire_types.target_account_id	LOWERCASE_RFC4122_UUID
endpoint_field_wire_types.target_participant_slot	INT64
endpoint_field_wire_types.ticket_id	LOWERCASE_RFC4122_UUID
session.access_token_ttl_s	1800
session.refresh_token_ttl_days	30
session.refresh_rotation	ONE_TIME_REFRESH_TOKEN; REUSE_REVOKES_TOKEN_FAMILY

工程字段	唯一值 / 执行语义
session.logout	REVOKE_CURRENT_SESSION_AND_PUSH_DISCONNECT
session.logout_all	REVOKE_ACCOUNT_SESSION_EPOCH
session.max_concurrent_devices	3
session.fourth_device_login	REQUIRE_USER_CHOOSE_ONE_EXISTING_SESSION_TO_REVOKE
settlement.outbox	TRANSACTIONAL_OUTBOX_WITH_MATCH_SETTLEMENT_COMMIT
settlement.retry	EXPONENTIAL_BACKOFF_1S_TO_300S_UNTIL_ACK; IDEMPOTENT_BY_MATCH_ID_ACCOUNT_ID
server_crash.recovery	RESTORE_LAST_COMMITTED_SNAPSHOT_AND_REPLAY_APPEND_ONLY_EVENTS; ELSE_VOID_ABORT
idempotency.record_ttl_days	7
session.max_devices	3; 第 4 设备登录返回现有会话列表（设备名/最近活跃/地区）
session.revoke_flow	用户明确选择撤销目标→被撤设备立即断线→新会话才签发；无自动踢除
session.revoke_race	同 tick 并发撤销按请求到达序原子处理；失败方收到 SESSION_SLOT_TAKEN

- 端点级错误矩阵：括号内为 retryable 及重试前置条件；未列出的状态码禁止返回。所有写端点必须携带幂等键。

写端点	通用错误（HTTP 语义 retryable）	端点特有错误	幂等冲突结果
POST /v2/session/guest	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_after) / 500 INTERNAL(yes,idempotent_replay) / 503 UNAVAILABLE(yes,backoff)	409 SESSION_SLOT_TAKEN(no,list_sessions_first)	幂等键重放返回首次结果，

			不重复执行副作用
POST /v2/account/bind	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_after) / 500 INTERNAL(yes,idempotent_replay) / 503 UNAVAILABLE(yes,backoff)	409 ALREADY_BOUND(no) / 409 IDENTITY_IN_USE(no,merge_preview_first)	幂等键重放返回首次结果，不重复执行副作用
POST /v2/account/merge/preview	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_after) / 500 INTERNAL(yes,idempotent_replay)	409 MERGE_CONFLICT(no,returns_conflict_report)	幂等键重放返

	) / 503 UNAVAILABLE(yes,backoff)		回 首 次 结 果 ， 不 重 复 执 行 副 作 用
POST /v2/account/merge/commit	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_after) / 500 INTERNAL(yes,idempotent_replay) / 503 UNAVAILABLE(yes,backoff)	409 PREVIEW_STALE(no,re_preview) / 409 MERGE_IN_PROGRESS(no,poll_status)	幂 等 键 重 放 返 回 首 次 结 果 ， 不 重 复 执 行 副 作 用

POST /v2/account/delete/request	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_after) / 500 INTERNAL(yes,idempotent_replay) / 503 UNAVAILABLE(yes,backoff)	409 DELETE_ALREADY_PENDING(no)	幂等键重放返回首次结果，不重复执行副作用
POST /v2/account/delete/cancel	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_after) / 500 INTERNAL(yes,idempotent_replay) / 503 UNAVAILABLE(yes,backoff)	409 NO_PENDING_DELETE(no)	幂等键重放返回首次结果，不重

			复 执 行 副 作 用
PATCH /v2/profile	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_aft er) / 500 INTERNAL(yes,idempotent_replay) / 503 UNAVAILABLE(yes,backoff)	409 NAME_TAKEN(no) / 422 NAME_POLICY_REJECTED(no)	幂 等 键 重 放 返 回 首 次 结 果 , 不 重 复 执 行 副 作 用
PUT /v2/settings	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_aft er) / 500 INTERNAL(yes,idempotent_replay) / 503 UNAVAILABLE(yes,backoff)	409 IDEMPOTENCY_CONFLICT(no,server_returns_first_res ult)	幂 等 键 重 放 返 回 首

			次结果，不重复执行副作用
POST /v2/like	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_after) / 500 INTERNAL(yes,idempotent_replay) / 503 UNAVAILABLE(yes,backoff)	409 IDEMPOTENCY_CONFLICT(no,server_returns_first_result)	幂等键重放返回首次结果，不重复执行副作用
POST /v2/matchmaking/enqueue	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403	409 IDEMPOTENCY_CONFLICT(no,server_returns_first_res	幂等

e	FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_after) / 500 INTERNAL(yes,idempotent_replay) / 503 UNAVAILABLE(yes,backoff)	ult)	键 重 放 返 回 首 次 结 果 , 不 重 复 执 行 副 作 用
DELETE /v2/matchmaking/enqueue	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_after) / 500 INTERNAL(yes,idempotent_replay) / 503 UNAVAILABLE(yes,backoff)	409 IDEMPOTENCY_CONFLICT(no,server_returns_first_result)	幂 等 键 重 放 返 回 首 次 结 果 , 不 重 复 执

			行 副 作 用
POST /v2/match/reconnect	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_aft er) / 500 INTERNAL(yes,idempotent_replay) / 503 UNAVAILABLE(yes,backoff)	409 IDEMPOTENCY_CONFLICT(no,server_returns_first_res ult)	幂 等 键 重 放 返 回 首 次 结 果 , 不 重 复 执 行 副 作 用
POST /v2/report	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_aft er) / 500 INTERNAL(yes,idempotent_replay) / 503 UNAVAILABLE(yes,backoff)	409 IDEMPOTENCY_CONFLICT(no,server_returns_first_res ult)	幂 等 键 重 放 返 回 首 次 结

			果，不重复执行副作用
POST /v2/payment/create	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_after) / 500 INTERNAL(yes,idempotent_replay) / 503 UNAVAILABLE(yes,backoff)	409 IDEMPOTENCY_CONFLICT(no,server_returns_first_result)	幂等键重放返回首次结果，不重复执行副作用
POST /v2/payment/verify	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_after)	409 IDEMPOTENCY_CONFLICT(no,server_returns_first_result)	幂等键重

	er) / 500 INTERNAL(yes,idempotent_replay) / 503 UNAVAILABLE(yes,backoff)		放 返 回 首 次 结 果 ， 不 重 复 执 行 副 作 用
POST /v2/payment/refund- webhook	400 INVALID_ARGUMENT(no) / 401 UNAUTHENTICATED(no) / 403 FORBIDDEN(no) / 429 RATE_LIMITED(yes,after_retry_aft er) / 500 INTERNAL(yes,idempotent_replay)) / 503 UNAVAILABLE(yes,backoff)	409 IDEMPOTENCY_CONFLICT(no,server_returns_first_res ult)	幂 等 键 重 放 返 回 首 次 结 果 ， 不 重 复 执 行 副

			作用
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15.3 对局事务合同

工程字段	唯一值 / 执行语义
common.transaction_id	MATCH_ID:ACCOUNT_ID:CLIENT_REQUEST_UUID
common.idempotent	是
common.server_receive_sequence	MONOTONIC_UINT64_PER_MATCH
common.lock_order	WORLD_ENTITY_ID_ASC_THEN_ACCOUNT_ID_ASC
pickup.steps	LOCK_ITEM_INSTANCE / VALIDATE_STILL_GROUND / VALIDATE_DISTANCE_LOS_LIFE_STATE / VALIDATE_DESTINATION_OR_REQUIRED_REPLACE_SELECTION / REMOVE_FROM_GROUND / WRITE_PLAYER_STATE / COMMIT_LEDGER / UNLOCK
pickup.concurrent_winner	LOWEST_SERVER_RECEIVE_SEQUENCE
pickup.los_required	是
pickup.distance_m	2
skill_book.empty_passive_slot	CONSUME_BOOK_AND_ADD_SAME_ID_LEVEL1
skill_book.same_id_not_level5	CONSUME_BOOK_AND_INCREMENT_ONE_LEVEL
skill_book.same_id_level5	REJECT; BOOK_REMAINS_GROUND
skill_book.different_id_all_four_slots_occupied	REQUIRE_EXPLICIT_OLD_SLOT; ATOMIC_REMOVE_OLD_SKILL_AND_ALL_ITS_TEMP_ENTITIES_AND_PERMANENT_LAYERS; CONSUME_NEW_BOOK; ADD_NEW_ID_LEVEL1
skill_book.cancel	BOOK_REMAINS_GROUND_PUBLIC
skill_book.book_stolen_before_confirm	FAIL_ITEM_GONE; KEEP_OLD_BUILD
active_replace.empty_slot	CONSUME_AND_SET_READY
active_replace.occupied	REQUIRE_CONFIRM; NEW_ACTIVE_STARTS_FULL_COOLDOWN
active_replace.true_death_after_replace	CURRENT_ACTIVE_BECOMES_READY
active_replace.taibai_new_exclusive	READY
active_replace.stolen_before_confirm	KEEP_OLD_ACTIVE
equipment.worn_slots	3
equipment.hand_slots_base	1
equipment.same_name_worn_unique	是
equipment.same_name_may_be_held_or_sold	是
equipment.full	REQUIRE_EXPLICIT_DROP_OR_REPLACE_SELECTION_IN_SAME_TRANSACTION
equipment.extra_hand_from_P8	1
equipment.capacity_recheck_after_removing_P8	IF_TWO_HAND_ITEMS, DROP_MOST_RECENTLY_ACQUIRED_NON_P8_INSTANCE_PUBLIC_AT_FEET; TIE_INSTANCE_ID_DESC
gem.goes_to_hud_bank	是
gem.ground_drop_allowed	否

工程字段	唯一值 / 执行语义
gem.all_passives_level5	REJECT_PICKUP_OR_PURCHASE
true_death_cancels_uncommitted_modal_transactions	是
sell.steps	LOCK_SHOP_PLAYER_AND_ITEM / VALIDATE_OPEN_DISTANCE_OUT_OF_PVP_AND_OWNERSHIP / SNAPSHOT_BASE_AND_B35_MODIFIER / RECONFIRM_IF_PURPLE_OR_GOLD / REMOVE_ITEM_FROM_SLOT / DESTROY_ITEM_INSTANCE / CREATE_GOLD / COMMIT_LEDGER
sell.cancel	NO_CHANGE
sell.disconnect_before_commit	ABORT
sell.disconnect_after_commit	IDEMPOTENT_RESULT
purchase.authority	§ 12.3 及 § E.17 商店规则
purchase.window	REQUEST_MUST_COMMIT_INSIDE_HALF_OPEN_INSTANCE_WINDOW
purchase.inventory_full	DROP_OR_REPLACE_SIDE_EFFECT_OCCURS_ONLY_INSIDE_SUCCESSFUL_PURCHASE_COMMIT
purchase.stale_or_sold_out	NO_GOLD_NO_ITEM_NO_OLD_ITEM_SIDE_EFFECT
taibai_swap.authority	§ 12.3 及 § E.17 商店规则
taibai_swap.channel_is_not_transaction_commit	是
taibai_swap.gold_and_service_cooldown_start_only_at_successful_swap_commit	是
hero_draft.authority	§ 2 及附录 A 英雄成长规则
hero_draft.reroll_and_timeout_race	SERVER_COMMIT_TICK_BEFORE_TIMEOUT_WINS; OTHERWISE_TIMEOUT_LEFTMOST

15.4 确定性与随机合同

工程字段	唯一值 / 执行语义
entity_id	SERVER_MONOTONIC_UINT64_PER_MATCH; NEVER_REUSED
event_id	MATCH_ID:SERVER_TICK:EVENT_SEQUENCE_UINT32
same_tick_order	EVENT_SEQUENCE_ASC
canonical_json	RFC8785_JCS_UTF8
replay_hash.algorithm	BLAKE3
replay_hash.domain	SANMING_WUCHANG_REPLAY_V3_0
replay_hash.input	LEN_PREFIX(DOMAIN) LEN_PREFIX(PREV_HASH_32) LEN_PREFIX(CANONICAL_EVENT_BYTES)
snapshot_hash.algorithm	BLAKE3
snapshot_hash.domain	SANMING_WUCHANG_SNAPSHOT_V3_0
snapshot_hash.entity_order	ENTITY_ID_ASC
snapshot_hash.component_order	COMPONENT_TYPE_ID_ASC
snapshot_hash.numeric_contract	本文对应章节
ruleset_compatibility	SERVER_LOADS_EXACT_RULESET_HASH_RECORDED_BY_MATCH; NO_LIVE_HOT_SWAP
content_mismatch	REJECT_JOIN_AND_REQUEST_PATCH
optimistic_version	ALL_MUTABLE_BACKEND_RESOURCES_USE_ETAG_OR_VERSION_UINT64

工程字段	唯一值 / 执行语义
idempotency_ttl_hours	48
simulation_numeric.position	MM_INT64
simulation_numeric.velocity	MM_PER_S_INT64
simulation_numeric.time	20HZ_TICK_UINT64
simulation_numeric.damage_formula	Q4_INT64
simulation_numeric.hp	INT64
simulation_numeric.probability	PPM_UINT32
simulation_numeric.floating_point_in_authoritative_simulation	否
domain_compat.legacy_v2_5_replays	只读解析器按 V2_5 旧 schema 独立命名空间加载；禁止与 V3_0 域混合校验或续写
domain_compat.v3_0_reader_rejects_v2_5	是；域字符串不匹配直接拒绝并报 DOMAIN_MISMATCH

15.5 随机流合同

工程字段	唯一值 / 执行语义
root_seed.algorithm	HMAC_SHA256
root_seed.key	SERVER_SECRET_32_BYTES
root_seed.message_fields	DOMAIN_UTF8=SANMING_WUCHANG_ROOT_V3_0 / MATCH_ID_UTF8 / RULESET_HASH_HEX_UTF8
root_seed.encoding	UINT32_BE_LENGTH_PREFIX_EACH_FIELD_THEN_RAW_BYTES
stream_seed.algorithm	HMAC_SHA256
stream_seed.key	ROOT_SEED_32_BYTES
stream_seed.message_fields	DOMAIN_UTF8=STREAM / STREAM_ID_UTF8
stream_seed.encoding	UINT32_BE_LENGTH_PREFIX_EACH_FIELD_THEN_RAW_BYTES
algorithm	PCG64_XSL_RR_128_64
streams	MAP_LAYOUT / MAP_FEATURE / SPAWN / REVIVE / SHOP / HERO_DRAFT / MONSTER / DROP / COMBAT_PROC / ACTIVE_SKILL / BOSS / BLACK_MOUNTAIN / AIR_DROP / STORM / BOT
consumption	ONE_DRAW_ONLY_WHEN_AUTHORITATIVE_EVENT_COMMITS; VALIDATION_FAILURE_CONSUMES_ZERO
shuffle	FISHER_YATES_WITH_STREAM
weighted_draw	POSITIVE_UINT64_WEIGHTS; SUM_MUST_FIT_UINT64; UNIFORM_INTEGER_HALF_OPEN_0_TO_SUM_MINUS_1_USING_REJECTION; FIRST_CUMULATIVE_GT_DRAW
replay_records	STREAM_ID / DRAW_ORDINAL_UINT64 / EVENT_ID / RESULT_OR_SHA256_RESULT
client_never_receives_future_stream_state	是
integer_uniform	REJECTION_SAMPLING_FROM_UINT64; REJECT_TAIL_ABOVE_LARGEST_MULTIPLE_OF_RANGE; MODULO_BIAS_FORBIDDEN

工程字段	唯一值 / 执行语义
validation_failure_consumes_draw	否
storm_consumption	AT_EACH_3S_EPOCH, ELIGIBLE_PLAYER_ENTITY_ID_ASC; ONE_DRAW_PER_ELIGIBLE_PLAYER; INELIGIBLE_PLAYER_CONSUMES_ZERO

15.6 回放、举报与反作弊

工程字段	唯一值 / 执行语义
server_tick_hz	20
rate_limits_per_s.movement	20
rate_limits_per_s.basic_attack	12
rate_limits_per_s.cast_cancel	10
rate_limits_per_s.interact	8
rate_limits_per_s.transaction	4
replay.normal_retention_days	7
replay.flagged_retention_days	180
replay.snapshot_interval_s	5
replay.report_pre_seconds	60
replay.report_post_seconds	15
replay.deterministic_hash_required_for_punishment	是
replay.hash_algorithm	BLAKE3_CHAINED_CANONICAL_EVENTS
replay.stores_ruleset_hash	是
replay.stores_rng_draw_ordinals	是
spectator.entry_delay_s	5
spectator.show_private_minimap	否
spectator.show_private_inventory	否
spectator.free_camera	否
preteam_review.minimum_repeat_matches	5
preteam_review.single_match_cooperation_legal	是
preteam_review.automatic_punishment	否
report_categories	CHEAT / REPEATED_PRETEAM / OFFENSIVE_NAME / EXPLOIT / OTHER
client_authoritative_fields	NONE
appeal_required	是
movement_tolerance.window_s	1
movement_tolerance.cumulative_extra_m	0.7
movement_tolerance.not_granted_per_tick	是
evidence_pack.contents	回放段(前 60s 后 15s) / 判定规则版本 / 检测器输出 / RNG 序数 / 服务器 tick

工程字段	唯一值 / 执行语义
	区间 / 处罚决定链
evidence_pack.retention_days	180; 与 flagged 回放同期
appeal.states	SUBMITTED → AUTO_RECHECK → HUMAN_REVIEW → UPHELD OVERTURNED; 每次迁移落档并通知
appeal.human_review_required_for	任何账号级处罚（封禁/降信誉）；对局内自动踢出可仅自动复核
appeal.overturn_transaction	撤销处罚原子恢复：信誉分 / 段位 / 被扣发奖励；恢复记录进不可变审计账本
appeal.deadline_days	30; 超期归档不再受理
residual_risk.spectator_voice_relay	已接受：5 秒观战延迟无法阻止场外语音报点；不因此限制玩家自由；不再追加限制

16 美术、怪物形象、音频、本地化与资产规范

- 怪物 38 套概念形象已按机械角色绑定：普通近战 10、普通远程 4、飞行 4、首领 4、五行肥猪 5、五行龙王 5、核心 BOSS6。形象皮肤不得复制或改写机械碰撞与数值。
- 红鳞大蟒、黄狮精、如意真仙、玉面狐狸均为普通/远程单位，不属于肥猪；五行肥猪仅限金肥猪、木肥猪、水肥猪、火肥猪、土肥猪。
- 3D 生产需为每个单位提供模型、骨架、动画、碰撞代理、受击体、攻击/技能帧、LOD 与材质预算。皮肤和概念图只决定外观，不决定碰撞。
- 重要预警必须视觉+音频双表达；迷雾外声音最多给模糊方向，不得泄露精确身份/坐标；伏石内普通移动脚步对圈外静音，攻击/施法/受击后恢复。
- 源音频 48kHz/24bit WAV；移动端最大 32 voice，PC 最大 48；致命预警、硬控、核心 BOSS 前摇和天劫提示不可被声道淘汰。
- 中文为首发权威文本；UI 字符串使用稳定 key；禁止运行时解析展示文案；任何本地化必须保持数值、单位、五行符号和预警等级一致。

16.1 怪物视觉映射

资产 ID	游戏名	机械角色	战斗模板	五行	状态
MVS_001	巴山虎	ORDINARY_MELEE	MEL_BY_ZONE	NONE	锁定
MVS_002	苍狼精	ORDINARY_MELEE	MEL_BY_ZONE	NONE	锁定
MVS_003	刁钻古怪	ORDINARY_MELEE	MEL_BY_ZONE	NONE	锁定
MVS_004	古怪刁钻	ORDINARY_MELEE	MEL_BY_ZONE	NONE	锁定
MVS_005	虎精小妖	ORDINARY_MELEE	MEL_BY_ZONE	NONE	锁定
MVS_006	黄狮精	ORDINARY_MELEE	MEL_BY_ZONE	NONE	锁定
MVS_007	蛇精小妖	ORDINARY_MELEE	MEL_BY_ZONE	NONE	锁定
MVS_008	树鬼	ORDINARY_MELEE	MEL_BY_ZONE	NONE	锁定

资产 ID	游戏名	机械角色	战斗模板	五行	状态
MVS_009	蜘蛛仔	ORDINARY_MELEE	MEL_BY_ZONE	NONE	锁定
MVS_010	红鳞大蟒	ORDINARY_MELEE	MEL_BY_ZONE	NONE	锁定
MVS_011	玉面狐狸	ORDINARY_RANGED	RNG_BY_ZONE	NONE	锁定
MVS_012	如意真仙	ORDINARY_RANGED	RNG_BY_ZONE	NONE	锁定
MVS_013	火鸦精	ORDINARY_RANGED	RNG_BY_ZONE	NONE	锁定
MVS_014	倚海龙	ORDINARY_RANGED	RNG_BY_ZONE	NONE	锁定
MVS_015	碧金睛兽	FLYING	FLY_BY_ZONE	NONE	锁定
MVS_016	晦月魔君	FLYING	FLY_BY_ZONE	NONE	锁定
MVS_017	混天大圣	FLYING	FLY_BY_ZONE	NONE	锁定
MVS_018	孔雀公主	FLYING	FLY_BY_ZONE	NONE	锁定
MVS_019	特处士	LEADER	LEADER_TANK	NONE	锁定
MVS_020	熊山君	LEADER	LEADER_TANK	NONE	锁定
MVS_021	南山大王	LEADER	LEADER_DPS	NONE	锁定
MVS_022	寅将军	LEADER	LEADER_DPS	NONE	锁定
MVS_023	金肥猪	PIG	PIG_METAL	METAL	锁定
MVS_024	木肥猪	PIG	PIG_WOOD	WOOD	锁定
MVS_025	水肥猪	PIG	PIG_WATER	WATER	锁定
MVS_026	火肥猪	PIG	PIG_FIRE	FIRE	锁定
MVS_027	土肥猪	PIG	PIG_EARTH	EARTH	锁定
MVS_028	金龙王	DRAGON_MINIBOSS	DRAGON_METAL	METAL	锁定
MVS_029	木龙王	DRAGON_MINIBOSS	DRAGON_WOOD	WOOD	锁定
MVS_030	水龙王	DRAGON_MINIBOSS	DRAGON_WATER	WATER	锁定
MVS_031	火龙王	DRAGON_MINIBOSS	DRAGON_FIRE	FIRE	锁定
MVS_032	土龙王	DRAGON_MINIBOSS	DRAGON_EARTH	EARTH	锁定
MVS_033	白鹿魔王	CORE_BOSS	CORE_V1_EARTHBREAK	NONE	锁定
MVS_034	辟暑大王	CORE_BOSS	CORE_V2_FIRELANE	NONE	锁定
MVS_035	地涌夫人	CORE_BOSS	CORE_V3_POISONPOOL	NONE	锁定
MVS_036	辟尘大王	CORE_BOSS	CORE_V4_WINDCHARGE	NONE	锁定
MVS_037	辟寒大王	CORE_BOSS	CORE_V5_THUNDERCHAIN	NONE	锁定
MVS_038	九灵元圣	CORE_BOSS	CORE_V6_MIRRORSHADOW	NONE	锁定

16.1.1 38 套怪物概念形象

以下概念图与 MVS 资产 ID 一一对应，只锁定轮廓、气质、五行辨识和角色归属；最终 3D 模型仍须服从碰撞体、攻击前摇、受击体与 LOD 预算。



MVS_001 巴山虎



MVS_002 苍狼精



MVS_003 刁钻古怪



MVS_004 古怪刁钻



MVS_005 虎精小妖



MVS_006 黄狮精



MVS_007 蛇精小妖



MVS_008 树鬼



MVS_009 蜘蛛仔



MVS_010 红鳞大蟒



MVS_011 玉面狐狸



MVS_012 如意真仙



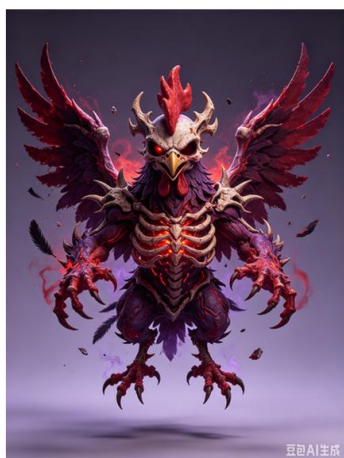
MVS_013 火鸦精



MVS_014 倚海龙



MVS_015 碧水金睛兽



MVS_016 晦月魔君



MVS_017 混天大圣



MVS_018 孔雀公主



MVS_019 特处士



MVS_020 熊山君



MVS_021 南山大王



MVS_022 寅将军



MVS_023 金肥猪



MVS_024 木肥猪



MVS_025 水肥猪



MVS_026 火肥猪



MVS_027 土肥猪



MVS_028 金龙王



MVS_029 木龙王



MVS_030 水龙王



MVS_031 火龙王



MVS_032 土龙王



MVS_033 白鹿魔王



MVS_034 辟暑大王



MVS_035 地涌夫人



MVS_036 辟尘大王



MVS_037 辟寒大王



MVS_038 九灵元圣

- 概念资产入库清单：38 套概念图必须补录源文件名与 SHA-256 并通过版权审查；G_CONCEPT_ASSET_MANIFEST 未闭合前禁止对外发布使用该形象的任何物料。

资产 ID	游戏名	概念源文件名	源文件 SHA-256	版权状态	派生 3D 资产
MVS_001	巴山虎	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_001_BUNDLE
MVS_002	苍狼精	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_002_BUNDLE
MVS_003	刁钻古怪	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_003_BUNDLE
MVS_004	古怪刁钻	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_004_BUNDLE
MVS_005	虎精小妖	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_005_BUNDLE
MVS_006	黄狮精	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_006_BUNDLE
MVS_007	蛇精小妖	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_007_BUNDLE
MVS_008	树鬼	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_008_BUNDLE
MVS_009	蜘蛛仔	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_009_BUNDLE

MVS_010	红鳞大蟒	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_010_BUNDLE
MVS_011	玉面狐狸	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_011_BUNDLE
MVS_012	如意真仙	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_012_BUNDLE
MVS_013	火鸦精	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_013_BUNDLE
MVS_014	倚海龙	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_014_BUNDLE
MVS_015	碧水金睛兽	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_015_BUNDLE
MVS_016	晦月魔君	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_016_BUNDLE
MVS_017	混天大圣	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_017_BUNDLE
MVS_018	孔雀公主	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_018_BUNDLE
MVS_019	特处士	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_019_BUNDLE
MVS_020	熊山君	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_020_BUNDLE
MVS_021	南山大王	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_021_BUNDLE
MVS_022	寅将军	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_022_BUNDLE
MVS_023	金肥猪	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_023_BUNDLE
MVS_024	木肥猪	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_024_BUNDLE
MVS_025	水肥猪	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_025_BUNDLE
MVS_026	火肥猪	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_026_BUNDLE
MVS_027	土肥猪	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_027_BUNDLE
MVS_028	金龙王	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_028_BUNDLE
MVS_029	木龙王	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_029_BUNDLE
MVS_030	水龙王	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_030_BUNDLE
MVS_031	火龙王	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_031_BUNDLE
MVS_032	土龙王	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_032_BUNDLE
MVS_033	白鹿魔王	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_033_BUNDLE
MVS_034	辟暑大王	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_034_BUNDLE
MVS_035	地涌夫人	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_035_BUNDLE
MVS_036	辟尘大王	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_036_BUNDLE
MVS_037	辟寒大王	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_037_BUNDLE

MVS_038	九灵元圣	NOT_INGESTED	PENDING_INGEST	PENDING_RIGHTS_REVIEW	MON_MVS_038_BUNDLE
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16.2 音频规则与台账

audio_id	category	owner_id	trigger	spatial	max_distance_m	max_instances	priority	duck_db	min_variants	status
SFX_ACT_H001_BUNDLE	ACTIVE_SKILL	H001	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H002_BUNDLE	ACTIVE_SKILL	H002	CAST/IMPACT_OR_ACTIVATE/END/LOOP	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H003_BUNDLE	ACTIVE_SKILL	H003	CAST/IMPACT_OR_ACTIVATE/END/LOOP	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H004_BUNDLE	ACTIVE_SKILL	H004	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H005_BUNDLE	ACTIVE_SKILL	H005	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H006_BUNDLE	ACTIVE_SKILL	H006	CAST/IMPACT_OR_ACTIVATE/END/LOOP	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H007_BUNDLE	ACTIVE_SKILL	H007	CAST/IMPACT_OR_ACTIVATE/END/LOOP	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H008_BUNDLE	ACTIVE_SKILL	H008	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H009_BUNDLE	ACTIVE_SKILL	H009	CAST/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H010_BUNDLE	ACTIVE_SKILL	H010	CAST	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H011_BUNDLE	ACTIVE_SKILL	H011	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H012_BUNDLE	ACTIVE_SKILL	H012	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H013_BUNDLE	ACTIVE_SKILL	H013	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H014_BUNDLE	ACTIVE_SKILL	H014	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H015_BUNDLE	ACTIVE_SKILL	H015	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE

audio_id	category	owner_id	trigger	spatial	max_distance_ m	max_instances	priority	duck_db	min_variants	status
SFX_ACT_H016_BUNDLE	ACTIVE_SKILL	H016	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H017_BUNDLE	ACTIVE_SKILL	H017	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H018_BUNDLE	ACTIVE_SKILL	H018	CAST/IMPACT_OR_ACTIVATE/END/LOOP	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H019_BUNDLE	ACTIVE_SKILL	H019	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H020_BUNDLE	ACTIVE_SKILL	H020	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H021_BUNDLE	ACTIVE_SKILL	H021	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H022_BUNDLE	ACTIVE_SKILL	H022	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H023_BUNDLE	ACTIVE_SKILL	H023	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H024_BUNDLE	ACTIVE_SKILL	H024	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H025_BUNDLE	ACTIVE_SKILL	H025	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H026_BUNDLE	ACTIVE_SKILL	H026	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H027_BUNDLE	ACTIVE_SKILL	H027	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H028_BUNDLE	ACTIVE_SKILL	H028	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H029_BUNDLE	ACTIVE_SKILL	H029	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H030_BUNDLE	ACTIVE_SKILL	H030	CAST/IMPACT_OR_ACTIVATE/END/LOOP	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H031_BUNDLE	ACTIVE_SKILL	H031	CAST/IMPACT_OR_ACTIVATE/END/LOOP	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H032_BUNDLE	ACTIVE_SKILL	H032	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE

audio_id	category	owner_id	trigger	spatial	max_distance_ m	max_instances	priority	duck_db	min_variants	status
SFX_ACT_H033_BUNDLE	ACTIVE_SKILL	H033	CAST/IMPACT_OR_ACTIVATE/END/LOOP	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H034_BUNDLE	ACTIVE_SKILL	H034	CAST/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H035_BUNDLE	ACTIVE_SKILL	H035	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H036_BUNDLE	ACTIVE_SKILL	H036	CAST/IMPACT_OR_ACTIVATE/END/LOOP	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H037_BUNDLE	ACTIVE_SKILL	H037	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_H038_BUNDLE	ACTIVE_SKILL	H038	CAST/IMPACT_OR_ACTIVATE/END/LOOP	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D1_BUNDLE	ACTIVE_SKILL	D1	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D3_BUNDLE	ACTIVE_SKILL	D3	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D4_BUNDLE	ACTIVE_SKILL	D4	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D6_BUNDLE	ACTIVE_SKILL	D6	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D7_BUNDLE	ACTIVE_SKILL	D7	CAST/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D9_BUNDLE	ACTIVE_SKILL	D9	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D10_BUNDLE	ACTIVE_SKILL	D10	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D11_BUNDLE	ACTIVE_SKILL	D11	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D12_BUNDLE	ACTIVE_SKILL	D12	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D13_BUNDLE	ACTIVE_SKILL	D13	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D14_BUNDLE	ACTIVE_SKILL	D14	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE

audio_id	category	owner_id	trigger	spatial	max_distance_ m	max_instances	priority	duck_db	min_variants	status
SFX_ACT_D15_BUNDLE	ACTIVE_SKILL	D15	CAST/IMPACT_OR_ACTIVATE	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D16_BUNDLE	ACTIVE_SKILL	D16	CAST	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D17_BUNDLE	ACTIVE_SKILL	D17	CAST	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D18_BUNDLE	ACTIVE_SKILL	D18	CAST/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D19_BUNDLE	ACTIVE_SKILL	D19	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D20_BUNDLE	ACTIVE_SKILL	D20	CAST/IMPACT_OR_ACTIVATE/END/LOOP	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D21_BUNDLE	ACTIVE_SKILL	D21	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_ACT_D22_BUNDLE	ACTIVE_SKILL	D22	CAST/IMPACT_OR_ACTIVATE/END	3D	35	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_MON_MEL_BUNDLE	MONSTER	MEL	DISCOVER/ATTACK/SKILL/HIT/DEATH	3D	20	4	MEDIUM	0	3	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_MON_RNG_BUNDLE	MONSTER	RNG	DISCOVER/ATTACK/SKILL/HIT/DEATH	3D	20	4	MEDIUM	0	3	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_MON_FLY_BUNDLE	MONSTER	FLY	DISCOVER/ATTACK/SKILL/HIT/DEATH	3D	20	4	MEDIUM	0	3	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_MON_PIG_BUNDLE	MONSTER	PIG	DISCOVER/ATTACK/SKILL/HIT/DEATH	3D	20	4	MEDIUM	0	3	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_MON_ELITE_TANK_BUNDLE	MONSTER	ELITE_TANK	DISCOVER/ATTACK/SKILL/HIT/DEATH	3D	45	4	MEDIUM	0	3	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_MON_ELITE_DPS_BUNDLE	MONSTER	ELITE_DPS	DISCOVER/ATTACK/SKILL/HIT/DEATH	3D	45	4	MEDIUM	0	3	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_MON_DRAGON_BUNDLE	MONSTER	DRAGON	DISCOVER/ATTACK/SKILL/HIT/DEATH	3D	45	4	MEDIUM	0	3	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_MON_CORE_BUNDLE	MONSTER	CORE	DISCOVER/ATTACK/SKILL/HIT/DEATH	3D	60	4	CRITICAL	-4	3	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
UI_CONFIRM	UI	SYSTEM	UI_CONFIRM	2D	0	2	HIGH	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE

audio_id	category	owner_id	trigger	spatial	max_distance_m	max_instances	priority	duck_db	min_variants	status
UI_CANCEL	UI	SYSTEM	UI_CANCEL	2D	0	2	HIGH	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
UI_ERROR	UI	SYSTEM	UI_ERROR	2D	0	2	HIGH	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
UI_PICKUP	UI	SYSTEM	UI_PICKUP	2D	0	2	HIGH	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
UI_BOOK	UI	SYSTEM	UI_BOOK	2D	0	2	HIGH	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
UI_EQUIP	UI	SYSTEM	UI_EQUIP	2D	0	2	HIGH	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
UI_SHOP_BUY	UI	SYSTEM	UI_SHOP_BUY	2D	0	2	HIGH	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
UI_SHOP_SELL	UI	SYSTEM	UI_SHOP_SELL	2D	0	2	HIGH	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
UI_RESPAWN	UI	SYSTEM	UI_RESPAWN	2D	0	2	HIGH	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
UI_RESULT	UI	SYSTEM	UI_RESULT	2D	0	2	HIGH	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
AMB_LOBBY	AMBIENCE	SYSTEM	REGION_STATE	2D/3D	30	2	LOW	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
AMB_MAP_GENERAL	AMBIENCE	SYSTEM	REGION_STATE	2D/3D	30	2	LOW	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
AMB_DRAGON_REGION	AMBIENCE	SYSTEM	REGION_STATE	2D/3D	30	2	LOW	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
AMB_THREE_COURTS	AMBIENCE	SYSTEM	REGION_STATE	2D/3D	30	2	LOW	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
AMB_STORM	AMBIENCE	SYSTEM	REGION_STATE	2D/3D	30	2	LOW	0	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
MUS_LOBBY	MUSIC	SYSTEM	GAME_PHASE	2D	0	1	LOW	-6	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
MUS_MATCH	MUSIC	SYSTEM	GAME_PHASE	2D	0	1	LOW	-6	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
MUS_FINAL	MUSIC	SYSTEM	GAME_PHASE	2D	0	1	LOW	-6	1	NOT_PRODUCED_EXTERNAL_AUDIO_GATE

audio_id	category	owner_id	trigger	spatial	max_distance_m	max_instances	priority	duck_db	min_variants	status
SFX_DRAGON_METAL_BREATH	MONSTER_SKILL	DRAGON_METAL	DRAGON_BREATH	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_METAL_TAIL	MONSTER_SKILL	DRAGON_METAL	DRAGON_TAIL	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_METAL_DASH	MONSTER_SKILL	DRAGON_METAL	DRAGON_DASH	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_METAL_HIT	MONSTER_SKILL	DRAGON_METAL	ANY_HIT	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_WOOD_BREATH	MONSTER_SKILL	DRAGON_WOOD	DRAGON_BREATH	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_WOOD_TAIL	MONSTER_SKILL	DRAGON_WOOD	DRAGON_TAIL	3D	45	2	HIGH	0	2	NOT_PRODUCED

audio_id	category	owner_id	trigger	spatial	max_distance_m	max_instances	priority	duck_db	min_variants	status
WOOD_TAIL										_EXTERNAL_AUDIO_GATE
SFX_DRAGON_WOOD_DASH	MONSTER_SKILL	DRAGON_WOOD	DRAGON_DASH	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_WOOD_HIT	MONSTER_SKILL	DRAGON_WOOD	ANY_HIT	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_WATER_BREATH	MONSTER_SKILL	DRAGON_WATER	DRAGON_BREATH	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_WATER_TAIL	MONSTER_SKILL	DRAGON_WATER	DRAGON_TAIL	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_WATER_DASH	MONSTER_SKILL	DRAGON_WATER	DRAGON_DASH	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO

audio_id	category	owner_id	trigger	spatial	max_distance_m	max_instances	priority	duck_db	min_variants	status
										DIO_GATE
SFX_DRAGON_WATER_HIT	MONSTER_SKILL	DRAGON_WATER	ANY_HIT	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_FIRE_BREATH	MONSTER_SKILL	DRAGON_FIRE	DRAGON_BREATH	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_FIRE_TAIL	MONSTER_SKILL	DRAGON_FIRE	DRAGON_TAIL	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_FIRE_DASH	MONSTER_SKILL	DRAGON_FIRE	DRAGON_DASH	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_FIRE_HIT	MONSTER_SKILL	DRAGON_FIRE	ANY_HIT	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE

audio_id	category	owner_id	trigger	spatial	max_distance_m	max_instances	priority	duck_db	min_variants	status
SFX_DRAGON_EARTH_BREATH	MONSTER_SKILL	DRAGON_EARTH	DRAGON_BREATH	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_EARTH_TAIL	MONSTER_SKILL	DRAGON_EARTH	DRAGON_TAIL	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_EARTH_DASH	MONSTER_SKILL	DRAGON_EARTH	DRAGON_DASH	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE
SFX_DRAGON_EARTH_HIT	MONSTER_SKILL	DRAGON_EARTH	ANY_HIT	3D	45	2	HIGH	0	2	NOT_PRODUCED_EXTERNAL_AUDIO_GATE

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asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
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asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
CHR_H001_BUNDLE	HERO_3D	H001	铁扇公主	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H002_BUNDLE	HERO_3D	H002	红孩儿	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H003_BUNDLE	HERO_3D	H003	蜘蛛精	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H004_BUNDLE	HERO_3D	H004	蝎子精	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H005_BUNDLE	HERO_3D	H005	多目怪	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
CHR_H006_BUNDLE	HERO_3D	H006	九头虫	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H007_BUNDLE	HERO_3D	H007	黄风怪	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H008_BUNDLE	HERO_3D	H008	太上老君	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H009_BUNDLE	HERO_3D	H009	孙悟空	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H010_BUNDLE	HERO_3D	H010	二郎神	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
CHR_H011_BUNDLE	HERO_3D	H011	哪吒	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H012_BUNDLE	HERO_3D	H012	六耳猕猴	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H013_BUNDLE	HERO_3D	H013	大鹏雕	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H014_BUNDLE	HERO_3D	H014	白骨精	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H015_BUNDLE	HERO_3D	H015	猪八戒	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
CHR_H016_BUNDLE	HERO_3D	H016	白龙马	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H017_BUNDLE	HERO_3D	H017	青狮怪	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H018_BUNDLE	HERO_3D	H018	牛魔王	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H019_BUNDLE	HERO_3D	H019	独角兕大王	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H020_BUNDLE	HERO_3D	H020	黄眉老祖	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
CHR_H021_BUNDLE	HERO_3D	H021	金角大王	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H022_BUNDLE	HERO_3D	H022	银角大王	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H023_BUNDLE	HERO_3D	H023	黄袍怪	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H024_BUNDLE	HERO_3D	H024	虎力大仙	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H025_BUNDLE	HERO_3D	H025	鹿力大仙	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
CHR_H026_BUNDLE	HERO_3D	H026	文殊菩萨	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H027_BUNDLE	HERO_3D	H027	普贤菩萨	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H028_BUNDLE	HERO_3D	H028	镇元大仙	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H029_BUNDLE	HERO_3D	H029	如来佛祖	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H030_BUNDLE	HERO_3D	H030	观音菩萨	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
CHR_H031_BUNDLE	HERO_3D	H031	托塔李天王	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H032_BUNDLE	HERO_3D	H032	唐僧	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H033_BUNDLE	HERO_3D	H033	沙和尚	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H034_BUNDLE	HERO_3D	H034	黑熊精	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H035_BUNDLE	HERO_3D	H035	白象精	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
CHR_H036_BUNDLE	HERO_3D	H036	灵感大王	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H037_BUNDLE	HERO_3D	H037	羊力大仙	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
CHR_H038_BUNDLE	HERO_3D	H038	赛太岁	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3500	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
ICO_ACT_H001	SKILL_ICON	H001	芭蕉风墙	PNG_ALPHA	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	512x512->Atlas128	P0	NOT_PRODUCED_EXTERNAL_ASSET_GATE
ICO_ACT_H002	SKILL_ICON	H002	三昧火墙	PNG_ALPHA	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	512x512->Atlas128	P0	NOT_PRODUCED_EXTERNAL_ASSET_GATE
ICO_ACT_H003	SKILL_ICON	H003	盘丝阵	PNG_ALPHA	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	512x512->Atlas128	P0	NOT_PRODUCED_EXTERNAL_ASSET_GATE
ICO_ACT_H004	SKILL_ICON	H004	倒马毒	PNG_ALPHA	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	512x512->Atlas128	P0	NOT_PRODUCED_EXTERNAL_ASSET_GATE

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
ICO_ACT_H005	SKILL_ICON	H005	金光	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H006	SKILL_ICON	H006	毒雾	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H007	SKILL_ICON	H007	三昧神风	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H008	SKILL_ICON	H008	八卦炉火	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H009	SKILL_ICON	H009	大闹天宫	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H010	SKILL_ICON	H010	天眼	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H011	SKILL_ICON	H011	风火轮	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H012	SKILL_ICON	H012	六耳幻身	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H013	SKILL_ICON	H013	金翅俯击	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H014	SKILL_ICON	H014	借尸还魂	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H015	SKILL_ICON	H015	九齿钉耙	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
ICO_ACT_H016	SKILL_ICON	H016	龙马奔腾	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H017	SKILL_ICON	H017	狮吼	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H018	SKILL_ICON	H018	旋风	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H019	SKILL_ICON	H019	金刚琢	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H020	SKILL_ICON	H020	石罗汉	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H021	SKILL_ICON	H021	紫金葫芦	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H022	SKILL_ICON	H022	砸钱	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H023	SKILL_ICON	H023	缠身	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H024	SKILL_ICON	H024	虎箭	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H025	SKILL_ICON	H025	鹿血	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H026	SKILL_ICON	H026	定慧诀	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
ICO_ACT_H027	SKILL_ICON	H027	行愿	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H028	SKILL_ICON	H028	袖里乾坤	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H029	SKILL_ICON	H029	五行山	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H030	SKILL_ICON	H030	甘露	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H031	SKILL_ICON	H031	宝塔	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H032	SKILL_ICON	H032	袈裟	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H033	SKILL_ICON	H033	流沙	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H034	SKILL_ICON	H034	黑风罩	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H035	SKILL_ICON	H035	象鼻缚	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H036	SKILL_ICON	H036	冻河	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_H037	SKILL_ICON	H037	羊灵之光	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
ICO_ACT_H038	SKILL_ICON	H038	迷烟	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D1	SKILL_ICON	D1	钩锁	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D3	SKILL_ICON	D3	雷击	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D4	SKILL_ICON	D4	变羊	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D6	SKILL_ICON	D6	闪现	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D7	SKILL_ICON	D7	潜行	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D9	SKILL_ICON	D9	箭雨	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D10	SKILL_ICON	D10	电链	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D11	SKILL_ICON	D11	轮盘	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D12	SKILL_ICON	D12	标记	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D13	SKILL_ICON	D13	换位	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
ICO_ACT_D 14	SKILL_ICON	D14	回溯	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D 15	SKILL_ICON	D15	妙手空空	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D 16	SKILL_ICON	D16	聚财	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D 17	SKILL_ICON	D17	连赏	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D 18	SKILL_ICON	D18	淘金狂热	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D 19	SKILL_ICON	D19	豆兵	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D 20	SKILL_ICON	D20	陷阱	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D 21	SKILL_ICON	D21	冰棺	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_ACT_D 22	SKILL_ICON	D22	铁布衫	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_ B01	SKILL_ICON	B01	被动 B01	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_ B02	SKILL_ICON	B02	被动 B02	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
ICO_PASS_B03	SKILL_ICON	B03	被动 B03	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B04	SKILL_ICON	B04	被动 B04	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B05	SKILL_ICON	B05	被动 B05	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B06	SKILL_ICON	B06	被动 B06	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B07	SKILL_ICON	B07	被动 B07	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B08	SKILL_ICON	B08	被动 B08	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B09	SKILL_ICON	B09	被动 B09	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B10	SKILL_ICON	B10	被动 B10	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B11	SKILL_ICON	B11	被动 B11	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B12	SKILL_ICON	B12	被动 B12	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B13	SKILL_ICON	B13	被动 B13	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
ICO_PASS_B14	SKILL_ICON	B14	被动 B14	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B15	SKILL_ICON	B15	被动 B15	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B16	SKILL_ICON	B16	被动 B16	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B17	SKILL_ICON	B17	被动 B17	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B18	SKILL_ICON	B18	被动 B18	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B19	SKILL_ICON	B19	被动 B19	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B20	SKILL_ICON	B20	被动 B20	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B21	SKILL_ICON	B21	被动 B21	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B22	SKILL_ICON	B22	被动 B22	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B23	SKILL_ICON	B23	被动 B23	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B24	SKILL_ICON	B24	被动 B24	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
ICO_PASS_B25	SKILL_ICON	B25	被动 B25	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B26	SKILL_ICON	B26	被动 B26	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B27	SKILL_ICON	B27	被动 B27	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B28	SKILL_ICON	B28	被动 B28	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B29	SKILL_ICON	B29	被动 B29	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B30	SKILL_ICON	B30	被动 B30	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B31	SKILL_ICON	B31	被动 B31	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B32	SKILL_ICON	B32	被动 B32	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B33	SKILL_ICON	B33	被动 B33	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B34	SKILL_ICON	B34	被动 B34	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B35	SKILL_ICON	B35	被动 B35	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
ICO_PASS_B36	SKILL_ICON	B36	被动 B36	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B37	SKILL_ICON	B37	被动 B37	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B38	SKILL_ICON	B38	被动 B38	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B39	SKILL_ICON	B39	被动 B39	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B40	SKILL_ICON	B40	被动 B40	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B41	SKILL_ICON	B41	被动 B41	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B42	SKILL_ICON	B42	被动 B42	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B43	SKILL_ICON	B43	被动 B43	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_PASS_B44	SKILL_ICON	B44	被动 B44	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_EQ_W1	EQUIPMENT_ICON	W1	精铁棍	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_EQ_W2	EQUIPMENT_ICON	W2	粗布战衣	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
ICO_EQ_W3	EQUIPMENT_ICON	W3	草鞋	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_W4	EQUIPMENT_ICON	W4	铜护腕	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_W5	EQUIPMENT_ICON	W5	轻甲背心	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_W6	EQUIPMENT_ICON	W6	行者束带	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_B1	EQUIPMENT_ICON	B1	狼牙棒	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_B2	EQUIPMENT_ICON	B2	虎皮裙	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_B3	EQUIPMENT_ICON	B3	避水珠	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_B4	EQUIPMENT_ICON	B4	定风珠	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_B5	EQUIPMENT_ICON	B5	药葫芦	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_B6	EQUIPMENT_ICON	B6	磨刀石	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_B7	EQUIPMENT_ICON	B7	铜镜	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
ICO_EQ_B8	EQUIPMENT_ICON	B8	聚宝袋	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_B9	EQUIPMENT_ICON	B9	夜巡灯	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_B10	EQUIPMENT_ICON	B10	驭兽鞭	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_B11	EQUIPMENT_ICON	B11	避雷木牌	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_B12	EQUIPMENT_ICON	B12	千里靴	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_B13	EQUIPMENT_ICON	B13	登云履	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_P1	EQUIPMENT_ICON	P1	七星剑	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_P2	EQUIPMENT_ICON	P2	火尖枪	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_P3	EQUIPMENT_ICON	P3	幌金绳	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_P4	EQUIPMENT_ICON	P4	人参果	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_P5	EQUIPMENT_ICON	P5	万年灵芝	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
ICO_EQ_P6	EQUIPMENT_ICON	P6	风袋	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_EQ_P7	EQUIPMENT_ICON	P7	招魂幡	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_EQ_P8	EQUIPMENT_ICON	P8	布袋	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_EQ_P9	EQUIPMENT_ICON	P9	照妖珠	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_EQ_P10	EQUIPMENT_ICON	P10	赌坊金牌	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_EQ_P11	EQUIPMENT_ICON	P11	判官笔	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_EQ_P12	EQUIPMENT_ICON	P12	荆棘甲	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_EQ_P13	EQUIPMENT_ICON	P13	夜行斗篷	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_EQ_P14	EQUIPMENT_ICON	P14	血玉葫芦	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_EQ_P15	EQUIPMENT_ICON	P15	疾风靴	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE
ICO_EQ_G1	EQUIPMENT_ICON	G1	九转金丹	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT RNAL_ASSE T_GATE

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
ICO_EQ_G2	EQUIPMENT_ICON	G2	混元珠	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_G3	EQUIPMENT_ICON	G3	渡劫铃	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_G4	EQUIPMENT_ICON	G4	筋斗云	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_G5	EQUIPMENT_ICON	G5	聚宝盆	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_G6	EQUIPMENT_ICON	G6	降魔杵	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_G7	EQUIPMENT_ICON	G7	顺风耳	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_G8	EQUIPMENT_ICON	G8	雷公凿	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_G9	EQUIPMENT_ICON	G9	噬魂戒	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
ICO_EQ_G10	EQUIPMENT_ICON	G10	金箍棒	PNG_ALPHA	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	NOT_A PPLICA BLE	512x512- >Atlas128	P0	NOT_PROD UCED_EXT ERNAL_AS SET_GATE
MON_MEL_BUNDLE	MONSTER_3D	MEL	近战小怪	FBX+LOD0-2+ANIM+COLLIDER	6000	3000	1000	35	1	1024x1024	P0	LOCKE D_ID_E XTERNA L_PRO DUCTIO N_REQ UIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_RNG_BUNDLE	MONSTER_3D	RNG	远程小怪	FBX+LOD0-2+ANIM+COLLIDER	6000	3000	1000	35	1	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
MON_FLY_BUNDLE	MONSTER_3D	FLY	飞行小怪	FBX+LOD0-2+ANIM+COLLIDER	6000	3000	1000	35	1	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
MON_PIG_BUNDLE	MONSTER_3D	PIG	五行肥猪	FBX+LOD0-2+ANIM+COLLIDER	8000	4000	1200	40	1	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
MON_ELITE_TANK_BUNDLE	MONSTER_3D	ELITE_TANK	肉盾首领	FBX+LOD0-2+ANIM+COLLIDER	12000	6000	2000	55	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
MON_ELITE_DPS_BUNDLE	MONSTER_3D	ELITE_DPS	输出首领	FBX+LOD0-2+ANIM+COLLIDER	12000	6000	2000	55	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_DRAGON_BUNDLE	MONSTER_3D	DRAGON	五行龙王	FBX+LOD0-2+ANIM+COLLIDER	18000	9000	3000	70	2	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
MON_CORE_BUNDLE	MONSTER_3D	CORE	核心 BOSS	FBX+LOD0-2+ANIM+COLLIDER	26000	13000	4500	90	3	1024x1024	P0	LOCKED_ID_EXTERNAL_PRODUCTION_REQUIRED
ICO_EQ_B14	EQUIPMENT_ICON	B14	蛭伏靴	PNG_ALPHA	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	512x512->Atlas128	P0	NOT_PRODUCED_EXTERNAL_ASSET_GATE
ICO_EQ_P16	EQUIPMENT_ICON	P16	连珠履	PNG_ALPHA	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	512x512->Atlas128	P0	NOT_PRODUCED_EXTERNAL_ASSET_GATE
ICO_EQ_P17	EQUIPMENT_ICON	P17	摘星靴	PNG_ALPHA	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	512x512->Atlas128	P0	NOT_PRODUCED_EXTERNAL_ASSET_GATE
ICO_EQ_P18	EQUIPMENT_ICON	P18	磐岩履	PNG_ALPHA	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	512x512->Atlas128	P0	NOT_PRODUCED_EXTERNAL_ASSET_GATE

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_MVS_01_BUNDLE	MONSTER_VISUAL_3D	MVS_01	巴山虎	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=ORDINARY_MELLY	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_02_BUNDLE	MONSTER_VISUAL_3D	MVS_02	苍狼精	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=ORDINARY_MELLY	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_03_BUNDLE	MONSTER_VISUAL_3D	MVS_03	刁钻古怪	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=ORDINARY_MELLY	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_MVS_004_BUNDLE	MONSTER_VISUAL_3D	MVS_004	古怪刁钻	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=ORDINARY_MELLY	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_005_BUNDLE	MONSTER_VISUAL_3D	MVS_005	虎精小妖	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=ORDINARY_MELLY	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_006_BUNDLE	MONSTER_VISUAL_3D	MVS_006	黄狮精	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=ORDINARY_MELLY	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_MVS_07_BUNDLE	MONSTER_VISUAL_3D	MVS_07	蛇精小妖	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=ORDINARY_MELLY	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_08_BUNDLE	MONSTER_VISUAL_3D	MVS_08	树鬼	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=ORDINARY_MELLY	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_09_BUNDLE	MONSTER_VISUAL_3D	MVS_09	蜘蛛仔	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=ORDINARY_MELLY	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_MVS_010_BUNDLE	MONSTER_VISUAL_3D	MVS_010	红鳞大蟒	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=ORDINARY_Melee	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_011_BUNDLE	MONSTER_VISUAL_3D	MVS_011	玉面狐狸	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=ORDINARY_Ranged	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_012_BUNDLE	MONSTER_VISUAL_3D	MVS_012	如意真仙	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=ORDINARY_Ranged	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_MVS_013_BUNDLE	MONSTER_VISUAL_3D	MVS_013	火鸦精	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=ORDINARY_RANGED	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_014_BUNDLE	MONSTER_VISUAL_3D	MVS_014	倚海龙	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=ORDINARY_RANGED	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_015_BUNDLE	MONSTER_VISUAL_3D	MVS_015	碧水金睛兽	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=FLYING	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_MVS_016_BUNDLE	MONSTER_VISUAL_3D	MVS_016	晦月魔君	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=FLYING	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_017_BUNDLE	MONSTER_VISUAL_3D	MVS_017	混天大圣	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=FLYING	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_018_BUNDLE	MONSTER_VISUAL_3D	MVS_018	孔雀公主	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=FLYING	6000	3000	1000	35	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_MVS_019_BUNDLE	MONSTER_VISUAL_3D	MVS_019	特处士	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=LEADER_TANK	12000	6000	2000	50	1	2048x2048	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_020_BUNDLE	MONSTER_VISUAL_3D	MVS_020	熊山君	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=LEADER_TANK	12000	6000	2000	50	1	2048x2048	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_021_BUNDLE	MONSTER_VISUAL_3D	MVS_021	南山大王	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=LEADER_DPS	12000	6000	2000	50	1	2048x2048	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_MVS_022_BUNDLE	MONSTER_VISUAL_3D	MVS_022	寅将军	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=LEADER_DPS	12000	6000	2000	50	1	2048x2048	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_023_BUNDLE	MONSTER_VISUAL_3D	MVS_023	金肥猪	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=PIG	8000	4000	1200	40	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_024_BUNDLE	MONSTER_VISUAL_3D	MVS_024	木肥猪	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=PIG	8000	4000	1200	40	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_MVS_025_BUNDLE	MONSTER_VISUAL_3D	MVS_025	水肥猪	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=PIG	8000	4000	1200	40	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_026_BUNDLE	MONSTER_VISUAL_3D	MVS_026	火肥猪	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=PIG	8000	4000	1200	40	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_027_BUNDLE	MONSTER_VISUAL_3D	MVS_027	土肥猪	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=PIG	8000	4000	1200	40	1	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_MVS_028_BUNDLE	MONSTER_VISUAL_3D	MVS_028	金龙王	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=DRAAGON	18000	9000	3000	70	2	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_029_BUNDLE	MONSTER_VISUAL_3D	MVS_029	木龙王	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=DRAAGON	18000	9000	3000	70	2	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_030_BUNDLE	MONSTER_VISUAL_3D	MVS_030	水龙王	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=DRAAGON	18000	9000	3000	70	2	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_MVS_031_BUNDLE	MONSTER_VISUAL_3D	MVS_031	火龙王	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=DRAAGON	18000	9000	3000	70	2	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_032_BUNDLE	MONSTER_VISUAL_3D	MVS_032	土龙王	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=DRAAGON	18000	9000	3000	70	2	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_033_BUNDLE	MONSTER_VISUAL_3D	MVS_033	白鹿魔王	FBX+LOD0-2+ANIMATION+COLLIDER;skeleton_family=CORE_BOSS	26000	13000	4500	90	3	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_MVS_034_BUNDLE	MONSTER_VISUAL_3D	MVS_034	辟暑大王	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=CORE_BOSS	26000	13000	4500	90	3	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_035_BUNDLE	MONSTER_VISUAL_3D	MVS_035	地涌夫人	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=CORE_BOSS	26000	13000	4500	90	3	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_036_BUNDLE	MONSTER_VISUAL_3D	MVS_036	辟尘大王	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=CORE_BOSS	26000	13000	4500	90	3	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_MVS_037_BUNDLE	MONSTER_VISUAL_3D	MVS_037	辟寒大王	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=CORE_BOSS	26000	13000	4500	90	3	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_MVS_038_BUNDLE	MONSTER_VISUAL_3D	MVS_038	九灵元圣	FBX+LOD0-2+ANIM+COLLIDER;skeleton_family=CORE_BOSS	26000	13000	4500	90	3	1024x1024	P0	LOCKED_CONCEPT_EXTERNAL_PRODUCTION_REQUIRED
MON_DRAGON_METAL_DECOR	MONSTER_VISUAL_VFX	DRAGON_METAL	金龙王模型装饰件	VFX_PREFAB+TEXTURE;attach=DRAGON_SKELETON	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	1	1024x1024	P0	NOT_PRODUCED_EXTERNAL_PRODUCTION_REQUIRED
MON_DRAGON_METAL_	MONSTER_VISUAL_	DRAGON_METAL	金龙王脚底阵纹	VFX_PREFAB+TEXTURE;	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	1	1024x1024	P0	NOT_PRODUCED_EXTERNAL

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
GROUNDRUNE	VFX			attach=DRAGON_SKELETON								AL_PRODUCTION_REQUIRED
MONDRAGON_WOOD_DECOR	MONSTER_VISUAL_VFX	DRAGON_WOOD	木龙王模型装饰件	VFX_P REFAB+TEXTURE; attach=DRAGON_SKELETON	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	1	1024x1024	P0	NOT_PRODUCED_EXTERNAL_PRODUCTION_REQUIRED
MONDRAGON_WOOD_GROUNDRUNE	MONSTER_VISUAL_VFX	DRAGON_WOOD	木龙王脚底阵纹	VFX_P REFAB+TEXTURE; attach=DRAGON_SKELETON	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	1	1024x1024	P0	NOT_PRODUCED_EXTERNAL_PRODUCTION_REQUIRED
MONDRAGON_WATER_DECOR	MONSTER_VISUAL_VFX	DRAGON_WATER	水龙王模型装饰件	VFX_P REFAB+TEXTURE; attach=DRAGON_SKELETON	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	1	1024x1024	P0	NOT_PRODUCED_EXTERNAL_PRODUCTION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
MON_DRAGON_WATER_GROUP_RUNE	MONSTER_VFX	DRAGON_WATER	水龙王脚底阵纹	VFX_P REFAB +TEXT URE; attach =DRAGON_SK ELETON	NOT_A PPLICABLE	NOT_A PPLICABLE	NOT_A PPLICABLE	NOT_A PPLICABLE	1	1024x1024	P0	NOT_PRODUCED_EXTENSION_REQUIRED
MON_DRAGON_FIRE_DECOR	MONSTER_VFX	DRAGON_FIRE	火龙王模型装饰件	VFX_P REFAB +TEXT URE; attach =DRAGON_SK ELETON	NOT_A PPLICABLE	NOT_A PPLICABLE	NOT_A PPLICABLE	NOT_A PPLICABLE	1	1024x1024	P0	NOT_PRODUCED_EXTENSION_REQUIRED
MON_DRAGON_FIRE_GROUND_RUNE	MONSTER_VFX	DRAGON_FIRE	火龙王脚底阵纹	VFX_P REFAB +TEXT URE; attach =DRAGON_SK ELETON	NOT_A PPLICABLE	NOT_A PPLICABLE	NOT_A PPLICABLE	NOT_A PPLICABLE	1	1024x1024	P0	NOT_PRODUCED_EXTENSION_REQUIRED
MON_DRAGON_EARTH_DECOR	MONSTER_VFX	DRAGON_EARTH	土龙王模型装饰件	VFX_P REFAB +TEXT URE; attach	NOT_A PPLICABLE	NOT_A PPLICABLE	NOT_A PPLICABLE	NOT_A PPLICABLE	1	1024x1024	P0	NOT_PRODUCED_EXTENSION_REQUIRED

asset_id	category	owner_id	display_name	deliverable	lod0_tri	lod1_tri	lod2_tri	bones_max	materials_max	source_texture	priority	status
				=DRAGON_SKELETON								ODUCTION_REQUIRED
MONDRAGON_EARTH_GROUND_RUNE	MONSTER_VISUAL_VFX	DRAGON_EARTH	土龙王脚底阵纹	VFX_PREFAB+TEXTURE; attach=DRAGON_SKELETON	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	NOT_APPLICABLE	1	1024x1024	P0	NOT_PRODUCED_EXTERNAL_PRODUCT_REQUIRED

17 AI 开发、性能与发布验收

AI 开发纪律 | AI 只能把本文转换为数据、代码、资源和测试；不得补写数值、发明默认值、把玩家文案当运行时规则，或为了实现方便添加绕过真实碰撞体的区域限制、终局显形等额外规则。

维度	固定要求
代码生成	闭合枚举、字段 Schema、ID 和单位先生成共享类型；服务端、客户端和测试向量使用同一规则快照；生成层禁止手改。
碰撞	玩家、怪物、召唤物、动态墙、高台、坡道、投射物统一 3D 体积；20Hz 连续扫描；同距离碰撞使用稳定实体 ID 排序。
性能最坏况	30 英雄+123 常驻怪+270 理论召唤+动态墙/陷阱/弹道/地面遗宝；服务端目标 500 持续实体+300 短时事件@20Hz。
客户端	低端移动设备持续≥30fps、P99 帧≤50ms、峰值内存≤850MB；中端目标 60fps；玩法信息不得随画质消失。
网络	20Hz；RTT P95≤80ms；无重复事务；断线/重连不复制金币、库存、掉落或随机消耗。
地图	首遇 P50 60~90 秒；出生互不直视；关键 POI≥3 独立接近路线；长廊≤60m；三庭和终局圈完整容纳。
战斗	五行误差≤1%；四人屠龙 TTK 17~26 秒；真伤、护盾、反伤、DoT、死亡链、未击和同 tick 事件全部通过向量测试。
跨端	触控误操作<3%，精瞄模式误判<2%；跨输入胜率指数 2000 局告警、10000 局才允许方向性调整；先修 UI 与辅助，不改数值。
发布	数据一致性、地图、战斗、输入、低端性能、网络、新手理解、资产权利、地区合规、经济漏洞、未劫、回放哈希、账号合并和事务完整性全部过门。

17.1 固定发布门槛

工程字段	唯一值 / 执行语义
statuses	DESIGN_PASS / BLOCKING_PRERELEASE_ENGINE / BLOCKING_PRERELEASE_DEVICE / BLOCKING_PRERELEASE_EXTERNAL / BLOCKING_PREDEVELOPMENT_FORMULA / MONITOR_ONLY / TOOL_PASS / TOOL_PENDING
gates[0].id	G_DATA_CONSISTENCY
gates[0].level	P0
gates[0].status	TOOL_PASS
gates[0].pass	unified_checker_fail_count==0
gates[1].id	G_MAP_RUNTIME
gates[1].level	L3
gates[1].status	BLOCKING_PRERELEASE_ENGINE
gates[1].pass	first_encounter_p50_s in [60,90] and spawn_face_violation==0
gates[2].id	G_COMBAT_RUNTIME
gates[2].level	L3
gates[2].status	BLOCKING_PRERELEASE_ENGINE
gates[2].pass	five_element_error<=0.01 and dragon_4p_ttk_s in [17,26]
gates[3].id	G_INPUT_RUNTIME
gates[3].level	L3
gates[3].status	BLOCKING_PRERELEASE_ENGINE
gates[3].pass	touch_error<0.03 and aim_mode_error<0.02
gates[4].id	G_LOW_MOBILE_PERF
gates[4].level	L3
gates[4].status	BLOCKING_PRERELEASE_ENGINE
gates[4].pass	fps_sustained>=30 and frame_p99_ms<=50 and memory_peak_mb<=850
gates[5].id	G_NETWORK
gates[5].level	L3
gates[5].status	BLOCKING_PRERELEASE_ENGINE
gates[5].pass	tick_hz==20 and rtt_p95_ms<=80 and duplication_count==0
gates[6].id	G_NEW_PLAYER_COMPREHENSION
gates[6].level	L4
gates[6].status	BLOCKING_PRERELEASE_DEVICE
gates[6].pass	core_controls_without_help>=0.85
gates[7].id	G_CROSS_INPUT_FAIRNESS
gates[7].level	LIVE

工程字段	唯一值 / 执行语义
gates[7].status	MONITOR_ONLY
gates[7].pass	adjusted_win_index in [0.45,0.55] with n>=2000 per input
gates[8].id	G_ASSET_RIGHTS
gates[8].level	EXTERNAL
gates[8].status	BLOCKING_PRERELEASE_EXTERNAL
gates[8].pass	owner_signoff==true
gates[9].id	G_REGION_COMPLIANCE
gates[9].level	EXTERNAL
gates[9].status	BLOCKING_PRERELEASE_EXTERNAL
gates[9].pass	regional_legal_signoff==true
gates[10].id	G_ECONOMY_EXPLOIT
gates[10].level	P0
gates[10].status	TOOL_PENDING
gates[10].pass	conservation_ledger_zero_diff and P10_P50_P90_in_band and extreme_build_sims_pass and shop_sellout_assert_pass and no_repeat_pickup_gold and no_recursive_attack_loop
gates[11].id	G_FINAL_STORM
gates[11].level	L1
gates[11].status	TOOL_PENDING
gates[11].pass	all_extreme_builds_have_finite_P99_and_no_hidden_reveal
gates[12].id	G_REPLAY_HASH
gates[12].level	L3
gates[12].status	BLOCKING_PRERELEASE_ENGINE
gates[12].pass	same_ruleset_replay_hash_exact
gates[13].id	G_ACCOUNT_MERGE
gates[13].level	L3
gates[13].status	BLOCKING_PRERELEASE_ENGINE
gates[13].pass	all_merge_cases_preserve_entitlements_and_progress
gates[14].id	G_TRANSACTION_INTEGRITY
gates[14].level	L3
gates[14].status	BLOCKING_PRERELEASE_ENGINE
gates[14].pass	zero_duplicate_or_partial_commit_under_disconnect_and_same_tick_contention
gameplay_monetization_enabled	否

工程字段	唯一值 / 执行语义
release_candidate_requires_all_blocking_prerelease_gates	是
cross_input_fairness.warning_min_matches_per_input	2000
cross_input_fairness.decision_min_matches_per_input	10000
predevelopment_blueprint_required	G_DATA_CONSISTENCY / G_ECONOMY_EXPLOIT / G_FINAL_STORM / G_NO_EMPTY_CELLS / G_PROJECTION_DRIFT
release_candidate_required	G_DATA_CONSISTENCY / G_MAP_RUNTIME / G_COMBAT_RUNTIME / G_INPUT_RUNTIME / G_LOW_MOBILE_PERF / G_NETWORK / G_NEW_PLAYER_COMPREHENSION / G_ASSET_RIGHTS / G_REGION_COMPLIANCE / G_ECONOMY_EXPLOIT / G_FINAL_STORM / G_REPLAY_HASH / G_ACCOUNT_MERGE / G_TRANSACTION_INTEGRITY
live_monitor_only	G_CROSS_INPUT_FAIRNESS
release_candidate_requires_all_in_release_candidate_required	是
blocking_predevelopment	DESIGN_PASS_REQUIRED / SCHEMA_AND_ENUM_GENERATION / MAP_PHYSICAL_GEOMETRY_VALIDATION / ECONOMY_CONSERVATION / CONTENT_REFERENCE_CLOSURE
blocking_prerelease	L3_ENGINE_TEST / L4_REAL_DEVICE / EXTERNAL_ASSET_DELIVERY / SECURITY_REPLAY / PERFORMANCE_BUDGET
monitor_only_after_launch	LIVE_BALANCE / INPUT_FAIRNESS / RETENTION / SHOP_SELL_THROUGH / ABUSE_DETECTION
release_candidate_rule	ALL_BLOCKING_PRERELEASE_ITEMS_PASS; MONITOR_ONLY_ITEMS_DO_NOT_BLOCK
gates[10].assertions.conservation	金币产生/转移/销毁三账本每 tick 守恒，全局差额==0
gates[10].assertions.distribution	18 分钟标准局 P10/P50/P90 个人金币在[2200,4200]/[4800,7800]/[9500,15500] 带内
gates[10].assertions.extreme_builds	B34 利息流/B31+D15 偷窃流/G5+D16 铸币流/B33 宝箱流四类极端构筑单独模拟，无指数增长
gates[10].assertions.shop_sellout	六商店全售罄情形下无死锁、无负库存、无重复扣款
gates[15].id	G_NO_EMPTY_CELLS
gates[15].level	P0
gates[15].status	TOOL_PASS
gates[15].pass	document_empty_cell_count==0
gates[16].id	G_PROJECTION_DRIFT

工程字段	唯一值 / 执行语义
gates[16].level	P0
gates[16].status	TOOL_PASS
gates[16].pass	prose_vs_payload_drift_count==0
gates[17].id	G_CONCEPT_ASSET_MANIFEST
gates[17].level	EXTERNAL
gates[17].status	BLOCKING_PRERELEASE_EXTERNAL
gates[17].pass	all_38_concepts_have_sha256_and_rights_cleared
gates[18].id	G_LOW_FLASH_ALTERNATES
gates[18].level	L3
gates[18].status	BLOCKING_PRERELEASE_ENGINE
gates[18].pass	every_strong_flash_telegraph_has_lowflash_variant
gate_evidence.binding	每个门槛必须绑定 evidence_id + checker_version + result_hash；状态只能由构建器写入，手工编辑状态即构建失败
gate_evidence.pending_value	未运行的检查器统一记 TOOL_PENDING；不允许手写 PASS
gates[0].evidence_id	CCK_V4_2_RUN_20260721
gates[0].checker_version	CCK_V4_2
gates[0].result_hash	见交付包 v4_final_check_report.json（哈希在报告尾部）
gates[10].evidence_id	PENDING_BUILD
gates[10].checker_version	PENDING_BUILD
gates[10].result_hash	PENDING_BUILD
gates[11].evidence_id	PENDING_BUILD
gates[11].checker_version	PENDING_BUILD
gates[11].result_hash	PENDING_BUILD
gates[12].evidence_id	PENDING_BUILD
gates[12].checker_version	PENDING_BUILD
gates[12].result_hash	PENDING_BUILD
gates[13].evidence_id	PENDING_BUILD

工程字段	唯一值 / 执行语义
gates[13].checker_version	PENDING_BUILD
gates[13].result_hash	PENDING_BUILD
gates[14].evidence_id	PENDING_BUILD
gates[14].checker_version	PENDING_BUILD
gates[14].result_hash	PENDING_BUILD
gates[15].evidence_id	CCK_V4_2_RUN_20260721
gates[15].checker_version	CCK_V4_2
gates[15].result_hash	同 gates[0]报告: no_empty_cells 断言
gates[16].evidence_id	CCK_V4_2_RUN_20260721
gates[16].checker_version	CCK_V4_2
gates[16].result_hash	同 gates[0]报告: projection_drift 断言
gates[17].evidence_id	PENDING_BUILD
gates[17].checker_version	PENDING_BUILD
gates[17].result_hash	PENDING_BUILD
gates[18].evidence_id	PENDING_BUILD
gates[18].checker_version	PENDING_BUILD
gates[18].result_hash	PENDING_BUILD
gates[1].evidence_id	PENDING_BUILD
gates[1].checker_version	PENDING_BUILD
gates[1].result_hash	PENDING_BUILD
gates[2].evidence_id	PENDING_BUILD
gates[2].checker_version	PENDING_BUILD
gates[2].result_hash	PENDING_BUILD
gates[3].evidence_id	PENDING_BUILD
gates[3].checker_version	PENDING_BUILD
gates[3].result_hash	PENDING_BUILD
gates[4].evidence_id	PENDING_BUILD

工程字段	唯一值 / 执行语义
gates[4].checker_version	PENDING_BUILD
gates[4].result_hash	PENDING_BUILD
gates[5].evidence_id	PENDING_BUILD
gates[5].checker_version	PENDING_BUILD
gates[5].result_hash	PENDING_BUILD
gates[6].evidence_id	PENDING_BUILD
gates[6].checker_version	PENDING_BUILD
gates[6].result_hash	PENDING_BUILD
gates[7].evidence_id	PENDING_BUILD
gates[7].checker_version	PENDING_BUILD
gates[7].result_hash	PENDING_BUILD
gates[8].evidence_id	PENDING_BUILD
gates[8].checker_version	PENDING_BUILD
gates[8].result_hash	PENDING_BUILD
gates[9].evidence_id	PENDING_BUILD
gates[9].checker_version	PENDING_BUILD
gates[9].result_hash	PENDING_BUILD

门槛状态释义 | 本文的玩法、数值、几何、交互和工程口径均已定案。BLOCKING_PRERELEASE_ENGINE、BLOCKING_PRERELEASE_EXTERNAL 与 MONITOR_ONLY 只表示实现后的验证阶段，不代表规则尚未确定，也不授权开发过程临时改写本文；若验证失败，必须形成可复现证据并回到同一真源版本化修订。

附录 A 全英雄与等级数值

A.1 38 英雄映射

ID	英雄	模板	移动类	五行	专属	L1 攻/血/速	L15 攻/血/速	射程/攻速
H001	铁扇公主	连弩	GROUND	火	H001	41/575/2.76	110/1650/3.4	20.0m/1.6s ⁻¹
H002	红孩儿	连弩	GROUND	火	H002	41/575/2.76	110/1650/3.4	20.0m/1.6s ⁻¹
H003	蜘蛛精	连弩	GROUND	木	H003	41/575/2.76	110/1650/3.4	20.0m/1.6s ⁻¹
H004	蝎子精	连弩	GROUND	土	H004	41/575/2.76	110/1650/3.4	20.0m/1.6s ⁻¹
H005	多目怪	长弓	GROUND	木	H005	50/540/2.6	134/1550/3.2	30.0m/0.9s ⁻¹
H006	九头虫	长弓 · 飞行	FLYING	水	H006	50/432/2.6	134/1240/3.2	30.0m/0.9s ⁻¹
H007	黄风怪	连弩 · 飞行	FLYING	木	H007	41/460/2.76	110/1320/3.4	20.0m/1.6s ⁻¹
H008	太上老君	长弓	GROUND	火	H008	50/540/2.6	134/1550/3.2	30.0m/0.9s ⁻¹
H009	孙悟空	刺客	GROUND	金	H009	60/488/3.01	160/1400/3.7	5.0m/1.4s ⁻¹
H010	二郎神	刺客	GROUND	金	H010	60/488/3.01	160/1400/3.7	5.0m/1.4s ⁻¹
H011	哪吒	刺客	GROUND	火	H011	60/488/3.01	160/1400/3.7	5.0m/1.4s ⁻¹
H012	六耳猕猴	刺客	GROUND	金	H012	60/488/3.01	160/1400/3.7	5.0m/1.4s ⁻¹
H013	大鹏雕	刺客 · 飞行	FLYING	金	H013	60/390/3.01	160/1120/3.7	5.0m/1.4s ⁻¹
H014	白骨精	刺客	GROUND	土	H014	60/488/3.01	160/1400/3.7	5.0m/1.4s ⁻¹
H015	猪八戒	战士	GROUND	水	H015	52/627/2.76	140/1800/3.4	5.0m/1.1s ⁻¹
H016	白龙马	战士	GROUND	水	H016	52/627/2.76	140/1800/3.4	5.0m/1.1s ⁻¹
H017	青狮怪	战士	GROUND	土	H017	52/627/2.76	140/1800/3.4	5.0m/1.1s ⁻¹
H018	牛魔王	战士	GROUND	火	H018	52/627/2.76	140/1800/3.4	5.0m/1.1s ⁻¹
H019	独角兕大王	战士	GROUND	金	H019	52/627/2.76	140/1800/3.4	5.0m/1.1s ⁻¹
H020	黄眉老祖	战士	GROUND	土	H020	52/627/2.76	140/1800/3.4	5.0m/1.1s ⁻¹
H021	金角大王	战士	GROUND	金	H021	52/627/2.76	140/1800/3.4	5.0m/1.1s ⁻¹
H022	银角大王	战士	GROUND	水	H022	52/627/2.76	140/1800/3.4	5.0m/1.1s ⁻¹
H023	黄袍怪	战士	GROUND	木	H023	52/627/2.76	140/1800/3.4	5.0m/1.1s ⁻¹
H024	虎力大仙	战士	GROUND	火	H024	52/627/2.76	140/1800/3.4	5.0m/1.1s ⁻¹
H025	鹿力大仙	战士	GROUND	木	H025	52/627/2.76	140/1800/3.4	5.0m/1.1s ⁻¹
H026	文殊菩萨	战士	GROUND	金	H026	52/627/2.76	140/1800/3.4	5.0m/1.1s ⁻¹
H027	普贤菩萨	战士	GROUND	土	H027	52/627/2.76	140/1800/3.4	5.0m/1.1s ⁻¹
H028	镇元大仙	战士	GROUND	木	H028	52/627/2.76	140/1800/3.4	5.0m/1.1s ⁻¹
H029	如来佛祖	坦克	GROUND	土	H029	37/767/2.44	100/2200/3.0	5.0m/0.9s ⁻¹
H030	观音菩萨	坦克	GROUND	木	H030	37/767/2.44	100/2200/3.0	5.0m/0.9s ⁻¹
H031	托塔李天王	坦克	GROUND	土	H031	37/767/2.44	100/2200/3.0	5.0m/0.9s ⁻¹
H032	唐僧	坦克	GROUND	金	H032	37/767/2.44	100/2200/3.0	5.0m/0.9s ⁻¹
H033	沙和尚	坦克	GROUND	水	H033	37/767/2.44	100/2200/3.0	5.0m/0.9s ⁻¹
H034	黑熊精	坦克	GROUND	木	H034	37/767/2.44	100/2200/3.0	5.0m/0.9s ⁻¹
H035	白象精	坦克	GROUND	水	H035	37/767/2.44	100/2200/3.0	5.0m/0.9s ⁻¹
H036	灵感大王	坦克	GROUND	水	H036	37/767/2.44	100/2200/3.0	5.0m/0.9s ⁻¹

ID	英雄	模板	移动类	五行	专属	L1 攻/血/速	L15 攻/血/速	射程/攻速
H037	羊力大仙	坦克	GROUND	木	H037	37/767/2.44	100/2200/3.0	5.0m/0.9s ⁻¹
H038	赛太岁	坦克	GROUND	火	H038	37/767/2.44	100/2200/3.0	5.0m/0.9s ⁻¹

A.2 八种属性模板 L1–L15

模板	等级	攻击	生命	移速	射程 m	攻速次每秒	该级 DPS
长弓	1	50	540	260	30	0.9	45.0
长弓	2	54	582	264	30	0.9	48.6
长弓	3	58	628	268	30	0.9	52.2
长弓	4	62	677	272	30	0.9	55.8
长弓	5	66	730	276	30	0.9	59.4
长弓	6	71	787	280	30	0.9	63.9
长弓	7	76	849	284	30	0.9	68.4
长弓	8	82	915	289	30	0.9	73.8
长弓	9	88	986	293	30	0.9	79.2
长弓	10	94	1064	297	30	0.9	84.6
长弓	11	101	1147	302	30	0.9	90.9
长弓	12	108	1237	306	30	0.9	97.2
长弓	13	116	1333	311	30	0.9	104.4
长弓	14	125	1438	315	30	0.9	112.5
长弓	15	134	1550	320	30	0.9	120.6
长弓 · 飞行	1	50	432	260	30	0.9	45.0
长弓 · 飞行	2	54	466	264	30	0.9	48.6
长弓 · 飞行	3	58	502	268	30	0.9	52.2
长弓 · 飞行	4	62	542	272	30	0.9	55.8
长弓 · 飞行	5	66	584	276	30	0.9	59.4
长弓 · 飞行	6	71	630	280	30	0.9	63.9
长弓 · 飞行	7	76	679	284	30	0.9	68.4
长弓 · 飞行	8	82	732	289	30	0.9	73.8
长弓 · 飞行	9	88	789	293	30	0.9	79.2
长弓 · 飞行	10	94	851	297	30	0.9	84.6
长弓 · 飞行	11	101	917	302	30	0.9	90.9
长弓 · 飞行	12	108	989	306	30	0.9	97.2
长弓 · 飞行	13	116	1067	311	30	0.9	104.4
长弓 · 飞行	14	125	1150	315	30	0.9	112.5
长弓 · 飞行	15	134	1240	320	30	0.9	120.6
连弩	1	41	575	276	20	1.6	65.6
连弩	2	44	620	281	20	1.6	70.4
连弩	3	47	668	285	20	1.6	75.2

模板	等级	攻击	生命	移速	射程 m	攻速次每秒	该级 DPS
连弩	4	51	721	289	20	1.6	81.6
连弩	5	54	777	293	20	1.6	86.4
连弩	6	58	838	298	20	1.6	92.8
连弩	7	63	903	302	20	1.6	100.8
连弩	8	67	974	307	20	1.6	107.2
连弩	9	72	1050	311	20	1.6	115.2
连弩	10	77	1132	316	20	1.6	123.2
连弩	11	83	1221	320	20	1.6	132.8
连弩	12	89	1316	325	20	1.6	142.4
连弩	13	96	1419	330	20	1.6	153.6
连弩	14	103	1530	335	20	1.6	164.8
连弩	15	110	1650	340	20	1.6	176.0
连弩·飞行	1	41	460	276	20	1.6	65.6
连弩·飞行	2	44	496	281	20	1.6	70.4
连弩·飞行	3	47	535	285	20	1.6	75.2
连弩·飞行	4	51	577	289	20	1.6	81.6
连弩·飞行	5	54	622	293	20	1.6	86.4
连弩·飞行	6	58	670	298	20	1.6	92.8
连弩·飞行	7	63	723	302	20	1.6	100.8
连弩·飞行	8	67	779	307	20	1.6	107.2
连弩·飞行	9	72	840	311	20	1.6	115.2
连弩·飞行	10	77	906	316	20	1.6	123.2
连弩·飞行	11	83	977	320	20	1.6	132.8
连弩·飞行	12	89	1053	325	20	1.6	142.4
连弩·飞行	13	96	1135	330	20	1.6	153.6
连弩·飞行	14	103	1224	335	20	1.6	164.8
连弩·飞行	15	110	1320	340	20	1.6	176.0
刺客	1	60	488	301	5	1.4	84.0
刺客	2	64	526	305	5	1.4	89.6
刺客	3	69	567	310	5	1.4	96.6
刺客	4	74	611	314	5	1.4	103.6
刺客	5	79	659	319	5	1.4	110.6
刺客	6	85	711	324	5	1.4	119.0
刺客	7	91	766	329	5	1.4	127.4
刺客	8	98	826	334	5	1.4	137.2
刺客	9	105	891	339	5	1.4	147.0
刺客	10	113	961	344	5	1.4	158.2
刺客	11	121	1036	349	5	1.4	169.4
刺客	12	130	1117	354	5	1.4	182.0

模板	等级	攻击	生命	移速	射程 m	攻速次每秒	该级 DPS
刺客	13	139	1204	359	5	1.4	194.6
刺客	14	149	1298	365	5	1.4	208.6
刺客	15	160	1400	370	5	1.4	224.0
刺客·飞行	1	60	390	301	5	1.4	84.0
刺客·飞行	2	64	421	305	5	1.4	89.6
刺客·飞行	3	69	454	310	5	1.4	96.6
刺客·飞行	4	74	489	314	5	1.4	103.6
刺客·飞行	5	79	527	319	5	1.4	110.6
刺客·飞行	6	85	569	324	5	1.4	119.0
刺客·飞行	7	91	613	329	5	1.4	127.4
刺客·飞行	8	98	661	334	5	1.4	137.2
刺客·飞行	9	105	713	339	5	1.4	147.0
刺客·飞行	10	113	769	344	5	1.4	158.2
刺客·飞行	11	121	829	349	5	1.4	169.4
刺客·飞行	12	130	894	354	5	1.4	182.0
刺客·飞行	13	139	963	359	5	1.4	194.6
刺客·飞行	14	149	1039	365	5	1.4	208.6
刺客·飞行	15	160	1120	370	5	1.4	224.0
战士	1	52	627	276	5	1.1	57.2
战士	2	56	676	281	5	1.1	61.6
战士	3	60	729	285	5	1.1	66.0
战士	4	65	786	289	5	1.1	71.5
战士	5	69	848	293	5	1.1	75.9
战士	6	74	914	298	5	1.1	81.4
战士	7	80	985	302	5	1.1	88.0
战士	8	86	1063	307	5	1.1	94.6
战士	9	92	1146	311	5	1.1	101.2
战士	10	98	1235	316	5	1.1	107.8
战士	11	106	1332	320	5	1.1	116.6
战士	12	113	1436	325	5	1.1	124.3
战士	13	122	1548	330	5	1.1	134.2
战士	14	130	1669	335	5	1.1	143.0
战士	15	140	1800	340	5	1.1	154.0
坦克	1	37	767	244	5	0.9	33.3
坦克	2	40	827	248	5	0.9	36.0
坦克	3	43	891	251	5	0.9	38.7
坦克	4	46	961	255	5	0.9	41.4
坦克	5	49	1036	259	5	0.9	44.1
坦克	6	53	1117	263	5	0.9	47.7

模板	等级	攻击	生命	移速	射程 m	攻速次每秒	该级 DPS
坦克	7	57	1204	267	5	0.9	51.3
坦克	8	61	1299	271	5	0.9	54.9
坦克	9	66	1400	275	5	0.9	59.4
坦克	10	70	1510	279	5	0.9	63.0
坦克	11	75	1628	283	5	0.9	67.5
坦克	12	81	1755	287	5	0.9	72.9
坦克	13	87	1892	291	5	0.9	78.3
坦克	14	93	2040	296	5	0.9	83.7
坦克	15	100	2200	300	5	0.9	90.0

附录 B 全技能、被动与装备数据

本附录补充玩家展示文本。实际机制以第 6-8 章的完整工程表为准；展示文本不得反向驱动运行时。

B.1 161 条中文展示文本

key	category	owner	skill_name	platform	text
equip_W1	EQUIPMENT	W1	精铁棍	ALL	攻击+15
equip_W2	EQUIPMENT	W2	粗布战衣	ALL	生命+220
equip_W3	EQUIPMENT	W3	草鞋	ALL	移速+18
equip_W4	EQUIPMENT	W4	铜护腕	ALL	攻击+10；生命+100
equip_W5	EQUIPMENT	W5	轻甲背心	ALL	生命+110；移速+10
equip_W6	EQUIPMENT	W6	行者束带	ALL	攻击+8；移速+10
equip_B1	EQUIPMENT	B1	狼牙棒	ALL	攻击+20；对野怪伤害+30%
equip_B2	EQUIPMENT	B2	虎皮裙	ALL	生命+280；受普攻伤害-8
equip_B3	EQUIPMENT	B3	避水珠	ALL	生命+200；受技能伤害-15%
equip_B4	EQUIPMENT	B4	定风珠	ALL	生命+200；被击退和拉扯距离减半
equip_B5	EQUIPMENT	B5	药葫芦	ALL	生命+150；脱战时每秒回复 8 点
equip_B6	EQUIPMENT	B6	磨刀石	ALL	攻击+12；暴击伤害+30%
equip_B7	EQUIPMENT	B7	铜镜	ALL	生命+200；受到的背刺/伏击加成伤害减半
equip_B8	EQUIPMENT	B8	聚宝袋	ALL	生命+120；击杀野怪+30 金
equip_B9	EQUIPMENT	B9	夜巡灯	ALL	攻击+10；视野范围+20%
equip_B10	EQUIPMENT	B10	驭兽鞭	ALL	攻击+12；召唤物攻击+30%
equip_B11	EQUIPMENT	B11	避雷木牌	ALL	生命+200；天劫伤害-30%
equip_B12	EQUIPMENT	B12	千里靴	ALL	移速+26；纯速度蓝鞋
equip_B13	EQUIPMENT	B13	登云履	ALL	移速+20；受到的减速效果减半（鞋）
equip_B14	EQUIPMENT	B14	蛰伏靴	ALL	移速+20；主动破隐突袭后最终移速×1.30，持续 2 秒；8 秒内部 CD
equip_P1	EQUIPMENT	P1	七星剑	ALL	攻击+35；对生命低于 30%的目标伤害+25%
equip_P2	EQUIPMENT	P2	火尖枪	ALL	攻击+28；普攻点燃 2 秒，每秒 20

key	category	owner	skill_name	platform	text
equip_P3	EQUIPMENT	P3	幌金绳	ALL	攻击+18; 你的硬控持续+0.5 秒
equip_P4	EQUIPMENT	P4	人参果	ALL	生命+400; 击杀英雄回复 30%最大生命
equip_P5	EQUIPMENT	P5	万年灵芝	ALL	生命+500; 每秒回复 8 点
equip_P6	EQUIPMENT	P6	风袋	ALL	生命+200; 主动技能 CD-20%
equip_P7	EQUIPMENT	P7	招魂幡	ALL	攻击+18; 召唤物生命与攻击+60%
equip_P8	EQUIPMENT	P8	布袋	ALL	无; 手牌格+1
equip_P9	EQUIPMENT	P9	照妖珠	ALL	生命+250; 20 米内隐身单位对你现形
equip_P10	EQUIPMENT	P10	赌坊金牌	ALL	生命+150; 黑山老妖大赢概率翻倍
equip_P11	EQUIPMENT	P11	判官笔	ALL	攻击+25; 击杀英雄使主动技能 READY。击杀发生在已提交施法期间时, 当前施法继续, 完成后主动 READY。
equip_P12	EQUIPMENT	P12	荆棘甲	ALL	生命+400; 反弹所受普攻伤害 20%
equip_P13	EQUIPMENT	P13	夜行斗篷	ALL	移速+25; 静止 2 秒后隐身。实际位移超过 0.1 米会破除, 转身或移动镜头不会。
equip_P14	EQUIPMENT	P14	血玉葫芦	ALL	攻击+25; 普攻吸血 10%
equip_P15	EQUIPMENT	P15	疾风靴	ALL	移速+30; 免疫减速 (鞋)
equip_P16	EQUIPMENT	P16	连珠履	ALL	移速+24; 普攻连续有效命中同一目标, 每层攻速+4%, 最多 6 层; 3 秒未续层或换目标清零
equip_P17	EQUIPMENT	P17	摘星靴	ALL	移速+22; 穿戴者普攻判定距离+0.7 米; 仍需直接视野, 不影响主动技能与召唤物
equip_P18	EQUIPMENT	P18	磐岩履	ALL	移速+24; 所受强制位移距离与硬控时长各×0.30; 同类抗性取最强, 硬控最低 0.30 秒
equip_G1	EQUIPMENT	G1	九转金丹	ALL	无; 致死链中原地满血复活并 2 秒无敌, 随后装备消失
equip_G2	EQUIPMENT	G2	混元珠	ALL	攻击+25; 五行克制加成由+50%提升为+100%
equip_G3	EQUIPMENT	G3	渡劫铃	ALL	生命+350; 天劫伤害-70%
equip_G4	EQUIPMENT	G4	筋斗云	ALL	移速+40; 脱战 5 秒升空。造成或受到实际伤害会落地, 拾取、开箱、交易、赌博或换英雄前自动落地 (鞋)。
equip_G5	EQUIPMENT	G5	聚宝盆	ALL	指定金币账本收入+50%; 不作用于起始金、退款、回滚、击杀钱包奖励、黑山毛回报及自身加成。
equip_G6	EQUIPMENT	G6	降魔杵	ALL	攻击+60; 对当前生命高于你的敌人伤害+25%
equip_G7	EQUIPMENT	G7	顺风耳	ALL	生命+300; 敌人施放主动时显示精确施法位置 1 秒; 高频时每 0.25 秒/10 米格合并, 最多显示最近 8 个标记。
equip_G8	EQUIPMENT	G8	雷公凿	ALL	攻击+20; 攻速+40%
equip_G9	EQUIPMENT	G9	噬魂戒	ALL	无; 每次击杀永久+2 攻击; 击杀英雄+10, 无上限
equip_G10	EQUIPMENT	G10	金箍棒	ALL	攻击+80; 普攻射程+3 米; 仅核心 BOSS 5%独立掉落
active_hero_01_铁扇公主	HERO_ACTIVE	铁扇公主	芭蕉风墙	ALL	创造长 12 米风墙持续 3 秒, 碰到的单位被弹飞 5 米。纯防御分割, 无伤害。克制远程弹道和突进。风墙不是障碍物, 单位可穿过但会被弹飞。(芭蕉扇一挥, 八百里风)

key	category	owner	skill_name	platform	text
active_hero_02_红孩儿	HERO_ACTIVE	红孩儿	三昧火墙	ALL	创造长 15 米火墙持续 5 秒，穿过伤害 100+攻击×0.4，点燃 3 秒每秒 30。地形封锁+伤害。不可破坏。
active_hero_03_蜘蛛精	HERO_ACTIVE	蜘蛛精	盘丝阵	ALL	目标区域 8 米半径蛛丝持续 4 秒，每秒 40+攻击×0.2，区域内减速 25%（蛛丝缠足）。与毒雾分家：原离场残留中毒删除（总伤下调约 30%作为加速的代价）——盘丝阵=控场网，毒雾=传染瘟疫，各归其位。（盘丝洞天罗地网）
active_hero_04_蝎子精	HERO_ACTIVE	蝎子精	倒马毒	ALL	对 20 米内可见敌人施毒：目标没有你施加的毒时叠 3 层；已有你的毒时消耗你施加的全部毒与毒层爆发。不会消耗其他玩家的毒。CD36 秒。
active_hero_05_多目怪	HERO_ACTIVE	多目怪	金光	ALL	石化一个敌人 1.5 秒，期间无敌但不能动。双刃剑控制，可借机调整位置或等 CD。（百眼魔君金光罩体）
active_hero_06_九头虫	HERO_ACTIVE	九头虫	毒雾	ALL	目标区域 8 米半径瘟疫 8 秒，每秒 40+攻击×0.2，离开残留 3 秒，接触传染。持续 DoT+传染，团战收益极高。
active_hero_07_黄风怪	HERO_ACTIVE	黄风怪	三昧神风	ALL	目标区域 10 米半径风沙持续 4 秒，每秒 60+攻击×0.3，减速 40%。大范围持续 AOE+强减速，远程唯一持续范围控制。（吹瞎过悟空火眼金睛的正是此风）
active_hero_08_太上老君	HERO_ACTIVE	太上老君	八卦炉火	ALL	目标区域 12 米半径炉火砸落，1.5 秒延迟后伤害 400+攻击×1.2。最高单发 AOE，有预警可躲，配合控制收益最大。
active_hero_09_孙悟空	HERO_ACTIVE	孙悟空	大闹天宫	ALL	5 秒内攻速+80%，被动的临时战斗效果量按白名单翻倍，触发概率不变，受伤+20%。不翻倍控制时长、CD、次数、经济、永久成长、致死保护或天劫概率。
active_hero_10_二郎神	HERO_ACTIVE	二郎神	天眼	ALL	下一刀必暴击+目标已损失生命 15%额外伤害，击杀重置 CD。收割型，残血目标斩杀增伤，满血目标收益仅必暴击。
active_hero_11_哪吒	HERO_ACTIVE	哪吒	风火轮	ALL	直线冲刺 25 米，撞到第一个敌人停下并眩晕 0.8 秒，伤害 100+攻击×0.5。位移+控制+伤害，冲刺可被打断。
active_hero_12_六耳猕猴	HERO_ACTIVE	六耳猕猴	六耳幻身	ALL	创造 2 个分身持续 8 秒，继承 30%属性，无伤害但无法分辨。迷惑型，30 人乱斗中混淆视听。（连如来座下都难辨真假）
active_hero_13_大鹏雕	HERO_ACTIVE	大鹏雕	金翅俯击	ALL	瞬移到 40 米内目标背后，下次攻击必暴击。突进+必暴击，需锁定目标不能空放。飞行+瞬移=任意地形贴脸。
active_hero_14_白骨精	HERO_ACTIVE	白骨精	借尸还魂	ALL	瞬移 15 米，原地留假身 1 秒后爆炸造成 120+攻击×0.5 伤害。位移+伤害，核心价值在位移。（三打白骨精，三次留下的都是假尸——魂走了，尸还在骗你）
active_hero_15_猪八戒	HERO_ACTIVE	猪八戒	九齿钉耙	ALL	前方扇形 60 度 AOE 伤害 160+攻击×0.7，减速 30%持续 2 秒。扇形 AOE+开团信号。
active_hero_16_白龙马	HERO_ACTIVE	白龙马	龙马奔腾	ALL	直线贯穿冲刺 25 米，不因碰撞停止，沿途所有敌人受 100+攻击×0.5 伤害并被击退 2 米（无眩晕）。与风火轮彻底分家：风火轮=单体撞停+眩晕，龙马奔腾=贯穿多段伤害+切入逃生两用。（八部天龙，一骑贵阵）
active_hero_17_青狮怪	HERO_ACTIVE	青狮怪	狮吼	ALL	砸地，周围 12 米敌人击退到范围边缘并眩晕 1 秒，伤害 150+攻击×0.7。群体击退+眩晕，背后是墙推不动但仍眩晕。
active_hero_18_牛魔王	HERO_ACTIVE	牛魔王	旋风	ALL	原地旋转 3 秒，每秒对周围 8 米伤害 100+攻击×0.5，可移动但移速减半。持续 AOE，配合溅射被动效果翻倍。
active_hero_19_独角兕大王	HERO_ACTIVE	独角兕大王	金刚琢	ALL	抛出金刚琢罩向目标区域，半径 10 米内敌人沉默 3 秒（无法使用主动技能）。（金刚琢专收兵器法宝——罩住你的法宝，让你放不出招）
active_hero_20_黄眉老祖	HERO_ACTIVE	黄眉老祖	石罗汉	ALL	召唤石罗汉作战 20 秒，500 生命 30 攻击。主要价值是扛伤和探路，不是输出。（小雷音寺的假罗汉）

key	category	owner	skill_name	platform	text
active_hero_21_金角大王	HERO_ACTIVE	金角大王	紫金葫芦	ALL	1 秒蓄力后把周围 15 米敌人吸到身边，伤害 120+攻击×0.6。群体拉取，有预警可跑，吸过来接自己的 AOE。（叫你名字，你敢答应吗）
active_hero_22_银角大王	HERO_ACTIVE	银角大王	砸钱	ALL	锁定 20 米内一个敌人，消耗当前金币 20%（最低 500、最高 5000），造成消耗金币 10%的真实伤害（50-500）。真伤无视比例和固定减伤、不吃五行，但可被护盾吸收。
active_hero_23_黄袍怪	HERO_ACTIVE	黄袍怪	缠身	ALL	小鬼附身目标 8 秒，每 2 秒 40 伤害+暴露视野。战争迷雾里极其致命，配合猎杀被动追杀。
active_hero_24_虎力大仙	HERO_ACTIVE	虎力大仙	虎箭	ALL	全图直线箭头，基础 150+攻击×1.0，飞行每 10 米+5%伤害，上限+50%（满伤 225+攻击×1.5）。全图狙击，满伤需 100 米距离，可被风墙冰墙阻挡。
active_hero_25_鹿力大仙	HERO_ACTIVE	鹿力大仙	鹿血	ALL	6 秒普攻吸血+50%，期间对周围 5 米敌人每秒 30 伤害并吸等量生命。站撸神技。
active_hero_26_文殊菩萨	HERO_ACTIVE	文殊菩萨	定慧诀	ALL	对 20 米内可见敌人造成 60+攻击×0.3 伤害并眩晕 2 秒；眩晕正常受韧性、免控和控制减免影响。CD30 秒。
active_hero_27_普贤菩萨	HERO_ACTIVE	普贤菩萨	行愿	ALL	为自己或 50 米内任意可见英雄瞬间回复 220 生命，不分敌我。没有合法目标时点按会治疗自己。
active_hero_28_镇元大仙	HERO_ACTIVE	镇元大仙	袖里乾坤	ALL	标记 30 米内可见敌人 8 秒，你对其伤害+30%；目标被你击杀后，转移到 20 米内最近可见敌人并刷新 8 秒，等距按实体 ID 裁决。
active_hero_29_如来佛祖	HERO_ACTIVE	如来佛祖	五行山	ALL	镇压目标区域——1.2 秒延迟后巨山砸落，半径 8 米，伤害 300+攻击×1.0，命中减速 40%持续 2 秒。坦克最强团战核弹，有预警可躲，配合控制/地形收益最大。（压了悟空五百年的那座山）
active_hero_30_观音菩萨	HERO_ACTIVE	观音菩萨	甘露	ALL	在 30 米内放置半径 8 米的甘露区域，持续 4 秒；区域内所有英雄不分敌我每秒回复 40 生命。
active_hero_31_托塔李天王	HERO_ACTIVE	托塔李天王	宝塔	ALL	创造半径 10 米环形石墙持续 5 秒，内外无法进出。最强地形控制，可困敌可保命。克制：带闪现/借尸还魂在墙形成前跑出。
active_hero_32_唐僧	HERO_ACTIVE	唐僧	袈裟	ALL	无敌 3 秒且可移动、普攻和施放主动；期间不能拾取、开箱、交易、赌博或换英雄。
active_hero_33_沙和尚	HERO_ACTIVE	沙和尚	流沙	ALL	目标区域半径 8 米流沙持续 4 秒，每秒 50+攻击×0.25，减速 60%，且区域内敌人无法使用位移技能（闪现/冲刺/瞬移全部失效——陷进流沙，拔不出腿）。全游戏唯一反位移领域（强制软控），克制闪现/借尸还魂/风火轮/金翅俯击。（流沙河，鹅毛飘不起）
active_hero_34_黑熊精	HERO_ACTIVE	黑熊精	黑风罩	ALL	获得 80%减伤并提高 15%移速，持续 4 秒。真实伤害不受该减伤影响，但仍可被护盾吸收。
active_hero_35_白象精	HERO_ACTIVE	白象精	象鼻缚	ALL	长鼻卷住 60 米内一个目标（弹道可躲），定住 2 秒——不可移动但可攻击可施法。（黄牙老象——鼻子卷人——挣得动手脚，挣不脱身位）
active_hero_36_灵感大王	HERO_ACTIVE	灵感大王	冻河	ALL	在 15 米内指定方向创造长 10 米冰墙，持续 40 秒，拥有 300 生命并可被攻击摧毁。每个施法者同时最多 1 面，放新墙时旧墙碎裂。
active_hero_37_羊力大仙	HERO_ACTIVE	羊力大仙	羊灵之光	ALL	双模式——对敌：石化目标 1.5 秒（无敌但不能动）；对己：护体 1.5 秒（无敌但不能动，微型冰棺救急）。与金光的差异：金光只能石化敌人，羊灵之光可敌可己——保命与控制二选一的抉择技。（下油锅时护住周身的那道灵光）
active_hero_38_赛太岁	HERO_ACTIVE	赛太岁	迷烟	ALL	投掷半径 8 米烟幕持续 4 秒，区域内敌人失去视野。战术价值：掩护撤退或遮挡视线抢装备。（紫金铃摇出的烟）

key	category	owner	skill_name	platform	text
active_generic_D1	GENERIC_ACTIVE	D1	钩锁	ALL	发射钩锁 80 米，命中后把敌人拉到面前。CD28 秒，伤害 80+攻击×0.4，弹道 60 米/秒。可拉普通野怪和首领；核心 BOSS 免疫强制位移。钩锁作为投射物：可穿过不阻挡投射物的宝塔，但会被风墙、冻河和实体地图墙阻挡。
active_generic_D3	GENERIC_ACTIVE	D3	雷击	ALL	锁定 70 米内目标，0.2 秒后落雷，造成 140+攻击×0.8 伤害并眩晕 1 秒。锁定后不再检查距离和视线，无敌可挡。CD24 秒。
active_generic_D4	GENERIC_ACTIVE	D4	变羊	ALL	目标变动物 2 秒，无法攻击但移速+30%。CD32 秒，射程 45 米，无伤害。边界：有弹道可躲；双刃剑——废对方输出但给逃跑能力；适合拆火不适合追杀；飞行单位变羊后仍飞行。
active_generic_D6	GENERIC_ACTIVE	D6	闪现	ALL	向指定方向尝试瞬移 15 米，CD15 秒，无伤害。全程使用玩家胶囊连续扫描：可穿过连续实体厚度不超过 1.5 米的可跨越障碍；宝塔墙 1.6 米，因此自然阻挡。若目标点非法，停在同方向最后合法点；不能越过地图边界，瞬移后无僵直。
active_generic_D7	GENERIC_ACTIVE	D7	潜行	ALL	进入隐身+30%加速，持续 5 秒。攻击或施法显形。CD32 秒，无伤害。边界：隐身期间对方看不到（除非照妖镜/照妖珠）；破隐一击无额外伤害；隐身可拉装备/技能书不显形。
active_generic_D9	GENERIC_ACTIVE	D9	箭雨	ALL	对 35 米内目标区域降下箭雨，圆形半径 8 米，范围内所有敌人受伤。CD24 秒，射程 35 米，伤害 130+攻击×0.6。边界：落箭区有 0.8 秒预警可走出；从天而降不可被风墙阻挡；清野效率高；与九齿钉耙（近战扇形+减速）形状彻底错开。
active_generic_D10	GENERIC_ACTIVE	D10	电链	ALL	对一个目标释放闪电，跳跃至 5 个敌人，每次跳跃衰减 15%。CD85 秒，初始伤害 300+攻击×1.0，跳跃范围：每次 15 米。边界：初始必中，后续自动寻 15 米内最近非友方；最多 6 目标；第 6 跳≈初始 44%；总伤害全吃约初始×4.3 倍；适合团战不适合单挑；范围内只 1 个目标只打 1 个。
active_generic_D11	GENERIC_ACTIVE	D11	轮盘	ALL	锁定 40 米内目标并造成 1—999 点均匀随机技能伤害，双方看见结果数字。伤害期望 500；不能暴击；五行、护盾和击杀归属按普通玩家技能处理。CD50 秒。
active_generic_D12	GENERIC_ACTIVE	D12	标记	ALL	标记 60 米内一个敌人 90 秒；只有施法者持续看见其精确位置。施法者击杀目标额外获得 800 金。CD70 秒。
active_generic_D13	GENERIC_ACTIVE	D13	换位	ALL	先选择 50 米内可见单位 A，再在 3 秒内选择不同单位 B 并原子换位；超时、取消或目标非法不进 CD。首领、龙王和核心 BOSS 免疫。CD62 秒。
active_generic_D14	GENERIC_ACTIVE	D14	回溯	ALL	回到 5 秒前的位置和血量（取两者中较高值）。CD95 秒，回溯时间 5 秒。边界：血量取较高值；位置若有障碍物回溯失败不穿墙；5 秒前在天劫区外现在区内可救命，反之会害死你；不重置技能 CD。
active_generic_D15	GENERIC_ACTIVE	D15	妙手空空	ALL	30 米内立即斩杀普通或飞行野怪；肥猪仅在生命不高于 30%时可斩杀。斩杀只把基础金币变为 3 倍，经验和掉落仍结算 1 次。对玩家转移 500 金，不足则全取。CD38 秒。
active_generic_D16	GENERIC_ACTIVE	D16	聚财	ALL	立即获得 450 金币。CD100 秒，不限次数；真死复活后主动按全局规则可用，已创造金币记录不会重置。

key	category	owner	skill_name	platform	text
active_generic_D17	GENERIC_ACTIVE	D17	连赏	ALL	开启赏金状态后，每次击杀玩家使赏金 500→1000→2000→4000→8000→16000→32000，封顶 32000；玩家击杀你获得当前赏金 50%。CD55 秒。
active_generic_D18	GENERIC_ACTIVE	D18	淘金狂热	ALL	扫描 80 米内落地超过 10 秒的装备并标记 20 秒；可穿 迷雾和墙，只向施法者显示，且不显示掉落者。CD16 秒。
active_generic_D19	GENERIC_ACTIVE	D19	豆兵	ALL	撒出 3 个豆兵持续 15 秒，每个 150 生命 15 攻击。CD50 秒。边界：豆兵自动攻击 附近敌人优先英雄；攻击力固定不受英雄加成；可被 AOE 清掉；不触发被动。
active_generic_D20	GENERIC_ACTIVE	D20	陷阱	ALL	放置可见陷阱，持续 90 秒、最多 3 个。任何实际胶囊轨 迹穿越都会触发，只有无敌或从触发体上方越过例外；短 按选最近陷阱引爆，长按选择。CD16 秒。
active_generic_D21	GENERIC_ACTIVE	D21	冰棺	ALL	冰封自己，无敌 4 秒但不能移动/攻击/施法。CD75 秒。边界：无敌但自缚；4 秒内 对方可守尸；适合拖过对方爆发窗口（大闹天宫 5 秒冰棺 4 秒拖掉大部分）；不能 用来逃生。
active_generic_D22	GENERIC_ACTIVE	D22	铁布衫	ALL	获得 600 护盾 5 秒，可吸收所有伤害类型。多个护盾并 存时先消耗最早到期者；到期相同按创建顺序。CD45 秒。
passive_B1	PASSIVE	B1	冰霜	ALL	1 级 10% 减速 30% 2 秒 2 级 13% 减速 40% 2 秒 3 级 16% 减速 50% 2.5 秒 4 级 20% 减速 60% 2.5 秒 5 级 25% 减速 80% 3 秒。质变：目标周围 5 米敌人也减速 35%
passive_B2	PASSIVE	B2	麻痹	ALL	1 级 8% 沉默 0.8 秒 2 级 10% 沉默 1.0 秒 3 级 12% 沉默 1.2 秒 4 级 15% 沉默 1.4 秒 5 级 18% 沉默 1.5 秒。质变：沉默结束时目标主动技能 CD+3 秒
passive_B3	PASSIVE	B3	击退	ALL	1 级 20% 击退 3 米 2 级 23% 3.5 米 3 级 26% 4 米 4 级 30% 4.5 米 5 级 35% 击 退 5 米。质变：暴击时目标周围 5 米敌人也击退 3 米
passive_B4	PASSIVE	B4	致盲	ALL	1 级 10% 目标普攻 50% miss 2 秒 2 级 12% 60% miss 2 秒 3 级 14% 70% miss 2.5 秒 4 级 17% 80% miss 2.5 秒 5 级 20% 80% miss 3 秒。质变：致盲期间目标无法暴 击
passive_B5	PASSIVE	B5	眩晕	ALL	1 级 5% 眩晕 0.4 秒 2 级 6% 0.5 秒 3 级 7% 0.6 秒 4 级 8% 0.7 秒 5 级 10% 眩 晕 0.8 秒。质变：触发时目标周围 3 米敌人也眩晕 0.3 秒
passive_B6	PASSIVE	B6	暴击	ALL	1 级 8% 暴击 180% 2 级 10% 190% 3 级 13% 200% 4 级 16% 210% 5 级 20% 暴 击 230%。质变：暴击时忽略目标 30% 护盾
passive_B7	PASSIVE	B7	溅射	ALL	1 级 10% 溅射 40% 2 级 13% 45% 3 级 16% 50% 4 级 20% 55% 5 级 25% 溅射 60%。质变：溅射范围扩大 50%，主目标也受溅射伤害
passive_B8	PASSIVE	B8	燃灰	ALL	1 级 叠 5 层引爆已损失 8% 2 级 10% 3 级 12% 4 级 叠 4 层引爆 12% 5 级 叠 4 层引爆 15%。质变：爆炸时传染 2 层给周围敌人
passive_B9	PASSIVE	B9	毒素	ALL	1 级 叠 1 层每秒 3 点最多 3 层 3 秒 2 级 4 点/秒最多 4 层 3 级 5 点/秒最多 4 层 4 秒 4 级 6 点/秒最多 5 层 4 秒 5 级 8 点/秒最多 5 层 4 秒。质变：叠满 5 层伤害+50%（12 点/秒）
passive_B10	PASSIVE	B10	斩杀	ALL	1 级 生命低于 20% 伤害+25% 2 级 22% +30% 3 级 25% +35% 4 级 28% +40% 5 级 30% +50%。质变：斩杀击杀 时回复自身 10% 最大生命

key	category	owner	skill_name	platform	text
passive_B11	PASSIVE	B11	连击	ALL	1 级 8% 追加 1 次普攻 2 级 10% 追加 1 次 3 级 12% 最多 2 次 4 级 15% 最多 2 次 5 级 20% 最多 3 次。质变：连击最后一刀 50%概率触发一个其他被动（从已装备其他 3 个被动中随机选一个，无视该被动原概率，无论只触发一次）。额外普攻不能再次触发连击；没有其他已拥有被动时，5 级附加触发不发生
passive_B12	PASSIVE	B12	狼灵	ALL	1 级 6% 召唤狼灵 15 秒 150 血 20 攻 2 级 8% 200/25 3 级 10% 250/30 4 级 12% 300/35 5 级 15% 350/40。质变：最多同时 2 只
passive_B13	PASSIVE	B13	火灵	ALL	1 级 6% 召唤火灵环绕 8 秒触碰 60 伤+点燃 3 秒 2 级 8% 80 伤 3 级 10% 100 伤 4 级 12% 120 伤 5 级 15% 150 伤。质变：最多 2 个火灵
passive_B14	PASSIVE	B14	冷箭	ALL	1 级 10% 自动射一箭 60 伤 2 级 12% 80 伤 3 级 14% 100 伤 4 级 16% 120 伤 5 级 20% 150 伤。质变：冷箭可暴击享受暴击加成
passive_B15	PASSIVE	B15	闪避	ALL	1 级 10% 躲避普攻 2 级 13% 3 级 16% 4 级 20% 5 级 25%。质变：成功闪避后 2 秒内移速+30%
passive_B16	PASSIVE	B16	铁皮	ALL	1 级 减 6 点普攻伤害 2 级 减 10 3 级 减 14 4 级 减 18 5 级 减 25。质变：10%概率完全格挡一次普攻
passive_B17	PASSIVE	B17	护盾	ALL	1 级 8% 护盾吸 60 伤 2 级 10% 80 伤 3 级 13% 100 伤 4 级 16% 130 伤 5 级 20% 160 伤。质变：护盾被打破时对周围造成 100 点 AOE 伤害
passive_B18	PASSIVE	B18	嗜血	ALL	1 级 每损 10%生命攻击+3% 2 级 +4% 3 级 +5% 4 级 +6% 5 级 +8%。质变：血量低于 30%时额外获得 10%吸血
passive_B19	PASSIVE	B19	假死	ALL	1 级 5% 致死免死回 10%血 2 级 6% 12% 3 级 7% 15% 4 级 8% 18% 5 级 10% 回 20%血。质变：触发后自动向随机方向闪现

key	category	owner	skill_name	platform	text
passive_B20	PASSIVE	B20	复活	ALL	1 级 死亡原地复活回 25%血（每局 1 次） 2 级 30% 3 级 35% 4 级 40% 5 级 50%。质变：复活后 5 秒内伤害 +30%且免控。这次充能属于整局，顶掉、重新获得或换英雄都不会恢复
passive_B21	PASSIVE	B21	回复	ALL	1 级 脱战 5 秒后每秒回 4 血 2 级 6 点 3 级 8 点脱战缩为 4 秒 4 级 10 点 5 级 15 点脱战缩为 3 秒。质变：脱战后第一次普攻伤害+30%
passive_B22	PASSIVE	B22	反伤	ALL	1 级 反弹所受普攻 10% 2 级 13% 3 级 16% 4 级 20% 5 级 25%。质变：反弹伤害可暴击
passive_B23	PASSIVE	B23	反击	ALL	1 级 6% 受击自动反击 80%伤害 2 级 8% 85% 3 级 10% 90% 4 级 13% 95% 5 级 16% 100%。质变：反击必定暴击。只对直接伤害反击，同一来源 0.5 秒内最多触发一次
passive_B24	PASSIVE	B24	化伤	ALL	受到技能伤害后，将其中一部分转为自己的回血；这是受伤后的恢复，不能防止致死。等级越高转化越多，满级时还会把转化量的一半反弹给对方。
passive_B25	PASSIVE	B25	暴怒	ALL	1 级 被暴击时下次普攻+20% 2 级 28% 3 级 36% 4 级 44% 5 级 55%。质变：触发后 2 秒内攻速+40%
passive_B26	PASSIVE	B26	背刺	ALL	1 级 从背后攻击+35% 2 级 42% 3 级 50% 4 级 58% 5 级 70%。质变：背后攻击必定暴击
passive_B27	PASSIVE	B27	疾跑	ALL	1 级 10% 受击移速+20% 2 秒 2 级 13% +25% 3 级 16% +30% 2.5 秒 4 级 20% +35% 5 级 25% +40% 3 秒。质变：加速期间免疫减速
passive_B28	PASSIVE	B28	猎杀	ALL	1 级 感知 30 米内残血敌人（低于 30%）对其移速+15% 2 级 40 米 +18% 3 级 50 米 +20% 4 级 60 米 +22% 5 级 80 米 +25%。质变：对残血敌人伤害+20%。只追踪当前可见或已被显形的残血敌人
passive_B29	PASSIVE	B29	伏击	ALL	1 级 对 5 秒内未受过你伤害的目标首击+25% 2 级 35% 3 级 45%时间缩为 4 秒 4 级 55% 5 级 70%时间缩为 3 秒。质变：触发伏击时额外暴露目标视野 3 秒

key	category	owner	skill_name	platform	text
passive_B30	PASSIVE	B30	残影	ALL	1 级 移动每 2 秒留一个残影 1.5 秒触碰减速 20% 2 级 减速 25%持续 2 秒 3 级 每 1.5 秒留一个减速 30% 4 级 每 1.5 秒留一个持续 2.5 秒 5 级 每秒留一个残影减速 35%持续 3 秒。质变：残影被触碰时爆炸 80 点 AOE
passive_B31	PASSIVE	B31	偷钱	ALL	1 级 6% 普攻偷 30 金 2 级 8% 40 金 3 级 10% 50 金 4 级 13% 60 金 5 级 16% 80 金。质变：偷金币时回复偷取 额 10%的生命
passive_B32	PASSIVE	B32	贪婪	ALL	1 级 击杀野怪额外+40 金 2 级 +60 3 级 +80 4 级 +100 5 级 +150。质变：击杀英雄也触发+250 金
passive_B33	PASSIVE	B33	寻宝	ALL	1 级 8%击杀野怪掉宝箱（100 金） 2 级 10%（120 金） 3 级 13%（120 金，10%含 1 宝石） 4 级 16%（120 金，15%含 1 宝石） 5 级 20%（120 金，20%含 1 宝石，5%含 G1-G9 随机金装）
passive_B34	PASSIVE	B34	利息	ALL	1 级 每分钟获持有金币 5%利息（上限 300 金） 2 级 6% 上限 400 3 级 7%上限 500 4 级 8%上限 600 5 级 10% 上限 800。质变：利息上限翻倍（1600 金）
passive_B35	PASSIVE	B35	拾荒	ALL	1 级 卖装备额外+10%金 2 级 13% 3 级 16% 4 级 20% 5 级 25%。质变：捡起地上装备时立刻获得该装备卖价的 20%金币
passive_B36	PASSIVE	B36	奔腾	ALL	1 级 移动每秒叠 1 层每层+1%移速最多 5 层停下清零 2 级 每层 1.2%最多 6 层 3 级 每层 1.5%最多 6 层 4 级 每层 1.8%最多 7 层 5 级 每层 2%最多 8 层。质变：叠满 8 层时普攻附带移速值 20%的额外伤害
passive_B37	PASSIVE	B37	共鸣	ALL	1 级 召唤物死亡回 10 血 2 级 15 血 3 级 20 血+30 点 AOE 4 级 25 血+40 点 AOE 5 级 30 血+50 点 AOE。质变：召唤物死亡效果提升 50%（45 血+75 点 AOE）。自然到期、替换或主人死亡清理的召唤物不触发共鸣
passive_B38	PASSIVE	B38	逆境	ALL	1 级 被控制时受伤-15% 2 级 20% 3 级 25% 4 级 30% 5 级 35%。质变：被控制时每秒回 2%最大生命

key	category	owner	skill_name	platform	text
passive_B39	PASSIVE	B39	石像	ALL	1 级 脱战 7 秒后自动召唤石像跟随 250 血 0 攻 2 级 300 血 3 级 350 血脱战缩为 6 秒 4 级 400 血 5 级 500 血脱战缩为 5 秒。质变：石像被摧毁时对周围造成 120 点 AOE 伤害
passive_B40	PASSIVE	B40	强韧	ALL	1 级 每击杀+2 生命无上限 2 级 +4 3 级 +6 4 级 +7 5 级 +10。质变：每击杀 50 个单位额外+160 生命（可多次触发）
passive_B41	PASSIVE	B41	赏金	ALL	1 级 你被击杀后击杀者头出现赏金标记 90 秒，任何人击杀该目标+400 金 2 级 +600 金 3 级 +800 金标记持续 120 秒 4 级 +1000 金 5 级 +1500 金。质变：标记期间所有玩家小地图可见赏金目标大致方位
passive_B42	PASSIVE	B42	悬赏	ALL	1 级 击杀带赏金标记敌人时主动技能 CD-10 秒 2 级 -15 秒 3 级 -20 秒 4 级 -25 秒 5 级 -30 秒。质变：击杀赏金目标后 5 秒内移速+40%
passive_B43	PASSIVE	B43	避雷	ALL	1 级 天劫区内被劈概率-20% (50%→40%) 2 级 -30% (→35%) 3 级 -40% (→30%) 天劫区内移速+10% 4 级 -50% (→25%) 5 级 -60% (→20%) 移速+15%。质变：天劫区内普攻伤害+10%
passive_B44	PASSIVE	B44	雷暴	ALL	1 级 8% 普攻触发落雷 80 点 AOE (半径 3 米) 2 级 10% 90 点 3 级 13% 100 点 AOE 范围 4 米 4 级 16% 110 点 5 级 20% 130 点 AOE 范围 5 米。质变：天劫区内触发概率 1.5 倍 (30%)
tip_control_pc	TIP	NOT_APPLICABLE	控制提示	PC	WASD 移动，左键普攻，右键或长按瞄准主动，E 键拾取与交互。
tip_control_mobile	TIP	NOT_APPLICABLE	控制提示	MOBILE	左摇杆移动，点普攻键攻击，点按或拖动主动键施法，靠近目标后点情境交互键。
tip_topic_01_技能书	TIP	NOT_APPLICABLE	技能书	ALL	技能书会学习或升级被动；四格已满时可选择顶替，满级同名书不可拾取。

key	category	owner	skill_name	platform	text
tip_topic_02_宝石	TIP	NOT_APPLICABLE	宝石	ALL	宝石进入独立宝石仓，可随时强化未满级被动，不占手牌格。
tip_topic_03_肥猪	TIP	NOT_APPLICABLE	肥猪	ALL	五行肥猪会顶撞后逃向窝底；克制它可更快拿到经济奖励，小心袋口埋伏。
tip_topic_04_伏石	TIP	NOT_APPLICABLE	伏石	ALL	进入伏石周围 2 米会隐身；攻击、施法或受击会显形，照妖道具可反隐。
tip_topic_05_商店	TIP	NOT_APPLICABLE	商店	ALL	商店通常每 3 分钟迁移并补货；最终剩余营业时间不足 90 秒时不会再补一轮。地图只提示大致区域，后期保留土地公和太白金星。
tip_topic_06_天劫	TIP	NOT_APPLICABLE	天劫	ALL	圈外每 3 秒有 50%概率遭雷击；尽快跑回安全区，安全圈归零后转为持续雷暴。
tip_topic_07_暴露	TIP	NOT_APPLICABLE	暴露	ALL	伤害玩家会在小地图暴露 3 秒；墙体仍会阻挡视野与锁定。
tip_topic_08_三命	TIP	NOT_APPLICABLE	三命	ALL	前两次真死会随机复活，第三次死亡才淘汰；复活点由系统在安全区内分配。
tip_topic_09_五行	TIP	NOT_APPLICABLE	五行	ALL	金克木、木克土、土克水、水克火、火克金；玩家互相及玩家与五行肥猪/龙王双向生效，普通怪无五行。
tip_topic_10_BOSS 公共掉落	TIP	NOT_APPLICABLE	BOSS 公共掉落	ALL	核心 BOSS 金币经验归尾刀，装备和主动技能公开落地，任何人都能抢。

附录 C 全怪物与定向掉落

C.1 技能书隐藏倾向权重

被动 ID	CORE	ELITE_DPS	ELITE_TANK	FLY	MEL	PIG	RNG
B01	1	3	1	1	1	1	6
B02	1	3	1	1	1	1	3

被动 ID	CORE	ELITE_DPS	ELITE_TANK	FLY	MEL	PIG	RNG
B03	1	3	1	3	1	1	1
B04	1	3	1	1	1	1	6
B05	1	3	1	1	1	1	3
B06	1	6	1	1	3	1	3
B07	1	3	1	1	1	1	3
B08	1	3	1	1	1	1	1
B09	1	3	1	1	3	1	1
B10	1	6	1	1	1	1	1
B11	1	6	1	1	6	1	1
B12	1	3	1	1	1	1	1
B13	1	3	1	3	1	1	1
B14	1	3	1	3	1	1	6
B15	1	1	3	1	1	1	1
B16	1	1	3	1	6	1	1
B17	1	1	6	1	1	1	1
B18	1	1	3	1	6	1	1
B19	1	1	3	1	1	1	1
B20	1	1	6	1	1	1	1
B21	1	1	3	1	1	1	1
B22	1	1	3	1	3	1	1
B23	1	1	1	1	3	1	1
B24	1	1	6	1	1	1	1
B25	1	1	3	1	1	1	1
B26	1	3	1	3	1	1	1
B27	1	1	1	6	3	1	1
B28	1	1	1	6	1	3	3
B29	1	3	1	1	1	3	3
B30	1	1	1	3	1	1	1
B31	1	1	1	1	1	6	1
B32	1	1	1	1	1	6	1
B33	1	1	1	1	1	6	1
B34	1	1	1	1	1	3	1
B35	1	1	1	1	1	6	1
B36	1	1	1	6	1	1	1
B37	1	1	1	1	1	1	1
B38	1	1	3	1	1	1	1
B39	1	1	3	1	1	1	1
B40	1	1	1	1	3	3	1
B41	1	1	1	1	1	3	1

被动 ID	CORE	ELITE_DPS	ELITE_TANK	FLY	MEL	PIG	RNG
B42	1	1	1	1	1	3	1
B43	1	1	1	1	1	1	1
B44	1	1	1	1	1	1	1

C.2 装备隐藏倾向权重

装备 ID	CORE	DRAGON_EARTH	DRAGON_FIRE	DRAGON_METAL	DRAGON_WATER	DRAGON_WOOD	ELITE_DPS	ELITE_TANK	PIG
W1	1	1	1	1	1	1	3	1	1
W2	1	1	1	1	1	1	1	3	1
W3	1	1	1	1	1	1	1	1	1
W4	1	1	1	1	1	1	3	3	1
W5	1	1	1	1	1	1	1	3	1
W6	1	1	1	1	1	1	3	1	1
B1	1	1	6	3	1	1	6	1	1
B2	1	6	1	1	1	1	1	6	1
B3	1	3	1	1	3	1	1	6	1
B4	1	3	1	1	6	1	1	3	1
B5	1	1	1	1	1	6	1	3	3
B6	1	1	3	6	1	1	6	1	1
B7	1	3	1	1	1	1	1	3	1
B8	1	1	1	1	1	3	1	1	6
B9	1	1	3	3	1	1	3	1	3
B10	1	1	1	3	1	3	3	1	1
B11	1	3	1	1	1	1	1	3	3
B12	1	1	1	1	1	1	1	1	1
B13	1	1	1	1	3	1	1	1	1
B14	1	1	3	1	3	2	3	1	3
P1	1	1	3	3	1	1	6	1	1
P2	1	1	6	1	1	1	3	1	1
P3	1	1	1	3	3	1	3	1	1
P4	1	3	1	1	1	3	1	3	1
P5	1	3	1	1	1	6	1	3	1
P6	1	1	1	1	6	1	3	1	1
P7	1	1	1	1	1	3	1	1	1
P8	1	1	1	1	1	1	1	1	6
P9	1	3	1	1	3	1	1	3	3
P10	1	1	1	1	1	1	1	1	6
P11	1	1	3	6	1	1	3	1	1
P12	1	6	1	1	1	1	1	6	1
P13	1	1	1	1	3	1	1	3	3
P14	1	1	3	3	1	3	3	1	3
P15	1	1	3	1	3	1	3	1	1
P16	1	1	6	3	1	3	6	1	1

装备 ID	CORE	DRAGON_EARTH	DRAGON_FIRE	DRAGON_METAL	DRAGON_WATER	DRAGON_WOOD	ELITE_DPS	ELITE_TANK	PIG
P17	1	1	2	2	2	1	3	1	1
P18	1	6	1	3	3	2	1	6	1
G1	1	3	1	1	1	1	1	1	1
G2	1	1	1	6	1	1	1	1	1
G3	1	6	1	1	1	1	1	1	1
G4	1	1	1	1	6	1	1	1	1
G5	1	1	1	1	1	6	1	1	1
G6	1	1	3	3	1	1	1	1	1
G7	1	1	1	1	3	3	1	1	1
G8	1	1	6	3	1	1	1	1	1
G9	1	1	1	1	1	3	1	1	1
G10	0	0	0	0	0	0	0	0	0

附录 D 地图坐标、路网、墙体与 POI

地图导入 | 以下坐标为米制设计坐标。运行时导入时转换为毫米整数；实体身份使用命名空间+ID，source_label 只作美术/关卡查找，不作运行时主键。

D.1 边界与节点

序号	边界点(x,y)m
1	-420 / 293.2
2	-340.1 / 333.1
3	-201.5 / 325.1
4	-110.9 / 343.8
5	38.4 / 322.5
6	161 / 341.1
7	302.2 / 317.1
8	400.8 / 245.2
9	416.8 / 111.9
10	384.8 / 10.7
11	419.5 / -106.6
12	368.8 / -245.2
13	256.9 / -322.5
14	129 / -306.5
15	11.7 / -341.1
16	-116.2 / -317.1
17	-233.5 / -341.1

序号	边界点(x,y)m
18	-353.4 / -271.8
19	-401.3 / -165.2
20	-374.7 / -50.6
21	-412 / 58.6

节点	X	Y	Z	属性
N000	-350.7	207.9	0	region_id=REGION_SPIDER_GORGE
N001	-254.8	207.9	0	region_id=REGION_SPIDER_GORGE
N002	-212.1	159.9	0	region_id=REGION_SPIDER_GORGE
N003	-110.9	159.9	0	region_id=REGION_BROKEN_GOLD
N004	-62.9	101.3	0	region_id=REGION_BROKEN_GOLD
N005	49	101.3	0	region_id=REGION_DRAGON_RIDGE
N006	97	154.6	0	region_id=REGION_DRAGON_RIDGE
N007	187.6	154.6	0	region_id=REGION_DRAGON_RIDGE
N008	246.2	199.9	0	region_id=REGION_DRAGON_RIDGE
N009	336.9	202.5	0	region_id=REGION_DRAGON_RIDGE
N010	-361.4	-29.3	0	region_id=REGION_SPIDER_GORGE
N011	-281.4	-29.3	0	region_id=REGION_SPIDER_GORGE
N012	-230.8	24	0	region_id=REGION_SPIDER_GORGE
N013	-153.5	24	0	region_id=REGION_SPIDER_GORGE
N014	-100.2	-21.3	0	region_id=REGION_SPIDER_GORGE
N015	-9.6	-21.3	0	region_id=REGION_LOST_SOUL_FIELD
N016	33	26.7	0	region_id=REGION_DRAGON_RIDGE
N017	134.3	26.7	0	region_id=REGION_BURNING_WATER
N018	187.6	-21.3	0	region_id=REGION_BURNING_WATER
N019	283.6	-21.3	0	region_id=REGION_BURNING_WATER
N020	339.5	26.7	0	region_id=REGION_BURNING_WATER
N021	-332.1	-181.2	0	region_id=REGION_LOST_SOUL_FIELD

节点	X	Y	Z	属性
N022	-241.4	-181.2	0	region_id=REGION_LOST_SOUL_FIELD
N023	-188.1	-127.9	0	region_id=REGION_LOST_SOUL_FIELD
N024	-105.5	-127.9	0	region_id=REGION_LOST_SOUL_FIELD
N025	-49.6	-186.6	0	region_id=REGION_LOST_SOUL_FIELD
N026	65	-186.6	0	region_id=REGION_CENTIPEDE_CITY
N027	118.3	-138.6	0	region_id=REGION_CENTIPEDE_CITY
N028	203.6	-138.6	0	region_id=REGION_CENTIPEDE_CITY
N029	267.6	-191.9	0	region_id=REGION_CENTIPEDE_CITY
N030	339.5	-191.9	0	region_id=REGION_BURNING_WATER
N031	-281.4	111.9	0	region_id=REGION_SPIDER_GORGE
N032	-158.8	226.5	0	region_id=REGION_BROKEN_GOLD
N033	-44.2	231.9	0	region_id=REGION_BROKEN_GOLD
N034	49	191.9	0	region_id=REGION_DRAGON_RIDGE
N035	-70.9	58.6	0	region_id=REGION_BROKEN_GOLD
N036	-49.6	-80	0	region_id=REGION_LOST_SOUL_FIELD
N037	211.6	72	0	region_id=REGION_BURNING_WATER
N038	-316.1	58.6	0	region_id=REGION_SPIDER_GORGE
N039	-180.2	111.9	0	region_id=REGION_SPIDER_GORGE
N040	-254.8	-69.3	0	region_id=REGION_SPIDER_GORGE
N041	-110.9	245.2	0	region_id=REGION_BROKEN_GOLD
N042	272.9	106.6	0	region_id=REGION_DRAGON_RIDGE
N043	320.9	-85.3	0	region_id=REGION_BURNING_WATER
N044	-150.8	-234.5	0	region_id=REGION_LOST_SOUL_FIELD
N045	-110.9	-314.5	0	region_id=REGION_LOST_SOUL_FIELD
N046	65	-274.5	0	region_id=REGION_CENTIPEDE_CITY
N047	115.7	-303.8	0	region_id=REGION_CENTIPEDE_CITY
N048	-217.5	74.6	0	region_id=REGION_SPIDER_GORGE

节点	X	Y	Z	属性
N049	75.7	-21.3	0	region_id=REGION_BURNING_WATER
N050	235.6	-74.6	0	region_id=REGION_BURNING_WATER
N051	-137.5	-69.3	0	region_id=REGION_SPIDER_GORGE
N052	22.4	109.3	0	region_id=REGION_DRAGON_RIDGE
N053	-16.1	105.6	0	region_id=REGION_BROKEN_GOLD
N054	54.3	128.7	0	region_id=REGION_DRAGON_RIDGE
N055	26.5	77.4	0	region_id=REGION_DRAGON_RIDGE
N056	-12.3	-58.6	0	region_id=REGION_LOST_SOUL_FIELD
N057	-43.3	-45.5	0	region_id=REGION_LOST_SOUL_FIELD
N058	26.7	-42.1	0	region_id=REGION_CENTIPEDE_CITY
N059	-46.2	-78	0	region_id=REGION_LOST_SOUL_FIELD
N060	150.3	-58.6	0	region_id=REGION_BURNING_WATER
N061	120.8	-43.9	0	region_id=REGION_BURNING_WATER
N062	189.4	-66	0	region_id=REGION_BURNING_WATER
N063	136.8	-92.3	0	region_id=REGION_CENTIPEDE_CITY
N064	-305.4	194.5	0	region_id=REGION_SPIDER_GORGE
N065	-188.1	48	0	region_id=REGION_SPIDER_GORGE
N066	-49.6	207.9	0	region_id=REGION_BROKEN_GOLD
N067	248.9	143.9	0	region_id=REGION_DRAGON_RIDGE
N068	320.9	-37.3	0	region_id=REGION_BURNING_WATER
N069	-302.7	-194.5	0	region_id=REGION_LOST_SOUL_FIELD
N070	-145.5	-266.5	0	region_id=REGION_LOST_SOUL_FIELD
N071	126.3	-255.8	0	region_id=REGION_CENTIPEDE_CITY
N072	302.2	-207.9	0	region_id=REGION_CENTIPEDE_CITY
N073	-353.4	58.6	0	region_id=REGION_SPIDER_GORGE
N074	368.8	69.3	0	region_id=REGION_BURNING_WATER
N075	51.7	-311.8	0	region_id=REGION_CENTIPEDE_CITY

节点	X	Y	Z	属性
N076	153	229.2	0	region_id=REGION_DRAGON_RIDGE
N077	288.9	106.6	0	region_id=REGION_DRAGON_RIDGE
N078	-228.1	-74.6	0	region_id=REGION_SPIDER_GORGE
N079	-100.2	-165.2	0	region_id=REGION_LOST_SOUL_FIELD
N080	200.9	-154.6	0	region_id=REGION_CENTIPEDE_CITY
N081	-158.8	111.9	0	region_id=REGION_SPIDER_GORGE
N082	-49.6	-119.9	0	region_id=REGION_LOST_SOUL_FIELD
N083	-251	212.7	0	region_id=REGION_BROKEN_GOLD
N084	-120.5	177.5	0	region_id=REGION_BROKEN_GOLD
N085	95.4	202	0	region_id=REGION_DRAGON_RIDGE
N086	345.9	140.2	0	region_id=REGION_DRAGON_RIDGE
N087	-275	6.9	0	region_id=REGION_SPIDER_GORGE
N088	-123.1	-57	0	region_id=REGION_SPIDER_GORGE
N089	231.3	-57	0	region_id=REGION_BURNING_WATER
N090	313.9	-110.3	0	region_id=REGION_BURNING_WATER
N091	-245.7	-153	0	region_id=REGION_LOST_SOUL_FIELD
N092	-59.2	-121	0	region_id=REGION_LOST_SOUL_FIELD
N093	76.8	-153	0	region_id=REGION_CENTIPEDE_CITY
N094	252.6	-183.9	0	region_id=REGION_CENTIPEDE_CITY
N095	-324.1	143.9	0	region_id=REGION_SPIDER_GORGE
N096	75.7	34.6	0	region_id=REGION_DRAGON_RIDGE
N097	363.5	26.7	0	region_id=REGION_BURNING_WATER
N098	-14.9	-165.2	0	region_id=REGION_LOST_SOUL_FIELD
N099	333.1	166.3	0	region_id=REGION_DRAGON_RIDGE
N100	315.5	141.2	0	region_id=REGION_DRAGON_RIDGE
N101	299.5	114.6	0	region_id=REGION_DRAGON_RIDGE
N102	360.8	181.2	0	region_id=REGION_DRAGON_RIDGE

节点	X	Y	Z	属性
N103	374.2	202.5	0	region_id=REGION_DRAGON_RIDGE
N104	325.7	36.2	0	region_id=REGION_BURNING_WATER
N105	299.5	32	0	region_id=REGION_BURNING_WATER
N106	283.6	10.7	0	region_id=REGION_BURNING_WATER
N107	350.2	48	0	region_id=REGION_BURNING_WATER
N108	368.8	64	0	region_id=REGION_BURNING_WATER
N109	-2.7	254.2	0	region_id=REGION_BROKEN_GOLD
N110	-1.6	231.9	0	region_id=REGION_BROKEN_GOLD
N111	-9.6	207.9	0	region_id=REGION_BROKEN_GOLD
N112	19.7	261.2	0	region_id=REGION_BROKEN_GOLD
N113	43.7	271.8	0	region_id=REGION_DRAGON_RIDGE
N114	-302.7	207.9	0	region_id=REGION_SPIDER_GORGE
N115	-233.5	183.9	0	region_id=REGION_SPIDER_GORGE
N116	-161.5	159.9	0	region_id=REGION_BROKEN_GOLD
N117	-86.9	130.6	0	region_id=REGION_BROKEN_GOLD
N118	-6.9	101.3	0	region_id=REGION_BROKEN_GOLD
N119	73	127.9	0	region_id=REGION_DRAGON_RIDGE
N120	142.3	154.6	0	region_id=REGION_DRAGON_RIDGE
N121	216.9	177.2	0	region_id=REGION_DRAGON_RIDGE
N122	291.6	201.2	0	region_id=REGION_DRAGON_RIDGE
N123	-321.4	-29.3	0	region_id=REGION_SPIDER_GORGE
N124	-256.1	-2.7	0	region_id=REGION_SPIDER_GORGE
N125	-192.1	24	0	region_id=REGION_SPIDER_GORGE
N126	-126.9	1.3	0	region_id=REGION_SPIDER_GORGE
N127	-54.9	-21.3	0	region_id=REGION_SPIDER_GORGE
N128	11.7	2.7	0	region_id=REGION_BURNING_WATER
N129	83.7	26.7	0	region_id=REGION_DRAGON_RIDGE

节点	X	Y	Z	属性
N130	161	2.7	0	region_id=REGION_BURNING_WATER
N131	235.6	-21.3	0	region_id=REGION_BURNING_WATER
N132	311.5	2.7	0	region_id=REGION_BURNING_WATER
N133	-286.8	-181.2	0	region_id=REGION_LOST_SOUL_FIELD
N134	-214.8	-154.6	0	region_id=REGION_LOST_SOUL_FIELD
N135	-146.8	-127.9	0	region_id=REGION_LOST_SOUL_FIELD
N136	-77.6	-157.2	0	region_id=REGION_LOST_SOUL_FIELD
N137	7.7	-186.6	0	region_id=REGION_CENTIPEDE_CITY
N138	91.7	-162.6	0	region_id=REGION_CENTIPEDE_CITY
N139	161	-138.6	0	region_id=REGION_CENTIPEDE_CITY
N140	235.6	-165.2	0	region_id=REGION_CENTIPEDE_CITY
N141	303.5	-191.9	0	region_id=REGION_CENTIPEDE_CITY
N142	-268.1	159.9	0	region_id=REGION_SPIDER_GORGE
N143	-256.1	68	0	region_id=REGION_SPIDER_GORGE
N144	-81.5	40	0	region_id=REGION_SPIDER_GORGE
N145	-102.9	-74.6	0	region_id=REGION_LOST_SOUL_FIELD
N146	115.7	90.6	0	region_id=REGION_DRAGON_RIDGE
N147	126.3	-56	0	region_id=REGION_BURNING_WATER
N148	264.9	89.3	0	region_id=REGION_BURNING_WATER
N149	275.6	-106.6	0	region_id=REGION_BURNING_WATER
N150	-185.5	193.2	0	region_id=REGION_BROKEN_GOLD
N151	-101.5	229.2	0	region_id=REGION_BROKEN_GOLD
N152	2.4	211.9	0	region_id=REGION_BROKEN_GOLD
N153	118.3	173.2	0	region_id=REGION_DRAGON_RIDGE
N154	-112.2	41.3	0	region_id=REGION_SPIDER_GORGE
N155	-18.9	42.6	0	region_id=REGION_BROKEN_GOLD
N156	-29.6	-50.6	0	region_id=REGION_LOST_SOUL_FIELD

节点	X	Y	Z	属性
N157	-49.6	-133.2	0	region_id=REGION_LOST_SOUL_FIELD
N158	173	49.3	0	region_id=REGION_BURNING_WATER
N159	247.6	25.3	0	region_id=REGION_BURNING_WATER
N160	-338.7	14.7	0	region_id=REGION_SPIDER_GORGE
N161	-298.7	85.3	0	region_id=REGION_SPIDER_GORGE
N162	-230.8	111.9	0	region_id=REGION_SPIDER_GORGE
N163	-166.8	68	0	region_id=REGION_SPIDER_GORGE
N164	-242.8	-22.7	0	region_id=REGION_SPIDER_GORGE
N165	-221.5	-98.6	0	region_id=REGION_SPIDER_GORGE
N166	-110.9	202.5	0	region_id=REGION_BROKEN_GOLD
N167	-77.6	238.5	0	region_id=REGION_BROKEN_GOLD
N168	49	146.6	0	region_id=REGION_DRAGON_RIDGE
N169	73	173.2	0	region_id=REGION_DRAGON_RIDGE
N170	199.6	113.3	0	region_id=REGION_DRAGON_RIDGE
N171	242.2	89.3	0	region_id=REGION_DRAGON_RIDGE
N172	304.9	154.6	0	region_id=REGION_DRAGON_RIDGE
N173	302.2	-53.3	0	region_id=REGION_BURNING_WATER
N174	330.2	-138.6	0	region_id=REGION_BURNING_WATER
N175	-128.2	-181.2	0	region_id=REGION_LOST_SOUL_FIELD
N176	-130.9	-274.5	0	region_id=REGION_LOST_SOUL_FIELD
N177	65	-230.5	0	region_id=REGION_CENTIPEDE_CITY
N178	-72.2	70.6	0	region_id=REGION_BROKEN_GOLD
N179	-90.9	9.3	0	region_id=REGION_SPIDER_GORGE
N180	106.3	122.6	0	region_id=REGION_DRAGON_RIDGE
N181	125	58.6	0	region_id=REGION_DRAGON_RIDGE
N182	130.3	-14.7	0	region_id=REGION_BURNING_WATER
N183	122.3	-97.3	0	region_id=REGION_CENTIPEDE_CITY

节点	X	Y	Z	属性
N184	255.6	144.5	0	region_id=REGION_DRAGON_RIDGE
N185	274.2	34	0	region_id=REGION_BURNING_WATER
N186	279.6	-64	0	region_id=REGION_BURNING_WATER
N187	271.6	-149.2	0	region_id=REGION_CENTIPEDE_CITY
N188	83.7	182.6	0	region_id=REGION_DRAGON_RIDGE
N189	80.1	169	0	region_id=REGION_DRAGON_RIDGE
N190	153	163.9	0	region_id=REGION_DRAGON_RIDGE
N191	149.3	150.4	0	region_id=REGION_DRAGON_RIDGE

D.2 路网边

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0001	N046	N047	SIDE	7	58.5	是	BIDIRECTIONAL
MAP_EDGE_0002	N038	N048	RISK	4.5	99.9	是	BIDIRECTIONAL
MAP_EDGE_0003	N048	N003	RISK	4.5	136.5	是	BIDIRECTIONAL
MAP_EDGE_0004	N016	N049	RISK	4.5	64.2	是	BIDIRECTIONAL
MAP_EDGE_0005	N049	N027	RISK	4.5	124.8	是	BIDIRECTIONAL
MAP_EDGE_0006	N018	N050	RISK	4.5	71.7	是	BIDIRECTIONAL
MAP_EDGE_0007	N050	N043	RISK	4.5	85.9	是	BIDIRECTIONAL
MAP_EDGE_0008	N023	N051	RISK	4.5	77.5	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0009	N051	N036	RISK	4.5	88.6	是	BIDIRECTIONAL
MAP_EDGE_0010	N052	N053	COURT	8	38.6	是	BIDIRECTIONAL
MAP_EDGE_0011	N053	N004	COURT	8	47	是	BIDIRECTIONAL
MAP_EDGE_0012	N052	N054	COURT	8	37.4	是	BIDIRECTIONAL
MAP_EDGE_0013	N054	N006	COURT	8	49.9	是	BIDIRECTIONAL
MAP_EDGE_0014	N052	N055	COURT	8	32.1	是	BIDIRECTIONAL
MAP_EDGE_0015	N055	N016	COURT	8	51.2	是	BIDIRECTIONAL
MAP_EDGE_0016	N056	N057	COURT	8	33.7	是	BIDIRECTIONAL
MAP_EDGE_0017	N057	N014	COURT	8	61.8	是	BIDIRECTIONAL
MAP_EDGE_0018	N056	N058	COURT	8	42.3	是	BIDIRECTIONAL
MAP_EDGE_0019	N058	N049	COURT	8	53.2	是	BIDIRECTIONAL
MAP_EDGE_0020	N056	N059	COURT	8	39.1	是	BIDIRECTIONAL
MAP_EDGE_0021	N059	N036	COURT	8	3.9	是	BIDIRECTIONAL
MAP_EDGE_0022	N060	N061	COURT	8	33	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0023	N061	N049	COURT	8	50.4	是	BIDIRECTIONAL
MAP_EDGE_0024	N060	N062	COURT	8	39.8	是	BIDIRECTIONAL
MAP_EDGE_0025	N062	N050	COURT	8	47	是	BIDIRECTIONAL
MAP_EDGE_0026	N060	N063	COURT	8	36.3	是	BIDIRECTIONAL
MAP_EDGE_0027	N063	N027	COURT	8	49.8	是	BIDIRECTIONAL
MAP_EDGE_0028	N064	N000	DEN	5	47.2	是	BIDIRECTIONAL
MAP_EDGE_0029	N065	N048	DEN	5	39.6	是	BIDIRECTIONAL
MAP_EDGE_0030	N066	N033	DEN	5	24.6	是	BIDIRECTIONAL
MAP_EDGE_0031	N067	N042	DEN	5	44.4	是	BIDIRECTIONAL
MAP_EDGE_0032	N068	N019	DEN	5	40.6	是	BIDIRECTIONAL
MAP_EDGE_0033	N069	N021	DEN	5	32.2	是	BIDIRECTIONAL
MAP_EDGE_0034	N070	N044	DEN	5	32.4	是	BIDIRECTIONAL
MAP_EDGE_0035	N071	N047	DEN	5	49.1	是	BIDIRECTIONAL
MAP_EDGE_0036	N072	N029	DEN	5	38.2	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0037	N073	N038	DEN	5	37.3	是	BIDIRECTIONAL
MAP_EDGE_0038	N074	N020	DEN	5	51.7	是	BIDIRECTIONAL
MAP_EDGE_0039	N075	N046	DEN	5	39.6	是	BIDIRECTIONAL
MAP_EDGE_0040	N076	N007	ARENA	8	82.3	是	BIDIRECTIONAL
MAP_EDGE_0041	N076	N006	ARENA	8	93.3	是	BIDIRECTIONAL
MAP_EDGE_0042	N076	N009	ARENA	8	185.8	是	BIDIRECTIONAL
MAP_EDGE_0043	N077	N042	ARENA	8	16	是	BIDIRECTIONAL
MAP_EDGE_0044	N077	N020	ARENA	8	94.6	是	BIDIRECTIONAL
MAP_EDGE_0045	N077	N008	ARENA	8	102.6	是	BIDIRECTIONAL
MAP_EDGE_0046	N078	N040	ARENA	8	27.2	是	BIDIRECTIONAL
MAP_EDGE_0047	N078	N023	ARENA	8	66.6	是	BIDIRECTIONAL
MAP_EDGE_0048	N078	N051	ARENA	8	90.8	是	BIDIRECTIONAL
MAP_EDGE_0049	N079	N024	ARENA	8	37.7	是	BIDIRECTIONAL
MAP_EDGE_0050	N079	N025	ARENA	8	54.9	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0051	N079	N044	ARENA	8	85.8	是	BIDIRECTIONAL
MAP_EDGE_0052	N080	N028	ARENA	8	16.2	是	BIDIRECTIONAL
MAP_EDGE_0053	N080	N029	ARENA	8	76.4	是	BIDIRECTIONAL
MAP_EDGE_0054	N080	N027	ARENA	8	84.1	是	BIDIRECTIONAL
MAP_EDGE_0055	N081	N039	ARENA	8	21.3	是	BIDIRECTIONAL
MAP_EDGE_0056	N081	N003	ARENA	8	67.8	是	BIDIRECTIONAL
MAP_EDGE_0057	N034	N006	ARENA	8	60.8	是	BIDIRECTIONAL
MAP_EDGE_0058	N034	N052	ARENA	8	86.8	是	BIDIRECTIONAL
MAP_EDGE_0059	N082	N036	ARENA	8	40	是	BIDIRECTIONAL
MAP_EDGE_0060	N082	N024	ARENA	8	56.5	是	BIDIRECTIONAL
MAP_EDGE_0061	N037	N042	ARENA	8	70.4	是	BIDIRECTIONAL
MAP_EDGE_0062	N037	N007	ARENA	8	86	是	BIDIRECTIONAL
MAP_EDGE_0063	N083	N001	SHOP	6	6.1	是	BIDIRECTIONAL
MAP_EDGE_0064	N083	N002	SHOP	6	65.6	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0065	N084	N003	SHOP	6	20	是	BIDIRECTIONAL
MAP_EDGE_0066	N084	N032	SHOP	6	62.3	是	BIDIRECTIONAL
MAP_EDGE_0067	N085	N006	SHOP	6	47.5	是	BIDIRECTIONAL
MAP_EDGE_0068	N085	N034	SHOP	6	47.5	是	BIDIRECTIONAL
MAP_EDGE_0069	N086	N009	SHOP	6	63	是	BIDIRECTIONAL
MAP_EDGE_0070	N086	N042	SHOP	6	80.4	是	BIDIRECTIONAL
MAP_EDGE_0071	N087	N011	SHOP	6	36.8	是	BIDIRECTIONAL
MAP_EDGE_0072	N087	N012	SHOP	6	47.4	是	BIDIRECTIONAL
MAP_EDGE_0073	N088	N051	SHOP	6	18.9	是	BIDIRECTIONAL
MAP_EDGE_0074	N088	N014	SHOP	6	42.4	是	BIDIRECTIONAL
MAP_EDGE_0075	N089	N050	SHOP	6	18.1	是	BIDIRECTIONAL
MAP_EDGE_0076	N089	N062	SHOP	6	42.9	是	BIDIRECTIONAL
MAP_EDGE_0077	N090	N043	SHOP	6	26	是	BIDIRECTIONAL
MAP_EDGE_0078	N090	N030	SHOP	6	85.5	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0079	N091	N022	SHOP	6	28.6	是	BIDIRECTIONAL
MAP_EDGE_0080	N091	N023	SHOP	6	62.8	是	BIDIRECTIONAL
MAP_EDGE_0081	N092	N036	SHOP	6	42.1	是	BIDIRECTIONAL
MAP_EDGE_0082	N092	N024	SHOP	6	46.9	是	BIDIRECTIONAL
MAP_EDGE_0083	N093	N026	SHOP	6	35.6	是	BIDIRECTIONAL
MAP_EDGE_0084	N093	N027	SHOP	6	44	是	BIDIRECTIONAL
MAP_EDGE_0085	N094	N029	SHOP	6	16.9	是	BIDIRECTIONAL
MAP_EDGE_0086	N094	N028	SHOP	6	66.8	是	BIDIRECTIONAL
MAP_EDGE_0087	N095	N031	SHOP	6	53.3	是	BIDIRECTIONAL
MAP_EDGE_0088	N095	N000	SHOP	6	69.3	是	BIDIRECTIONAL
MAP_EDGE_0089	N096	N016	SHOP	6	43.4	是	BIDIRECTIONAL
MAP_EDGE_0090	N096	N049	SHOP	6	56	是	BIDIRECTIONAL
MAP_EDGE_0091	N097	N020	SHOP	6	24	是	BIDIRECTIONAL
MAP_EDGE_0092	N097	N009	SHOP	6	177.9	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0093	N098	N025	SHOP	6	40.7	是	BIDIRECTIONAL
MAP_EDGE_0094	N098	N026	SHOP	6	82.7	是	BIDIRECTIONAL
MAP_EDGE_0095	N000	N001	SPAWN_LINK	6	95.9	是	BIDIRECTIONAL
MAP_EDGE_0096	N000	N010	SPAWN_LINK	6	237.4	是	BIDIRECTIONAL
MAP_EDGE_0097	N030	N029	SPAWN_LINK	6	72	是	BIDIRECTIONAL
MAP_EDGE_0098	N030	N043	SPAWN_LINK	6	108.2	是	BIDIRECTIONAL
MAP_EDGE_0099	N044	N045	SPAWN_LINK	6	89.4	是	BIDIRECTIONAL
MAP_EDGE_0100	N044	N022	SPAWN_LINK	6	105.1	是	BIDIRECTIONAL
MAP_EDGE_0101	N006	N054	SPAWN_LINK	6	49.9	是	BIDIRECTIONAL
MAP_EDGE_0102	N006	N007	SPAWN_LINK	6	90.6	是	BIDIRECTIONAL
MAP_EDGE_0103	N013	N014	SPAWN_LINK	6	70	是	BIDIRECTIONAL
MAP_EDGE_0104	N013	N012	SPAWN_LINK	6	77.3	是	BIDIRECTIONAL
MAP_EDGE_0105	N047	N046	SPAWN_LINK	6	58.5	是	BIDIRECTIONAL
MAP_EDGE_0106	N047	N028	SPAWN_LINK	6	187.2	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0107	N009	N008	SPAWN_LINK	6	90.6	是	BIDIRECTIONAL
MAP_EDGE_0108	N009	N020	SPAWN_LINK	6	175.9	是	BIDIRECTIONAL
MAP_EDGE_0109	N041	N032	SPAWN_LINK	6	51.5	是	BIDIRECTIONAL
MAP_EDGE_0110	N041	N033	SPAWN_LINK	6	67.9	是	BIDIRECTIONAL
MAP_EDGE_0111	N060	N061	SPAWN_LINK	6	33	是	BIDIRECTIONAL
MAP_EDGE_0112	N060	N063	SPAWN_LINK	6	36.3	是	BIDIRECTIONAL
MAP_EDGE_0113	N010	N011	SPAWN_LINK	6	80	是	BIDIRECTIONAL
MAP_EDGE_0114	N010	N000	SPAWN_LINK	6	237.4	是	BIDIRECTIONAL
MAP_EDGE_0115	N020	N019	SPAWN_LINK	6	73.7	是	BIDIRECTIONAL
MAP_EDGE_0116	N020	N042	SPAWN_LINK	6	104.1	是	BIDIRECTIONAL
MAP_EDGE_0117	N056	N057	SPAWN_LINK	6	33.7	是	BIDIRECTIONAL
MAP_EDGE_0118	N056	N058	SPAWN_LINK	6	42.3	是	BIDIRECTIONAL
MAP_EDGE_0119	N021	N022	SPAWN_LINK	6	90.6	是	BIDIRECTIONAL
MAP_EDGE_0120	N021	N010	SPAWN_LINK	6	154.7	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0121	N037	N042	SPAWN_LINK	6	70.4	是	BIDIRECTIONAL
MAP_EDGE_0122	N037	N017	SPAWN_LINK	6	89.6	是	BIDIRECTIONAL
MAP_EDGE_0123	N002	N039	SPAWN_LINK	6	57.7	是	BIDIRECTIONAL
MAP_EDGE_0124	N002	N001	SPAWN_LINK	6	64.2	是	BIDIRECTIONAL
MAP_EDGE_0125	N026	N027	SPAWN_LINK	6	71.7	是	BIDIRECTIONAL
MAP_EDGE_0126	N026	N046	SPAWN_LINK	6	87.9	是	BIDIRECTIONAL
MAP_EDGE_0127	N053	N052	SPAWN_LINK	6	38.7	是	BIDIRECTIONAL
MAP_EDGE_0128	N053	N004	SPAWN_LINK	6	47	是	BIDIRECTIONAL
MAP_EDGE_0129	N024	N051	SPAWN_LINK	6	66.8	是	BIDIRECTIONAL
MAP_EDGE_0130	N024	N025	SPAWN_LINK	6	81.1	是	BIDIRECTIONAL
MAP_EDGE_0131	N040	N011	SPAWN_LINK	6	48	是	BIDIRECTIONAL
MAP_EDGE_0132	N040	N023	SPAWN_LINK	6	88.7	是	BIDIRECTIONAL
MAP_EDGE_0133	N043	N019	SPAWN_LINK	6	74	是	BIDIRECTIONAL
MAP_EDGE_0134	N043	N030	SPAWN_LINK	6	108.2	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0135	N038	N031	SPAWN_LINK	6	63.6	是	BIDIRECTIONAL
MAP_EDGE_0136	N038	N011	SPAWN_LINK	6	94.5	是	BIDIRECTIONAL
MAP_EDGE_0137	N028	N050	SPAWN_LINK	6	71.5	是	BIDIRECTIONAL
MAP_EDGE_0138	N028	N063	SPAWN_LINK	6	81.2	是	BIDIRECTIONAL
MAP_EDGE_0139	N016	N055	SPAWN_LINK	6	51.2	是	BIDIRECTIONAL
MAP_EDGE_0140	N016	N015	SPAWN_LINK	6	64.2	是	BIDIRECTIONAL
MAP_EDGE_0141	N008	N007	SPAWN_LINK	6	74.1	是	BIDIRECTIONAL
MAP_EDGE_0142	N008	N009	SPAWN_LINK	6	90.6	是	BIDIRECTIONAL
MAP_EDGE_0143	N022	N023	SPAWN_LINK	6	75.4	是	BIDIRECTIONAL
MAP_EDGE_0144	N022	N021	SPAWN_LINK	6	90.6	是	BIDIRECTIONAL
MAP_EDGE_0145	N045	N044	SPAWN_LINK	6	89.4	是	BIDIRECTIONAL
MAP_EDGE_0146	N045	N046	SPAWN_LINK	6	180.4	是	BIDIRECTIONAL
MAP_EDGE_0147	N017	N018	SPAWN_LINK	6	71.7	是	BIDIRECTIONAL
MAP_EDGE_0148	N017	N049	SPAWN_LINK	6	75.8	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0149	N003	N004	SPAWN_LINK	6	75.8	是	BIDIRECTIONAL
MAP_EDGE_0150	N003	N032	SPAWN_LINK	6	82.1	是	BIDIRECTIONAL
MAP_EDGE_0151	N048	N012	SPAWN_LINK	6	52.4	是	BIDIRECTIONAL
MAP_EDGE_0152	N048	N039	SPAWN_LINK	6	52.8	是	BIDIRECTIONAL
MAP_EDGE_0153	N025	N024	SPAWN_LINK	6	81.1	是	BIDIRECTIONAL
MAP_EDGE_0154	N025	N026	SPAWN_LINK	6	114.6	是	BIDIRECTIONAL
MAP_EDGE_0155	N023	N022	SPAWN_LINK	6	75.4	是	BIDIRECTIONAL
MAP_EDGE_0156	N023	N051	SPAWN_LINK	6	77.5	是	BIDIRECTIONAL
MAP_EDGE_0157	N007	N008	SPAWN_LINK	6	74.1	是	BIDIRECTIONAL
MAP_EDGE_0158	N007	N037	SPAWN_LINK	6	86	是	BIDIRECTIONAL
MAP_EDGE_0159	N019	N050	SPAWN_LINK	6	71.7	是	BIDIRECTIONAL
MAP_EDGE_0160	N019	N020	SPAWN_LINK	6	73.7	是	BIDIRECTIONAL
MAP_EDGE_0161	N035	N004	SPAWN_LINK	6	43.4	是	BIDIRECTIONAL
MAP_EDGE_0162	N035	N014	SPAWN_LINK	6	85.2	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0163	N029	N030	SPAWN_LINK	6	72	是	BIDIRECTIONAL
MAP_EDGE_0164	N029	N028	SPAWN_LINK	6	83.3	是	BIDIRECTIONAL
MAP_EDGE_0165	N027	N063	SPAWN_LINK	6	49.8	是	BIDIRECTIONAL
MAP_EDGE_0166	N027	N026	SPAWN_LINK	6	71.7	是	BIDIRECTIONAL
MAP_EDGE_0167	N050	N062	SPAWN_LINK	6	47	是	BIDIRECTIONAL
MAP_EDGE_0168	N050	N019	SPAWN_LINK	6	71.7	是	BIDIRECTIONAL
MAP_EDGE_0169	N042	N037	SPAWN_LINK	6	70.4	是	BIDIRECTIONAL
MAP_EDGE_0170	N042	N008	SPAWN_LINK	6	97	是	BIDIRECTIONAL
MAP_EDGE_0171	N014	N051	SPAWN_LINK	6	60.8	是	BIDIRECTIONAL
MAP_EDGE_0172	N014	N057	SPAWN_LINK	6	61.8	是	BIDIRECTIONAL
MAP_EDGE_0173	N033	N041	SPAWN_LINK	6	67.9	是	BIDIRECTIONAL
MAP_EDGE_0174	N033	N034	SPAWN_LINK	6	101.5	是	BIDIRECTIONAL
MAP_EDGE_0175	N099	N100	RAMP	5	30.6	是	BIDIRECTIONAL
MAP_EDGE_0176	N100	N101	RAMP	5	31.1	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0177	N099	N102	RAMP	5	31.5	是	BIDIRECTIONAL
MAP_EDGE_0178	N102	N103	RAMP	5	25.1	是	BIDIRECTIONAL
MAP_EDGE_0179	N101	N077	SIDE	5	13.3	是	BIDIRECTIONAL
MAP_EDGE_0180	N104	N105	RAMP	5	26.5	是	BIDIRECTIONAL
MAP_EDGE_0181	N105	N106	RAMP	5	26.7	是	BIDIRECTIONAL
MAP_EDGE_0182	N104	N107	RAMP	5	27.2	是	BIDIRECTIONAL
MAP_EDGE_0183	N107	N108	RAMP	5	24.6	是	BIDIRECTIONAL
MAP_EDGE_0184	N106	N105	SIDE	5	26.7	是	BIDIRECTIONAL
MAP_EDGE_0185	N109	N110	RAMP	5	22.4	是	BIDIRECTIONAL
MAP_EDGE_0186	N110	N111	RAMP	5	25.3	是	BIDIRECTIONAL
MAP_EDGE_0187	N109	N112	RAMP	5	23.4	是	BIDIRECTIONAL
MAP_EDGE_0188	N112	N113	RAMP	5	26.2	是	BIDIRECTIONAL
MAP_EDGE_0189	N111	N110	SIDE	5	25.3	是	BIDIRECTIONAL
MAP_EDGE_0190	N000	N114	MAIN	12	48	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0191	N114	N001	MAIN	12	48	是	BIDIRECTIONAL
MAP_EDGE_0192	N114	N083	SIDEDOOR	4.5	51.9	是	BIDIRECTIONAL
MAP_EDGE_0193	N001	N115	MAIN	12	32.1	是	BIDIRECTIONAL
MAP_EDGE_0194	N115	N002	MAIN	12	32.1	是	BIDIRECTIONAL
MAP_EDGE_0195	N115	N083	SIDEDOOR	4.5	33.7	是	BIDIRECTIONAL
MAP_EDGE_0196	N002	N116	MAIN	12	50.6	是	BIDIRECTIONAL
MAP_EDGE_0197	N116	N003	MAIN	12	50.6	是	BIDIRECTIONAL
MAP_EDGE_0198	N116	N084	SIDEDOOR	4.5	44.7	是	BIDIRECTIONAL
MAP_EDGE_0199	N003	N117	MAIN	12	37.9	是	BIDIRECTIONAL
MAP_EDGE_0200	N117	N004	MAIN	12	37.9	是	BIDIRECTIONAL
MAP_EDGE_0201	N117	N084	SIDEDOOR	4.5	57.7	是	BIDIRECTIONAL
MAP_EDGE_0202	N004	N118	MAIN	12	56	是	BIDIRECTIONAL
MAP_EDGE_0203	N118	N005	MAIN	12	56	是	BIDIRECTIONAL
MAP_EDGE_0204	N118	N052	SIDEDOOR	4.5	30.4	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0205	N005	N119	MAIN	12	35.9	是	BIDIRECTIONAL
MAP_EDGE_0206	N119	N006	MAIN	12	35.9	是	BIDIRECTIONAL
MAP_EDGE_0207	N119	N054	SIDEDOOR	4.5	18.7	是	BIDIRECTIONAL
MAP_EDGE_0208	N006	N120	MAIN	12	45.3	是	BIDIRECTIONAL
MAP_EDGE_0209	N120	N007	MAIN	12	45.3	是	BIDIRECTIONAL
MAP_EDGE_0210	N120	N085	SIDEDOOR	4.5	66.7	是	BIDIRECTIONAL
MAP_EDGE_0211	N007	N121	MAIN	12	37	是	BIDIRECTIONAL
MAP_EDGE_0212	N121	N008	MAIN	12	37	是	BIDIRECTIONAL
MAP_EDGE_0213	N121	N120	SIDEDOOR	4.5	78	是	BIDIRECTIONAL
MAP_EDGE_0214	N008	N122	MAIN	12	45.3	是	BIDIRECTIONAL
MAP_EDGE_0215	N122	N009	MAIN	12	45.3	是	BIDIRECTIONAL
MAP_EDGE_0216	N122	N121	SIDEDOOR	4.5	78.4	是	BIDIRECTIONAL
MAP_EDGE_0217	N010	N123	MAIN	12	40	是	BIDIRECTIONAL
MAP_EDGE_0218	N123	N011	MAIN	12	40	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0219	N123	N040	SIDEDOOR	4.5	77.7	是	BIDIRECTIONAL
MAP_EDGE_0220	N011	N124	MAIN	12	36.8	是	BIDIRECTIONAL
MAP_EDGE_0221	N124	N012	MAIN	12	36.8	是	BIDIRECTIONAL
MAP_EDGE_0222	N124	N087	SIDEDOOR	4.5	21.2	是	BIDIRECTIONAL
MAP_EDGE_0223	N012	N125	MAIN	12	38.6	是	BIDIRECTIONAL
MAP_EDGE_0224	N125	N013	MAIN	12	38.6	是	BIDIRECTIONAL
MAP_EDGE_0225	N125	N048	SIDEDOOR	4.5	56.6	是	BIDIRECTIONAL
MAP_EDGE_0226	N013	N126	MAIN	12	35	是	BIDIRECTIONAL
MAP_EDGE_0227	N126	N014	MAIN	12	35	是	BIDIRECTIONAL
MAP_EDGE_0228	N126	N088	SIDEDOOR	4.5	58.5	是	BIDIRECTIONAL
MAP_EDGE_0229	N014	N127	MAIN	12	45.3	是	BIDIRECTIONAL
MAP_EDGE_0230	N127	N015	MAIN	12	45.3	是	BIDIRECTIONAL
MAP_EDGE_0231	N127	N057	SIDEDOOR	4.5	26.8	是	BIDIRECTIONAL
MAP_EDGE_0232	N015	N128	MAIN	12	32.1	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0233	N128	N016	MAIN	12	32.1	是	BIDIRECTIONAL
MAP_EDGE_0234	N128	N058	SIDEDOOR	4.5	47.2	是	BIDIRECTIONAL
MAP_EDGE_0235	N016	N129	MAIN	12	50.6	是	BIDIRECTIONAL
MAP_EDGE_0236	N129	N017	MAIN	12	50.6	是	BIDIRECTIONAL
MAP_EDGE_0237	N129	N049	SIDEDOOR	4.5	48.6	是	BIDIRECTIONAL
MAP_EDGE_0238	N017	N130	MAIN	12	35.9	是	BIDIRECTIONAL
MAP_EDGE_0239	N130	N018	MAIN	12	35.9	是	BIDIRECTIONAL
MAP_EDGE_0240	N130	N061	SIDEDOOR	4.5	61.5	是	BIDIRECTIONAL
MAP_EDGE_0241	N018	N131	MAIN	12	48	是	BIDIRECTIONAL
MAP_EDGE_0242	N131	N019	MAIN	12	48	是	BIDIRECTIONAL
MAP_EDGE_0243	N131	N089	SIDEDOOR	4.5	36	是	BIDIRECTIONAL
MAP_EDGE_0244	N019	N132	MAIN	12	36.9	是	BIDIRECTIONAL
MAP_EDGE_0245	N132	N020	MAIN	12	36.9	是	BIDIRECTIONAL
MAP_EDGE_0246	N132	N106	SIDEDOOR	4.5	29.1	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0247	N021	N133	MAIN	12	45.3	是	BIDIRECTIONAL
MAP_EDGE_0248	N133	N022	MAIN	12	45.3	是	BIDIRECTIONAL
MAP_EDGE_0249	N133	N091	SIDEDOOR	4.5	49.8	是	BIDIRECTIONAL
MAP_EDGE_0250	N022	N134	MAIN	12	37.7	是	BIDIRECTIONAL
MAP_EDGE_0251	N134	N023	MAIN	12	37.7	是	BIDIRECTIONAL
MAP_EDGE_0252	N134	N091	SIDEDOOR	4.5	31	是	BIDIRECTIONAL
MAP_EDGE_0253	N023	N135	MAIN	12	41.3	是	BIDIRECTIONAL
MAP_EDGE_0254	N135	N024	MAIN	12	41.3	是	BIDIRECTIONAL
MAP_EDGE_0255	N135	N051	SIDEDOOR	4.5	59.4	是	BIDIRECTIONAL
MAP_EDGE_0256	N024	N136	MAIN	12	40.5	是	BIDIRECTIONAL
MAP_EDGE_0257	N136	N025	MAIN	12	40.5	是	BIDIRECTIONAL
MAP_EDGE_0258	N136	N079	SIDEDOOR	4.5	24	是	BIDIRECTIONAL
MAP_EDGE_0259	N025	N137	MAIN	12	57.3	是	BIDIRECTIONAL
MAP_EDGE_0260	N137	N026	MAIN	12	57.3	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0261	N137	N098	SIDEDOOR	4.5	31.1	是	BIDIRECTIONAL
MAP_EDGE_0262	N026	N138	MAIN	12	35.9	是	BIDIRECTIONAL
MAP_EDGE_0263	N138	N027	MAIN	12	35.9	是	BIDIRECTIONAL
MAP_EDGE_0264	N138	N093	SIDEDOOR	4.5	17.7	是	BIDIRECTIONAL
MAP_EDGE_0265	N027	N139	MAIN	12	42.6	是	BIDIRECTIONAL
MAP_EDGE_0266	N139	N028	MAIN	12	42.6	是	BIDIRECTIONAL
MAP_EDGE_0267	N139	N080	SIDEDOOR	4.5	43.1	是	BIDIRECTIONAL
MAP_EDGE_0268	N028	N140	MAIN	12	41.6	是	BIDIRECTIONAL
MAP_EDGE_0269	N140	N029	MAIN	12	41.6	是	BIDIRECTIONAL
MAP_EDGE_0270	N140	N094	SIDEDOOR	4.5	25.3	是	BIDIRECTIONAL
MAP_EDGE_0271	N029	N141	MAIN	12	36	是	BIDIRECTIONAL
MAP_EDGE_0272	N141	N030	MAIN	12	36	是	BIDIRECTIONAL
MAP_EDGE_0273	N141	N094	SIDEDOOR	4.5	51.5	是	BIDIRECTIONAL
MAP_EDGE_0274	N001	N142	MAIN	12	49.8	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0275	N142	N031	MAIN	12	49.8	是	BIDIRECTIONAL
MAP_EDGE_0276	N142	N115	SIDEDOOR	4.5	42.1	是	BIDIRECTIONAL
MAP_EDGE_0277	N031	N143	MAIN	12	50.7	是	BIDIRECTIONAL
MAP_EDGE_0278	N143	N012	MAIN	12	50.7	是	BIDIRECTIONAL
MAP_EDGE_0279	N143	N048	SIDEDOOR	4.5	39.2	是	BIDIRECTIONAL
MAP_EDGE_0280	N144	N035	SIDEDOOR	4.5	21.5	是	BIDIRECTIONAL
MAP_EDGE_0281	N014	N145	MAIN	12	53.4	是	BIDIRECTIONAL
MAP_EDGE_0282	N145	N024	MAIN	12	53.4	是	BIDIRECTIONAL
MAP_EDGE_0283	N145	N088	SIDEDOOR	4.5	26.8	是	BIDIRECTIONAL
MAP_EDGE_0284	N146	N119	SIDEDOOR	4.5	56.7	是	BIDIRECTIONAL
MAP_EDGE_0285	N147	N060	SIDEDOOR	4.5	24.1	是	BIDIRECTIONAL
MAP_EDGE_0286	N148	N042	SIDEDOOR	4.5	19.1	是	BIDIRECTIONAL
MAP_EDGE_0287	N149	N043	SIDEDOOR	4.5	50.1	是	BIDIRECTIONAL
MAP_EDGE_0288	N002	N150	MAIN	12	42.7	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0289	N150	N032	MAIN	12	42.7	是	BIDIRECTIONAL
MAP_EDGE_0290	N150	N116	SIDEDOOR	4.5	41	是	BIDIRECTIONAL
MAP_EDGE_0291	N032	N151	MAIN	12	57.4	是	BIDIRECTIONAL
MAP_EDGE_0292	N151	N033	MAIN	12	57.4	是	BIDIRECTIONAL
MAP_EDGE_0293	N151	N041	SIDEDOOR	4.5	18.5	是	BIDIRECTIONAL
MAP_EDGE_0294	N033	N152	MAIN	12	50.7	是	BIDIRECTIONAL
MAP_EDGE_0295	N152	N034	MAIN	12	50.7	是	BIDIRECTIONAL
MAP_EDGE_0296	N152	N110	SIDEDOOR	4.5	20.4	是	BIDIRECTIONAL
MAP_EDGE_0297	N153	N006	SIDEDOOR	4.5	28.3	是	BIDIRECTIONAL
MAP_EDGE_0298	N013	N154	MAIN	12	44.8	是	BIDIRECTIONAL
MAP_EDGE_0299	N154	N035	MAIN	12	44.8	是	BIDIRECTIONAL
MAP_EDGE_0300	N154	N144	SIDEDOOR	4.5	30.7	是	BIDIRECTIONAL
MAP_EDGE_0301	N035	N155	MAIN	12	54.4	是	BIDIRECTIONAL
MAP_EDGE_0302	N155	N016	MAIN	12	54.4	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0303	N155	N055	SIDEDOOR	4.5	57.2	是	BIDIRECTIONAL
MAP_EDGE_0304	N015	N156	MAIN	12	35.5	是	BIDIRECTIONAL
MAP_EDGE_0305	N156	N036	MAIN	12	35.5	是	BIDIRECTIONAL
MAP_EDGE_0306	N156	N056	SIDEDOOR	4.5	19.1	是	BIDIRECTIONAL
MAP_EDGE_0307	N036	N157	MAIN	12	53.3	是	BIDIRECTIONAL
MAP_EDGE_0308	N157	N025	MAIN	12	53.3	是	BIDIRECTIONAL
MAP_EDGE_0309	N157	N092	SIDEDOOR	4.5	15.6	是	BIDIRECTIONAL
MAP_EDGE_0310	N017	N158	MAIN	12	44.8	是	BIDIRECTIONAL
MAP_EDGE_0311	N158	N037	MAIN	12	44.8	是	BIDIRECTIONAL
MAP_EDGE_0312	N158	N130	SIDEDOOR	4.5	48.2	是	BIDIRECTIONAL
MAP_EDGE_0313	N037	N159	MAIN	12	58.9	是	BIDIRECTIONAL
MAP_EDGE_0314	N159	N019	MAIN	12	58.9	是	BIDIRECTIONAL
MAP_EDGE_0315	N159	N106	SIDEDOOR	4.5	38.8	是	BIDIRECTIONAL
MAP_EDGE_0316	N010	N160	SIDE	7	49.5	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0317	N160	N038	SIDE	7	49.5	是	BIDIRECTIONAL
MAP_EDGE_0318	N160	N123	BREACH	4	47.3	是	BIDIRECTIONAL
MAP_EDGE_0319	N038	N161	SIDE	7	31.8	是	BIDIRECTIONAL
MAP_EDGE_0320	N161	N031	SIDE	7	31.8	是	BIDIRECTIONAL
MAP_EDGE_0321	N161	N143	SIDEDOOR	4.5	46	是	BIDIRECTIONAL
MAP_EDGE_0322	N031	N162	SIDE	7	50.6	是	BIDIRECTIONAL
MAP_EDGE_0323	N162	N039	SIDE	7	50.6	是	BIDIRECTIONAL
MAP_EDGE_0324	N162	N143	SIDEDOOR	4.5	50.7	是	BIDIRECTIONAL
MAP_EDGE_0325	N039	N163	SIDE	7	45.9	是	BIDIRECTIONAL
MAP_EDGE_0326	N163	N013	SIDE	7	45.9	是	BIDIRECTIONAL
MAP_EDGE_0327	N163	N125	SIDEDOOR	4.5	50.7	是	BIDIRECTIONAL
MAP_EDGE_0328	N012	N164	SIDE	7	48.2	是	BIDIRECTIONAL
MAP_EDGE_0329	N164	N040	SIDE	7	48.2	是	BIDIRECTIONAL
MAP_EDGE_0330	N164	N124	SIDEDOOR	4.5	24	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0331	N040	N165	SIDE	7	44.4	是	BIDIRECTIONAL
MAP_EDGE_0332	N165	N023	SIDE	7	44.4	是	BIDIRECTIONAL
MAP_EDGE_0333	N165	N078	SIDEDOOR	4.5	24.9	是	BIDIRECTIONAL
MAP_EDGE_0334	N003	N166	SIDE	7	42.6	是	BIDIRECTIONAL
MAP_EDGE_0335	N166	N041	SIDE	7	42.6	是	BIDIRECTIONAL
MAP_EDGE_0336	N166	N084	SIDEDOOR	4.5	26.8	是	BIDIRECTIONAL
MAP_EDGE_0337	N041	N167	SIDE	7	34	是	BIDIRECTIONAL
MAP_EDGE_0338	N167	N033	SIDE	7	34	是	BIDIRECTIONAL
MAP_EDGE_0339	N167	N151	SIDEDOOR	4.5	25.7	是	BIDIRECTIONAL
MAP_EDGE_0340	N005	N168	SIDE	7	45.3	是	BIDIRECTIONAL
MAP_EDGE_0341	N168	N034	SIDE	7	45.3	是	BIDIRECTIONAL
MAP_EDGE_0342	N168	N054	SIDEDOOR	4.5	18.7	是	BIDIRECTIONAL
MAP_EDGE_0343	N034	N169	SIDE	7	30.4	是	BIDIRECTIONAL
MAP_EDGE_0344	N169	N006	SIDE	7	30.4	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0345	N169	N168	SIDEDOOR	4.5	35.9	是	BIDIRECTIONAL
MAP_EDGE_0346	N007	N170	SIDE	7	43	是	BIDIRECTIONAL
MAP_EDGE_0347	N170	N037	SIDE	7	43	是	BIDIRECTIONAL
MAP_EDGE_0348	N170	N121	SIDEDOOR	4.5	66.3	是	BIDIRECTIONAL
MAP_EDGE_0349	N037	N171	SIDE	7	35.2	是	BIDIRECTIONAL
MAP_EDGE_0350	N171	N042	SIDE	7	35.2	是	BIDIRECTIONAL
MAP_EDGE_0351	N171	N148	SIDEDOOR	4.5	22.7	是	BIDIRECTIONAL
MAP_EDGE_0352	N042	N172	SIDE	7	57.7	是	BIDIRECTIONAL
MAP_EDGE_0353	N172	N009	SIDE	7	57.7	是	BIDIRECTIONAL
MAP_EDGE_0354	N172	N100	SIDEDOOR	4.5	17.1	是	BIDIRECTIONAL
MAP_EDGE_0355	N019	N173	SIDE	7	37	是	BIDIRECTIONAL
MAP_EDGE_0356	N173	N043	SIDE	7	37	是	BIDIRECTIONAL
MAP_EDGE_0357	N173	N132	SIDEDOOR	4.5	56.7	是	BIDIRECTIONAL
MAP_EDGE_0358	N043	N174	SIDE	7	54.1	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0359	N174	N030	SIDE	7	54.1	是	BIDIRECTIONAL
MAP_EDGE_0360	N174	N090	SIDEDOOR	4.5	32.6	是	BIDIRECTIONAL
MAP_EDGE_0361	N024	N175	SIDE	7	57.9	是	BIDIRECTIONAL
MAP_EDGE_0362	N175	N044	SIDE	7	57.9	是	BIDIRECTIONAL
MAP_EDGE_0363	N175	N079	SIDEDOOR	4.5	32.2	是	BIDIRECTIONAL
MAP_EDGE_0364	N044	N176	SIDE	7	44.7	是	BIDIRECTIONAL
MAP_EDGE_0365	N176	N045	SIDE	7	44.7	是	BIDIRECTIONAL
MAP_EDGE_0366	N176	N175	BREACH	4	93.3	是	BIDIRECTIONAL
MAP_EDGE_0367	N026	N177	SIDE	7	44	是	BIDIRECTIONAL
MAP_EDGE_0368	N177	N046	SIDE	7	44	是	BIDIRECTIONAL
MAP_EDGE_0369	N177	N137	SIDEDOOR	4.5	72.2	是	BIDIRECTIONAL
MAP_EDGE_0370	N004	N178	MAIN	12	32	是	BIDIRECTIONAL
MAP_EDGE_0371	N178	N144	MAIN	12	32	是	BIDIRECTIONAL
MAP_EDGE_0372	N178	N154	SIDEDOOR	4.5	49.6	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0373	N144	N179	MAIN	12	32	是	BIDIRECTIONAL
MAP_EDGE_0374	N179	N014	MAIN	12	32	是	BIDIRECTIONAL
MAP_EDGE_0375	N179	N126	SIDEDOOR	4.5	36.9	是	BIDIRECTIONAL
MAP_EDGE_0376	N006	N180	MAIN	12	33.3	是	BIDIRECTIONAL
MAP_EDGE_0377	N180	N146	MAIN	12	33.3	是	BIDIRECTIONAL
MAP_EDGE_0378	N180	N119	SIDEDOOR	4.5	33.7	是	BIDIRECTIONAL
MAP_EDGE_0379	N146	N181	MAIN	12	33.3	是	BIDIRECTIONAL
MAP_EDGE_0380	N181	N017	MAIN	12	33.3	是	BIDIRECTIONAL
MAP_EDGE_0381	N181	N158	SIDEDOOR	4.5	48.9	是	BIDIRECTIONAL
MAP_EDGE_0382	N017	N182	MAIN	12	41.5	是	BIDIRECTIONAL
MAP_EDGE_0383	N182	N147	MAIN	12	41.5	是	BIDIRECTIONAL
MAP_EDGE_0384	N182	N061	SIDEDOOR	4.5	30.7	是	BIDIRECTIONAL
MAP_EDGE_0385	N147	N183	MAIN	12	41.5	是	BIDIRECTIONAL
MAP_EDGE_0386	N183	N027	MAIN	12	41.5	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0387	N183	N063	SIDEDOOR	4.5	15.3	是	BIDIRECTIONAL
MAP_EDGE_0388	N008	N184	MAIN	12	56.1	是	BIDIRECTIONAL
MAP_EDGE_0389	N184	N148	MAIN	12	56.1	是	BIDIRECTIONAL
MAP_EDGE_0390	N184	N042	SIDEDOOR	4.5	41.7	是	BIDIRECTIONAL
MAP_EDGE_0391	N148	N185	MAIN	12	56.1	是	BIDIRECTIONAL
MAP_EDGE_0392	N185	N019	MAIN	12	56.1	是	BIDIRECTIONAL
MAP_EDGE_0393	N185	N106	SIDEDOOR	4.5	25.1	是	BIDIRECTIONAL
MAP_EDGE_0394	N019	N186	MAIN	12	42.8	是	BIDIRECTIONAL
MAP_EDGE_0395	N186	N149	MAIN	12	42.8	是	BIDIRECTIONAL
MAP_EDGE_0396	N186	N173	SIDEDOOR	4.5	25	是	BIDIRECTIONAL
MAP_EDGE_0397	N149	N187	MAIN	12	42.8	是	BIDIRECTIONAL
MAP_EDGE_0398	N187	N029	MAIN	12	42.8	是	BIDIRECTIONAL
MAP_EDGE_0399	N187	N140	SIDEDOOR	4.5	39.4	是	BIDIRECTIONAL
MAP_EDGE_0400	N034	N188	MAIN	12	35.9	是	BIDIRECTIONAL

边 ID	A	B	类型	宽 m	长 m	Nav	方向
MAP_EDGE_0401	N188	N153	MAIN	12	35.9	是	BIDIRECTIONAL
MAP_EDGE_0402	N188	N189	POCKET	4	14	是	BIDIRECTIONAL
MAP_EDGE_0403	N153	N190	MAIN	12	35.9	是	BIDIRECTIONAL
MAP_EDGE_0404	N190	N007	MAIN	12	35.9	是	BIDIRECTIONAL
MAP_EDGE_0405	N190	N191	POCKET	4	14	是	BIDIRECTIONAL

D.3 实体墙多边形

墙 ID	类别	Z0	高 m	顶点数	碰撞
MAP_WALL_001	BOUND	0	6	5	实体挤出多边形
MAP_WALL_002	VAULT	0	2.5	5	实体挤出多边形
MAP_WALL_003	VAULT	0	2.5	3	实体挤出多边形
MAP_WALL_004	VAULT	0	2.5	4	实体挤出多边形
MAP_WALL_005	VAULT	0	2.5	4	实体挤出多边形
MAP_WALL_006	VAULT	0	2.5	4	实体挤出多边形
MAP_WALL_007	VAULT	0	2.5	5	实体挤出多边形
MAP_WALL_008	VAULT	0	2.5	6	实体挤出多边形
MAP_WALL_009	VAULT	0	2.5	5	实体挤出多边形
MAP_WALL_010	VAULT	0	2.5	5	实体挤出多边形
MAP_WALL_011	VAULT	0	2.5	5	实体挤出多边形
MAP_WALL_012	VAULT	0	2.5	3	实体挤出多边形
MAP_WALL_013	VAULT	0	2.5	5	实体挤出多边形
MAP_WALL_014	VAULT	0	2.5	3	实体挤出多边形

墙 ID	类别	Z0	高 m	顶点数	碰撞
MAP_WALL_015	BOUND	0	6	5	实体挤出多边形
MAP_WALL_016	BOUND	0	6	7	实体挤出多边形
MAP_WALL_017	BOUND	0	6	3	实体挤出多边形
MAP_WALL_018	VAULT	0	2.5	3	实体挤出多边形
MAP_WALL_019	VAULT	0	2.5	7	实体挤出多边形
MAP_WALL_020	VAULT	0	2.5	6	实体挤出多边形
MAP_WALL_021	VAULT	0	2.5	3	实体挤出多边形
MAP_WALL_022	VAULT	0	2.5	7	实体挤出多边形
MAP_WALL_023	VAULT	0	2.5	7	实体挤出多边形
MAP_WALL_024	VAULT	0	2.5	4	实体挤出多边形
MAP_WALL_025	VAULT	0	2.5	3	实体挤出多边形
MAP_WALL_026	VAULT	0	2.5	3	实体挤出多边形
MAP_WALL_027	VAULT	0	2.5	5	实体挤出多边形
MAP_WALL_028	VAULT	0	2.5	5	实体挤出多边形
MAP_WALL_029	VAULT	0	2.5	4	实体挤出多边形
MAP_WALL_030	VAULT	0	2.5	6	实体挤出多边形
MAP_WALL_031	BOUND	0	6	5	实体挤出多边形
MAP_WALL_032	BOUND	0	6	5	实体挤出多边形
MAP_WALL_033	BOUND	0	6	4	实体挤出多边形
MAP_WALL_034	BOUND	0	6	5	实体挤出多边形
MAP_WALL_035	BOUND	0	6	3	实体挤出多边形
MAP_WALL_036	BOUND	0	6	3	实体挤出多边形
MAP_WALL_037	BOUND	0	6	3	实体挤出多边形
MAP_WALL_038	VAULT	0	2.5	5	实体挤出多边形
MAP_WALL_039	VAULT	0	2.5	3	实体挤出多边形
MAP_WALL_040	VAULT	0	2.5	5	实体挤出多边形
MAP_WALL_041	VAULT	0	2.5	5	实体挤出多边形

墙 ID	类别	Z0	高 m	顶点数	碰撞
MAP_WALL_042	VAULT	0	2.5	5	实体挤出多边形
MAP_WALL_043	VAULT	0	2.5	5	实体挤出多边形
MAP_WALL_044	VAULT	0	2.5	4	实体挤出多边形
MAP_WALL_045	BOUND	0	6	5	实体挤出多边形
MAP_WALL_046	BOUND	0	6	4	实体挤出多边形
MAP_WALL_047	BOUND	0	6	5	实体挤出多边形

墙 ID	顶点序号	有序顶点(x,y)m
MAP_WALL_001	1-5	V1=-348.00 / 269.20; V2=-252.10 / 285.20; V3=-225.50 / 247.80; V4=-260.10 / 226.50; V5=-340.10 / 234.50
MAP_WALL_002	1-5	V1=-214.80 / 285.20; V2=-150.80 / 295.80; V3=-129.50 / 261.20; V4=-164.20 / 237.20; V5=-217.50 / 250.50
MAP_WALL_003	1-7	V1=-288.30 / 110.40; V2=-288.20 / 110.40; V3=-288.20 / 110.00; V4=-288.10 / 109.90; V5=-293.90 / 101.10; V6=-340.10 / 106.60; V7=-337.40 / 149.20
MAP_WALL_003	1-6	V1=-288.30 / 110.70; V2=-288.40 / 111.00; V3=-288.40 / 111.10; V4=-288.40 / 111.40; V5=-288.40 / 111.70; V6=-288.40 / 111.80
MAP_WALL_003	1-5	V1=-326.80 / 141.00; V2=-326.50 / 140.80; V3=-326.50 / 140.70; V4=-288.40 / 112.20; V5=-288.40 / 112.10
MAP_WALL_003	1-4	V1=-327.30 / 141.60; V2=-327.30 / 141.50; V3=-327.20 / 141.50; V4=-326.90 / 141.10
MAP_WALL_003	1-7	V1=-327.40 / 141.70; V2=-327.40 / 141.80; V3=-327.50 / 141.90; V4=-327.60 / 142.00; V5=-327.60 / 142.10; V6=-327.70 / 142.20; V7=-327.70 / 142.30
MAP_WALL_003	1-3	V1=-337.40 / 149.20; V2=-331.20 / 150.70; V3=-327.80 / 142.40
MAP_WALL_004	1-4	V1=-274.10 / 164.40; V2=-286.30 / 120.60; V3=-320.80 / 146.50; V4=-323.40 / 152.60
MAP_WALL_005	1-4	V1=-259.70 / 116.40; V2=-272.90 / 116.40; V3=-262.40 / 154.10; V4=-246.80 / 133.20
MAP_WALL_006	1-4	V1=-195.30 / 127.40; V2=-188.00 / 116.40; V3=-228.30 / 116.40; V4=-228.10 / 119.90
MAP_WALL_007	1-5	V1=-160.60 / 124.30; V2=-169.80 / 116.90; V3=-178.70 / 116.90; V4=-186.90 / 129.30; V5=-169.50 / 133.20
MAP_WALL_008	1-4	V1=-157.40 / 107.10; V2=-155.30 / 108.40; V3=-150.00 / 113.70; V4=-137.50 / 101.30

墙 ID	顶点序号	有序顶点(x,y)m
MAP_WALL_008	1-6	V1=-137.50 / 101.30; V2=-163.10 / 71.10; V3=-173.50 / 105.60; V4=-171.90 / 106.90; V5=-158.80 / 106.90; V6=-157.90 / 107.00
MAP_WALL_009	1-5	V1=-211.70 / 74.10; V2=-212.00 / 74.40; V3=-210.00 / 76.40; V4=-181.10 / 99.50; V5=- 171.40 / 67.40
MAP_WALL_010	1-5	V1=-228.90 / 107.40; V2=-190.30 / 107.40; V3=-220.30 / 77.50; V4=-226.10 / 76.50; V5=-230.80 / 77.30
MAP_WALL_011	1-5	V1=-100.20 / 301.10; V2=-36.20 / 311.80; V3=-1.60 / 277.20; V4=-22.90 / 245.20; V5=- 89.50 / 250.50
MAP_WALL_012	1-8	V1=42.60 / 275.20; V2=42.30 / 275.00; V3=33.50 / 271.10; V4=30.40 / 287.80; V5=91.70 / 309.10; V6=131.70 / 266.50; V7=99.70 / 231.90; V8=38.40 / 245.20
MAP_WALL_012	1-8	V1=42.90 / 275.20; V2=43.30 / 275.30; V3=43.60 / 275.30; V4=44.00 / 275.30; V5=44.30 / 275.30; V6=44.60 / 275.20; V7=45.00 / 275.10; V8=45.30 / 275.00
MAP_WALL_012	1-3	V1=45.60 / 274.80; V2=46.90 / 273.30; V3=47.00 / 272.90
MAP_WALL_012	1-4	V1=47.10 / 272.60; V2=47.20 / 272.30; V3=47.20 / 271.20; V4=46.80 / 270.30
MAP_WALL_012	1-3	V1=46.70 / 270.00; V2=45.10 / 268.60; V3=34.80 / 264.10
MAP_WALL_012	41-41	V41=38.4 / 245.2
MAP_WALL_013	1-5	V1=244.90 / 220.90; V2=161.80 / 233.00; V3=166.30 / 285.20; V4=235.60 / 277.20; V5=262.20 / 234.50
MAP_WALL_014	1-3	V1=167.00 / 222.10; V2=234.10 / 212.40; V3=224.90 / 205.20
MAP_WALL_015	1-5	V1=338.70 / 209.30; V2=325.70 / 209.20; V3=283.30 / 215.40; V4=288.90 / 245.20; V5=347.50 / 229.20
MAP_WALL_015	1-5	V1=339.30 / 209.10; V2=340.60 / 208.50; V3=341.70 / 207.60; V4=342.60 / 206.60; V5=342.90 / 206.00
MAP_WALL_015	1-6	V1=343.20 / 205.40; V2=343.50 / 204.80; V3=343.90 / 202.70; V4=343.80 / 202.10; V5=343.80 / 201.40; V6=343.60 / 200.70
MAP_WALL_015	1-5	V1=356.40 / 182.80; V2=344.80 / 176.60; V3=341.70 / 197.50; V4=342.80 / 198.80; V5=343.40 / 200.10
MAP_WALL_015	33-33	V33=347.5 / 229.2
MAP_WALL_016	1-3	V1=330.60 / 168.80; V2=330.30 / 168.30; V3=328.90 / 166.30

墙 ID	顶点序号	有序顶点(x,y)m
MAP_WALL_016	1-6	V1=330.70 / 168.80; V2=330.80 / 168.90; V3=330.90 / 169.00; V4=331.00 / 169.10; V5=331.10 / 169.10; V6=331.20 / 169.20
MAP_WALL_016	1-7	V1=328.90 / 166.30; V2=316.10 / 163.30; V3=333.10 / 188.70; V4=333.30 / 170.40; V5=331.40 / 169.30; V6=331.20 / 169.30; V7=331.20 / 169.20
MAP_WALL_017	1-3	V1=287.70 / 194.10; V2=326.60 / 195.20; V3=308.50 / 168.10
MAP_WALL_018	1-3	V1=-333.00 / 35.60; V2=-352.90 / -3.10; V3=-350.70 / 34.60
MAP_WALL_019	1-7	V1=-311.70 / 36.70; V2=-290.50 / -17.10; V3=-295.70 / -22.30; V4=-304.40 / -22.30; V5=-322.60 / -18.00; V6=-334.70 / 12.60; V7=-322.60 / 36.10
MAP_WALL_020	1-6	V1=-278.90 / 7.70; V2=-279.00 / 7.60; V3=-281.80 / -8.40; V4=-284.40 / -10.90; V5=- 303.30 / 37.10; V6=-300.10 / 37.30
MAP_WALL_021	1-3	V1=-347.40 / -12.10; V2=-338.40 / 5.50; V3=-329.80 / -16.30
MAP_WALL_022	1-7	V1=-127.50 / -39.30; V2=-127.40 / -40.50; V3=-154.50 / -65.30; V4=-175.50 / -66.50; V5=-222.80 / -58.60; V6=-220.10 / -16.00; V7=-158.80 / -5.30
MAP_WALL_023	1-7	V1=-41.50 / 42.30; V2=-21.00 / 35.90; V3=-0.90 / 29.80; V4=11.70 / 16.00; V5=-14.90 / - 10.70; V6=-76.20 / -2.70; V7=-73.60 / 34.60
MAP_WALL_024	1-4	V1=54.70 / -2.60; V2=69.00 / -18.70; V3=53.00 / -25.50; V4=54.40 / -2.70
MAP_WALL_025	1-3	V1=71.70 / -0.90; V2=71.70 / -11.90; V3=62.70 / -1.80
MAP_WALL_026	1-3	V1=97.40 / 1.60; V2=80.50 / -12.20; V3=82.60 / 0.20
MAP_WALL_027	1-5	V1=123.30 / -14.20; V2=121.80 / -30.40; V3=119.40 / -37.60; V4=83.80 / -19.80; V5=109.20 / 1.00
MAP_WALL_028	1-5	V1=93.70 / -61.20; V2=82.40 / -30.30; V3=120.00 / -49.00; V4=119.80 / -51.20; V5=102.30 / -64.00
MAP_WALL_029	1-4	V1=52.40 / -36.60; V2=74.40 / -27.30; V3=85.80 / -58.80; V4=51.70 / -48.00
MAP_WALL_030	1-6	V1=228.60 / 38.40; V2=261.70 / -4.40; V3=253.40 / -14.30; V4=190.30 / -14.30; V5=180.80 / -5.70; V6=177.00 / 26.70
MAP_WALL_030	9-9	V9=177 / 26.7
MAP_WALL_031	1-5	V1=334.70 / 82.60; V2=335.20 / 44.70; V3=334.30 / 44.30; V4=309.80 / 82.90; V5=310.20 / 85.30

墙 ID	顶点序号	有序顶点(x,y)m
MAP_WALL_032	1-5	V1=351.30 / 80.70; V2=354.90 / 56.60; V3=348.30 / 50.90; V4=343.20 / 48.50; V5=342.70 / 81.70
MAP_WALL_033	1-4	V1=366.70 / 32.00; V2=363.30 / 54.60; V3=366.60 / 57.40; V4=374.20 / 37.30
MAP_WALL_034	1-5	V1=333.80 / 12.50; V2=344.10 / 21.30; V3=345.30 / 22.60; V4=353.60 / 22.60; V5=336.90 / 10.70
MAP_WALL_035	1-5	V1=326.90 / 33.00; V2=328.40 / 33.70; V3=331.30 / 30.20; V4=331.90 / 29.30; V5=321.30 / 20.20
MAP_WALL_035	1-3	V1=326.90 / 33.00; V2=326.40 / 32.80; V3=306.20 / 29.50
MAP_WALL_035	17-17	V17=321.3 / 20.2
MAP_WALL_036	1-3	V1=306.40 / 60.10; V2=323.60 / 39.50; V3=302.80 / 36.10
MAP_WALL_037	1-3	V1=346.50 / 30.70; V2=356.60 / 45.40; V3=358.90 / 30.70
MAP_WALL_038	1-5	V1=-334.60 / -146.80; V2=-344.00 / -98.10; V3=-284.10 / -90.60; V4=-254.80 / -125.30; V5=-286.80 / -157.20
MAP_WALL_039	1-3	V1=-348.00 / -119.50; V2=-343.10 / -145.00; V3=-348.00 / -143.90
MAP_WALL_040	1-5	V1=-220.00 / -198.50; V2=-159.80 / -233.90; V3=-177.50 / -253.20; V4=-236.10 / - 239.90; V5=-233.50 / -202.50
MAP_WALL_041	1-5	V1=-149.50 / -219.80; V2=-151.70 / -225.00; V3=-154.30 / -227.90; V4=-209.50 / - 195.40; V5=-180.20 / -186.60
MAP_WALL_042	1-5	V1=-94.90 / -223.90; V2=-36.20 / -210.50; V3=-4.30 / -247.80; V4=-36.20 / -279.80; V5=-94.90 / -263.80
MAP_WALL_043	1-5	V1=56.60 / -228.20; V2=60.50 / -231.20; V3=60.50 / -271.40; V4=51.50 / -273.50; V5=49.00 / -229.20
MAP_WALL_044	1-7	V1=122.90 / -255.10; V2=116.40 / -284.40; V3=113.00 / -287.80; V4=90.20 / -283.80; V5=69.50 / -271.90; V6=69.50 / -226.40; V7=107.70 / -221.20
MAP_WALL_044	1-3	V1=123.00 / -254.70; V2=123.90 / -253.30; V3=124.40 / -252.90
MAP_WALL_044	1-6	V1=124.70 / -252.70; V2=125.70 / -252.40; V3=126.10 / -252.40; V4=126.40 / -252.30; V5=126.70 / -252.40; V6=127.10 / -252.40
MAP_WALL_044	1-3	V1=127.40 / -252.50; V2=128.90 / -253.40; V3=129.30 / -254.00
MAP_WALL_044	1-4	V1=129.40 / -254.30; V2=129.80 / -255.20; V3=129.80 / -256.30; V4=129.70 / -256.60

墙 ID	顶点序号	有序顶点(x,y)m
MAP_WALL_044	1-4	V1=107.70 / -221.20; V2=139.40 / -250.70; V3=127.20 / -273.60; V4=125.60 / -275.20
MAP_WALL_045	1-5	V1=198.30 / -218.50; V2=262.20 / -207.90; V3=307.50 / -239.90; V4=272.90 / -274.50; V5=208.90 / -261.20
MAP_WALL_046	1-4	V1=310.00 / -111.10; V2=327.20 / -166.10; V3=288.90 / -154.60; V4=288.90 / -106.60
MAP_WALL_046	1-4	V1=309.90 / -110.40; V2=310.00 / -110.60; V3=310.00 / -110.80; V4=310.00 / -111.00
MAP_WALL_046	1-4	V1=310.00 / -109.60; V2=310.00 / -110.00; V3=309.90 / -110.00; V4=309.90 / -110.20
MAP_WALL_046	1-4	V1=288.90 / -106.60; V2=311.30 / -104.80; V3=310.10 / -109.30; V4=310.00 / -109.40
MAP_WALL_047	1-5	V1=340.30 / -170.00; V2=328.60 / -103.40; V3=355.50 / -101.30; V4=374.20 / -143.90; V5=342.20 / -170.60

D.4 高台与坡道

高台 ID	名称	顶高	眺望 m	顶面	侧壁	职责
MAP_HIGHLAND_01	龙脊东台	4	45	WALKABLE_HORIZONTAL_P LANE	SOLID_POLYGON_PERIMETER_EXCEPT _RAMP_PORTALS	INFORMATION_ADVANTAGE_WITH OUT_HIGH_VALUE_PERMANENT_R ESOURCE
MAP_HIGHLAND_02	龙脊南台	3.5	40	WALKABLE_HORIZONTAL_P LANE	SOLID_POLYGON_PERIMETER_EXCEPT _RAMP_PORTALS	INFORMATION_ADVANTAGE_WITH OUT_HIGH_VALUE_PERMANENT_R ESOURCE
MAP_HIGHLAND_03	蛛丝观星台	4	45	WALKABLE_HORIZONTAL_P LANE	SOLID_POLYGON_PERIMETER_EXCEPT _RAMP_PORTALS	INFORMATION_ADVANTAGE_WITH OUT_HIGH_VALUE_PERMANENT_R ESOURCE

高台 ID	有序顶面顶点(x,y)m
MAP_HIGHLAND_01	V1=310.2 / 202.5; V2=360.8 / 189.2; V3=368.8 / 143.9; V4=326.2 / 133.2; V5=299.5 / 162.6
MAP_HIGHLAND_02	V1=299.5 / 64; V2=347.5 / 58.6; V3=360.8 / 26.7; V4=326.2 / 5.3; V5=294.2 / 26.7
MAP_HIGHLAND_03	V1=-30.9 / 271.8; V2=11.7 / 279.8; V3=30.4 / 250.5; V4=3.7 / 229.2; V5=-28.2 / 239.9

坡道 ID	高台	宽	Z0	Z1	中心线	插值	可走
MAP_RAMP_01_01	MAP_HIGHLAND_01	5	0	4	315.5 / 141.2; 299.5 / 114.6	LINEAR_BY_CENTERLINE_PROGRESS	是
MAP_RAMP_01_02	MAP_HIGHLAND_01	5	0	4	360.8 / 181.2; 374.2 / 202.5	LINEAR_BY_CENTERLINE_PROGRESS	是
MAP_RAMP_02_01	MAP_HIGHLAND_02	5	0	3.5	299.5 / 32; 283.6 / 10.7	LINEAR_BY_CENTERLINE_PROGRESS	是
MAP_RAMP_02_02	MAP_HIGHLAND_02	5	0	3.5	350.2 / 48; 368.8 / 64	LINEAR_BY_CENTERLINE_PROGRESS	是
MAP_RAMP_03_01	MAP_HIGHLAND_03	5	0	4	-1.6 / 231.9; -9.6 / 207.9	LINEAR_BY_CENTERLINE_PROGRESS	是
MAP_RAMP_03_02	MAP_HIGHLAND_03	5	0	4	19.7 / 261.2; 43.7 / 271.8	LINEAR_BY_CENTERLINE_PROGRESS	是

D.5 三庭完整设施

庭 ID	名称	中心	有序边界	入口	region_id
MAP_COURT_01	B1_惊蛰庭	22.4 / 109.3	-13.2 / 123.2; 25.5 / 144.8; 64.1 / 123.2; 56.4 / 83; 14.7 / 75.3; -17.8 / 95.4	-16.1 / 105.7; 54.3 / 128.7; 26.5 / 77.5	REGION_AGON_RIE
MAP_COURT_02	B2_玄冥庭	-12.3 / -58.6	-44.7 / -46.3; -4.5 / -23.1; 31 / -44.7; 21.7 / -86.5; -23.1 / -94.2; -54 / -72.5	-43.3 / -45.5; 26.7 / -42.1; -46.2 / -78	REGION_ST_SOULELD
MAP_COURT_03	B3_焚天庭	150.3 / -58.6	116.3 / -46.3; 156.5 / -24.6; 193.6 / -47.8; 184.3 / -88; 139.5 / -94.2; 108.6 / -72.5	120.8 / -43.9; 189.4 / -66; 136.8 / -92.3	REGION_RNING_WER

庭 ID	设施类型	设施 ID	坐标 m	半径 m
MAP_COURT_01	终局商位	MAP_FINAL_SHOP_C1_1	49.3 / 111.2	不适用
MAP_COURT_01	终局商位	MAP_FINAL_SHOP_C1_2	22 / 82.3	不适用
MAP_COURT_01	复活位	MAP_REVIVE_C1_1	56.6 / 92.7	不适用
MAP_COURT_01	复活位	MAP_REVIVE_C1_2	25.1 / 71.4	不适用
MAP_COURT_01	复活位	MAP_REVIVE_C1_3	-9.1 / 88	不适用
MAP_COURT_01	复活位	MAP_REVIVE_C1_4	-11.8 / 125.9	不适用
MAP_COURT_01	复活位	MAP_REVIVE_C1_5	19.7 / 147.2	不适用
MAP_COURT_01	复活位	MAP_REVIVE_C1_6	53.9 / 130.6	不适用
MAP_COURT_01	伏石	MAP_FUSHI_COURT_C1_1	-6.7 / 102	2
MAP_COURT_01	伏石	MAP_FUSHI_COURT_C1_2	15.1 / 138.4	2
MAP_COURT_01	宝箱备点	MAP_FINAL_CHEST_C1_1	34.4 / 109.3	不适用
MAP_COURT_01	宝箱备点	MAP_FINAL_CHEST_C1_2	10.4 / 109.3	不适用
MAP_COURT_01	宝箱备点	MAP_FINAL_CHEST_C1_3	22.4 / 97.3	不适用
MAP_COURT_01	宝箱备点	MAP_FINAL_CHEST_C1_4	22.4 / 121.3	不适用
MAP_COURT_02	终局商位	MAP_FINAL_SHOP_C2_1	12.7 / -68.8	不适用
MAP_COURT_02	终局商位	MAP_FINAL_SHOP_C2_2	-24.6 / -82.6	不适用
MAP_COURT_02	复活位	MAP_REVIVE_C2_1	11 / -88.7	不适用
MAP_COURT_02	复活位	MAP_REVIVE_C2_2	-26.6 / -93.8	不适用
MAP_COURT_02	复活位	MAP_REVIVE_C2_3	-49.9 / -63.8	不适用
MAP_COURT_02	复活位	MAP_REVIVE_C2_4	-35.6 / -28.6	不适用
MAP_COURT_02	复活位	MAP_REVIVE_C2_5	2.1 / -23.4	不适用
MAP_COURT_02	复活位	MAP_REVIVE_C2_6	25.4 / -53.5	不适用
MAP_COURT_02	伏石	MAP_FUSHI_COURT_C2_1	-41.6 / -52.3	2

庭 ID	设施类型	设施 ID	坐标 m	半径 m
MAP_COURT_02	伏石	MAP_FUSHI_COURT_C2_2	-5.9 / -29.3	2
MAP_COURT_02	宝箱备点	MAP_FINAL_CHEST_C2_1	-0.3 / -58.6	不适用
MAP_COURT_02	宝箱备点	MAP_FINAL_CHEST_C2_2	-24.3 / -58.6	不适用
MAP_COURT_02	宝箱备点	MAP_FINAL_CHEST_C2_3	-12.3 / -70.6	不适用
MAP_COURT_02	宝箱备点	MAP_FINAL_CHEST_C2_4	-12.3 / -46.6	不适用
MAP_COURT_03	终局商位	MAP_FINAL_SHOP_C3_1	130.8 / -77.3	不适用
MAP_COURT_03	终局商位	MAP_FINAL_SHOP_C3_2	133.4 / -37.5	不适用
MAP_COURT_03	复活位	MAP_REVIVE_C3_1	113.4 / -67.6	不适用
MAP_COURT_03	复活位	MAP_REVIVE_C3_2	124.1 / -31.1	不适用
MAP_COURT_03	复活位	MAP_REVIVE_C3_3	161 / -22.2	不适用
MAP_COURT_03	复活位	MAP_REVIVE_C3_4	187.2 / -49.7	不适用
MAP_COURT_03	复活位	MAP_REVIVE_C3_5	176.5 / -86.1	不适用
MAP_COURT_03	复活位	MAP_REVIVE_C3_6	139.6 / -95.1	不适用
MAP_COURT_03	伏石	MAP_FUSHI_COURT_C3_1	168.1 / -34.5	2
MAP_COURT_03	伏石	MAP_FUSHI_COURT_C3_2	174.5 / -76.4	2
MAP_COURT_03	宝箱备点	MAP_FINAL_CHEST_C3_1	162.3 / -58.6	不适用
MAP_COURT_03	宝箱备点	MAP_FINAL_CHEST_C3_2	138.3 / -58.6	不适用
MAP_COURT_03	宝箱备点	MAP_FINAL_CHEST_C3_3	150.3 / -70.6	不适用
MAP_COURT_03	宝箱备点	MAP_FINAL_CHEST_C3_4	150.3 / -46.6	不适用



惊蛰庭决赛圈与设施



玄冥庭决赛圈与设施



梵天庭决赛圈与设施

D.6 出生宏区与 80 微位

宏区 ID	标签	微位 ID
MAP_SPAWN_MACRO_01	R01	MAP_SPAWN_POINT:MAP_SPAWN_POINT_01; MAP_SPAWN_POINT:MAP_SPAWN_POINT_02
MAP_SPAWN_MACRO_02	R02	MAP_SPAWN_POINT:MAP_SPAWN_POINT_03; MAP_SPAWN_POINT:MAP_SPAWN_POINT_04
MAP_SPAWN_MACRO_03	R03	MAP_SPAWN_POINT:MAP_SPAWN_POINT_05; MAP_SPAWN_POINT:MAP_SPAWN_POINT_06
MAP_SPAWN_MACRO_04	R04	MAP_SPAWN_POINT:MAP_SPAWN_POINT_07; MAP_SPAWN_POINT:MAP_SPAWN_POINT_08
MAP_SPAWN_MACRO_05	R05	MAP_SPAWN_POINT:MAP_SPAWN_POINT_09; MAP_SPAWN_POINT:MAP_SPAWN_POINT_10
MAP_SPAWN_MACRO_06	R06	MAP_SPAWN_POINT:MAP_SPAWN_POINT_11; MAP_SPAWN_POINT:MAP_SPAWN_POINT_12
MAP_SPAWN_MACRO_07	R07	MAP_SPAWN_POINT:MAP_SPAWN_POINT_13; MAP_SPAWN_POINT:MAP_SPAWN_POINT_14
MAP_SPAWN_MACRO_08	R08	MAP_SPAWN_POINT:MAP_SPAWN_POINT_15; MAP_SPAWN_POINT:MAP_SPAWN_POINT_16
MAP_SPAWN_MACRO_09	R09	MAP_SPAWN_POINT:MAP_SPAWN_POINT_17; MAP_SPAWN_POINT:MAP_SPAWN_POINT_18
MAP_SPAWN_MACRO_10	R10	MAP_SPAWN_POINT:MAP_SPAWN_POINT_19; MAP_SPAWN_POINT:MAP_SPAWN_POINT_20
MAP_SPAWN_MACRO_11	R11	MAP_SPAWN_POINT:MAP_SPAWN_POINT_21; MAP_SPAWN_POINT:MAP_SPAWN_POINT_22
MAP_SPAWN_MACRO_12	R12	MAP_SPAWN_POINT:MAP_SPAWN_POINT_23; MAP_SPAWN_POINT:MAP_SPAWN_POINT_24
MAP_SPAWN_MACRO_13	R13	MAP_SPAWN_POINT:MAP_SPAWN_POINT_25; MAP_SPAWN_POINT:MAP_SPAWN_POINT_26

宏区 ID	标签	微位 ID
MAP_SPAWN_MACRO_14	R14	MAP_SPAWN_POINT:MAP_SPAWN_POINT_27; MAP_SPAWN_POINT:MAP_SPAWN_POINT_28
MAP_SPAWN_MACRO_15	R15	MAP_SPAWN_POINT:MAP_SPAWN_POINT_29; MAP_SPAWN_POINT:MAP_SPAWN_POINT_30
MAP_SPAWN_MACRO_16	R16	MAP_SPAWN_POINT:MAP_SPAWN_POINT_31; MAP_SPAWN_POINT:MAP_SPAWN_POINT_32
MAP_SPAWN_MACRO_17	R17	MAP_SPAWN_POINT:MAP_SPAWN_POINT_33; MAP_SPAWN_POINT:MAP_SPAWN_POINT_34
MAP_SPAWN_MACRO_18	R18	MAP_SPAWN_POINT:MAP_SPAWN_POINT_35; MAP_SPAWN_POINT:MAP_SPAWN_POINT_36
MAP_SPAWN_MACRO_19	R19	MAP_SPAWN_POINT:MAP_SPAWN_POINT_37; MAP_SPAWN_POINT:MAP_SPAWN_POINT_38
MAP_SPAWN_MACRO_20	R20	MAP_SPAWN_POINT:MAP_SPAWN_POINT_39; MAP_SPAWN_POINT:MAP_SPAWN_POINT_40
MAP_SPAWN_MACRO_21	R21	MAP_SPAWN_POINT:MAP_SPAWN_POINT_41; MAP_SPAWN_POINT:MAP_SPAWN_POINT_42
MAP_SPAWN_MACRO_22	R22	MAP_SPAWN_POINT:MAP_SPAWN_POINT_43; MAP_SPAWN_POINT:MAP_SPAWN_POINT_44
MAP_SPAWN_MACRO_23	R23	MAP_SPAWN_POINT:MAP_SPAWN_POINT_45; MAP_SPAWN_POINT:MAP_SPAWN_POINT_46
MAP_SPAWN_MACRO_24	R24	MAP_SPAWN_POINT:MAP_SPAWN_POINT_47; MAP_SPAWN_POINT:MAP_SPAWN_POINT_48
MAP_SPAWN_MACRO_25	R25	MAP_SPAWN_POINT:MAP_SPAWN_POINT_49; MAP_SPAWN_POINT:MAP_SPAWN_POINT_50
MAP_SPAWN_MACRO_26	R26	MAP_SPAWN_POINT:MAP_SPAWN_POINT_51; MAP_SPAWN_POINT:MAP_SPAWN_POINT_52
MAP_SPAWN_MACRO_27	R27	MAP_SPAWN_POINT:MAP_SPAWN_POINT_53; MAP_SPAWN_POINT:MAP_SPAWN_POINT_54
MAP_SPAWN_MACRO_28	R28	MAP_SPAWN_POINT:MAP_SPAWN_POINT_55; MAP_SPAWN_POINT:MAP_SPAWN_POINT_56
MAP_SPAWN_MACRO_29	R29	MAP_SPAWN_POINT:MAP_SPAWN_POINT_57; MAP_SPAWN_POINT:MAP_SPAWN_POINT_58
MAP_SPAWN_MACRO_30	R30	MAP_SPAWN_POINT:MAP_SPAWN_POINT_59; MAP_SPAWN_POINT:MAP_SPAWN_POINT_60
MAP_SPAWN_MACRO_31	R31	MAP_SPAWN_POINT:MAP_SPAWN_POINT_61; MAP_SPAWN_POINT:MAP_SPAWN_POINT_62
MAP_SPAWN_MACRO_32	R32	MAP_SPAWN_POINT:MAP_SPAWN_POINT_63; MAP_SPAWN_POINT:MAP_SPAWN_POINT_64
MAP_SPAWN_MACRO_33	R33	MAP_SPAWN_POINT:MAP_SPAWN_POINT_65; MAP_SPAWN_POINT:MAP_SPAWN_POINT_66
MAP_SPAWN_MACRO_34	R34	MAP_SPAWN_POINT:MAP_SPAWN_POINT_67; MAP_SPAWN_POINT:MAP_SPAWN_POINT_68
MAP_SPAWN_MACRO_35	R35	MAP_SPAWN_POINT:MAP_SPAWN_POINT_69; MAP_SPAWN_POINT:MAP_SPAWN_POINT_70
MAP_SPAWN_MACRO_36	R36	MAP_SPAWN_POINT:MAP_SPAWN_POINT_71; MAP_SPAWN_POINT:MAP_SPAWN_POINT_72
MAP_SPAWN_MACRO_37	R37	MAP_SPAWN_POINT:MAP_SPAWN_POINT_73; MAP_SPAWN_POINT:MAP_SPAWN_POINT_74
MAP_SPAWN_MACRO_38	R38	MAP_SPAWN_POINT:MAP_SPAWN_POINT_75; MAP_SPAWN_POINT:MAP_SPAWN_POINT_76
MAP_SPAWN_MACRO_39	R39	MAP_SPAWN_POINT:MAP_SPAWN_POINT_77; MAP_SPAWN_POINT:MAP_SPAWN_POINT_78
MAP_SPAWN_MACRO_40	R40	MAP_SPAWN_POINT:MAP_SPAWN_POINT_79; MAP_SPAWN_POINT:MAP_SPAWN_POINT_80

微位 ID	标签	坐标 m	朝向°	宏区	出口节点	region_id
MAP_SPAWN_POINT_01	R01_1	-322.7 / 210.9	-2.5	MAP_SPAWN_MACRO_01	N001	REGION_1 DER_GORGE
MAP_SPAWN_POINT_02	R01_2	-355 / 180	-91.8	MAP_SPAWN_MACRO_01	N010	REGION_1 DER_GORGE
MAP_SPAWN_POINT_03	R02_1	311.5 / -194.9	176.1	MAP_SPAWN_MACRO_02	N029	REGION_2 NTIPEDE TY

微位 ID	标签	坐标 m	朝向°	宏区	出口节点	region_id
MAP_SPAWN_POINT_04	R02_2	337.6 / -163.8	102.1	MAP_SPAWN_MACRO_02	N043	REGION_02 RNING_W ER
MAP_SPAWN_POINT_05	R03_1	-135.6 / -258.2	-66.2	MAP_SPAWN_MACRO_03	N045	REGION_03 ST_SOUL ELD
MAP_SPAWN_POINT_06	R03_2	-173.5 / -217.7	151.8	MAP_SPAWN_MACRO_03	N022	REGION_03 ST_SOUL ELD
MAP_SPAWN_POINT_07	R04_1	75.1 / 137.8	-156.4	MAP_SPAWN_MACRO_04	N054	REGION_04 AGON_RI E
MAP_SPAWN_POINT_08	R04_2	125 / 151.6	2.7	MAP_SPAWN_MACRO_04	N007	REGION_04 AGON_RI E
MAP_SPAWN_POINT_09	R05_1	-130.2 / 8.1	-44.5	MAP_SPAWN_MACRO_05	N014	REGION_05 DER_GOF
MAP_SPAWN_POINT_10	R05_2	-181.5 / 27	-176.5	MAP_SPAWN_MACRO_05	N012	REGION_05 DER_GOF
MAP_SPAWN_POINT_11	R06_1	89.9 / -292.4	144.3	MAP_SPAWN_MACRO_06	N046	REGION_06 NTIPEDE TY
MAP_SPAWN_POINT_12	R06_2	131.5 / -280.5	63.1	MAP_SPAWN_MACRO_06	N028	REGION_06 NTIPEDE TY
MAP_SPAWN_POINT_13	R07_1	309 / 198.7	178.9	MAP_SPAWN_MACRO_07	N008	REGION_07 AGON_RI E

微位 ID	标签	坐标 m	朝向°	宏区	出口节点	region_id
MAP_SPAWN_POINT_14	R07_2	334.3 / 174.5	-88	MAP_SPAWN_MACRO_07	N020	REGION_AGON_RIE
MAP_SPAWN_POINT_15	R08_1	-135.9 / 232.2	-166	MAP_SPAWN_MACRO_08	N032	REGION_OKEN_G
MAP_SPAWN_POINT_16	R08_2	-84 / 236.7	-7	MAP_SPAWN_MACRO_08	N033	REGION_OKEN_G
MAP_SPAWN_POINT_17	R09_1	132.7 / -53.2	142	MAP_SPAWN_MACRO_09	N061	REGION_RNING_WER
MAP_SPAWN_POINT_18	R09_2	140.1 / -76	-101.4	MAP_SPAWN_MACRO_09	N063	REGION_NTIPEDITY
MAP_SPAWN_POINT_19	R10_1	-363.1 / -1.2	86.6	MAP_SPAWN_MACRO_10	N000	REGION_DER_GOF
MAP_SPAWN_POINT_20	R10_2	-333.4 / -32.3	3.3	MAP_SPAWN_MACRO_10	N011	REGION_DER_GOF
MAP_SPAWN_POINT_21	R11_1	336.1 / 54.6	89.7	MAP_SPAWN_MACRO_11	N009	REGION_RNING_WER
MAP_SPAWN_POINT_22	R11_2	316.3 / 10.7	-135.6	MAP_SPAWN_MACRO_11	N019	REGION_RNING_WER
MAP_SPAWN_POINT_23	R12_1	-30.5 / -54.2	145.9	MAP_SPAWN_MACRO_12	N057	REGION_ST_SOULELD
MAP_SPAWN_POINT_24	R12_2	10.3 / -52.3	31.9	MAP_SPAWN_MACRO_12	N058	REGION_NTIPEDITY

微位 ID	标签	坐标 m	朝向°	宏区	出口节点	region_id
MAP_SPAWN_POINT_25	R13_1	-304.1 / -178.2	-2.7	MAP_SPAWN_MACRO_13	N022	REGION_ ST_SOUL ELD
MAP_SPAWN_POINT_26	R13_2	-334.4 / -153.2	102.3	MAP_SPAWN_MACRO_13	N010	REGION_ ST_SOUL ELD
MAP_SPAWN_POINT_27	R14_1	234.5 / 88.3	25.4	MAP_SPAWN_MACRO_14	N042	REGION_ AGON_RI E
MAP_SPAWN_POINT_28	R14_2	185.9 / 60.4	-146.8	MAP_SPAWN_MACRO_14	N017	REGION_ RNING_W ER
MAP_SPAWN_POINT_29	R15_1	-194.1 / 138.3	-62.1	MAP_SPAWN_MACRO_15	N039	REGION_ DER_GOF
MAP_SPAWN_POINT_30	R15_2	-228.5 / 182.8	136.4	MAP_SPAWN_MACRO_15	N001	REGION_ OKEN_GO
MAP_SPAWN_POINT_31	R16_1	83.8 / -165.6	38.1	MAP_SPAWN_MACRO_16	N027	REGION_ NTIPEDE TY
MAP_SPAWN_POINT_32	R16_2	62 / -214.6	-87.1	MAP_SPAWN_MACRO_16	N046	REGION_ NTIPEDE TY
MAP_SPAWN_POINT_33	R17_1	4.8 / 110.6	-4.4	MAP_SPAWN_MACRO_17	N052	REGION_ OKEN_GO
MAP_SPAWN_POINT_34	R17_2	-42.1 / 106.2	-166.6	MAP_SPAWN_MACRO_17	N004	REGION_ OKEN_GO
MAP_SPAWN_POINT_35	R18_1	-121.6 / -104.8	114.2	MAP_SPAWN_MACRO_18	N051	REGION_ ST_SOUL ELD

微位 ID	标签	坐标 m	朝向°	宏区	出口节点	region_id
MAP_SPAWN_POINT_36	R18_2	-88.4 / -150.2	-43.1	MAP_SPAWN_MACRO_18	N025	REGION_ ST_SOUL ELD
MAP_SPAWN_POINT_37	R19_1	-271.9 / -49	115.8	MAP_SPAWN_MACRO_19	N011	REGION_ DER_GOF
MAP_SPAWN_POINT_38	R19_2	-235.7 / -90	-38.5	MAP_SPAWN_MACRO_19	N023	REGION_ DER_GOF
MAP_SPAWN_POINT_39	R20_1	328.6 / -112.3	-82.2	MAP_SPAWN_MACRO_20	N030	REGION_ RNING_W ER
MAP_SPAWN_POINT_40	R20_2	309.3 / -59.6	124	MAP_SPAWN_MACRO_20	N019	REGION_ RNING_W ER
MAP_SPAWN_POINT_41	R21_1	-303.3 / 83.7	52.2	MAP_SPAWN_MACRO_21	N031	REGION_ DER_GOF
MAP_SPAWN_POINT_42	R21_2	-308.6 / 31.5	-65.9	MAP_SPAWN_MACRO_21	N011	REGION_ DER_GOF
MAP_SPAWN_POINT_43	R22_1	193.1 / -164.7	-119.1	MAP_SPAWN_MACRO_22	N047	REGION_ NTIPEDE TY
MAP_SPAWN_POINT_44	R22_2	218.8 / -114.9	67.4	MAP_SPAWN_MACRO_22	N050	REGION_ NTIPEDE TY
MAP_SPAWN_POINT_45	R23_1	26.5 / 54	90	MAP_SPAWN_MACRO_23	N055	REGION_ AGON_RI E
MAP_SPAWN_POINT_46	R23_2	12.2 / 7.7	-126.9	MAP_SPAWN_MACRO_23	N015	REGION_ AGON_RI E

微位 ID	标签	坐标 m	朝向°	宏区	出口节点	region_id
MAP_SPAWN_POINT_47	R24_1	274.1 / 203.7	-1.1	MAP_SPAWN_MACRO_24	N009	REGION_ AGON_RI E
MAP_SPAWN_POINT_48	R24_2	222.3 / 185.1	-138.6	MAP_SPAWN_MACRO_24	N007	REGION_ AGON_RI E
MAP_SPAWN_POINT_49	R25_1	-215.8 / -192.8	-32.7	MAP_SPAWN_MACRO_25	N044	REGION_ ST_SOUL ELD
MAP_SPAWN_POINT_50	R25_2	-269.4 / -178.2	-177.3	MAP_SPAWN_MACRO_25	N021	REGION_ ST_SOUL ELD
MAP_SPAWN_POINT_51	R26_1	-126.1 / -290.8	113.8	MAP_SPAWN_MACRO_26	N044	REGION_ ST_SOUL ELD
MAP_SPAWN_POINT_52	R26_2	-120.7 / -288.1	119.4	MAP_SPAWN_MACRO_26	N044	REGION_ ST_SOUL ELD
MAP_SPAWN_POINT_53	R27_1	157 / 43.4	27.6	MAP_SPAWN_MACRO_27	N037	REGION_ RNING_W ER
MAP_SPAWN_POINT_54	R27_2	153.1 / 5.7	-38.1	MAP_SPAWN_MACRO_27	N018	REGION_ RNING_W ER
MAP_SPAWN_POINT_55	R28_1	-90.8 / 140.1	-54.3	MAP_SPAWN_MACRO_28	N004	REGION_ OKEN_GO
MAP_SPAWN_POINT_56	R28_2	-124.8 / 184.4	128.9	MAP_SPAWN_MACRO_28	N032	REGION_ OKEN_GO
MAP_SPAWN_POINT_57	R29_1	-221.7 / 46.8	-111.8	MAP_SPAWN_MACRO_29	N012	REGION_ DER_GO

微位 ID	标签	坐标 m	朝向°	宏区	出口节点	region_id
MAP_SPAWN_POINT_58	R29_2	-195.5 / 92.3	51.9	MAP_SPAWN_MACRO_29	N039	REGION_ DER_GOF
MAP_SPAWN_POINT_59	R30_1	-71.1 / -168.4	130.4	MAP_SPAWN_MACRO_30	N024	REGION_ ST_SOUL ELD
MAP_SPAWN_POINT_60	R30_2	-66.7 / -164.2	136.9	MAP_SPAWN_MACRO_30	N024	REGION_ ST_SOUL ELD
MAP_SPAWN_POINT_61	R31_1	-211.2 / -111.7	135.8	MAP_SPAWN_MACRO_31	N040	REGION_ ST_SOUL ELD
MAP_SPAWN_POINT_62	R31_2	-210.1 / -145.6	-131.4	MAP_SPAWN_MACRO_31	N022	REGION_ ST_SOUL ELD
MAP_SPAWN_POINT_63	R32_1	159.6 / 151.6	177.3	MAP_SPAWN_MACRO_32	N006	REGION_ AGON_RI E
MAP_SPAWN_POINT_64	R32_2	211.6 / 169.3	41.4	MAP_SPAWN_MACRO_32	N008	REGION_ AGON_RI E
MAP_SPAWN_POINT_65	R33_1	302.9 / -0.8	36.8	MAP_SPAWN_MACRO_33	N020	REGION_ RNING_W ER
MAP_SPAWN_POINT_66	R33_2	295.1 / -47	-56	MAP_SPAWN_MACRO_33	N043	REGION_ RNING_W ER
MAP_SPAWN_POINT_67	R34_1	-69.4 / 82.6	70.6	MAP_SPAWN_MACRO_34	N004	REGION_ OKEN_GO
MAP_SPAWN_POINT_68	R34_2	-83.3 / 33.4	-107.1	MAP_SPAWN_MACRO_34	N014	REGION_ DER_GOF

微位 ID	标签	坐标 m	朝向°	宏区	出口节点	region_id
MAP_SPAWN_POINT_69	R35_1	295.6 / -188.9	-3.9	MAP_SPAWN_MACRO_35	N030	REGION_ NTIPEDE TY
MAP_SPAWN_POINT_70	R35_2	295.6 / -194.9	3.9	MAP_SPAWN_MACRO_35	N030	REGION_ NTIPEDE TY
MAP_SPAWN_POINT_71	R36_1	99.5 / -159.5	-141.9	MAP_SPAWN_MACRO_36	N026	REGION_ NTIPEDE TY
MAP_SPAWN_POINT_72	R36_2	131.3 / -114.2	75.8	MAP_SPAWN_MACRO_36	N063	REGION_ NTIPEDE TY
MAP_SPAWN_POINT_73	R37_1	225.7 / -101	-120.5	MAP_SPAWN_MACRO_37	N028	REGION_ RNING_W ER
MAP_SPAWN_POINT_74	R37_2	256.5 / -55.8	51.9	MAP_SPAWN_MACRO_37	N019	REGION_ RNING_W ER
MAP_SPAWN_POINT_75	R38_1	293.1 / 87	-52.5	MAP_SPAWN_MACRO_38	N020	REGION_ RNING_W ER
MAP_SPAWN_POINT_76	R38_2	247 / 95.4	-146.5	MAP_SPAWN_MACRO_38	N037	REGION_ AGON_RI E
MAP_SPAWN_POINT_77	R39_1	-123.5 / -5.5	135.5	MAP_SPAWN_MACRO_39	N013	REGION_ DER_GOF
MAP_SPAWN_POINT_78	R39_2	-87.7 / 3.9	72.9	MAP_SPAWN_MACRO_39	N035	REGION_ DER_GOF
MAP_SPAWN_POINT_79	R40_1	-72.3 / 234.4	164.4	MAP_SPAWN_MACRO_40	N041	REGION_ OKEN_GO

微位 ID	标签	坐标 m	朝向°	宏区	出口节点	region_id
MAP_SPAWN_POINT_80	R40_2	-71.1 / 240.3	173	MAP_SPAWN_MACRO_40	N041	REGION_1 OKEN_G0

D.7 商店宏区与 48 微锚点

宏区 ID	标签	微点 ID
MAP_SHOP_MACRO_01	S01	MAP_SHOP_POINT:MAP_SHOP_POINT_01; MAP_SHOP_POINT:MAP_SHOP_POINT_02; MAP_SHOP_POINT:MAP_SHOP_POINT_03
MAP_SHOP_MACRO_02	S02	MAP_SHOP_POINT:MAP_SHOP_POINT_04; MAP_SHOP_POINT:MAP_SHOP_POINT_05; MAP_SHOP_POINT:MAP_SHOP_POINT_06
MAP_SHOP_MACRO_03	S03	MAP_SHOP_POINT:MAP_SHOP_POINT_07; MAP_SHOP_POINT:MAP_SHOP_POINT_08; MAP_SHOP_POINT:MAP_SHOP_POINT_09
MAP_SHOP_MACRO_04	S04	MAP_SHOP_POINT:MAP_SHOP_POINT_10; MAP_SHOP_POINT:MAP_SHOP_POINT_11; MAP_SHOP_POINT:MAP_SHOP_POINT_12
MAP_SHOP_MACRO_05	S05	MAP_SHOP_POINT:MAP_SHOP_POINT_13; MAP_SHOP_POINT:MAP_SHOP_POINT_14; MAP_SHOP_POINT:MAP_SHOP_POINT_15
MAP_SHOP_MACRO_06	S06	MAP_SHOP_POINT:MAP_SHOP_POINT_16; MAP_SHOP_POINT:MAP_SHOP_POINT_17; MAP_SHOP_POINT:MAP_SHOP_POINT_18
MAP_SHOP_MACRO_07	S07	MAP_SHOP_POINT:MAP_SHOP_POINT_19; MAP_SHOP_POINT:MAP_SHOP_POINT_20; MAP_SHOP_POINT:MAP_SHOP_POINT_21
MAP_SHOP_MACRO_08	S08	MAP_SHOP_POINT:MAP_SHOP_POINT_22; MAP_SHOP_POINT:MAP_SHOP_POINT_23; MAP_SHOP_POINT:MAP_SHOP_POINT_24
MAP_SHOP_MACRO_09	S09	MAP_SHOP_POINT:MAP_SHOP_POINT_25; MAP_SHOP_POINT:MAP_SHOP_POINT_26; MAP_SHOP_POINT:MAP_SHOP_POINT_27
MAP_SHOP_MACRO_10	S10	MAP_SHOP_POINT:MAP_SHOP_POINT_28; MAP_SHOP_POINT:MAP_SHOP_POINT_29; MAP_SHOP_POINT:MAP_SHOP_POINT_30
MAP_SHOP_MACRO_11	S11	MAP_SHOP_POINT:MAP_SHOP_POINT_31; MAP_SHOP_POINT:MAP_SHOP_POINT_32; MAP_SHOP_POINT:MAP_SHOP_POINT_33
MAP_SHOP_MACRO_12	S12	MAP_SHOP_POINT:MAP_SHOP_POINT_34; MAP_SHOP_POINT:MAP_SHOP_POINT_35; MAP_SHOP_POINT:MAP_SHOP_POINT_36
MAP_SHOP_MACRO_13	S13	MAP_SHOP_POINT:MAP_SHOP_POINT_37; MAP_SHOP_POINT:MAP_SHOP_POINT_38; MAP_SHOP_POINT:MAP_SHOP_POINT_39
MAP_SHOP_MACRO_14	S14	MAP_SHOP_POINT:MAP_SHOP_POINT_40; MAP_SHOP_POINT:MAP_SHOP_POINT_41; MAP_SHOP_POINT:MAP_SHOP_POINT_42
MAP_SHOP_MACRO_15	S15	MAP_SHOP_POINT:MAP_SHOP_POINT_43; MAP_SHOP_POINT:MAP_SHOP_POINT_44; MAP_SHOP_POINT:MAP_SHOP_POINT_45
MAP_SHOP_MACRO_16	S16	MAP_SHOP_POINT:MAP_SHOP_POINT_46; MAP_SHOP_POINT:MAP_SHOP_POINT_47; MAP_SHOP_POINT:MAP_SHOP_POINT_48

微点 ID	标签	坐标 m	宏区	region_id
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微点 ID	标签	坐标 m	宏区	region_id
MAP_SHOP_POINT_01	S01_1	-251 / 212.7	MAP_SHOP_MACRO_01	REGION_01 OKEN_G01
MAP_SHOP_POINT_02	S01_2	-255.2 / 203.6	MAP_SHOP_MACRO_01	REGION_01 DER_G01
MAP_SHOP_POINT_03	S01_3	-255.9 / 221.4	MAP_SHOP_MACRO_01	REGION_01 OKEN_G01
MAP_SHOP_POINT_04	S02_1	-120.5 / 177.5	MAP_SHOP_MACRO_02	REGION_02 OKEN_G02
MAP_SHOP_POINT_05	S02_2	-124.6 / 168.4	MAP_SHOP_MACRO_02	REGION_02 OKEN_G02
MAP_SHOP_POINT_06	S02_3	-125.4 / 186.2	MAP_SHOP_MACRO_02	REGION_02 OKEN_G02
MAP_SHOP_POINT_07	S03_1	95.4 / 202	MAP_SHOP_MACRO_03	REGION_03 AGON_R03 E
MAP_SHOP_POINT_08	S03_2	91.2 / 192.9	MAP_SHOP_MACRO_03	REGION_03 AGON_R03 E
MAP_SHOP_POINT_09	S03_3	90.5 / 210.7	MAP_SHOP_MACRO_03	REGION_03 AGON_R03 E
MAP_SHOP_POINT_10	S04_1	345.9 / 140.2	MAP_SHOP_MACRO_04	REGION_04 AGON_R04 E
MAP_SHOP_POINT_11	S04_2	341.8 / 131.1	MAP_SHOP_MACRO_04	REGION_04 AGON_R04 E
MAP_SHOP_POINT_12	S04_3	341 / 148.9	MAP_SHOP_MACRO_04	REGION_04 AGON_R04 E

微点 ID	标签	坐标 m	宏区	region_id
MAP_SHOP_POINT_13	S05_1	-275 / 6.9	MAP_SHOP_MACRO_05	REGION_13 DER_GOR
MAP_SHOP_POINT_14	S05_2	-279.2 / -2.2	MAP_SHOP_MACRO_05	REGION_14 DER_GOR
MAP_SHOP_POINT_15	S05_3	-279.9 / 15.6	MAP_SHOP_MACRO_05	REGION_15 DER_GOR
MAP_SHOP_POINT_16	S06_1	-123.1 / -57	MAP_SHOP_MACRO_06	REGION_16 DER_GOR
MAP_SHOP_POINT_17	S06_2	-127.3 / -66.1	MAP_SHOP_MACRO_06	REGION_17 DER_GOR
MAP_SHOP_POINT_18	S06_3	-128 / -48.3	MAP_SHOP_MACRO_06	REGION_18 DER_GOR
MAP_SHOP_POINT_19	S07_1	231.3 / -57	MAP_SHOP_MACRO_07	REGION_19 RNING_W ER
MAP_SHOP_POINT_20	S07_2	227.2 / -66.1	MAP_SHOP_MACRO_07	REGION_20 RNING_W ER
MAP_SHOP_POINT_21	S07_3	226.4 / -48.3	MAP_SHOP_MACRO_07	REGION_21 RNING_W ER
MAP_SHOP_POINT_22	S08_1	313.9 / -110.3	MAP_SHOP_MACRO_08	REGION_22 RNING_W ER
MAP_SHOP_POINT_23	S08_2	309.8 / -119.4	MAP_SHOP_MACRO_08	REGION_23 RNING_W ER
MAP_SHOP_POINT_24	S08_3	309 / -101.6	MAP_SHOP_MACRO_08	REGION_24 RNING_W ER

微点 ID	标签	坐标 m	宏区	region_id
MAP_SHOP_POINT_25	S09_1	-245.7 / -153	MAP_SHOP_MACRO_09	REGION_09 ST_SOUTHWEST ELD
MAP_SHOP_POINT_26	S09_2	-249.9 / -162.1	MAP_SHOP_MACRO_09	REGION_09 ST_SOUTHWEST ELD
MAP_SHOP_POINT_27	S09_3	-250.6 / -144.3	MAP_SHOP_MACRO_09	REGION_09 ST_SOUTHWEST ELD
MAP_SHOP_POINT_28	S10_1	-59.2 / -121	MAP_SHOP_MACRO_10	REGION_10 ST_SOUTHWEST ELD
MAP_SHOP_POINT_29	S10_2	-63.3 / -130.1	MAP_SHOP_MACRO_10	REGION_10 ST_SOUTHWEST ELD
MAP_SHOP_POINT_30	S10_3	-64.1 / -112.3	MAP_SHOP_MACRO_10	REGION_10 ST_SOUTHWEST ELD
MAP_SHOP_POINT_31	S11_1	76.8 / -153	MAP_SHOP_MACRO_11	REGION_11 NTIPEDE TY
MAP_SHOP_POINT_32	S11_2	72.6 / -162.1	MAP_SHOP_MACRO_11	REGION_11 NTIPEDE TY
MAP_SHOP_POINT_33	S11_3	71.8 / -144.3	MAP_SHOP_MACRO_11	REGION_11 NTIPEDE TY
MAP_SHOP_POINT_34	S12_1	252.6 / -183.9	MAP_SHOP_MACRO_12	REGION_12 NTIPEDE TY

微点 ID	标签	坐标 m	宏区	region_id
MAP_SHOP_POINT_35	S12_2	248.5 / -193	MAP_SHOP_MACRO_12	REGION_ NTIPEDE TY
MAP_SHOP_POINT_36	S12_3	247.7 / -175.2	MAP_SHOP_MACRO_12	REGION_ NTIPEDE TY
MAP_SHOP_POINT_37	S13_1	-324.1 / 143.9	MAP_SHOP_MACRO_13	REGION_ DER_GOF
MAP_SHOP_POINT_38	S13_2	-328.2 / 134.8	MAP_SHOP_MACRO_13	REGION_ DER_GOF
MAP_SHOP_POINT_39	S13_3	-329 / 152.6	MAP_SHOP_MACRO_13	REGION_ DER_GOF
MAP_SHOP_POINT_40	S14_1	75.7 / 34.6	MAP_SHOP_MACRO_14	REGION_ AGON_RI E
MAP_SHOP_POINT_41	S14_2	71.5 / 25.6	MAP_SHOP_MACRO_14	REGION_ AGON_RI E
MAP_SHOP_POINT_42	S14_3	70.8 / 43.4	MAP_SHOP_MACRO_14	REGION_ AGON_RI E
MAP_SHOP_POINT_43	S15_1	363.5 / 26.7	MAP_SHOP_MACRO_15	REGION_ RNING_W ER
MAP_SHOP_POINT_44	S15_2	359.3 / 17.6	MAP_SHOP_MACRO_15	REGION_ RNING_W ER
MAP_SHOP_POINT_45	S15_3	358.6 / 35.4	MAP_SHOP_MACRO_15	REGION_ RNING_W ER

微点 ID	标签	坐标 m	宏区	region_id
MAP_SHOP_POINT_46	S16_1	-14.9 / -165.2	MAP_SHOP_MACRO_16	REGION_16 ST_SOULDIER ELD
MAP_SHOP_POINT_47	S16_2	-19.1 / -174.3	MAP_SHOP_MACRO_16	REGION_16 ST_SOULDIER ELD
MAP_SHOP_POINT_48	S16_3	-19.8 / -156.5	MAP_SHOP_MACRO_16	REGION_16 ST_SOULDIER ELD

D.8 怪物槽、巢穴与迁移位

槽 ID	源标签	类型	圈层	巢穴	主坐标	3 迁移位	region_id
MAP_MONSTER_SLOT_001	MEL000	MEL	OUTER	MAP_NEST_032	-350.7 / 207.9	-322.6 / 192.5; -386.7 / 186.2; -351.4 / 259.9	REGION_16 DER_GOD
MAP_MONSTER_SLOT_002	MEL001	MEL	OUTER	MAP_NEST_033	339.5 / -191.9	367.6 / -207.2; 303.5 / -213.5; 338.9 / -139.9	REGION_16 RNING_W ER
MAP_MONSTER_SLOT_003	MEL002	MEL	OUTER	MAP_NEST_034	-150.8 / -234.5	-122.7 / -249.9; -186.8 / -256.2; -151.5 / -182.5	REGION_16 ST_SOULDIER ELD
MAP_MONSTER_SLOT_004	MEL003	MEL	MIDDLE	MAP_NEST_035	97 / 154.6	125.1 / 139.2; 61 / 132.9; 96.4 / 206.6	REGION_16 AGON_R E
MAP_MONSTER_SLOT_005	MEL004	MEL	MIDDLE	MAP_NEST_036	-153.5 / 24	-125.4 / 8.6; -189.5 / 2.3; -154.1 / 76	REGION_16 DER_GOD
MAP_MONSTER_SLOT_006	MEL005	MEL	OUTER	MAP_NEST_037	115.7 / -303.8	143.8 / -319.2; 79.7 / -325.4; 115 / -251.8	REGION_16 NTIPEDE TY

槽 ID	源标签	类型	圈层	巢穴	主坐标	3 迁移位	region_id
MAP_MONSTER_SLOT_007	MEL006	MEL	OUTER	MAP_NEST_038	336.9 / 202.5	364.9 / 187.2; 300.9 / 180.9; 336.2 / 254.6	REGION_AGON_RIVER
MAP_MONSTER_SLOT_008	MEL007	MEL	OUTER	MAP_NEST_039	-110.9 / 245.2	-82.8 / 229.8; -146.8 / 223.5; -111.5 / 297.2	REGION_TOKEN_GOLD
MAP_MONSTER_SLOT_009	MEL008	MEL	MIDDLE	MAP_NEST_040	150.3 / -58.6	178.4 / -74; 114.3 / -80.3; 149.7 / -6.6	REGION_BURNING_WOODS
MAP_MONSTER_SLOT_010	MEL009	MEL	OUTER	MAP_NEST_041	-361.4 / -29.3	-333.3 / -44.7; -397.4 / -51; -362 / 22.7	REGION_DERGON
MAP_MONSTER_SLOT_011	MEL010	MEL	OUTER	MAP_NEST_042	339.5 / 26.7	367.6 / 11.3; 303.5 / 5; 338.9 / 78.7	REGION_BURNING_WOODS
MAP_MONSTER_SLOT_012	MEL011	MEL	INNER	MAP_NEST_043	-12.3 / -58.6	15.8 / -74; -48.2 / -80.3; -12.9 / -6.6	REGION_ST_SOULFIELD
MAP_MONSTER_SLOT_013	MEL012	MEL	OUTER	MAP_NEST_044	-332.1 / -181.2	-304 / -196.6; -368 / -202.9; -332.7 / -129.2	REGION_ST_SOULFIELD
MAP_MONSTER_SLOT_014	MEL013	MEL	MIDDLE	MAP_NEST_045	211.6 / 72	239.7 / 56.6; 175.6 / 50.3; 211 / 124	REGION_BURNING_WOODS
MAP_MONSTER_SLOT_015	MEL014	MEL	OUTER	MAP_NEST_046	-212.1 / 159.9	-184 / 144.5; -248.1 / 138.3; -212.8 / 211.9	REGION_DERGON
MAP_MONSTER_SLOT_016	MEL015	MEL	MIDDLE	MAP_NEST_047	65 / -186.6	93.1 / -201.9; 29 / -208.2; 64.4 / -134.5	REGION_INTIPEDE
MAP_MONSTER_SLOT_017	MEL016	MEL	INNER	MAP_NEST_048	-16.1 / 105.6	12 / 90.3; -52.1 / 84; -16.7 / 157.7	REGION_TOKEN_GOLD

槽 ID	源标签	类型	圈层	巢穴	主坐标	3 迁移位	region_id
MAP_MONSTER_SLOT_018	MEL017	MEL	MIDDLE	MAP_NEST_049	-105.5 / -127.9	-77.4 / -143.3; -141.5 / -149.6; -106.2 / -75.9	REGION_018 ST_SOUL ELD
MAP_MONSTER_SLOT_019	MEL018	MEL	OUTER	MAP_NEST_050	-254.8 / -69.3	-226.7 / -84.6; -290.8 / -90.9; -255.4 / -17.3	REGION_019 DER_GOF
MAP_MONSTER_SLOT_020	MEL019	MEL	OUTER	MAP_NEST_051	320.9 / -85.3	349 / -100.6; 284.9 / -106.9; 320.2 / -33.3	REGION_020 RNING_W ER
MAP_MONSTER_SLOT_021	MEL020	MEL	OUTER	MAP_NEST_052	-316.1 / 58.6	-288 / 43.3; -352 / 37; -316.7 / 110.7	REGION_021 DER_GOF
MAP_MONSTER_SLOT_022	MEL021	MEL	MIDDLE	MAP_NEST_053	203.6 / -138.6	231.7 / -153.9; 167.6 / -160.2; 203 / -86.6	REGION_022 NTIPEDE TY
MAP_MONSTER_SLOT_023	MEL022	MEL	INNER	MAP_NEST_054	33 / 26.7	61.1 / 11.3; -2.9 / 5; 32.4 / 78.7	REGION_023 AGON_RI E
MAP_MONSTER_SLOT_024	MEL023	MEL	OUTER	MAP_NEST_055	246.2 / 199.9	274.3 / 184.5; 210.3 / 178.2; 245.6 / 251.9	REGION_024 AGON_RI E
MAP_MONSTER_SLOT_025	MEL024	MEL	OUTER	MAP_NEST_056	-241.4 / -181.2	-213.4 / -196.6; -277.4 / -202.9; -242.1 / -129.2	REGION_025 ST_SOUL ELD
MAP_MONSTER_SLOT_026	MEL025	MEL	OUTER	MAP_NEST_057	-110.9 / -314.5	-82.8 / -329.8; -146.8 / -336.1; -111.5 / -262.4	REGION_026 ST_SOUL ELD
MAP_MONSTER_SLOT_027	MEL026	MEL	MIDDLE	MAP_NEST_058	134.3 / 26.7	162.4 / 11.3; 98.3 / 5; 133.7 / 78.7	REGION_027 RNING_W ER
MAP_MONSTER_SLOT_028	MEL027	MEL	MIDDLE	MAP_NEST_059	-110.9 / 159.9	-82.8 / 144.5; -146.8 / 138.3; -111.5 / 211.9	REGION_028 OKEN_GO

槽 ID	源标签	类型	圈层	巢穴	主坐标	3 迁移位	region_id
MAP_MONSTER_SLOT_029	MEL028	MEL	MIDDLE	MAP_NEST_060	-217.5 / 74.6	-189.4 / 59.3; -253.4 / 53; -218.1 / 126.6	REGION_029 DER_GORGE
MAP_MONSTER_SLOT_030	MEL029	MEL	MIDDLE	MAP_NEST_061	-49.6 / -186.6	-21.5 / -201.9; -85.5 / -208.2; -50.2 / -134.5	REGION_030 ST_SOUL ELD
MAP_MONSTER_SLOT_031	MEL030	MEL	MIDDLE	MAP_NEST_062	-188.1 / -127.9	-160.1 / -143.3; -224.1 / -149.6; -188.8 / -75.9	REGION_031 ST_SOUL ELD
MAP_MONSTER_SLOT_032	MEL031	MEL	MIDDLE	MAP_NEST_063	187.6 / 154.6	215.7 / 139.2; 151.6 / 132.9; 187 / 206.6	REGION_032 AGON_RIVER
MAP_MONSTER_SLOT_033	MEL032	MEL	OUTER	MAP_NEST_064	283.6 / -21.3	311.6 / -36.7; 247.6 / -43; 282.9 / 30.7	REGION_033 RNING_W ER
MAP_MONSTER_SLOT_034	MEL033	MEL	INNER	MAP_NEST_065	-70.9 / 58.6	-42.8 / 43.3; -106.9 / 37; -71.5 / 110.7	REGION_034 OKEN_GORGE
MAP_MONSTER_SLOT_035	MEL034	MEL	OUTER	MAP_NEST_066	267.6 / -191.9	295.7 / -207.2; 231.6 / -213.5; 266.9 / -139.9	REGION_035 NTIPEDE TY
MAP_MONSTER_SLOT_036	MEL035	MEL	MIDDLE	MAP_NEST_067	118.3 / -138.6	146.4 / -153.9; 82.3 / -160.2; 117.7 / -86.6	REGION_036 NTIPEDE TY
MAP_MONSTER_SLOT_037	MEL036	MEL	MIDDLE	MAP_NEST_068	235.6 / -74.6	263.7 / -90; 199.6 / -96.3; 234.9 / -22.6	REGION_037 RNING_W ER
MAP_MONSTER_SLOT_038	MEL037	MEL	OUTER	MAP_NEST_069	272.9 / 106.6	301 / 91.2; 236.9 / 85; 272.3 / 158.6	REGION_038 AGON_RIVER
MAP_MONSTER_SLOT_039	MEL038	MEL	INNER	MAP_NEST_070	-100.2 / -21.3	-72.1 / -36.7; -136.2 / -43; -100.8 / 30.7	REGION_039 DER_GORGE

槽 ID	源标签	类型	圈层	巢穴	主坐标	3 迁移位	region_id
MAP_MONSTER_SLOT_040	MEL039	MEL	MIDDLE	MAP_NEST_071	-44.2 / 231.9	-16.1 / 216.5; -80.2 / 210.2; -44.9 / 283.9	REGION_MONSTER_SLOT_040
MAP_MONSTER_SLOT_041	MEL040	MEL	MIDDLE	MAP_NEST_001	-125.8 / -66.7	-156.1 / -90.3; -155 / -58.3; -140.9 / -86.8	REGION_MONSTER_SLOT_041
MAP_MONSTER_SLOT_042	MEL041	MEL	MIDDLE	MAP_NEST_002	-164.4 / 117.6	-107.9 / 119.3; -150.6 / 158.6; -193.7 / 140	REGION_MONSTER_SLOT_042
MAP_MONSTER_SLOT_043	MEL042	MEL	MIDDLE	MAP_NEST_002	-165.1 / 112.7	-192.1 / 151.3; -200.9 / 138; -173.1 / 146	REGION_MONSTER_SLOT_043
MAP_MONSTER_SLOT_044	MEL043	MEL	MIDDLE	MAP_NEST_002	-150.4 / 104.7	-181.2 / 80.4; -191.8 / 108; -190.5 / 142.7	REGION_MONSTER_SLOT_044
MAP_MONSTER_SLOT_045	MEL044	MEL	MIDDLE	MAP_NEST_003	-231.1 / 102.5	-271.4 / 121.8; -269.9 / 103.3; -261.9 / 76.1	REGION_MONSTER_SLOT_045
MAP_MONSTER_SLOT_046	MEL045	MEL	MIDDLE	MAP_NEST_003	-236.2 / 100.7	-188.6 / 136.9; -246 / 155.6; -264.5 / 117.2	REGION_MONSTER_SLOT_046
MAP_MONSTER_SLOT_047	MEL046	MEL	MIDDLE	MAP_NEST_003	-242.7 / 105.7	-178.8 / 121.9; -244.7 / 75.1; -248.2 / 86.9	REGION_MONSTER_SLOT_047
MAP_MONSTER_SLOT_048	MEL047	MEL	MIDDLE	MAP_NEST_004	-220 / 15.9	-192.3 / 33.6; -195.1 / -16.3; -261 / 20.4	REGION_MONSTER_SLOT_048
MAP_MONSTER_SLOT_049	MEL048	MEL	INNER	MAP_NEST_005	-60.8 / -19	-62 / 19.8; -75.6 / 16.1; -68.4 / -53.8	REGION_MONSTER_SLOT_049
MAP_MONSTER_SLOT_050	MEL049	MEL	INNER	MAP_NEST_005	-48.6 / -19.3	-10.8 / -43.7; -46.2 / 14; -69.5 / 25.5	REGION_MONSTER_SLOT_050
MAP_MONSTER_SLOT_051	MEL050	MEL	INNER	MAP_NEST_006	75.5 / -28.6	114.9 / -52.3; 68.2 / 10; 70.5 / 23.2	REGION_MONSTER_SLOT_051
MAP_MONSTER_SLOT_052	MEL051	MEL	INNER	MAP_NEST_006	80.6 / -24.9	54.5 / 15.6; 66.4 / -54; 46.5 / -13.5	REGION_MONSTER_SLOT_052

槽 ID	源标签	类型	圈层	巢穴	主坐标	3 迁移位	region_id
MAP_MONSTER_SLOT_053	MEL052	MEL	INNER	MAP_NEST_006	79.5 / -12.7	106.7 / -62.6; 96.6 / 22.6; 82.3 / -64.7	REGION_001 RNING_WORLD
MAP_MONSTER_SLOT_054	MEL053	MEL	INNER	MAP_NEST_007	-9.9 / 50.5	5.8 / 7.1; -37.9 / 4.9; -55.9 / 52.7	REGION_001 OKEN_GOLD
MAP_MONSTER_SLOT_055	MEL054	MEL	INNER	MAP_NEST_007	-31.4 / 37.4	-52.9 / 14.1; -48.9 / 48.9; -63.3 / 43.1	REGION_001 OKEN_GOLD
MAP_MONSTER_SLOT_056	MEL055	MEL	INNER	MAP_NEST_008	76.2 / 44.9	76.8 / -12.5; 114.9 / 32; 41.9 / 65.4	REGION_001 AGON_RIVER
MAP_MONSTER_SLOT_057	MEL056	MEL	INNER	MAP_NEST_008	76.6 / 48.1	112.6 / 19.4; 41.2 / 25.8; 69 / -19.8	REGION_001 AGON_RIVER
MAP_MONSTER_SLOT_058	MEL057	MEL	INNER	MAP_NEST_008	73.7 / 43.7	79.3 / 4.4; 59.9 / 6.8; 112.1 / 24.5	REGION_001 AGON_RIVER
MAP_MONSTER_SLOT_059	RNG000	RNG	OUTER	MAP_NEST_009	311.3 / 152.8	307.1 / 195.7; 298.8 / 194; 275 / 149	REGION_001 AGON_RIVER
MAP_MONSTER_SLOT_060	RNG001	RNG	OUTER	MAP_NEST_009	302.2 / 167.2	336.5 / 184.6; 313.6 / 198.9; 313 / 191.6	REGION_001 AGON_RIVER
MAP_MONSTER_SLOT_061	RNG002	RNG	OUTER	MAP_NEST_010	209.4 / 185.6	174.8 / 192.7; 186.3 / 147.8; 260.9 / 174.9	REGION_001 AGON_RIVER
MAP_MONSTER_SLOT_062	RNG003	RNG	OUTER	MAP_NEST_010	205.5 / 176.5	253.8 / 187; 186.9 / 138.6; 251.3 / 158.5	REGION_001 AGON_RIVER

槽 ID	源标签	类型	圈层	巢穴	主坐标	3 迁移位	region_id
MAP_MONSTER_SLOT_063	RNG004	RNG	OUTER	MAP_NEST_010	204.8 / 178.5	227.1 / 226.8; 194.9 / 156.3; 257 / 177.5	REGION_AGON_RIE
MAP_MONSTER_SLOT_064	RNG005	RNG	OUTER	MAP_NEST_011	223 / -167.5	254.2 / -115.4; 181.8 / -166.8; 185.6 / -150	REGION_NTIPEDET
MAP_MONSTER_SLOT_065	RNG006	RNG	OUTER	MAP_NEST_011	247.6 / -159.1	183.3 / -159.6; 253.9 / -118.7; 284.9 / -172.7	REGION_NTIPEDET
MAP_MONSTER_SLOT_066	RNG007	RNG	OUTER	MAP_NEST_011	236.4 / -171.4	215.5 / -191.4; 285.5 / -154.4; 196 / -179.1	REGION_NTIPEDET
MAP_MONSTER_SLOT_067	RNG008	RNG	OUTER	MAP_NEST_012	265.7 / 35.6	320.5 / 43.8; 252.7 / -9.6; 239.9 / 57	REGION_RNING_WER
MAP_MONSTER_SLOT_068	RNG009	RNG	OUTER	MAP_NEST_012	265.1 / 35.9	299.9 / 61.6; 304.5 / 54.2; 234.6 / 65.3	REGION_RNING_WER
MAP_MONSTER_SLOT_069	RNG010	RNG	OUTER	MAP_NEST_012	285.8 / 33.2	250.1 / 76.5; 231.5 / 16.3; 277.7 / 69.7	REGION_RNING_WER
MAP_MONSTER_SLOT_070	RNG011	RNG	OUTER	MAP_NEST_013	-252.6 / 77.8	-215.8 / 49.8; -292.4 / 46.4; -307.1 / 78.3	REGION_DER_GOF
MAP_MONSTER_SLOT_071	RNG012	RNG	OUTER	MAP_NEST_013	-257.3 / 58.6	-218.9 / 88.6; -289.3 / 66; -307.6 / 86.6	REGION_DER_GOF
MAP_MONSTER_SLOT_072	RNG013	RNG	OUTER	MAP_NEST_013	-266.7 / 67.4	-215.9 / 51; -296.1 / 71.3; -289.5 / 70.5	REGION_DER_GOF
MAP_MONSTER_SLOT_073	RNG014	RNG	OUTER	MAP_NEST_014	346.4 / 147.5	347.6 / 173.7; 311 / 99.8; 341.3 / 176	REGION_AGON_RIE

槽 ID	源标签	类型	圈层	巢穴	主坐标	3 迁移位	region_id
MAP_MONSTER_SLOT_074	RNG015	RNG	OUTER	MAP_NEST_014	341.9 / 130.7	350.1 / 186.3; 303.1 / 137.3; 308.3 / 129	REGION_014 AGON_RING E
MAP_MONSTER_SLOT_075	RNG016	RNG	OUTER	MAP_NEST_015	270.1 / 147.2	281.4 / 129; 255.5 / 103.6; 209.1 / 126	REGION_015 AGON_RING E
MAP_MONSTER_SLOT_076	RNG017	RNG	OUTER	MAP_NEST_015	252.9 / 132.9	280.7 / 125; 247.1 / 99.1; 227.4 / 188.5	REGION_015 AGON_RING E
MAP_MONSTER_SLOT_077	RNG018	RNG	OUTER	MAP_NEST_016	257.1 / 146.4	224.7 / 98.1; 250.2 / 180.6; 198.3 / 132.8	REGION_016 AGON_RING E
MAP_MONSTER_SLOT_078	RNG019	RNG	OUTER	MAP_NEST_016	247.4 / 133.7	270.6 / 191; 247.7 / 90.9; 235.4 / 100.3	REGION_016 AGON_RING E
MAP_MONSTER_SLOT_079	RNG020	RNG	MIDDLE	MAP_NEST_017	-84.4 / -158.1	-24.6 / -154.8; -87 / -192.1; -64.8 / -128.9	REGION_017 ST_SOULE ELD
MAP_MONSTER_SLOT_080	RNG021	RNG	MIDDLE	MAP_NEST_017	-81.8 / -161.5	-122.8 / -172.2; -101.5 / -126.3; -96.6 / -127.1	REGION_017 ST_SOULE ELD
MAP_MONSTER_SLOT_081	RNG022	RNG	MIDDLE	MAP_NEST_017	-70.5 / -153.8	-119.4 / -141.8; -26.9 / -136.6; -39.1 / -148.5	REGION_017 ST_SOULE ELD
MAP_MONSTER_SLOT_082	RNG023	RNG	MIDDLE	MAP_NEST_018	-95.9 / -172.5	-84.1 / -202.9; -80 / -213.8; -126 / -211.1	REGION_018 ST_SOULE ELD
MAP_MONSTER_SLOT_083	RNG024	RNG	MIDDLE	MAP_NEST_018	-101.5 / -172.5	-118.4 / -122.3; -79 / -209.6; -154.6 / -170.8	REGION_018 ST_SOULE ELD

槽 ID	源标签	类型	圈层	巢穴	主坐标	3 迁移位	region_id
MAP_MONSTER_SLOT_084	RNG025	RNG	MIDDLE	MAP_NEST_018	-92 / -161.5	-148.1 / -181.3; -148.6 / -149.7; -75.7 / -202.2	REGION_01 ST_SOUL ELD
MAP_MONSTER_SLOT_085	RNG026	RNG	MIDDLE	MAP_NEST_019	136.8 / -24.6	96 / 5.6; 166.6 / 12; 169.4 / 19	REGION_01 RNING_W ER
MAP_MONSTER_SLOT_086	RNG027	RNG	MIDDLE	MAP_NEST_019	143.8 / -11	87.7 / -48.8; 114 / -47; 119.4 / 15.9	REGION_01 RNING_W ER
MAP_MONSTER_SLOT_087	RNG028	RNG	MIDDLE	MAP_NEST_019	125.6 / -5.4	100.2 / -24.7; 134.6 / -45.3; 77.7 / -20.6	REGION_01 RNING_W ER
MAP_MONSTER_SLOT_088	RNG029	RNG	MIDDLE	MAP_NEST_020	245.8 / -16	197.7 / -20.6; 203.1 / -64; 274.9 / 2.7	REGION_01 RNING_W ER
MAP_MONSTER_SLOT_089	RNG030	RNG	MIDDLE	MAP_NEST_020	226.7 / -33.4	279.9 / -11.7; 284.1 / 4.4; 196.6 / -40.3	REGION_01 RNING_W ER
MAP_MONSTER_SLOT_090	RNG031	RNG	MIDDLE	MAP_NEST_020	224.4 / -23.8	217.5 / -53.7; 207.9 / -53.1; 183.8 / -30.4	REGION_01 RNING_W ER
MAP_MONSTER_SLOT_091	RNG032	RNG	INNER	MAP_NEST_021	-41.8 / -116.8	-3 / -133.7; -1.9 / -102.8; -84.6 / -109.4	REGION_01 ST_SOUL ELD
MAP_MONSTER_SLOT_092	RNG033	RNG	INNER	MAP_NEST_021	-36 / -114.9	-67.1 / -81.2; -60.2 / -167.2; -73.2 / -156.2	REGION_01 ST_SOUL ELD
MAP_MONSTER_SLOT_093	RNG034	RNG	INNER	MAP_NEST_021	-55 / -132.1	-8.4 / -117.4; -61.4 / -166.2; -96.3 / -135.1	REGION_01 ST_SOUL ELD

槽 ID	源标签	类型	圈层	巢穴	主坐标	3 迁移位	region_id
MAP_MONSTER_SLOT_094	RNG035	RNG	INNER	MAP_NEST_022	84.7 / 44.5	87.1 / 5.9; 42.5 / 31.7; 101.5 / -4.4	REGION_001 AGON_RIVER E
MAP_MONSTER_SLOT_095	RNG036	RNG	INNER	MAP_NEST_022	85.8 / 43.4	41.1 / 37.7; 67.7 / 66.2; 39.5 / 30.6	REGION_001 AGON_RIVER E
MAP_MONSTER_SLOT_096	RNG037	RNG	INNER	MAP_NEST_022	65.8 / 25.4	44.4 / 27.9; 79 / -8.6; 119.9 / 61.8	REGION_001 AGON_RIVER E
MAP_MONSTER_SLOT_097	FLY000	FLY	MIDDLE	MAP_NEST_023	2.1 / 202.4	-8.1 / 181.9; -49 / 218.3; -4.7 / 168.9	REGION_002 OKEN_GOLDEN G
MAP_MONSTER_SLOT_098	FLY001	FLY	MIDDLE	MAP_NEST_024	-170.3 / 167.1	-208.1 / 180.6; -151.2 / 131.1; -118.2 / 147.6	REGION_002 OKEN_GOLDEN G
MAP_MONSTER_SLOT_099	FLY002	FLY	MIDDLE	MAP_NEST_025	-123.5 / -44.6	-138.4 / -16.1; -121.6 / -7; -86.6 / -33.4	REGION_002 DER_GOLDEN G
MAP_MONSTER_SLOT_100	FLY003	FLY	MIDDLE	MAP_NEST_025	-127.5 / -63.6	-159.9 / -76.9; -158.5 / -63.5; -99.7 / -12	REGION_002 DER_GOLDEN G
MAP_MONSTER_SLOT_101	FLY004	FLY	MIDDLE	MAP_NEST_026	131.4 / -99.6	88.4 / -94.2; 80.6 / -105.2; 89.2 / -99.8	REGION_003 NTIPEDE TY
MAP_MONSTER_SLOT_102	FLY005	FLY	MIDDLE	MAP_NEST_026	109.4 / -93	89.9 / -112.6; 138.6 / -123; 147.4 / -119.1	REGION_003 NTIPEDE TY
MAP_MONSTER_SLOT_103	FLY006	FLY	MIDDLE	MAP_NEST_027	-114 / 192.2	-87.6 / 251.2; -67.6 / 211.4; -91.7 / 244	REGION_003 OKEN_GOLDEN G
MAP_MONSTER_SLOT_104	FLY007	FLY	MIDDLE	MAP_NEST_028	-39.3 / -134.3	-27 / -169; -78.4 / -168.1; -28.1 / -181.6	REGION_003 ST_SOUL ELD
MAP_MONSTER_SLOT_105	FLY008	FLY	OUTER	MAP_NEST_029	-275.2 / -36.8	-266.6 / -62.2; -282.4 / -78.5; -315.7 / -54	REGION_003 DER_GOLDEN G

槽 ID	源标签	类型	圈层	巢穴	主坐标	3 迁移位	region_id
MAP_MONSTER_SLOT_106	FLY009	FLY	OUTER	MAP_NEST_030	293.8 / -64.8	255.9 / -53.4; 327.9 / -22.2; 331.9 / -45	REGION_001 RNING_W ER
MAP_MONSTER_SLOT_107	FLY010	FLY	OUTER	MAP_NEST_030	292.3 / -48.9	335.1 / -55.7; 343.6 / -89.2; 311.7 / -84.1	REGION_001 RNING_W ER
MAP_MONSTER_SLOT_108	FLY011	FLY	OUTER	MAP_NEST_031	-266.3 / 0.5	-279.6 / -41.2; -227.3 / 13.3; -292.9 / -37.3	REGION_001 DER_GOP

巢穴 ID	标签	类型	圈层	坐标	槽	起步区
MAP_NEST_001	MEL_中_017	MEL	中	-123.1 / -57	MEL040	否
MAP_NEST_002	MEL_中_018	MEL	中	-158.8 / 111.9	MEL041 / MEL042 / MEL043	否
MAP_NEST_003	MEL_中_019	MEL	中	-230.8 / 111.9	MEL044 / MEL045 / MEL046	否
MAP_NEST_004	MEL_中_020	MEL	中	-230.8 / 24	MEL047	否
MAP_NEST_005	MEL_内_021	MEL	内	-54.9 / -21.3	MEL048 / MEL049	否
MAP_NEST_006	MEL_内_022	MEL	内	75.7 / -21.3	MEL050 / MEL051 / MEL052	否
MAP_NEST_007	MEL_内_023	MEL	内	-18.9 / 42.6	MEL053 / MEL054	否
MAP_NEST_008	MEL_内_024	MEL	内	75.7 / 34.6	MEL055 / MEL056 / MEL057	否
MAP_NEST_009	RNG_外_025	RNG	外	304.9 / 154.6	RNG000 / RNG001	否
MAP_NEST_010	RNG_外_026	RNG	外	216.9 / 177.2	RNG002 / RNG003 / RNG004	否
MAP_NEST_011	RNG_外_027	RNG	外	235.6 / -165.2	RNG005 / RNG006 / RNG007	否
MAP_NEST_012	RNG_外_028	RNG	外	274.2 / 34	RNG008 / RNG009 / RNG010	否
MAP_NEST_013	RNG_外_029	RNG	外	-256.1 / 68	RNG011 / RNG012 / RNG013	否
MAP_NEST_014	RNG_外_030	RNG	外	345.9 / 140.2	RNG014 / RNG015	否
MAP_NEST_015	RNG_外_031	RNG	外	255.6 / 144.5	RNG016 / RNG017	否
MAP_NEST_016	RNG_外_032	RNG	外	248.9 / 143.9	RNG018 / RNG019	否
MAP_NEST_017	RNG_中_033	RNG	中	-77.6 / -157.2	RNG020 / RNG021 / RNG022	否
MAP_NEST_018	RNG_中_034	RNG	中	-100.2 / -165.2	RNG023 / RNG024 / RNG025	否
MAP_NEST_019	RNG_中_035	RNG	中	130.3 / -14.7	RNG026 / RNG027 / RNG028	否
MAP_NEST_020	RNG_中_036	RNG	中	235.6 / -21.3	RNG029 / RNG030 / RNG031	否
MAP_NEST_021	RNG_内_037	RNG	内	-49.6 / -119.9	RNG032 / RNG033 / RNG034	否
MAP_NEST_022	RNG_内_038	RNG	内	75.7 / 34.6	RNG035 / RNG036 / RNG037	否
MAP_NEST_023	FLY_中_039	FLY	中	2.4 / 211.9	FLY000	否
MAP_NEST_024	FLY_中_040	FLY	中	-161.5 / 159.9	FLY001	否
MAP_NEST_025	FLY_中_041	FLY	中	-123.1 / -57	FLY002 / FLY003	否
MAP_NEST_026	FLY_中_042	FLY	中	122.3 / -97.3	FLY004 / FLY005	否
MAP_NEST_027	FLY_中_043	FLY	中	-110.9 / 202.5	FLY006	否
MAP_NEST_028	FLY_中_044	FLY	中	-49.6 / -133.2	FLY007	否

巢穴 ID	标签	类型	圈层	坐标	槽	起步区
MAP_NEST_029	FLY_外_045	FLY	外	-281.4 / -29.3	FLY008	否
MAP_NEST_030	FLY_外_046	FLY	外	302.2 / -53.3	FLY009 / FLY010	否
MAP_NEST_031	FLY_外_047	FLY	外	-275 / 6.9	FLY011	否
MAP_NEST_032	START_R01	MEL	外	-350.7 / 207.9	MEL000	R01
MAP_NEST_033	START_R02	MEL	外	339.5 / -191.9	MEL001	R02
MAP_NEST_034	START_R03	MEL	外	-150.8 / -234.5	MEL002	R03
MAP_NEST_035	START_R04	MEL	中	97 / 154.6	MEL003	R04
MAP_NEST_036	START_R05	MEL	中	-153.5 / 24	MEL004	R05
MAP_NEST_037	START_R06	MEL	外	115.7 / -303.8	MEL005	R06
MAP_NEST_038	START_R07	MEL	外	336.9 / 202.5	MEL006	R07
MAP_NEST_039	START_R08	MEL	外	-110.9 / 245.2	MEL007	R08
MAP_NEST_040	START_R09	MEL	中	150.3 / -58.6	MEL008	R09
MAP_NEST_041	START_R10	MEL	外	-361.4 / -29.3	MEL009	R10
MAP_NEST_042	START_R11	MEL	外	339.5 / 26.7	MEL010	R11
MAP_NEST_043	START_R12	MEL	内	-12.3 / -58.6	MEL011	R12
MAP_NEST_044	START_R13	MEL	外	-332.1 / -181.2	MEL012	R13
MAP_NEST_045	START_R14	MEL	中	211.6 / 72	MEL013	R14
MAP_NEST_046	START_R15	MEL	外	-212.1 / 159.9	MEL014	R15
MAP_NEST_047	START_R16	MEL	中	65 / -186.6	MEL015	R16
MAP_NEST_048	START_R17	MEL	内	-16.1 / 105.6	MEL016	R17
MAP_NEST_049	START_R18	MEL	中	-105.5 / -127.9	MEL017	R18
MAP_NEST_050	START_R19	MEL	外	-254.8 / -69.3	MEL018	R19
MAP_NEST_051	START_R20	MEL	外	320.9 / -85.3	MEL019	R20
MAP_NEST_052	START_R21	MEL	外	-316.1 / 58.6	MEL020	R21
MAP_NEST_053	START_R22	MEL	中	203.6 / -138.6	MEL021	R22
MAP_NEST_054	START_R23	MEL	内	33 / 26.7	MEL022	R23
MAP_NEST_055	START_R24	MEL	外	246.2 / 199.9	MEL023	R24
MAP_NEST_056	START_R25	MEL	外	-241.4 / -181.2	MEL024	R25
MAP_NEST_057	START_R26	MEL	外	-110.9 / -314.5	MEL025	R26
MAP_NEST_058	START_R27	MEL	中	134.3 / 26.7	MEL026	R27
MAP_NEST_059	START_R28	MEL	中	-110.9 / 159.9	MEL027	R28
MAP_NEST_060	START_R29	MEL	中	-217.5 / 74.6	MEL028	R29
MAP_NEST_061	START_R30	MEL	中	-49.6 / -186.6	MEL029	R30
MAP_NEST_062	START_R31	MEL	中	-188.1 / -127.9	MEL030	R31
MAP_NEST_063	START_R32	MEL	中	187.6 / 154.6	MEL031	R32
MAP_NEST_064	START_R33	MEL	外	283.6 / -21.3	MEL032	R33
MAP_NEST_065	START_R34	MEL	内	-70.9 / 58.6	MEL033	R34
MAP_NEST_066	START_R35	MEL	外	267.6 / -191.9	MEL034	R35
MAP_NEST_067	START_R36	MEL	中	118.3 / -138.6	MEL035	R36
MAP_NEST_068	START_R37	MEL	中	235.6 / -74.6	MEL036	R37
MAP_NEST_069	START_R38	MEL	外	272.9 / 106.6	MEL037	R38

巢穴 ID	标签	类型	圈层	坐标	槽	起步区
MAP_NEST_070	START_R39	MEL	内	-100.2 / -21.3	MEL038	R39
MAP_NEST_071	START_R40	MEL	中	-44.2 / 231.9	MEL039	R40

联动 A	联动 B
MAP_NEST_001	MAP_NEST_025
MAP_NEST_008	MAP_NEST_022
MAP_NEST_015	MAP_NEST_016
MAP_NEST_017	MAP_NEST_018
MAP_NEST_021	MAP_NEST_028

D.9 猪窝、龙宫、首领场、伏石、关口与宝箱

猪窝 ID	入口	初始	袋底	宽	长	逃跑路径	region_id
MAP_PIG_DEN_01	-350.7 / 207.9	-334.9 / 203.2	-305.4 / 194.5	5	47.2	MAP_NODE:N000; MAP_NODE:N064	REGION_01 DER_GORGE
MAP_PIG_DEN_02	-217.5 / 74.6	-207.2 / 65.3	-188.1 / 48	5	39.6	MAP_NODE:N048; MAP_NODE:N065	REGION_02 DER_GORGE
MAP_PIG_DEN_03	-44.2 / 231.9	-46.1 / 223.5	-49.6 / 207.9	5	24.6	MAP_NODE:N033; MAP_NODE:N066	REGION_03 OKEN_GORGE
MAP_PIG_DEN_04	272.9 / 106.6	264.5 / 119.7	248.9 / 143.9	5	44.4	MAP_NODE:N042; MAP_NODE:N067	REGION_04 AGON_RIVER E
MAP_PIG_DEN_05	283.6 / -21.3	296.6 / -26.9	320.9 / -37.3	5	40.6	MAP_NODE:N019; MAP_NODE:N068	REGION_05 RNING_W ER
MAP_PIG_DEN_06	-332.1 / -181.2	-321.8 / -185.9	-302.7 / -194.5	5	32.2	MAP_NODE:N021; MAP_NODE:N069	REGION_06 ST_SOUL ELD
MAP_PIG_DEN_07	-150.8 / -234.5	-149 / -245.7	-145.5 / -266.5	5	32.4	MAP_NODE:N044; MAP_NODE:N070	REGION_07 ST_SOUL ELD

猪窝 ID	入口	初始	袋底	宽	长	逃跑路径	region_id
MAP_PIG_DEN_08	115.7 / -303.8	119.4 / -287	126.3 / -255.8	5	49.1	MAP_NODE:N047; MAP_NODE:N071	REGION_08 NTIPEDE TY
MAP_PIG_DEN_09	267.6 / -191.9	279.7 / -197.5	302.2 / -207.9	5	38.2	MAP_NODE:N029; MAP_NODE:N072	REGION_09 NTIPEDE TY
MAP_PIG_DEN_10	-316.1 / 58.6	-329.1 / 58.6	-353.4 / 58.6	5	37.3	MAP_NODE:N038; MAP_NODE:N073	REGION_10 DER_GOF
MAP_PIG_DEN_11	339.5 / 26.7	349.8 / 41.6	368.8 / 69.3	5	51.7	MAP_NODE:N020; MAP_NODE:N074	REGION_11 RNING_W ER
MAP_PIG_DEN_12	65 / -274.5	60.4 / -287.6	51.7 / -311.8	5	39.6	MAP_NODE:N046; MAP_NODE:N075	REGION_12 NTIPEDE TY

龙宫 ID	标签	坐标	region_id
MAP_DRAGON_ARENA_01	DK1	153 / 229.2	REGION_01 AGON_RI E
MAP_DRAGON_ARENA_02	DK2	288.9 / 106.6	REGION_02 AGON_RI E
MAP_DRAGON_ARENA_03	DK3	-228.1 / -74.6	REGION_03 DER_GOF
MAP_DRAGON_ARENA_04	DK4	-100.2 / -165.2	REGION_04 ST_SOUL ELD
MAP_DRAGON_ARENA_05	DK5	200.9 / -154.6	REGION_05 NTIPEDE TY

首领场 ID	标签	坐标	region_id
MAP_LEADER_FIELD_01	E1	-158.8 / 111.9	REGION_01 DER_GORGE
MAP_LEADER_FIELD_02	E2	49 / 191.9	REGION_02 AGON_RIVER E
MAP_LEADER_FIELD_03	E3	-49.6 / -119.9	REGION_03 ST_SOUL ELD
MAP_LEADER_FIELD_04	E4	211.6 / 72	REGION_04 RNING_W ER

伏石 ID	标签	坐标	半径	region_id
MAP_FUSHI_CANDIDATE_01	H01	-286.8 / 194.5	2	REGION_01 DER_GORGE
MAP_FUSHI_CANDIDATE_02	H02	-169.5 / 48	2	REGION_02 DER_GORGE
MAP_FUSHI_CANDIDATE_03	H03	-30.9 / 207.9	2	REGION_03 OKEN_GORGE
MAP_FUSHI_CANDIDATE_04	H04	267.6 / 143.9	2	REGION_04 AGON_RIVER E
MAP_FUSHI_CANDIDATE_05	H05	302.2 / -37.3	2	REGION_05 RNING_W ER
MAP_FUSHI_CANDIDATE_06	H06	-284.1 / -194.5	2	REGION_06 ST_SOUL ELD

伏石 ID	标签	坐标	半径	region_id
MAP_FUSHI_CANDIDATE_07	H07	-126.9 / -261.2	2	REGION_ ST_SOUL ELD
MAP_FUSHI_CANDIDATE_08	H08	145 / -255.8	2	REGION_ NTIPEDE TY
MAP_FUSHI_CANDIDATE_09	H09	283.6 / -207.9	2	REGION_ NTIPEDE TY
MAP_FUSHI_CANDIDATE_10	H10	-334.7 / 58.6	2	REGION_ DER_GOF
MAP_FUSHI_CANDIDATE_11	H11	350.2 / 69.3	2	REGION_ RNING_W ER
MAP_FUSHI_CANDIDATE_12	H12	51.7 / -293.2	2	REGION_ NTIPEDE TY
MAP_FUSHI_CANDIDATE_13	H13	-44.2 / 111.9	2	REGION_ OKEN_GO
MAP_FUSHI_CANDIDATE_14	H14	142.3 / 143.9	2	REGION_ AGON_RI E
MAP_FUSHI_CANDIDATE_15	H15	42.6 / -29.8	2	REGION_ NTIPEDE TY
MAP_FUSHI_CANDIDATE_16	H16	178 / -147.1	2	REGION_ NTIPEDE TY
MAP_FUSHI_CANDIDATE_17	H17	-234.5 / 181.2	2	REGION_ DER_GOF

伏石 ID	标签	坐标	半径	region_id
MAP_FUSHI_CANDIDATE_18	H18	-146.6 / 143.9	2	REGION_18 OKEN_G
MAP_FUSHI_CANDIDATE_19	H19	74.6 / 64	2	REGION_19 AGON_RI E
MAP_FUSHI_CANDIDATE_20	H20	274.5 / 58.6	2	REGION_20 RNING_W ER
MAP_FUSHI_CANDIDATE_21	H21	-224.4 / -53.3	2	REGION_21 DER_GOF
MAP_FUSHI_CANDIDATE_22	H22	-133.2 / -80	2	REGION_22 ST_SOUL ELD
MAP_FUSHI_CANDIDATE_23	H23	93.3 / -85.3	2	REGION_23 NTIPEDE TY
MAP_FUSHI_CANDIDATE_24	H24	231.3 / -114.6	2	REGION_24 NTIPEDE TY

关口 ID	坐标	region_id
MAP_CHOKE_01	-62.9 / 111.9	REGION_01 OKEN_G
MAP_CHOKE_02	123.7 / 143.9	REGION_02 AGON_RI E
MAP_CHOKE_03	24 / -29.8	REGION_03 NTIPEDE TY

关口 ID	坐标	region_id
MAP_CHOKE_04	159.4 / -147.1	REGION_04 NTIPEDE TY

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_001	-375 / 298.8	REGION_01 DER_GOR
MAP_CHEST_POINT_002	-375 / 253.8	REGION_01 DER_GOR
MAP_CHEST_POINT_003	-375 / 208.8	REGION_01 DER_GOR
MAP_CHEST_POINT_004	-375 / 163.8	REGION_01 DER_GOR
MAP_CHEST_POINT_005	-375 / 118.8	REGION_01 DER_GOR
MAP_CHEST_POINT_006	-375 / 73.8	REGION_01 DER_GOR
MAP_CHEST_POINT_007	-375 / 28.8	REGION_01 DER_GOR
MAP_CHEST_POINT_008	-375 / -16.2	REGION_01 DER_GOR
MAP_CHEST_POINT_009	-375 / -61.2	REGION_01 DER_GOR
MAP_CHEST_POINT_010	-375 / -106.2	REGION_01 DER_GOR
MAP_CHEST_POINT_011	-375 / -151.2	REGION_01 DER_GOR

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_012	-375 / -196.2	REGION_01 ST_SOUL ELD
MAP_CHEST_POINT_013	-330 / 298.8	REGION_01 DER_GOR
MAP_CHEST_POINT_014	-330 / 208.8	REGION_01 DER_GOR
MAP_CHEST_POINT_015	-330 / 73.8	REGION_01 DER_GOR
MAP_CHEST_POINT_016	-330 / 28.8	REGION_01 DER_GOR
MAP_CHEST_POINT_017	-330 / -61.2	REGION_01 DER_GOR
MAP_CHEST_POINT_018	-330 / -151.2	REGION_01 ST_SOUL ELD
MAP_CHEST_POINT_019	-330 / -196.2	REGION_01 ST_SOUL ELD
MAP_CHEST_POINT_020	-330 / -241.2	REGION_01 ST_SOUL ELD
MAP_CHEST_POINT_021	-285 / 298.8	REGION_01 OKEN_GOR
MAP_CHEST_POINT_022	-285 / 208.8	REGION_01 DER_GOR
MAP_CHEST_POINT_023	-285 / 163.8	REGION_01 DER_GOR
MAP_CHEST_POINT_024	-285 / 118.8	REGION_01 DER_GOR

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_025	-285 / 73.8	REGION_025 DER_GORGE
MAP_CHEST_POINT_026	-285 / -61.2	REGION_026 DER_GORGE
MAP_CHEST_POINT_027	-285 / -241.2	REGION_027 ST_SOUL ELD
MAP_CHEST_POINT_028	-285 / -286.2	REGION_028 ST_SOUL ELD
MAP_CHEST_POINT_029	-240 / 298.8	REGION_029 OKEN_GORGE
MAP_CHEST_POINT_030	-240 / 163.8	REGION_030 DER_GORGE
MAP_CHEST_POINT_031	-240 / 118.8	REGION_031 DER_GORGE
MAP_CHEST_POINT_032	-240 / 73.8	REGION_032 DER_GORGE
MAP_CHEST_POINT_033	-240 / 28.8	REGION_033 DER_GORGE
MAP_CHEST_POINT_034	-240 / -16.2	REGION_034 DER_GORGE
MAP_CHEST_POINT_035	-240 / -61.2	REGION_035 DER_GORGE
MAP_CHEST_POINT_036	-240 / -106.2	REGION_036 DER_GORGE
MAP_CHEST_POINT_037	-240 / -196.2	REGION_037 ST_SOUL ELD

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_038	-240 / -241.2	REGION_001 ST_SOULDIER ELD
MAP_CHEST_POINT_039	-240 / -286.2	REGION_001 ST_SOULDIER ELD
MAP_CHEST_POINT_040	-240 / -331.2	REGION_001 ST_SOULDIER ELD
MAP_CHEST_POINT_041	-195 / 298.8	REGION_002 OKEN_GOLD
MAP_CHEST_POINT_042	-195 / 208.8	REGION_002 OKEN_GOLD
MAP_CHEST_POINT_043	-195 / 163.8	REGION_002 OKEN_GOLD
MAP_CHEST_POINT_044	-195 / 28.8	REGION_002 DER_GOLD
MAP_CHEST_POINT_045	-195 / -106.2	REGION_001 ST_SOULDIER ELD
MAP_CHEST_POINT_046	-195 / -151.2	REGION_001 ST_SOULDIER ELD
MAP_CHEST_POINT_047	-195 / -286.2	REGION_001 ST_SOULDIER ELD
MAP_CHEST_POINT_048	-195 / -331.2	REGION_001 ST_SOULDIER ELD

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_049	-150 / 298.8	REGION_049 OKEN_GOLD
MAP_CHEST_POINT_050	-150 / 208.8	REGION_050 OKEN_GOLD
MAP_CHEST_POINT_051	-150 / 163.8	REGION_051 OKEN_GOLD
MAP_CHEST_POINT_052	-150 / 118.8	REGION_052 OKEN_GOLD
MAP_CHEST_POINT_053	-150 / 73.8	REGION_053 DER_GOLD
MAP_CHEST_POINT_054	-150 / 28.8	REGION_054 DER_GOLD
MAP_CHEST_POINT_055	-150 / -106.2	REGION_055 ST_SOUL ELD
MAP_CHEST_POINT_056	-150 / -151.2	REGION_056 ST_SOUL ELD
MAP_CHEST_POINT_057	-150 / -196.2	REGION_057 ST_SOUL ELD
MAP_CHEST_POINT_058	-150 / -241.2	REGION_058 ST_SOUL ELD
MAP_CHEST_POINT_059	-150 / -286.2	REGION_059 ST_SOUL ELD
MAP_CHEST_POINT_060	-105 / 298.8	REGION_060 OKEN_GOLD

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_061	-105 / 253.8	REGION_061 OKEN_GOLD
MAP_CHEST_POINT_062	-105 / 208.8	REGION_062 OKEN_GOLD
MAP_CHEST_POINT_063	-105 / 118.8	REGION_063 OKEN_GOLD
MAP_CHEST_POINT_064	-105 / 73.8	REGION_064 DER_GOLD
MAP_CHEST_POINT_065	-105 / 28.8	REGION_065 DER_GOLD
MAP_CHEST_POINT_066	-105 / -16.2	REGION_066 DER_GOLD
MAP_CHEST_POINT_067	-105 / -106.2	REGION_067 ST_SOUL ELD
MAP_CHEST_POINT_068	-105 / -151.2	REGION_068 ST_SOUL ELD
MAP_CHEST_POINT_069	-105 / -196.2	REGION_069 ST_SOUL ELD
MAP_CHEST_POINT_070	-105 / -241.2	REGION_070 ST_SOUL ELD
MAP_CHEST_POINT_071	-105 / -286.2	REGION_071 ST_SOUL ELD
MAP_CHEST_POINT_072	-60 / 163.8	REGION_072 OKEN_GOLD

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_073	-60 / 118.8	REGION_073 OKEN_GOLD
MAP_CHEST_POINT_074	-60 / 73.8	REGION_074 OKEN_GOLD
MAP_CHEST_POINT_075	-60 / -16.2	REGION_075 DER_GOLD
MAP_CHEST_POINT_076	-60 / -151.2	REGION_076 ST_SOUL ELD
MAP_CHEST_POINT_077	-60 / -196.2	REGION_077 ST_SOUL ELD
MAP_CHEST_POINT_078	-60 / -286.2	REGION_078 ST_SOUL ELD
MAP_CHEST_POINT_079	-15 / 298.8	REGION_079 OKEN_GOLD
MAP_CHEST_POINT_080	-15 / 253.8	REGION_080 OKEN_GOLD
MAP_CHEST_POINT_081	-15 / 208.8	REGION_081 OKEN_GOLD
MAP_CHEST_POINT_082	-15 / 163.8	REGION_082 OKEN_GOLD
MAP_CHEST_POINT_083	-15 / -196.2	REGION_083 ST_SOUL ELD
MAP_CHEST_POINT_084	-15 / -286.2	REGION_084 ST_SOUL ELD

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_085	-15 / -331.2	REGION_001 ST_SOUL ELD
MAP_CHEST_POINT_086	30 / 298.8	REGION_001 OKEN_GOLD
MAP_CHEST_POINT_087	30 / 253.8	REGION_001 OKEN_GOLD
MAP_CHEST_POINT_088	30 / 208.8	REGION_001 AGON_RIVER E
MAP_CHEST_POINT_089	30 / 28.8	REGION_001 AGON_RIVER E
MAP_CHEST_POINT_090	30 / -16.2	REGION_001 NTIPEDE TY
MAP_CHEST_POINT_091	30 / -106.2	REGION_001 NTIPEDE TY
MAP_CHEST_POINT_092	30 / -151.2	REGION_001 NTIPEDE TY
MAP_CHEST_POINT_093	30 / -196.2	REGION_001 NTIPEDE TY
MAP_CHEST_POINT_094	30 / -241.2	REGION_001 NTIPEDE TY

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_095	30 / -286.2	REGION_01 NTIPEDE TY
MAP_CHEST_POINT_096	30 / -331.2	REGION_01 NTIPEDE TY
MAP_CHEST_POINT_097	75 / 163.8	REGION_02 AGON_RI E
MAP_CHEST_POINT_098	75 / 73.8	REGION_02 AGON_RI E
MAP_CHEST_POINT_099	75 / -16.2	REGION_02 RNING_W ER
MAP_CHEST_POINT_100	75 / -61.2	REGION_01 NTIPEDE TY
MAP_CHEST_POINT_101	75 / -106.2	REGION_01 NTIPEDE TY
MAP_CHEST_POINT_102	75 / -196.2	REGION_01 NTIPEDE TY
MAP_CHEST_POINT_103	75 / -286.2	REGION_01 NTIPEDE TY
MAP_CHEST_POINT_104	120 / 298.8	REGION_02 AGON_RI E

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_105	120 / 253.8	REGION_1 AGON_RIVER E
MAP_CHEST_POINT_106	120 / 163.8	REGION_1 AGON_RIVER E
MAP_CHEST_POINT_107	120 / 118.8	REGION_1 AGON_RIVER E
MAP_CHEST_POINT_108	120 / 73.8	REGION_1 AGON_RIVER E
MAP_CHEST_POINT_109	120 / 28.8	REGION_1 RNING_W ER
MAP_CHEST_POINT_110	120 / -106.2	REGION_1 NTIPEDE TY
MAP_CHEST_POINT_111	120 / -151.2	REGION_1 NTIPEDE TY
MAP_CHEST_POINT_112	120 / -196.2	REGION_1 NTIPEDE TY
MAP_CHEST_POINT_113	120 / -286.2	REGION_1 NTIPEDE TY
MAP_CHEST_POINT_114	165 / 298.8	REGION_1 AGON_RIVER E

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_115	165 / 208.8	REGION_1 AGON_RIVER E
MAP_CHEST_POINT_116	165 / 163.8	REGION_1 AGON_RIVER E
MAP_CHEST_POINT_117	165 / 118.8	REGION_1 AGON_RIVER E
MAP_CHEST_POINT_118	165 / 73.8	REGION_1 AGON_RIVER E
MAP_CHEST_POINT_119	165 / 28.8	REGION_1 RNING_W ER
MAP_CHEST_POINT_120	165 / -151.2	REGION_1 NTIPEDE TY
MAP_CHEST_POINT_121	165 / -196.2	REGION_1 NTIPEDE TY
MAP_CHEST_POINT_122	165 / -241.2	REGION_1 NTIPEDE TY
MAP_CHEST_POINT_123	165 / -286.2	REGION_1 NTIPEDE TY
MAP_CHEST_POINT_124	210 / 298.8	REGION_1 AGON_RIVER E

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_125	210 / 208.8	REGION_1 AGON_RIVER E
MAP_CHEST_POINT_126	210 / 163.8	REGION_1 AGON_RIVER E
MAP_CHEST_POINT_127	210 / 118.8	REGION_1 AGON_RIVER E
MAP_CHEST_POINT_128	210 / 73.8	REGION_1 AGON_RIVER E
MAP_CHEST_POINT_129	210 / -16.2	REGION_1 RNING_W ER
MAP_CHEST_POINT_130	210 / -106.2	REGION_1 NTIPEDE TY
MAP_CHEST_POINT_131	210 / -151.2	REGION_1 NTIPEDE TY
MAP_CHEST_POINT_132	210 / -196.2	REGION_1 NTIPEDE TY
MAP_CHEST_POINT_133	210 / -286.2	REGION_1 NTIPEDE TY
MAP_CHEST_POINT_134	255 / 298.8	REGION_1 AGON_RIVER E

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_135	255 / 253.8	REGION_135 AGON_RIVER
MAP_CHEST_POINT_136	255 / 208.8	REGION_136 AGON_RIVER
MAP_CHEST_POINT_137	255 / 163.8	REGION_137 AGON_RIVER
MAP_CHEST_POINT_138	255 / 118.8	REGION_138 AGON_RIVER
MAP_CHEST_POINT_139	255 / 73.8	REGION_139 RNING_WATER
MAP_CHEST_POINT_140	255 / 28.8	REGION_140 RNING_WATER
MAP_CHEST_POINT_141	255 / -16.2	REGION_141 RNING_WATER
MAP_CHEST_POINT_142	255 / -106.2	REGION_142 RNING_WATER
MAP_CHEST_POINT_143	255 / -151.2	REGION_143 NTIPEDE TY
MAP_CHEST_POINT_144	255 / -286.2	REGION_144 NTIPEDE TY

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_145	300 / 298.8	REGION_145 AGON_RIVER
MAP_CHEST_POINT_146	300 / 253.8	REGION_146 AGON_RIVER
MAP_CHEST_POINT_147	300 / 208.8	REGION_147 AGON_RIVER
MAP_CHEST_POINT_148	300 / 163.8	REGION_148 AGON_RIVER
MAP_CHEST_POINT_149	300 / 118.8	REGION_149 AGON_RIVER
MAP_CHEST_POINT_150	300 / 73.8	REGION_150 RNING_WATER
MAP_CHEST_POINT_151	300 / 28.8	REGION_151 RNING_WATER
MAP_CHEST_POINT_152	300 / -16.2	REGION_152 RNING_WATER
MAP_CHEST_POINT_153	300 / -61.2	REGION_153 RNING_WATER
MAP_CHEST_POINT_154	300 / -286.2	REGION_154 NTIPEDE TY

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_155	345 / 253.8	REGION_155 AGON_RIVER E
MAP_CHEST_POINT_156	345 / -16.2	REGION_156 RNING_W ER
MAP_CHEST_POINT_157	345 / -61.2	REGION_157 RNING_W ER
MAP_CHEST_POINT_158	345 / -196.2	REGION_158 RNING_W ER
MAP_CHEST_POINT_159	345 / -241.2	REGION_159 NTIPEDE TY
MAP_CHEST_POINT_160	390 / 208.8	REGION_160 AGON_RIVER E
MAP_CHEST_POINT_161	390 / 163.8	REGION_161 AGON_RIVER E
MAP_CHEST_POINT_162	390 / 118.8	REGION_162 AGON_RIVER E
MAP_CHEST_POINT_163	390 / 73.8	REGION_163 RNING_W ER
MAP_CHEST_POINT_164	390 / -16.2	REGION_164 RNING_W ER

宝箱点 ID	坐标	region_id
MAP_CHEST_POINT_165	390 / -61.2	REGION_1 RNING_W ER
MAP_CHEST_POINT_166	390 / -106.2	REGION_1 RNING_W ER
MAP_CHEST_POINT_167	390 / -151.2	REGION_1 RNING_W ER

附录 E 运行时工程字段、枚举、事件与资产台账

以下表把正文不宜展开的执行字段收进同一文档，构成本文内部的规范化工程附录。

E.1 单局与致死总合同

工程字段	唯一值 / 执行语义
product_id	SANMING_WUCHANG
title_zh_cn	三命无常
match.players	30
match.mode	SOLO
match.lives_per_player	3
match.winner_condition	EXACTLY_ONE_PARTICIPANT_NOT_IN_ELIMINATED
match.alive_body_count_is_not_winner_condition	是
match.simultaneous_tie_requires_all_candidates_eliminated_after_lethal_chain	是
match.co_champion.placement	1
match.co_champion.next_placement_skips_tied_places	是
match.co_champion.mmr_placement_value	AVERAGE_ORDINAL_OF_TIED_SLOTS
match.co_champion.each_gets_full_out_of_match_champion_reward	是
match.co_champion.does_not_duplicate_in_match_economy	是
match.server_anomaly_insurance_s	1800
match.insurance_is_not_player_facing_sudden_death	是
match.insurance_result	ABORT_AND_MANUAL_REVIEW_IF_NO_WINNER

工程字段	唯一值 / 执行语义
life_transaction.idempotency_key	MATCH_ID+PLAYER_ID+TRUE_DEATH_ORDINAL
life_transaction.commit_order	RESOLVE_LETHAL_PROTECTION / CONSUME_ONE_LIFE_IF_AVAILABLE / COMMIT_DEATH_CREDIT_AND_DROPS / RESET_TRUE_DEATH_STATE / ASSIGN_REVIVE_POINT / ENTER_SOUL_FLIGHT_OR_ELIMINATED
life_transaction.projectile_owner_eliminated	CONTINUE_AND_CREDIT_OWNER_WITH_ALL_NORMAL_KILL_REWARDS
life_transaction.monster_final_blow.credit	MONSTER
life_transaction.monster_final_blow.player_kill_rewards	否
life_transaction.monster_final_blow.break_victim_kill_streak	是
life_transaction.monster_final_blow.B41_without_player_target	DESTROY_WITHOUT_REWARD
life_transaction.monster_final_blow.D12_on_target_true_death	CLEAR
life_transaction.victim_wallet_snapshot	LAST_COMMITTED_BALANCE_BEFORE_LETHAL_DAMAGE_APPLICATION
true_death_reset.authority	§ 2 及附录 A 英雄成长规则
aborted_match.mmr	NO_CHANGE
aborted_match.xiuwei	FIXED_PARTICIPATION_COMPENSATION_20
aborted_match.tasks_progress	否
aborted_match.in_match_gold_carries_out	否
aborted_match.diagnostic_replay_retained	是
aborted_match.report_entry_available	是
storm.authority	§ 10 天劫规则
storm.freedom_invariant	APOCALYPSE_DOES_NOT_DISABLE_STEALTH_OR_FORCE_REVEAL
lethal_resolution.atomic_scope	ONE_LETHAL_DAMAGE_INSTANCE_PER_TARGET_PER_SERVER_TICK_ORDER
lethal_resolution.priority	DAMAGE_IMMUNITY_OR_DODGE / SHIELD_ABSORPTION / HP_DAMAGE_COMMIT / B19_FEIGN_DEATH_ROLL / G1_NINE_TURN_PILL_IF_OWNED / B20_PASSIVE_REVIVE_IF_CHARGE_AVAILABLE / CONSUME_ONE_LIFE_AND_COMMIT_TRUE_DEATH / ELIMINATE_IF_NO_LIFE_REMAINS
lethal_resolution.H032_KASAYA	PREVENTS_DAMAGE_BEFORE_LETHAL_CHAIN; IT_IS_NOT_A_LETHAL_PROTECTION
lethal_resolution.B19_success	SET_HP_BY_LEVEL; DO_NOT_CONSUME_LIFE; DO_NOT_DROP; DO_NOT_RESET_COOLDOWNS
lethal_resolution.G1_success	CONSUME_ITEM_INSTANCE; SET_FULL_HP; INVULNERABLE_2S; DO_NOT_CONSUME_LIFE
lethal_resolution.B20_success	CONSUME_MATCH_CHARGE; SET_HP_BY_LEVEL; DO_NOT_CONSUME_LIFE; DO_NOT_DROP; DO_NOT_RESET_COOLDOWNS
lethal_resolution.true_death_only_after_all_protections_fail	是
lethal_resolution.same_tick_new_damage_after_protection	NEW_DAMAGE_INSTANCE_REENTERS_CHAIN; INVULNERABILITY_APPLIES_IF_ACTIVE
machine_authorities.heroes	§ 2 及附录 A 英雄成长规则
machine_authorities.active_skills	§ 6 主动技能规则

工程字段	唯一值 / 执行语义
machine_authorities.passives	§ 7 被动技能规则
machine_authorities.equipment	§ 8 装备规则
machine_authorities.combat	§ 5 战斗规则
machine_authorities.elements	§ 5 战斗规则
machine_authorities.storm	§ 10 天劫规则
machine_authorities.map	§ 9 及附录 D 地图规则
machine_authorities.monsters	§ 11 怪物规则
machine_authorities.monster_ai	§ 11 怪物规则
machine_authorities.monster_visuals	§ 11 怪物规则
machine_authorities.directed_books	§ 12 及附录 C 经济掉落规则
machine_authorities.directed_equipment	§ 8 装备规则
machine_authorities.shops	§ 12.3 及 § E.17 商店规则
machine_authorities.shop_bags	§ 12.3 及 § E.17 商店规则
machine_authorities.black_mountain	§ 12.5 及 § E.18 黑山规则
machine_authorities.economy	§ 12 及附录 C 经济掉落规则
machine_authorities.transactions	§ 15 工程运行规则
machine_authorities.ground_loot	§ 12 地面物与空投规则
machine_authorities.movement	RULE-MAP-001
machine_authorities.dynamic_solids	RULE-MAP-001
machine_authorities.summons	RULE-CORE-001
machine_authorities.input	§ 3 输入与界面规则
machine_authorities.ui_flow	§ 3 输入与界面规则
machine_authorities.rng	§ 15 工程运行规则
machine_authorities.determinism	§ 15 工程运行规则
machine_authorities.network	§ 15 工程运行规则
machine_authorities.connection	附录 E 工程契约
machine_authorities.backend	§ 15 工程运行规则
machine_authorities.bots	RULE-MATCH-001
machine_authorities.matchmaking	RULE-MATCH-001
machine_authorities.release_gates	§ 16 至 § 17 资产与发布规则
machine_authorities.units_ids	RULE-CORE-001
machine_authorities.authority_schema	RULE-CORE-001
machine_authorities.player_combat	§ 5 战斗规则
machine_authorities.life_progression	§ 2 及附录 A 英雄成长规则

工程字段	唯一值 / 执行语义
machine_authorities.vision_information	RULE-COMBAT-001
machine_authorities.economy_drops	§ 12 及附录 C 经济掉落规则
machine_authorities.monster_combat	§ 5 战斗规则
machine_authorities.airdrop	§ 12 及附录 C 经济掉落规则
machine_authorities.state_machines	附录 E 工程契约
machine_authorities.event_order	附录 E 工程契约
machine_authorities.closed_enums	附录 E 工程契约
machine_authorities.analytics	附录 E 工程契约
machine_authorities.security_replay	§ 15 工程运行规则
machine_authorities.settings	附录 E 工程契约
machine_authorities.metagame	§ 14 局外规则
machine_authorities.collections	§ 14 局外规则
machine_authorities.ui_layout	§ 3 输入与界面规则
machine_authorities.monetization	§ 14 局外规则
machine_authorities.completion_matrix	RULE-RELEASE-001
machine_authorities.codegen	RULE-CORE-001
machine_authorities.audio_slots	§ 16 至 § 17 资产与发布规则
machine_authorities.content_acceptance_vectors	RULE-RELEASE-001
machine_authorities.requirements_traceability	RULE-RELEASE-001
machine_authorities.equipment_catalog	§ 8 装备规则
natural_language_is_non_authoritative	是
authority_loading.order	VERIFY_MANIFEST_DELIVERY_HASH / VERIFY_RUNTIME_RULESET_HASH / LOAD_CLOSED_ENUMS_AND_SCHEMA / LOAD_MACHINE_AUTHORITIES / RESOLVE_TYPED_CONTENT_REFS / RUN_CROSS_FILE_ASSERTIONS
authority_loading.any_failure	REFUSE_SERVER_START_OR_CLIENT_JOIN
authority_loading.hot_swap_during_match	否
authority_classes.MATCH_RULESET	units_ids / authority_schema / heroes / active_skills / passives / equipment / player_combat / combat / elements / storm / map / monsters / monster_ai / monster_combat / directed_books / directed_equipment / shops / shop_bags / black_mountain / economy / economy_drops / transactions / ground_loot / movement / dynamic_solids / summons / input / ui_flow / rng / determinism / bots / matchmaking / life_progression / vision_information / airdrop / state_machines / event_order / closed_enums
authority_classes.SERVICE_CONTRACT	network / connection / backend / analytics / security_replay / settings / metagame / collections / ui_layout / monetization
authority_classes.DEVELOPMENT_ONLY	monster_visuals / release_gates / completion_matrix / codegen / audio_slots / content_acceptance_vectors / requirements_traceability / equipment_catalog
authority_classes.rule	MATCH_RULESET_HASH_ONLY_USES_MATCH_RULESET; SERVICE_HASH_USES_SERVICE_CONTRACT; DELIVERY_HASH_USES_ALL_DELIVERED_FILES

工程字段	唯一值 / 执行语义
content_counts.heroes	38
content_counts.active	57
content_counts.passive	44
content_counts.equipment	48
content_counts.monster_archetypes	22
round22_clean_rebuild_and_3d_closure_integrated	是
release_pipeline_must_be_single_writer_and_rebuild_artifacts	是
ruleset_semver	3.0.0
service_contract_version	3.0.0
ruleset_hash_algorithm	SHA256_UTF8_CANONICAL_RULE_EXPORT_EXCLUDING_RULESET_HASH_FIELD
compatibility_policy	MATCH_BINDS_EXACT_RULESET_HASH; NO_HOT_SWAP; REPLAY_REQUIRES_EXACT_HASH
replay.genesis_prev_hash_32	771807ee94b4c493fb7cfb4eec41e8e3a87847542140eec4e9cbda5b7ceff7d3
replay.genesis_domain	SANMING_WUCHANG REPLAY GENESIS 3.0.0
rule_projection.authority	APPENDIX_E_FULLY_NAMESPACED_ENGINEERING_KEYS
rule_projection.body_and_player_text	GENERATED_READ_ONLY_PROJECTION; NEVER_RUNTIME_INPUT
empty_field_policy	FORBID_BLANK; USE_NONE_OR_EMPTY_ARRAY_OR_NOT_APPLICABLE_OR_INHERIT_DEFAULT
aborted_match.client_result_code	VOID_MATCH_SERVER_INTEGRITY_OR_30M_TIMEOUT
aborted_match.player_message_key	settlement_void_match_retry
aborted_match.requeue_action	ONE_TAP_AVAILABLE_AFTER_RESULT_ACK
aborted_match.settlement_idempotent	是
map.wall_polygon_simplification_tolerance_m	0.05
map.wall_polygon_orientation	CLOCKWISE
map.wall_polygon_explicit_closing_vertex	否
map.wall_polygon_geometry_sha256	73cb6bbe7de36c9b371924bdaff6d5a518295448c707b4dd50defab159e04573
ruleset_hash	97e97296640c35d9c9e2ed0962d30246fa66dd3b1a41b86201562022873c49cb
map.region_polygon_sha256	11918269d7f585be1b829e38100d194028e80e9692e9868814996690a7339d8d
active.runtime_authority	E.26_STRONG_TYPED_PAYLOAD_TABLE
active.natural_language_runtime_parse	FORBIDDEN

工程字段	唯一值 / 执行语义
H009.runtime_authority	E.27_EXACT_FIELD_WHITELIST
passive.runtime_authority	E.31_STRONG_TYPED_PAYLOAD_TABLE
equipment.effect_runtime_authority	E.32_STRONG_TYPED_PAYLOAD_TABLE
monster_skill.runtime_authority	E.33_STRONG_TYPED_PAYLOAD_TABLE
summon.runtime_authority	E.34_STRONG_TYPED_PAYLOAD_TABLE

E.2 生命、等级与复活

工程字段	唯一值 / 执行语义
hero_offer.start_gold	500
hero_offer.offer_count	3
hero_offer.pool	38_HEROES_EQUAL_WEIGHT
hero_offer.within_offer_unique	是
hero_offer.duration_s	15
hero_offer.timeout	LEFTMOST
hero_offer.reroll_cost_gold	250
hero_offer.reroll_while_balance_gte_cost	是
hero_offer.across_rerolls_repeat_allowed	是
hero_offer.pity	NONE
hero_offer.rng_stream	HERO_DRAFT
hero_offer.window_semantics	SELECTION_REQUEST_COMMIT_TICK_MUST_BE_BEFORE_TIMEOUT_TICK
hero_offer.transaction_id	MATCH_ID:ACCOUNT_ID:HERO_DRAFT_REQUEST_UUID
hero_offer.reroll_atomic	LOCK_PLAYER; REVALIDATE_WINDOW_AND_GOLD; DEDUCT_COST; CONSUME_ONE_HERO_DRAFT_RNG_DRAW_SEQUENCE; REPLACE_ALL_THREE_OFFERS; COMMIT
hero_offer.timeout_vs_inflight	ONLY_ALREADY_COMMITTED_REROLL_OR_SELECTION_COUNTS; OTHERWISE_LEFTMOST_CURRENT_COMMITTED_OFFER
hero_offer.duplicate_request	RETURN_IDEMPOTENT_RESULT_WITHOUT_SECOND_GOLD_OR_RNG_CONSUMPTION
level.min	1
level.max	15
level.curve[0].from_level	1
level.curve[0].to_level	2
level.curve[0].xp_required	50
level.curve[0].cumulative_xp	50
level.curve[1].from_level	2
level.curve[1].to_level	3
level.curve[1].xp_required	70

工程字段	唯一值 / 执行语义
level.curve[1].cumulative_xp	120
level.curve[2].from_level	3
level.curve[2].to_level	4
level.curve[2].xp_required	90
level.curve[2].cumulative_xp	210
level.curve[3].from_level	4
level.curve[3].to_level	5
level.curve[3].xp_required	110
level.curve[3].cumulative_xp	320
level.curve[4].from_level	5
level.curve[4].to_level	6
level.curve[4].xp_required	130
level.curve[4].cumulative_xp	450
level.curve[5].from_level	6
level.curve[5].to_level	7
level.curve[5].xp_required	155
level.curve[5].cumulative_xp	605
level.curve[6].from_level	7
level.curve[6].to_level	8
level.curve[6].xp_required	180
level.curve[6].cumulative_xp	785
level.curve[7].from_level	8
level.curve[7].to_level	9
level.curve[7].xp_required	210
level.curve[7].cumulative_xp	995
level.curve[8].from_level	9
level.curve[8].to_level	10
level.curve[8].xp_required	240
level.curve[8].cumulative_xp	1235
level.curve[9].from_level	10
level.curve[9].to_level	11
level.curve[9].xp_required	275
level.curve[9].cumulative_xp	1510
level.curve[10].from_level	11
level.curve[10].to_level	12
level.curve[10].xp_required	315
level.curve[10].cumulative_xp	1825

工程字段	唯一值 / 执行语义
level.curve[11].from_level	12
level.curve[11].to_level	13
level.curve[11].xp_required	450
level.curve[11].cumulative_xp	2275
level.curve[12].from_level	13
level.curve[12].to_level	14
level.curve[12].xp_required	550
level.curve[12].cumulative_xp	2825
level.curve[13].from_level	14
level.curve[13].to_level	15
level.curve[13].xp_required	700
level.curve[13].cumulative_xp	3525
level.max_cumulative_xp	3525
level.overflow_multi_level	是
level.at_level_15	DISCARD_FUTURE_XP
level.passive_trickle.xp	10
level.passive_trickle.interval_s	10
level.passive_trickle.eligible_states	ALIVE / TRUE_DEATH_RESOLVING / SOUL_FLIGHT / REVIVE_PROTECTION
player_kill_rewards.gold_created	$500 + 100 * \text{VICTIM_LEVEL} + \text{MIN}(\text{FLOOR}(0.10 * \text{VICTIM_WALLET_SNAPSHOT}), 800)$
player_kill_rewards.victim_wallet_not_deducted	是
player_kill_rewards.third_death_finisher_bonus_gold	500
player_kill_rewards.xp_created	$\text{MIN}(180, 60 + 8 * \text{VICTIM_LEVEL})$
player_kill_rewards.assist_gold	0
player_kill_rewards.assist_xp	$\text{MIN}(60, 20 + 3 * \text{VICTIM_LEVEL})$
player_kill_rewards.monster_or_world_final_blow	0
life.initial_lives	3
life.first_second_true_death	CONSUME_ONE_AND_REVIVE
life.third_true_death	CONSUME_LAST_AND_ELIMINATE
life.settlement_s	1
life.true_death_reset.active	CURRENT_ACTIVE_READY
life.true_death_reset.passive_internal_cooldowns	CONTINUE_REMAINING; NEVER_RESET_BY_TRUE_DEATH
life.true_death_reset.temporary_status_dot_control	CLEAR
life.true_death_reset.owner_entities	CLEAR
life.true_death_reset.preserve	GOLD / LEVEL / XP / PASSIVE_LEVELS / EQUIPMENT / PERMANENT_LAYERS
life.true_death_reset.equipment_temporary_effect_state	CLEAR_STACKS_TARGETS_INTERNAL_COOLDOWNS_STATIONARY_TIMERS_AND_OWNER_BOUND_ENTITIES

工程字段	唯一值 / 执行语义
life.true_death_reset.equipment_permanent_layers	PRESERVE_ONLY_WHEN_ITEM_EFFECT_EXPLICITLY_DECLARES_PERMANENT_LAYER
life.true_death_reset.running_equipment_created_independent_dot	CONTINUES_ONLY_IF_EFFECT_CONTRACT_EXPLICITLY_SAYS_INDEPENDENT_AFTER_APPLICATION
death_drop.first_second.drop	ALL_HAND_EQUIPMENT
death_drop.first_second.worn	KEEP
death_drop.third.drop	ALL_HAND_AND_ALL_THREE_WORN_EQUIPMENT
death_drop.gold	NEVER_DROP
death_drop.placement	RADIAL_8_DIRECTIONS_1_2_TO_2_5M; PROJECT_TO_GROUND_NAV; THEN_NEAREST_LEGAL_NAV_WITHIN_6M; ELSE NEAREST_GLOBAL_LEGAL_NAV_POINT_BY_NAV_DISTANCE_THEN_POINT_ID; NEVER_SPAWN_INSIDE_SOLID_OR_OFF_NAV
death_drop.public_immediately	是
death_drop.transaction_snapshot_before_scatter	是
revive.soul_speed_mps	18
revive.flight_duration_s	CLAMP(3,STRAIGHT_DISTANCE_M/18,10)
revive.soul_collision	否
revive.soul_invulnerable	是
revive.soul_can_interact_or_cast	否
revive.soul_vision_m	30
revive.soul_does_not_reveal_hidden	是
revive.revalidate_on_arrival	是
revive.reroute_extra_cap_s	3
revive.no_candidate_fallback	SAFE_ZONE_NAV_POINT_MAXIMIZING_MIN_NAV_DISTANCE_TO_ALL_LIVING_PLAYERS
revive.hp_percent	100
revive.protection_s	3
revive.protection.damage_immunity	是
revive.protection.cannot_basic_or_cast	是
revive.protection.cannot_pickup_shop_open_chest_or_gamble	是
revive.protection.movement_allowed	是
revive.protection.enter_fushi_allowed	是
revive.protection.ends_early_on_offensive_or_interaction_intent	是
revive.protection.storm_not_target	是
revive.protection.early_end_request_semantics	REMOVE_PROTECTION_AT_INPUT_VALIDATION_THEN_REVALIDATE_AND_EXECUTE_THE_SAME_REQUEST_WITHOUT_IMMUNITY; IF_REQUEST_OTHERWISE_ILLEGAL_IT_STILL_FAILS
revive.protection.damage_event_same_tick_as_voluntary_break	PROTECTION_REMOVAL_COMMITS_BEFORE_CAST_OR_INTERACTION_COMMIT

工程字段	唯一值 / 执行语义
revive.soul_scouting.reveals_normal_nonhidden_world_information_inside_30M	是
revive.soul_scouting.updates_personal_fog_memory	是
revive.soul_scouting.can_mark_personally_seen_shop_until_that_shop_moves	是
revive.soul_scouting.cannot_reveal_stealth_or_interact	是
revive.arrival_timeout	BASE_FLIGHT_CLAMP_PLUS_AT_MOST_3S_REROUTE; THEN_USE_DETERMINISTIC_FALLBACK_AND_ARRIVE
simultaneous_final.resolution	FINISH_ALL_COMMITTED_LETHAL_CHAINS_IN_TICK
simultaneous_final.tie_if_all_remaining_eliminated	是
simultaneous_final.tie_rank	ALL_PLACEMENT_1
simultaneous_final.next_rank_skips_tie_count	是
simultaneous_final.mmr_placement	AVERAGE_ORDINAL_OF_TIED_SLOTS
simultaneous_final.prelethal_tiebreak_not_used	是
player_kill_rewards.wallet_bonus_cap_gold	800
player_kill_rewards.wallet_bonus_G5_eligible	否
player_kill_rewards.assist_eligibility	NON_FINAL_BLOW_PLAYER_OWNER_WITH_ACTUAL_HP_OR_SHIELD_DAMAGE_OR_HARD_CONTROL_WITHIN_PREVIOUS_8S; UNIQUE_ONCE_PER_TRUE_DEATH
revive.pool.before_12m	ALL_80_MAP_SPAWN_POINT_IDS / ALL_192_MAP_NODE_IDS; FILTER_CURRENT_SAFE_CIRCLE_AND_PHYSICAL_LEGALITY
revive.pool.12m_to_15m	ALL_192_MAP_NODE_IDS; FILTER_CURRENT_SAFE_CIRCLE_AND_PHYSICAL_LEGALITY
revive.pool.at_or_after_15m	SELECTED_FINAL_COURT_SIX_AUTHORED_REVIVE_POINT_IDS
revive.strict_filters	OUTSIDE_STORM_AND_10S_PREDICTED_STORM / NAV_REACHABLE / NOT_IN_SOLID / NOT_IN_SHOP_PAD_BOSS_BODY_DYNAMIC_SOLID / DISTANCE_FROM_LIVING_PLAYER_GTE_18M / NO_DIRECT_LOS_FROM_LIVING_PLAYER_WITHIN_30M
revive.score_lexicographic	LOWEST_DEATH_HEAT_30S / MAX_MIN_NAV_DISTANCE_TO_LIVING_PLAYERS / MAX_DISTANCE_TO_LAST_DEATH / POINT_ID_ASC
revive.degraded_after_no_strict_candidates	3
revive.degraded_filters	KEEP_PHYSICAL_STORM_AND_GTE_12M; DROP_LOS_REQUIREMENT; SAME_SCORE

工程字段	唯一值 / 执行语义
revive.fallback	SAFE_ZONE_NAV_POINT_MAXIMIZING_MIN_NAV_DISTANCE_THEN_POINT_ID_ASC
revive.concurrent_assignment	TRUE_DEATH_COMMIT_ORDER; RESERVED_POINT_BLOCKS_LATER_ASSIGNMENT_WITHIN_6M_UNTIL_ARRIVAL_OR_REROUTE

E.3 战斗与伤害

工程字段	唯一值 / 执行语义
fixed_step_hz	20
basic_attack.legal_targets	OTHER_ALIVE_PLAYER / HOSTILE_MONSTER / DAMAGEABLE_HOSTILE_SUMMON
basic_attack.target_selection.touch_tap	LOCKED_TARGET_IF_LEGAL_ELSE_VISIBLE_TARGET_IN_FACING_SECTOR
basic_attack.target_selection.windows_left_click	VISIBLE_LEGAL_TARGET_UNDER_POINTER_ELSE_VISIBLE_TARGET_IN_FACING_SECTOR
basic_attack.target_selection.facing_sector_full_angle_deg	24
basic_attack.target_selection.tie_break	MOST_RECENT_VALID_DAMAGE_TO_SELF / SMALLEST_SCREEN_ANGLE / SHORTEST_NAV_DISTANCE / ENTITY_ID
basic_attack.movement_during_windup	FULL_SPEED_ALLOWED
basic_attack.cancel_before_release	CAST_ACTIVE / HARD_CONTROL / TRUE_DEATH
basic_attack.move_input_does_not_cancel	是
basic_attack.validate_range_and_direct_visibility_again_at_release	是
basic_attack.attack_interval_s	1 / CURRENT_ATTACKS_PER_SECOND; STARTS_AT_RELEASE_TICK
basic_attack.base_crit_chance_percent	0
basic_attack.base_crit_damage_multiplier	2
basic_attack.monster_can_crit	否
basic_attack.templates.长弓.windup_s	0.35
basic_attack.templates.长弓.backswing_visual_s	0.15
basic_attack.templates.长弓.delivery	PROJECTILE
basic_attack.templates.长弓.projectile_speed_mps	45
basic_attack.templates.长弓 · 飞行.windup_s	0.35
basic_attack.templates.长弓 · 飞行.backswing_visual_s	0.15
basic_attack.templates.长弓 · 飞行.delivery	PROJECTILE
basic_attack.templates.长弓 · 飞行.projectile_speed_mps	45
basic_attack.templates.连弩.windup_s	0.18
basic_attack.templates.连弩.backswing_visual_s	0.1

工程字段	唯一值 / 执行语义
basic_attack.templates.连弩.delivery	PROJECTILE
basic_attack.templates.连弩.projectile_speed_mps	55
basic_attack.templates.连弩 · 飞行.windup_s	0.18
basic_attack.templates.连弩 · 飞行.backswing_visual_s	0.1
basic_attack.templates.连弩 · 飞行.delivery	PROJECTILE
basic_attack.templates.连弩 · 飞行.projectile_speed_mps	55
basic_attack.templates.刺客.windup_s	0.22
basic_attack.templates.刺客.backswing_visual_s	0.12
basic_attack.templates.刺客.delivery	MELEE_RELEASE_HIT
basic_attack.templates.刺客.projectile_speed_mps	0
basic_attack.templates.刺客 · 飞行.windup_s	0.22
basic_attack.templates.刺客 · 飞行.backswing_visual_s	0.12
basic_attack.templates.刺客 · 飞行.delivery	MELEE_RELEASE_HIT
basic_attack.templates.刺客 · 飞行.projectile_speed_mps	0
basic_attack.templates.战士.windup_s	0.3
basic_attack.templates.战士.backswing_visual_s	0.15
basic_attack.templates.战士.delivery	MELEE_RELEASE_HIT
basic_attack.templates.战士.projectile_speed_mps	0
basic_attack.templates.坦克.windup_s	0.38
basic_attack.templates.坦克.backswing_visual_s	0.2
basic_attack.templates.坦克.delivery	MELEE_RELEASE_HIT
basic_attack.templates.坦克.projectile_speed_mps	0
basic_attack.projectile_tags	BLOCKED_BY_STATIC_SOLID / BLOCKABLE_BY_WIND_WALL / OWNER_CREDIT_PERSISTS_AFTER_OWNER_DEATH
basic_attack.ranged_projectile.tracking	HOMING_TO_COMMITTED_TARGET_CENTER
basic_attack.ranged_projectile.vision_loss_after_release	CONTINUE
basic_attack.ranged_projectile.blocked_by	STATIC_SOLID / WIND_WALL / FROZEN_RIVER
basic_attack.ranged_projectile.target_eliminated_before_impact	DESTROY_PROJECTILE
basic_attack.ranged_projectile.owner_true_death_or_elimination	CONTINUE_WITH_ORIGINAL_CREDIT
basic_attack.ranged_projectile.collision_radius_m	0.12
basic_attack.ranged_projectile.continuous_sweep_each_tick	是
basic_attack.proc_generated_basic_classification.B11_EXTRA_BASIC	DamageForm=BASIC; OwnerCredit=CASTER; COUNTS_AS_BASIC_HIT_FOR_P16_AND_OTHER_ELIGIBLE_PASSIVES

工程字段	唯一值 / 执行语义
basic_attack.proc_generated_basic_classification.SPLASH_DOT_REFLECT_EQUIPMENT_PROC	NOT_BASIC_EVEN_IF_ORIGINATED_FROM_BASIC
attributes.order	BASE_LEVEL_ROW / FLAT_MODIFIERS_SUM / PERCENT_MODIFIERS_MULTIPLY / TEMPORARY_MULTIPLIERS
attributes.move_speed_flat_unit	GAME_SPEED_UNIT_0_01_MPS
attributes.movement_speed_cap	NONE
attributes.attack_speed_cap	NONE
attributes.max_hp_change_preserves_hp_ratio	是
attributes.same_named_equipment_worn_unique	是
attributes.different_shoes_stack_normally	是
control_stacking.hard_control_same_kind	KEEP_LATER_END_TIME; DO_NOT_ADD_DURATIONS
control_stacking.different_hard_controls	COEXIST; EFFECTIVE_RESTRICTIONS_ARE_UNION
control_stacking.slow	ONLY_STRONGEST_PERCENT_APPLIES; DURATION_TRACKED_PER_SOURCE
control_stacking.damage_reduction	MULTIPLICATIVE_REMAINING_UNLESS_EXPLICIT_FIXED
control_stacking.silence	BLOCK_ACTIVE_CAST_START; DOES_NOT_CANCEL_ALREADY_COMMITTED_INSTANT_CAST
control_stacking.root	BLOCK_TRANSLATION_AND_DASH; BASIC_AND_ACTIVE_CASTS_STILL_ALLOWED_UNLESS_ALSO_SILENCED
control_stacking.invulnerable	DAMAGE_ZERO; TARGETABILITY_PRESERVED_UNLESS_SKILL_SAYS_OTHERWISE
control_stacking.forced_movement_into_solid	STOP_AT_LAST_LEGAL_CAPSULE_POSITION
control_stacking.hard_control_taxonomy.hard_reducible	STUN / FREEZE / PETRIFY / ROOT / FEAR / TRANSFORM / KNOCKUP
control_stacking.hard_control_taxonomy.soft_not_reduced_by_P18	SLOW / SILENCE
control_stacking.hard_control_taxonomy.immunity_and_cleansed_are_not_duration_resistance	是
control_stacking.target_duration_resistance	SAME_CATEGORY_TAKE_STRONGEST_LOWEST_MULTIPLIER; NEVER_MULTIPLY
control_stacking.duration_resolution_order	IF_IMMUNE_OR_CLEANSSED_THEN_ZERO_ELSE_MAX(EFFECT_MINIMUM,(BASE_PLUS_SOURCE_FLAT_EXTENSION)*TARGET_STRONGEST_MULTIPLIER)
control_stacking.forced_movement_resistance	SAME_CATEGORY_TAKE_STRONGEST_LOWEST_MULTIPLIER; B4_AND_P18_NEVER_MULTIPLY
control_stacking.control_application_snapshot	SOURCE_EXTENSION_AND_TARGET_RESISTANCE_SNAPSHOT_AT_APPLICATION; UNEQUIP_DOES_NOT_RETROACTIVELY_CHANGE_RUNNING_CONTROL
damage_pipeline	VALIDATE_TYPED_SOURCE_TARGET_LIFE_AND_CAST_OR_IMPACT_REQUIREMENTS / RESOLVE_PROJECTILE_OR_SHAPE_COLLISION / RESOLVE_FULL_IMMUNITY_DODGE_AND_BLOCK_IF_ELIGIBLE / CALCULATE_FIXED_PLUS_CURRENT_ATTRIBUTE_COEFFICIENT_Q4 / APPLY_CRITICAL_IF_ELIGIBLE / APPLY_SOURCE_CONDITIONAL_AMPLIFIERS / APPLY_BIDIRECTIONAL_ELEMENT_IF_RULE_REQUIRES / APPLY_TARGET_PERCENT_REDUCTION_UNLESS_DAMAGE_FORM_BYPASSES / APPLY_TARGET_FIXED_REDUCTION_UNLESS_DAMAGE_FORM_BYPASSES / CLAMP_POSITIVE_NONIMMUNE_DAMAGE_TO_MINIMUM_ONE / ABSORB_SHIELDS_NEWEST_FIRST / COMMIT_INTEGER_HP_DAMAGE / RUN_POST_DAMAGE_AND_LETHAL_TRANSACTION / APPEND_LEDGER_REPLAY_ANALYTICS
damage_forms.BASIC.crit	是
damage_forms.BASIC.element	是

工程字段	唯一值 / 执行语义
damage_forms.BASIC.percent_reduction	是
damage_forms.BASIC.fixed_reduction	是
damage_forms.BASIC.shieldable	是
damage_forms.BASIC.invulnerability_blocks	是
damage_forms.BASIC.positive_nonimmune_minimum_one	是
damage_forms.SKILL.crit	否
damage_forms.SKILL.element	是
damage_forms.SKILL.percent_reduction	是
damage_forms.SKILL.fixed_reduction	是
damage_forms.SKILL.shieldable	是
damage_forms.SKILL.invulnerability_blocks	是
damage_forms.SKILL.positive_nonimmune_minimum_one	是
damage_forms.DOT.crit	否
damage_forms.DOT.element	是
damage_forms.DOT.percent_reduction	是
damage_forms.DOT.fixed_reduction	否
damage_forms.DOT.shieldable	是
damage_forms.DOT.invulnerability_blocks	是
damage_forms.DOT.positive_nonimmune_minimum_one	是
damage_forms.PERCENT.crit	否
damage_forms.PERCENT.element	否
damage_forms.PERCENT.percent_reduction	是
damage_forms.PERCENT.fixed_reduction	否
damage_forms.PERCENT.shieldable	是
damage_forms.PERCENT.invulnerability_blocks	是
damage_forms.PERCENT.positive_nonimmune_minimum_one	是
damage_forms.REFLECT.crit	否
damage_forms.REFLECT.element	否
damage_forms.REFLECT.percent_reduction	是
damage_forms.REFLECT.fixed_reduction	否
damage_forms.REFLECT.shieldable	是
damage_forms.REFLECT.invulnerability_blocks	是
damage_forms.REFLECT.positive_nonimmune_minimum_one	是

工程字段	唯一值 / 执行语义
damage_forms.TRUE.crit	否
damage_forms.TRUE.element	否
damage_forms.TRUE.percent_reduction	否
damage_forms.TRUE.fixed_reduction	否
damage_forms.TRUE.shieldable	是
damage_forms.TRUE.invulnerability_blocks	是
damage_forms.TRUE.positive_nonimmune_minimum_one	是
damage_forms.STORM.crit	否
damage_forms.STORM.element	否
damage_forms.STORM.only_reductions	B11 / G3
damage_forms.STORM.shieldable	是
damage_forms.STORM.percent_reduction	否
damage_forms.STORM.fixed_reduction	否
damage_forms.STORM.invulnerability_blocks	是
damage_forms.STORM.positive_nonimmune_minimum_one	是
rounding.formula_internal	SIGNED_INT64_Q4
rounding.hp_commit	ROUND_HALF_AWAY_FROM_ZERO_TO_INTEGER
rounding.gold_formula	FLOOR_TO_INTEGER
rounding.position	ROUND_TO_NEAREST_MILLIMETER_TIES_AWAY_FROM_ZERO
damage_form_required_fields	crit / element / percent_reduction / fixed_reduction / shieldable / invulnerability_blocks / positive_nonimmune_minimum_one
element_rule_resolution.attacker_player_to_target_player	OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT
element_rule_resolution.attacker_player_to_target_pig_or_dragon	OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT
element_rule_resolution.attacker_pig_or_dragon_to_target_player	BIDIRECTIONAL_FIVE_ELEMENT
element_rule_resolution.all_other_monster_or_world_sources	NONE
element_rule_resolution.forms_forced_none	TRUE / PERCENT / REFLECT / STORM
element_rule_resolution.forms_allowed	BASIC / SKILL / DOT
element_rule_resolution.order	DAMAGE_FORM_FORCED_NONE_FIRST_ELSE_SOURCE_TARGET_CLASS_RULE
damage_event_default_fields_by_origin.PLAYER_BASIC.DamageForm	BASIC
damage_event_default_fields_by_origin.PLAYER_BASIC.SourceFaction	PLAYER_OWNED
damage_event_default_fields_by_origin.PLAYER_BASIC.ElementRule	DERIVE_BY_SOURCE_TARGET

工程字段	唯一值 / 执行语义
damage_event_default_fields_by_origin.PLAYER_BASIC.HitKind	DIRECT
damage_event_default_fields_by_origin.PLAYER_BASIC.OwnerCredit	CASTER
damage_event_default_fields_by_origin.PLAYER_ACTIVE.DamageForm	SKILL
damage_event_default_fields_by_origin.PLAYER_ACTIVE.SourceFaction	PLAYER_OWNED
damage_event_default_fields_by_origin.PLAYER_ACTIVE.ElementRule	DERIVE_BY_SOURCE_TARGET
damage_event_default_fields_by_origin.PLAYER_ACTIVE.HitKind	ABILITY
damage_event_default_fields_by_origin.PLAYER_ACTIVE.OwnerCredit	CASTER
damage_event_default_fields_by_origin.PLAYER_PASSIVE_OR_EQUIPMENT.DamageForm	SKILL
damage_event_default_fields_by_origin.PLAYER_PASSIVE_OR_EQUIPMENT.SourceFaction	PLAYER_OWNED
damage_event_default_fields_by_origin.PLAYER_PASSIVE_OR_EQUIPMENT.ElementRule	DERIVE_BY_SOURCE_TARGET
damage_event_default_fields_by_origin.PLAYER_PASSIVE_OR_EQUIPMENT.HitKind	ABILITY
damage_event_default_fields_by_origin.PLAYER_PASSIVE_OR_EQUIPMENT.OwnerCredit	CASTER
damage_event_default_fields_by_origin.PLAYER_OWNE_D_SUMMON.DamageForm	SKILL
damage_event_default_fields_by_origin.PLAYER_OWNE_D_SUMMON.SourceFaction	PLAYER_OWNED
damage_event_default_fields_by_origin.PLAYER_OWNE_D_SUMMON.ElementRule	DERIVE_BY_SOURCE_TARGET
damage_event_default_fields_by_origin.PLAYER_OWNE_D_SUMMON.HitKind	ABILITY
damage_event_default_fields_by_origin.PLAYER_OWNE_D_SUMMON.OwnerCredit	SUMMON_OWNER
damage_event_default_fields_by_origin.MONSTER_BASIC.DamageForm	BASIC
damage_event_default_fields_by_origin.MONSTER_BASIC.SourceFaction	MONSTER
damage_event_default_fields_by_origin.MONSTER_BASIC.ElementRule	ARCHETYPE_BASIC_DAMAGE_SPEC
damage_event_default_fields_by_origin.MONSTER_BASIC.HitKind	DIRECT

工程字段	唯一值 / 执行语义
damage_event_default_fields_by_origin.MONSTER_BASIC.OwnerCredit	MONSTER
damage_event_default_fields_by_origin.MONSTER_ABILITY.DamageForm	SKILL
damage_event_default_fields_by_origin.MONSTER_ABILITY.SourceFaction	MONSTER
damage_event_default_fields_by_origin.MONSTER_ABILITY.ElementRule	ABILITY_DAMAGE_SPEC
damage_event_default_fields_by_origin.MONSTER_ABILITY.HitKind	ABILITY
damage_event_default_fields_by_origin.MONSTER_ABILITY.OwnerCredit	MONSTER
damage_event_default_fields_by_origin.WORLD_STORM.DamageForm	STORM
damage_event_default_fields_by_origin.WORLD_STORM.SourceFaction	WORLD
damage_event_default_fields_by_origin.WORLD_STORM.ElementRule	NONE
damage_event_default_fields_by_origin.WORLD_STORM.HitKind	ENVIRONMENT
damage_event_default_fields_by_origin.WORLD_STORM.OwnerCredit	WORLD
basic_attack.scheduler	FIXED_POINT_TIME_ACCUMULATOR_Q16_MICROSECONDS
basic_attack.hold_input	STATEFUL_HELD_COMMAND; NOT_ONE_NETWORK_REQUEST_PER_ATTACK
basic_attack.when_interval_lt_tick	ACCUMULATE_AND_COMMIT_MULTIPLE_RELEASES_IN_STABLE_SEQUENCE_WITHIN_TICK
basic_attack.per_tick_event_order	PLAYER_ENTITY_ID / LOCAL_ATTACK_ORDINAL
basic_attack.player_facing_attack_speed_cap	NONE
basic_attack.content_build_assertion	THEORETICAL_MAX_RELEASES_PER_ENTITY_PER_TICK_LTE_8; OTHERWISE CONTENT_BUILD_FAILS

E.4 战斗解析

工程字段	唯一值 / 执行语义
server_authoritative	是
reflect_and_dot_guards.REFLECT_CANNOT_TRIGGER_REFLECT	是

工程字段	唯一值 / 执行语义
reflect_and_dot_guards.DOT_CANNOT_TRIGGER_BASIC_ATTACK_PROC	是
reflect_and_dot_guards.SPLASH_CANNOT_SPLASH_AGIN	是
reflect_and_dot_guards.PLAYER_OWNED_SUMMON_CREATE_OWNER_BUT_CANNOT_FARM_PERMANENT_KILL_LAYERS	是
positive_nonimmune_minimum_hp_or_shield_damage	1
periodic.first_tick	START_TICK_PLUS_INTERVAL
periodic.last_tick	TICK_TIME_LESS_THAN_OR_EQUAL_TO_EFFECT_END_TICK
periodic.tick_count	FLOOR(DURATION_S / INTERVAL_S)
periodic.zero_or_partial_interval_at_end	NO_EXTRA_TICK
periodic.same_tick_order	SOURCE_ENTITY_ID_ASC / EFFECT_INSTANCE_ID_ASC / TICK_ORDINAL_ASC
child_damage.default_proc_policy	NO_TRIGGER_SOURCE_DEFINING_PROC_AND_NO_RECURSIVE_CHILD_CHAIN_UNLESS_EXPLICIT_ALLOWLIST
child_damage.default_credit	INHERIT_PARENT_OWNER_CREDIT
child_damage.default_rng	NO_NEW_PROC_DRAW_UNLESS_EXPLICIT
child_damage.explicit_allowlist_authority	PER_EFFECT_TYPED_PAYLOAD

E.5 五行

工程字段	唯一值 / 执行语义
cycle.METAL	WOOD
cycle.WOOD	EARTH
cycle.EARTH	WATER
cycle.WATER	FIRE
cycle.FIRE	METAL
multiplier_if_attacker_counters_target	1.5
multiplier_otherwise	1
no_defender_side_reduction	是
mixed_origin_pearl_G2_if_owner_attacks_and_counters	2
G2_does_not_amplify_incoming_counter_damage	是
player_vs_player	ALL_NORMAL_BASIC_SKILL_DOT_DAMAGE_INHERITS_ATTACKER_ELEMENT; TRUE_PERCENT_STORM_REFLECT_USE_NONE

工程字段	唯一值 / 执行语义
monster_classes_with_element	PIG / DRAGON
monster_classes_without_element	ORDINARY_MELEE / ORDINARY_RANGED / FLYING / LEADER / CORE_BOSS
dragon_and_pig_outgoing	BASIC_AND_ALL_ABILITIES_INHERIT_MONSTER_ELEMENT
bidirectional_example	WATER_PLAYER_TO_FIRE_DRAGON_X1_5; FIRE_DRAGON_TO_WATER_PLAYER_X1_0

E.6 输入

工程字段	唯一值 / 执行语义
logical_view.aspect	16:9
logical_view.world_information_equal_across_platforms	是
mobile.move	LEFT_STICK
mobile.basic	BASIC_BUTTON
mobile.active_tap_AIM_POINT	NEAREST_VISIBLE_LEGAL_ENEMY_FEET_ELSE_FACING_DEFAULT_RANGE
mobile.active_drag	PRECISION_AIM_RELEASE_TO_CAST
mobile.explicit_lock	DRAG_BASIC_BUTTON_TOWARD_TARGET_LOCK_2S
mobile.context_tap	ACTIVATE_HIGHLIGHTED_OR_CYCLE_OVERLAP_CANDIDATES
mobile.context_hold	OPEN_LOOT_CLUSTER_LIST
windows.move	WASD
windows.basic	MOUSE_LEFT
windows.active	MOUSE_RIGHT
windows.precision_hold_s	0.22
windows.precision_pointer_threshold_px	4
windows.interact	E
windows.ui_pointer_events_consume_world_cast	是
windows.right_hold_drag	AFTER_0_22S_SHOW_PREVIEW; RELEASE_CAST; SHORT_CLICK_SMART_CAST
facing_sector_full_angle_deg	24
target_tie_break	MOST_RECENT_VALID_DAMAGE_TO_SELF / SMALLEST_SCREEN_ANGLE / SHORTEST_NAV_DISTANCE / ENTITY_ID
unit_target_requires_direct_visibility	是
unit_target_range_cannot_exceed_legal_direct_vision	是
long_range_aim.presentation_camera_lead_max_m	20
long_range_aim.grants_extra_enemy_information	否
long_range_aim.minimap_ray_allowed	是
chest_channels.NORMAL_B33_s	0.35
chest_channels.AIRDROP_s	3
app_focus_lost	ZERO_MOVE_CANCEL_UNCOMMITTED_CHANNEL_AND_AFTER_3S_USE_DISCONNECT_BOT

工程字段	唯一值 / 执行语义
supported_platforms	ANDROID / IOS / WINDOWS
controller_supported	否
basic_hold.touch	HOLD_BASIC_BUTTON
basic_hold.windows	HOLD_MOUSE_LEFT
basic_hold.repeat	START_NEXT_WINDUP_WHEN_ATTACK_INTERVAL_READY
basic_hold.retargate_each_windup	是
basic_hold.release	STOP_STARTING_NEW_WINDUP; COMMITTED_RELEASE_CONTINUES
modal_damage_close_policy.SHOP	POSITIVE_HP_OR_SHIELD_DAMAGE_FROM_PLAYER_OWNED_SOURCE_ONLY
modal_damage_close_policy.EQUIPMENT_SWAP_CHANNEL	ANY_POSITIVE_HP_OR_SHIELD_DAMAGE
modal_damage_close_policy.ITEM_REPLACE_CONFIRM	ANY_POSITIVE_HP_OR_SHIELD_DAMAGE
modal_damage_close_policy.CHEST_CHANNEL	READ_AIRDROP_OR_NORMAL_CHEST_INTERRUPT_CONTRACT
modal_damage_close_policy.FULL_IMMUNE_OR_DODGE	DOES_NOT_CLOSE_OR_INTERRUPT
windows.interact_tap	E; ACTIVATE_HIGHEST_PRIORITY_OR_CYCLE_OVERLAP_CANDIDATES
windows.interact_hold_s	0.35
windows.interact_hold	OPEN_NEARBY_LOOT_CLUSTER_LIST
windows.inventory_panel_hold	TAB
windows.build_system	NONE
two_target.first_confirm	ENTER_WAIT_SECOND_TARGET_AND_HIGHLIGHT_FIRST
two_target.second_timeout_s	3
two_target.cancel_timeout_or_first_invalid	CANCEL_NO_COOLDOWN
two_target.mobile	FIRST_TAP_CONFIRM_A_THEN_SECOND_TAP_CONFIRM_B; CANCEL_BUTTON_VISIBLE
two_target.windows	LEFT_CLICK_A_THEN_LEFT_CLICK_B; RIGHT_CLICK_OR_ESCAPE_CANCEL
trap.mobile.tap	DETONATE_NEAREST_OWN_TRAP_TO_AIM_VECTOR
trap.mobile.hold	OPEN_MAX_THREE_TRAP_SELECTOR
trap.windows.tap	RIGHT_CLICK_OWN_TRAP_OR_ACTIVE_TAP_DETONATES_CURSOR_NEAREST
trap.windows.hold	RIGHT_HOLD_OPENS_NUMBERED_TRAP_SELECTOR
aim.ground_point_resolution	相机射线与首个合法可走面交点（自上而下逐层）；无交点时钳制到最近合法面边缘
aim.direction_plane	方向技能以角色瞄准高度（胸口 1.2m）水平面解算方向；不随地面坡度扭曲
aim.projectile_3d	弹道一律真实 3D capsule 扫掠；高台上下层按真实高度差命中或越过

工程字段	唯一值 / 执行语义
aim.pc_mobile_identical_solver	是；PC 鼠标与手机拖拽共用同一解算器，仅输入采样不同
multi_interact.d20_press	短按：有己方陷阱→引爆瞄准向量最近陷阱；无陷阱→放置。长按：始终打开陷阱选择器

E.7 视野与信息态

工程字段	唯一值 / 执行语义
base_direct_vision_m	30
direct_vision_requires_line_of_sight	是
wall_blocks_direct_vision_and_target_lock	是
damage_player_reveal.duration_s	3
damage_player_reveal.refresh	是
damage_player_reveal.replication_rate_cap_hz	1
damage_player_reveal.shows_exact_minimap_position	是
fushi.ring_width_m	2
fushi.outside_cannot_see_inside	是
fushi.same_ring_mutual_vision	是
fushi.break_on	BASIC_RELEASE / ACTIVE_CAST_COMMIT / ACTUAL_HP_OR_SHIELD_DAMAGE
fushi.pickup_does_not_break	是
fushi.aoe_blind_cast_allowed	是
stealth.direct_targeting_forbidden_unless_revealed	是
stealth.collision_and_blind_aoe_still_apply	是
stealth.walls_still_block_reveal	是
stealth.episode_id	MONOTONIC_UINT32_PER_PLAYER_INCREMENT_ON_EACH_ENTER_STEALTH
stealth.break_event.fields	player_entity_id / stealth_episode_id / cause / server_tick
stealth.break_event.hostile_action_causes	OWNER_BASIC_RELEASE_AGAINST_HOSTILE / OWNER_HOSTILE_ACTIVE_CAST_COMMIT
stealth.break_event.reveal_is_not_hostile_break	是
stealth.break_event.one_equipment_trigger_per_episode	是
stealth.break_event.hostile_active_definition	TARGETS_HOSTILE_OR_CREATES_PLAYER_OWNED_DAMAGE_CONTROL_TRAP_OR_HOSTILE_SUMMON
stealth.break_event.only_emit_on_actual_state_transition	是
stealth.source_kinds.FUSHI_HIDDEN	COUNTS_AS_STEALTH
stealth.source_kinds.ACTIVE_STEALTH	COUNTS_AS_STEALTH
stealth.source_kinds.EQUIPMENT_STEALTH	COUNTS_AS_STEALTH
stealth.source_kinds.FOG_OF_WAR_ONLY	NOT_STEALTH
stealth.source_kinds.SMOKE_OCCLUSION_ONLY	NOT_STEALTH

工程字段	唯一值 / 执行语义
reveals.DEMON_MIRROR.radius_m	20
reveals.DEMON_MIRROR.duration_s	3
reveals.DEMON_MIRROR.wall_blocked	是
reveals.DEMON_MIRROR.fushi_stone_not_blocker	是
reveals.DEMON_PEARL.radius_m	20
reveals.DEMON_PEARL.duration	WHILE_WORN
reveals.DEMON_PEARL.wall_blocked	是
reveals.CORE_BOSS_EYE.radius_m	10
reveals.CORE_BOSS_EYE.source	LIVING_CORE_BOSS_OR_DEATH_ANCHOR
reveals.CORE_BOSS_EYE.shared_to_all_players_inside_radius	是
reveals.CORE_BOSS_EYE.reveals_all_stealth_and_fushi	是
reveals.CORE_BOSS_EYE.after_boss_death_s	10
reveals.CORE_BOSS_EYE.anchor_at_death_position	是
smoke_H038.enemy_inside_direct_vision_cap_m	2
smoke_H038.blocks_vision_across_boundary	是
smoke_H038.caster_vision_unchanged	是
smoke_H038.true_sight_overrides	是
final_storm.does_not_force_reveal	是
final_storm.does_not_disable_stealth_fushi_or_healing	是
information_state_machine.states	UNKNOWN / VISIBLE / DIRECTION / POSITION_SNAPSHOT / REVEALED
information_state_machine.targetable_states	VISIBLE / REVEALED
information_state_machine.UNKNOWN	NO_WORLD_OR_MINIMAP_ENEMY_POSITION
information_state_machine.VISIBLE	DIRECT_LOS_AND_RANGE; WORLD_RENDER_AND_LEGAL_TARGETING
information_state_machine.DIRECTION	FUZZY_REGION_OR_DIRECTION_ONLY; NO_TARGET_LOCK
information_state_machine.POSITION_SNAPSHOT	MINIMAP_POINT_AT_SNAPSHOT_TICK; NO_TARGET_LOCK; DOES_NOT_FOLLOW
information_state_machine.REVEALED	EXACT_TRACKED_POSITION_FOR_EFFECT_DURATION; TARGETING_STILL_REQUIRES_SKILL_RANGE_AND_WALL_RULE
information_state_machine.no_generic_last_known_ghost	是
information_state_machine.direct_visibility_loss	VISIBLE_TO_UNKNOWN_IMMEDIATELY_UNLESS_AN_EXPLICIT_EFFECT_SEPARATELY_GRANTS_DIRECTION_SNAPSHOT_OR_REVEAL
damage_player_reveal.source	ATTACKER_ENTITY
damage_player_reveal.recipient	EACH_ACTUALLY_DAMAGED_VICTIM_ONLY
damage_player_reveal.information_state	POSITION_SNAPSHOT_STREAM_ON_MINIMAP; NEVER_GRANTS_TARGET_LOCK_OR_WALL_BYPASS

工程字段	唯一值 / 执行语义
damage_player_reveal.refresh_semantics	NEW_ACTUAL_DAMAGE_RESETS_END_TICK_TO_NOW_PLUS_3S
fushi.volume_id	MAP_FUSHI_CANDIDATE_ID
fushi.overlap_mutual_vision	TWO_ENTITIES_MUTUALLY_VISIBLE_IF_THEIR_ACTIVE_FUSHI_VOLUME_ID_SETS_INTERSECT
fushi.overlap_no_common_id	OUTSIDE_CANNOT_SEE_INSIDE_REMAINS
stealth.sources_independent	是；ACTIVE_STEALTH / EQUIPMENT_STEALTH / FUSHI_HIDDEN 各自独立计时与破隐，不互相消耗
stealth.hostile_action_clears_active_and_equipment	敌对普攻释放或敌对主动提交清除 D7 / P13，并使伏石内单位显形 1.5 秒
stealth.fushi_rehide_after_reveal_s	1.5；显形结束时仍在伏石半径内才重新隐藏，否则保持可见
stealth.fushi_not_cleared_by_leaving_combat	伏石隐藏只由位置决定；战斗状态不改变判定
reveal.d12_overrides_stealth_and_fushi	是；D12 精确显形覆盖一切隐身与伏石隐藏
reveal.d12_no_through_wall_lock	是；墙体阻挡直接视线时仅提供位置情报（小地图与轮廓），不可锁定
reveal.smoke_interaction	显形不穿透 H038 烟雾的锁定禁止；烟内目标同样仅位置情报

E.8 移动、飞行、高台与动态实体

工程字段	唯一值 / 执行语义
player_capsule.radius_m	0.45
player_capsule.height_m	1.8
blink.semantic_block_tags	否
blink.maximum_continuous_solid_chord_m	1.5
blink.swept_capsule	是
blink.blocked_result	LAST_LEGAL_POINT
blink.solid_chord_rule	CAPSULE_MAY_CROSS_ONLY_WHEN_CONTINUOUS_INTERSECTION_CHORD_M_LTE_CONFIGURED_MAXIMUM; OTHERWISE_STOP_LAST_LEGAL_POINT
blink.no_map_object_semantic_blink_ban	是
flight_capsule.bottom_height_m	3
flight.whole_capsule_swept	是
flight.forced_land_search_m	6
flight.forced_land_fallback	LAST_LEGAL_GROUND_POINT_IN_THIS_FLIGHT
flight_collision_rule	WHOLE_CAPSULE_PHYSICAL_SWEEP_AGAINST_POLYGON_EXTRUSION_AT_CURRENT_Z

工程字段	唯一值 / 执行语义
flight.no_map_object_semantic_flight_ban	是
movement_class_profiles.GROUND.locomotion_z	FOLLOW_WALKABLE_SURFACE
movement_class_profiles.GROUND.uses_navmesh	是
movement_class_profiles.GROUND.can_use_authored_ramps	是
movement_class_profiles.FLYING.innate	是
movement_class_profiles.FLYING.capsule_bottom_height_above_local_ground_m	3
movement_class_profiles.FLYING.horizontal_motion	PLAYER_INPUT_WITH_CONTINUOUS_3D_CAPSULE_SWEEP
movement_class_profiles.FLYING.can_cross_solid_only_when_entire_capsule_clears_extruded_height	是
movement_class_profiles.FLYING.does_not_require_ramp	是
movement_class_profiles.FLYING.ground_aoe_and_unit_targeting	NORMAL_UNLESS_EFFECT_EXPLICITLY_EXCLUDES_FLYING
movement_class_profiles.FLYING.ground_interaction_distance	HORIZONTAL_SURFACE_GAP; VERTICAL_OFFSET_IGNORED_FOR_LOOT_SHOP_CHEST
movement_class_profiles.FLYING.forced_land_from_G4_damage_rule	否
movement_class_profiles.TEMPORARY_EQUIPMENT_FLIGHT.eligible_owner_class	GROUND_ONLY
movement_class_profiles.TEMPORARY_EQUIPMENT_FLIGHT.capsule_bottom_height_above_local_ground_m	3
movement_class_profiles.TEMPORARY_EQUIPMENT_FLIGHT.enter_after_out_of_combat_s	5
movement_class_profiles.TEMPORARY_EQUIPMENT_FLIGHT.forced_land_on_actual_damage_dealt_or_received	是
movement_class_profiles.TEMPORARY_EQUIPMENT_FLIGHT.G4_on_innate_flying_owner	MOVE_SPEED_STAT_ONLY; NO_SECOND_FLIGHT_STATE
combat_volume_conventions.unit_target_range	HORIZONTAL_CAPSULE_SURFACE_GAP_LTE_RANGE_M
combat_volume_conventions.ground_point_range	HORIZONTAL_DISTANCE_FROM_CASTER_CAPSULE_CENTER_PROJECTION_LTE_RANGE_M
combat_volume_conventions.circle_hit	TARGET_HORIZONTAL_CAPSULE_OVERLAPS_CIRCLE; BOUNDARY_INCLUSIVE
combat_volume_conventions.line_width	FULL_WIDTH_METERS_CENTERED_ON_LINE; CAPSULE_OVERLAP; BOUNDARY_INCLUSIVE
combat_volume_conventions.cone_angle	FULL_INCLUDED_ANGLE_DEGREES; CAPSULE_OVERLAP; BOUNDARY_INCLUSIVE
combat_volume_conventions.ring_radius	CENTERLINE_RADIUS; WALL_THICKNESS_EXTENDS_EQUALLY_INWARD_AND_OUTWARD
combat_volume_conventions.default_shape_vertical_span	FROM_LOCAL_GROUND_Z_MINUS_0_5M_TO_LOCAL_GROUND_Z_PLUS_6_0M
combat_volume_conventions.flying_is_hit_by_default_shapes	是

工程字段	唯一值 / 执行语义
combat_volume_conventions.direct_los	3D_SEGMENT_EYE_OR_PROJECTILE_ORIGIN_TO_TARGET_CAPSULE_AIM_POINT_AGAINST_EXTRUDED_SOLIDS
projectile_collision.continuous_sweep_each_20hz_tick	是
projectile_collision.basic_projectile_radius_m	0.12
projectile_collision.active_projectile_default_radius_m	0.2
projectile_collision.monster_projectile_default_radius_m	0.2
projectile_collision.first_contact_order	DISTANCE_ALONG_SWEEP_ASC_THEN_STATIC_SOLID_BEFORE_ACTOR_THEN_ENTITY_ID_ASC
projectile_collision.tunneling_through_thin_wall_or_actor	FORBIDDEN
map3d.base_ground_surface	TRIANGULATED_PLAYABLE_BOUNDARY_AT_Z0_EXCLUDING_STATIC_SOLID_FOOTPRINTS
map3d.highland_top_surfaces	AUTHORED_POLYGONS_AT_TABLE_D69_Z
map3d.ramp_surfaces	AUTHORED_RAMP_STRIPPS_WITH_LINEAR_Z_BY_CENTERLINE_PROGRESS
map3d.minimum_vertical_clearance_m	2.4
map3d.static_wall_thickness	FOOTPRINT_POLYGON_AREA_IS_THE_THICKNESS; NO_ZERO_WIDTH_WALLS
map3d.occluder_and_collider_source	SAME_EXTRUDED_VOLUME; NO_SEPARATE_ART_ONLY_OCCLUDER
map3d.ground_material_default	STONE_DUST
map3d.ground_material_region_override	REGION_ID_TO_FOOTSTEP_MATERIAL_CLOSED_TABLE; COSMETIC_ONLY
map3d.dressing_exclusion	NAV_CORRIDOR_PLUS_0_75M / INTERACTION_PAD / SPAWN_AND_REVIVE_6M / BOSS_LANDING / DYNAMIC_SOLID_RESERVATION
region.authority	AUTHORED_SHARED_EDGE_POLYGON_TABLE_9_1A
region.point_query	POINT_IN_POLYGON; SHARED_EDGE_TIE=REGION_ID_ASC
region.coverage_assertion	UNION_EQUALS_PLAYABLE_BOUNDARY_AND_PAIRWISE_INTERIOR_OVERLAP_AREA_ZERO
dynamic_wall.trapped_repath	墙体生成/销毁的下一 tick 强制全体受影响实体重新寻路
dynamic_wall.no_path_1s_fallback	连续 1 秒无合法路径→回退到最后合法 Nav 点（沿历史轨迹反向逐点验证）
dynamic_wall.never_teleport_through	是；任何回退不得穿越实体碰撞，逐段 capsule sweep 验证
nav_agent.PLAYER	radius=0.5m; height=1.8m; step=0.4m; slope=45deg; turn=INSTANT
nav_agent.ORDINARY_MONSTER	radius=0.5m; height=1.8m; step=0.4m; slope=45deg; turn=360deg/s
nav_agent.FLYING	radius=0.4m; height=1.4m; step=IGNORED_FLIGHT; slope=IGNORED_FLIGHT; turn=360deg/s

工程字段	唯一值 / 执行语义
nav_agent.PIG	radius=0.8m; height=1.6m; step=0.4m; slope=45deg; turn=300deg/s
nav_agent.LEADER	radius=1.2m; height=2.4m; step=0.5m; slope=40deg; turn=240deg/s
nav_agent.DRAGON	radius=1.5m; height=2.8m; step=0.6m; slope=40deg; turn=200deg/s
nav_agent.CORE_BOSS	radius=2.0m; height=3.6m; step=0.6m; slope=35deg; turn=160deg/s
nav_agent.SUMMON	radius=各原型碰撞半径 (E.34) ; height=各原型碰撞高; step=0.4m; slope=45deg; turn=360deg/s
nav_agent.bake_rule	每代理半径独立烘焙 NavMesh 层; 运行时禁止用半径修正代替独立层

E.9 动态实体放置

工程字段	唯一值 / 执行语义
applies_to	H031_PAGODA / H036_FROZEN_RIVER / AIR_DROP_RESERVED_VOLUME
query_order	CLAMP_OR_CANCEL_BY_SKILL_RANGE_RULE / BUILD_PROPOSED_3D_SOLID / REJECT_MAP_BOUNDARY_OR_EXISTING_SOLID_PENETRATION / REJECT_ANY_LIVING_ACTOR_CAPSULE_OVERLAP / REJECT_SHOP_INTERACTION_PAD / REJECT_CHEST_INTERACTION_PAD_OR_AIRDROP_RESERVATION / REJECT_BOSS_SPAWN_BODY_AND_MIGRATION_LANDING_VOLUME / REJECT_IF_CLOSES_LAST_NAV_PATH_OF_ANY_OCCUPIED_ENCLOSURE / COMMIT_ENTITY
invalid_cast	NO_ENTITY; NO_COOLDOWN; NO_RESOURCE_COST; CLIENT_SHOWS_INVALID_PREVIEW
H031_overlap_at_formation.telegraph_s	0.6
H031_overlap_at_formation.cast_commit	RESERVE_RING_VOLUME_WITHOUT_SOLID_COLLISION
H031_overlap_at_formation.formation_revalidate	AT_TELEGRAPH_END_REJECT_IF_LAST_NAV_PATH_WOULD_CLOSE
H031_overlap_at_formation.actor_resolution	FOR_EACH_OVERLAPPING_CAPSULE_ENTITY_ID_ASC_PROJECT_TO_NEAREST_LEGAL_POINT_ON_INNER_O R_OUTER_SIDE
H031_overlap_at_formation.distance_tie	INNER_SIDE_WINS
H031_overlap_at_formation.no_legal_point_within_3m	CANCEL_FORMATION_AND_REFUND_COOLDOWN
H036.semantic_blink_block	否
H036.semantic_flight_block	否
H036.physical_collision_only	是
destroyed_or_expired	REBUILD_LOCAL_NAV_AND_REVALIDATE_PENDING_PATHS_NEXT_TICK
D20.solid	否
D20.visible_trigger_entity	是
D20.uses_solid_placement_query	否

E.10 召唤物

工程字段	唯一值 / 执行语义
global.inherit	OWNER_FACTION / OWNER_ELEMENT / OWNER_FATAL_CREDIT
global.does_not_inherit_by_default	CRIT / BASIC_ATTACK_PASSIVES / OWNER_EQUIPMENT_STATS / EXTRA_FOG_VISION
global.no_gold_or_xp_on_death	是
global.monster_legal_target	是
global.target_selection.recompute_s	0.25
global.target_selection.must_be_directly_visible	是
global.target_selection.must_be_nav_reachable_for_m elee	是
global.target_selection.within_aggro_range	是
global.target_selection.priority_then_tie_break	PRIORITY_INDEX_ASC / NAV_PATH_DISTANCE_MM_ASC / ENTITY_ID_ASC
global.hit_and_movement_volume_required	是
STONE_ARHAT.hp	500
STONE_ARHAT.attack	30
STONE_ARHAT.attack_speed_per_s	0.7
STONE_ARHAT.range_m	1.5
STONE_ARHAT.move_speed_mps	2.2
STONE_ARHAT.aggro_range_m	8
STONE_ARHAT.leash_m	12
STONE_ARHAT.threat_bonus	300
STONE_ARHAT.target_priority	MONSTER_ATTACKING_OWNER / NEAREST_MONSTER / NEAREST_ENEMY_PLAYER
STONE_ARHAT.collision_radius_m	0.5
STONE_ARHAT.collision_height_m	1.6
STONE_ARHAT.movement_collision	是
BEAN_SOLDIER.hp	150
BEAN_SOLDIER.attack	15
BEAN_SOLDIER.attack_speed_per_s	1
BEAN_SOLDIER.range_m	1.5
BEAN_SOLDIER.move_speed_mps	2.8
BEAN_SOLDIER.aggro_range_m	8
BEAN_SOLDIER.leash_m	15
BEAN_SOLDIER.target_priority	VISIBLE_ENEMY_PLAYER / MONSTER_ATTACKING_OWNER / NEAREST_LEGAL_ENEMY
BEAN_SOLDIER.collision_radius_m	0.35
BEAN_SOLDIER.collision_height_m	1.2
BEAN_SOLDIER.movement_collision	是
SIX_EAR_DECOY.attack	0
SIX_EAR_DECOY.damage_enabled	否

工程字段	唯一值 / 执行语义
SIX_EAR_DECOY.ai.owner_history_delay_s_by_spawn_index	0.2 / 0.35
SIX_EAR_DECOY.ai.lateral_offset_m_by_spawn_index	-1.2 / 1.2
SIX_EAR_DECOY.ai.position_target	OWNER_HISTORY_POSITION_PLUS_NAV_PROJECTED_LATERAL_OFFSET
SIX_EAR_DECOY.ai.movement_speed_mps	OWNER_CURRENT_MOVE_SPEED_MPS
SIX_EAR_DECOY.ai.fake_basic_interval_s	1.25
SIX_EAR_DECOY.ai.fake_basic_target	OWNER_COMMITTED_BASIC_TARGET_IF_VISIBLE_ELSE_NONE
SIX_EAR_DECOY.ai.fake_basic_deals_damage	否
SIX_EAR_DECOY.ai.rng_draws	0
SIX_EAR_DECOY.collision_radius_m	0.45
SIX_EAR_DECOY.collision_height_m	1.8
SIX_EAR_DECOY.movement_collision	否
SIX_EAR_DECOY.damageable_hit_volume	是
B37.eligible_destroy_causes	HOSTILE_PLAYER_DAMAGE / HOSTILE_MONSTER_DAMAGE
B37.ineligible_removal_causes	WORLD_DAMAGE / NATURAL_EXPIRY / REPLACEMENT / OWNER_TRUE_DEATH_CLEANUP / OWNER_ELIMINATION_CLEANUP
B39.move_speed_mps	3
B39.collision	RADIUS_0_6M_HEIGHT_1_8M
B39.aggro	NONE
B39.targetable	YES_NO_REWARD
summon.runtime_authority	E.34_STRONG_TYPED_PAYLOAD_TABLE
summon.stat_duplication_rule	第 7 章/E.26/11.3 中出现的召唤物数值仅为投影；构建器逐字段比对其与 E.34 (B12↔WOLF_SPIRIT / B13↔FIRE_SPIRIT / B39↔STONE_STATUE / H020↔STONE_ARHAT / D19↔BEAN_SOLDIER / CORE_V6↔CORE_MIRROR) ， 任何不一致即构建失败

E.11 天劫

工程字段	唯一值 / 执行语义
normal.before_s	1200
normal.check_interval_s	3
normal.base_hit_probability	0.5
normal.base_damage_max_hp_percent	20
normal.stun_s	0.5
normal.targets	ALIVE_PLAYER_OUTSIDE_SAFE_CIRCLE

工程字段	唯一值 / 执行语义
normal.does_not_target	MONSTER / PLAYER_OWNED_SUMMON / SOUL / REVIVE_PROTECTED
normal.B43_probability_by_level	0.4 / 0.35 / 0.3 / 0.25 / 0.2
normal.B11_remaining_damage	0.7
normal.G3_remaining_damage	0.3
normal.B11_G3_stack	MULTIPLICATIVE_REMAINING=0.21
normal.other_percent_or_fixed_reduction	否
normal.shieldable	是
normal.invulnerability_blocks	是
normal.active_interval_s_half_open	0 / 1200
normal.event_at_exact_end_s	NONE
apocalypse.start_s	1200
apocalypse.tick_s	1
apocalypse.initial_max_hp_percent	2
apocalypse.add_percentage_points_every_s	5
apocalypse.add_percentage_points	1
apocalypse.no_hit_probability	是
apocalypse.stun_s	0
apocalypse.B43_affects	否
apocalypse.B11_remaining_damage	0.7
apocalypse.G3_remaining_damage	0.3
apocalypse.B11_G3_stack	MULTIPLICATIVE_REMAINING=0.21
apocalypse.other_percent_or_fixed_reduction	否
apocalypse.shieldable	是
apocalypse.invulnerability_blocks	是
apocalypse.does_not_reveal_or_disable_stealth	是
apocalypse.targets	ALL_NON_ELIMINATED_PLAYER_BODIES_NOT_SOUL_OR_REVIVE_PROTECTED / LIVING_CORE_BOSS
apocalypse.does_not_target	ORDINARY_MONSTER / FLYING_MONSTER / PIG / LEADER / DRAGON / PLAYER_OWNED_SUMMON
apocalypse.core_boss_damage_multiplier_of_player_value	0.5
apocalypse.starts_globally_regardless_of_position	是
apocalypse.core_boss_damage_formula	$\text{FLOOR}(\text{CURRENT_EVENT_PERCENT}/100 * \text{CORE_BOSS_MAX_HP_SNAPSHOT} * 0.5)$
apocalypse.phase_enter_s	1200
apocalypse.first_damage_event_s	1201
apocalypse.damage_event_times_s	1201_PLUS_NONNEGATIVE_INTEGER
apocalypse.percent_at_event	$2 + \text{FLOOR}((\text{EVENT_TIME_S}-1200)/5)$
position_sample	PLAYER_CAPSULE_CENTER_AT_STORM_EVENT_TICK

工程字段	唯一值 / 执行语义
boundary_counts_as_safe	是
safe_radius_zero_semantics	NO_GEOMETRIC_SAFE_POINT_EXISTS_FOR_PLAYER_TARGETING_OR_GROUND_ITEM_DOOM; BOUNDARY_COUNTS_AS_SAFE_ONLY_WHEN_RADIUS_GT_ZERO
apocalypse.core_boss_max_hp_snapshot_at_event	CURRENT_COMMITTED_MAX_HP_BEFORE_EVENT
apocalypse.core_boss_damage_form	STORM; SHIELDABLE=YES; ONLY_B11_G3_REMAINING_MULTIPLIERS_APPLY

E.12 怪物 AI

工程字段	唯一值 / 执行语义
monster.machine.states	IDLE_PATROL / ALERT_TELEGRAPH / CHASE / ATTACK_OR_CAST / RETURN_HOME / SPAWN_PROTECTED / STORM_EXIT_PENDING
monster.machine.transitions[0].from_states	IDLE_PATROL
monster.machine.transitions[0].to	ALERT_TELEGRAPH
monster.machine.transitions[0].guard_id	LEGAL_TARGET_ENTERS_DETECTION_OR_DEALS_DAMAGE
monster.machine.transitions[1].from_states	ALERT_TELEGRAPH
monster.machine.transitions[1].to	CHASE
monster.machine.transitions[1].guard_id	ALERT_FINISHED_FOR_ARCHETYPE
monster.machine.transitions[2].from_states	CHASE
monster.machine.transitions[2].to	ATTACK_OR_CAST
monster.machine.transitions[2].guard_id	TARGET_IN_RANGE_LOS_AND_ACTION_READY
monster.machine.transitions[3].from_states	ATTACK_OR_CAST
monster.machine.transitions[3].to	CHASE
monster.machine.transitions[3].guard_id	ACTION_COMMIT_OR_CANCEL_FINISHED
monster.machine.transitions[4].from_states	ALERT_TELEGRAPH / CHASE / ATTACK_OR_CAST
monster.machine.transitions[4].to	RETURN_HOME

工程字段	唯一值 / 执行语义
monster.machine.transitions[4].guard_id	TARGET_INVALID_OR_LEASH_BROKEN_OR_UNREACHABLE_2S
monster.machine.transitions[5].from_states	RETURN_HOME
monster.machine.transitions[5].to	IDLE_PATROL
monster.machine.transitions[5].guard_id	WITHIN_1M_HOME_AND_FULL_HP
monster.machine.transitions[6].from_states	SPAWN_PROTECTED
monster.machine.transitions[6].to	IDLE_PATROL
monster.machine.transitions[6].guard_id	SPAWN_PROTECTION_1S_FINISHED
monster.machine.transitions[7].from_states	IDLE_PATROL / ALERT_TELEGRAPH / CHASE / ATTACK_OR_CAST / RETURN_HOME / SPAWN_PROTECTED
monster.machine.transitions[7].to	STORM_EXIT_PENDING
monster.machine.transitions[7].guard_id	CLASS_STORM_COVERAGE_EXIT_CONDITION_STARTED
monster.machine.transitions[8].from_states	STORM_EXIT_PENDING
monster.machine.transitions[8].to	IDLE_PATROL
monster.machine.transitions[8].guard_id	CLASS_ALLOWS_REENTRY_AND_REENTRY_RESET_BEFORE_EXIT
return_home.hard_control_immune	是
return_home.damage_reduction_percent	60
return_home.heal_delay_s	1.5
return_home.heal_max_hp_percent_per_s	8
return_home.can_be_killed	是
group_scheduler.first_basic_uniform_s	0 / 0.5
group_scheduler.same_group_impact_gap_s	0.15
group_scheduler.ready_priority	SIGNATURE / MOBILITY / COMMON_AOE / BASIC
group_scheduler.six_melee_slots_per_player	是
sound_aggro.same_or_linked_nest_radius_m	12
sound_aggro.requires_nav_reachable	是
sound_aggro.does_not_cross_more_than_one_junction	是
all_non_pig_monsters_proactively_attack	是
pig_proactively_warns_charges_then_fights_at_nest_end	是

工程字段	唯一值 / 执行语义
storm_exit.enter	PER_CLASS_COVERAGE_CONDITION_FROM_MONSTER_COMBAT_CONTRACT
storm_exit.during	KEEP_CURRENT_AI_UNTIL_EXIT_COMMIT_UNLESS_DRAGON_SPECIAL_NONINTERACTIVE_WINDOW
storm_exit.cancel	LEGAL_REENTRY_BEFORE_EXIT_COMMIT_WHEN_CLASS_ALLOWS_REENTRY_RESET
storm_exit.commit	REMOVE_ENTITY_NO_DEATH_NO_DROP_NO_GOLD_NO_XP_NO_KILL_OR_PERMANENT_LAYER
guard_registry.ALERT_FINISHED_FOR_ARCHETYPE	DEFAULT_0_35S; PIG_0_6S_CHARGE_WARNING
guard_registry.CLASS_STORM_COVERAGE_EXIT_CONDITION_STARTED	§ 5 战斗规则
guard_registry.CLASS_ALLOWS_REENTRY_AND_REENTRY_RESET_BEFORE_EXIT	ORDINARY_FLYING_LEADER_PIG_ONLY
transition_encoding.from_states	NONEMPTY_ARRAY_OF_DECLARED_STATE_IDS
transition_encoding.to	ONE_DECLARED_STATE_ID
transition_encoding.guard_id	ONE_CLOSED_GUARD_ID
transition_encoding.wildcards	FORBIDDEN
monster.visual_selection.pool	MATCH_MECHANICAL_ROLE_AND_ALLOWED_REGION_THEME
monster.visual_selection.algorithm	PER_ROLE_PER_REGION_SHUFFLE_BAG
monster.visual_selection.candidate_sort	MVS_ID_ASC_BEFORE_FISHER_YATES
monster.visual_selection.ordinary_and_flying_draw_at	SPAWN_COMMIT
monster.visual_selection.no_immediate_repeat_when_pool_gt_1	是
monster.visual_selection.pig_and_dragon	ELEMENT_LOCKED_VISUAL_VARIANT
monster.visual_selection.core_boss	MATCH_LOCKED_VARIANT_DRAWN_ONCE

E.13 怪物战斗

工程字段	唯一值 / 执行语义
catalog_signature_cd_is_authoritative	是
scheduler_cd_only_default_when_signature_cd_missing	是
respawn.avoid_all_living_player_direct_vision_first	是
respawn.degraded_after_no_point_s	5
respawn.degraded_requires	AVOID_KILLER_DIRECT_VISION / DISTANCE_FROM_ANY_PLAYER_AT_LEAST_12M
respawn.covered_node_cancels_pending_respawn_immediately	是
respawn.scheduled_respawn_query	AT_DUE_TICK_BUILD_SORTED_LEGAL_CANDIDATE_SET_ONCE

工程字段	唯一值 / 执行语义
respawn.no_candidate_retry_interval_s	1
respawn.no_candidate_first_5s	NO_RNG_DRAW;REQUERY_STRICT_CANDIDATES_EACH_SECOND
respawn.after_5s	REQUERY_DEGRADED_CANDIDATES_EACH_SECOND;DRAW_ONCE_ONLY_WHEN_NONEMPTY
respawn.candidate_sort	POINT_ID_ASC_BEFORE_ONE_UNIFORM_MONSTER_RNG_DRAW
threat.damage_weight	1
threat.proximity_score_range	0 / 200
threat.reachable_score	100
threat.unreachable_score	0
threat.decay_percent_per_s	10
threat.recompute_s	0.25
threat.tie_break	NEAREST_NAV_DISTANCE / ENTITY_ID
threat.summon_inherits_owner_threat_rate	0.6
threat.stone_arhat_fixed_taut	300
legal_targets	PLAYER / PLAYER_OWNED_DAMAGEABLE_SUMMON
friendly_fire	否
damage_taxonomy.required_fields	DamageForm / SourceFaction / ElementRule / HitKind / OwnerCredit
damage_taxonomy.ordinary_leader_core_element_rule	NONE
damage_taxonomy.dragon_and_pig_element_rule	BIDIRECTIONAL_FIVE_ELEMENT
core_boss.active_count	1
core_boss.first_spawn_s	300
core_boss.respawn_s	300
core_boss.court_rotation	SHUFFLE_3_COURTS_NO_IMMEDIATE_REPEAT
core_boss.variant	DRAW_ONE_OF_SIX_AT_MATCH_CREATION_AND_KEEP_FOR_ALL_RESPAWNS
core_boss.variant_changes_on_respawn	否
core_boss.eye_reveal_radius_m	10
core_boss.eye_shared_to_players_in_radius	是
core_boss.death_eye_anchor_s	10
core_boss.final_court_migration	AT_720_REVEAL_FINAL_COURT; ALIVE_NONFINAL_BOSS_REMAINS_KILLABLE_UNTIL_ITS_COURT_STORM_COVER;THEN_MIGRATE; DEAD_TIMER_DUE_BEFORE_1140_RESPAWNS_FINAL_COURT
core_boss.final_respawn_due_at_or_after_s	1140
core_boss.final_respawn_result	NO_RESPAWN
core_boss.forced_final_migration.eligible	LIVING_CORE_BOSS_IN_NONFINAL_COURT_AFTER_FINAL_COURT_REVEAL
core_boss.forced_final_migration.coverage_test_hz	20
core_boss.forced_final_migration.coverage_test	EVERY_VERTEX_OF_CURRENT_COURT_NAV_POLYGON_DISTANCE_TO_SAFE_CENTER_MM_GT_SAFE_RADIUS_MM
core_boss.forced_final_migration.trigger	FIRST_TICK_COVERAGE_TEST_TRUE
core_boss.forced_final_migration.same_tick_priority	PLAYER_AND_MONSTER_HITS_AND_LETHAL_COMMIT_BEFORE_SCHEDULED_MIGRATION;DEATH_WINS

工程字段	唯一值 / 执行语义
core_boss.forced_final_migration.on_trigger	CLEAR_THREAT; BECOME_UNTARGETABLE_AND_NONCOLLIDING; START_1S_SKY_TRANSIT
core_boss.forced_final_migration.transit_s	1
core_boss.forced_final_migration.landing	FINAL_COURT_CENTER_AUTHORED_BOSS_POINT
core_boss.forced_final_migration.landing_invulnerable_s	1
core_boss.forced_final_migration.preserve	CURRENT_HP_INTEGER / MAX_HP_SNAPSHOT / REMAINING_DEBUFF_TIMERS / REMAINING_ABILITY_COOLDOWNS / VARIANT
core_boss.forced_final_migration.does_not_drop	是
core_boss.forced_final_migration.storm_during_transit	NOT_A_BODY; NOT_TARGETED
core_boss.random_targeting.candidate_order	THREAT_DESC_THEN_ENTITY_ID_ASC
core_boss.random_targeting.random_candidate_draw	UNIFORM_INDEX_USING_BOSS_STREAM_AFTER_CANDIDATE_LIST_COMMIT
core_boss.random_targeting.cast_position_snapshot	TARGET_CAPSULE_CENTER_AT_CAST_COMMIT
core_boss.random_targeting.target_invalid_before_commit	NO_CAST_NO_COOLDOWN_NO_RNG_DRAW
core_boss.random_targeting.target_invalid_after_commit	EFFECT_CONTINUES_AT_SNAPSHOT_POSITION
core_boss.final_respawn_due_window	DUE_TICK_LT_1140_RESPAWNS_FINAL; DUE_TICK_GTE_1140_NO_RESPAWN
core_boss.death_cutoff_window	DEATH_TICK_LT_840_STARTS_300S_TIMER; DUE_TICK_LT_1140_RESPAWNS_FINAL; OTHERWISE_NO_TIMER_OR_RESPAWN
core_boss.death_starts_respawn_timer_only_if_death_tick_lt_s	840
gold_dragon.effect	REDUCE_TARGET_PERCENT_REDUCTION_BY_10_PERCENTAGE_POINTS_MIN_0
core_dust.outside_cannot_see_or_lock_inside	是
core_dust.inside_to_inside_uses_normal_wall_vision	是
core_dust.public_boss_hp_remains	是
pig_element_distribution	FISHER_YATES_FIVE_ELEMENTS_WITH_MONSTER_RNG; FIRST_THREE_GET_TWO_TOKENS_EACH; LAST_TWO_GET_ONE_EACH
leader_fields.field_count	4
leader_fields.initial_roles	TANK / TANK / DAMAGE / DAMAGE
leader_fields.assignment	SHUFFLE_FIELDS_ONCE_AT_MATCH_CREATION
leader_fields.role_locked_for_match	是
leader_fields.respawn_s	180
leader_fields.visual.draw_at	EACH_SPAWN_COMMIT
leader_fields.visual.pool	MONSTER_VISUAL_REGISTRY_ROWS_WITH_MECHANICAL_TIER_MATCHING_ASSIGNED_ROLE
leader_fields.visual.method	UNIFORM_USING_MONSTER_STREAM_SORTED_BY_ASSET_ID
leader_fields.visual.mechanical_role_never_changes	是
leader_fields.element	NONE
first_ability_delay_s.leader_uniform	1.5 / 2.5
first_ability_delay_s.dragon_uniform	2 / 3

工程字段	唯一值 / 执行语义
first_ability_delay_s.core_ring	5
first_ability_delay_s.core_signature	10
ability_ready_priority	SIGNATURE / MOBILITY / COMMON_AOE
ready_but_not_cast_gets_next_priority	是
same_group_impact_gap_s	0.15
pig.active_count	8
pig.nest_count	12
pig.first_spawn_s	0
pig.respawn_s	120
pig.element_multiset	2_2_2_1_1; SHUFFLE_FIVE_ELEMENTS; TOKEN_RETAINS_ELEMENT_ACROSS_NEST_MOVE
pig.nest_assignment	8_DISTINCT_NESTS_UNIFORM_WITHOUT_REPLACEMENT
pig.on_death	TOKEN_TIMER_THEN_UNIFORM_INACTIVE_LEGAL_NEST; OLD_NEST_BECOMES_INACTIVE
pig.no_immediate_same_nest	是
pig.escape	ON_FIRST_DAMAGE_RUN_ALONG_AUTHORED_ESCAPE_PATH_TO_NEST_END_THEN_TURN_AND_FIGHT
pig.proactive_aggro	是
pig.element_multiset_algorithm	SORT_ELEMENTS_BY_ENUM; FISHER_YATES_MONSTER_RNG; FIRST_THREE_COUNT2_LAST_TWO_COUNT1; ASSIGN_TOKENS_BY_TOKEN_ID_ASC; TOKEN_RETAINS_ELEMENT_ACROSS_NEST_MOVE
pig.respawn_no_legal_nest	RETRY_USING_GLOBAL_RESPAWN_QUERY; IF_ALL_12_NESTS_PERMANENTLY_STORM_COVERED_CANCEL_TOKEN_FOR_MATCH
dragon.active_count	2
dragon.arena_count	5
dragon.window_start_s	300
dragon.respawn_s	240
dragon.active_arenas_distinct	是
dragon.on_death_arena	UNIFORM_INACTIVE_ARENA; NEVER_SAME_AS_PREVIOUS
dragon.element	INDEPENDENT_UNIFORM_FIVE_ELEMENTS_ON_EACH_SPAWN; REPEAT_ALLOWED
dragon.alive_at_900	REMAIN_KILLABLE_UNTIL_ARENA_FULLY_STORM_COVERED_THEN_5S_NONINTERACTIVE_NO_DROP_EXIT
dragon.death_starts_timer_only_if_death_s_lt	660
dragon.respawn_commit_requires_due_s_lt	900
dragon.pending_due_at_or_after_900	CANCEL_WITHOUT_RNG_DRAW
dragon.spawn_commit_requires_legal_inactive_arena_not_fully_storm_covered	是
dragon.respawn_timer_rule	START_240S_TIMER_IFF_DEATH_TICK_LT_660_SO_DUE_TICK_IS_STRICTLY_LT_900
spawn_protection.duration_s	1
spawn_protection.untargetable	是
spawn_protection.cannot_attack	是
spawn_protection.vfx_required	是

工程字段	唯一值 / 执行语义
storm_coverage_lifecycle.ordinary_flying_leader_pig	FULL_BODY_OUTSIDE_SAFE_CIRCLE_CONTINUOUS_60S_THEN_NONINTERACTIVE_EXIT_NO_DROP_NO_CREDIT
storm_coverage_lifecycle.reentry_before_60s	RESET_TIMER_TO_ZERO
storm_coverage_lifecycle.dragon_override	ARENA_FULLY_COVERED_THEN_5S_NONINTERACTIVE_EXIT_NO_DROP_NO_CREDIT
storm_coverage_lifecycle.core_override	USE_CORE_BOSS_FINAL_MIGRATION_CONTRACT
storm_coverage_lifecycle.normal_storm_never_directly_damages_monsters	是
storm_coverage_lifecycle.no_cross_region_migration_for_ordinary_monsters	是
collision_and_projectile.ranged_basic_projectile_radius_m	0.2
collision_and_projectile.continuous_sweep_each_server_tick	是
collision_and_projectile.melee_range_metric	HORIZONTAL_CAPSULE_SURFACE_GAP
collision_and_projectile.aoe_geometry	§ 5 战斗规则
stage_multiplier.time_map[0]	[0,300)S_INDEX_0
stage_multiplier.time_map[1]	[300,600)S_INDEX_1
stage_multiplier.time_map[2]	[600,900)S_INDEX_2
stage_multiplier.time_map[3]	[900,MATCH_END)S_INDEX_3
stage_multiplier.snapshot	READ_CURRENT_MATCH_TIME_AT_DAMAGE_OR_MAX_HP_RECALC_EVENT
PIG_METAL.basic_attack_enabled	否
PIG_METAL.charge_damage_authority	PIG_CHARGE.damage=80
PIG_WOOD.basic_attack_enabled	否
PIG_WOOD.charge_damage_authority	PIG_CHARGE.damage=80
PIG_WATER.basic_attack_enabled	否
PIG_WATER.charge_damage_authority	PIG_CHARGE.damage=80
PIG_FIRE.basic_attack_enabled	否
PIG_FIRE.charge_damage_authority	PIG_CHARGE.damage=80
PIG_EARTH.basic_attack_enabled	否
PIG_EARTH.charge_damage_authority	PIG_CHARGE.damage=80
phase_transition.max_hp_rebase	阶段切换 tick 重算最大生命=基础 HP×新阶段倍率；当前生命按切换前百分比等比重设，FLOOR 落地，至少 1
phase_transition.dead_entity_spawn	已死亡实体重生时按重生提交 tick 所处阶段生成
phase_transition.mid_combat_allowed	是；战斗中同样在整分钟 tick 执行，不延迟
skill_fixed_damage_phase_scaling	怪物技能固定伤害×该原型攻击阶段倍率（11.2 攻击阶段倍率列）；Q4 乘法后

工程字段	唯一值 / 执行语义
	FLOOR
skill_cd_phase_multiplier.LEADER	1 / 1 / 0.95 / 0.90
skill_cd_phase_multiplier.DRAGON_MINI_BOSS	1 / 1 / 0.90 / 0.85
skill_cd_phase_multiplier.CORE_BOSS	1 / 1 / 0.90 / 0.85
skill_cd_phase_multiplier.applies_at	施放完成进入 CD 的 tick 按当时阶段倍率一次性折算；已在走的 CD 不回溯

E.14 经济掉落

工程字段	唯一值 / 执行语义
last_hit.monster_gold_xp	FINAL_DAMAGE_OWNER_ONLY
last_hit.core_boss_gold_xp	FINAL_DAMAGE_OWNER_ONLY
last_hit.monster_final_blow_on_player	MONSTER_CREDIT_NO_PLAYER_REWARD
last_hit.world_kills_monster	NO_GOLD_XP_BUT_PUBLIC_ITEM_DROPS_STILL_ROLL
monster_tables.ORDINARY_OUTER.gold	90
monster_tables.ORDINARY_OUTER.xp	28
monster_tables.ORDINARY_OUTER.book_chance_percent	10
monster_tables.ORDINARY_OUTER.gem_chance_percent	30
monster_tables.ORDINARY_MIDDLE.gold	140
monster_tables.ORDINARY_MIDDLE.xp	40
monster_tables.ORDINARY_MIDDLE.book_chance_percent	10
monster_tables.ORDINARY_MIDDLE.gem_chance_percent	30
monster_tables.ORDINARY_INNER.gold	180
monster_tables.ORDINARY_INNER.xp	56
monster_tables.ORDINARY_INNER.book_chance_percent	10
monster_tables.ORDINARY_INNER.gem_chance_percent	30
monster_tables.FLYING_OUTER.gold	140
monster_tables.FLYING_OUTER.xp	40
monster_tables.FLYING_OUTER.book_chance_percent	20
monster_tables.FLYING_OUTER.gem_chance_percent	30
monster_tables.FLYING_MIDDLE.gold	200

工程字段	唯一值 / 执行语义
monster_tables.FLYING_MIDDLE.xp	56
monster_tables.FLYING_MIDDLE.book_chance_percent	20
monster_tables.FLYING_MIDDLE.gem_chance_percent	30
monster_tables.FLYING_INNER.gold	260
monster_tables.FLYING_INNER.xp	78
monster_tables.FLYING_INNER.book_chance_percent	20
monster_tables.FLYING_INNER.gem_chance_percent	30
monster_tables.PIG.gold	700
monster_tables.PIG.xp	80
monster_tables.PIG.book_chance_percent	55
monster_tables.PIG.gem_chance_percent	60
monster_tables.PIG.equipment.chance_percent	25
monster_tables.PIG.equipment.quality_percent.WHITE	60
monster_tables.PIG.equipment.quality_percent.BLUE	30
monster_tables.PIG.equipment.quality_percent.PURPLE	10
monster_tables.LEADER.gold	1400
monster_tables.LEADER.xp	200
monster_tables.LEADER.book_count	1
monster_tables.LEADER.gem_count	2
monster_tables.LEADER.equipment.chance_percent	75
monster_tables.LEADER.equipment.quality_percent.WH TE	35
monster_tables.LEADER.equipment.quality_percent.BLU E	40
monster_tables.LEADER.equipment.quality_percent.PUR PLE	25
monster_tables.LEADER.generic_active_chance_percent	10
monster_tables.DRAGON.gold	2000
monster_tables.DRAGON.xp	300
monster_tables.DRAGON.book_count	0
monster_tables.DRAGON.gem_count	3
monster_tables.DRAGON.equipment.count	1
monster_tables.DRAGON.equipment.quality_percent.BL UE	55
monster_tables.DRAGON.equipment.quality_percent.PU RPLE	35
monster_tables.DRAGON.equipment.quality_percent.GO LD	10

工程字段	唯一值 / 执行语义
monster_tables.DRAGON.equipment.gold_pool	G1_G9
monster_tables.DRAGON.counter_last_hit_upgrade.BLUE	PURPLE
monster_tables.DRAGON.counter_last_hit_upgrade.PURPLE	GOLD
monster_tables.DRAGON.counter_last_hit_upgrade.GOLD	GOLD
monster_tables.DRAGON.counter_last_hit_upgrade.redraw_within_upgraded_element_theme	是
monster_tables.CORE_BOSS.gold	3000
monster_tables.CORE_BOSS.xp	400
monster_tables.CORE_BOSS.book_count	2
monster_tables.CORE_BOSS.gem_count	3
monster_tables.CORE_BOSS.equipment.count	1
monster_tables.CORE_BOSS.equipment.quality_percent.PURPLE	75
monster_tables.CORE_BOSS.equipment.quality_percent.GOLD	25
monster_tables.CORE_BOSS.equipment.gold_pool	G1_G9
monster_tables.CORE_BOSS.independent_G10_chance_percent	5
monster_tables.CORE_BOSS.generic_active_count	1
directed_selection.category_tendency_chance_percent.BOOK	50
directed_selection.category_tendency_chance_percent.EQUIPMENT	35
directed_selection.procedure	ROLL_CATEGORY_TENDENCY_ONCE / SELECT_TENDENCY_OR_GLOBAL_LEGAL_POOL / FILTER_BY_ALREADY_FIXED_QUALITY_AND_ITEM_GENERATION_CONTEXT / READ_FINAL_PER_ID_WEIGHTS_FROM_DIRECTED_CSV / ONE_INTEGER_WEIGHTED_DRAW
directed_selection.never_apply_50_or_35_twice	是
directed_selection.quality_probability_never_changed_by_tendency	是
directed_selection.empty_pool	FALLBACK_GLOBAL_LEGAL_SAME_CATEGORY_AND_QUALITY
directed_selection.weights_are_hidden_from_client	是
directed_selection.item_generation_context_filter.apply_before_weighted_draw	是
directed_selection.item_generation_context_filter.P10	ELIGIBLE_ONLY_IF_HEISHAN_SERVICE_OPEN_AT_REWARD_COMMIT_TICK
directed_selection.item_generation_context_filter.empty_after_filter	FALLBACK_GLOBAL_LEGAL_SAME_CATEGORY_AND_QUALITY_WITH_SAME_CONTEXT_FILTER

工程字段	唯一值 / 执行语义
directed_selection.item_generation_context_filter.existing_ground_or_owned_P10_is_never_destroyed_by_this_filter	是
ground_ownership	PUBLIC_IMMEDIATELY_NO_LAST_HIT_PROTECTION
absent_optional_drop_category	ZERO_ITEMS_AND_ZERO_ROLLS
G5.eligible_ledger_classes	MONSTER_BASE / PLAYER_KILL_BASE_LEVEL / FINAL_LIFE_BONUS / EQUIPMENT_SELL_BASE / B8_MONSTER_BONUS / B31_STEAL_TRANSFER / B32_KILL_CREATE / B33_CHEST / B34_INTEREST / B35_SELL_BONUS / B35_PICKUP_CREATE / D15_EXECUTE_BASE / D15_PLAYER_STEAL_TRANSFER / D16_CREATE / D17_BOUNTY / MAP_CHEST / AIRDROP / BLACK_MOUNTAIN_POSITIVE_NET_PROFIT
G5.explicit_exclusions	STARTING_GOLD / REFUND / ROLLBACK / PLAYER_KILL_WALLET_BONUS / BLACK_MOUNTAIN_GROSS_RETURN / G5_OWN_BONUS
G5.new_ledger_policy	CONTENT_BUILD_FAIL_UNTIL_ELIGIBILITY_DECLARED
B20.charge_scope	MATCH_ID / PLAYER_ID / B20
B20.reacquire_or_hero_swap	DOES_NOT_RESTORE_CHARGE

E.15 经济效果

工程字段	唯一值 / 执行语义
B35.pickup_level5_percent_base_sell	20
B35.reward_once_per_item_instance_per_account_per_match	否
B35.owner_dropped_item_never_rewards_same_owner	是
B35.item_field	B35_pickup_reward_consumed
B35.reward_once_per_item_instance_globally	是
B35.first_eligible_B35_pickup_consumes_reward	是
B40_G9.hard_cap	NONE
B40_G9.must_validate_20min_P99_layers	是
B40_G9.eligible_kills	ORDINARY_MONSTER / FLYING_MONSTER / PIG / LEADER / DRAGON / CORE_BOSS / ENEMY_PLAYER_TRUE_DEATH
B40_G9.player_owned_summon_kills_grant_layers	否
storm_resistance.authority	§ 10 天劫规则
equipment_swap.pvp_out_of_combat_s	3
equipment_swap.channel_s	0.5
equipment_swap.move_allowed	是
equipment_swap.any_actual_damage_interrupts	是

工程字段	唯一值 / 执行语义
equipment_swap.cooldown_equip_change	PRESERVE_REMAINING_PERCENT
equipment_swap.attack_speed_equip_change	CURRENT_COMMITTED_ATTACK_UNCHANGED; NEXT_ATTACK_NEW_SPEED
equipment_swap.max_hp_equip_change	PRESERVE_HP_PERCENT_FLOOR_MIN_1; UNEQUIP_CANNOT_KILL
passive_replace.clear_owner_bound_entities_and_future_timers	是
passive_replace.clear_permanent_layers_of_replaced_passive	是
passive_replace.existing_independent_dot_continues	是
passive_replace.new_book_level	1
gem_sink.land_sell_gold_each	100
gem_sink.ground_drop_allowed	否
core_boss_drop.regular_equipment_count	1
core_boss_drop.regular_quality.PURPLE	75
core_boss_drop.regular_quality.GOLD	25
core_boss_drop.G10_independent_extra_chance	0.05
D12_clear_on	TARGET_TRUE_DEATH / CASTER_TRUE_DEATH / ACTIVE_REPLACED / DURATION_END
D12_disconnect_does_not_clear	是
B41.instances_stack_by_source	否
B41.player_final_blow_collects_all	是
B41.monster_final_blow_destroys_all	是
B41.merge_same_target	PAYOUT_MAX; DURATION_MAX; KEEP_ALL_SOURCE_IDS_FOR_AUDIT
B41.maximum_active_payout_per_target_gold	1500
B33_gold_pool	G1 / G2 / G3 / G4 / G5 / G6 / G7 / G8 / G9
G10_only_core_boss	是
ground_permanent_items.authority	§ 12 地面物与空投规则
ground_permanent_items.economy_invariant	NEVER_DELETE_FOR_CAPACITY_PRESSURE

E.16 地面物

工程字段	唯一值 / 执行语义
expiry.SKILL_BOOK.mode	NEVER_IN_SAFE_ZONE
expiry.WHITE_EQUIPMENT.mode	NEVER_IN_SAFE_ZONE
expiry.BLUE_EQUIPMENT.mode	NEVER_IN_SAFE_ZONE
expiry.PURPLE_EQUIPMENT.mode	NEVER_IN_SAFE_ZONE
expiry.GOLD_EQUIPMENT.mode	NEVER_IN_SAFE_ZONE
expiry.GENERIC_ACTIVE.mode	NEVER_IN_SAFE_ZONE
expiry.BOSS_DROP.mode	NEVER_IN_SAFE_ZONE

工程字段	唯一值 / 执行语义
expiry.GEM.mode	AFTER_S
expiry.GEM.seconds	60
expiry.NORMAL_CHEST.mode	AFTER_S
expiry.NORMAL_CHEST.seconds	60
expiry.AIR_DROP_CHEST.mode	AFTER_S
expiry.AIR_DROP_CHEST.seconds	120
storm_doom.continuous_outside_safe_zone_s	60
storm_doom.reenter_safe_zone_before_60s	CANCEL_AND_RESET_TIMER_TO_ZERO
storm_doom.covered_again	START_NEW_60S_TIMER
storm_doom.applies_to_all_ground_items_including_permanent_classes	是
storm_doom.same_tick_pickup_vs_destroy	PICKUP_STAGE_PRECEDES_STORM_DOOM_STAGE; LEGALLY_COMMITTED_PICKUP_WINS
storm_doom.safe_radius_zero	ALL_GROUND_ITEMS_START_OR_CONTINUE_DOOM_TIMER_EVEN_AT_EXACT_CIRCLE_CENTER
storm_doom.center_point_is_not_a_safe_reentry_when_radius_zero	是
storm_doom.timer_resolution	CONTINUOUS_SERVER_TIME; DESTROY_WHEN_ACCUMULATED_CURRENT_OUTSIDE_SEGMENT_GTE_60S
virtual_loot_cluster.grid_cell_m	4
virtual_loot_cluster.minimum_entries	2
virtual_loot_cluster.server_container_not_item_merge	是
virtual_loot_cluster.preserves_each_item_instance	是
virtual_loot_cluster.preserves_each_item_instance_id_and_B35_consumed_flag	是
virtual_loot_cluster.client_entries_per_page	12
virtual_loot_cluster.sort	QUALITY_DESC / TYPE_ENUM_ASC / GROUND_SPAWN_TICK_ASC / ITEM_INSTANCE_ID_ASC
virtual_loot_cluster.beam	HIGHEST_QUALITY_IN_CLUSTER
virtual_loot_cluster.network_entity_one_per_nonempty_cell	是
ground_entity_capacity.authoritative_item_record_hard_cap	NONE
ground_entity_capacity.performance_alarm_records	8192
ground_entity_capacity.replicated_cluster_soft_cap	256
ground_entity_capacity.overflow	KEEP_ALL_SERVER_RECORDS; INCREASE_CLUSTER_PAGING_AND_LOD; NEVER_DELETE_OR_REFUSE_LEGAL_DROP
ground_entity_capacity.integrity	ITEM_LEDGER_COUNT_MUST_EQUAL_LIVE_RECORDS_PLUS_CONSUMED_OR_DESTROYED_RECORDS
legal_ground_placement.must_be_on_nav_ground	是
legal_ground_placement.must_not_overlap_static_or_dynamic_solid	是
legal_ground_placement.fallback_order	LOCAL_8_RAYS / NEAREST_NAV_WITHIN_6M / NEAREST_GLOBAL_LEGAL_NAV_BY_DISTANCE_THEN_ID

工程字段	唯一值 / 执行语义
legal_ground_placement.unreachable_or_inside_solid_item_spawn	FORBIDDEN
ground_capacity.allocated_item_records	16384
ground_capacity.allocated_virtual_clusters	1024
ground_capacity.content_build_worst_case_window_s	1800
ground_capacity.content_build_max_legal_concurrent_records	12288
ground_capacity.build_assertion	COMPUTED_WORST_CASE_LTE_12288; FAIL_CONTENT_BUILD_OTHERWISE
ground_capacity.runtime_overflow_is_integrity_abort_not_item_DELETION	是
ground_capacity.replication	SPATIAL_INTEREST_CELLS / PAGED_CLUSTER_CONTENT / FULL_SERVER_LEDGER
interact_priority.order	1 致命安全提示 → 2 英雄目标交互 → 3 商店/太白/黑山 → 4 空投 → 5 宝箱 → 6 遗宝堆 → 7 普通拾取
interact_priority.tie_break	同级多目标按世界距离 mm 升序，再按 instance_id 升序
interact_priority.short_press_cycles_same_level	是；再次短按在同级目标间轮换
relic_pile.default_focus	打开列表默认焦点第一项（页面序号 1）
relic_pile.page_keeps_ordinal	翻页后焦点保持同一序号位
relic_pile.focus_item_removed	焦点物被取走/销毁→焦点移到下一项；无下一项→上一项；列表空→关闭

E.17 商店

工程字段	唯一值 / 执行语义
instances[0].id	LAND_A
instances[0].type	LAND
instances[0].windows_s_half_open[0].open_inclusive_s	30
instances[0].windows_s_half_open[0].end_exclusive_s	210
instances[0].windows_s_half_open[1].open_inclusive_s	210
instances[0].windows_s_half_open[1].end_exclusive_s	390

工程字段	唯一值 / 执行语义
instances[0].windows_s_half_open[2].open_inclusive_s	390
instances[0].windows_s_half_open[2].end_exclusive_s	570
instances[0].windows_s_half_open[3].open_inclusive_s	570
instances[0].windows_s_half_open[3].end_exclusive_s	750
instances[0].windows_s_half_open[4].open_inclusive_s	750
instances[0].windows_s_half_open[4].end_exclusive_s	900
instances[0].windows_s_half_open[5].open_inclusive_s	900
instances[0].windows_s_half_open[5].end_exclusive_s	1200
instances[0].window_event	RELOCATE_AND_ROLL_NEW_LISTING_AT_OPEN
instances[1].id	SHOE_A
instances[1].type	SHOE
instances[1].windows_s_half_open[0].open_inclusive_s	45
instances[1].windows_s_half_open[0].end_exclusive_s	225
instances[1].windows_s_half_open[1].open_inclusive_s	225
instances[1].windows_s_half_open[1].end_exclusive_s	405
instances[1].windows_s_half_open[2].open_inclusive_s	405
instances[1].windows_s_half_open[2].end_exclusive_s	585
instances[1].windows_s_half_open[3].open_inclusive_s	585
instances[1].windows_s_half_open[3].end_exclusive_s	795
instances[1].window_event	RELOCATE_AND_ROLL_NEW_LISTING_AT_OPEN
instances[2].id	TAIBAI
instances[2].type	TAIBAI
instances[2].windows_s_half_open[0].open_inclusive_s	60
instances[2].windows_s_half_open[0].end_exclusive_s	240
instances[2].windows_s_half_open[1].open_inclusive_s	240
instances[2].windows_s_half_open[1].end_exclusive_s	420
instances[2].windows_s_half_open[2].open_inclusive_s	420
instances[2].windows_s_half_open[2].end_exclusive_s	600
instances[2].windows_s_half_open[3].open_inclusive_s	600
instances[2].windows_s_half_open[3].end_exclusive_s	780
instances[2].windows_s_half_open[4].open_inclusive_s	780
instances[2].windows_s_half_open[4].end_exclusive_s	900
instances[2].windows_s_half_open[5].open_inclusive_s	900
instances[2].windows_s_half_open[5].end_exclusive_s	1200
instances[2].window_event	RELOCATE_SERVICE_AT_OPEN
instances[3].id	HEISHAN

工程字段	唯一值 / 执行语义
instances[3].type	HEISHAN
instances[3].windows_s_half_open[0].open_inclusive_s	75
instances[3].windows_s_half_open[0].end_exclusive_s	255
instances[3].windows_s_half_open[1].open_inclusive_s	255
instances[3].windows_s_half_open[1].end_exclusive_s	435
instances[3].windows_s_half_open[2].open_inclusive_s	435
instances[3].windows_s_half_open[2].end_exclusive_s	615
instances[3].windows_s_half_open[3].open_inclusive_s	615
instances[3].windows_s_half_open[3].end_exclusive_s	750
instances[3].window_event	RELOCATE_SERVICE_AT_OPEN
instances[4].id	LAND_B
instances[4].type	LAND
instances[4].windows_s_half_open[0].open_inclusive_s	90
instances[4].windows_s_half_open[0].end_exclusive_s	270
instances[4].windows_s_half_open[1].open_inclusive_s	270
instances[4].windows_s_half_open[1].end_exclusive_s	450
instances[4].windows_s_half_open[2].open_inclusive_s	450
instances[4].windows_s_half_open[2].end_exclusive_s	630
instances[4].windows_s_half_open[3].open_inclusive_s	630
instances[4].windows_s_half_open[3].end_exclusive_s	885
instances[4].window_event	RELOCATE_AND_ROLL_NEW_LISTING_AT_OPEN
instances[5].id	SHOE_B
instances[5].type	SHOE
instances[5].windows_s_half_open[0].open_inclusive_s	105
instances[5].windows_s_half_open[0].end_exclusive_s	285
instances[5].windows_s_half_open[1].open_inclusive_s	285
instances[5].windows_s_half_open[1].end_exclusive_s	465
instances[5].windows_s_half_open[2].open_inclusive_s	465
instances[5].windows_s_half_open[2].end_exclusive_s	645
instances[5].windows_s_half_open[3].open_inclusive_s	645
instances[5].windows_s_half_open[3].end_exclusive_s	840
instances[5].window_event	RELOCATE_AND_ROLL_NEW_LISTING_AT_OPEN
land_inventory.white_non_shoe_slots	5
land_inventory.white_draw	LIST_ALL_FIVE_ONCE_EACH
land_inventory.blue_non_shoe_slots	3
land_inventory.blue_draw	FISHER_YATES_SHOP_RNG_OVER_ID_ASC_THEN_TAKE_FIRST_3_WITHOUT_REPLACEMENT

工程字段	唯一值 / 执行语义
land_inventory.purple_non_shoe_slots_max	1
land_inventory.purple_slot_probability	0.7
land_inventory.purple_draw	ON_SUCCESS_FILTER_BY_ROTATION_CONTEXT_THEN_FISHER_YATES_SHOP_RNG_OVER_ID_ASC_AND_TAKE_FIRST_1
land_inventory.gem_count	3
land_inventory.purple_context_filter.P10	ELIGIBLE_ONLY_IF_HEISHAN_SERVICE_OPEN_AT_ROTATION_OPEN_TICK
land_inventory.purple_context_filter.filter_before_uniform_draw	是
land_inventory.purple_context_filter.excluded_item_does_not_cancel_successful_purple_slot	是
shoe_inventory.white_shoe_slots	1
shoe_inventory.white_draw	W3_ONCE
shoe_inventory.blue_shoe_slots	2
shoe_inventory.blue_draw	FISHER_YATES_SHOP_RNG_OVER_B12_B13_B14_THEN_TAKE_FIRST_2_WITHOUT_REPLACEMENT
shoe_inventory.purple_shoe_slots_max	1
shoe_inventory.purple_slot_probability	0.4
shoe_inventory.purple_draw	ON_SUCCESS_UNIFORM_ONE_OF_P15_P16_P17_P18_USING_SHOP_RNG
shoe_inventory.equipment_shelf_only_sells_shoes	是
shoe_inventory.instant_consumables_are_separate_shelf	是
consumables.CLAIRVOYANCE.duration_s	10
consumables.CLAIRVOYANCE.base_vision_m	30
consumables.CLAIRVOYANCE.boosted_vision_m	45
consumables.CLAIRVOYANCE.exact_active_shop_locations	是
consumables.CLAIRVOYANCE.reveal_invisibility	否
consumables.CLAIRVOYANCE.walls_block	是
consumables.DEMON_MIRROR.duration_s	3
consumables.DEMON_MIRROR.radius_m	20
consumables.DEMON_MIRROR.reveal_invisibility	是
consumables.DEMON_MIRROR.walls_block	是
consumables.DEMON_MIRROR.fushi_does_not_block	是
taibai.price	1500
taibai.service_cooldown_s	120
taibai.pvp_out_of_combat_s	5
taibai.channel_s	3
taibai.new_exclusive_ready	是
taibai.same_hero_allowed	是

工程字段	唯一值 / 执行语义
taibai.choice_locked_at_channel_start	是
taibai.movement_allowed_within_shop_tether	是
taibai.tether_radius_m	3
taibai.leave_tether	CANCEL_NO_GOLD_NO_COOLDOWN
taibai.player_owned_positive_damage_interrupts	是
taibai.monster_or_storm_damage_does_not_interrupt	是
taibai.commit_order	REVALIDATE_ALIVE_SHOP_OPEN_TETHER_OUT_OF_PVP_AND_COOLDOWN / DEDUCT_CONFIGURED_TAIBAI_PRICE / START_120S_SERVICE_COOLDOWN / CLEAR_OLD_ACTIVE_OWNER_BOUND_ENTITIES / SWAP_HERO_BODY_AND_ELEMENT / PRESERVE_HP_PERCENT_FLOOR_MIN_ONE / PRESERVE_LEVEL_XP_GOLD_PASSIVES_GEMS_EQUIPMENT_PERMANENT_LAYERS_AND_ALL_RUNNING_STA TUS_TIMERS / REPLACE_CURRENT_ACTIVE_WITH_NEW_HERO_EXCLUSIVE_AND_SET_READY / COMMIT_LEDGER
taibai.interrupted_before_commit	NO_GOLD_NO_COOLDOWN_NO_SWAP
taibai.duplicate_request	RETURN_IDEMPOTENT_RESULT
terminal_shops	LAND_A / TAIBAI
close_all_when_safe_radius_zero	是
placement_policy.rng_stream	SHOP
placement_policy.same_type_min_nav_path_distance_m	100
placement_policy.different_active_shop_macros	REQUIRED_EXCEPT_TERMINAL_LAND_A_AND_TAIBAI_PAIR
placement_policy.dynamic_solid_cannot_occupy_inter action_pad	是
placement_policy.fallback	KEEP_OLD_ONLY_IF_STILL_FULLY_LEGAL / OTHERWISE_CLOSE_AND_SHOW_RELOCATING / RETRY_EVERY_5S
placement_policy.terminal_side_assignment	RANDOM_PERMUTATION_OF_TWO_FINAL_SHOP_POINTS_USING_SHOP_RNG
placement_policy.at_safe_radius_zero	PERMANENT_CLOSE_PRE_TICK_GATE
placement_policy.terminal_pair_exception.eligible_inst ances	LAND_A / TAIBAI
placement_policy.terminal_pair_exception.starts_at	OPEN_OF_THEIR_FINAL_WINDOW
placement_policy.terminal_pair_exception.must_use_t wo_distinct_points_in_same_final_court	是
placement_policy.terminal_pair_exception.point_nav_d istance_m_range	35 / 45
placement_policy.terminal_pair_exception.boss_center _min_distance_m	25
placement_policy.terminal_pair_exception.clear_radius _m	6
placement_policy.terminal_pair_exception.same_type_ distance_rule_still_applies	是
rotation_transaction.window_semantics	HALF_OPEN_OPEN_INCLUSIVE_END_EXCLUSIVE

工程字段	唯一值 / 执行语义
rotation_transaction.at_end_tick	PRE_TICK_CLOSE_OLD_INSTANCE_VERSION_BEFORE_INPUT_VALIDATION
rotation_transaction.uncommitted_request_at_or_after_end_tick	REJECT_STALE_VERSION_NO_GOLD_OR_INVENTORY_CHANGE
rotation_transaction.server_receive_before_end_does_not_reserve_or_guarantee_commit	是
rotation_transaction.new_location_and_listing	ONE_ATOMIC_SCHEDULED_WORLD_EVENT_FOR_NEW_WINDOW
concurrency.maximum_concurrent	6
concurrency.open_count_is_derived_from_half_open_windows	是
concurrency.no_gap_between_adjacent_windows_of_same_instance	是
type_eligible_macros.LAND	ALL_LEGAL_SHOP_MACROS
type_eligible_macros.SHOE	ALL_LEGAL_SHOP_MACROS
type_eligible_macros.TAIBAI[0].namespace	MAP_SHOP_MACRO
type_eligible_macros.TAIBAI[0].id	MAP_SHOP_MACRO_02
type_eligible_macros.TAIBAI[1].namespace	MAP_SHOP_MACRO
type_eligible_macros.TAIBAI[1].id	MAP_SHOP_MACRO_03
type_eligible_macros.TAIBAI[2].namespace	MAP_SHOP_MACRO
type_eligible_macros.TAIBAI[2].id	MAP_SHOP_MACRO_06
type_eligible_macros.TAIBAI[3].namespace	MAP_SHOP_MACRO
type_eligible_macros.TAIBAI[3].id	MAP_SHOP_MACRO_07
type_eligible_macros.TAIBAI[4].namespace	MAP_SHOP_MACRO
type_eligible_macros.TAIBAI[4].id	MAP_SHOP_MACRO_10
type_eligible_macros.TAIBAI[5].namespace	MAP_SHOP_MACRO
type_eligible_macros.TAIBAI[5].id	MAP_SHOP_MACRO_11
type_eligible_macros.TAIBAI[6].namespace	MAP_SHOP_MACRO
type_eligible_macros.TAIBAI[6].id	MAP_SHOP_MACRO_14
type_eligible_macros.HEISHAN	ALL_LEGAL_NON_FINAL_SHOP_MACROS
placement_draw.legal_filters	NOT_IN_STORM_NOW / SAFE_FOR_MIN_WINDOW_REMAINING_OR_120S / DIFFERENT_MACRO_FROM_OTHER_ACTIVE_SHOP_UNLESS_TERMINAL_PAIR_EXCEPTION / SAME_TYPE_NAV_DISTANCE_GTE_100M / INTERACTION_PAD_CLEAR
placement_draw.draw	ONE_INTEGER_WEIGHTED_DRAW_SHOP_STREAM_OVER_MICRO_POINT_ID_ASC
placement_draw.no_legal	KEEP_OLD_ONLY_IF_LEGAL_ELSE_CLOSE_RELOCATING_AND_RETRY_EVERY_5S
placement_draw.no_hidden_retry_draw_consumption	是
placement_draw.weight_formula.base_weight	1000
placement_draw.weight_formula.living_player_within_radius_m	120
placement_draw.weight_formula.weight_per_living_player	250

工程字段	唯一值 / 执行语义
placement_draw.weight_formula.maximum_weight	3000
sell_service.equipment_shop_types	LAND / SHOE
sell_service.equipment_accepted	ANY_OWNED_EQUIPMENT_FROM_WORN_OR_HAND
sell_service.equipment_base_price	READ_ITEM_INSTANCE_EQUIPMENT_REF_SELL_PRICE_FROM_EQUIPMENT_RUNTIME
sell_service.equipment_final_price	APPLY_SCAVENGER_PASSIVE_TO_BASE_ONCE
sell_service.skill_gem.accepted_shop_types	LAND
sell_service.skill_gem.price_each	100
sell_service.skill_gem.deduct_from_hud_bank	1
sell_service.sold_equipement_instance	DESTROY; NEVER_RESTOCK
sell_service.purple_gold_confirm	是
sell_service.atomic	是
terminal_land_restock.inventory	ROLL_ONE_FULL_LAND_INVENTORY_USING_NORMAL_PROBABILITIES
terminal_land_restock.periodic_rotation_after	否
terminal_land_restock.no_global_quota_reset_because_no_additional_global_quota_exists	是
terminal_land_restock.at	LAND_A_FINAL_WINDOW_OPEN_INCLUSIVE
terminal_shop_close	SAFE_RADIUS_ZERO_PRE_TICK_GATE; NO_LITERAL_TIME_AUTHORITY
inventory_draw_contract.authority	§ 12.3 及 § E.17 商店规则
inventory_draw_contract.independent_per_instance_and_rotation	是
inventory_draw_contract.duplicate_within_one_listing	否
inventory_draw_contract.duplicate_across_instances_or_rotations	是
inventory_draw_contract.all_draw_candidates_sorted_by_id_before_rng	是
inventory_draw_contract.one_probability_draw_before_optional_purple_selection	是
inventory_draw_contract.no_additional_global_item_gem_or_quality_caps	是
stock_transaction.one_slot_is_one_unique_virtual_listing_instance	是
stock_transaction.opening_ui_does_not_reserve	是
stock_transaction.purchase_commit	ATOMIC_VALIDATE_HALF_OPEN_WINDOW_INSTANCE_VERSION_LISTING_STOCK_GOLD_LIFE_DISTANCE_LOSS_PVP_AND_DESTINATION_THEN_APPLY
stock_transaction.simultaneous_purchase	SERVER_RECEIVE_SEQUENCE_WINS
stock_transaction.sold_out	REMAIN_EMPTY_UNTIL_NEXT_ROTATION
stock_transaction.unsold_on_rotation	DESTROY_VIRTUAL_LISTING_ONLY; NEVER_DESTROY_PLAYER_OR_GROUND_EQUIPMENT
stock_transaction.listing_version_fields	shop_instance_id / instance_version / rotation_open_server_tick / listing_instance_id / remaining_stock

工程字段	唯一值 / 执行语义
stock_transaction.disconnect_or_reconnect	CLIENT_RELOADS_AUTHORITATIVE_LISTING_VERSION; STALE_PURCHASE_REQUEST_REJECTED_WITH_NO_GOLD_CHANGE
stock_transaction.inventory_full	REQUIRE_EXPLICIT_REPLACE_OR_GROUND_DROP_SELECTION; LISTING_NOT_RESERVED_WHILE_MODAL_OPEN
stock_transaction.inventory_full_commit_order	LOCK_SHOP_LISTING_PLAYER_AND_SELECTED_OLD_ITEM_IF_ANY / REVALIDATE_LISTING_AND_GOLD / REVALIDATE_SELECTED_DROP_OR_REPLACE_TARGET_STILL_OWNED / DEDUCT_GOLD_AND_DECREMENT_LISTING_IN_SAME_COMMIT / ADD_PURCHASED_ITEM_TO_SELECTED_DESTINATION / DROP_OLD_ITEM_PUBLIC_AT_NEAREST_LEGAL_GROUND_IF_DROP_SELECTED_OR_DESTROY_ONLY_IF_EX PLICIT_REPLACE_SELECTED / COMMIT_LEDGER
stock_transaction.listing_lost_while_modal_open	ABORT_WITHOUT_DROPPING_OR_DESTROYING_OLD_ITEM_AND_WITHOUT_GOLD_CHANGE
stock_transaction.skill_gem_purchase_at_all_passives_level5	REJECT_AND_LEAVE_STOCK_UNCHANGED
consumable_stock_per_equipment_shop_rotation.CLAIRVOYANCE	1
consumable_stock_per_equipment_shop_rotation.DEMON_MIRROR	1
full_match_listing_capacity.white_equipment	58
full_match_listing_capacity.blue_equipment	46
full_match_listing_capacity.purple_equipment_expected	10.2
full_match_listing_capacity.purple_equipment_minimum	0
full_match_listing_capacity.purple_equipment_maximum	18
full_match_listing_capacity.skill_gems	30
full_match_listing_capacity.clairvoyance	18
full_match_listing_capacity.demon_mirror	18
full_match_listing_capacity.capacity_is_listing_not_guaranteed_purchase	是
full_match_listing_capacity.non_operational_derived_summary	是
price_authorities.equipment_buy_and_sell	§ 8 装备规则*.buy_price_and_sell_price
price_authorities.hero_draft_reroll	§ 2 及附录 A 英雄成长规则
price_authorities.taibai_swap	§ 12.3 及 § E.17 商店规则
price_authorities.skill_gem_buy	250
price_authorities.clairvoyance_buy	300
price_authorities.demon_mirror_buy	500
price_authorities.skill_gem_sell_at_land_only	100

工程字段	唯一值 / 执行语义
deterministic_listing_draw_contract.rng_consumer_order_for_same_tick	LAND_A / LAND_B / SHOE_A / SHOE_B
deterministic_listing_draw_contract.candidate_sort	ITEM_ID_ASC_BEFORE_FISHER_YATES
deterministic_listing_draw_contract.independent_finite_bag_between_instances	否
deterministic_listing_draw_contract.anti_repeat_across_windows	否
deterministic_listing_draw_contract.duplicates_across_instances_or_windows	是
terminal_migration_event.event_id	TERMINAL_SHOP_PAIR_WINDOW_OPEN
interaction.radius_m	2.5
interaction.direct_line_of_sight_required	是
interaction.legal_life_state	ALIVE
interaction.generic_shop_pvp_out_of_combat_s	3
interaction.pvp_combat_timestamp_event	POSITIVE_HP_OR_SHIELD_DAMAGE_DEALT_OR_RECEIVED_FROM_PLAYER_OWNED_SOURCE
interaction.monster_or_storm_damage_does_not_set_pvp_combat_timestamp	是
interaction.opening_or_staying_in_ui_grants_no_safety	是
taibai.same_hero_swap.active_result	KEEP_CURRENT_ACTIVE_ID_AND_REMAINING_COOLDOWN; CLEAR_UNCOMMITTED_CAST_PREVIEW_ONLY
taibai.different_hero_swap.active_result	REPLACE_WITH_NEW_HERO_EXCLUSIVE_ACTIVE_AND_SET_READY
listing_refresh.minimum_remaining_open_time_s	90
listing_refresh.short_final_remainder	MERGE_WITH_PREVIOUS_WINDOW; KEEP_LOCATION_AND_UNSOLD_STOCK; NO_NEW_RNG_DRAW

E.18 黑山

工程字段	唯一值 / 执行语义
base_probabilities.PASSIVE.big	0.15
base_probabilities.PASSIVE.flat	0.5
base_probabilities.PASSIVE.loss	0.35
base_probabilities.EQUIPMENT.big	0.15
base_probabilities.EQUIPMENT.flat	0.4
base_probabilities.EQUIPMENT.loss	0.45
base_probabilities.ACTIVE.big	0.1
base_probabilities.ACTIVE.flat	0.5

工程字段	唯一值 / 执行语义
base_probabilities.ACTIVE.loss	0.4
base_probabilities.GOLD.big	0.1
base_probabilities.GOLD.flat	0.4
base_probabilities.GOLD.loss	0.5
gambling_badge_probabilities.PASSIVE.big	0.3
gambling_badge_probabilities.PASSIVE.flat	0.5
gambling_badge_probabilities.PASSIVE.loss	0.2
gambling_badge_probabilities.EQUIPMENT.big	0.3
gambling_badge_probabilities.EQUIPMENT.flat	0.4
gambling_badge_probabilities.EQUIPMENT.loss	0.3
gambling_badge_probabilities.ACTIVE.big	0.2
gambling_badge_probabilities.ACTIVE.flat	0.5
gambling_badge_probabilities.ACTIVE.loss	0.3
gambling_badge_probabilities.GOLD.big	0.2
gambling_badge_probabilities.GOLD.flat	0.4
gambling_badge_probabilities.GOLD.loss	0.4
equipment_gold_pool	G1 / G2 / G3 / G4 / G5 / G6 / G7 / G8 / G9
rng_stream	BLACK_MOUNTAIN
server_atomic	是
active_big_excludes.candidate_pool	ALL_38_EXCLUSIVE_ACTIVE_REFS_SORTED_BY_TYPED_REF
active_big_excludes.excluded_ref	CURRENT_ACTIVE_IF_EXCLUSIVE_ELSE_CURRENT_HERO_DEFAULT_EXCLUSIVE
active_big_excludes.draw	UNIFORM_FROM_REMAINING_37_USING_BLACK_MOUNTAIN_STREAM
passive_candidate_empty	REJECT_AND_LOG_CONFIGURATION_ERROR
P8_capacity_loss_order	LOCK_INPUT / RESOLVE_P8_EFFECT / DROP_MOST_RECENT_NON_OUTPUT_HAND_INSTANCE_IF_OVER_CAP / PLACE_OUTPUT
regional_non_gambling_refinery.uses_per_player_per_match	3
regional_non_gambling_refinery.recipes	3_WHITE_TO_UNIFORM_BLUE / 3_BLUE_TO_UNIFORM_PURPLE / 2_PURPLE_PLUS_2000_GOLD_TO_UNIFORM_G1_G9
regional_non_gambling_refinery.G10_allowed	否
regional_non_gambling_refinery.multi_select_required	是
regional_non_gambling_refinery.input_sources	WORN_SLOTS / HAND_SLOTS
regional_non_gambling_refinery.output	HAND_OR_WORN_IF_EXPLICITLY_SELECTED_AND_LEGAL_ELSE_PUBLIC_GROUND
regional_non_gambling_refinery.partial_commit_allowed	否
attempts_per_player_per_match	3

工程字段	唯一值 / 执行语义
P10_probability_snapshot	WORN_AT_CONFIRMATION_COMMIT;CHANGES_AFTER_COMMIT_DO_NOT_AFFECT_RESULT
passive.input	ONE_NONEMPTY_OWNED_PASSIVE_SLOT
passive.big	REPLACE_WITH_RANDOM_NOT_CURRENTLY_OWNED; LEVEL=MIN(5,OLD_LEVEL+1)
passive.flat	REPLACE_WITH_RANDOM_NOT_CURRENTLY_OWNED; LEVEL=OLD_LEVEL
passive.loss	REMOVE_INPUT_PASSIVE_AND_ALL_ITS_ENTITIES_AND_PERMANENT_LAYERS
passive.candidate_excludes	OTHER_THREE_SLOTS_AND_INPUT_ID
passive.empty_candidate_fallback	RETURN_INPUT_WITH_LEVEL_PLUS_ONE_CAPPED_5
equipment.input_sources	WORN_3 / HAND_1_OR_2
equipment.eligible_qualities	WHITE / BLUE / PURPLE
equipment.big	WHITE_TO_RANDOM_BLUE; BLUE_TO_RANDOM_PURPLE; PURPLE_TO_RANDOM_G1_G9
equipment.flat	RANDOM_DIFFERENT_ID_SAME_QUALITY; IF_EMPTY_RETURN_INPUT
equipment.loss	DESTROY_INPUT
equipment.output	ORIGINAL_SLOT_IF_LEGAL_ELSE_HAND_ELSE_PUBLIC_GROUND_AT_SHOP
equipment.G10_allowed	否
active.input	CURRENT_ACTIVE
active.big	ONE_OF_37_EXCLUSIVE_BY_ACTIVE_BIG_EXCLUDES
active.flat	UNIFORM_19_GENERIC_EXCLUDING_CURRENT_IF_GENERIC
active.loss	CURRENT_HERO_DEFAULT_EXCLUSIVE
active.all_results_start_full_cooldown	是
gold_modes.DOUBLE.bet_min	500
gold_modes.DOUBLE.bet_max	5000
gold_modes.DOUBLE.step	100
gold_modes.DOUBLE.cannot_exceed_balance	是
gold_modes.DOUBLE.deduct_at_confirm	是
gold_modes.DOUBLE.big_return_multiplier	2
gold_modes.DOUBLE.flat_return_multiplier	1
gold_modes.DOUBLE.loss_return_multiplier	0.5
gold_modes.DOUBLE.fraction_round	FLOOR_GOLD
gold_modes.PURPLE.bet	2000
gold_modes.PURPLE.deduct_at_confirm	是
gold_modes.PURPLE.big	CREATE_ONE_UNIFORM_PURPLE_EQUIPMENT
gold_modes.PURPLE.flat_return_gold	2000
gold_modes.PURPLE.loss_return_gold	1000
gold_modes.PURPLE.output	HAND_ELSE_PUBLIC_GROUND
transaction_order	LOCK_PLAYER_GAMBLE_COUNT_AND_INPUTS / REVALIDATE_SHOP_OPEN_BALANCE_OWNERSHIP_P10 / SNAPSHOT_PROBABILITIES / CONSUME_ONE_ATTEMPT / DEDUCT_OR_DESTROY_INPUT / DRAW_BLACK_MOUNTAIN_RNG / CREATE_RESULT / PLACE_RESULT / COMMIT_LEDGER_AND_REPLAY

工程字段	唯一值 / 执行语义
gold_modes.income_multiplier_application	G5_DOES_NOT_MULTIPLY_GROSS_RETURN; ONLY_NET_PROFIT_IF_POSITIVE_IS_ELIGIBLE_ONCE
gold_modes.expected_return.base	0.85
gold_modes.expected_return.P10	1.00

E.19 空投

工程字段	唯一值 / 执行语义
times_s	360 / 720 / 1080
warning_s	15
reservation.radius_m	3
reservation.blocks_dynamic_solid_placement_query	是
reservation.blocks_player_movement	否
player_overlap.nav_search_m	4
player_overlap.if_no_point	MOVE_AIRDROP_TO_NEXT_LEGAL_CANDIDATE
player_overlap.never_push_through_wall_or_storm	是
fallback_priority	NO_SOLID_OVERLAP / OPENABLE_SPACE / OTHER_CHEST_DISTANCE / SHOP_DISTANCE / MONSTER_DISTANCE
final_chest.authored_backup_points	4
final_chest.boss_body_has_priority	CHEST_RELOCATES_TO_NEXT_BACKUP; BOSS_NEVER_MOVES
final_chest.choose	FIRST_LEGAL_POINT_AFTER_BOSS_BODY_AND_SOLID_CHECK_IN_ARRAY_ORDER_ROTATED_BY_AIR_DROP _RNG
channel_s	3
interrupts	ACTUAL_NON_STORM_DAMAGE / FORCED_MOVEMENT / STUN / KNOCKUP / PULL / MOVE_OVER_0_25M / TRUE_DEATH
does_not_interrupt_alone	SLOW / SILENCE / ROOT / NORMAL_STORM_DAMAGE_WITHOUT_STUN
drop_placement	SEARCH_8_DIRECTIONS_ON_RING_1_2_TO_2_0M_ELSE_LOOT_CLUSTER_AT_CHEST
reward.opener_gold	1200
reward.equipment_count	1
reward.equipment_quality	GOLD
reward.equipment_pool	G1 / G2 / G3 / G4 / G5 / G6 / G7 / G8 / G9
reward.selection	UNIFORM_AIR_DROP_RNG
reward.equipment_public_ground	是
reward.gold_to_first_successful_channeler	是
reward.single_commit	是
normal_B33_chest.eligible_final_blow_targets	ORDINARY_MONSTER / FLYING_MONSTER / PIG / LEADER / DRAGON / CORE_BOSS

工程字段	唯一值 / 执行语义
normal_B33_chest.summon_final_blow_uses_owner_passive	否
normal_B33_chest.chest_trigger_percent_by_level	8 / 10 / 13 / 16 / 20
normal_B33_chest.chest_gold_by_level	100 / 120 / 120 / 120 / 120
normal_B33_chest.gem_chance_percent_by_level	0 / 0 / 10 / 15 / 20
normal_B33_chest.gold_equipment_chance_percent_by_level	0 / 0 / 0 / 0 / 5
normal_B33_chest.gold_equipment_pool	G1 / G2 / G3 / G4 / G5 / G6 / G7 / G8 / G9
normal_B33_chest.spawn_at	MONSTER_DEATH_POINT_ELSE_NEAREST_LEGAL_WITHIN_3M
normal_B33_chest.ground_expiry_s	60
normal_B33_chest.open_channel_s	0.35
normal_B33_chest.public_first_COMMIT_wins	是
normal_B33_chest.nested_roll_order	ROLL_CHEST / CREATE_CHEST_WITH_GOLD / ROLL_GEM_IF_LEVEL_AT_LEAST_3 / ROLL_GOLD_EQUIPMENT_IF_LEVEL_5
fallback_min_distances_m.OTHER_CHEST	12
fallback_min_distances_m.SHOP_INTERACTION_PAD	15
fallback_min_distances_m.LIVING_MONSTER_BODY	12
fallback_min_distances_m.LIVING_PLAYER_CAPSULE	3

E.20 UI 流程

工程字段	唯一值 / 执行语义
modals.GENERIC_ACTIVE_REPLACE.opens_when	ACTIVE_SLOT_OCCUPIED_AND_NEW_ACTIVE_PICKED
modals.EQUIPMENT_REPLACE.required_when_hand_full	是
modals.LOOT_CLUSTER.mobile_hold_opens	是
modals.LOOT_CLUSTER.pc_interact_then_list	是
modals.NORMAL_CHEST_CHANNEL.duration_s	0.35
modals.AIRDROP_CHANNEL.duration_s	3
item_disappeared_during_modal	CLOSE_AND_SHOW_ALREADY_PICKED
pointer_click_through	否
spectator.delay_s	5
spectator.invalid_target	NEAREST_NOT_ELIMINATED_ELSE_PUBLIC_FINAL_COURT_CAMERA
spectator.free_camera	否
shop.open_gate	ALIVE_AND_PVP_OUT_3S_AND_WITHIN_2_5M_DIRECT_LOS_AND_INSTANCE_WINDOW_OPEN
shop.authoritative_fields	shop_instance_id / instance_version / window_open_tick / window_end_tick / listing_instance_id / item_ref / price / remaining_stock
shop.sold_out_slot	VISIBLE_DISABLED_UNTIL_ROTATION; NEVER_HIDE_AS_IF_NOT_ROLLED

工程字段	唯一值 / 执行语义
shop.rotation_warning_s	10
shop.permanent_close_warning_s	30
shop.purchase_pending	DISABLE_ONLY_SUBMITTED_LISTING_ON_THIS_CLIENT; SERVER_COMMIT_DECIDES
shop.purchase_race_loser	SHOW_SOLD_OUT_NO_GOLD_CHANGE_AND_REFRESH_LISTING
shop.inventory_full	OPEN_EXPLICIT_REPLACE_OR_DROP_MODAL_WITHOUT_RESERVING_LISTING
shop.incoming_player_damage	POSITIVE_HP_OR_SHIELD_DAMAGE_CLOSES_UI_AND_CANCELS_UNCOMMITTED_REQUEST
shop.reconnect	DISCARD_CACHED_LISTING_AND_FETCH_CURRENT_INSTANCE_VERSION
shop.P10_after_heishan_close	SHOW_SPECIAL_EFFECT_DORMANT_BUT_MAX_HP_AND_SELL_VALUE_REMAIN
shop.taibai_gate	ALIVE_AND_PVP_OUT_5S_AND_WITHIN_2_5M_DIRECT_LOS_AND_INSTANCE_WINDOW_OPEN
shop.remaining_time_display	DERIVE_FROM_WINDOW_END_TICK_MINUS_CURRENT_SERVER_TICK; NEVER_ACCEPT_CLIENT_COUNTDOWN_AS_AUTHORITY
shop.monster_or_storm_damage	DOES_NOT_CLOSE_SHOP_UI_BY_ITSELF
shop.at_window_end	CLOSE_UI_BEFORE_INPUT_VALIDATION_AND_SHOW_SHOP_MOVED_OR_CLOSED
elevation.minimap_arrow	可见敌人处于不同可走层时小地图图标附 ↑ / ↓ 箭头；同层无箭头
elevation.lock_indicator	锁定目标高度差 $\geq 2m$ 时目标框显示高度差图标
elevation.no_fog_leak	箭头与图标只对已可见/已显形目标生成；不因此泄露迷雾信息

E.21 状态机

工程字段	唯一值 / 执行语义
machines.MATCH.states	LOBBY / PICKING / LOADING / PLAYING / FINAL_STORM / SETTLED / ABORTED
machines.MATCH.winner_guard	count(participant.life_state != ELIMINATED) == 1
machines.MATCH.tie_guard	all_remaining_candidates_commit_ELIMINATED_in_same_tick
machines.MATCH.terminal	SETTLED / ABORTED
machines.MATCH.transitions[0].to	PICKING
machines.MATCH.transitions[0].guard	PARTICIPANT_SLOTS_COMMITTED_30
machines.MATCH.transitions[0].from_states	LOBBY
machines.MATCH.transitions[1].to	LOADING
machines.MATCH.transitions[1].guard	ALL_CONFIRMED_OR_PICK_TIMER_15S_EXPIRED
machines.MATCH.transitions[1].from_states	PICKING
machines.MATCH.transitions[2].to	PLAYING
machines.MATCH.transitions[2].guard	SERVER_WORLD_READY; CLIENT_LATE_LOAD_DOES_NOT_DELAY_MATCH
machines.MATCH.transitions[2].from_states	LOADING
machines.MATCH.transitions[3].to	FINAL_STORM
machines.MATCH.transitions[3].guard	PRE_TICK_FINAL_STORM_GATE_AT_SAFE_RADIUS_ZERO
machines.MATCH.transitions[3].from_states	PLAYING

工程字段	唯一值 / 执行语义
machines.MATCH.transitions[4].to	SETTLED
machines.MATCH.transitions[4].guard	WINNER_OR_COCHAMPION_GUARD_AFTER_ALL_COMMITTED_LETHAL_CHAINS
machines.MATCH.transitions[4].from_states	PLAYING / FINAL_STORM
machines.MATCH.transitions[5].to	ABORTED
machines.MATCH.transitions[5].guard	SERVER_INTEGRITY_ABORT
machines.MATCH.transitions[5].from_states	LOBBY / PICKING / LOADING / PLAYING / FINAL_STORM
machines.PLAYER_LIFE.states	ALIVE / TRUE_DEATH_RESOLVING / SOUL_FLIGHT / REVIVE_PROTECTION / ELIMINATED
machines.PLAYER_LIFE.remaining_lives_decrement_at	LIFE_CONSUME_COMMIT
machines.PLAYER_LIFE.soul_or_protection_counts_as_not_eliminated	是
machines.PLAYER_LIFE.transitions[0].to	TRUE_DEATH_RESOLVING
machines.PLAYER_LIFE.transitions[0].guard	LETHAL_CHAIN_REACHES_CONSUME_LIFE
machines.PLAYER_LIFE.transitions[0].from_states	ALIVE / REVIVE_PROTECTION
machines.PLAYER_LIFE.transitions[1].to	SOUL_FLIGHT
machines.PLAYER_LIFE.transitions[1].guard	AFTER_1S_AND_REMAINING_LIVES_GT_0
machines.PLAYER_LIFE.transitions[1].from_states	TRUE_DEATH_RESOLVING
machines.PLAYER_LIFE.transitions[2].to	ELIMINATED
machines.PLAYER_LIFE.transitions[2].guard	REMAINING_LIVES_EQ_0
machines.PLAYER_LIFE.transitions[2].from_states	TRUE_DEATH_RESOLVING
machines.PLAYER_LIFE.transitions[3].to	REVIVE_PROTECTION
machines.PLAYER_LIFE.transitions[3].guard	ARRIVE_REVALIDATED_REVIVE_POINT
machines.PLAYER_LIFE.transitions[3].from_states	SOUL_FLIGHT
machines.PLAYER_LIFE.transitions[4].to	ALIVE
machines.PLAYER_LIFE.transitions[4].guard	3S_EXPIRES_OR_OFFENSIVE_OR_INTERACTION_INTENT
machines.PLAYER_LIFE.transitions[4].from_states	REVIVE_PROTECTION
machines.PLAYER_LIFE.revive_protection_input_transition	EARLY_END_AND_SAME_REQUEST_REVALIDATION_IS_ONE_ATOMIC_INPUT_VALIDATION_STEP
machines.ACTIVE_SLOT.states	READY / CASTING / CHANNELING / COOLDOWN
machines.ACTIVE_SLOT.true_death	CLEAR_OWNER_ENTITIES_THEN_CURRENT_ID_READY
machines.ACTIVE_SLOT.transitions[0].to	CASTING
machines.ACTIVE_SLOT.transitions[0].guard	INPUT_AND_CONTEXT_LEGAL
machines.ACTIVE_SLOT.transitions[0].from_states	READY
machines.ACTIVE_SLOT.transitions[1].to	READY
machines.ACTIVE_SLOT.transitions[1].guard	CANCEL_BEFORE_COMMIT; NO_CD
machines.ACTIVE_SLOT.transitions[1].from_states	CASTING
machines.ACTIVE_SLOT.transitions[2].to	CHANNELING
machines.ACTIVE_SLOT.transitions[2].guard	CHANNEL_CAST_COMMITS; CD_STARTS

工程字段	唯一值 / 执行语义
machines.ACTIVE_SLOT.transitions[2].from_states	CASTING
machines.ACTIVE_SLOT.transitions[3].to	COOLDOWN
machines.ACTIVE_SLOT.transitions[3].guard	INSTANT_OR_WINDUP_CAST_COMMITS; CD_STARTS
machines.ACTIVE_SLOT.transitions[3].from_states	CASTING
machines.ACTIVE_SLOT.transitions[4].to	COOLDOWN
machines.ACTIVE_SLOT.transitions[4].guard	ENDS_OR_INTERRUPTED; EXISTING_CD_CONTINUES
machines.ACTIVE_SLOT.transitions[4].from_states	CHANNELING
machines.ACTIVE_SLOT.transitions[5].to	READY
machines.ACTIVE_SLOT.transitions[5].guard	REMAINING_CD_REACHES_ZERO_OR_TRUE_DEATH_RESET
machines.ACTIVE_SLOT.transitions[5].from_states	COOLDOWN
machines.ACTIVE_SLOT.true_death_global_transition.from_states	READY / CASTING / CHANNELING / COOLDOWN
machines.ACTIVE_SLOT.true_death_global_transition.when_remaining_lives_gt_zero	CLEAR_OWNER_ENTITIES_AND_SET_READY
machines.ACTIVE_SLOT.true_death_global_transition.when_remaining_lives_eq_zero	DESTROY_RUNTIME_SLOT_WITH_NO_FURTHER_CAST_OR_COOLDOWN
machines.ACTIVE_SLOT.true_death_global_transition.priority	AFTER_LETHAL_COMMIT_BEFORE_OTHER_ACTIVE_TRANSITIONS
machines.MODAL.states	CLOSED / OPEN_VALID / SUBMITTING / FAILED_ITEM_GONE / COMMITTED
machines.MODAL.world_not_paused	是
machines.MODAL.transitions[0].from_states	CLOSED
machines.MODAL.transitions[0].to	OPEN_VALID
machines.MODAL.transitions[0].guard	SERVER_VALIDATED_ITEM_INSTANCE_AND_CONTEXT
machines.MODAL.transitions[1].from_states	OPEN_VALID
machines.MODAL.transitions[1].to	SUBMITTING
machines.MODAL.transitions[1].guard	ONE_CONFIRM_INPUT_COMMITS_REQUEST_ID
machines.MODAL.transitions[2].from_states	OPEN_VALID
machines.MODAL.transitions[2].to	CLOSED
machines.MODAL.transitions[2].guard	CANCEL_INPUT_OR_DAMAGE_CLOSE_RULE
machines.MODAL.transitions[3].from_states	OPEN_VALID / SUBMITTING
machines.MODAL.transitions[3].to	FAILED_ITEM_GONE
machines.MODAL.transitions[3].guard	ITEM_INSTANCE_NO_LONGER_AVAILABLE
machines.MODAL.transitions[4].from_states	SUBMITTING
machines.MODAL.transitions[4].to	COMMITTED
machines.MODAL.transitions[4].guard	ATOMIC_TRANSACTION_COMMITTED
machines.MODAL.transitions[5].from_states	FAILED_ITEM_GONE / COMMITTED
machines.MODAL.transitions[5].to	CLOSED
machines.MODAL.transitions[5].guard	ONE_UI_FRAME_RESULT_PRESENTED

工程字段	唯一值 / 执行语义
machines.CORE_BOSS.states	ACTIVE_COURT / DEAD_TIMER / MIGRATION_WARNING / MIGRATING / LANDING_INVULNERABLE / FINAL_ACTIVE / FINAL_DEAD
machines.CORE_BOSS.migration_warning_killed	DEAD_TIMER
machines.CORE_BOSS.respawn_due_before_19_00	FINAL_COURT
machines.CORE_BOSS.respawn_due_at_or_after_19_00	FINAL_DEAD
machines.CORE_BOSS.death_before_14_00	START_300S_TIMER
machines.CORE_BOSS.death_at_or_after_14_00	NO_RESPAWN_TIMER
transition_encoding.from_states	NONEMPTY_ARRAY_OF_DECLARED_STATE_IDS
transition_encoding.to	ONE_DECLARED_STATE_ID
transition_encoding.guard	ONE_CLOSED_GUARD_ID_OR_SEMICOLON_AND_OF_GUARD_IDS
transition_encoding.wildcard_state_tokens	FORBIDDEN

E.22 同 tick 事件顺序

工程字段	唯一值 / 执行语义
pre_tick_world_gates	SAFE_RADIUS_ZERO_AND_FINAL_STORM_STATE_TRANSITION / PERMANENT_OR_WINDOW_END_SHOP_CLOSE / MATCH_ABORT_GATE
same_tick_priority	APPLY_PRE_TICK_WORLD_GATES / VALIDATE_INPUT_AND_CONTEXT_INCLUDING_REVIVE_PROTECTION_VOLUNTARY_BREAK / APPLY_MOVEMENT_AND_SWEEP_COLLISION / COMMIT_CASTS_AND_SPAWN_PROJECTILES_OR_DYNAMIC_SOLIDS / RESOLVE_HITS_DAMAGE_AND_LETHAL_PROTECTION / COMMIT_LIFE_DEATH_DROPS_AND_CREDITS_AS_SNAPSHOT_SET / COMMIT_PICKUPS_BY_SERVER_RECEIVE_SEQUENCE / COMMIT_SHOP_AND_GAMBLE_TRANSACTIONS / APPLY_PHASE_APPROPRIATE_STORM_DAMAGE_AND_ITEM_DOOM / COMMIT_SCHEDULED_WORLD_EVENTS_EXCEPT_PRE_TICK_GATES / RUN_AI_TRANSITIONS / REPLICATE_AND_APPEND_REPLAY
ice_wall_same_tick	MOVEMENT_FIRST_WALL_BLOCKS_FROM_NEXT_TICK
at_20_00_shop_requests	REJECT_BEFORE_VALIDATION
at_safe_radius_zero_tick.normal_storm_check	DISABLED
at_safe_radius_zero_tick.apocalypse_phase	ENTERED_PRE_INPUT
at_safe_radius_zero_tick.shop_requests	REJECT
at_safe_radius_zero_tick.ground_item_center_point_exception	NONE_ALL_POINTS_ARE_STORM_COVERED

E.23 闭合枚举

工程字段	唯一值 / 执行语义
EffectType	CONTROL / DAMAGE / ECONOMY_CREATE_GOLD / HEAL / MOVEMENT / SCRIPTED / SHIELD / SUMMON

工程字段	唯一值 / 执行语义
DamageForm	BASIC / SKILL / DOT / PERCENT / REFLECT / STORM / TRUE
SourceFaction	PLAYER_OWNED / MONSTER / WORLD
ElementRule	NONE / OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT / BIDIRECTIONAL_FIVE_ELEMENT
HitKind	DIRECT / ABILITY / AREA / PERIODIC / REFLECTED / ENVIRONMENT
OwnerCredit	CASTER / SUMMON_OWNER / MONSTER / WORLD
LifeState	ALIVE / TRUE_DEATH_RESOLVING / SOUL_FLIGHT / REVIVE_PROTECTION / ELIMINATED
RuleStatus	DESIGN_PASS / BLOCKING_PRERELEASE_ENGINE / BLOCKING_PRERELEASE_DEVICE / MONITOR_ONLY / BLOCKING_PRERELEASE_EXTERNAL / EXECUTION_REQUIRED
unknown_enum_policy	REJECT_CONTENT_BUILD; NEVER_FALLBACK_TO_STRING_OR_DEFAULT
Platform	ANDROID / IOS / WINDOWS
InputMode	TOUCH / KEYBOARD_MOUSE
ContentNamespace	HERO / ACTIVE_EXCLUSIVE / ACTIVE_GENERIC / PASSIVE / EQUIPMENT / MONSTER_ARCHETYPE / MONSTER_VISUAL / MAP_NODE / MAP_PIG_DEN / MAP_DRAGON_ARENA / MAP_LEADER_FIELD / MAP_COURT / MAP_SHOP_POINT / SHOP_INSTANCE / GROUND_ITEM_INSTANCE / PLAYER_ENTITY / MONSTER_ENTITY
Targeting	SELF / DIRECTION / GROUND_POINT / VISIBLE_ENEMY / SELF_FORWARD / VISIBLE_ANY_HERO_OR_SELF / GROUND_POINT_OR_SELF / VISIBLE_ENEMY_OR_SELF / TWO_VISIBLE_UNITS / VISIBLE_MONSTER_OR_ENEMY / GROUND_POINT_OR_OWN_TRAP
CastType	INSTANT / WINDUP / CHANNEL / TOGGLE
Shape	CIRCLE / CONE / LINE / LINE_WALL / NONE / POINT / PROJECTILE / RING_WALL
EquipmentEffectType	ACTIVE_COOLDOWN_RATE / ALL_GOLD_INCOME_MULTIPLIER / BASIC_ATTACK_RANGE_FLAT / BASIC_LIFESTEAL / BASIC_ON_HIT_DOT / BASIC_SAME_TARGET_ATTACK_SPEED_STACK / BLACK_MOUNTAIN_OUTCOME_OVERRIDE / CONDITIONAL_BONUS_DAMAGE_REDUCTION / CONDITIONAL_DAMAGE_MULTIPLIER / CRIT_DAMAGE_BONUS / DAMAGE_MULTIPLIER / DIRECT_VISION_RADIUS_MULTIPLIER / ELEMENT_COUNTER_MULTIPLIER_OVERRIDE / EXTRA_HAND_SLOT / FIXED_DAMAGE_REDUCTION / FLIGHT_OUT_OF_COMBAT / FORCED_MOVEMENT_DISTANCE_MULTIPLIER / GLOBAL_ACTIVE_CAST_PING / HARD_CONTROL_AND_FORCED_MOVEMENT_RESIST / LETHAL_PROTECTION_CONSUMABLE / NONE / OFFENSIVE_STEALTH_BREAK_MOVE_SPEED / ON_ENEMY_PLAYER_TRUE_DEATH_ACTIVE_READY / ON_ENEMY_PLAYER_TRUE_DEATH_HEAL / ON_MONSTER_KILL_GOLD_CREATE / OWNER_HARD_CONTROL_DURATION_ADD / PERCENT_DAMAGE_REDUCTION / PERMANENT_ATTACK_ON_KILL / REFLECT / REGEN / SHOE_ONLY / SLOW_IMMUNITY / SLOW_MAGNITUDE_MULTIPLIER / STATIONARY_STEALTH / STAT_ONLY / STORM_DAMAGE_REDUCTION / SUMMON_ATTACK_MULTIPLIER / SUMMON_HP_AND_ATTACK_MULTIPLIER / TRUE_SIGHT

工程字段	唯一值 / 执行语义
PassiveTrigger	BASIC_HIT_DOT / BASIC_HIT_PROC / BASIC_HIT_PROC_TRANSFER_GOLD / BASIC_HIT_STACK / CONTINUOUS_MOVEMENT_STACK / DIRECT_DAMAGE_CONDITIONAL / DIRECT_DAMAGE_FROM_BACK_ARC / FIRST_DIRECT_HIT_AFTER_GAP / INCOMING_ACTUAL_DAMAGE_PROC / INCOMING_BASIC_FIXED_REDUCTION / INCOMING_BASIC_PRE_HIT_PROC / INCOMING_BASIC_PROC_PRE_DAMAGE / LETHAL_PROC / LETHAL_PROTECTION_ONCE_MATCH / LOW_HP_HUNT / MISSING_HP_SCALING / MOVING_AFTERIMAGE / NORMAL_STORM_ZONE_MODIFIER / ON_ELIGIBLE_KILL_PERMANENT_MAX_HP / ON_ENEMY_BOUNTY_TARGET_TRUE_DEATH / ON_KILL_GOLD_CREATE / ON_MONSTER_KILL_NESTED_CHEST / ON_OWNER_TRUE_DEATH_MARK_KILLER / OUT_OF_ANY_COMBAT_REGEN / OUT_OF_ANY_COMBAT_SUMMON / OWNED_SUMMON_DEATH / POST_INCOMING_BASIC_REFLECT / POST_SKILL_DAMAGE_IF_SURVIVES / SELL_AND_PICKUP_ECONOMY / SERVER_MINUTE_INTEREST / WHEN_CRITICALLY_HIT / WHILE_HARD_CONTROLLED
MapMonsterKind	FLY / MEL / RNG
MapBand	INNER / MIDDLE / OUTER
MapEdgeClass	ARENA / BREACH / COURT / DEN / MAIN / POCKET / RAMP / RISK / SHOP / SIDE / SIDEDOOR / SPAWN_LINK
MapWallClass	BOUND / VAULT
MapStormPhaseKind	HOLD / SHRINK / STORM
ControlType	SLOW / ROOT / SILENCE / BLIND / KNOCKBACK / PULL / STUN / PETRIFY / POLYMORPH / KNOCKUP
ForcedMovementType	KNOCKBACK / KNOCKUP / PULL / THROW
StealthSourceKind	ACTIVE_STEALTH / EQUIPMENT_STEALTH / FOG_OF_WAR_ONLY / FUSHI_HIDDEN / SMOKE_OCCLUSION_ONLY
StealthBreakCause	ACTUAL_DAMAGE_TAKEN / OWNER_BASIC_RELEASE_AGAINST_HOSTILE / OWNER_HOSTILE_ACTIVE_CAST_COMMIT / REVEALED_BY_ENEMY / STEALTH_TIMEOUT / VOLUNTARY_CANCEL
ActiveGenericID	D1 / D3 / D4 / D6 / D7 / D9 / D10 / D11 / D12 / D13 / D14 / D15 / D16 / D17 / D18 / D19 / D20 / D21 / D22
ActiveGenericReservedID	D2 / D5 / D8

工程字段	唯一值 / 执行语义
ActiveEffectType	AERIAL_DELAYED_AREA_DAMAGE / ANY_HERO_HEAL_ZONE / ARM_NEXT_BASIC / ATOMIC_SWAP_TWO_VISIBLE_UNITS / BASIC_LIFESTEAL_AND_DAMAGE_AURA / BLINK_LEAVE_DELAYED_DECOY / CAPSULE_SWEEP_BLINK / CHAIN_DAMAGE / CHARGED_RADIAL_PULL / CONDITIONAL_POISON_DETONATE / CONE_DAMAGE_SLOW / CONTAGIOUS_DAMAGE_ZONE / CREATE_GOLD / DAMAGE_SLOW_ANTIMOBILITY_ZONE / DAMAGE_SLOW_ZONE / DASH_STOP_FIRST / DELAYED_AREA_DAMAGE / DELAYED_AREA_DAMAGE_SLOW / DELAYED_SILENCE_ZONE / DESTRUCTIBLE_ICE_WALL / DUAL_PETRIFY / FIRE_WALL / GLOBAL_LINE_PROJECTILE / LOCKED_DAMAGE_STUN / MOBILE_CHANNEL_AOE / MOBILE_INVULNERABILITY / MONSTER_EXECUTE_OR_PLAYER_GOLD_STEAL / PAGODA_RING_SOLID / PETRIFY_TARGET / PIERCING_DASH / PROJECTILE_POLYMORPH / PROJECTILE_PULL / PROJECTILE_ROOT / RADIAL_DAMAGE_KNOCK_TO_EDGE / SCAN_OLD_EQUIPMENT / SELF_COMBAT_BUFF / SELF_LOCK_INVULNERABILITY / SELF_REDUCTION_SPEED_BUFF / SELF_REWIND / SELF_SHIELD / SELF_STEALTH_SPEED / SMOKE_VISION_ZONE / SPAWN_DECOYS / SPAWN_SUMMON / SPEND_GOLD_TRUE_DAMAGE / TARGET_BEHIND_BLINK_AND_ARM_CRIT / TARGET_DAMAGE_AMP_MARK / TARGET_DAMAGE_HARD_CONTROL / TARGET_DOT_REVEAL / TARGET_REVEAL_BOUNTY_MARK / TOGGLE_PLAYER_BOUNTY / UNIFORM_RANDOM_DAMAGE / VISIBLE_ANY_HERO_HEAL / VISIBLE_TRAP_OR_REMOTE_DETONATE / WIND_WALL
ActiveEffectSchema.family_authority	E.30 逐类型字段矩阵与 BASE/DAMAGE/HEAL/SUMMON 族声明为唯一 schema 权威；common_required 已废止
ActiveEffectSchema.damage_events_required	含伤害载荷必须使用 damage_events[]；每个子事件独立声明 damage_form / hit_kind / element_rule / source_faction / owner_credit / trigger
ActiveEffectSchema.omitted_damage_fields	USE_EXPLICIT_ACTIVE_DEFAULTS_ONLY; NEVER_INFER_FROM_EFFECT_TYPE_NAME
ActiveEffectDefaults.player_normal_damage	DamageForm=SKILL; SourceFaction=PLAYER_OWNED; ElementRule=OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT; HitKind=ABILITY; OwnerCredit=CASTER

工程字段	唯一值 / 执行语义
MapPhysicalObjectClass	STATIC_BOUND / STATIC_VAULT / H031_PAGODA / H036_FROZEN_RIVER / FUSHI_STONE / SMOKE_ZONE / RAMP_WALKABLE
MobilityTag	BLINK / DASH / CHARGE / FORCED_SELF_MOVEMENT / FLIGHT_ASCEND / FLIGHT_DESCEND
MonsterSkillShape	CIRCLE / CONE / LINE / LINE_CAPSULE / THREE_PARALLEL_LINES / THREE_RADIAL_LANES_WITH_ONE_GAP / SINGLE_TARGET_CHAIN / NONE
CollisionQuery	3D_CAPSULE_SWEEP
MobilityTagAuthority	MobilityTag_CLOSED_ENUM_AND_ACTIVE_REGISTRY
BlockerClass	STATIC_SOLID / WIND_WALL / FROZEN_RIVER
H009ExcludedFieldClass	PROC_CHANCE / DURATION / COOLDOWN / COUNT / ECONOMY / PERMANENT_GROWTH / LETHAL_PROTECTION / STORM_PROBABILITY
DamageEventId	MAIN / CROSS / BURN / STACK_DOT / DETONATION / SPLASH / POISON_DOT / CONTACT / CONTACT_BURN / ARROW / BREAK_AOE / REFLECT / COUNTER / CONVERT_REFLECT / AFTERIMAGE_EXPLOSION / EXTRA_BASIC_BONUS / RESONANCE_AOE / STATUE_DESTROY_AOE / LIGHTNING
SummonArchetype	STONE_ARHAT / BEAN_SOLDIER / SIX_EAR_DECOY / WOLF_SPIRIT / FIRE_SPIRIT / STONE_STATUE / CORE_MIRROR
ValueContainerType	INT64 / Q4 / BOOL / ENUM / SET / ARRAY / FORMULA / HASH / REF
EquipmentQuality	WHITE / BLUE / PURPLE / GOLD
StatKind	ATTACK_FLAT / MAX_HP_FLAT / MOVE_SPEED_FLAT / ATTACK_SPEED_PERCENT

E.24 设置

工程字段	唯一值 / 执行语义
cloud[0].id	smart_cast
cloud[0].type	bool
cloud[0].default	是
cloud[1].id	tutorial_tips
cloud[1].type	bool
cloud[1].default	是
cloud[2].id	colorblind_mode

工程字段	唯一值 / 执行语义
cloud[2].type	enum
cloud[2].values	OFF / DEUTERANOPIA / PROTANOPIA / TRITANOPIA
cloud[2].default	OFF
cloud[2].legacy_value_mapping.A	DEUTERANOPIA
cloud[2].legacy_value_mapping.B	PROTANOPIA
cloud[2].legacy_value_mapping.C	TRITANOPIA
cloud[3].id	high_contrast_indicators
cloud[3].type	bool
cloud[3].default	否
cloud[4].id	reduced_flashes
cloud[4].type	bool
cloud[4].default	否
cloud[5].id	camera_shake
cloud[5].type	bool
cloud[5].default	是
cloud[6].id	damage_numbers
cloud[6].type	bool
cloud[6].default	是
cloud[7].id	battle_history_public
cloud[7].type	bool
cloud[7].default	是
cloud[8].id	frequent_heroes_public
cloud[8].type	bool
cloud[8].default	是
cloud[9].id	region
cloud[9].type	enum
cloud[9].values	AUTO / CN / APAC / EU / NA / MENA
cloud[9].default	AUTO
device_local[0].id	quality
device_local[0].type	enum
device_local[0].values	AUTO / LOW / MEDIUM / HIGH
device_local[0].default	AUTO
device_local[1].id	fps_cap
device_local[1].type	enum
device_local[1].values	30 / 45 / 60
device_local[1].default	30

工程字段	唯一值 / 执行语义
device_local[2].id	resolution_scale
device_local[2].type	enum
device_local[2].values	0.7 / 0.85 / 1
device_local[2].default	0.85
device_local[3].id	master_volume
device_local[3].type	range
device_local[3].min	0
device_local[3].max	100
device_local[3].default	80
device_local[4].id	music_volume
device_local[4].type	range
device_local[4].min	0
device_local[4].max	100
device_local[4].default	60
device_local[5].id	sfx_volume
device_local[5].type	range
device_local[5].min	0
device_local[5].max	100
device_local[5].default	80
device_local[6].id	ui_volume
device_local[6].type	range
device_local[6].min	0
device_local[6].max	100
device_local[6].default	70
device_local[7].id	mobile_control_scale
device_local[7].type	range
device_local[7].min	80
device_local[7].max	120
device_local[7].default	100
device_local[8].id	mobile_control_opacity
device_local[8].type	range
device_local[8].min	80
device_local[8].max	120
device_local[8].default	100
device_local[9].id	pc_aim_sensitivity
device_local[9].type	range
device_local[9].min	50

工程字段	唯一值 / 执行语义
device_local[9].max	200
device_local[9].default	100
device_local[10].id	show_network_stats
device_local[10].type	bool
device_local[10].default	否
gameplay_information_must_not_change_with_quality	是
controller_supported	否
color_is_never_sole_gameplay_channel	是
element_shape_symbols.METAL	DIAMOND
element_shape_symbols.WOOD	LEAF
element_shape_symbols.WATER	WAVE
element_shape_symbols.FIRE	FLAME
element_shape_symbols.EARTH	SQUARE
cloud_conflict	LATEST_SERVER_UPDATED_AT_PER_FIELD; TIE_HIGHER_RESOURCE_VERSION
device_local_not_cloud_synced	是
reduced_flashes.alternate_required	是；每个强闪光预警（五行爆发/雷暴落点/硬控预警）必须提供形状脉冲+边框+文字三合一低闪替代资产
reduced_flashes.asset_naming	原资产 ID 后缀_LOWFLASH；缺失低闪变体时构建失败
reduced_flashes.identical_timing	低闪替代的预警时长与判定完全一致，仅表现不同

E.25 埋点

工程字段	唯一值 / 执行语义
common_required	event_id / event_time_server / session_id / account_hash / region / client_platform / build_version
match_required	match_id / population_mode / human_count / bot_count / mmr_bucket
events[0].id	session_start
events[0].authority	CLIENT
events[0].fields	device_tier / input_mode
events[1].id	matchmaking_start
events[1].authority	SERVER
events[1].fields	search_radius / new_player_match_index
events[2].id	match_created
events[2].authority	SERVER
events[2].fields	human_count / bot_mix / population_mode
events[3].id	match_load_complete
events[3].authority	CLIENT

工程字段	唯一值 / 执行语义
events[3].fields	load_ms / resource_retry_count
events[4].id	hero_selected
events[4].authority	SERVER
events[4].fields	hero_id / reroll_count
events[4].required_fields	reroll_used / offer_set_index
events[5].id	spawn_assigned
events[5].authority	SERVER
events[5].fields	spawn_macro / spawn_micro / life_index
events[6].id	first_encounter
events[6].authority	SERVER
events[6].fields	other_type / elapsed_s / distance_m / los
events[7].id	monster_aggro
events[7].authority	SERVER
events[7].fields	monster_type / group_size / source_region
events[8].id	monster_killed
events[8].authority	SERVER
events[8].fields	monster_type / last_hit_source / kill_s / drop_ids
events[9].id	player_true_death
events[9].authority	SERVER
events[9].fields	life_index / killer_type / killer_id_hash / region_id / nearby_monsters / nearby_players
events[10].id	player_respawn
events[10].authority	SERVER
events[10].fields	life_index / soul_flight_s / respawn_region
events[11].id	loot_spawned
events[11].authority	SERVER
events[11].fields	item_id / source_type / region_id
events[12].id	loot_picked
events[12].authority	SERVER
events[12].fields	item_id / ground_age_s / result
events[13].id	skillbook_resolved
events[13].authority	SERVER
events[13].fields	book_id / action / replaced_id / result
events[14].id	gem_allocated
events[14].authority	SERVER
events[14].fields	passive_id / before_level / after_level
events[15].id	active_replaced
events[15].authority	SERVER

工程字段	唯一值 / 执行语义
events[15].fields	old_active_id / new_active_id / source
events[16].id	equipment_changed
events[16].authority	SERVER
events[16].fields	item_id / action / slot / result
events[17].id	shop_discovered
events[17].authority	SERVER
events[17].fields	shop_type / region_id / method
events[18].id	shop_transaction
events[18].authority	SERVER
events[18].fields	transaction_id / shop_type / action / item_id / gold_delta / result / latency_ms
events[19].id	boss_engagement
events[19].authority	SERVER
events[19].fields	boss_type / element / participants / elapsed_s
events[20].id	boss_killed
events[20].authority	SERVER
events[20].fields	boss_type / element / last_hit_element / upgrade_triggered / participant_count / drop_ids
events[21].id	storm_stage
events[21].authority	SERVER
events[21].fields	stage / safe_radius_m / center_target
events[22].id	storm_hit
events[22].authority	SERVER
events[22].fields	stage / damage / stunned / escaped_after_s
events[23].id	match_settled
events[23].authority	SERVER
events[23].fields	settlement_id / placement / kills_human / kills_bot / mmr_delta_internal / xiuwei / ledger_created / ledger_transferred / ledger_destroyed
events[24].id	disconnect_state
events[24].authority	SERVER
events[24].fields	state / duration_s / bot_takeover / device_switch
events[25].id	performance_sample
events[25].authority	CLIENT
events[25].fields	frame_ms / memory_mb / rtt_ms / packet_loss / position_correction_m
events[26].id	ui_input
events[26].authority	CLIENT
events[26].fields	action / control / hold_ms / cancelled / drawer_state
events[27].required_fields	match_id / participant_slot / server_tick
events[27].id	airdrop_warned

工程字段	唯一值 / 执行语义
events[27].authority	SERVER
events[27].fields	match_id / participant_slot / server_tick
events[28].required_fields	match_id / participant_slot / server_tick
events[28].id	airdrop_arrived
events[28].authority	SERVER
events[28].fields	match_id / participant_slot / server_tick
events[29].required_fields	match_id / participant_slot / server_tick
events[29].id	airdrop_opened
events[29].authority	SERVER
events[29].fields	match_id / participant_slot / server_tick
events[30].required_fields	match_id / participant_slot / server_tick
events[30].id	black_mountain_committed
events[30].authority	SERVER
events[30].fields	match_id / participant_slot / server_tick
events[31].required_fields	match_id / participant_slot / server_tick
events[31].id	taibai_swap_committed
events[31].authority	SERVER
events[31].fields	match_id / participant_slot / server_tick
events[32].required_fields	match_id / participant_slot / server_tick
events[32].id	revive_protection_removed
events[32].authority	SERVER
events[32].fields	match_id / participant_slot / server_tick
events[33].required_fields	match_id / participant_slot / server_tick
events[33].id	B35_item_instance_reward
events[33].authority	SERVER
events[33].fields	match_id / participant_slot / server_tick
events[34].required_fields	match_id / participant_slot / server_tick
events[34].id	co_champion_settled
events[34].authority	SERVER
events[34].fields	match_id / participant_slot / server_tick
privacy.raw_account_id_forbidden	是
privacy.account_hash	HMAC_SHA256_DAILY_ROTATING_ANALYTICS_KEY_ACCOUNT_UUID
privacy.participant_slot	UINT8_0_TO_29
privacy.ip_address_not_event_field	是
time.event_time_server	UTC_RFC3339_MILLISECONDS
time.server_tick	UINT64_20HZ
time.client_time	OPTIONAL_DIAGNOSTIC_NOT_ORDER_AUTHORITY

工程字段	唯一值 / 执行语义
field_types.event_id	UUID
field_types.session_id	UUID
field_types.account_hash	HEX_SHA256
field_types.match_id	UUID
field_types.participant_slot	UINT8
field_types.server_tick	UINT64
field_types.elapsed_s	DECIMAL_SECONDS
field_types.distance_m	DECIMAL_METERS
field_types.gold_delta	INT64
field_types.result	CLOSED_ENUM
retention_days.raw_event	30
retention_days.aggregated_anonymous	400
retention_days.security_events	SECURITY_REPLAY_CONFIG
unknown_field	DROP_AND_COUNT_SCHEMA_ERROR; NEVER_COERCE_TYPE
field_types_complete.account_hash	HEX_SHA256
field_types_complete.action	UTF8_NFC_1_TO_64
field_types_complete.after_level	INT64
field_types_complete.before_level	INT64
field_types_complete.book_id	TYPED_CONTENT_REF
field_types_complete.boss_type	UTF8_NFC_1_TO_64
field_types_complete.bot_count	INT64
field_types_complete.bot_mix	MAP_BOT_DIFFICULTY_TO_UINT8
field_types_complete.bot_takeover	BOOLEAN
field_types_complete.build_version	UTF8_NFC_1_TO_64
field_types_complete.cancelled	BOOLEAN
field_types_complete.center_target	UTF8_NFC_1_TO_64
field_types_complete.client_platform	ENUM_ANDROID_IOS_WINDOWS
field_types_complete.control	UTF8_NFC_1_TO_64
field_types_complete.damage	INT64
field_types_complete.device_switch	BOOLEAN
field_types_complete.device_tier	UTF8_NFC_1_TO_64
field_types_complete.distance_m	DECIMAL_NUMBER
field_types_complete.drawer_state	UTF8_NFC_1_TO_64
field_types_complete.drop_ids	CONTENT_REF_ARRAY
field_types_complete.duration_s	DECIMAL_NUMBER
field_types_complete.elapsed_s	DECIMAL_NUMBER
field_types_complete.element	ENUM_METAL_WOOD_WATER_FIRE_EARTH_NONE

工程字段	唯一值 / 执行语义
field_types_complete.escaped_after_s	DECIMAL_NUMBER
field_types_complete.event_id	UUID
field_types_complete.event_time_server	UTC_RFC3339_MILLISECONDS
field_types_complete.frame_ms	DECIMAL_NUMBER
field_types_complete.gold_delta	INT64
field_types_complete.ground_age_s	DECIMAL_NUMBER
field_types_complete.group_size	INT64
field_types_complete.hero_id	TYPED_CONTENT_REF
field_types_complete.hold_ms	INT64
field_types_complete.human_count	INT64
field_types_complete.input_mode	UTF8_NFC_1_TO_64
field_types_complete.item_id	TYPED_CONTENT_REF
field_types_complete.kill_s	DECIMAL_NUMBER
field_types_complete.killer_id_hash	HEX_SHA256
field_types_complete.killer_type	UTF8_NFC_1_TO_64
field_types_complete.kills_bot	INT64
field_types_complete.kills_human	INT64
field_types_complete.last_hit_element	ENUM_METAL_WOOD_WATER_FIRE_EARTH_NONE
field_types_complete.last_hit_source	UTF8_NFC_1_TO_64
field_types_complete.latency_ms	INT64
field_types_complete.ledger_created	INT64
field_types_complete.ledger_destroyed	INT64
field_types_complete.ledger_transferred	INT64
field_types_complete.life_index	INT64
field_types_complete.load_ms	INT64
field_types_complete.los	BOOLEAN
field_types_complete.match_id	UUID
field_types_complete.memory_mb	UINT32
field_types_complete.method	UTF8_NFC_1_TO_64
field_types_complete.mmr_bucket	UTF8_NFC_1_TO_64
field_types_complete.mmr_delta_internal	INT32
field_types_complete.monster_type	UTF8_NFC_1_TO_64
field_types_complete.nearby_monsters	INT64
field_types_complete.nearby_players	INT64
field_types_complete.new_active_id	TYPED_CONTENT_REF
field_types_complete.new_player_match_index	INT64
field_types_complete.offer_set_index	INT64

工程字段	唯一值 / 执行语义
field_types_complete.old_active_id	TYPED_CONTENT_REF
field_types_complete.other_type	UTF8_NFC_1_TO_64
field_types_complete.packet_loss	DECIMAL_0_TO_1
field_types_complete.participant_count	INT64
field_types_complete.participant_slot	UINT8_0_TO_29
field_types_complete.participants	PARTICIPANT_SLOT_ARRAY
field_types_complete.passive_id	TYPED_CONTENT_REF
field_types_complete.placement	INT64
field_types_complete.population_mode	UTF8_NFC_1_TO_64
field_types_complete.position_correction_m	DECIMAL_NUMBER
field_types_complete.region	UTF8_NFC_1_TO_64
field_types_complete.region_id	UTF8_NFC_1_TO_64
field_types_complete.replaced_id	TYPED_CONTENT_REF
field_types_complete.reroll_count	INT64
field_types_complete.reroll_used	BOOLEAN
field_types_complete.resource_retry_count	INT64
field_types_complete.respawn_region	UTF8_NFC_1_TO_64
field_types_complete.result	UTF8_NFC_1_TO_64
field_types_complete.rtt_ms	INT64
field_types_complete.safe_radius_m	DECIMAL_NUMBER
field_types_complete.search_radius	INT64
field_types_complete.server_tick	UINT64_20HZ
field_types_complete.session_id	UUID
field_types_complete.settlement_id	UUID
field_types_complete.shop_type	UTF8_NFC_1_TO_64
field_types_complete.slot	UTF8_NFC_1_TO_64
field_types_complete.soul_flight_s	DECIMAL_NUMBER
field_types_complete.source	UTF8_NFC_1_TO_64
field_types_complete.source_region	UTF8_NFC_1_TO_64
field_types_complete.source_type	UTF8_NFC_1_TO_64
field_types_complete.spawn_macro	UTF8_NFC_1_TO_64
field_types_complete.spawn_micro	UTF8_NFC_1_TO_64
field_types_complete.stage	UTF8_NFC_1_TO_64
field_types_complete.state	UTF8_NFC_1_TO_64
field_types_complete.stunned	BOOLEAN
field_types_complete.transaction_id	UUID
field_types_complete.upgrade_triggered	BOOLEAN

工程字段	唯一值 / 执行语义
field_types_complete.xiuwei	INT64
field_type_coverage_count	98
untyped_field_policy	REJECT_EVENT_AND_INCREMENT_TELEMETRY_SCHEMA_ERROR
bounded_utf8_is_not_enum	是

E.26 主动技能强类型运行时载荷

本表是 57 项主动的运行时权威。runtime_payload_json 中的每个值均带显式类型标签；内容编译器只读取本表，禁止解析 § 6 的自然语言或分号字符串。未知字段、重复字段、类型不符或缺少必填字段均使内容构建失败。

active_id	runtime_payload_json
H001	{ "effect_type": { "type": "ENUM", "value": "WIND_WALL" }, "targeting": { "type": "ENUM", "value": "DIRECTION" }, "cast_type": { "type": "ENUM", "value": "INSTANT" }, "range_m": { "type": "INT64", "value": 15, "unit": "m" }, "shape": { "type": "ENUM", "value": "LINE_WALL" }, "radius_m": { "type": "INT64", "value": 0, "unit": "m" }, "width_m": { "type": "INT64", "value": 12, "unit": "m" }, "duration_s": { "type": "INT64", "value": 3, "unit": "s" }, "length_m": { "type": "INT64", "value": 12, "unit": "m" }, "knockback_m": { "type": "INT64", "value": 5, "unit": "m" }, "solid": { "type": "BOOL", "value": false }, "blocks_projectile_tag": { "type": "ENUM", "value": "BLOCKABLE_BY_WIND_WALL" }, "schema_version": { "type": "ENUM", "value": "V3_0" }, "payload_sha256": { "type": "HASH", "value": "018287f28dd6d30f5c433bd53a332248dfb30354ddafcd1917884afda4c64436" } }

ac tiv e_ id	runtime_payload_json
H0 02	<pre>{ "effect_type": { "type": "ENUM", "value": "FIRE_WALL", "targeting": { "type": "ENUM", "value": "DIRECTION", "cast_type": { "type": "ENUM", "value": "INSTANT", "range_m": { "type": "INT64", "value": 15, "unit": "m", "shape": { "type": "ENUM", "value": "LINE_WALL", "radius_m": { "type": "INT64", "value": 0, "unit": "m", "width_m": { "type": "INT64", "value": 15, "unit": "m", "duration_s": { "type": "INT64", "value": 5, "unit": "s", "length_m": { "type": "INT64", "value": 15, "unit": "m", "thickness_m": { "type": "INT64", "value": 1, "unit": "m", "source_faction": { "type": "ENUM", "value": "PLAYER_OWNED", "element_rule": { "type": "ENUM", "value": "OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT", "owner_credit": { "type": "ENUM", "value": "CASTER", "damage_events": { "type": "ARRAY", "element": "DamageEventSpec", "ordered": true, "value": [{ "event_id": { "type": "ENUM", "value": "CROSS", "damage": { "type": "FORMULA", "src": "fixed=100", "ast": { "q4": 1000000 }, "attack_coef": { "type": "Q4", "value_scaled_1e4": 4000, "human": 0.4, "unit": "ratio", "damage_form": { "type": "ENUM", "value": "SKILL", "hit_kind": { "type": "ENUM", "value": "ABILITY", "element_rule": { "type": "ENUM", "value": "OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT", "source_faction": { "type": "ENUM", "value": "PLAYER_OWNED", "owner_credit": { "type": "ENUM", "value": "CASTER", "trigger": { "type": "ENUM", "value": "FULL_OUTSIDE_TO_INSIDE_OR_INSIDE_TO_OUTSIDE", "reentry_lock_per_target_s": { "type": "Q4", "value_scaled_1e4": 5000, "human": 0.5, "unit": "s", "standing_overlap_does_not_retrigger": { "type": "BOOL", "value": true }, "event_id": { "type": "ENUM", "value": "BURN", "damage_per_s": { "type": "INT64", "value": 30, "unit": "hp/s", "duration_s": { "type": "INT64", "value": 3, "unit": "s", "refresh": { "type": "BOOL", "value": true, "damage_form": { "type": "ENUM", "value": "DOT", "hit_kind": { "type": "ENUM", "value": "PERIODIC", "element_rule": { "type": "ENUM", "value": "OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT", "source_faction": { "type": "ENUM", "value": "PLAYER_OWNED", "owner_credit": { "type": "ENUM", "value": "CASTER", "trigger": { "type": "ENUM", "value": "APPLIED_ON_CROSS_COMMIT" } } } } } } } } } } } } } } } } } } } }] } } } } } } } } } } } } } } } }</pre>
H0 03	<pre>{ "effect_type": { "type": "ENUM", "value": "DAMAGE_SLOW_ZONE", "targeting": { "type": "ENUM", "value": "GROUND_POINT", "cast_type": { "type": "ENUM", "value": "INSTANT", "range_m": { "type": "INT64", "value": 30, "unit": "m", "shape": { "type": "ENUM", "value": "CIRCLE", "radius_m": { "type": "INT64", "value": 8, "unit": "m", "width_m": { "type": "INT64", "value": 0, "unit": "m", "duration_s": { "type": "INT64", "value": 4, "unit": "s", "tick_s": { "type": "INT64", "value": 1, "unit": "s", "damage": { "type": "FORMULA", "src": "fixed=40", "ast": { "q4": 400000 }, "attack_coef": { "type": "Q4", "value_scaled_1e4": 2000, "human": 0.2, "unit": "ratio", "slow_percent": { "type": "INT64", "value": 25, "unit": "pct", "damage_form": { "type": "ENUM", "value": "SKILL", "source_faction": { "type": "ENUM", "value": "PLAYER_OWNED", "element_rule": { "type": "ENUM", "value": "OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT", "hit_kind": { "type": "ENUM", "value": "PERIODIC", "owner_credit": { "type": "ENUM", "value": "CASTER", "schema_version": { "type": "ENUM", "value": "V3_0", "payload_sha256": { "type": "HASH", "value": "4299cb8041ab1a1fe32dbc31cddf8fce1ffb463192a4761130b8b293f56675df" } } } } } } } } } } } } } } } } } } } }</pre>

ac tiv e_ id	runtime_payload_json
H0 04	{ "effect_type":{"type":"ENUM","value":"CONDITIONAL_POISON_DETONATE"}, "targeting":{"type":"ENUM","value":"VISIBLE_ENEMY"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":20,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":4,"unit":"s"}, "max_count":{"type":"INT64","value":10,"unit":"scalar"}, "stack_scope":{"type":"ENUM","value":"CASTER_OWNED_DOT_AND_TOXIN_ONLY"}, "when_no_owned_dot_add_stacks":{"type":"INT64","value":3,"unit":"scalar"}, "when_owned_dot_consume_all_owned_dot_and_toxin_stacks":{"type":"BOOL","value":true}, "other_owner_stacks_unchanged":{"type":"BOOL","value":true}, "damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true, "value":[{"event_id":{"type":"ENUM","value":"STACK_DOT"}, "damage_per_stack_per_s":{"type":"INT64","value":5,"unit":"hp/s"}, "duration_s":{"type":"INT64","value":4,"unit":"s"}, "damage_form":{"type":"ENUM","value":"DOT"}, "hit_kind":{"type":"ENUM","value":"PERIODIC"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "trigger":{"type":"ENUM","value":"WHEN_NO_OWNED_DOT_STACKS_APPLIED"}]}, {"event_id":{"type":"ENUM","value":"DETONATION"}, "damage_target_max_hp_percent_per_stack":{"type":"INT64","value":3,"unit":"hp"}, "damage_form":{"type":"ENUM","value":"PERCENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "trigger":{"type":"ENUM","value":"WHEN_OWNED_DOT_CONSUME_ALL_STACKS"}}]}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"4c0ffbc7b787854156cd4c4a37901eec1888d6374984e660d1eac8547b30a06f"}} }
H0 05	{ "effect_type":{"type":"ENUM","value":"PETRIFY_TARGET"}, "targeting":{"type":"ENUM","value":"VISIBLE_ENEMY"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":20,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"s"}, "target_invulnerable":{"type":"BOOL","value":true}, "target_cannot_act":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"90700a6db2699b4caf735b59bbb82319f1c2cd742351917f325d254c8d7187b2"}} }

ac tiv e_ id	runtime_payload_json
H0 06	{ "effect_type":{"type":"ENUM","value":"CONTAGIOUS_DAMAGE_ZONE"}, "targeting":{"type":"ENUM","value":"GROUND_POINT"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":30,"unit":"m"}, "shape":{"type":"ENUM","value":"CIRCLE"}, "radius_m":{"type":"INT64","value":8,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":8,"unit":"s"}, "tick_s":{"type":"INT64","value":1,"unit":"s"}, "damage":{"type":"FORMULA","src":"fixed=40", "ast":{"q4":400000}}, "attack_coef":{"type":"Q4","value_scaled_1e4":2000, "human":"0.2","unit":"ratio"}, "outside_residual_s":{"type":"INT64","value":3,"unit":"s"}, "infection_radius_m":{"type":"INT64","value":2,"unit":"m"}, "check_interval_s":{"type":"Q4","value_scaled_1e4":5000, "human":"0.5","unit":"s"}, "max_generation":{"type":"INT64","value":2,"unit":"scalar"}, "stack":{"type":"BOOL","value":false}, "refresh":{"type":"BOOL","value":true}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"PERIODIC"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "infection_instance_scope":{"type":"ENUM","value":"PER_CASTER_PER_ROOT_EFFECT"}, "generation_origin":{"type":"ENUM","value":"ROOT_EFFECT_INSTANCE_ID"}, "same_caster_reinfection":{"type":"ENUM","value":"REFRESH_DURATION_NO_STACK"}, "different_caster_infection":{"type":"ENUM","value":"INDEPENDENT"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"99808230cc9c9eee2f369c073edea417116b667735fdbb83ac8404156edfebb0"}} }
H0 07	{ "effect_type":{"type":"ENUM","value":"DAMAGE_SLOW_ZONE"}, "targeting":{"type":"ENUM","value":"GROUND_POINT"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":30,"unit":"m"}, "shape":{"type":"ENUM","value":"CIRCLE"}, "radius_m":{"type":"INT64","value":10,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":4,"unit":"s"}, "tick_s":{"type":"INT64","value":1,"unit":"s"}, "damage":{"type":"FORMULA","src":"fixed=60", "ast":{"q4":600000}}, "attack_coef":{"type":"Q4","value_scaled_1e4":3000, "human":"0.3","unit":"ratio"}, "slow_percent":{"type":"INT64","value":40,"unit":"pct"}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"PERIODIC"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"70b4018c0e10e8ae29dd652af0484830cb90f98914802b59ccc69638e7530b5f"}} }

ac tiv e_ id	runtime_payload_json
H0 08	{ "effect_type":{"type":"ENUM","value":"DELAYED_AREA_DAMAGE"}, "targeting":{"type":"ENUM","value":"GROUND_POINT"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":30,"unit":"m"}, "shape":{"type":"ENUM","value":"CIRCLE"}, "radius_m":{"type":"INT64","value":12,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "telegraph_s":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"s"}, "damage":{"type":"FORMULA","src":"fixed=400","ast":{"q4":4000000}}, "attack_coef":{"type":"Q4","value_scaled_1e4":12000,"human":"1.2","unit":"ratio"}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OW_NED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"23b5370ed295b9a36a74751917e41257c11cbd04bee511380aaa07b97afa49d3"} }
H0 09	{ "effect_type":{"type":"ENUM","value":"SELF_COMBAT_BUFF"}, "targeting":{"type":"ENUM","value":"SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":0,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":5,"unit":"s"}, "attack_speed_add_percent":{"type":"INT64","value":80,"unit":"pct"}, "passive_effect_magnitude_multiplier":{"type":"INT64","value":2,"unit":"ratio"}, "passive_proc_probability_multiplier":{"type":"INT64","value":1,"unit":"ratio"}, "incoming_damage_multiplier":{"type":"Q4","value_scaled_1e4":12000,"human":"1.2","unit":"ratio"}, "passive_magnitude_authority":{"type":"ENUM","value":"H009_EXACT_FIELD_WHITELIST"}, "excluded_field_classes":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["COOLDOWN","COUNT","DURATION","ECONOMY","LETHAL_PROTECTION","PERMANENT_GROWTH","PROC_CHANCE","STORM_PROBABILITY"]}, "transform_additive":{"type":"FORMULA","src":"VALUE*2","ast":{"op":"MUL","args":[{"var":"VALUE"},{"q4":20000}]}}, "transform_delta_from_one":{"type":"FORMULA","src":"1+(VALUE-1)*2","ast":{"op":"ADD","args":[{"q4":10000},{op:"MUL",args:[{"op":"SUB",args:[{"var":"VALUE"},{"q4":10000}]}]}]}}, "transform_remaining_from_one":{"type":"FORMULA","src":"MAX(0,1-(1-VALUE)*2)","ast":{"op":"MAX",args:[{"q4":0},{op:"SUB",args:[{"q4":10000},{op:"MUL",args:[{"op":"SUB",args:[{"q4":10000},{var":"VALUE"}]}]}]}]}}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"7f94e7c35b0e797775b19bb9d6be7dd6b4294a390a5ef852201c670e252e5674"} }

ac tiv e_ id	runtime_payload_json
H0 10	{ "effect_type":{"type":"ENUM","value":"ARM_NEXT_BASIC"}, "targeting":{"type":"ENUM","value":"SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":0,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "consume_on":{"type":"ENUM","value":"VALID_BASIC_HIT_RESOLUTION"}, "guaranteed_crit":{"type":"BOOL","value":true}, "extra_target_missing_hp_percent":{"type":"INT64","value":15,"unit":"pct"}, "kill_resets_active_cd":{"type":"BOOL","value":true}, "consume_if_damage_zero_due_to_dodge_or_immunity":{"type":"BOOL","value":true}, "do_not_consume_if_no_hit_entity":{"type":"BOOL","value":true}, "kill_reset_timing":{"type":"ENUM","value":"AFTER_LETHAL_COMMIT"}, "arm_persistence":{"type":"ENUM","value":"UNTIL_NEXT_LEGAL_BASIC_NO_TIMEOUT"}, "cleared_on":{"type":"SET","element":{"type":"ENUM","unique":true,"ordered":false,"value":["ACTIVE_REPLACED","HERO_SWAP","TRUE_DEATH"]}}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"5d446207f556b804518450f466cce01a84537e9ca9093f6e9f3566200fc5f76c"}} }
H0 11	{ "effect_type":{"type":"ENUM","value":"DASH_STOP_FIRST"}, "targeting":{"type":"ENUM","value":"DIRECTION"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":25,"unit":"m"}, "shape":{"type":"ENUM","value":"LINE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":3,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "distance_m":{"type":"INT64","value":25,"unit":"m"}, "damage":{"type":"FORMULA","src":"fixed=100","ast":{"q4":1000000}}, "attack_coef":{"type":"Q4","value_scaled_1e4":5000,"human":0.5,"unit":"ratio"}, "stun_s":{"type":"Q4","value_scaled_1e4":8000,"human":0.8,"unit":"s"}, "interruptible_by_hard_control":{"type":"BOOL","value":true}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"e8e686fd8b2f035bee08b2f4f6d63d18bb40bbde501cfc9803ea3e9b0c71d176"}} }

ac tiv e_ id	runtime_payload_json
H0 12	{ "effect_type":{"type":"ENUM","value":"SPAWN_DECOYS"}, "targeting":{"type":"ENUM","value":"SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":0,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":8,"unit":"s"}, "count":{"type":"INT64","value":2,"unit":"scalar"}, "hp_inherit_percent":{"type":"INT64","value":30,"unit":"pct"}, "move_speed_inherit_percent":{"type":"INT64","value":30,"unit":"pct"}, "attack":{"type":"INT64","value":0,"unit":"scalar"}, "unit_collision":{"type":"BOOL","value":false}, "copy_owner_visual_and_hp_percent":{"type":"BOOL","value":true}, "decoy_max_hp":{"type":"FORMULA","src":"FLOOR(OWNER_CURRENT_MAX_HP*0.3)","ast":{"op":"FLOOR","args":[{"op":"MUL","args":[{"var":"OWNER_CURRENT_MAX_HP"},{"q4":3000}]}]}}, "decoy_current_hp":{"type":"FORMULA","src":"FLOOR(DECOY_MAX_HP*OWNER_CURRENT_HP/OWNER_CURRENT_MAX_HP)","ast":{"op":"FLOOR","args":[{"op":"DIV_FLOOR","args":[{"op":"MUL","args":[{"var":"DECOY_MAX_HP"},{"var":"OWNER_CURRENT_HP"}]}],"var":"OWNER_CURRENT_MAX_HP"}]}}, "decoy_death_credit":{"type":"ENUM","value":"NONE"}, "decoy_gold_xp_drop":{"type":"ENUM","value":"NONE"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"c16e3ff15a26d88860c4900e7b5048bd5c95f33b3168540382a51eac9b04c214"}}
H0 13	{ "effect_type":{"type":"ENUM","value":"TARGET_BEHIND_BLINK_AND_ARM_CRIT"}, "targeting":{"type":"ENUM","value":"VISIBLE_ENEMY"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":40,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "behind_offset_m":{"type":"Q4","value_scaled_1e4":15000,"human":1.5,"unit":"m"}, "arm_next_basic_crit":{"type":"BOOL","value":true}, "candidate_directions":{"type":"ENUM","value":"REAR_THE_N_CLOCKWISE_45_DEG_EIGHT_POINTS"}, "candidate_radius_m":{"type":"Q4","value_scaled_1e4":15000,"human":1.5,"unit":"m"}, "candidate_query":{"type":"ENUM","value":"PLAYER_CAPSULE_SWEEP_AND_DESTINATION_LEGALITY"}, "no_candidate":{"type":"ENUM","value":"FAIL_NO_COOLDOWN"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"ba4291838d09ed0f567673add44df57330fbbcc39960ee3f33ea6b5ecb9224db"}}

ac tiv e_ id	runtime_payload_json
H0 14	{ "effect_type":{"type":"ENUM","value":"BLINK_LEAVE_DELAYED_DECOY"}, "targeting":{"type":"ENUM","value":"DIRECTION"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":15,"unit":"m"}, "shape":{"type":"ENUM","value":"POINT"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":1,"unit":"s"}, "distance_m":{"type":"INT64","value":15,"unit":"m"}, "delay_s":{"type":"INT64","value":1,"unit":"s"}, "damage":{"type":"FORMULA","src":"fixed=120","ast":{"q4":1200000}}, "attack_coef":{"type":"Q4","value_scaled_1e4":5000,"human":"0.5","unit":"ratio"}, "blink_uses_capsule_sweep":{"type":"BOOL","value":true}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"1a935165279092479a3b7341d316040932eadea4722e1105384a9325f028d1dd"} }
H0 15	{ "effect_type":{"type":"ENUM","value":"CONE_DAMAGE_SLOW"}, "targeting":{"type":"ENUM","value":"DIRECTION"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":8,"unit":"m"}, "shape":{"type":"ENUM","value":"CONE"}, "radius_m":{"type":"INT64","value":8,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":2,"unit":"s"}, "angle_deg":{"type":"INT64","value":60,"unit":"deg"}, "damage":{"type":"FORMULA","src":"fixed=160","ast":{"q4":1600000}}, "attack_coef":{"type":"Q4","value_scaled_1e4":7000,"human":"0.7","unit":"ratio"}, "slow_percent":{"type":"INT64","value":30,"unit":"pct"}, "slow_s":{"type":"INT64","value":2,"unit":"s"}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"6148e591c50675095cf9e529958eeca05ed4ba4bcab8a8152a4c81d9e005e6e"} }
H0 16	{ "effect_type":{"type":"ENUM","value":"PIERCING_DASH"}, "targeting":{"type":"ENUM","value":"DIRECTION"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":25,"unit":"m"}, "shape":{"type":"ENUM","value":"LINE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":3,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "distance_m":{"type":"INT64","value":25,"unit":"m"}, "damage":{"type":"FORMULA","src":"fixed=100","ast":{"q4":1000000}}, "attack_coef":{"type":"Q4","value_scaled_1e4":5000,"human":"0.5","unit":"ratio"}, "knockback_m":{"type":"INT64","value":2,"unit":"m"}, "hit_each_target_once":{"type":"BOOL","value":true}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"a45d0cf6bcf9e591be3d5112cee0a03969fa1547b3b5d43aff15bfdbf4457f61"} }

ac tiv e_ id	runtime_payload_json
H0 17	{ "effect_type":{"type":"ENUM","value":"RADIAL_DAMAGE_KNOCK_TO_EDGE"}, "targeting":{"type":"ENUM","value":"SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":0,"unit":"m"}, "shape":{"type":"ENUM","value":"CIRCLE"}, "radius_m":{"type":"INT64","value":12,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "damage":{"type":"FORMULA","src":"fixed=150","ast":{"q4":1500000}}, "attack_coef":{"type":"Q4","value_scaled_1e4":7000,"human":"0.7","unit":"ratio"}, "stun_s":{"type":"INT64","value":1,"unit":"s"}, "blocked_knock_still_stuns":{"type":"BOOL","value":true}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "knock_direction":{"type":"ENUM","value":"CASTER_TO_TARGET_RADIAL"}, "knock_destination":{"type":"ENUM","value":"SKILL_EDGE_OR_LAST_LEGAL_SWEEP_POINT"}, "actor_collision":{"type":"ENUM","value":"STOP_LAST_LEGAL_POINT"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"be34fa50d9203f6a5b908d1076022730f32eff122f9019da0ffe7f810fc730bb"}}
H0 18	{ "effect_type":{"type":"ENUM","value":"MOBILE_CHANNEL_AOE"}, "targeting":{"type":"ENUM","value":"SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":0,"unit":"m"}, "shape":{"type":"ENUM","value":"CIRCLE"}, "radius_m":{"type":"INT64","value":8,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":3,"unit":"s"}, "tick_s":{"type":"INT64","value":1,"unit":"s"}, "damage_per_tick":{"type":"FORMULA","src":"fixed=100","ast":{"q4":1000000}}, "attack_coef":{"type":"Q4","value_scaled_1e4":5000,"human":"0.5","unit":"ratio"}, "self_move_multiplier":{"type":"Q4","value_scaled_1e4":5000,"human":"0.5","unit":"ratio"}, "hard_control_interrupts":{"type":"BOOL","value":true}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"PERIODIC"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"1db016ef2e6846b760cb2b7004c460f37ac5276c6aa7e49707260b94a95a1ee7"}}
H0 19	{ "effect_type":{"type":"ENUM","value":"DELAYED_SILENCE_ZONE"}, "targeting":{"type":"ENUM","value":"GROUND_POINT"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":25,"unit":"m"}, "shape":{"type":"ENUM","value":"CIRCLE"}, "radius_m":{"type":"INT64","value":10,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":3,"unit":"s"}, "telegraph_s":{"type":"Q4","value_scaled_1e4":4000,"human":"0.4","unit":"s"}, "silence_s":{"type":"INT64","value":3,"unit":"s"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"001e3aa1a02203854c11f4ba1f4f5634bdb825bd56816a96cdcc5e24fc990127"}}

ac tiv e_ id	runtime_payload_json
H0 20	{ "effect_type": { "type": "ENUM", "value": "SPAWN_SUMMON" }, "targeting": { "type": "ENUM", "value": "SELF_FORWARD" }, "cast_type": { "type": "ENUM", "value": "INSTANT" }, "range_m": { "type": "INT64", "value": 3, "unit": "m" }, "shape": { "type": "ENUM", "value": "POINT" }, "radius_m": { "type": "INT64", "value": 0, "unit": "m" }, "width_m": { "type": "INT64", "value": 0, "unit": "m" }, "duration_s": { "type": "INT64", "value": 20, "unit": "s" }, "archetype": { "type": "ENUM", "value": "STONE_ARHAT" }, "count": { "type": "INT64", "value": 1, "unit": "scalar" }, "hp": { "type": "INT64", "value": 500, "unit": "hp" }, "attack": { "type": "INT64", "value": 30, "unit": "scalar" }, "schema_version": { "type": "ENUM", "value": "V3_0" }, "payload_sha256": { "type": "HASH", "value": "3ef5fa101ae02dae46779333902e621a6aa846c785a9c37d698827bd1d60ee0f" } }
H0 21	{ "effect_type": { "type": "ENUM", "value": "CHARGED_RADIAL_PULL" }, "targeting": { "type": "ENUM", "value": "SELF" }, "cast_type": { "type": "ENUM", "value": "CHANNEL" }, "range_m": { "type": "INT64", "value": 0, "unit": "m" }, "shape": { "type": "ENUM", "value": "CIRCLE" }, "radius_m": { "type": "INT64", "value": 15, "unit": "m" }, "width_m": { "type": "INT64", "value": 0, "unit": "m" }, "duration_s": { "type": "INT64", "value": 0, "unit": "s" }, "charge_s": { "type": "INT64", "value": 1, "unit": "s" }, "max_targets": { "type": "INT64", "value": 8, "unit": "scalar" }, "damage": { "type": "FORMULA", "src": "fixed=120", "ast": { "q4": 1200000 } }, "attack_coef": { "type": "Q4", "value_scaled_1e4": 6000, "human": "0.6", "unit": "ratio" }, "landing_ring_m": { "type": "INT64", "value": 2, "unit": "m" }, "wall_blocks": { "type": "BOOL", "value": true }, "immune_tiers": { "type": "SET", "element": "ENUM", "unique": true, "ordered": false, "value": ["CORE_BOSS", "DRAGON", "LEADER"] }, "damage_form": { "type": "ENUM", "value": "SKILL" }, "source_faction": { "type": "ENUM", "value": "PLAYER_OWNED" }, "element_rule": { "type": "ENUM", "value": "OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT" }, "hit_kind": { "type": "ENUM", "value": "ABILITY" }, "owner_credit": { "type": "ENUM", "value": "CASTER" }, "max_target_selection_order": { "type": "SET", "element": "ENUM", "unique": true, "ordered": false, "value": ["DISTANCE_MM_ASC", "ENTITY_ID_ASC", "THREAT_DESC"] }, "landing_slot_layout": { "type": "ENUM", "value": "EIGHT_DISTINCT_RING_SLOTS_AROUND_CASTER" }, "landing_slot_order": { "type": "ENUM", "value": "CLOCKWISE_FROM_CASTER_FORWARD" }, "occupied_slot": { "type": "ENUM", "value": "NEXT_LEGAL_SLOT_ELSE_LAST_LEGAL_SWEEP_POINT" }, "schema_version": { "type": "ENUM", "value": "V3_0" }, "payload_sha256": { "type": "HASH", "value": "c4ca1d1fcd5d2cfb362d63ef63c1ab307025d7d9971e0439f8af7c9a22750259" } }

ac tiv e_ id	runtime_payload_json
H0 22	{ "effect_type":{"type":"ENUM","value":"SPEND_GOLD_TRUE_DAMAGE"}, "targeting":{"type":"ENUM","value":"VISIBLE_ENEMY"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":20,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "minimum_balance_to_cast":{"type":"INT64","value":500,"unit":"scalar"}, "cost":{"type":"FORMULA","src":"FLOOR(CLAMP(CURRENT_GOLD*0.20,500,5000))","ast":{"op":"FLOOR","args":[{"op":"CLAMP","args":[{"op":"MUL","args":[{"var":"CURRENT_GOLD"},{"q4":2000}]}],{"q4":5000000}]}}, "damage":{"type":"FORMULA","src":"FLOOR(COST*0.10)","ast":{"op":"FLOOR","args":[{"op":"MUL","args":[{"var":"COST"},{"q4":1000}]}}, "shieldable":{"type":"BOOL","value":true}, "atomic_deduct_and_cast":{"type":"BOOL","value":true}, "damage_form":{"type":"ENUM","value":"TRUE"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"NONE"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"374524b25c4262e8807976531b8486296bcc8fe315c2681fa0f06dd3d5fa0678"}} }
H0 23	{ "effect_type":{"type":"ENUM","value":"TARGET_DOT_REVEAL"}, "targeting":{"type":"ENUM","value":"VISIBLE_ENEMY"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":30,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":8,"unit":"s"}, "tick_s":{"type":"INT64","value":2,"unit":"s"}, "damage_per_tick":{"type":"INT64","value":40,"unit":"hp"}, "reveal":{"type":"ENUM","value":"EXACT_WORLD_AND_MINIMAP"}, "clear_on_caster_true_death":{"type":"BOOL","value":true}, "damage_form":{"type":"ENUM","value":"DOT"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"PERIODIC"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"d7ddb76d249cd9af0c7379a15a6b8ba0bc148d006d61edb831fdb9996fb7a269"}} }

ac tiv e_ id	runtime_payload_json
H0 24	{ "effect_type":{"type":"ENUM","value":"GLOBAL_LINE_PROJECTILE"},"targeting":{"type":"ENUM","value":"DIRECTION"},"cast_type":{"type":"ENUM","value":"INSTANT"},"range_m":{"type":"INT64","value":1040,"unit":"m"},"shape":{"type":"ENUM","value":"LINE"},"radius_m":{"type":"INT64","value":0,"unit":"m"},"width_m":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"m"},"duration_s":{"type":"INT64","value":0,"unit":"s"},"projectile_speed_mps":{"type":"INT64","value":45,"unit":"m/s"},"max_lifetime_s":{"type":"INT64","value":24,"unit":"s"},"damage":{"type":"FORMULA","src":"fixed=150","ast":{"q4":1500000}},"attack_coef":{"type":"INT64","value":1,"unit":"ratio"},"distance_bonus_percent_per_10m":{"type":"INT64","value":5,"unit":"scalar"},"distance_bonus_cap_percent":{"type":"INT64","value":50,"unit":"pct"},"blocked_by":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["FROZEN_RIVER","STATIC_SOLID","WIND_WALL"]},"damage_form":{"type":"ENUM","value":"SKILL"},"source_faction":{"type":"ENUM","value":"PLAYER_OWNED"},"element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"owner_credit":{"type":"ENUM","value":"CASTER"},"distance_bonus_orign":{"type":"ENUM","value":"PROJECTILE_RELEASE_POSITION"},"distance_bonus_measure":{"type":"ENUM","value":"ACCUMULATED_PROJECTILE_TRAVEL_DISTANCE_TO_HIT_POINT"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"b6e8349f41406b4806a3382a43db3a036a01568affe507ef29ac187c3b6e3b3a"}}
H0 25	{ "effect_type":{"type":"ENUM","value":"BASIC_LIFESTEAL_AND_DAMAGE_AURA"},"targeting":{"type":"ENUM","value":"SELF"},"cast_type":{"type":"ENUM","value":"INSTANT"},"range_m":{"type":"INT64","value":0,"unit":"m"},"shape":{"type":"ENUM","value":"CIRCLE"},"radius_m":{"type":"INT64","value":5,"unit":"m"},"width_m":{"type":"INT64","value":0,"unit":"m"},"duration_s":{"type":"INT64","value":6,"unit":"s"},"basic_lifesteal_add_percent":{"type":"INT64","value":50,"unit":"pct"},"tick_s":{"type":"INT64","value":1,"unit":"s"},"damage_per_tick":{"type":"INT64","value":30,"unit":"hp"},"aura_heal_equals_actual_hp_damage":{"type":"BOOL","value":true},"shield_damage_does_not_heal":{"type":"BOOL","value":true},"damage_form":{"type":"ENUM","value":"DOT"},"source_faction":{"type":"ENUM","value":"PLAYER_OWNED"},"element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"},"hit_kind":{"type":"ENUM","value":"PERIODIC"},"owner_credit":{"type":"ENUM","value":"CASTER"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"6de2ef3eb1470eb88d6b43cee3534ea736ff892307bea08d0fe4b8bf532ed2a4"}}

ac tiv e_ id	runtime_payload_json
H0 26	{ "effect_type":{"type":"ENUM","value":"TARGET_DAMAGE_HARD_CONTROL"}, "targeting":{"type":"ENUM","value":"VISIBLE_ENEMY"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":20,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":2,"unit":"s"}, "damage":{"type":"FORMULA","src":"fixed=60","ast":{"q4":600000}}, "attack_coef":{"type":"Q4","value_scaled_1e4":3000,"human":"0.3","unit":"ratio"}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "control_type":{"type":"ENUM","value":"STUN"}, "control_duration_s":{"type":"INT64","value":2,"unit":"s"}, "control_uses_normal_tenacity":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"f0c92433e85a3a6005408d01495a66dda79cff5e433cff251db255daa4cd0377"}}
H0 27	{ "effect_type":{"type":"ENUM","value":"VISIBLE_ANY_HERO_HEAL"}, "targeting":{"type":"ENUM","value":"VISIBLE_ANY_HERO_OR_SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":50,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "heal":{"type":"INT64","value":220,"unit":"hp"}, "faction_filter":{"type":"ENUM","value":"ANY_HERO_INCLUDING_ENEMY_OR_SELF"}, "quick_cast_no_target":{"type":"ENUM","value":"SELF"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"f96261152703b2d7c4574159339d4ec33a69de0e6cc9a15af57326be4ebd2e1a"}}
H0 28	{ "effect_type":{"type":"ENUM","value":"TARGET_DAMAGE_AMP_MARK"}, "targeting":{"type":"ENUM","value":"VISIBLE_ENEMY"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":30,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":8,"unit":"s"}, "caster_damage_multiplier":{"type":"Q4","value_scaled_1e4":13000,"human":"1.3","unit":"ratio"}, "on_credited_true_death_transfer":{"type":"ENUM","value":"NEAREST_VISIBLE_ENEMY_WITHIN_20M"}, "transfer_refresh_s":{"type":"INT64","value":8,"unit":"s"}, "no_legal_transfer":{"type":"ENUM","value":"END"}, "transfer_distance_metric":{"type":"ENUM","value":"DIRECT_WORLD_DISTANCE_MM"}, "equal_distance_tie_break":{"type":"ENUM","value":"ENTITY_ID_ASC"}, "caster_true_death":{"type":"ENUM","value":"CLEAR_MARK"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"25a832ab6e263ed3a2a756c48c05aef92343a37c50c590e20013f08b75162884"}}

ac tiv e_ id	runtime_payload_json
H0 29	{ "effect_type":{"type":"ENUM","value":"DELAYED_AREA_DAMAGE_SLOW"}, "targeting":{"type":"ENUM","value":"GROUND_POINT_OR_SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":30,"unit":"m"}, "shape":{"type":"ENUM","value":"CIRCLE"}, "radius_m":{"type":"INT64","value":8,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "telegraph_s":{"type":"Q4","value_scaled_1e4":12000,"human":"1.2","unit":"s"}, "damage":{"type":"FORMULA","src":"fixed=300","ast":{"q4":3000000}}, "attack_coef":{"type":"INT64","value":1,"unit":"ratio"}, "slow_percent":{"type":"INT64","value":40,"unit":"pct"}, "slow_s":{"type":"INT64","value":2,"unit":"s"}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"bc39f753be7bba7e02d49a0c3aac11df038c0b494a1d1d7cb7163a0cbf428aa3"}} }
H0 30	{ "effect_type":{"type":"ENUM","value":"ANY_HERO_HEAL_ZONE"}, "targeting":{"type":"ENUM","value":"GROUND_POINT_OR_SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":30,"unit":"m"}, "shape":{"type":"ENUM","value":"CIRCLE"}, "radius_m":{"type":"INT64","value":8,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":4,"unit":"s"}, "tick_s":{"type":"INT64","value":1,"unit":"s"}, "heal_per_tick":{"type":"INT64","value":40,"unit":"hp"}, "faction_filter":{"type":"ENUM","value":"ANY_HERO"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"bed807eeb9b2eaf1f53deaf692ab24848c76330feb8da3a5cea15450df804ed"}} }
H0 31	{ "effect_type":{"type":"ENUM","value":"PAGODA_RING_SOLID"}, "targeting":{"type":"ENUM","value":"SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":0,"unit":"m"}, "shape":{"type":"ENUM","value":"RING_WALL"}, "radius_m":{"type":"INT64","value":10,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":5,"unit":"s"}, "telegraph_s":{"type":"Q4","value_scaled_1e4":6000,"human":"0.6","unit":"s"}, "wall_thickness_m":{"type":"Q4","value_scaled_1e4":16000,"human":"1.6","unit":"m"}, "height_m":{"type":"INT64","value":4,"unit":"m"}, "segments":{"type":"INT64","value":32,"unit":"scalar"}, "destructible":{"type":"BOOL","value":false}, "instance_limit":{"type":"INT64","value":1,"unit":"scalar"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"5a85adfa01c78d5d293a79b03a37ae2e5194e12159b4d3f97f2f5a8dbddf6824"}} }

ac tiv e_ id	runtime_payload_json
H0 32	{ "effect_type": { "type": "ENUM", "value": "MOBILE_INVULNERABILITY" }, "targeting": { "type": "ENUM", "value": "SELF" }, "cast_type": { "type": "ENUM", "value": "INSTANT" }, "range_m": { "type": "INT64", "value": 0, "unit": "m" }, "shape": { "type": "ENUM", "value": "NONE" }, "radius_m": { "type": "INT64", "value": 0, "unit": "m" }, "width_m": { "type": "INT64", "value": 0, "unit": "m" }, "duration_s": { "type": "INT64", "value": 3, "unit": "s" }, "can_move": { "type": "BOOL", "value": true }, "can_basic": { "type": "BOOL", "value": true }, "can_cast": { "type": "BOOL", "value": true }, "world_interactions_during_invulnerability": { "type": "ENUM", "value": "FORBID_PICKUP_SHOP_CHEST_GAMBLE_TAIBAI" }, "combat_actions_remain_as_DECLARED": { "type": "BOOL", "value": true }, "schema_version": { "type": "ENUM", "value": "V3_0" }, "payload_sha256": { "type": "HASH", "value": "93b81cad5b17b9874e8c10e4edbbf02044df829137a9d2012736acc0280f849e" } }
H0 33	{ "effect_type": { "type": "ENUM", "value": "DAMAGE_SLOW_ANTIMOBILITY_ZONE" }, "targeting": { "type": "ENUM", "value": "GROUND_POINT" }, "cast_type": { "type": "ENUM", "value": "INSTANT" }, "range_m": { "type": "INT64", "value": 30, "unit": "m" }, "shape": { "type": "ENUM", "value": "CIRCLE" }, "radius_m": { "type": "INT64", "value": 8, "unit": "m" }, "width_m": { "type": "INT64", "value": 0, "unit": "m" }, "duration_s": { "type": "INT64", "value": 4, "unit": "s" }, "tick_s": { "type": "INT64", "value": 1, "unit": "s" }, "damage": { "type": "FORMULA", "src": "fixed=50", "ast": { "q4": 500000 } }, "attack_coef": { "type": "Q4", "value_scaled_1e4": 2500, "human": "0.25", "unit": "ratio" }, "slow_percent": { "type": "INT64", "value": 60, "unit": "pct" }, "blocks_mobility_cast_start": { "type": "BOOL", "value": true }, "does_not_cancel_committed_mobility": { "type": "BOOL", "value": true }, "damage_form": { "type": "ENUM", "value": "SKILL" }, "source_faction": { "type": "ENUM", "value": "PLAYER_OWNED" }, "element_rule": { "type": "ENUM", "value": "OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT" }, "hit_kind": { "type": "ENUM", "value": "PERIODIC" }, "owner_credit": { "type": "ENUM", "value": "CASTER" }, "mobility_tag_authority": { "type": "ENUM", "value": "MobilityTag_CLOSED_ENUM_AND_ACTIVE_REGISTRY" }, "zone_checks": { "type": "ENUM", "value": "CAST_START_POSITION_INSIDE_VOLUME" }, "schema_version": { "type": "ENUM", "value": "V3_0" }, "payload_sha256": { "type": "HASH", "value": "b0359babd8c7c1e6934ee217b87c1cfaacee0ec52c439fc546a27967684ede69" } }
H0 34	{ "effect_type": { "type": "ENUM", "value": "SELF_REDUCTION_SPEED_BUFF" }, "targeting": { "type": "ENUM", "value": "SELF" }, "cast_type": { "type": "ENUM", "value": "INSTANT" }, "range_m": { "type": "INT64", "value": 0, "unit": "m" }, "shape": { "type": "ENUM", "value": "NONE" }, "radius_m": { "type": "INT64", "value": 0, "unit": "m" }, "width_m": { "type": "INT64", "value": 0, "unit": "m" }, "duration_s": { "type": "INT64", "value": 4, "unit": "s" }, "incoming_non_true_remaining_multiplier": { "type": "Q4", "value_scaled_1e4": 2000, "human": "0.2", "unit": "ratio" }, "move_speed_add_percent": { "type": "INT64", "value": 15, "unit": "pct" }, "schema_version": { "type": "ENUM", "value": "V3_0" }, "payload_sha256": { "type": "HASH", "value": "cb0450cc4f5c48deadd686ad46577654ca68410e8b43ec7004862fdf31c6a016" } }

ac tiv e_ id	runtime_payload_json
H0 35	{ "effect_type":{"type":"ENUM","value":"PROJECTILE_ROOT"}, "targeting":{"type":"ENUM","value":"VISIBLE_ENEMY"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":60,"unit":"m"}, "shape":{"type":"ENUM","value":"PROJECTILE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":1,"unit":"m"}, "duration_s":{"type":"INT64","value":2,"unit":"s"}, "projectile_speed_mps":{"type":"INT64","value":30,"unit":"m/s"}, "root_s":{"type":"INT64","value":2,"unit":"s"}, "root_allows_basic_and_active":{"type":"BOOL","value":true}, "blocked_by":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["FROZEN_RIVER","STATIC_SOLID","WIND_WALL"]}, "projectile_radius_m":{"type":"Q4","value_scaled_1e4":5000,"human":"0.5","unit":"m"}, "homing":{"type":"BOOL","value":false}, "collision_query":{"type":"ENUM","value":"3D_CAPSULE_SWEEP"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"c366c4ed854c170e7a99a5745f3f72be1e5c50b8e1f5e8eaf1aa3190d1271c55"}}
H0 36	{ "effect_type":{"type":"ENUM","value":"DESTRUCTIBLE_ICE_WALL"}, "targeting":{"type":"ENUM","value":"DIRECTION"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":15,"unit":"m"}, "shape":{"type":"ENUM","value":"LINE_WALL"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":10,"unit":"m"}, "duration_s":{"type":"INT64","value":40,"unit":"s"}, "length_m":{"type":"INT64","value":10,"unit":"m"}, "thickness_m":{"type":"INT64","value":1,"unit":"m"}, "height_m":{"type":"Q4","value_scaled_1e4":25000,"human":"2.5","unit":"m"}, "hp":{"type":"INT64","value":300,"unit":"hp"}, "instance_limit":{"type":"INT64","value":1,"unit":"scalar"}, "new_replaces_old":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"eb64d0a5a9fac8bd21674c58e70621a5282b1dc5ee1adb854a2ab44e78b340df"}}
H0 37	{ "effect_type":{"type":"ENUM","value":"DUAL_PETRIFY"}, "targeting":{"type":"ENUM","value":"VISIBLE_ENEMY_OR_SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":20,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"s"}, "enemy_invulnerable":{"type":"BOOL","value":true}, "enemy_cannot_act":{"type":"BOOL","value":true}, "self_invulnerable":{"type":"BOOL","value":true}, "self_cannot_act":{"type":"BOOL","value":true}, "cooldowns_and_buffs_continue":{"type":"BOOL","value":true}, "dot_ticks_attempt_and_resolve_zero_due_to_invulnerability":{"type":"BOOL","value":true}, "timers_do_not_pause":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"7c6187fb3de313acf387f3e4ca64034d72b3273ca51d329e658d4cf9b07caab8"}}

ac tiv e_ id	runtime_payload_json
H0 38	{ "effect_type":{"type":"ENUM","value":"SMOKE_VISION_ZONE"}, "targeting":{"type":"ENUM","value":"GROUND_POINT"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":30,"unit":"m"}, "shape":{"type":"ENUM","value":"CIRCLE"}, "radius_m":{"type":"INT64","value":8,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":4,"unit":"s"}, "enemy_inside_direct_vision_cap_m":{"type":"INT64","value":2,"unit":"m"}, "blocks_vision_across_boundary":{"type":"BOOL","value":true}, "caster_vision_unchanged":{"type":"BOOL","value":true}, "true_sight_overrides":{"type":"BOOL","value":true}, "area_skills_can_blind_cast":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"0c68f63413322d66270073ae44cddd234676be9336fb28338fc8b800e82a262c"}}
D1	{ "effect_type":{"type":"ENUM","value":"PROJECTILE_PULL"}, "targeting":{"type":"ENUM","value":"DIRECTION"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":80,"unit":"m"}, "shape":{"type":"ENUM","value":"PROJECTILE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":1,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "projectile_speed_mps":{"type":"INT64","value":60,"unit":"m/s"}, "damage":{"type":"FORMULA","src":"fixed=80","ast":{"q4":800000}}, "attack_coef":{"type":"Q4","value_scaled_1e4":4000,"human":"0.4","unit":"ratio"}, "pull_to_caster_front_m":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"m"}, "blocked_by":{"type":"SET","element":{"type":"ENUM","unique":true,"ordered":false,"value":["FROZEN_RIVER","STATIC_SOLID","WIND_WALL"]}, "immune_tiers":{"type":"SET","element":{"type":"ENUM","unique":true,"ordered":false,"value":["CORE_BOSS","DRAGON"]}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "pull_destination":{"type":"ENUM","value":"NEAREST_LEGAL_POINT_IN_FRONT_OF_CASTER"}, "no_legal_pull_point":{"type":"ENUM","value":"DAMAGE_COMMITS_PULL_FAILS"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"0b65247c15fa4b84a62a3243a5198c3dcd8595edce45f22af0ff1c496809598e"}}

ac tiv e_ id	runtime_payload_json
D3	{ "effect_type":{"type":"ENUM","value":"LOCKED_DAMAGE_STUN"}, "targeting":{"type":"ENUM","value":"VISIBLE_ENEMY"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":70,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "damage":{"type":"FORMULA","src":"fixed=140","ast":{"q4":1400000}}, "attack_coef":{"type":"Q4","value_scaled_1e4":8000,"human":0.8,"unit":"ratio"}, "stun_s":{"type":"INT64","value":1,"unit":"s"}, "miss":{"type":"ENUM","value":"ONLY_IF_TARGET_INVULNERABLE_AT_IMPACT"}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_factor":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "impact_delay_s":{"type":"Q4","value_scaled_1e4":2000,"human":0.2,"unit":"s"}, "after_lock_revalidation":{"type":"ENUM","value":"INVULNERABILITY_ONLY"}, "range_los_not_rechecked":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"b9445d443992dc992721f07dccec3ec8651cdfa1e5460ce132ebe20b9b3ad9f3"} }
D4	{ "effect_type":{"type":"ENUM","value":"PROJECTILE_POLYMORPH"}, "targeting":{"type":"ENUM","value":"VISIBLE_ENEMY"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":45,"unit":"m"}, "shape":{"type":"ENUM","value":"PROJECTILE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":1,"unit":"m"}, "duration_s":{"type":"INT64","value":2,"unit":"s"}, "projectile_speed_mps":{"type":"INT64","value":30,"unit":"m/s"}, "disable_basic":{"type":"BOOL","value":true}, "disable_active":{"type":"BOOL","value":true}, "move_speed_multiplier":{"type":"Q4","value_scaled_1e4":13000,"human":1.3,"unit":"ratio"}, "preserve_movement_class":{"type":"BOOL","value":true}, "on_polymorph":{"type":"ENUM","value":"INTERRUPT_ACTIVE_WINDUP_AND_CHANNEL"}, "existing_owned_entities":{"type":"ENUM","value":"PERSIST_NORMALLY"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"d89dae103ea79578c00082ca36f5995e4cc258338c309ea5cb7d76756dd3ca11"} }
D6	{ "effect_type":{"type":"ENUM","value":"CAPSULE_SWEEP_BLINK"}, "targeting":{"type":"ENUM","value":"DIRECTION"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":15,"unit":"m"}, "shape":{"type":"ENUM","value":"POINT"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "distance_m":{"type":"INT64","value":15,"unit":"m"}, "max_continuous_solid_chord_m":{"type":"Q4","value_scaled_1e4":15000,"human":1.5,"unit":"m"}, "illegal_endpoint":{"type":"ENUM","value":"LAST_LEGAL_ALONG_DIRECTION"}, "post_cast_lock_s":{"type":"INT64","value":0,"unit":"s"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"f9aac0b68daaca818e71e5892a516eff489d7664858fdc9d598fccbe548e958"} }

ac tiv e_ id	runtime_payload_json
D7	{ "effect_type":{"type":"ENUM","value":"SELF_STEALTH_SPEED"}, "targeting":{"type":"ENUM","value":"SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":0,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":5,"unit":"s"}, "move_speed_multiplier":{"type":"Q4","value_scaled_1e4":13000,"human":"1.3","unit":"ratio"}, "break_on":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["ACTIVE_CAST_COMMIT","BASIC_RELEASE"]}, "pickup_does_not_break":{"type":"BOOL","value":true}, "break_attack_bonus":{"type":"INT64","value":0,"unit":"scalar"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"831aa3863f9940ac65fd009fa9075a7c9663cf44b347c43ef78b2b4e0ef7208c"}}
D9	{ "effect_type":{"type":"ENUM","value":"AERIAL_DELAYED_AREA_DAMAGE"}, "targeting":{"type":"ENUM","value":"GROUND_POINT"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":35,"unit":"m"}, "shape":{"type":"ENUM","value":"CIRCLE"}, "radius_m":{"type":"INT64","value":8,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "telegraph_s":{"type":"Q4","value_scaled_1e4":8000,"human":"0.8","unit":"s"}, "damage":{"type":"FORMULA","src":"fixed=130","ast":{"q4":1300000}}, "attack_coef":{"type":"Q4","value_scaled_1e4":6000,"human":"0.6","unit":"ratio"}, "wind_wall_blocked":{"type":"BOOL","value":false}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"a25a2234e12dde93395a59a696ad7c201eb29ae51ab65ecd106fef9d36c65152"}}
D10	{ "effect_type":{"type":"ENUM","value":"CHAIN_DAMAGE"}, "targeting":{"type":"ENUM","value":"VISIBLE_ENEMY"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":30,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "initial_range_m":{"type":"INT64","value":30,"unit":"m"}, "chain_range_m":{"type":"INT64","value":15,"unit":"m"}, "max_targets":{"type":"INT64","value":6,"unit":"scalar"}, "damage":{"type":"FORMULA","src":"fixed=300","ast":{"q4":3000000}}, "attack_coef":{"type":"INT64","value":1,"unit":"ratio"}, "each_jump_multiplier":{"type":"Q4","value_scaled_1e4":8500,"human":"0.85","unit":"ratio"}, "target_repeat":{"type":"BOOL","value":false}, "next_target":{"type":"ENUM","value":"NEAREST_LEGAL_THEN_ENTITY_ID"}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"1e09f529fb883013556a267908d5d0a88d2bcbb17e9716a78d7892114c8113fc"}}

ac tiv e_ id	runtime_payload_json
D1 1	{ "effect_type":{"type":"ENUM","value":"UNIFORM_RANDOM_DAMAGE"}, "targeting":{"type":"ENUM","value":"VISIBLE_ENEMY"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":40,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "integer_inclusive":{"type":"ARRAY","element":"INT64","ordered":true,"unit":"scalar","value":[1,999]}, "rng_stream":{"type":"ENUM","value":"ACTIVE_SKILL"}, "crit":{"type":"BOOL","value":false}, "rejected_cast_consumes_rng":{"type":"BOOL","value":false}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "can_crit":{"type":"BOOL","value":false}, "shieldable":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"c640686c0a37f381a169b240db05c9942806e1f39f18c506ef6bddf2eb189b46"}}
D1 2	{ "effect_type":{"type":"ENUM","value":"TARGET_REVEAL_BOUNTY_MARK"}, "targeting":{"type":"ENUM","value":"VISIBLE_ENEMY"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":60,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":90,"unit":"s"}, "caster_kill_bonus_created_gold":{"type":"INT64","value":800,"unit":"gold"}, "clear_on":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["CASTER_TRUE_DEATH","DURATION_END","RECAST_REPLACE","TARGET_TRUE_DEATH"]}, "exact_reveal_recipient":{"type":"ENUM","value":"CASTER_ONLY"}, "other_players_receive_no_information":{"type":"BOOL","value":true}, "B41_independent":{"type":"BOOL","value":true}, "reveal_does_not_grant_through_wall_lock":{"type":"BOOL","value":true}, "wall_blocked_target":{"type":"ENUM","value":"POSITION_INTEL_ONLY"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"0f76e9bf866f3e48aac5c511cae2e9a51dec8916b10cdb305501af61a05b27f4"}}

ac tiv e_ id	runtime_payload_json
D1 3	{ "effect_type":{"type":"ENUM","value":"ATOMIC_SWAP_TWO_VISIBLE_UNITS"}, "targeting":{"type":"ENUM","value":"TWO_VISIBLE_UNITS"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":50,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "legal_types":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["DAMAGEABLE_SUMMON","FLYING_MONSTER","ORDINARY_MONSTER","PIG","PLAYER"]}, "immune_types":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["CORE_BOSS","DRAGON","LEADER"]}, "either_destination_illegal":{"type":"ENUM","value":"FAIL_NO_CD"}, "selection_state":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["COMMIT_OR_CANCEL","WAIT_FIRST_TARGET","WAIT_SECOND_TARGET"]}, "second_target_timeout_s":{"type":"INT64","value":3,"unit":"s"}, "cancel_or_timeout":{"type":"ENUM","value":"NO_COOLDOWN"}, "second_invalid":{"type":"ENUM","value":"KEEP_WAITING_UNTIL_TIMEOUT"}, "second_target_same_as_first":{"type":"ENUM","value":"INVALID_KEEP_WAITING"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"148a3ef8f6d4038856a620ba32f09659d59c479c5bb361483a7b361a1dc3420a"}}
D1 4	{ "effect_type":{"type":"ENUM","value":"SELF_REWIND"}, "targeting":{"type":"ENUM","value":"SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":0,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "lookback_s":{"type":"INT64","value":5,"unit":"s"}, "hp_result":{"type":"FORMULA","src":"MAX(CURRENT_HP,HISTORICAL_HP_CLAMPED_TO_CURRENT_MAX_HP)","ast":{"op":"MAX","args":[{"var":"CURRENT_HP"},{"var":"HISTORICAL_HP_CLAMPED_TO_CURRENT_MAX_HP"}]}}, "position_result":{"type":"ENUM","value":"HISTORICAL_IF_LEGAL_ELSE_CURRENT"}, "position_illegal_does_not_cancel_hp_restore":{"type":"BOOL","value":true}, "cooldowns_rewind":{"type":"BOOL","value":false}, "gold_inventory_status_rewind":{"type":"BOOL","value":false}, "history_sample_hz":{"type":"INT64","value":20,"unit":"hz"}, "history_buffer_s":{"type":"INT64","value":6,"unit":"s"}, "sample_choice":{"type":"ENUM","value":"LATEST_VALID_ALIVE_SAMPLE_AT_OR_BEFORE_NOW_MINUS_5S"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"e6fc0030098c286c416052c5ed4281d0dec24135c36c33c8f9ebc47a6561cae1"}}

ac tiv e_ id	runtime_payload_json
D1 5	{ "effect_type":{"type":"ENUM","value":"MONSTER_EXECUTE_OR_PLAYER_GOLD_STEAL"}, "targeting":{"type":"ENUM","value":"VISIBLE_MONSTER_OR_ENEMY"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":30,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "execute_types":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["FLYING_MONSTER","ORDINARY_MONSTER"]}, "xp_and_drops":{"type":"ENUM","value":"NORMAL_ONCE"}, "immune_types":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["CORE_BOSS","DRAGON","LEADER","PLAYER_OWNED_SUMMON"]}, "player_transfer_gold":{"type":"INT64","value":500,"unit":"gold"}, "insufficient":{"type":"ENUM","value":"TRANSFER_ALL_REMAINING"}, "never_creates_gold_on_player":{"type":"BOOL","value":true}, "pig_execute_current_hp_lte_percent":{"type":"INT64","value":30,"unit":"pct"}, "execute_base_gold_multiplier":{"type":"INT64","value":3,"unit":"ratio"}, "gold_multiplier_scope":{"type":"ENUM","value":"BASE_MONSTER_GOLD_ONLY"}, "xp_and_drop_multiplier":{"type":"INT64","value":1,"unit":"ratio"}, "kill_credit":{"type":"ENUM","value":"CASTER_NORMAL_MONSTER_KILL"}, "on_kill_triggers_once":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["B32","B33","B40","EQUIP_B8","G9"]}, "does_not_trigger_basic_hit_chain":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"d310b991b40fa03d8d8767e9fb5f48f03c634d6676d4b4f3411677d24ae25c4e"}}
D1 6	{ "effect_type":{"type":"ENUM","value":"CREATE_GOLD"}, "targeting":{"type":"ENUM","value":"SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":0,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":0,"unit":"s"}, "amount":{"type":"INT64","value":450,"unit":"scalar"}, "uses_per_match":{"type":"ENUM","value":"UNLIMITED"}, "ledger":{"type":"ENUM","value":"GOLD_CREATE_ACTIVE_D16"}, "true_death_ready_rule":{"type":"ENUM","value":"ACTIVE_READY_BUT_MATCH_GOLD_LEDGER_REMAINS"}, "economy_sim_source":{"type":"ENUM","value":"GOLD_CREATE_ACTIVE_D16"}, "uses_per_life_not_capped":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"9d4a5af696d49af08b34d164c5f2a67f2f31c63a299240a0df23474222d6ebfa"}}

ac tiv e_ id	runtime_payload_json
D1 7	{ "effect_type": { "type": "ENUM", "value": "TOGGLE_PLAYER_BOUNTY", "targeting": { "type": "ENUM", "value": "SELF", "cast_type": { "type": "ENUM", "value": "INSTANT", "range_m": { "type": "INT64", "value": 0, "unit": "m", "shape": { "type": "ENUM", "value": "NONE", "radius_m": { "type": "INT64", "value": 0, "unit": "m", "width_m": { "type": "INT64", "value": 0, "unit": "m", "duration_s": { "type": "INT64", "value": 0, "unit": "s", "toggle_cd_s": { "type": "INT64", "value": 55, "unit": "s", "base_bounty": { "type": "INT64", "value": 500, "unit": "scalar", "on_each_credited_player_true_death": { "type": "FORMULA", "src": "MIN(500*2^COUNT,32000)", "ast": { "op": "MIN", "args": { "op": "MUL", "args": [{ "q4": 50000000 }, { "op": "POW_INT", "args": [{ "q4": 20000 }, { "var": "COUNT" }] }] }, { "q4": 320000000 }] }, "assign_to": "BOUNTY", "cap": { "type": "INT64", "value": 32000, "unit": "scalar", "player_killer_gets_created_gold_percent": { "type": "INT64", "value": 50, "unit": "pct", "monster_final_blow": { "type": "ENUM", "value": "DESTROY_WITHOUT_PAYOUT", "manual_off": { "type": "ENUM", "value": "RESET_AND_OFF", "true_death": { "type": "ENUM", "value": "PAY_IF_PLAYER_KILLER_THEN_RESET_AND_OFF", "numeric_storage": { "type": "ENUM", "value": "SIGNED_INT64_SATURATING", "head_marker_shows_capped_value": { "type": "BOOL", "value": true }, "schema_version": { "type": "ENUM", "value": "V3_0", "payload_sha256": { "type": "HASH", "value": "18474b61926189967e1f579b7e06b8fa1cab49ac8f2c9691e2719f0d7ce4b3c1" } } } } } } } } } } } } } } }
D1 8	{ "effect_type": { "type": "ENUM", "value": "SCAN_OLD_EQUIPMENT", "targeting": { "type": "ENUM", "value": "SELF", "cast_type": { "type": "ENUM", "value": "INSTANT", "range_m": { "type": "INT64", "value": 0, "unit": "m", "shape": { "type": "ENUM", "value": "CIRCLE", "radius_m": { "type": "INT64", "value": 80, "unit": "m", "width_m": { "type": "INT64", "value": 0, "unit": "m", "duration_s": { "type": "INT64", "value": 20, "unit": "s", "minimum_ground_age_s": { "type": "INT64", "value": 10, "unit": "s", "mark_duration_s": { "type": "INT64", "value": 20, "unit": "s", "snapshot_at_cast": { "type": "BOOL", "value": true, "newly_aging_items_not_added": { "type": "BOOL", "value": true, "scan_ignores_fog_and_walls": { "type": "BOOL", "value": true, "recipient": { "type": "ENUM", "value": "CASTER_ONLY", "marker_reveals_owner_or_source": { "type": "BOOL", "value": false, "schema_version": { "type": "ENUM", "value": "V3_0", "payload_sha256": { "type": "HASH", "value": "df0d1e9ff3a5a749857e221662bcf2a10fab92d8a6c1972963bab7262ab068ff" } } } } } } } } } } } } } }
D1 9	{ "effect_type": { "type": "ENUM", "value": "SPAWN_SUMMON", "targeting": { "type": "ENUM", "value": "SELF_FORWARD", "cast_type": { "type": "ENUM", "value": "INSTANT", "range_m": { "type": "INT64", "value": 3, "unit": "m", "shape": { "type": "ENUM", "value": "POINT", "radius_m": { "type": "INT64", "value": 0, "unit": "m", "width_m": { "type": "INT64", "value": 0, "unit": "m", "duration_s": { "type": "INT64", "value": 15, "unit": "s", "archetype": { "type": "ENUM", "value": "BEAN_SOLDIER", "count": { "type": "INT64", "value": 3, "unit": "scalar", "hp": { "type": "INT64", "value": 150, "unit": "hp", "attack": { "type": "INT64", "value": 15, "unit": "scalar", "attack_attribute_inherit": { "type": "INT64", "value": 0, "unit": "scalar", "triggers_owner_passives": { "type": "BOOL", "value": false, "schema_version": { "type": "ENUM", "value": "V3_0", "payload_sha256": { "type": "HASH", "value": "4a561ebc5ae6caed8630019f7bd4b9fc4aad077ed3a0a5d7d9aeb640a7e1812a" } } } } } } } } } } } } }

ac tiv e_ id	runtime_payload_json
D2 0	{ "effect_type":{"type":"ENUM","value":"VISIBLE_TRAP_OR_REMOTE_DETONATE"}, "targeting":{"type":"ENUM","value":"GROUND_POINT_OR_OWN_TRAP"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":3,"unit":"m"}, "shape":{"type":"ENUM","value":"CIRCLE"}, "radius_m":{"type":"INT64","value":6,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":90,"unit":"s"}, "placement_range_m":{"type":"INT64","value":3,"unit":"m"}, "instance_limit":{"type":"INT64","value":3,"unit":"scalar"}, "fourth_removes_oldest":{"type":"BOOL","value":true}, "arming_s":{"type":"Q4","value_scaled_1e4":5000,"human":"0.5","unit":"s"}, "body_blocks":{"type":"BOOL","value":false}, "trigger_radius_m":{"type":"Q4","value_scaled_1e4":7500,"human":"0.75","unit":"m"}, "trigger_damage":{"type":"FORMULA","src":"fixed=180","ast":{"q4":1800000}}, "trigger_attack_coef":{"type":"Q4","value_scaled_1e4":8000,"human":"0.8","unit":"ratio"}, "root_s":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"s"}, "reveal_s":{"type":"INT64","value":8,"unit":"s"}, "remote_range":{"type":"ENUM","value":"GLOBAL"}, "remote_damage":{"type":"FORMULA","src":"fixed=100","ast":{"q4":1000000}}, "remote_attack_coef":{"type":"Q4","value_scaled_1e4":6000,"human":"0.6","unit":"ratio"}, "remote_radius_m":{"type":"INT64","value":6,"unit":"m"}, "damage_form":{"type":"ENUM","value":"SKILL"}, "source_faction":{"type":"ENUM","value":"PLAYER_OWNED"}, "element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "hit_kind":{"type":"ENUM","value":"ABILITY"}, "owner_credit":{"type":"ENUM","value":"CASTER"}, "all_capsule_trajectory_cross_triggers":{"type":"BOOL","value":true}, "exceptions":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["INVULNERABLE","PHYSICALLY_ABOVE_TRIGGER_VOLUME"]}, "remote_selection_tap":{"type":"ENUM","value":"NEAREST_TRAP_TO_AIM_VECTOR"}, "remote_selection_hold":{"type":"ENUM","value":"OPEN_OWN_TRAP_SELECTOR"}, "remote_selection_tie":{"type":"ENUM","value":"TRAP_INSTANCE_ID_ASC"}, "trap_hp":{"type":"INT64","value":250,"unit":"hp"}, "destructible_by_enemy":{"type":"BOOL","value":true}, "destroy_reward":{"type":"ENUM","value":"NONE"}, "destroy_explosion":{"type":"BOOL","value":false}, "blocks_movement":{"type":"BOOL","value":false}, "no_own_trap_press":{"type":"ENUM","value":"PLACEMENT_MODE"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"a73c6c26eca525583f64a58dcf3bb8cfa9eefa12156828e8431d1b80841e42e0"} }
D2 1	{ "effect_type":{"type":"ENUM","value":"SELF_LOCK_INVULNERABILITY"}, "targeting":{"type":"ENUM","value":"SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":0,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":4,"unit":"s"}, "can_move":{"type":"BOOL","value":false}, "can_basic":{"type":"BOOL","value":false}, "can_cast":{"type":"BOOL","value":false}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"a9f5753da1ca809d60d961adee500182903af5fbf24c83d6f90376cfe32d4463"} }

active_id	runtime_payload_json
D22	{ "effect_type":{"type":"ENUM","value":"SELF_SHIELD"}, "targeting":{"type":"ENUM","value":"SELF"}, "cast_type":{"type":"ENUM","value":"INSTANT"}, "range_m":{"type":"INT64","value":0,"unit":"m"}, "shape":{"type":"ENUM","value":"NONE"}, "radius_m":{"type":"INT64","value":0,"unit":"m"}, "width_m":{"type":"INT64","value":0,"unit":"m"}, "duration_s":{"type":"INT64","value":5,"unit":"s"}, "amount":{"type":"INT64","value":600,"unit":"scalar"}, "absorbs":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["BASIC","DOT","PERCENT","REFLECT","SKILL","STORM","TRUE"]}, "shield_absorption_order":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["CREATION_SEQUENCE_ASC","EARLIEST_EXPIRY_TICK_FIRST"]}, "same_source_new_cast":{"type":"ENUM","value":"INDEPENDENT_INSTANCE"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"8220a33fb2339cfd6884f4b937e826a840ac8fa13759c1c4a4e26cbdeb231d6"} }

E.27 H009 被动效果倍率精确白名单

被动 ID	H009 允许翻倍字段	变换类型
B01	slow_percent / l5_aoe_slow_percent	ADDITIVE_MAGNITUDE
B02	NONE	EXCLUDED_CONTROL_DURATION
B03	knockback_m / l5_aoe_knockback_m	ADDITIVE_MAGNITUDE
B04	basic_miss_percent	ADDITIVE_MAGNITUDE
B05	NONE	EXCLUDED_CONTROL_DURATION_AND_PROC
B06	critical_damage_percent / l5_ignore_shield_percent	ADDITIVE_MAGNITUDE
B07	splash_damage_percent	ADDITIVE_MAGNITUDE
B08	lost_hp_damage_percent	ADDITIVE_MAGNITUDE
B09	damage_per_stack_s / l5_full_stack_damage_multiplier	ADDITIVE_OR_DELTA_FROM_ONE
B10	damage_multiplier / l5_kill_heal_max_hp_percent	DELTA_FROM_ONE_OR_ADDITIVE
B11	NONE	EXCLUDED_CHANCE_AND_COUNT
B12	summon_hp / summon_attack	ADDITIVE_MAGNITUDE
B13	contact_damage	ADDITIVE_MAGNITUDE
B14	damage	ADDITIVE_MAGNITUDE
B15	l5_move_speed_percent	ADDITIVE_MAGNITUDE
B16	reduction	ADDITIVE_MAGNITUDE
B17	shield / l5_break_aoe_damage	ADDITIVE_MAGNITUDE
B18	attack_percent_per_missing_10pct_hp / l5_lifesteal_percent	ADDITIVE_MAGNITUDE

被动 ID	H009 允许翻倍字段	变换类型
B19	NONE	EXCLUDED_LETHAL_PROTECTION
B20	NONE	EXCLUDED_LETHAL_PROTECTION
B21	hp_per_s / l5_first_basic_damage_multiplier	ADDITIVE_OR_DELTA_FROM_ONE
B22	actual_hp_plus_shield_damage_percent	ADDITIVE_MAGNITUDE
B23	counter_damage_attack_coefficient	ADDITIVE_MAGNITUDE
B24	heal_percent_actual_hp_plus_shield_damage / l5_reflect_pre_overheal_conversion_percent	ADDITIVE_MAGNITUDE
B25	next_basic_damage_multiplier / l5_attack_speed_percent	DELTA_FROM_ONE_OR_ADDITIVE
B26	damage_multiplier	DELTA_FROM_ONE
B27	move_speed_percent	ADDITIVE_MAGNITUDE
B28	move_speed_toward_target_percent / l5_damage_multiplier	ADDITIVE_OR_DELTA_FROM_ONE
B29	damage_multiplier	DELTA_FROM_ONE
B30	slow_percent / l5_explosion_damage	ADDITIVE_MAGNITUDE
B31	l5_heal_percent_transferred_gold	ADDITIVE_MAGNITUDE
B32	NONE	EXCLUDED_ECONOMY
B33	NONE	EXCLUDED_ECONOMY_AND_CHANCE
B34	NONE	EXCLUDED_ECONOMY
B35	NONE	EXCLUDED_ECONOMY
B36	move_speed_percent_per_stack / l5_full_stack_basic_extra_damage_move_speed_value_percent	ADDITIVE_MAGNITUDE
B37	heal / aoe_damage	ADDITIVE_MAGNITUDE
B38	incoming_damage_remaining_multiplier / l5_heal_max_hp_percent_per_s	REMAINING_FROM_ONE_OR_ADDITIVE
B39	statue_hp / l5_destroy_aoe_damage	ADDITIVE_MAGNITUDE
B40	NONE	EXCLUDED_PERMANENT_GROWTH
B41	NONE	EXCLUDED_ECONOMY_AND_DURATION
B42	l5_move_speed_percent	ADDITIVE_MAGNITUDE
B43	move_speed_percent / l5_basic_damage_multiplier	ADDITIVE_OR_DELTA_FROM_ONE
B44	damage	ADDITIVE_MAGNITUDE

E.28 数值类型、单位与容器系统

工程字段	唯一值 / 执行语义
type_system.closed_types	INT64 / Q4 / BOOL / ENUM / SET / ARRAY / FORMULA / HASH

	/ REF
type_system.REF	只读章节交叉引用（含 \$ 的规则指针）；运行时不解析 REF 求值，仅供构建器与人读定位权威条目
type_system.INT64	64 位有符号整数；饱和运算 SIGNED_INT64_SATURATING；构建阶段拒绝任何浮点字面量
type_system.Q4	定点数：value_scaled_1e4=实际值×10000 后取整存为 INT64；权威模拟只使用缩放整数，human 字段仅供人读
type_system.float_forbidden_runtime	是；出现 FLOAT/NUMBER/STRING 类型即构建失败
type_system.unit_required	是；每个数值字段必须携带 unit：m / mm / s / m/s / pct / deg / hz / gold / hp / ratio / scalar
type_system.unit_source	字段名单向推导，优先级：速率 (attack_speed_per_s→1/s；*_per_s 与 *_per_stack_s→hp/s) > 后缀 (_m→m, _s→s, _mps→m/s, _percent→pct, _deg→deg, _gold→gold, _hp→hp) > 乘数/系数→ratio>概率→pct>伤害/治疗/护盾量→hp>纯计数→scalar
type_system.ARRAY	有序同质容器；必须声明 element 与 ordered=是；per_level=是表示按 L1-L5 索引
type_system.SET	无序去重容器；必须声明 element=ENUM 与 unique=是；序列化时按字典序排序保证回放哈希稳定
type_system.q4_mul	MUL(a,b)=FLOOR(a*b/10000)；逐步截断，禁止先聚合后取整
type_system.q4_div	DIV_FLOOR(a,b)=FLOOR(a*10000/b)；b=0 时构建期公式直接拒绝，运行期饱和为 0 并记录断言事件
type_system.final_rounding	任何伤害/治疗/护盾落地前最后一步 FLOOR 到整数 HP；中间过程保持 Q4

E.29 公式 DSL 与 AST 合同

工程字段	唯一值 / 执行语义
formula_dsl.grammar	expr := term (('+' '-') term)* ; term := pow (('*' '/') pow)* ; pow := factor ('^' factor)? ; factor := 数字 变量 函数(args) (expr) -factor

formula_dsl.functions	FLOOR / CEIL / MAX / MIN / CLAMP / MOD / ATAN2_DEG / ABS; 白名单外函数构建失败
formula_dsl.variables_whitelist	BASE_DURATION / COST / COUNT / CURRENT_GAME_MOVE_SPEED / CURRENT_GOLD / CURRENT_HP / DECOY_MAX_HP / HISTORICAL_HP_CLAMPED_TO_CURRENT_MAX_HP / MISSING_HP_PERCENT / OWNER_CURRENT_HP / OWNER_CURRENT_MAX_HP / PRE_OVERHEAL_AMOUNT / SOURCE_FLAT_EXTENSION / TARGET_STRONGEST_MULTIPLIER / VALUE / target_x / target_y / viewer_x / viewer_y
formula_dsl.number_literal	编译为 Q4 缩放整数 ({q4:n}) ; 运行时不解析源文本
formula_dsl.operator_semantics	ADD/SUB/MUL 按 Q4 定点; '/'一律 DIV_FLOOR; '^'仅允许非负整数指数 POW_INT, 逐乘饱和
formula_dsl.evaluation	只执行载荷内已编译 ast; src 字段仅为审计对照; ast 与 src 不一致时以 ast 为准并触发 G_DATA_CONSISTENCY 失败
formula_dsl.overflow	SIGNED_INT64_SATURATING; 饱和事件计入埋点 formula_saturation_count
formula_dsl.assign_to	带 assign_to 的公式将结果写入该命名槽位 (如 D17 的 BOUNTY) , 不产生其他副作用
formula_dsl.variable_case	变量名大小写敏感; 白名单为闭合集, 大写为全局状态量、小写为几何流量 (如 target_x) ; 白名单外标识符构建失败

E.30 主动效果逐类型字段矩阵

- 基础结构族替代单一 common_required: 所有类型必含 BASE 族; 含伤害者必含 damage_events[]或单事件 DAMAGE 族。下表为 57 种 effect_type 的闭合字段集, 三段语义: 未标注者必填; 标注'允许缺省'者可省略; 矩阵外字段一律禁止出现。违反任一条即构建失败。

effect_type	闭合字段集 (除 schema_version/payload_sha256 外)
AERIAL_DELAYED_AREA_DAMAGE	attack_coef / cast_type / damage / damage_form / duration_s / effect_type / element_rule / hit_kind / owner_credit / radius_m / range_m / shape / source_faction / targeting / telegraph_s / width_m / wind_wall_blocked

ANY_HERO_HEAL_ZONE	cast_type / duration_s / effect_type / faction_filter / heal_per_tick / radius_m / range_m / shape / targeting / tick_s / width_m
ARM_NEXT_BASIC	arm_persistence / cast_type / cleared_on / consume_if_damage_zero_due_to_dodge_or_immunity / consume_on / do_not_consume_if_no_hit_entity / duration_s / effect_type / extra_target_missing_hp_percent / guaranteed_crit / kill_reset_timing / kill_resets_active_cd / radius_m / range_m / shape / targeting / width_m
ATOMIC_SWAP_TWO_VISIBLE_UNITS	cancel_or_timeout / cast_type / duration_s / effect_type / either_destination_illegal / immune_types / legal_types / radius_m / range_m / second_invalid / second_target_same_as_first / second_target_timeout_s / selection_state / shape / targeting / width_m
BASIC_LIFESTEAL_AND_DAMAGE_AURA	aura_heal_equals_actual_hp_damage / basic_lifesteal_add_percent / cast_type / damage_form / damage_per_tick / duration_s / effect_type / element_rule / hit_kind / owner_credit / radius_m / range_m / shape / shield_damage_does_not_heal / source_faction / targeting / tick_s / width_m
BLINK_LEAVE_DELAYED_DECOY	attack_coef / blink_uses_capsule_sweep / cast_type / damage / damage_form / delay_s / distance_m / duration_s / effect_type / element_rule / hit_kind / owner_credit / radius_m / range_m / shape / source_faction / targeting / width_m
CAPSULE_SWEEP_BLINK	cast_type / distance_m / duration_s / effect_type / illegal_endpoint / max_continuous_solid_chord_m / post_cast_lock_s / radius_m / range_m / shape / targeting / width_m
CHAIN_DAMAGE	attack_coef / cast_type / chain_range_m / damage / damage_form / duration_s / each_jump_multiplier / effect_type / element_rule / hit_kind / initial_range_m / max_targets / next_target / owner_credit / radius_m / range_m / shape / source_faction / target_repeat / targeting / width_m
CHARGED_RADIAL_PULL	attack_coef / cast_type / charge_s / damage / damage_form / duration_s / effect_type / element_rule / hit_kind / immune_tiers / landing_ring_m / landing_slot_layout / landing_slot_order / max_target_selection_order / max_targets / occupied_slot / owner_credit / radius_m / range_m / shape / source_faction / targeting / wall_blocks / width_m

CONDITIONAL_POISON_DETONATE	cast_type / damage_events / duration_s / effect_type / max_count / other_owner_stacks_unchanged / radius_m / range_m / shape / stack_scope / targeting / when_no_owned_dot_add_stacks / when_owned_dot_consume_all_owned_dot_and_toxin_stacks / width_m
CONE_DAMAGE_SLOW	angle_deg / attack_coef / cast_type / damage / damage_form / duration_s / effect_type / element_rule / hit_kind / owner_credit / radius_m / range_m / shape / slow_percent / slow_s / source_faction / targeting / width_m
CONTAGIOUS_DAMAGE_ZONE	attack_coef / cast_type / check_interval_s / damage / damage_form / different_caster_infection / duration_s / effect_type / element_rule / generation_origin / hit_kind / infection_instance_scope / infection_radius_m / max_generation / outside_residual_s / owner_credit / radius_m / range_m / refresh / same_caster_reinfection / shape / source_faction / stack / targeting / tick_s / width_m
CREATE_GOLD	amount / cast_type / duration_s / economy_sim_source / effect_type / ledger / radius_m / range_m / shape / targeting / true_death_ready_rule / uses_per_life_not_capped / uses_per_match / width_m
DAMAGE_SLOW_ANTIMOBILITY_ZONE	attack_coef / blocks_mobility_cast_start / cast_type / damage / damage_form / does_not_cancel_committed_mobility / duration_s / effect_type / element_rule / hit_kind / mobility_tag_authority / owner_credit / radius_m / range_m / shape / slow_percent / source_faction / targeting / tick_s / width_m / zone_checks
DAMAGE_SLOW_ZONE	attack_coef / cast_type / damage / damage_form / duration_s / effect_type / element_rule / hit_kind / owner_credit / radius_m / range_m / shape / slow_percent / source_faction / targeting / tick_s / width_m
DASH_STOP_FIRST	attack_coef / cast_type / damage / damage_form / distance_m / duration_s / effect_type / element_rule / hit_kind / interruptible_by_hard_control / owner_credit / radius_m / range_m / shape / source_faction / stun_s / targeting / width_m
DELAYED_AREA_DAMAGE	attack_coef / cast_type / damage / damage_form / duration_s / effect_type / element_rule / hit_kind / owner_credit / radius_m / range_m / shape / source_faction / targeting / telegraph_s / width_m
DELAYED_AREA_DAMAGE_SLOW	attack_coef / cast_type / damage / damage_form / duration_s / effect_type / element_rule / hit_kind / owner_credit / radius_m / range_m / shape /

	slow_percent / slow_s / source_faction / targeting / telegraph_s / width_m
DELAYED_SILENCE_ZONE	cast_type / duration_s / effect_type / radius_m / range_m / shape / silence_s / targeting / telegraph_s / width_m
DESTRUCTIBLE_ICE_WALL	cast_type / duration_s / effect_type / height_m / hp / instance_limit / length_m / new_replaces_old / radius_m / range_m / shape / targeting / thickness_m / width_m
DUAL_PETRIFY	cast_type / cooldowns_and_buffs_continue / dot_ticks_attempt_and_resolve_zero_due_to_invulnerability / duration_s / effect_type / enemy_cannot_act / enemy_invulnerable / radius_m / range_m / self_cannot_act / self_invulnerable / shape / targeting / timers_do_not_pause / width_m
FIRE_WALL	cast_type / damage_events / duration_s / effect_type / element_rule / length_m / owner_credit / radius_m / range_m / shape / source_faction / targeting / thickness_m / width_m
GLOBAL_LINE_PROJECTILE	attack_coef / blocked_by / cast_type / damage / damage_form / distance_bonus_cap_percent / distance_bonus_measure / distance_bonus_origin / distance_bonus_percent_per_10m / duration_s / effect_type / element_rule / hit_kind / max_lifetime_s / owner_credit / projectile_speed_mps / radius_m / range_m / shape / source_faction / targeting / width_m
LOCKED_DAMAGE_STUN	after_lock_revalidation / attack_coef / cast_type / damage / damage_form / duration_s / effect_type / element_rule / hit_kind / impact_delay_s / miss / owner_credit / radius_m / range_los_not_rechecked / range_m / shape / source_faction / stun_s / targeting / width_m
MOBILE_CHANNEL_AOE	attack_coef / cast_type / damage_form / damage_per_tick / duration_s / effect_type / element_rule / hard_control_interrupts / hit_kind / owner_credit / radius_m / range_m / self_move_multiplier / shape / source_faction / targeting / tick_s / width_m
MOBILE_INVULNERABILITY	can_basic / can_cast / can_move / cast_type / combat_actions_remain_as_DECLARED / duration_s / effect_type / radius_m / range_m / shape / targeting / width_m / world_interactions_during_invulnerability

MONSTER_EXECUTE_OR_PLAYER_GOLD_STEAL	cast_type / does_not_trigger_basic_hit_chain / duration_s / effect_type / execute_base_gold_multiplier / execute_types / gold_multiplier_scope / immune_types / insufficient / kill_credit / never_creates_gold_on_player / on_kill_triggers_once / pig_execute_current_hp_lte_percent / player_transfer_gold / radius_m / range_m / shape / targeting / width_m / xp_and_drop_multiplier / xp_and_drops
PAGODA_RING_SOLID	cast_type / destructible / duration_s / effect_type / height_m / instance_limit / radius_m / range_m / segments / shape / targeting / telegraph_s / wall_thickness_m / width_m
PETRIFY_TARGET	cast_type / duration_s / effect_type / radius_m / range_m / shape / target_cannot_act / target_invulnerable / targeting / width_m
PIERCING_DASH	attack_coef / cast_type / damage / damage_form / distance_m / duration_s / effect_type / element_rule / hit_each_target_once / hit_kind / knockback_m / owner_credit / radius_m / range_m / shape / source_faction / targeting / width_m
PROJECTILE_POLYMORPH	cast_type / disable_active / disable_basic / duration_s / effect_type / existing_owned_entities / move_speed_multiplier / on_polymorph / preserve_movement_class / projectile_speed_mps / radius_m / range_m / shape / targeting / width_m
PROJECTILE_PULL	attack_coef / blocked_by / cast_type / damage / damage_form / duration_s / effect_type / element_rule / hit_kind / immune_tiers / no_legal_pull_point / owner_credit / projectile_speed_mps / pull_destination / pull_to_caster_front_m / radius_m / range_m / shape / source_faction / targeting / width_m
PROJECTILE_ROOT	blocked_by / cast_type / collision_query / duration_s / effect_type / homing / projectile_radius_m / projectile_speed_mps / radius_m / range_m / root_allows_basic_and_active / root_s / shape / targeting / width_m
RADIAL_DAMAGE_KNOCK_TO_EDGE	actor_collision / attack_coef / blocked_knock_still_stuns / cast_type / damage / damage_form / duration_s / effect_type / element_rule / hit_kind / knock_destination / knock_direction / owner_credit / radius_m / range_m / shape / source_faction / stun_s / targeting / width_m
SCAN_OLD_EQUIPMENT	cast_type / duration_s / effect_type / mark_duration_s / marker_reveals_owner_or_source / minimum_ground_age_s /

	newly_aging_items_not_added / radius_m / range_m / recipient / scan_ignores_fog_and_walls / shape / snapshot_at_cast / targeting / width_m
SELF_COMBAT_BUFF	attack_speed_add_percent / cast_type / duration_s / effect_type / excluded_field_classes / incoming_damage_multiplier / passive_effect_magnitude_multiplier / passive_magnitude_authority / passive_proc_probability_multiplier / radius_m / range_m / shape / targeting / transform_additive / transform_delta_from_one / transform_remaining_from_one / width_m
SELF_LOCK_INVULNERABILITY	can_basic / can_cast / can_move / cast_type / duration_s / effect_type / radius_m / range_m / shape / targeting / width_m
SELF_REDUCTION_SPEED_BUFF	cast_type / duration_s / effect_type / incoming_non_true_remaining_multiplier / move_speed_add_percent / radius_m / range_m / shape / targeting / width_m
SELF_REWIND	cast_type / cooldowns_rewind / duration_s / effect_type / gold_inventory_status_rewind / history_buffer_s / history_sample_hz / hp_result / lookback_s / position_illegal_does_not_cancel_hp_restore / position_result / radius_m / range_m / sample_choice / shape / targeting / width_m
SELF_SHIELD	absorbs / amount / cast_type / duration_s / effect_type / radius_m / range_m / same_source_new_cast / shape / shield_absorption_order / targeting / width_m
SELF_STEALTH_SPEED	break_attack_bonus / break_on / cast_type / duration_s / effect_type / move_speed_multiplier / pickup_does_not_break / radius_m / range_m / shape / targeting / width_m
SMOKE_VISION_ZONE	area_skills_can_blind_cast / blocks_vision_across_boundary / cast_type / caster_vision_unchanged / duration_s / effect_type / enemy_inside_direct_vision_cap_m / radius_m / range_m / shape / targeting / true_sight_overrides / width_m
SPAWN_DECOYS	attack / cast_type / copy_owner_visual_and_hp_percent / count / decoy_current_hp / decoy_death_credit / decoy_gold_xp_drop / decoy_max_hp / duration_s / effect_type / hp_inherit_percent / move_speed_inherit_percent / radius_m / range_m / shape / targeting /

	unit_collision / width_m
SPAWN_SUMMON	archetype / attack / cast_type / count / duration_s / effect_type / hp / radius_m / range_m / shape / targeting / width_m 允许缺省： attack_attribute_inherit / triggers_owner_passives
SPEND_GOLD_TRUE_DAMAGE	atomic_deduct_and_cast / cast_type / cost / damage / damage_form / duration_s / effect_type / element_rule / hit_kind / minimum_balance_to_cast / owner_credit / radius_m / range_m / shape / shieldable / source_faction / targeting / width_m
TARGET_BEHIND_BLINK_AND_ARM_CRIT	arm_next_basic_crit / behind_offset_m / candidate_directions / candidate_query / candidate_radius_m / cast_type / duration_s / effect_type / no_candidate / radius_m / range_m / shape / targeting / width_m
TARGET_DAMAGE_AMP_MARK	cast_type / caster_damage_multiplier / caster_true_death / duration_s / effect_type / equal_distance_tie_break / no_legal_transfer / on_credited_true_death_transfer / radius_m / range_m / shape / targeting / transfer_distance_metric / transfer_refresh_s / width_m
TARGET_DAMAGE_HARD_CONTROL	attack_coef / cast_type / control_duration_s / control_type / control_uses_normal_tenacity / damage / damage_form / duration_s / effect_type / element_rule / hit_kind / owner_credit / radius_m / range_m / shape / source_faction / targeting / width_m
TARGET_DOT_REVEAL	cast_type / clear_on_caster_true_death / damage_form / damage_per_tick / duration_s / effect_type / element_rule / hit_kind / owner_credit / radius_m / range_m / reveal / shape / source_faction / targeting / tick_s / width_m
TARGET_REVEAL_BOUNTY_MARK	B41_independent / cast_type / caster_kill_bonus_created_gold / clear_on / duration_s / effect_type / exact_reveal_recipient / other_players_receive_no_information / radius_m / range_m / reveal_does_not_grant_through_wall_lock / shape / targeting / wall_blocked_target / width_m
TOGGLE_PLAYER_BOUNTY	base_bounty / cap / cast_type / duration_s / effect_type / head_marker_shows_capped_value / manual_off / monster_final_blow / numeric_storage / on_each_credited_player_true_death / player_killer_gets_created_gold_percent / radius_m / range_m / shape /

	targeting / toggle_cd_s / true_death / width_m
UNIFORM_RANDOM_DAMAGE	can_crit / cast_type / crit / damage_form / duration_s / effect_type / element_rule / hit_kind / integer_inclusive / owner_credit / radius_m / range_m / rejected_cast_consumes_rng / rng_stream / shape / shieldable / source_faction / targeting / width_m
VISIBLE_ANY_HERO_HEAL	cast_type / duration_s / effect_type / faction_filter / heal / quick_cast_no_target / radius_m / range_m / shape / targeting / width_m
VISIBLE_TRAP_OR_REMOTE_DETONATE	all_capsule_trajectory_cross_triggers / arming_s / blocks_movement / body_blocks / cast_type / damage_form / destroy_explosion / destroy_reward / destructible_by_enemy / duration_s / effect_type / element_rule / exceptions / fourth_removes_oldest / hit_kind / instance_limit / no_own_trap_press / owner_credit / placement_range_m / radius_m / range_m / remote_attack_coef / remote_damage / remote_radius_m / remote_range / remote_selection_hold / remote_selection_tap / remote_selection_tie / reveal_s / root_s / shape / source_faction / targeting / trap_hp / trigger_attack_coef / trigger_damage / trigger_radius_m / width_m
WIND_WALL	blocks_projectile_tag / cast_type / duration_s / effect_type / knockback_m / length_m / radius_m / range_m / shape / solid / targeting / width_m
工程字段	唯一值 / 执行语义
ActiveEffectSchema.BASE_required	effect_type / targeting / cast_type / range_m / shape / duration_s
ActiveEffectSchema.DAMAGE_family	含任何伤害的载荷必须提供 damage_events[]（每事件独立 damage_form/hit_kind/element_rule/source_faction/owner_credit/trigger ）或等价单事件字段组
ActiveEffectSchema.HEAL_family	heal 类字段 + faction_filter；owner_credit 仅伤害/治疗事件必填
ActiveEffectSchema.SUMMON_family	archetype / count / duration_s；战斗数值一律引用 E.34
ActiveEffectSchema.owner_credit_scope	只在产生伤害/治疗/经济事件的载荷必填；纯位移/视野/状态载荷禁止携带
ActiveEffectSchema.implicit_defaults_forbidde n	是；任何字段禁止从 effect_type 名称或全局默认推断

E.31 被动技能强类型运行时载荷（44 项）

- 运行时唯一权威。第 7 章工程数值列为本表的单向投影，运行时禁止解析。

passive_ id	runtime_payload_json
B01	{ "passive_id":{"type":"ENUM","value":"B01"}, "category":{"type":"ENUM","value":"控制类"}, "trigger":{"type":"ENUM","value":"BASIC_HIT_PROC"}, "chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[10,13,16,20,25]}, "slow_percent":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[30,40,50,60,80]}, "duration_s":{"type":"ARRAY","element":"Q4","unit":"s","per_level":true,"value_scaled_1e4":[2000,20000,25000,25000,30000]}, "human":["2","2","2.5","2.5","3"], "l5_aoe_radius_m":{"type":"INT64","value":5,"unit":"m"}, "l5_aoe_slow_percent":{"type":"INT64","value":35,"unit":"pct"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"875802916ce7996ca0393652e012a4d02bf57bd03309805463d6f6660f45d08c"} }
B02	{ "passive_id":{"type":"ENUM","value":"B02"}, "category":{"type":"ENUM","value":"控制类"}, "trigger":{"type":"ENUM","value":"BASIC_HIT_PROC"}, "chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[8,10,12,15,18]}, "silence_s":{"type":"ARRAY","element":"Q4","unit":"s","per_level":true,"value_scaled_1e4":[8000,10000,12000,14000,15000]}, "human":["0.8","1","1.2","1.4","1.5"], "l5_on_end_active_cd_add_s":{"type":"INT64","value":3,"unit":"s"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"6b9b481b1f59a23a947add30be748ddbc5557827ab5838549e092abaff049655"} }
B03	{ "passive_id":{"type":"ENUM","value":"B03"}, "category":{"type":"ENUM","value":"控制类"}, "trigger":{"type":"ENUM","value":"BASIC_HIT_PROC"}, "chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[20,23,26,30,35]}, "knockback_m":{"type":"ARRAY","element":"Q4","unit":"m","per_level":true,"value_scaled_1e4":[30000,35000,40000,45000,50000]}, "human":["3","3.5","4","4.5","5"], "l5_on_critical_aoe_radius_m":{"type":"INT64","value":5,"unit":"m"}, "l5_aoe_knockback_m":{"type":"INT64","value":3,"unit":"m"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"5b89032fb1e24d6531552f4e9a76fc10755d0440cd8a904bf77567adc2211cd8"} }
B04	{ "passive_id":{"type":"ENUM","value":"B04"}, "category":{"type":"ENUM","value":"控制类"}, "trigger":{"type":"ENUM","value":"BASIC_HIT_PROC"}, "chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[10,12,14,17,20]}, "basic_miss_percent":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[50,60,70,80,80]}, "duration_s":{"type":"ARRAY","element":"Q4","unit":"s","per_level":true,"value_scaled_1e4":[20000,20000,25000,25000,30000]}, "human":["2","2","2.5","2.5","3"], "l5_disable_critical":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"be6dadf1a20a4bb7e98363876f99d35ede51a21a1bf357dd10b90faf78ef7441"} }
B05	{ "passive_id":{"type":"ENUM","value":"B05"}, "category":{"type":"ENUM","value":"控制类"}, "trigger":{"type":"ENUM","value":"BASIC_HIT_PROC"}, "chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[5,6,7,8,10]}, "stun_s":{"type":"ARRAY","element":"Q4","unit":"s","per_level":true,"value_scaled_1e4":[4000,5000,6000,7000,8000]}, "human":["0.4","0.5","0.6","0.7","0.8"], "internal_cd_s":{"type":"INT64","value":2,"unit":"s"}, }

	<pre>internal_cd_scope":{"type":"ENUM","value":"PER_TARGET"},"l5_aoe_radius_m":{"type":"INT64","value":3,"unit":"m"},"l5_aoe_stun_s":{"type":"Q4","value_scaled_1e4":3000,"human":"0.3","unit":"s"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"bf9f346fec70592cdbd895075ae9c7775facfb60c179dbe7fe6fdf6225944b21"}}}</pre>
B06	<pre>{"passive_id":{"type":"ENUM","value":"B06"},"category":{"type":"ENUM","value":"伤害类"},"trigger":{"type":"ENUM","value":"BASIC_HIT_PROC"},"chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[8,10,13,16,20]},"critical_damage_percent":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[180,190,200,210,230]},"l5_ignore_shield_percent":{"type":"INT64","value":30,"unit":"pct"},"damage_event_model":{"type":"ENUM","value":"ONE_PARENT_WITH_SHIELD_CHILD_AND_HP_CHILD"},"proc_and_lifesteal_count":{"type":"ENUM","value":"ONCE_PER_PARENT"},"l5_split_rounding":{"type":"ENUM","value":"TOTAL_CRIT_FIRST_HP_CHILD_FLOOR_REMAINDER_TO_SHIELD_CHILD"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"435e219f28afab542d4761de7d049893522b0d70635bdcd865466c8a89fad0a"}}}</pre>
B07	<pre>{"passive_id":{"type":"ENUM","value":"B07"},"category":{"type":"ENUM","value":"伤害类"},"trigger":{"type":"ENUM","value":"BASIC_HIT_PROC"},"chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[10,13,16,20,25]},"splash_damage_percent":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[40,45,50,55,60]},"base_radius_m":{"type":"INT64","value":3,"unit":"m"},"l5_radius_multiplier":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"ratio"},"l5_main_target_receives_splash":{"type":"BOOL","value":true},"splash_cannot_splash":{"type":"BOOL","value":true},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"SPLASH"},"magnitude_field":{"type":"ENUM","value":"splash_damage_percent"},"damage_form":{"type":"ENUM","value":"SKILL"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"},"source_faction":{"type":"ENUM","value":"PLAYER_OWNED"},"owner_credit":{"type":"ENUM","value":"OWNER"},"trigger":{"type":"ENUM","value":"ON_PROC_COMMIT"}]}],"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"bff9ab0b66c7452480bb991c3f5f75b0f551f54f97147befe429244564e5f00e"}}}</pre>
B08	<pre>{"passive_id":{"type":"ENUM","value":"B08"},"category":{"type":"ENUM","value":"伤害类"},"trigger":{"type":"ENUM","value":"BASIC_HIT_STACK"},"stacks_required":{"type":"ARRAY","element":"INT64","unit":"scalar","per_level":true,"value":[5,5,5,4,4]},"lost_hp_damage_percent":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[8,10,12,12,15]},"damage_form":{"type":"ENUM","value":"PERCENT"},"l5_spread_stacks":{"type":"INT64","value":2,"unit":"scalar"},"spread_radius_m":{"type":"INT64","value":5,"unit":"m"},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"DETONATION"},"magnitude_field":{"type":"ENUM","value":"lost_hp_damage_percent"},"damage_form":{"type":"ENUM","value":"PERCENT"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"},"source_faction":{"type":"ENUM","value":"PLAYER_OWNED"},"owner_credit":{"type":"ENUM","value":"OWNER"},"trigger":{"type":"ENUM","value":"ON_STACKS_REACHED"}]}],"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"435e219f28afab542d4761de7d049893522b0d70635bdcd865466c8a89fad0a"}}}</pre>

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B09	{"passive_id":{"type":"ENUM","value":"B09"},"category":{"type":"ENUM","value":"伤害类"}, "trigger":{"type":"ENUM","value":"BASIC_HIT_DOT"},"damage_per_stack_s":{"type":"ARRAY","element":"INT64","unit":"hp/s","per_level":true,"value":[3,4,5,6,8]}, "max_stacks":{"type":"ARRAY","element":"INT64","unit":"scalar","per_level":true,"value":[3,4,4,5]}, "duration_s":{"type":"ARRAY","element":"INT64","unit":"s","per_level":true,"value":[3,3,4,4]}, "refresh_duration":{"type":"BOOL","value":true}, "l5_full_stack_damage_multiplier":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"ratio"}, "damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"POISON_DOT"},"magnitude_field":{"type":"ENUM","value":"damage_per_stack_s"},"damage_form":{"type":"ENUM","value":"DOT"},"hit_kind":{"type":"ENUM","value":"PERIODIC"},"element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"},"source_faction":{"type":"ENUM","value":"PLAYER_OWNED"},"owner_credit":{"type":"ENUM","value":"OWNER"},"trigger":{"type":"ENUM","value":"PER_STACK_PER_SECOND"}]}}, "schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"5177e3c979d9d547caca8478800b105d86c1c85f7dc2a8ef217ba300fb4fba44"}}}
B10	{"passive_id":{"type":"ENUM","value":"B10"},"category":{"type":"ENUM","value":"伤害类"}, "trigger":{"type":"ENUM","value":"DIRECT_DAMAGE_CONDITIONAL"},"hp_threshold_percent":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[20,22,25,28,30]}, "damage_multiplier":{"type":"ARRAY","element":"Q4","unit":"ratio","per_level":true,"value_scaled_1e4":[12500,13000,13500,14000,15000]}, "human":["1.25","1.3","1.35","1.4","1.5"]}, "l5_kill_heal_max_hp_percent":{"type":"INT64","value":10,"unit":"pct"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"aa2566caff9679490f418c952d835a25740b3653c17b5dd6d13b083cca3d39d9"}}}
B11	{"passive_id":{"type":"ENUM","value":"B11"},"category":{"type":"ENUM","value":"伤害类"}, "trigger":{"type":"ENUM","value":"BASIC_HIT_PROC"},"chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[8,10,12,15,20]}, "max_extra_basics":{"type":"ARRAY","element":"INT64","unit":"scalar","per_level":true,"value":[1,1,2,2,3]}, "extra_basic_may_trigger_other_passives":{"type":"BOOL","value":true}, "cannot_trigger_self":{"type":"BOOL","value":true}, "l5_last_hit_other_passive_chance":{"type":"INT64","value":50,"unit":"scalar"}, "l5_other_passive_selection":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["TRIGGER_ONCE_IGNORE_PROC_CHANCE","UNIFORM_FROM_OTHER_THREE"]}, "extra_basic_cannot_trigger_B11":{"type":"BOOL","value":true}, "l5_candidate_pool":{"type":"ENUM","value":"B11_CLOSED_POOL_INTERSECT_OWNED_OTHERS"},"no_candidate":{"type":"ENUM","value":"NO_EXTRA_PASSIVE_TRIGGER"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"0e4654df12a8fbecb9dbb7c184fb07f7c4f4ded91fd5292d5c5e3a5244a0909a"}}}
B12	{"passive_id":{"type":"ENUM","value":"B12"},"category":{"type":"ENUM","value":"伤害类"}, "trigger":{"type":"ENUM","value":"BASIC_HIT_PROC"},"chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_

	<pre>level":true,"value":[6,8,10,12,15]],"summon_hp":{"type":"ARRAY","element":"INT64","unit":"hp","per_level":true,"value": :[150,200,250,300,350]],"summon_attack":{"type":"ARRAY","element":"INT64","unit":"scalar","per_level":true,"value":[2 0,25,30,35,40]],"duration_s":{"type":"INT64","value":15,"unit":"s"},"max_instances":{"type":"ARRAY","element":"INT64"," unit":"scalar","per_level":true,"value":[1,1,1,1,2]],"summon_archetype":{"type":"ENUM","value":"WOLF_SPIRIT"},"sum mon_stat_authority":{"type":"REF","value":"E.34 唯一权威；本载荷内数值须与 E.34 字节级一致 "},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"a484ef15b1eacf19c43 97d21afac40828a2e86208d2356e60aeaaf3ec213cf22"}}</pre>
B13	<pre>{"passive_id":{"type":"ENUM","value":"B13"},"category":{"type":"ENUM","value":"伤害类 "},"trigger":{"type":"ENUM","value":"BASIC_HIT_PROC"},"chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_ level":true,"value":[6,8,10,12,15]],"contact_damage":{"type":"ARRAY","element":"INT64","unit":"hp","per_level":true,"va lue":[60,80,100,120,150]],"duration_s":{"type":"INT64","value":8,"unit":"s"},"contact_internal_cd_s":{"type":"INT64","val ue":1,"unit":"s"},"contact_internal_cd_scope":{"type":"ENUM","value":"PER_TARGET"},"burn_duration_s":{"type":"INT 64","value":3,"unit":"s"},"max_instances":{"type":"ARRAY","element":"INT64","unit":"scalar","per_level":true,"value":[1,1 ,1,1,2]],"burn_damage_per_s":{"type":"ARRAY","element":"INT64","unit":"hp/s","per_level":true,"value":[6,8,10,12,15]]," damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM"," value":"CONTACT"},"magnitude_field":{"type":"ENUM","value":"contact_damage"},"damage_form":{"type":"ENUM","v alue":"SKILL"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"element_rule":{"type":"ENUM","value":"OWNER_ELEMEN T_IF_TARGET_HAS_ELEMENT"},"source_faction":{"type":"ENUM","value":"PLAYER_OWNED"},"owner_credit":{"type":" ENUM","value":"OWNER"},"trigger":{"type":"ENUM","value":"ON_CONTACT_INTERNAL_CD"}},{"event_id":{"type":"ENU M","value":"CONTACT_BURN"},"magnitude_field":{"type":"ENUM","value":"burn_damage_per_s"},"damage_form":{"t ype":"ENUM","value":"DOT"},"hit_kind":{"type":"ENUM","value":"PERIODIC"},"element_rule":{"type":"ENUM","value":"O WNER_ELEMENT_IF_TARGET_HAS_ELEMENT"},"source_faction":{"type":"ENUM","value":"PLAYER_OWNED"},"owner_ credit":{"type":"ENUM","value":"OWNER"},"trigger":{"type":"ENUM","value":"APPLIED_ON_CONTACT_COMMIT"}]}},{"su mmon_archetype":{"type":"ENUM","value":"FIRE_SPIRIT"},"summon_stat_authority":{"type":"REF","value":"E.34 唯一 权威；本载荷内数值须与 E.34 字节级一致 "},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"f0cba28d9fcfc079a4d 792d7369dbf38cf9ee66ddccda982a65ba51775a50063"}}</pre>
B14	<pre>{"passive_id":{"type":"ENUM","value":"B14"},"category":{"type":"ENUM","value":"伤害类 "},"trigger":{"type":"ENUM","value":"BASIC_HIT_PROC"},"chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_ level":true,"value":[10,12,14,16,20]],"damage":{"type":"ARRAY","element":"INT64","unit":"hp","per_level":true,"value":[6 0,80,100,120,150]],"projectile_speed_mps":{"type":"INT64","value":30,"unit":"m/s"},"l5_can_crit":{"type":"BOOL","valu e":true},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type": "ENUM","value":"ARROW"},"magnitude_field":{"type":"ENUM","value":"damage"},"damage_form":{"type":"ENUM","val ue":"SKILL"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_</pre>

	IF_TARGET_HAS_ELEMENT"},"source_faction":{"type":"ENUM","value":"PLAYER_OWNED"},"owner_credit":{"type":"ENUM","value":"OWNER"},"trigger":{"type":"ENUM","value":"ON_PROC_COMMIT"}}},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"0ad39697e58f7b5edfd20490948db120e9d6aef7f5eaa38b681b1496762a45a3"}}
B15	{"passive_id":{"type":"ENUM","value":"B15"},"category":{"type":"ENUM","value":"生存类"},"trigger":{"type":"ENUM","value":"INCOMING_BASIC_PRE_HIT_PROC"},"chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[10,13,16,20,25]},"result":{"type":"ENUM","value":"DODGE_ZERO_DAMAGE"},"l5_move_speed_percent":{"type":"INT64","value":30,"unit":"pct"},"l5_duration_s":{"type":"INT64","value":2,"unit":"s"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"0aa523744640773c8ade8da04e834d858177d2cceab29a32152acee2892c2eac"}}
B16	{"passive_id":{"type":"ENUM","value":"B16"},"category":{"type":"ENUM","value":"生存类"},"trigger":{"type":"ENUM","value":"INCOMING_BASIC_FIXED_REDUCTION"},"reduction":{"type":"ARRAY","element":"INT64","unit":"scalar","per_level":true,"value":[6,10,14,18,25]},"minimum_final_damage":{"type":"INT64","value":1,"unit":"hp"},"l5_full_block_chance":{"type":"INT64","value":10,"unit":"scalar"},"l5_block_order":{"type":"ENUM","value":"AFTER_B15_BEFORE_B17"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"577abb686d4d965820ec480fcf7423022b5f4226dd17ce71332e10bc5ac0303c"}}
B17	{"passive_id":{"type":"ENUM","value":"B17"},"category":{"type":"ENUM","value":"生存类"},"trigger":{"type":"ENUM","value":"INCOMING_BASIC_PROC_PRE_DAMAGE"},"chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[8,10,13,16,20]},"shield":{"type":"ARRAY","element":"INT64","unit":"hp","per_level":true,"value":[60,80,100,130,160]},"duration_s":{"type":"INT64","value":10,"unit":"s"},"absorbs_triggering_hit":{"type":"BOOL","value":true},"l5_break_aoe_damage":{"type":"INT64","value":100,"unit":"hp"},"l5_break_aoe_radius_m":{"type":"INT64","value":3,"unit":"m"},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"BREAK_AOE"},"magnitude_field":{"type":"ENUM","value":"l5_break_aoe_damage"},"damage_form":{"type":"ENUM","value":"SKILL"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"},"source_faction":{"type":"ENUM","value":"PLAYER_OWNED"},"owner_credit":{"type":"ENUM","value":"OWNER"},"trigger":{"type":"ENUM","value":"ON_SHIELD_BREAK_L5"}}]},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"b4457ce4e567ad1ac49d0ddfc7c36ad20c36d65268d5054e679018b1657d720"}}
B18	{"passive_id":{"type":"ENUM","value":"B18"},"category":{"type":"ENUM","value":"生存类"},"trigger":{"type":"ENUM","value":"MISSING_HP_SCALING"},"attack_percent_per_missing_10pct_hp":{"type":"ARRAY","element":"INT64","unit":"hp","per_level":true,"value":[3,4,5,6,8]},"step_rounding":{"type":"FORMULA","src":"FLOOR(MISSING_HP_PERCENT/10)","ast":{"op":"FLOOR","args":{"op":"DIV_FLOOR","args":{"var":"MISSING_HP_PERCENT"},"q4":100000}}}},"l5_below_hp_percent":{"type":"INT64","value":30,"unit":"pct"},"l5_lifesteal_percent":{"type":"INT64",

	"value":10,"unit":"pct"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"0ac1c9e9cc886a86cf020ea601c34da06db5cfe9be72f1fd2d88910657ff3cdd"}}}
B19	{ "passive_id":{"type":"ENUM","value":"B19"}, "category":{"type":"ENUM","value":"生存类"}, "trigger":{"type":"ENUM","value":"LETHAL_PROC"}, "chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[5,6,7,8,10]}, "heal_max_hp_percent":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[10,12,15,18,20]}, "l5_random_blink":{"type":"BOOL","value":true}, "random_blink_distance_m":{"type":"INT64","value":8,"unit":"m"}, "blink_rng_stream":{"type":"ENUM","value":"COMBAT_PROC"}, "blink_direction_draw":{"type":"ENUM","value":"SINGLE_DRAW_EIGHT_DIRECTIONS"}, "blink_destination":{"type":"ENUM","value":"FARTHEST_LEGAL_POINT_ALONG_DIRECTION"}, "all_points_illegal":{"type":"ENUM","value":"NO_BLINK_NO_REDRAW"}, "post_success_trigger_lock_s":{"type":"INT64","value":1,"unit":"s"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"21fc2b2ba28e45961ba7c66c6c06d0709c04773eda85903823627de89a0b743f"} }
B20	{ "passive_id":{"type":"ENUM","value":"B20"}, "category":{"type":"ENUM","value":"生存类"}, "trigger":{"type":"ENUM","value":"LETHAL_PROTECTION_ONCE_MATCH"}, "heal_max_hp_percent":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[25,30,35,40,50]}, "charge":{"type":"INT64","value":1,"unit":"scalar"}, "consume_life":{"type":"BOOL","value":false}, "l5_buff_s":{"type":"INT64","value":5,"unit":"s"}, "l5_damage_multiplier":{"type":"Q4","value_scaled_1e4":13000,"human":"1.3","unit":"ratio"}, "l5_control_immune":{"type":"BOOL","value":true}, "charge_ledger_key":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["B20","MATCH_ID","PLAYER_ID"]}, "removed_or_reacquired_does_not_restore":{"type":"BOOL","value":true}, "hero_swap_does_not_restore":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"4815fee33780f4a1d7ea5883b572a4c116ce6ef5cbf2f48aa03548ff431498c0"} }
B21	{ "passive_id":{"type":"ENUM","value":"B21"}, "category":{"type":"ENUM","value":"生存类"}, "trigger":{"type":"ENUM","value":"OUT_OF_ANY_COMBAT_REGEN"}, "delay_s":{"type":"ARRAY","element":"INT64","unit":"s","per_level":true,"value":[5,5,4,4,3]}, "hp_per_s":{"type":"ARRAY","element":"INT64","unit":"hp/s","per_level":true,"value":[4,6,8,10,15]}, "l5_first_basic_damage_multiplier":{"type":"Q4","value_scaled_1e4":13000,"human":"1.3","unit":"ratio"}, "first_basic_consumes_bonus":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"47ea709c1b9e49ce18ec62014446b11c6d136a7faaf46f95658e4dc3c5cefd a1"} }
B22	{ "passive_id":{"type":"ENUM","value":"B22"}, "category":{"type":"ENUM","value":"反击类"}, "trigger":{"type":"ENUM","value":"POST_INCOMING_BASIC_REFLECT"}, "actual_hp_plus_shield_damage_percent":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[10,13,16,20,25]}, "cannot_trigger_reflect":{"type":"BOOL","value":true}, "l5_can_crit":{"type":"BOOL","value":true}, "l5_crit_uses_owner_crit_snapshot_without_triggering_proc":{"type":"BOOL","value":true}, "damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"REFLECT"},"magnitude_field":{"type":"ENUM","value":"actual_hp_plu

	s_shield_damage_percent"},"damage_form":{"type":"ENUM","value":"REFLECT"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"element_rule":{"type":"ENUM","value":"NONE"},"source_faction":{"type":"ENUM","value":"PLAYER_OWNED"},"owner_credit":{"type":"ENUM","value":"OWNER"},"trigger":{"type":"ENUM","value":"POST_INCOMING_BASIC}}},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"f65e19874d2bfa2c045511c02455d7d26dfe6d6881b9ef5c9458c26f5f66aae"}}
B23	{"passive_id":{"type":"ENUM","value":"B23"},"category":{"type":"ENUM","value":"反击类"},"trigger":{"type":"ENUM","value":"INCOMING_ACTUAL_DAMAGE_PROC"},"chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[6,8,10,13,16]},"counter_damage_attack_coefficient":{"type":"ARRAY","element":"Q4","unit":"ratio","per_level":true,"value_scaled_1e4":[8000,8500,9000,9500,10000],"human":["0.8","0.85","0.9","0.95","1"]},"damage_form":{"type":"ENUM","value":"REFLECT"},"cannot_trigger_basic_passives":{"type":"BOOL","value":true},"l5_forced_critical":{"type":"BOOL","value":true},"eligible_damage":{"type":"ENUM","value":"DIRECT_ACTUAL_HP_OR_SHIELD_DAMAGE_ONLY"},"dot_and_periodic_ineligible":{"type":"BOOL","value":true},"internal_cd_s":{"type":"Q4","value_scaled_1e4":5000,"human":"0.5","unit":"s"},"internal_cd_scope":{"type":"ENUM","value":"PER_SOURCE_ENTITY"},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"COUNTER"},"magnitude_field":{"type":"ENUM","value":"counter_damage_attack_coefficient"},"damage_form":{"type":"ENUM","value":"REFLECT"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"element_rule":{"type":"ENUM","value":"NONE"},"source_faction":{"type":"ENUM","value":"PLAYER_OWNED"},"owner_credit":{"type":"ENUM","value":"OWNER"},"trigger":{"type":"ENUM","value":"ON_PROC_COMMIT}}]},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"e70fa77816a1bfc530d418c26fa8cf5af793c8b3303f78d6df92570be09fd31"}}
B24	{"passive_id":{"type":"ENUM","value":"B24"},"category":{"type":"ENUM","value":"反击类"},"trigger":{"type":"ENUM","value":"POST_SKILL_DAMAGE_IF_SURVIVES"},"heal_percent_actual_hp_plus_shield_damage":{"type":"ARRAY","element":"INT64","unit":"hp","per_level":true,"value":[10,13,16,20,25]},"cannot_prevent_lethal":{"type":"BOOL","value":true},"overheal_discarded":{"type":"BOOL","value":true},"l5_reflect_pre_overheal_conversion_percent":{"type":"INT64","value":50,"unit":"pct"},"cannot_trigger_reflect":{"type":"BOOL","value":true},"l5_reflect_amount":{"type":"FORMULA","src":"FLOOR(PRE_OVERHEAL_AMOUNT*0.5)","ast":{"op":"FLOOR","args":[{"op":"MUL","args":[{"var":"PRE_OVERHEAL_AMOUNT"},{"q4":5000}]}}},"l5_reflect_target":{"type":"ENUM","value":"ATTACKER_OF_TRIGGERING_DAMAGE"},"l5_reflect_damage_form":{"type":"ENUM","value":"REFLECT"},"l5_reflect_cannot_recurse":{"type":"BOOL","value":true},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"CONVERT_REFLECT"},"magnitude_field":{"type":"ENUM","value":"l5_reflect_amount"},"damage_form":{"type":"ENUM","value":"REFLECT"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"element_rule":{"type":"ENUM","value":"NONE"},"source_faction":{"type":"ENUM","value":"PLAYER_OWNED"},"owner_credit":{"type":"ENUM","value":"OWNER"},"trigger":{"type":"ENUM","value":"ON_CONVERT_L5}}]},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"aa626f4100179c4a1db83ae9d5402fa6be397c2631828d3525

	3a666984e3ea57"}}}
B25	{ "passive_id":{"type":"ENUM","value":"B25"}, "category":{"type":"ENUM","value":"反击类"}, "trigger":{"type":"ENUM","value":"WHEN_CRITICALLY_HIT"}, "next_basic_damage_multiplier":{"type":"ARRAY","element":"Q4","unit":"ratio","per_level":true,"value_scaled_1e4":[12000,12800,13600,14400,15500],"human":["1.2","1.28","1.36","1.44","1.55"]}, "one_charge":{"type":"BOOL","value":true}, "l5_attack_speed_percent":{"type":"INT64","value":40,"unit":"pct"}, "l5_duration_s":{"type":"INT64","value":2,"unit":"s"}, "monster_cannot_crit":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"714d831c2ff8c905b9dcd5f96a17dff1e903e2826a337718af4af6223498f51a"}}}
B26	{ "passive_id":{"type":"ENUM","value":"B26"}, "category":{"type":"ENUM","value":"战术类"}, "trigger":{"type":"ENUM","value":"DIRECT_DAMAGE_FROM_BACK_ARC"}, "back_arc_deg":{"type":"INT64","value":120,"unit":"deg"}, "damage_multiplier":{"type":"ARRAY","element":"Q4","unit":"ratio","per_level":true,"value_scaled_1e4":[13500,14200,15000,15800,17000],"human":["1.35","1.42","1.5","1.58","1.7"]}, "eligible_damage_forms":{"type":"ENUM","value":"BASIC"}, "l5_forced_critical":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"637476f8632b7d8dd15d07cacadff80b48fe2b66ca512746ac1c72df296a93bb"}}}
B27	{ "passive_id":{"type":"ENUM","value":"B27"}, "category":{"type":"ENUM","value":"战术类"}, "trigger":{"type":"ENUM","value":"INCOMING_ACTUAL_DAMAGE_PROC"}, "chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[10,13,16,20,25]}, "move_speed_percent":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[20,25,30,35,40]}, "duration_s":{"type":"ARRAY","element":"Q4","unit":"s","per_level":true,"value_scaled_1e4":[20000,20000,25000,25000,30000],"human":["2","2","2.5","2.5","3"]}, "l5_slow_immune":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"bfbfeefddc93af36abc75bbc537e0db6c46cb9a3ed1e40826c15b3def23dbd910"}}}
B28	{ "passive_id":{"type":"ENUM","value":"B28"}, "category":{"type":"ENUM","value":"战术类"}, "trigger":{"type":"ENUM","value":"LOW_HP_HUNT"}, "detect_range_m":{"type":"ARRAY","element":"INT64","unit":"m","per_level":true,"value":[30,40,50,60,80]}, "target_hp_below_percent":{"type":"INT64","value":30,"unit":"pct"}, "move_speed_toward_target_percent":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[15,18,20,22,25]}, "does_not_grant_vision":{"type":"BOOL","value":true}, "l5_damage_multiplier":{"type":"Q4","value_scaled_1e4":12000,"human":"1.2","unit":"ratio"}, "eligible_target_information_state":{"type":"ENUM","value":"VISIBLE_OR_REVEALED"}, "hidden_unrevealed_target_grants_no_speed_or_direction_leak":{"type":"BOOL","value":true}, "multi_candidate_selection":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["ENTITY_ID_ASC","NAV_PATH_DISTANCE_MM_ASC"]}, "reselect_interval_s":{"type":"Q4","value_scaled_1e4":2500,"human":"0.25","unit":"s"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"3037125982fd0de9bf69371313e84c38276673b1c981af63f6659f66fb1768a4"}}}

B29	{ "passive_id":{"type":"ENUM","value":"B29"},"category":{"type":"ENUM","value":"战术类"}, "trigger":{"type":"ENUM","value":"FIRST_DIRECT_HIT_AFTER_GAP"},"qualification_gap_s":{"type":"ARRAY","element":"INT64","unit":"s","per_level":true,"value":[5,5,4,4,3],"semantic":"OWNER_OUTGOING_DIRECT_DAMAGE_GAP_S"},"damage_multiplier":{"type":"ARRAY","element":"Q4","unit":"ratio","per_level":true,"value_scaled_1e4":[12500,13500,14500,15500,17000],"human":["1.25","1.35","1.45","1.55","1.7"]},"eligible":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["BASIC","SKILL"]},"dot_does_not_trigger":{"type":"BOOL","value":true},"l5_reveal_target_s":{"type":"INT64","value":3,"unit":"s"},"trigger_requires_positive_hp_or_shield_damage":{"type":"BOOL","value":true},"zero_damage_hit_does_not_consume_or_reset":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"73538b99fd08618da502dffa1cd102929085704a658dac49722ed09d17ddfbf4"}}
B30	{ "passive_id":{"type":"ENUM","value":"B30"},"category":{"type":"ENUM","value":"战术类"}, "trigger":{"type":"ENUM","value":"MOVING_AFTERIMAGE"},"spawn_interval_s":{"type":"ARRAY","element":"Q4","unit":"s","per_level":true,"value_scaled_1e4":[20000,20000,15000,15000,10000],"human":["2","2","1.5","1.5","1"]},"afterimage_lifetime_s":{"type":"Q4","value_scaled_1e4":15000,"human":1.5,"unit":"s"},"slow_percent":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[20,25,30,30,35]},"slow_duration_s":{"type":"ARRAY","element":"Q4","unit":"s","per_level":true,"value_scaled_1e4":[15000,20000,20000,25000,30000],"human":["1.5","2","2","2.5","3"]},"l5_explosion_damage":{"type":"INT64","value":80,"unit":"hp"},"l5_radius_m":{"type":"INT64","value":3,"unit":"m"},"each_enemy_triggers_each_afterimage_once":{"type":"BOOL","value":true},"afterimage_visible_to":{"type":"ENUM","value":"ALL"},"afterimage_solid":{"type":"BOOL","value":false},"spawn_at":{"type":"ENUM","value":"OWNER_HISTORICAL_PATH_POSITION"},"trigger_radius_m":{"type":"Q4","value_scaled_1e4":8000,"human":0.8,"unit":"m"},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"AFTERIMAGE_EXPLOSION"},"magnitude_field":{"type":"ENUM","value":"l5_explosion_damage"},"damage_form":{"type":"ENUM","value":"SKILL"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"element_rule":{"type":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"},"source_faction":{"type":"ENUM","value":"PLAYER_OWNED"},"owner_credit":{"type":"ENUM","value":"OWNER"},"trigger":{"type":"ENUM","value":"ON_AFTERIMAGE_TOUCHED_L5"}]},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"7b269c714f907c57016d18fcfacd8d06843774b20fe96f11421b3c4cd9057878"}}
B31	{ "passive_id":{"type":"ENUM","value":"B31"},"category":{"type":"ENUM","value":"经济类"}, "trigger":{"type":"ENUM","value":"BASIC_HIT_PROC_TRANSFER_GOLD"},"chance":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[6,8,10,13,16]},"gold":{"type":"ARRAY","element":"INT64","unit":"scalar","per_level":true,"value":[30,40,50,60,80]},"insufficient":{"type":"ENUM","value":"TRANSFER_REMAINING"},"not_create":{"type":"BOOL","value":true},"l5_heal_percent_transferred_gold":{"type":"INT64","value":10,"unit":"gold"},"eligible_basic":{"type":"ENUM","value":"MAIN_BASIC_ONLY"},"extra_basic_ineligible":{"type":"BOOL","value":true},"internal_cd_s":{"type":"Q4","value_scaled_1e4":4000,"human":0.4,"unit":"s"},"internal_cd_scope":{"type":"ENUM","value":"PER_TARGET"},"

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B32	{"passive_id":{"type":"ENUM","value":"B32"},"category":{"type":"ENUM","value":"经济类"},"trigger":{"type":"ENUM","value":"ON_KILL_GOLD_CREATE"},"monster_gold":{"type":"ARRAY","element":"INT64","unit":"gold","per_level":true,"value":[40,60,80,100,150]},"eligible_monsters":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["CORE_BOSS","DRAGON","FLYING_MONSTER","LEADER","ORDINARY_MONSTER","PIG"]},"l5_enemy_player_true_death_gold":{"type":"INT64","value":250,"unit":"gold"},"summon_kill":{"type":"BOOL","value":false},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"fe8768aa63e3d53ec3a968a764675d26812f338e15d0431b889eab51aeee8484"}}
B33	{"passive_id":{"type":"ENUM","value":"B33"},"category":{"type":"ENUM","value":"经济类"},"trigger":{"type":"ENUM","value":"ON_MONSTER_KILL_NESTED_CHEST"},"chest_chance":{"type":"ARRAY","element":"INT64","unit":"scalar","per_level":true,"value":[8,10,13,16,20]},"chest_gold":{"type":"ARRAY","element":"INT64","unit":"gold","per_level":true,"value":[100,120,120,120,120]},"gem_chance":{"type":"ARRAY","element":"INT64","unit":"scalar","per_level":true,"value":[0,0,10,15,20]},"gold_equipment_chance":{"type":"ARRAY","element":"INT64","unit":"scalar","per_level":true,"value":[0,0,0,0,5]},"gold_pool":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["G1","G2","G3","G4","G5","G6","G7","G8","G9"]},"summon_kill":{"type":"BOOL","value":false},"nested_contents_roll_at":{"type":"ENUM","value":"CHEST_SPAWN_COMMIT"},"contents_snapshot_immutable":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"74b748f2041731875aa5f4196d559edee41c80fb56f9ac0c93c3637139a612d5"}}
B34	{"passive_id":{"type":"ENUM","value":"B34"},"category":{"type":"ENUM","value":"经济类"},"trigger":{"type":"ENUM","value":"SERVER_MINUTE_INTEREST"},"percent":{"type":"ARRAY","element":"INT64","unit":"scalar","per_level":true,"value":[5,6,7,8,10]},"cap_gold":{"type":"ARRAY","element":"INT64","unit":"gold","per_level":true,"value":[300,400,500,600,1600]},"snapshot":{"type":"ENUM","value":"COMMITTED_WALLET_AT_EXACT_60S_BOUNDARY_BEFORE_INTEREST"},"compound":{"type":"BOOL","value":true},"eligible_life_states":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["ALIVE","REVIVE_PROTECTION","SOUL_FLIGHT","TRUE_DEATH_RESOLVING"]},"eliminated_ineligible":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"3173ddbdc4fe659c81b60f04312ce1fb74efedc8ee0014317ab59eb4c3738918"}}
B35	{"passive_id":{"type":"ENUM","value":"B35"},"category":{"type":"ENUM","value":"经济类"},"trigger":{"type":"ENUM","value":"SELL_AND_PICKUP_ECONOMY"},"sell_bonus_percent":{"type":"ARRAY","element":"INT64","unit":"pct","per_level":true,"value":[10,13,16,20,25]},"l5_pickup_base_sell_percent":{"type":"INT64","value":20,"unit":"pct"},"l5_reward_once_per_item_instance_globally":{"type":"BOOL","value":true},"owner_dropped_never_rewards_same_owner":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"071bb120ffcae1101a763c5a838582fb2eb78b70a97e33df1b230bd112574f44"}}

B36	<pre>{ "passive_id": {"type": "ENUM", "value": "B36"}, "category": {"type": "ENUM", "value": "改规则类"}, "trigger": {"type": "ENUM", "value": "CONTINUOUS_MOVEMENT_STACK"}, "stack_interval_s": {"type": "INT64", "value": 1, "unit": "s"}, "move_speed_percent_per_stack": {"type": "ARRAY", "element": "Q4", "unit": "scalar", "per_level": true, "value_scaled_1e4": [10000, 12000, 15000, 18000, 20000], "human": ["1", "1.2", "1.5", "1.8", "2"]}, "max_stacks": {"type": "ARRAY", "element": "INT64", "unit": "scalar", "per_level": true, "value": [5, 6, 6, 7, 8]}, "clear_on": {"type": "ENUM", "value": "MOVE_INPUT_ZERO_OR_ACTUAL_SPEED_LT_0_1_FOR_0_2S"}, "I5_full_stack_basic_extra_damage_move_speed_value_percent": {"type": "INT64", "value": 20, "unit": "pct"}, "extra_can_crit": {"type": "BOOL", "value": true}, "move_speed_value_unit": {"type": "ENUM", "value": "GAME_MOVE_SPEED_INTEGER"}, "I5_extra_damage_formula": {"type": "FORMULA", "src": "FLOOR(CURRENT_GAME_MOVE_SPEED*0.20)", "ast": {"op": "FLOOR", "args": [{"op": "MUL", "args": [{"var": "CURRENT_GAME_MOVE_SPEED"}, {"q4": 2000}]}]}}, "damage_events": {"type": "ARRAY", "element": "DamageEventSpec", "ordered": true, "value": [{"event_id": {"type": "ENUM", "value": "EXTRA_BASIC_BONUS"}, "magnitude_field": {"type": "ENUM", "value": "I5_extra_damage_formula"}, "damage_form": {"type": "ENUM", "value": "SKILL"}, "hit_kind": {"type": "ENUM", "value": "ABILITY"}, "element_rule": {"type": "ENUM", "value": "OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "source_faction": {"type": "ENUM", "value": "PLAYER_OWNED"}, "owner_credit": {"type": "ENUM", "value": "OWNER"}, "trigger": {"type": "ENUM", "value": "ON_FULL_STACK_BASIC_HIT_L5"}]}]}, "schema_version": {"type": "ENUM", "value": "V3_0"}, "payload_sha256": {"type": "HASH", "value": "e30222b8fc0825c29a2baa58a44f0fc70902b5e8de49c0de6a5901b0153954de"} }</pre>
B37	<pre>{ "passive_id": {"type": "ENUM", "value": "B37"}, "category": {"type": "ENUM", "value": "改规则类"}, "trigger": {"type": "ENUM", "value": "OWNED_SUMMON_DEATH"}, "heal": {"type": "ARRAY", "element": "INT64", "unit": "hp", "per_level": true, "value": [10, 15, 20, 25, 45]}, "aoe_damage": {"type": "ARRAY", "element": "INT64", "unit": "hp", "per_level": true, "value": [0, 0, 30, 40, 75]}, "aoe_radius_m": {"type": "INT64", "value": 4, "unit": "m"}, "I5_values_already_include_50pct_bonus": {"type": "BOOL", "value": true}, "eligible_death_causes": {"type": "SET", "element": "ENUM", "unique": true, "ordered": false, "value": ["HOSTILE_MONSTER_DAMAGE", "HOSTILE_PLAYER_DAMAGE"]}, "ineligible_removal_causes": {"type": "SET", "element": "ENUM", "unique": true, "ordered": false, "value": ["NATURAL_EXPIRY", "OWNER_ELIMINATION_CLEANUP", "OWNER_TRUE_DEATH_CLEANUP", "REPLACEMENT", "WORLD_DAMAGE"]}, "damage_events": {"type": "ARRAY", "element": "DamageEventSpec", "ordered": true, "value": [{"event_id": {"type": "ENUM", "value": "RESONANCE_AOE"}, "magnitude_field": {"type": "ENUM", "value": "aoe_damage"}, "damage_form": {"type": "ENUM", "value": "SKILL"}, "hit_kind": {"type": "ENUM", "value": "ABILITY"}, "element_rule": {"type": "ENUM", "value": "OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"}, "source_faction": {"type": "ENUM", "value": "PLAYER_OWNED"}, "owner_credit": {"type": "ENUM", "value": "OWNER"}, "trigger": {"type": "ENUM", "value": "ON_ELIGIBLE_SUMMON_DEATH"}]}]}, "schema_version": {"type": "ENUM", "value": "V3_0"}, "payload_sha256": {"type": "HASH", "value": "9df70258a29cfb59ee9221a1ab0d0376135606865e82c1659a5e78a3b7bffc2d"} }</pre>
B38	<pre>{ "passive_id": {"type": "ENUM", "value": "B38"}, "category": {"type": "ENUM", "value": "改规则类"}, "trigger": {"type": "ENUM", "value": "WHILE_HARD_CONTROLLED"}, "incoming_damage_remaining_multiplier": {"type": "ARRAY", "element": "Q4", "unit": "ratio", "per_level": true, "value_scaled_1e4": [8500, 8000, 7500, 7000, 6500], "human": ["0.85", "0.8", "0.75", "0.7", "0.65"]}, "I5_heal_max_hp_percent_per_s": {"type": "INT64", "value": 2, "unit": "hp/s"}, "soft_control_n }</pre>

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B39	{"passive_id":{"type":"ENUM","value":"B39"},"category":{"type":"ENUM","value":"改规则类 "},"trigger":{"type":"ENUM","value":"OUT_OF_ANY_COMBAT_SUMMON"},"delay_s":{"type":"ARRAY","element":"INT64", "unit":"s","per_level":true,"value":[7,7,6,6,5]},"statue_hp":{"type":"ARRAY","element":"INT64","unit":"hp","per_level":tru e,"value":[250,300,350,400,500]},"statue_attack":{"type":"INT64","value":0,"unit":"scalar"},"follow_owner":{"type":"BOO L","value":true},"one_instance":{"type":"BOOL","value":true},"l5_destroy_aoe_damage":{"type":"INT64","value":120,"u nit":"hp"},"l5_radius_m":{"type":"INT64","value":4,"unit":"m"},"statue_move_speed_mps":{"type":"INT64","value":3,"u nit":"m/s"},"statue_collision_radius_m":{"type":"Q4","value_scaled_1e4":6000,"human":"0.6","unit":"m"},"statue_colli sion_height_m":{"type":"Q4","value_scaled_1e4":18000,"human":"1.8","unit":"m"},"statue_aggro":{"type":"ENUM","val ue":"NONE"},"statue_targetable":{"type":"BOOL","value":true},"statue_destroy_credit":{"type":"ENUM","value":"ATTAC KER_ONLY_NO_REWARD"},"forbidden_block_ole_corridor":{"type":"BOOL","value":true},"owner_block_yield_after_ s":{"type":"Q4","value_scaled_1e4":5000,"human":"0.5","unit":"s"},"yield_fail":{"type":"ENUM","value":"TEMPORARY_N O_COLLISION"},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id" ":{"type":"ENUM","value":"STATUE_DESTROY_AOE"},"magnitude_field":{"type":"ENUM","value":"l5_destroy_aoe_dam age"},"damage_form":{"type":"ENUM","value":"SKILL"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"element_rule":{"t ype":"ENUM","value":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"},"source_faction":{"type":"ENUM","value":"PLA YER_OWNED"},"owner_credit":{"type":"ENUM","value":"OWNER"},"trigger":{"type":"ENUM","value":"ON_STATUE_DES TROYED_L5"}]},"summon_archetype":{"type":"ENUM","value":"STONE_STATUE"},"summon_stat_authority":{"type":" REF","value":"E.34 唯一权威；本载荷内数值须与 E.34 字节级一致 "},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"1d2f285212e5e3aaf22 6b2dfe618b6c0cb0eda9de3581951397b777fe0685487f5}}
B40	{"passive_id":{"type":"ENUM","value":"B40"},"category":{"type":"ENUM","value":"养成类 "},"trigger":{"type":"ENUM","value":"ON_ELIGIBLE_KILL_PERMANENT_MAX_HP"},"hp_per_kill":{"type":"ARRAY","eleme nt":"INT64","unit":"hp","per_level":true,"value":[2,4,6,7,10]},"eligible":{"type":"SET","element":"ENUM","unique":true,"o rdered":false,"value":["CORE_BOSS","DRAGON","ENEMY_PLAYER_TRUE_DEATH","FLYING_MONSTER","LEADER","OR DINARY_MONSTER","PIG"]},"summon_kill":{"type":"BOOL","value":false},"l5_every_50_kills_bonus_hp":{"type":"INT64 ","value":160,"unit":"hp"},"no_cap":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"pa yload_sha256":{"type":"HASH","value":"0ee7ca2de6d9f5bfcd6f46911cdda8bfe255e6796dd373d868d95175485546b6f5"} }
B41	{"passive_id":{"type":"ENUM","value":"B41"},"category":{"type":"ENUM","value":"赏金体系 "},"trigger":{"type":"ENUM","value":"ON_OWNER_TRUE_DEATH_MARK_KILLER"},"payout_gold":{"type":"ARRAY","elem

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B42	{"passive_id":{"type":"ENUM","value":"B42"},"category":{"type":"ENUM","value":"赏金体系"},"trigger":{"type":"ENUM","value":"ON_ENEMY_BOUNTY_TARGET_TRUE_DEATH"},"active_cd_reduce_s":{"type":"ARRAY","element":{"INT64","unit":"s","per_level":true,"value":[10,15,20,25,30]},"minimum_remaining_s":{"type":"INT64","value":0,"unit":"s"},"l5_move_speed_percent":{"type":"INT64","value":40,"unit":"pct"},"l5_duration_s":{"type":"INT64","value":5,"unit":"s"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"85f05e7f9e79efb886e911e3dff82a934658cf8b6177ab896e4b9e0b206fe7e5"}}
B43	{"passive_id":{"type":"ENUM","value":"B43"},"category":{"type":"ENUM","value":"天劫体系"},"trigger":{"type":"ENUM","value":"NORMAL_STORM_ZONE_MODIFIER"},"normal_hit_probability":{"type":"ARRAY","element":{"Q4","unit":"pct","per_level":true,"value_scaled_1e4":[4000,3500,3000,2500,2000],"human":["0.4","0.35","0.3","0.25","0.2"]},"move_speed_percent":{"type":"ARRAY","element":{"INT64","unit":"pct","per_level":true,"value":[0,0,10,10,15]},"l5_basic_damage_multiplier":{"type":"Q4","value_scaled_1e4":11000,"human":"1.1","unit":"ratio"},"does_not_change_apocalypse_probability_or_damage":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"67e501767bf82b4a46c2bf20e7018155aad9dab210feee5d241d8891f8fe049e"}}
B44	{"passive_id":{"type":"ENUM","value":"B44"},"category":{"type":"ENUM","value":"天劫体系"},"trigger":{"type":"ENUM","value":"BASIC_HIT_PROC"},"chance":{"type":"ARRAY","element":{"INT64","unit":"pct","per_level":true,"value":[8,10,13,16,20]},"damage":{"type":"ARRAY","element":{"INT64","unit":"hp","per_level":true,"value":[80,90,100,110,130]},"radius_m":{"type":"ARRAY","element":{"INT64","unit":"m","per_level":true,"value":[3,3,4,4,5]},"damage_form":{"type":"ENUM","value":"SKILL"},"inherits_element":{"type":"BOOL","value":true},"l5_normal_storm_zone_chance_multiplier":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"ratio"},"l5_zone_chance":{"type":"INT64"

	4","value":30,"unit":"scalar"},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value": [{"event_id":{"type":"ENUM","value":"LIGHTNING"},"magnitude_field":{"type":"ENUM","value":"damage"},"damage_for m":{"type":"ENUM","value":"SKILL"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"element_rule":{"type":"ENUM","valu e":"OWNER_ELEMENT_IF_TARGET_HAS_ELEMENT"},"source_faction":{"type":"ENUM","value":"PLAYER_OWNED"},"o wner_credit":{"type":"ENUM","value":"OWNER"},"trigger":{"type":"ENUM","value":"ON_PROC_COMMIT"}]}},{"schema_v ersion":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"2db33b2946cbb117836afdb8102cf 7fc84d7f8e48cfa1f4ba69276f24a6ffe35"}]}
工程字 段	唯一值 / 执行语义
B11.clos ed_cand idate_ pool	B01 / B02 / B03 / B04 / B05 / B07 / B08 / B09 / B12 / B13 / B14 / B44
B11.poo l_rule	入池标准：原生触发为出手普攻事件且效果可在连击最后一刀命中时刻作为一次原子事件结算
B11.excl uded_w ith_reas on	B06=作用于命中本身已结算 / B31=仅主普攻合格 / B19 B20=致死上下文 / B34=分钟利息 / B40=永久成长 / 其余=受 击、状态机、经济或位置上下文
B11.dra w_sema ntics	候选=已装备其他 3 被动 $\cap$ closed_candidate_pool；空集则不触发；命中候选后无视其概率、按当前等级结算一 次；内置 CD 仍生效，CD 中则本次触发落空且不重抽
B11.rng _stream	COMBAT_PROC

E.32 装备强类型运行时载荷（48 项）

- 运行时唯一权威。第 8 章属性与效果列为本表的单向投影，运行时禁止解析。

equi pm ent_ id	runtime_payload_json
W1	{"equipment_id":{"type":"ENUM","value":"W1"},"quality":{"type":"ENUM","value":"WHITE"},"stats":{"type":"ARRAY","element ":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"ATTACK_FLAT"},"amount":{"type":"INT64","value":15,"

	unit":"scalar"}]]], "effect":{"type":{"type":"ENUM", "value":"NONE"}}, "buy_gold":{"type":"INT64", "value":600, "unit":"gold"}, "sell_gold":{"type":"INT64", "value":240, "unit":"gold"}, "in_shop":{"type":"BOOL", "value":true}, "schema_version":{"type":"ENUM", "value":"V3_0"}, "payload_sha256":{"type":"HASH", "value":"82c9da2713112a93d3b163dfb058616fdc21fbcf6e2508197097661bcc9ae4d8"}}}
W2	{"equipment_id":{"type":"ENUM", "value":"W2"}, "quality":{"type":"ENUM", "value":"WHITE"}, "stats":{"type":"ARRAY", "element":{"StatGrant", "ordered":true, "value":[{"stat":{"type":"ENUM", "value":"MAX_HP_FLAT"}, "amount":{"type":"INT64", "value":220, "unit":"hp"}]}], "effect":{"type":{"type":"ENUM", "value":"NONE"}}, "buy_gold":{"type":"INT64", "value":600, "unit":"gold"}, "sell_gold":{"type":"INT64", "value":240, "unit":"gold"}, "in_shop":{"type":"BOOL", "value":true}, "schema_version":{"type":"ENUM", "value":"V3_0"}, "payload_sha256":{"type":"HASH", "value":"d941476c9d7b0a59d631af558f7e2e7894b58ad341c54f0ab143dc4b8de2f660"}}
W3	{"equipment_id":{"type":"ENUM", "value":"W3"}, "quality":{"type":"ENUM", "value":"WHITE"}, "stats":{"type":"ARRAY", "element":{"StatGrant", "ordered":true, "value":[{"stat":{"type":"ENUM", "value":"MOVE_SPEED_FLAT"}, "amount":{"type":"INT64", "value":18, "unit":"scalar"}]}], "effect":{"type":{"type":"ENUM", "value":"SHOE_ONLY"}}, "buy_gold":{"type":"INT64", "value":600, "unit":"gold"}, "sell_gold":{"type":"INT64", "value":240, "unit":"gold"}, "in_shop":{"type":"BOOL", "value":true}, "schema_version":{"type":"ENUM", "value":"V3_0"}, "payload_sha256":{"type":"HASH", "value":"187af42f838c78bb4d7a934fb6d39923cc609347c2a4a3c15b0044d03e6838b7"}}
W4	{"equipment_id":{"type":"ENUM", "value":"W4"}, "quality":{"type":"ENUM", "value":"WHITE"}, "stats":{"type":"ARRAY", "element":{"StatGrant", "ordered":true, "value":[{"stat":{"type":"ENUM", "value":"ATTACK_FLAT"}, "amount":{"type":"INT64", "value":10, "unit":"scalar"}}, {"stat":{"type":"ENUM", "value":"MAX_HP_FLAT"}, "amount":{"type":"INT64", "value":100, "unit":"hp"}]}], "effect":{"type":{"type":"ENUM", "value":"NONE"}}, "buy_gold":{"type":"INT64", "value":600, "unit":"gold"}, "sell_gold":{"type":"INT64", "value":240, "unit":"gold"}, "in_shop":{"type":"BOOL", "value":true}, "schema_version":{"type":"ENUM", "value":"V3_0"}, "payload_sha256":{"type":"HASH", "value":"83e16827f2fc5b8c16f472ffbc2c3795f11aa3ae5737a9a5d81be4c2a9db343f"}}
W5	{"equipment_id":{"type":"ENUM", "value":"W5"}, "quality":{"type":"ENUM", "value":"WHITE"}, "stats":{"type":"ARRAY", "element":{"StatGrant", "ordered":true, "value":[{"stat":{"type":"ENUM", "value":"MAX_HP_FLAT"}, "amount":{"type":"INT64", "value":110, "unit":"hp"}}, {"stat":{"type":"ENUM", "value":"MOVE_SPEED_FLAT"}, "amount":{"type":"INT64", "value":10, "unit":"scalar"}]}], "effect":{"type":{"type":"ENUM", "value":"NONE"}}, "buy_gold":{"type":"INT64", "value":600, "unit":"gold"}, "sell_gold":{"type":"INT64", "value":240, "unit":"gold"}, "in_shop":{"type":"BOOL", "value":true}, "schema_version":{"type":"ENUM", "value":"V3_0"}, "payload_sha256":{"type":"HASH", "value":"90bbf6a2e8d35c8c0da65909d8866d2e2dd810be14148cfdba7159aab88b8aee"}}
W6	{"equipment_id":{"type":"ENUM", "value":"W6"}, "quality":{"type":"ENUM", "value":"WHITE"}, "stats":{"type":"ARRAY", "element":{"StatGrant", "ordered":true, "value":[{"stat":{"type":"ENUM", "value":"ATTACK_FLAT"}, "amount":{"type":"INT64", "value":8, "unit":"scalar"}}, {"stat":{"type":"ENUM", "value":"MOVE_SPEED_FLAT"}, "amount":{"type":"INT64", "value":10, "unit":"scalar"}]}], "effect":{"type":{"type":"ENUM", "value":"NONE"}}, "buy_gold":{"type":"INT64", "value":600, "unit":"gold"}, "sell_gold":{"type":"I

	NT64","value":240,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"4d1ae835f3a6bd2ffc2125b5b60039bcfcfb65490226f3f1c39f10df9d47feef"}}}
B1	{"equipment_id":{"type":"ENUM","value":"B1"},"quality":{"type":"ENUM","value":"BLUE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"ATTACK_FLAT"},"amount":{"type":"INT64","value":20,"unit":"scalar"}]}]},"effect":{"type":{"type":"ENUM","value":"DAMAGE_MULTIPLIER"},"target_classes":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["CORE_BOSS","DRAGON","FLYING_MONSTER","LEADER","ORDINARY_MONSTER","PIG"]},"multiplier":{"type":"Q4","value_scaled_1e4":13000,"human":"1.3","unit":"scalar"},"buy_gold":{"type":"INT64","value":2000,"unit":"gold"},"sell_gold":{"type":"INT64","value":800,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"b1f9b98bf42a850c04b5e32fc7d883fef1ca4df2154ce873af3ca5caed219496"}}}
B2	{"equipment_id":{"type":"ENUM","value":"B2"},"quality":{"type":"ENUM","value":"BLUE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MAX_HP_FLAT"},"amount":{"type":"INT64","value":280,"unit":"hp"}]}]},"effect":{"type":{"type":"ENUM","value":"FIXED_DAMAGE_REDUCTION"},"eligible_damage_forms":{"type":"ENUM","value":"BASIC"},"amount":{"type":"INT64","value":8,"unit":"scalar"},"minimum_final_damage":{"type":"INT64","value":1,"unit":"hp"},"buy_gold":{"type":"INT64","value":2000,"unit":"gold"},"sell_gold":{"type":"INT64","value":800,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"7875b987c9fef5c5e0ea07d51f9bf8e0741fa5ec31a6042b284fb41bf717f6ae"}}}
B3	{"equipment_id":{"type":"ENUM","value":"B3"},"quality":{"type":"ENUM","value":"BLUE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MAX_HP_FLAT"},"amount":{"type":"INT64","value":200,"unit":"hp"}]}]},"effect":{"type":{"type":"ENUM","value":"PERCENT_DAMAGE_REDUCTION"},"eligible_damage_forms":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["DOT","SKILL"]},"remaining_multiplier":{"type":"Q4","value_scaled_1e4":8500,"human":"0.85","unit":"ratio"},"buy_gold":{"type":"INT64","value":2000,"unit":"gold"},"sell_gold":{"type":"INT64","value":800,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"0bca9776c7eb2cfb8d412dbed1277e6c4936978175adba9c454061e2f01a57e0"}}}
B4	{"equipment_id":{"type":"ENUM","value":"B4"},"quality":{"type":"ENUM","value":"BLUE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MAX_HP_FLAT"},"amount":{"type":"INT64","value":200,"unit":"hp"}]}]},"effect":{"type":{"type":"ENUM","value":"FORCED_MOVEMENT_DISTANCE_MULTIPLIER"},"eligible":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["KNOCKBACK","PULL"]},"multiplier":{"type":"Q4","value_scaled_1e4":5000,"human":"0.5","unit":"scalar"},"buy_gold":{"type":"INT64","value":2000,"unit":"gold"},"sell_gold":{"type":"INT64","value":800,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"4a40b314ac5c3a8ec69f2a107c64ae395b183876dd01ff067d21f350a200b288"}}}
B5	{"equipment_id":{"type":"ENUM","value":"B5"},"quality":{"type":"ENUM","value":"BLUE"},"stats":{"type":"ARRAY","element":

	"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MAX_HP_FLAT"},"amount":{"type":"INT64","value":150,"unit":"hp"}]}],{"effect":{"type":{"type":"ENUM","value":"REGEN"},"out_of_any_combat_s":{"type":"INT64","value":5,"unit":"s"},"hp_per_s":{"type":"INT64","value":8,"unit":"hp/s"},"tick_s":{"type":"INT64","value":1,"unit":"s"},"buy_gold":{"type":"INT64","value":2000,"unit":"gold"},"sell_gold":{"type":"INT64","value":800,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"aaa33c84b2438f8c19c5868826fcf27595017aa404cbf6e941c4921ad07484eb"}}}
B6	{"equipment_id":{"type":"ENUM","value":"B6"},"quality":{"type":"ENUM","value":"BLUE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"ATTACK_FLAT"},"amount":{"type":"INT64","value":12,"unit":"scalar"}]}],{"effect":{"type":{"type":"ENUM","value":"CRIT_DAMAGE_BONUS"},"add_percentage_points":{"type":"INT64","value":30,"unit":"scalar"},"buy_gold":{"type":"INT64","value":2000,"unit":"gold"},"sell_gold":{"type":"INT64","value":800,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"89efd894c4141a90fe556fd2e2bcb32d0d7d01473480b1c7947fcc04985bb432"}}}
B7	{"equipment_id":{"type":"ENUM","value":"B7"},"quality":{"type":"ENUM","value":"BLUE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MAX_HP_FLAT"},"amount":{"type":"INT64","value":200,"unit":"hp"}]}],{"effect":{"type":{"type":"ENUM","value":"CONDITIONAL_BONUS_DAMAGE_REDUCTION"},"bonus_sources":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["AMBUSH_B29","BACKSTAB_B26"]},"bonus_portion_multiplier":{"type":"Q4","value_scaled_1e4":5000,"human":"0.5","unit":"ratio"},"buy_gold":{"type":"INT64","value":2000,"unit":"gold"},"sell_gold":{"type":"INT64","value":800,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"b4c61080ca8013e8b95f875e06216d080f14ade80eab0d171286b4725fe9b102"}}}
B8	{"equipment_id":{"type":"ENUM","value":"B8"},"quality":{"type":"ENUM","value":"BLUE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MAX_HP_FLAT"},"amount":{"type":"INT64","value":120,"unit":"hp"}]}],{"effect":{"type":{"type":"ENUM","value":"ON_MONSTER_KILL_GOLD_CREATE"},"amount":{"type":"INT64","value":30,"unit":"scalar"},"eligible":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["CORE_BOSS","DRAGON","FLYING_MONSTER","LEADER","ORDINARY_MONSTER","PIG"]},"summon_kill":{"type":"BOOL","value":false},"buy_gold":{"type":"INT64","value":2000,"unit":"gold"},"sell_gold":{"type":"INT64","value":800,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"257b622230c0dd15eab7d31c642702af0c9c791e208c8a5cb60819f9250d9d0b"}}}
B9	{"equipment_id":{"type":"ENUM","value":"B9"},"quality":{"type":"ENUM","value":"BLUE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"ATTACK_FLAT"},"amount":{"type":"INT64","value":10,"unit":"scalar"}]}],{"effect":{"type":{"type":"ENUM","value":"DIRECT_VISION_RADIUS_MULTIPLIER"},"multiplier":{"type":"Q4","value_scaled_1e4":12000,"human":"1.2","unit":"scalar"},"does_not_ignore_walls_or_stealth":{"type":"BOOL","value":true},"buy_gold":{"type":"INT64","value":2000,"unit":"gold"},"sell_gold":{"type":"INT64","value":800,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"257b622230c0dd15eab7d31c642702af0c9c791e208c8a5cb60819f9250d9d0b"}}}

	e":{"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"314cc2607adcfd5cca8195557fa69b2013eb0a7bdd3cabc11f96c8293e43a4"}}
B10	{"equipment_id":{"type":"ENUM","value":"B10"},"quality":{"type":"ENUM","value":"BLUE"},"stats":{"type":"ARRAY","element":{"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"ATTACK_FLAT"},"amount":{"type":"INT64","value":12,"unit":"scalar"}]},"effect":{"type":{"type":"ENUM","value":"SUMMON_ATTACK_MULTIPLIER"},"multiplier":{"type":"Q4","value_scaled_1e4":13000,"human":"1.3","unit":"scalar"},"buy_gold":{"type":"INT64","value":2000,"unit":"gold"},"sell_gold":{"type":"INT64","value":800,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"15db782b63869359f4993c3d8c1399a73929cf7863867d40bb6d5ee5973ea9cd"}}
B11	{"equipment_id":{"type":"ENUM","value":"B11"},"quality":{"type":"ENUM","value":"BLUE"},"stats":{"type":"ARRAY","element":{"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MAX_HP_FLAT"},"amount":{"type":"INT64","value":200,"unit":"hp"}]},"effect":{"type":{"type":"ENUM","value":"STORM_DAMAGE_REDUCTION"},"remaining_multiplier":{"type":"Q4","value_scaled_1e4":7000,"human":"0.7","unit":"ratio"},"applies":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["APOCALYPSE_STORM","NORMAL_STORM"]},"buy_gold":{"type":"INT64","value":2000,"unit":"gold"},"sell_gold":{"type":"INT64","value":800,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"916de6515f9570376b8f5d4630b1ddb9df6d361134ab2821ff440fa3cb85c0e3"}}
B12	{"equipment_id":{"type":"ENUM","value":"B12"},"quality":{"type":"ENUM","value":"BLUE"},"stats":{"type":"ARRAY","element":{"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MOVE_SPEED_FLAT"},"amount":{"type":"INT64","value":26,"unit":"scalar"}]},"effect":{"type":{"type":"ENUM","value":"SHOE_ONLY"},"buy_gold":{"type":"INT64","value":2000,"unit":"gold"},"sell_gold":{"type":"INT64","value":800,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"7fd284a8206dbc2268d45dbb6af22e4f99e688a5960345fc0ee8de2898c79f81"}}
B13	{"equipment_id":{"type":"ENUM","value":"B13"},"quality":{"type":"ENUM","value":"BLUE"},"stats":{"type":"ARRAY","element":{"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MOVE_SPEED_FLAT"},"amount":{"type":"INT64","value":20,"unit":"scalar"}]},"effect":{"type":{"type":"ENUM","value":"SLOW_MAGNITUDE_MULTIPLIER"},"multiplier":{"type":"Q4","value_scaled_1e4":5000,"human":"0.5","unit":"scalar"},"is_shoe":{"type":"BOOL","value":true},"buy_gold":{"type":"INT64","value":2000,"unit":"gold"},"sell_gold":{"type":"INT64","value":800,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"f3a580715d5eea975a04610e2be7ac7a0b17ff276740e92e939987342952d5f8"}}
B14	{"equipment_id":{"type":"ENUM","value":"B14"},"quality":{"type":"ENUM","value":"BLUE"},"stats":{"type":"ARRAY","element":{"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MOVE_SPEED_FLAT"},"amount":{"type":"INT64","value":20,"unit":"scalar"}]},"effect":{"type":{"type":"ENUM","value":"OFFENSIVE_STEALTH_BREAK_MOVE_SPEED"},"is_shoe":{"

	<pre> type":"BOOL","value":true},"eligible_transition":{"type":"ENUM","value":"STEALTH_TO_VISIBLE"},"eligible_causes":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["OWNER_BASIC_RELEASE_AGAINST_HOSTILE","OWNER_HOSTILE_ACTIVE_CAST_COMMIT"]},"excluded_causes":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["ACTUAL_DAMAGE_TAKEN","REVEALED_BY_ENEMY","STEALTH_TIMEOUT","VOLUNTARY_CANCEL"]},"final_move_speed_multiplier":{"type":"Q4","value_scaled_1e4":13000,"human":"1.3","unit":"ratio"},"duration_s":{"type":"INT64","value":2,"unit":"s"},"internal_cooldown_s":{"type":"INT64","value":8,"unit":"s"},"once_per_stealth_episode":{"type":"BOOL","value":true},"stacking":{"type":"ENUM","value":"REFRESH_DURATION_NO_MAGNITUDE_STACK"},"eligible_hidden_sources":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["ACTIVE_STEALTH","EQUIPMENT_STEALTH","FUSHI_HIDDEN"]},"excluded_hidden_sources":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["FOG_OF_WAR_ONLY","SMOKE_OCCLUSION_ONLY"]},"hostile_active_definition":{"type":"ENUM","value":"ACTIVE_COMMIT_THAT_TARGETS_HOSTILE_OR_CREATES_PLAYER_OWNED_DAMAGE_CONTROL_TRAP_OR_HOSTILE_SUMMON"},"clear_speed_buff_on":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["ITEM_INSTANCE_DESTROYED","OWNER_TRUE_DEATH","UNEQUIP_B14"]},"internal_cooldown_on_unequip_or_true_death":{"type":"ENUM","value":"CLEAR_WITH_ITEM_EFFECT_STATE"},"buy_gold":{"type":"INT64","value":2000,"unit":"gold"},"sell_gold":{"type":"INT64","value":800,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"943760e707f184acece7498e70149dfd66a730477d7ba3f9a011f88561edaee8"}} </pre>
P1	<pre> {"equipment_id":{"type":"ENUM","value":"P1"},"quality":{"type":"ENUM","value":"PURPLE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"ATTACK_FLAT"},"amount":{"type":"INT64","value":35,"unit":"scalar"}]}]},"effect":{"type":{"type":"ENUM","value":"CONDITIONAL_DAMAGE_MULTIPLIER"},"condition":{"type":"ENUM","value":"TARGET_CURRENT_HP_PERCENT_LT_30"},"multiplier":{"type":"Q4","value_scaled_1e4":12500,"human":"1.25","unit":"scalar"},"buy_gold":{"type":"INT64","value":6000,"unit":"gold"},"sell_gold":{"type":"INT64","value":2400,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"78b9d836c8a6108deb7ef79426cb1a2c0dc5d66e7dc1ce84aeb16bdb780a9096"}} </pre>
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P4	<pre> {"equipment_id":{"type":"ENUM","value":"P4"},"quality":{"type":"ENUM","value":"PURPLE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MAX_HP_FLAT"},"amount":{"type":"INT64","value":400,"unit":"hp"}]}}, "effect":{"type":{"type":"ENUM","value":"ON_ENEMY_PLAYER_TRUE_DEATH_HEAL"},"max_hp_percent":{"type":"INT64","value":30,"unit":"pct"}, "overheal_discarded":{"type":"BOOL","value":true},"buy_gold":{"type":"INT64","value":6000,"unit":"gold"},"sell_gold":{"type":"INT64","value":2400,"unit":"gold"}, "in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"abc25423c2fa0b9b9c6fa9840900a8d03e870f1f28eee33e00b14af0ccf88376"}} </pre>
P5	<pre> {"equipment_id":{"type":"ENUM","value":"P5"},"quality":{"type":"ENUM","value":"PURPLE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MAX_HP_FLAT"},"amount":{"type":"INT64","value":500,"unit":"hp"}]}}, "effect":{"type":{"type":"ENUM","value":"REGEN"},"out_of_combat_required":{"type":"BOOL","value":false},"hp_per_s":{"type":"INT64","value":8,"unit":"hp/s"}, "tick_s":{"type":"INT64","value":1,"unit":"s"},"buy_gold":{"type":"INT64","value":6000,"unit":"gold"},"sell_gold":{"type":"INT64","value":2400,"unit":"gold"}, "in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"435729da1b284665bc75d28d517a83f8ad5436c9e5ff1e26eb24665a0b5b77fd"}} </pre>
P6	<pre> {"equipment_id":{"type":"ENUM","value":"P6"},"quality":{"type":"ENUM","value":"PURPLE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MAX_HP_FLAT"},"amount":{"type":"INT64","value":200,"unit":"hp"}]}}, "effect":{"type":{"type":"ENUM","value":"ACTIVE_COOLDOWN_RATE"},"remaining_time_multiplier_on_equip":{"type":"Q4","value_scaled_1e4":8000,"human":"0.8","unit":"ratio"}, "cooldown_duration_multiplier":{"type":"Q4","value_scaled_1e4":8000,"human":"0.8","unit":"ratio"},"preserve_remaining_percent":{"type":"BOOL","value":true}}, "buy_gold":{"type":"INT64","value":6000,"unit":"gold"},"sell_gold":{"type":"INT64","value":2400,"unit":"gold"},"in_shop":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"799066c6e35ec9f2f905471c46a759df27f5cd66229008bec60bf9e45c465265"}} </pre>
P7	<pre> {"equipment_id":{"type":"ENUM","value":"P7"},"quality":{"type":"ENUM","value":"PURPLE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"ATTACK_FLAT"},"amount":{"type":"INT64","value":18,"unit":"scalar"}]}}, "effect":{"type":{"type":"ENUM","value":"SUMMON_HP_AND_ATTACK_MULTIPLIER"},"multiplier":{"type":"Q4","value_scaled_1e4":16000,"human":"1.6","unit":"scalar"}}, "buy_gold":{"type":"INT64","value":6000,"unit":"gold"},"sell_gold":{"type":"INT64","value":2400,"unit":"gold"},"in_shop":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"c30730be8e3ddbffd2c1020cd3464ee19dae1409f0fba46420fde1752459f773"}} </pre>

P8	{ "equipment_id":{ "type": "ENUM", "value": "P8", "quality":{ "type": "ENUM", "value": "PURPLE", "stats":{ "type": "ENUM", "value": "NONE", "effect":{ "type":{ "type": "ENUM", "value": "EXTRA_HAND_SLOT", "amount":{ "type": "INT64", "value": 1, "unit": "scalar" }, "buy_gold":{ "type": "INT64", "value": 6000, "unit": "gold", "sell_gold":{ "type": "INT64", "value": 2400, "unit": "gold", "in_shop":{ "type": "BOOL", "value": true, "schema_version":{ "type": "ENUM", "value": "V3_0", "payload_sha256":{ "type": "HASH", "value": "082078611dfe478bd3fd7eac77fb3858c97a01bb2cf57b81493085a23b45ca4c" } } } } } } }
P9	{ "equipment_id":{ "type": "ENUM", "value": "P9", "quality":{ "type": "ENUM", "value": "PURPLE", "stats":{ "type": "ARRAY", "element": "StatGrant", "ordered": true, "value":{ "stat":{ "type": "ENUM", "value": "MAX_HP_FLAT", "amount":{ "type": "INT64", "value": 250, "unit": "hp" } } }, "effect":{ "type":{ "type": "ENUM", "value": "TRUE_SIGHT", "radius_m":{ "type": "INT64", "value": 20, "unit": "m", "walls_block":{ "type": "BOOL", "value": true, "shared_with_others":{ "type": "BOOL", "value": false }, "buy_gold":{ "type": "INT64", "value": 6000, "unit": "gold", "sell_gold":{ "type": "INT64", "value": 2400, "unit": "gold", "in_shop":{ "type": "BOOL", "value": true, "schema_version":{ "type": "ENUM", "value": "V3_0", "payload_sha256":{ "type": "HASH", "value": "d797d8df53658455e64dd6eed96ce2ec15b2b51ff83edaf088d695cbf420e58d" } } } } } }
P10	{ "equipment_id":{ "type": "ENUM", "value": "P10", "quality":{ "type": "ENUM", "value": "PURPLE", "stats":{ "type": "ARRAY", "element": "StatGrant", "ordered": true, "value":{ "stat":{ "type": "ENUM", "value": "MAX_HP_FLAT", "amount":{ "type": "INT64", "value": 150, "unit": "hp" } } }, "effect":{ "type":{ "type": "ENUM", "value": "BLACK_MOUNTAIN_OUTCOME_OVERRIDE", "probability_authority":{ "type": "REF", "value": " § 12.5 及 § E.18 黑山规则", "rule":{ "type": "SET", "element": "ENUM", "unique": true, "ordered": false, "value": ["BIG_WIN_DOUBLES_FROM_BASE", "FLAT_UNCHANGED", "INCREASE_DEDUCTED_FROM_LOSS"], "service_active_condition":{ "type": "ENUM", "value": "HEISHAN_SERVICE_OPEN_AT_EFFECT_EVALUATION_TICK", "after_service_close":{ "type": "SET", "element": "ENUM", "unique": true, "ordered": false, "value": ["EXISTING_ITEM_RETAINS_MAX_HP_150_AND_CAN_BE_WORN_HELD_DROPPED_OR_SOLD", "SPECIAL_EFFECT_DORMANT"] }, "new_random_generation_eligibility":{ "type": "ENUM", "value": "HEISHAN_SERVICE_OPEN_AT_GENERATION_COMMIT_TICK", "time_authority":{ "type": "REF", "value": " § 12.3 及 § E.17 商店规则 =HEISHAN].windows_s_half_open" }, "buy_gold":{ "type": "INT64", "value": 6000, "unit": "gold", "sell_gold":{ "type": "INT64", "value": 2400, "unit": "gold", "in_shop":{ "type": "BOOL", "value": true, "schema_version":{ "type": "ENUM", "value": "V3_0", "payload_sha256":{ "type": "HASH", "value": "3aa1db1cdf4a2261fd6494a82470ecabcf9184047b64138c1339be86fb11a" } } } } } }
P11	{ "equipment_id":{ "type": "ENUM", "value": "P11", "quality":{ "type": "ENUM", "value": "PURPLE", "stats":{ "type": "ARRAY", "element": "StatGrant", "ordered": true, "value":{ "stat":{ "type": "ENUM", "value": "ATTACK_FLAT", "amount":{ "type": "INT64", "value": 25, "unit": "scalar" } } }, "effect":{ "type":{ "type": "ENUM", "value": "ON_ENEMY_PLAYER_TRUE_DEATH_ACTIVE_READY", "ready":{ "type": "BOOL", "value": true, "kill_during_committed_cast":{ "type": "ENUM", "value": "CAST_CONTINUES", "post_cast_cool_down":{ "type": "ENUM", "value": "READY", "kill_during_cooldown":{ "type": "ENUM", "value": "READY", "buy_gold":{ "type": "INT64", "value": 6000, "unit": "gold", "sell_gold":{ "type": "INT64", "value": 2400, "unit": "gold", "in_shop":{ "type": "BOOL", "value": true, "schema_version":{ "type": "ENUM", "value": "V3_0", "payload_sha256":{ "type": "HASH", "value": "758b414250d7e6bfedd0f821067d254da5e352e5c407fd87501d546227b01d95" } } } } } } }

P12	{ "equipment_id":{ "type": "ENUM", "value": "P12"}, "quality":{ "type": "ENUM", "value": "PURPLE"}, "stats":{ "type": "ARRAY", "element": "StatGrant", "ordered": true, "value": { "stat":{ "type": "ENUM", "value": "MAX_HP_FLAT"}, "amount":{ "type": "INT64", "value": 400, "unit": "hp"} }}, "effect":{ "type":{ "type": "ENUM", "value": "REFLECT"}, "eligible_damage_forms":{ "type": "ENUM", "value": "BASIC"}, "actual_post_mitigation_damage_percent":{ "type": "INT64", "value": 20, "unit": "pct"}, "cannot_trigger_reflect":{ "type": "BOOL", "value": true}}, "buy_gold":{ "type": "INT64", "value": 6000, "unit": "gold"}, "sell_gold":{ "type": "INT64", "value": 2400, "unit": "gold"}, "in_shop":{ "type": "BOOL", "value": true}, "schema_version":{ "type": "ENUM", "value": "V3_0"}, "payload_sha256":{ "type": "HASH", "value": "363f8964a05bb55ff5390f86f105f3d4a2792b45cc984a0221df2dff770e1ef8"} }
P13	{ "equipment_id":{ "type": "ENUM", "value": "P13"}, "quality":{ "type": "ENUM", "value": "PURPLE"}, "stats":{ "type": "ARRAY", "element": "StatGrant", "ordered": true, "value": { "stat":{ "type": "ENUM", "value": "MOVE_SPEED_FLAT"}, "amount":{ "type": "INT64", "value": 25, "unit": "scalar"} }}, "effect":{ "type":{ "type": "ENUM", "value": "STATIONARY_STEALTH"}, "stationary_required_s":{ "type": "INT64", "value": 2, "unit": "s"}, "breaks_on":{ "type": "SET", "element": "ENUM", "unique": true, "ordered": false, "value": { "ACTUAL_DAMAGE_TAKEN", "ATTACK", "CAST", "MOVE_OVER_0_1M"} }, "walls_normal":{ "type": "BOOL", "value": true}, "forced_displacement_over_m":{ "type": "ENUM", "value": "0.1_BREAKS"}, "camera_or_facing_rotation":{ "type": "ENUM", "value": "DOES_NOT_BREAK"}, "position_noise_under_or_equal_m":{ "type": "ENUM", "value": "0.1_DOES_NOT_BREAK"}, "buy_gold":{ "type": "INT64", "value": 6000, "unit": "gold"}, "sell_gold":{ "type": "INT64", "value": 2400, "unit": "gold"}, "in_shop":{ "type": "BOOL", "value": true}, "schema_version":{ "type": "ENUM", "value": "V3_0"}, "payload_sha256":{ "type": "HASH", "value": "b25ca1e69c748f6ccf763c05596607af633c2d9429e99f6ab7a3765f8056ff5a"} }
P14	{ "equipment_id":{ "type": "ENUM", "value": "P14"}, "quality":{ "type": "ENUM", "value": "PURPLE"}, "stats":{ "type": "ARRAY", "element": "StatGrant", "ordered": true, "value": { "stat":{ "type": "ENUM", "value": "ATTACK_FLAT"}, "amount":{ "type": "INT64", "value": 25, "unit": "scalar"} }}, "effect":{ "type":{ "type": "ENUM", "value": "BASIC_LIFESTEAL"}, "actual_hp_damage_percent":{ "type": "INT64", "value": 10, "unit": "pct"}, "does_not_heal_from_shield_damage":{ "type": "BOOL", "value": true}}, "buy_gold":{ "type": "INT64", "value": 6000, "unit": "gold"}, "sell_gold":{ "type": "INT64", "value": 2400, "unit": "gold"}, "in_shop":{ "type": "BOOL", "value": true}, "schema_version":{ "type": "ENUM", "value": "V3_0"}, "payload_sha256":{ "type": "HASH", "value": "a9afe1973bf3f598f11c042380cfdaf20067a9417af1ad828b19c7401daf4b85"} }
P15	{ "equipment_id":{ "type": "ENUM", "value": "P15"}, "quality":{ "type": "ENUM", "value": "PURPLE"}, "stats":{ "type": "ARRAY", "element": "StatGrant", "ordered": true, "value": { "stat":{ "type": "ENUM", "value": "MOVE_SPEED_FLAT"}, "amount":{ "type": "INT64", "value": 30, "unit": "scalar"} }}, "effect":{ "type":{ "type": "ENUM", "value": "SLOW_IMMUNITY"}, "is_shoe":{ "type": "BOOL", "value": true}}, "buy_gold":{ "type": "INT64", "value": 6000, "unit": "gold"}, "sell_gold":{ "type": "INT64", "value": 2400, "unit": "gold"}, "in_shop":{ "type": "BOOL", "value": true}, "schema_version":{ "type": "ENUM", "value": "V3_0"}, "payload_sha256":{ "type": "HASH", "value": "2e24819bd4e9ca5b63e6f75af637d2edd18d2c1704c68b67782e7e3212bdf502"} }
P16	{ "equipment_id":{ "type": "ENUM", "value": "P16"}, "quality":{ "type": "ENUM", "value": "PURPLE"}, "stats":{ "type": "ARRAY", "element": "StatGrant", "ordered": true, "value": { "stat":{ "type": "ENUM", "value": "MOVE_SPEED_FLAT"}, "amount":{ "type": "INT64", "value": 24, "unit": "scalar"} }}, "effect":{ "type":{ "type": "ENUM", "value": "BASIC_SAME_TARGET_ATTACK_SPEED_STACK"}, "is_sho

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P17	<pre>{"equipment_id":{"type":"ENUM","value":"P17"},"quality":{"type":"ENUM","value":"PURPLE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MOVE_SPEED_FLAT"},"amount":{"type":"INT64","value":22,"unit":"scalar"}]},"effect":{"type":{"type":"ENUM","value":"BASIC_ATTACK_RANGE_FLAT"},"is_shoe":{"type":"BOOL","value":true},"meters":{"type":"Q4","value_scaled_1e4":7000,"human":"0.7","unit":"scalar"},"owner_basic_only":{"type":"BOOL","value":true},"active_and_summon_unchanged":{"type":"BOOL","value":true},"direct_visibility_and_release_revalidation_still_required":{"type":"BOOL","value":true},"applies_at":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["BASIC_RELEASE_RANGE_REVALIDATION","BASIC_TARGET_ACQUIRE"]},"does_not_extend_projectile_lifetime_or_homing_distance":{"type":"BOOL","value":true},"buy_gold":{"type":"INT64","value":6000,"unit":"gold"},"sell_gold":{"type":"INT64","value":2400,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"dee67c4503691ef2a28bd4d63f8de364291c880ea2c733d437dbe3dd5d8e6007"}}</pre>
P18	<pre>{"equipment_id":{"type":"ENUM","value":"P18"},"quality":{"type":"ENUM","value":"PURPLE"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MOVE_SPEED_FLAT"},"amount":{"type":"INT64","value":24,"unit":"scalar"}]},"effect":{"type":{"type":"ENUM","value":"HARD_CONTROL_AND_FORCED_MOVEMENT_RESIST"},"is_shoe":{"type":"BOOL","value":true},"forced_movement_distance_multiplier":{"type":"Q4","value_scaled_1e4":3000,"human":"0.3","unit":"ratio"},"eligible_forced_movement":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["KNOCKBACK","KNOCKUP","PULL","THROW"]},"hard_control_duration_multiplier":{"type":"Q4","value_scaled_1e4":3000,"human":"0.3","unit":"ratio"},"eligible_hard_control":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["FEAR","FREEZE","KNOCKUP","PETRIFY","ROOT","STUN","TRANSFORM"]},"excluded_soft_control":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["SILENCE","SLOW"]},"hard_control_minimum_duration_s":{"type":"Q4","value_scaled_1e4":3000,"human":"0.3","unit":"s"},"same_category_resistance_stacking":{"type":"ENUM","value":"T</pre>

	AKE_STRONGEST_LOWEST_MULTIPLIER"},"duration_order":{"type":"FORMULA","src":"MAX(0.3,(BASE_DURATION + SOURCE_FLAT_EXTENSION) * TARGET_STRONGEST_MULTIPLIER)","ast":{"op":"MAX","args":[{"q4":3000},{op:"MUL","args":[{"op":"ADD","args":[{"var":"BASE_DURATION"},{"var":"SOURCE_FLAT_EXTENSION"}]},{var":"TARGET_STRONGEST_MULTIPLIER"}]}]},"minimum_applies_only_if_pre_resistance_duration_gt_zero":{"type":"BOOL","value":true},"immunity_or_cleanses_zero_stays_zero":{"type":"BOOL","value":true},"forced_movement_minimum_distance_m":{"type":"INT64","value":0,"unit":"m"},"airborne_control_uses":{"type":"ENUM","value":"KNOCKUP_CONTROL_DURATION_AND_KNOCKUP_FORCED_MOVEMENT_DISTANCE_BOTH_RESOLVE_ONCE"},"clear_state_on_unequip":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["NO_PERSISTENT_STATE","REEVALUATE_ONLY_FUTURE_CONTROL_APPLICATIONS"]},"buy_gold":{"type":"INT64","value":6000,"unit":"gold"},"sell_gold":{"type":"INT64","value":2400,"unit":"gold"},"in_shop":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"d43c2cac747ed5249d48a629b81a4c59188dd601f66bbda538596f62be9cf1c7"}}
G1	{"equipment_id":{"type":"ENUM","value":"G1"},"quality":{"type":"ENUM","value":"GOLD"},"stats":{"type":"ENUM","value":"NONE"},"effect":{"type":{"type":"ENUM","value":"LETHAL_PROTECTION_CONSUMABLE"},"priority_after":{"type":"ENUM","value":"B20"},"hp_percent":{"type":"INT64","value":100,"unit":"pct"},"invulnerable_s":{"type":"INT64","value":2,"unit":"s"},"consume_item":{"type":"BOOL","value":true},"consume_life":{"type":"BOOL","value":false},"buy_gold":{"type":"ENUM","value":"NOT_SOLD"},"sell_gold":{"type":"INT64","value":4000,"unit":"gold"},"in_shop":{"type":"BOOL","value":false},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"fee108cd01cef218b9c8731b7bd0734493ed239cbc5cc06f11f39a1a57f447c8"}}
G2	{"equipment_id":{"type":"ENUM","value":"G2"},"quality":{"type":"ENUM","value":"GOLD"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"ATTACK_FLAT"},"amount":{"type":"INT64","value":25,"unit":"scalar"}]}],"effect":{"type":{"type":"ENUM","value":"ELEMENT_COUNTER_MULTIPLIER_OVERRIDE"},"normal":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"scalar"},"owner_countersing_target":{"type":"INT64","value":2,"unit":"scalar"},"incoming_unchanged":{"type":"BOOL","value":true},"buy_gold":{"type":"ENUM","value":"NOT_SOLD"},"sell_gold":{"type":"INT64","value":4000,"unit":"gold"},"in_shop":{"type":"BOOL","value":false},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"9c563a05c412e7f15381cf5c6a1a44c1e66c2bf57ef8910ec6c9281047271dbf"}}
G3	{"equipment_id":{"type":"ENUM","value":"G3"},"quality":{"type":"ENUM","value":"GOLD"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MAX_HP_FLAT"},"amount":{"type":"INT64","value":350,"unit":"hp"}]}],"effect":{"type":{"type":"ENUM","value":"STORM_DAMAGE_REDUCTION"},"remaining_multiplier":{"type":"Q4","value_scaled_1e4":3000,"human":"0.3","unit":"ratio"},"applies":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["APOCALYPSE_STORM","NORMAL_STORM"]},"buy_gold":{"type":"ENUM","value":"NOT_SOLD"},"sell_gold":{"type":"INT64","value":4000,"unit":"gold"},"in_shop":{"type":"BOOL","value":false},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"ca0a1db183d76ef2c58d169543d91b2850fa682e027543c554b1d60e"}}

	131f01db"}}}
G4	{ "equipment_id":{"type":"ENUM","value":"G4"},"quality":{"type":"ENUM","value":"GOLD"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MOVE_SPEED_FLAT"},"amount":{"type":"INT64","value":40,"unit":"scalar"}]}]},"effect":{"type":{"type":"ENUM","value":"FLIGHT_OUT_OF_COMBAT"},"out_of_any_combat_s":{"type":"INT64","value":5,"unit":"s"},"capsule_bottom_height_m":{"type":"INT64","value":3,"unit":"m"},"land_on_any_actual_damage_dealt_or_received":{"type":"BOOL","value":true},"is_shoe":{"type":"BOOL","value":true},"world_interaction":{"type":"ENUM","value":"AUTO_LAND_BEFORE_PICKUP_SHOP_CHEST_GAMBLE_TAIBAI"},"combat_active_cast_does_not_auto_land_unless_damage_dealt":{"type":"BOOL","value":true},"no_legal_landing_within_6m":{"type":"ENUM","value":"INTERACTION_FAILS_KEEP_FLYING_NO_COST_NO_UI"},"buy_gold":{"type":"ENUM","value":"NOT_SOLD"},"sell_gold":{"type":"INT64","value":4000,"unit":"gold"},"in_shop":{"type":"BOOL","value":false},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"e2f93b2683338fcae1ecd7c47caf4799a6e70e97a754f4f778d67e53d96bdc77"}} }
G5	{ "equipment_id":{"type":"ENUM","value":"G5"},"quality":{"type":"ENUM","value":"GOLD"},"stats":{"type":"ENUM","value":"NONE"},"effect":{"type":{"type":"ENUM","value":"ALL_GOLD_INCOME_MULTIPLIER"},"multiplier":{"type":"Q4","value_scaled_1e4":15000,"human":1.5,"unit":"scalar"},"eligible_ledger_classes":{"type":"ENUM","value":"G5_EXACT_LEDGER_WHITE_LIST"},"bonus_ledger_class":{"type":"ENUM","value":"GOLD_CREATE_G5_BONUS"},"does_not_reapply_to_own_bonus":{"type":"BOOL","value":true},"excludes":{"type":"SET","element":{"type":"ENUM","unique":true,"ordered":false,"value":["BLACK_MOUNTAIN_GROSS_RETURN","PLAYER_KILL_WALLET_BONUS","REFUND","ROLLBACK"]},"black_mountain_only_positive_net_profit_eligible_once":{"type":"BOOL","value":true},"buy_gold":{"type":"ENUM","value":"NOT_SOLD"},"sell_gold":{"type":"INT64","value":4000,"unit":"gold"},"in_shop":{"type":"BOOL","value":false},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"dc1601f473d6d32265e09ecef4f493e7eef1e59311f4aac5902c6cb2360a0b3"}} }
G6	{ "equipment_id":{"type":"ENUM","value":"G6"},"quality":{"type":"ENUM","value":"GOLD"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"ATTACK_FLAT"},"amount":{"type":"INT64","value":60,"unit":"scalar"}]}]},"effect":{"type":{"type":"ENUM","value":"CONDITIONAL_DAMAGE_MULTIPLIER"},"condition":{"type":"ENUM","value":"TARGET_CURRENT_HP_GREATER_THAN_OWNER_CURRENT_HP_AT_DAMAGE_SNAPSHOT"},"multiplier":{"type":"Q4","value_scaled_1e4":12500,"human":1.25,"unit":"scalar"},"buy_gold":{"type":"ENUM","value":"NOT_SOLD"},"sell_gold":{"type":"INT64","value":4000,"unit":"gold"},"in_shop":{"type":"BOOL","value":false},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"0e5bc2a3d8ca03ba190454a22cf1c1de6a7ab60470918139c15bd8937560ce75"}} }
G7	{ "equipment_id":{"type":"ENUM","value":"G7"},"quality":{"type":"ENUM","value":"GOLD"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"MAX_HP_FLAT"},"amount":{"type":"INT64","value":300,"unit":"hp"}]}]},"effect":{"type":{"type":"ENUM","value":"GLOBAL_ACTIVE_CAST_PING"},"enemy_active_cast_only":{"type":" }

	<pre> BOOL","value":true},"minimap_exact_cast_position_snapshot_s":{"type":"INT64","value":1,"unit":"s"},"does_not_grant_vision_or_lock":{"type":"BOOL","value":true},"coalesce_window_s":{"type":"Q4","value_scaled_1e4":2500,"human":"0.25","unit":"s"},"coalesce_spatial_cell_m":{"type":"INT64","value":10,"unit":"m"},"max_simultaneous_markers":{"type":"INT64","value":8,"unit":"scalar"},"overflow_policy":{"type":"ENUM","value":"KEEP_MOST_RECENT_THEN_CAST_EVENT_ID_ASC"},"buy_gold":{"type":"ENUM","value":"NOT_SOLD"},"sell_gold":{"type":"INT64","value":4000,"unit":"gold"},"in_shop":{"type":"BOOL","value":false},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"f936807cf0138a70c9c4dcd087512015ee86072ffbcfc446e4d978cf8e7a4757"}} </pre>
G8	<pre> {"equipment_id":{"type":"ENUM","value":"G8"},"quality":{"type":"ENUM","value":"GOLD"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"ATTACK_FLAT"},"amount":{"type":"INT64","value":20,"unit":"scalar"}},{"stat":{"type":"ENUM","value":"ATTACK_SPEED_PERCENT"},"amount":{"type":"INT64","value":40,"unit":"percent"}]}]},"effect":{"type":{"type":"ENUM","value":"STAT_ONLY"},"buy_gold":{"type":"ENUM","value":"NOT_SOLD"},"sell_gold":{"type":"INT64","value":4000,"unit":"gold"},"in_shop":{"type":"BOOL","value":false},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"72e9bb190362fb725538b763d4e0df028b6e4f57b0da3f8a597b0d52a6495e34"}} </pre>
G9	<pre> {"equipment_id":{"type":"ENUM","value":"G9"},"quality":{"type":"ENUM","value":"GOLD"},"stats":{"type":"ENUM","value":"NONE"},"effect":{"type":{"type":"ENUM","value":"PERMANENT_ATTACK_ON_KILL"},"ordinary_amount":{"type":"INT64","value":2,"unit":"scalar"},"enemy_player_true_death_amount":{"type":"INT64","value":10,"unit":"scalar"},"eligible":{"type":"SET","element":"ENUM","unique":true,"ordered":false,"value":["CORE_BOSS","DRAGON","ENEMY_PLAYER_TRUE_DEATH","FLYING_MONSTER","LEADER","ORDINARY_MONSTER","PIG"]},"summon_kill":{"type":"BOOL","value":false},"no_cap":{"type":"BOOL","value":true},"buy_gold":{"type":"ENUM","value":"NOT_SOLD"},"sell_gold":{"type":"INT64","value":4000,"unit":"gold"},"in_shop":{"type":"BOOL","value":false},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"00ecd009f2b0d921112e831d05407c5af41e835c7473897a48b305590bb48653"}} </pre>
G10	<pre> {"equipment_id":{"type":"ENUM","value":"G10"},"quality":{"type":"ENUM","value":"GOLD"},"stats":{"type":"ARRAY","element":"StatGrant","ordered":true,"value":[{"stat":{"type":"ENUM","value":"ATTACK_FLAT"},"amount":{"type":"INT64","value":80,"unit":"scalar"}]}]},"effect":{"type":{"type":"ENUM","value":"BASIC_ATTACK_RANGE_FLAT"},"meters":{"type":"INT64","value":3,"unit":"scalar"},"buy_gold":{"type":"ENUM","value":"NOT_SOLD"},"sell_gold":{"type":"INT64","value":4000,"unit":"gold"},"in_shop":{"type":"BOOL","value":false},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"e0d035ebfc9b61ad04e75fbf0117ade61ac87f7dfaffe097ac8f6c65d3c5991a"}} </pre>

E.33 怪物技能强类型运行时载荷

工程字段	唯一值 / 执行语义
MonsterSkillShape. closed_enum	CIRCLE / CONE / LINE / LINE_CAPSULE / THREE_PARALLEL_LINES / THREE_RADIAL_LANES_WITH_ONE_GAP / SINGLE_TARGET_CHAIN / NONE

MonsterSkillShape. CIRCLE.required	radius_m
MonsterSkillShape. CONE.required	range_m / angle_deg
MonsterSkillShape. LINE.required	length_m / width_m
MonsterSkillShape. LINE_CAPSULE.req uired	length_m / width_m / height_m
MonsterSkillShape. THREE_PARALLEL_ LINES.required	length_m / width_m / gap_m
MonsterSkillShape. THREE_RADIAL_LA NES_WITH_ONE_G AP.required	length_m / width_m / lane_count=3 / gap_lane_random_stream=MONSTER_AI
MonsterSkillShape. SINGLE_TARGET_C HAIN.required	jumps / jump_range_m / decay_percent_per_jump
monster_skill.cd_p hase_multiplier.LE ADER	1 / 1 / 0.95 / 0.90 (0/5/10/15 分钟阶段；Q4 存储)
monster_skill.cd_p hase_multiplier.DR AGON_MINIBOSS	1 / 1 / 0.90 / 0.85
monster_skill.cd_p hase_multiplier.CO RE_BOSS	1 / 1 / 0.90 / 0.85
monster_skill.fixed _damage_phase_s caling	技能固定伤害×该原型攻击阶段倍率（11.2 表攻击阶段倍率列）；四舍五入禁止，FLOOR 落地
monster_basic.DR	0.45 / 0.25

AGON.windup_recover_s	
monster_basic.CORE_BOSS.windup_recover_s	0.60 / 0.35
monster_basic.runtime_authority	11.2 机械原型两表（数值/物理与形态列）为怪物普攻唯一权威；本表仅补充龙王与核心前后摇
MonsterSkillShape.NONE.required	无几何字段（召唤/状态类技能）
CORE_V6.body_damage_spec	已删除；本体施法不产生伤害事件，普攻伤害规格由 E.34 CORE_MIRROR 载荷唯一声明
monster_skill_id	runtime_payload_json
PIG_CHARGE	<pre>{ "monster_skill_id": {"type": "ENUM", "value": "PIG_CHARGE"}, "user_archetype": {"type": "ENUM", "value": "五行肥猪"}, "cooldown_s": {"type": "INT64", "value": 6, "unit": "s"}, "windup_s": {"type": "Q4", "value_scaled_1e4": 6000, "human": "0.6", "unit": "s"}, "shape": {"type": "ENUM", "value": "LINE"}, "length_m": {"type": "INT64", "value": 3, "unit": "m"}, "width_m": {"type": "Q4", "value_scaled_1e4": 16000, "human": "1.6", "unit": "m"}, "damage_events": {"type": "ARRAY", "element": "DamageEventSpec", "ordered": true, "value": [{"event_id": {"type": "ENUM", "value": "MAIN"}, "damage": {"type": "INT64", "value": 80, "unit": "hp"}, "damage_form": {"type": "ENUM", "value": "SKILL"}, "element_rule": {"type": "ENUM", "value": "BIDIRECTIONAL_FIVE_ELEMENT"}, "hit_kind": {"type": "ENUM", "value": "ABILITY"}, "source_faction": {"type": "ENUM", "value": "MONSTER"}, "owner_credit": {"type": "ENUM", "value": "MONSTER"}, "fixed_damage_phase_scaling": {"type": "BOOL", "value": true}]}, "knockback_m": {"type": "INT64", "value": 2, "unit": "m"}, "schema_version": {"type": "ENUM", "value": "V3_0"}, "payload_sha256": {"type": "HASH", "value": "f049570bf125a1c02fd4fbaf214c26606f6324e6e436bb93c5ff875748d6981e"}} }</pre>
LEADER_SMASH	<pre>{ "monster_skill_id": {"type": "ENUM", "value": "LEADER_SMASH"}, "user_archetype": {"type": "ENUM", "value": "肉盾首领"}, "cooldown_s": {"type": "INT64", "value": 8, "unit": "s"}, "windup_s": {"type": "INT64", "value": 1, "unit": "s"}, "shape": {"type": "ENUM", "value": "CIRCLE"}, "radius_m": {"type": "INT64", "value": 5, "unit": "m"}, "damage_events": {"type": "ARRAY", "element": "DamageEventSpec", "ordered": true, "value": [{"event_id": {"type": "ENUM", "value": "MAIN"}, "damage": {"type": "INT64", "value": 150, "unit": "hp"}, "damage_form": {"type": "ENUM", "value": "SKILL"}, "element_rule": {"type": "ENUM", "value": "NONE"}, "hit_kind": {"type": "ENUM", "value": "ABILITY"}, "source_faction": {"type": "ENUM", "value": "MONSTER"}, "owner_credit": {"type": "ENUM", "value": "MONSTER"}, "fixed_damage_phase_scaling": {"type": "BOOL", "value": true}]}, "slow_percent": {"type": "INT64", "value": 30, "unit": "pct"}, "slow }</pre>

	<pre>_s":{"type":"INT64","value":2,"unit":"s"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"461d20a71cc3340cc020385cd6e55ec6a84a44079c49b1131ca69f3d6758699e"}}</pre>
LEADER_CONE	<pre>{"monster_skill_id":{"type":"ENUM","value":"LEADER_CONE"},"user_archetype":{"type":"ENUM","value":"输出首领"}, "cooldown_s":{"type":"INT64","value":6,"unit":"s"},"windup_s":{"type":"INT64","value":1,"unit":"s"},"shape":{"type":"ENUM","value":"CONE"},"range_m":{"type":"INT64","value":8,"unit":"m"},"angle_deg":{"type":"INT64","value":60,"unit":"deg"},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"MAIN"},"damage":{"type":"INT64","value":200,"unit":"hp"},"damage_form":{"type":"ENUM","value":"SKILL"},"element_rule":{"type":"ENUM","value":"NONE"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"source_faction":{"type":"ENUM","value":"MONSTER"},"owner_credit":{"type":"ENUM","value":"MONSTER"},"fixed_damage_phase_scaling":{"type":"BOOL","value":true}]}},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"a37e9459bde87990f0acc3708f6bbfb5cdb535d58f7a37a8c2deae4c5ba6fcd0"}}</pre>
DRAGON_BREATH	<pre>{"monster_skill_id":{"type":"ENUM","value":"DRAGON_BREATH"},"user_archetype":{"type":"ENUM","value":"五行龙王"}, "cooldown_s":{"type":"INT64","value":7,"unit":"s"},"windup_s":{"type":"Q4","value_scaled_1e4":12000,"human":"1.2","unit":"s"},"shape":{"type":"ENUM","value":"CONE"},"range_m":{"type":"INT64","value":10,"unit":"m"},"angle_deg":{"type":"INT64","value":70,"unit":"deg"},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"MAIN"},"damage":{"type":"INT64","value":280,"unit":"hp"},"damage_form":{"type":"ENUM","value":"SKILL"},"element_rule":{"type":"ENUM","value":"BIDIRECTIONAL_FIVE_ELEMENT"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"source_faction":{"type":"ENUM","value":"MONSTER"},"owner_credit":{"type":"ENUM","value":"MONSTER"},"fixed_damage_phase_scaling":{"type":"BOOL","value":true}]},"inherits_element":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"677cf9b395fab5fb3b8e715626917de12c1a71b6ae46d"}}</pre>
DRAGON_TAIL	<pre>{"monster_skill_id":{"type":"ENUM","value":"DRAGON_TAIL"},"user_archetype":{"type":"ENUM","value":"五行龙王"}, "cooldown_s":{"type":"INT64","value":10,"unit":"s"},"windup_s":{"type":"Q4","value_scaled_1e4":6000,"human":"0.6","unit":"s"},"shape":{"type":"ENUM","value":"CIRCLE"},"radius_m":{"type":"INT64","value":5,"unit":"m"},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"MAIN"},"damage":{"type":"INT64","value":180,"unit":"hp"},"damage_form":{"type":"ENUM","value":"SKILL"},"element_rule":{"type":"ENUM","value":"BIDIRECTIONAL_FIVE_ELEMENT"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"source_faction":{"type":"ENUM","value":"MONSTER"},"owner_credit":{"type":"ENUM","value":"MONSTER"},"fixed_damage_phase_scaling":{"type":"BOOL","value":true}]},"kno</pre>

	ckback_m":{"type":"INT64","value":3,"unit":"m"},"inherits_element":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"ecb5c611a8a0a1a63c9497637e824907a66ee549ba179fa79f6ce6d5c1e58b4b"}}
DRAGON_DASH	{"monster_skill_id":{"type":"ENUM","value":"DRAGON_DASH"},"user_archetype":{"type":"ENUM","value":"五行龙王"}, "cooldown_s":{"type":"INT64","value":12,"unit":"s"},"windup_s":{"type":"INT64","value":1,"unit":"s"},"shape":{"type":"ENUM","value":"LINE_CAPSULE"},"length_m":{"type":"INT64","value":8,"unit":"m"},"width_m":{"type":"Q4","value_scaled_1e4":24000,"human":"2.4","unit":"m"},"height_m":{"type":"INT64","value":3,"unit":"m"},"telegraph_s":{"type":"INT64","value":1,"unit":"s"},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"MAIN"},"damage":{"type":"INT64","value":120,"unit":"hp"},"damage_form":{"type":"ENUM","value":"SKILL"},"element_rule":{"type":"ENUM","value":"BIDIRECTIONAL_FIVE_ELEMENT"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"source_faction":{"type":"ENUM","value":"MONSTER"},"owner_credit":{"type":"ENUM","value":"MONSTER"},"fixed_damage_phase_scaling":{"type":"BOOL","value":true}}],"condition":{"type":"ENUM","value":"TOP_THREAT_AT_LEAST_12M_FOR_CONTINUOUS_2S_AND_REACHABLE"},"max_distance_m":{"type":"INT64","value":8,"unit":"m"},"slow_percent":{"type":"INT64","value":25,"unit":"pct"},"slow_s":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"s"},"inherits_element":{"type":"BOOL","value":true},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"28bded4a974106e22e4a0c47e17b985b0a3db0995d25d3032f97247848246398"}}
CORE_RING	{"monster_skill_id":{"type":"ENUM","value":"CORE_RING"},"user_archetype":{"type":"ENUM","value":"核心 BOSS 公共骨架"}, "cooldown_s":{"type":"INT64","value":10,"unit":"s"},"windup_s":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"s"},"shape":{"type":"ENUM","value":"CIRCLE"},"radius_m":{"type":"INT64","value":10,"unit":"m"},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"MAIN"},"damage":{"type":"INT64","value":400,"unit":"hp"},"damage_form":{"type":"ENUM","value":"SKILL"},"element_rule":{"type":"ENUM","value":"NONE"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"source_faction":{"type":"ENUM","value":"MONSTER"},"owner_credit":{"type":"ENUM","value":"MONSTER"},"fixed_damage_phase_scaling":{"type":"BOOL","value":true}}],"first_cast_delay_s":{"type":"INT64","value":5,"unit":"s"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"3ba29f9d763fcb6e01390341d4e9d55c76bb56fc7846244d4f7bc938b736bff5"}}
CORE_METEOR	{"monster_skill_id":{"type":"ENUM","value":"CORE_METEOR"},"user_archetype":{"type":"ENUM","value":"核心 BOSS 公共骨架"}, "cooldown_s":{"type":"INT64","value":20,"unit":"s"},"windup_s":{"type":"INT64","value":2,"unit":"s"},"shape":{"type":"ENUM","value":"CIRCLE"},"radius_m":{"type":"INT64","value":6,"unit":"m"},"damage_events":{"ty

	pe":{"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"MAIN"},"damage":{"type":"INT64","value":500,"unit":"hp"},"damage_form":{"type":"ENUM","value":"SKILL"},"element_rule":{"type":"ENUM","value":"NONE"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"source_faction":{"type":"ENUM","value":"MONSTER"},"owner_credit":{"type":"ENUM","value":"MONSTER"},"fixed_damage_phase_scaling":{"type":"BOOL","value":true}}],"first_cast_delay_s":{"type":"INT64","value":10,"unit":"s"},"target":{"type":"ENUM","value":"RANDOM_LIVING_THREAT_TARGET_CAST_POSITION"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"a19185a0b1064cb97dd5a2f6f763fdd527c75c8e9686d7d86f6394503227afbe"}}}
CORE_V1_EARTHBREAK	{"monster_skill_id":{"type":"ENUM","value":"CORE_V1_EARTHBREAK"},"user_archetype":{"type":"ENUM","value":"CORE_COMMON"},"cooldown_s":{"type":"INT64","value":16,"unit":"s"},"windup_s":{"type":"Q4","value_scaled_1e4":12000,"human":"1.2","unit":"s"},"shape":{"type":"ENUM","value":"THREE_PARALLEL_LINES"},"length_m":{"type":"INT64","value":20,"unit":"m"},"width_m":{"type":"INT64","value":2,"unit":"m"},"gap_m":{"type":"INT64","value":3,"unit":"m"},"damage":{"type":"INT64","value":350,"unit":"hp"},"knockup_s":{"type":"Q4","value_scaled_1e4":6000,"human":"0.6","unit":"s"},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"MAIN"},"magnitude_field":{"type":"ENUM","value":"damage"},"damage_form":{"type":"ENUM","value":"SKILL"},"source_faction":{"type":"ENUM","value":"MONSTER"},"element_rule":{"type":"ENUM","value":"NONE"},"hit_kind":{"type":"ENUM","value":"ABILITY"},"owner_credit":{"type":"ENUM","value":"MONSTER"},"trigger":{"type":"ENUM","value":"ON_SKILL_RESOLVE"},"fixed_damage_phase_scaling":{"type":"BOOL","value":true}}],"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"f81f28ed5befde218f5e7d09c368a760e5de0bd793211a74789dab075e7230eb"}}}
CORE_V2_FIRELANE	{"monster_skill_id":{"type":"ENUM","value":"CORE_V2_FIRELANE"},"user_archetype":{"type":"ENUM","value":"CORE_COMMON"},"cooldown_s":{"type":"INT64","value":18,"unit":"s"},"windup_s":{"type":"Q4","value_scaled_1e4":12000,"human":"1.2","unit":"s"},"shape":{"type":"ENUM","value":"THREE_RADIAL_LANES_WITH_ONE_GAP"},"length_m":{"type":"INT64","value":18,"unit":"m"},"width_m":{"type":"INT64","value":3,"unit":"m"},"duration_s":{"type":"INT64","value":6,"unit":"s"},"damage_per_s":{"type":"INT64","value":100,"unit":"hp/s"},"lane_count":{"type":"INT64","value":3,"unit":"scalar"},"gap_lane_random_stream":{"type":"ENUM","value":"MONSTER_AI"},"damage_events":{"type":"ARRAY","element":"DamageEventSpec","ordered":true,"value":[{"event_id":{"type":"ENUM","value":"MAIN"},"magnitude_field":{"type":"ENUM","value":"damage_per_s"},"damage_form":{"type":"ENUM","value":"DOT"},"source_faction":{"type":"ENUM","value":"MONSTER"},"element_rule":{"type":"ENUM","value":"NONE"},"hit_kind":{"type":"ENUM","value":"PERIODIC"},"owner_credit":{"type":"ENUM","value":"MONSTER"},"trigger":{"type":"ENUM","value":"ON_SKILL_RESOLVE"},"fixed_damage_phase_scaling":{"type":"BOOL","value":true}}],"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"c25b63f8c952417c728544fbd9fee3b4116dc6b576a0f7e8e2c68c33fdd527c75c8e9686d7d86f6394503227afbe"}}}

	a6daa9c4d"}}
CORE_V3_POISON POOL	{ "monster_skill_id": {"type": "ENUM", "value": "CORE_V3_POISONPOOL"}, "user_archetype": {"type": "ENUM", "value": "CORE_COMMON"}, "cooldown_s": {"type": "INT64", "value": 20, "unit": "s"}, "windup_s": {"type": "Q4", "value_scaled_1e4": 15000, "human": 1.5, "unit": "s"}, "target_count": {"type": "INT64", "value": 3, "unit": "scalar"}, "shape": {"type": "ENUM", "value": "CIRCLE"}, "radius_m": {"type": "INT64", "value": 5, "unit": "m"}, "duration_s": {"type": "INT64", "value": 8, "unit": "s"}, "damage_per_s": {"type": "INT64", "value": 70, "unit": "hp/s"}, "damage_events": {"type": "ARRAY", "element": "DamageEventSpec", "ordered": true, "value": [{"event_id": {"type": "ENUM", "value": "MAIN"}, "magnitude_field": {"type": "ENUM", "value": "damage_per_s"}, "damage_form": {"type": "ENUM", "value": "DOT"}, "source_faction": {"type": "ENUM", "value": "MONSTER"}, "element_rule": {"type": "ENUM", "value": "NONE"}, "hit_kind": {"type": "ENUM", "value": "PERIODIC"}, "owner_credit": {"type": "ENUM", "value": "MONSTER"}, "trigger": {"type": "ENUM", "value": "ON_SKILL_RESOLVE"}, "fixed_damage_phase_scaling": {"type": "BOOL", "value": true}]}], "schema_version": {"type": "ENUM", "value": "V3_0"}, "payload_sha256": {"type": "HASH", "value": "bd5bd1c46d95417bcb0824e665d245bbe762521d58f3998e500d5005cb6e89df"} }
CORE_V4_WINDCH ARGE	{ "monster_skill_id": {"type": "ENUM", "value": "CORE_V4_WINDCHARGE"}, "user_archetype": {"type": "ENUM", "value": "CORE_COMMON"}, "cooldown_s": {"type": "INT64", "value": 16, "unit": "s"}, "windup_s": {"type": "Q4", "value_scaled_1e4": 12000, "human": 1.2, "unit": "s"}, "shape": {"type": "ENUM", "value": "LINE"}, "length_m": {"type": "INT64", "value": 18, "unit": "m"}, "width_m": {"type": "INT64", "value": 3, "unit": "m"}, "damage": {"type": "INT64", "value": 350, "unit": "hp"}, "knockback_m": {"type": "INT64", "value": 5, "unit": "m"}, "stops_at_solid": {"type": "BOOL", "value": true}, "damage_events": {"type": "ARRAY", "element": "DamageEventSpec", "ordered": true, "value": [{"event_id": {"type": "ENUM", "value": "MAIN"}, "magnitude_field": {"type": "ENUM", "value": "damage"}, "damage_form": {"type": "ENUM", "value": "SKILL"}, "source_faction": {"type": "ENUM", "value": "MONSTER"}, "element_rule": {"type": "ENUM", "value": "NONE"}, "hit_kind": {"type": "ENUM", "value": "ABILITY"}, "owner_credit": {"type": "ENUM", "value": "MONSTER"}, "trigger": {"type": "ENUM", "value": "ON_SKILL_RESOLVE"}, "fixed_damage_phase_scaling": {"type": "BOOL", "value": true}]}], "schema_version": {"type": "ENUM", "value": "V3_0"}, "payload_sha256": {"type": "HASH", "value": "8488a608df1ddd6edb1ceeadfd5b424483de52ec24d8c417014219bc810f046b"} }
CORE_V5_THUNDE RCHAIN	{ "monster_skill_id": {"type": "ENUM", "value": "CORE_V5_THUNDERCHAIN"}, "user_archetype": {"type": "ENUM", "value": "CORE_COMMON"}, "cooldown_s": {"type": "INT64", "value": 18, "unit": "s"}, "windup_s": {"type": "Q4", "value_scaled_1e4": 15000, "human": 1.5, "unit": "s"}, "target": {"type": "ENUM", "value": "RANDOM_LIVING_THREAT_TARGET"}, "damage": {"type": "INT64", "value": 300, "unit": "hp"}, "jumps": {"type": "INT64", "value": 4, "unit": "scalar"}, "jump_range_m": {"type": "INT64", "value": 6, "unit": "m"}, "decay_percent_per_jump": {"type": "INT64", "value": 20, "unit": "scalar"}, "shape": {"type": "ENUM", "value": "SINGLE_TARGET_CHAIN"}, "damage_events": {"type": "ARRAY", "element": "DamageEventSpec", "ordered": true, "value": [{"event_id": {"type": "ENUM", "value": "MAIN"}, "magnitude_field": {"type": "ENUM", "value": "damage"}, "damage_form": {"type": "ENUM", "value": "SKILL"}, "source_faction": {"type": "ENUM", "value": "MONSTER"}, "element_rule": {"type": "ENUM", "value": "NONE"}, "hit

	<pre>_kind":{"type":"ENUM","value":"ABILITY"},"owner_credit":{"type":"ENUM","value":"MONSTER"},"trigger":{"type":"ENUM","value":"ON_SKILL_RESOLVE"},"fixed_damage_phase_scaling":{"type":"BOOL","value":true}}}, "schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"56b8cdba816e756eb83c75f7a712c296cf951df8a7bde0857458030c1e215940"}}</pre>
CORE_V6_MIRROR SHADOW	<pre>{"monster_skill_id":{"type":"ENUM","value":"CORE_V6_MIRRORSHADOW"},"user_archetype":{"type":"ENUM","value":"CORE_COMMON"},"cooldown_s":{"type":"INT64","value":24,"unit":"s"},"windup_s":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"s"},"summon_count":{"type":"INT64","value":2,"unit":"scalar"},"summon_hp":{"type":"INT64","value":2000,"unit":"hp"},"summon_attack":{"type":"INT64","value":60,"unit":"scalar"},"summon_attack_speed_per_s":{"type":"Q4","value_scaled_1e4":8000,"human":"0.8","unit":"1/s"},"duration_s":{"type":"INT64","value":10,"unit":"s"},"decoy_cannot_drop_or_credit":{"type":"BOOL","value":true},"summon_move_speed_mps":{"type":"Q4","value_scaled_1e4":26000,"human":"2.6","unit":"m/s"},"summon_attack_range_m":{"type":"Q4","value_scaled_1e4":18000,"human":"1.8","unit":"m"},"summon_detection_range_m":{"type":"INT64","value":12,"unit":"m"},"summon_collision_radius_m":{"type":"Q4","value_scaled_1e4":6500,"human":"0.65","unit":"m"},"summon_collision_height_m":{"type":"INT64","value":2,"unit":"m"},"summon_targeting":{"type":"ENUM","value":"HIGHEST_CORE_BOSS_THREAT_VISIBLE_AND_REACHABLE_ELSE_NEAREST_VISIBLE_PLAYER"},"summon_leash":{"type":"ENUM","value":"WHOLE_CURRENT_COURT"},"summon_movement_collision":{"type":"BOOL","value":true},"mirror_basic_damage_spec":{"type":"REF","value":"E.34_CORE_MIRROR 载荷"},"shape":{"type":"ENUM","value":"NONE"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"40ed77b925594b921af74cac4918c01152d265dbc69527af71f6b9d4a9c7a91e"}}</pre>

E.34 召唤物强类型运行时载荷

- 运行时唯一权威；E.10 保留全局继承与 AI 规则，逐原型战斗数值以本表为准。

summon_archetype	runtime_payload_json
STONE_ARHAT	<pre>{"summon_archetype":{"type":"ENUM","value":"STONE_ARHAT"},"source":{"type":"ENUM","value":"H020"},"hp":{"type":"INT64","value":500,"unit":"hp"},"attack":{"type":"INT64","value":30,"unit":"scalar"},"duration_s":{"type":"INT64","value":20,"unit":"s"},"attack_speed_per_s":{"type":"Q4","value_scaled_1e4":7000,"human":"0.7","unit":"1/s"},"range_m":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"m"},"move_speed_mps":{"type":"Q4","value_scaled_1e4":22000,"human":"2.2","unit":"m/s"},"aggro_range_m":{"type":"INT64","value":8,"unit":"m"},"leash_m":{"type":"INT64","value":12,"unit":"m"},"threat_bonus":{"type":"INT64","value":300,"unit":"scalar"},"collision_radius_m":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"m"},"collision_height_m":{"type":"INT64","value":2,"unit":"m"},"summon_targeting":{"type":"ENUM","value":"HIGHEST_CORE_BOSS_THREAT_VISIBLE_AND_REACHABLE_ELSE_NEAREST_VISIBLE_PLAYER"},"summon_leash":{"type":"ENUM","value":"WHOLE_CURRENT_COURT"},"summon_movement_collision":{"type":"BOOL","value":true},"mirror_basic_damage_spec":{"type":"REF","value":"E.34_CORE_MIRROR 载荷"},"shape":{"type":"ENUM","value":"NONE"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"40ed77b925594b921af74cac4918c01152d265dbc69527af71f6b9d4a9c7a91e"}}</pre>

	":5000,"human":"0.5","unit":"m"},"collision_height_m":{"type":"Q4","value_scaled_1e4":16000,"human":"1.6","unit":"m"},"movement_collision":{"type":"BOOL","value":true},"target_priority":{"type":"ENUM","value":"MONSTER_ATTACKING_OWNER_THEN_NEAREST_MONSTER_THEN_NEAREST_ENEMY_PLAYER"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"926bd91bdc2ac4b422017ad22cd6e51932f3b4bf228b4a7307348cf45bf5d00a"}}}
BEAN _SOL DIER	{"summon_archetype":{"type":"ENUM","value":"BEAN_SOLDIER"},"source":{"type":"ENUM","value":"D19"},"hp":{"type":"INT64","value":150,"unit":"hp"},"attack":{"type":"INT64","value":15,"unit":"scalar"},"attack_speed_per_s":{"type":"INT64","value":1,"unit":"1/s"},"range_m":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"m"},"move_speed_mps":{"type":"Q4","value_scaled_1e4":28000,"human":"2.8","unit":"m/s"},"aggro_range_m":{"type":"INT64","value":8,"unit":"m"},"leash_m":{"type":"INT64","value":15,"unit":"m"},"collision_radius_m":{"type":"Q4","value_scaled_1e4":3500,"human":"0.35","unit":"m"},"collision_height_m":{"type":"Q4","value_scaled_1e4":12000,"human":"1.2","unit":"m"},"movement_collision":{"type":"BOOL","value":true},"triggers_owner_passives":{"type":"BOOL","value":false},"target_priority":{"type":"ENUM","value":"VISIBLE_ENEMY_PLAYER_THEN_MONSTER_ATTACKING_OWNER_THEN_NEAREST_LEGAL_ENEMY"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"ec2ca98231428b1dc3e7bc5acab09f4d898f46d6595f5d1499ab34bcfca46fed"}}}
SIX_E AR_D ECOY	{"summon_archetype":{"type":"ENUM","value":"SIX_EAR_DECOY"},"source":{"type":"ENUM","value":"H012"},"attack":{"type":"INT64","value":0,"unit":"scalar"},"damage_enabled":{"type":"BOOL","value":false},"movement_collision":{"type":"BOOL","value":false},"damageable_hit_volume":{"type":"BOOL","value":true},"collision_radius_m":{"type":"Q4","value_scaled_1e4":4500,"human":"0.45","unit":"m"},"collision_height_m":{"type":"Q4","value_scaled_1e4":18000,"human":"1.8","unit":"m"},"fake_basic_interval_s":{"type":"Q4","value_scaled_1e4":12500,"human":"1.25","unit":"s"},"rng_draws":{"type":"INT64","value":0,"unit":"scalar"},"hp_rule":{"type":"FORMULA","src":"FLOOR(OWNER_CURRENT_MAX_HP*0.3)","ast":{"op":"FLOOR","args":[{"op":"MUL","args":[{"var":"OWNER_CURRENT_MAX_HP"},{"q4":3000}]}}},"assign_to":"DECOY_MAX_HP"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_sha256":{"type":"HASH","value":"c1ae83e6513c581a1409cefe95a5ae4bbfc3a12214011e12de667c38599b11e3"}}}
WOLF _SPIR IT	{"summon_archetype":{"type":"ENUM","value":"WOLF_SPIRIT"},"source":{"type":"ENUM","value":"B12"},"hp":{"type":"ARRAY","element":"INT64","ordered":true,"per_level":true,"unit":"hp","value":[150,200,250,300,350]},"attack":{"type":"ARRAY","element":"INT64","ordered":true,"per_level":true,"unit":"scalar","value":[20,25,30,35,40]},"attack_speed_per_s":{"type":"INT64","value":1,"unit":"1/s"},"range_m":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"m"},"move_speed_mps":{"type":"INT64","value":3,"unit":"m/s"},"aggro_range_m":{"type":"INT64","value":8,"unit":"m"},"leash_m":{"type":"INT64","value":12,"unit":"m"},"duration_s":{"type":"INT64","value":15,"unit":"s"},"max_instances":{"type":"ARRAY","element":"INT64","ordered":true,"per_level":true,"unit":"scalar","value":[1,1,1,1,2]},"collision_radius_m":{"type":"Q4","value_scaled_1e4":4000,"human":"0.4","unit":"m"},"collision_height_m":{"type":"Q4","value_scaled_1e4":12000,"human":"1.2","unit":"m"},"movement_collision":{"type":"BOOL","value":true},"target_priority":{"type":"ENUM","value":"OWNER_TRIGGERING_BASIC_TARGET_THEN_NEAREST_LEGAL_ENEMY"},"schema_version":{"type":"ENUM","value":"V3_0"},"payload_s

	ha256":{"type":"HASH","value":"f8d466ba1451594259f94f83d0d766c778aee648e9ef1faecf890e242a320907"}}
FIRE_SPIRIT	{ "summon_archetype":{"type":"ENUM","value":"FIRE_SPIRIT"}, "source":{"type":"ENUM","value":"B13"}, "contact_damage":{"type":"ARRAY","element":"INT64","ordered":true,"per_level":true,"unit":"hp","value":[60,80,100,120,150]}, "duration_s":{"type":"INT64","value":8,"unit":"s"}, "orbit_radius_m":{"type":"Q4","value_scaled_1e4":15000,"human":"1.5","unit":"m"}, "orbit_angular_speed_deg_s":{"type":"INT64","value":180,"unit":"s"}, "contact_internal_cd_s":{"type":"INT64","value":1,"unit":"s"}, "contact_internal_cd_scope":{"type":"ENUM","value":"PER_TARGET"}, "burn_duration_s":{"type":"INT64","value":3,"unit":"s"}, "max_instances":{"type":"ARRAY","element":"INT64","ordered":true,"per_level":true,"unit":"scalar","value":[1,1,1,1,2]}, "movement_collision":{"type":"BOOL","value":false}, "damageable_hit_volume":{"type":"BOOL","value":false}, "contact_radius_m":{"type":"Q4","value_scaled_1e4":5000,"human":"0.5","unit":"m"}, "untargetable":{"type":"BOOL","value":true}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"5de014c2a88ddc97b57d80ca96433c369ceaa6daa8196011e27c9c628457a4e9"}}
STONE_STATUE	{ "summon_archetype":{"type":"ENUM","value":"STONE_STATUE"}, "source":{"type":"ENUM","value":"B39"}, "hp":{"type":"ARRAY","element":"INT64","ordered":true,"per_level":true,"unit":"hp","value":[250,300,350,400,500]}, "attack":{"type":"INT64","value":0,"unit":"scalar"}, "move_speed_mps":{"type":"INT64","value":3,"unit":"m/s"}, "collision_radius_m":{"type":"Q4","value_scaled_1e4":6000,"human":"0.6","unit":"m"}, "collision_height_m":{"type":"Q4","value_scaled_1e4":18000,"human":"1.8","unit":"m"}, "movement_collision":{"type":"BOOL","value":true}, "aggro":{"type":"ENUM","value":"NONE"}, "targetable":{"type":"BOOL","value":true}, "destroy_credit":{"type":"ENUM","value":"ATTACKER_ONLY_NO_REWARD"}, "forbidden_block_ole_corridor":{"type":"BOOL","value":true}, "owner_block_yield_after_s":{"type":"Q4","value_scaled_1e4":5000,"human":"0.5","unit":"s"}, "yield_fail":{"type":"ENUM","value":"TEMPORARY_NO_COLLISION"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"03ff66b62d278d08218f5c1dd13746fcf643670734c07503e2ce828b666b72bd"}}
CORE_MIRROR	{ "summon_archetype":{"type":"ENUM","value":"CORE_MIRROR"}, "source":{"type":"ENUM","value":"CORE_V6_MIRRORS_HADROW"}, "hp":{"type":"INT64","value":2000,"unit":"hp"}, "attack":{"type":"INT64","value":60,"unit":"scalar"}, "attack_speed_per_s":{"type":"Q4","value_scaled_1e4":8000,"human":"0.8","unit":"1/s"}, "range_m":{"type":"Q4","value_scaled_1e4":18000,"human":"1.8","unit":"m"}, "move_speed_mps":{"type":"Q4","value_scaled_1e4":26000,"human":"2.6","unit":"m/s"}, "detection_range_m":{"type":"INT64","value":12,"unit":"m"}, "collision_radius_m":{"type":"Q4","value_scaled_1e4":6500,"human":"0.65","unit":"m"}, "collision_height_m":{"type":"INT64","value":2,"unit":"m"}, "movement_collision":{"type":"BOOL","value":true}, "duration_s":{"type":"INT64","value":10,"unit":"s"}, "cannot_drop_or_credit":{"type":"BOOL","value":true}, "leash":{"type":"ENUM","value":"WHOLE_CURRENT_COURT"}, "basic_damage_form":{"type":"ENUM","value":"BASIC"}, "basic_hit_kind":{"type":"ENUM","value":"DIRECT"}, "basic_element_rule":{"type":"ENUM","value":"NONE"}, "basic_source_faction":{"type":"ENUM","value":"MONSTER"}, "basic_owner_credit":{"type":"ENUM","value":"MONSTER"}, "targeting":{"type":"ENUM","value":"HIGHEST_CORE_BOSS_THREAT_VISIBLE_AND_REACHABLE_ELSE_NEAREST_VISIBLE_PLAYER"}, "schema_version":{"type":"ENUM","value":"V3_0"}, "payload_sha256":{"type":"HASH","value":"44a587eb13505bf36c601"}

	98260c718ae33e79216ec80467b7467a8c4ee92e47a""]}}
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E.35 H009 快照语义、字段饱和与区域叠加

工程字段	唯一值 / 执行语义
H009.instance_snapshot	被动效果在触发瞬间按当时倍率快照；H009 结束不回退已生成的护盾/DOT/召唤物/永久层
H009.new_instance_reads_live	是；H009 结束后新触发实例读取实时值
H009.permanent_layers	PERMANENT_GROWTH 类字段被排除，永不受 H009 影响 (E.27 白名单已排除)
H009.saturation.pct_chance_cap	任何概率/命中类 pct 字段翻倍后 CLAMP 到[0,100]
H009.saturation.remaining_multiplier_floor	任何 remaining_multiplier 类字段按 transform_remaining_from_one 后 CLAMP 到[0,1]
H009.saturation.slow_percent_cap	减速百分比翻倍后 CLAMP 到[0,90]；90 为全局减速硬上限
H009.saturation.knockback_cap_m	击退距离翻倍后 CLAMP 到[0,10]
H009.saturation.damage_heal_no_cap	伤害/治疗幅度字段不设额外上限，仅受 SIGNED_INT64_SATURATING 保护
H009.saturation.per_field_authority	E.27 列出的每个允许字段按其 unit 自动匹配上述饱和规则；未匹配者默认无上限
periodic_refresh.extends_end_time_only	是；刷新只延长结束时间
periodic_refresh.tick_phase_preserved	是；不重置下一跳时间，刷新瞬间禁止双跳
zone_stacking.same_caster_same_skill	REFRESH_DURATION_NO_MAGNITUDE_STACK (H003/H006/H007/H025/H030/H033)
zone_stacking.different_casters.damage_heal	独立并行结算
zone_stacking.different_casters.slow	只取最强减速 (TAKE_STRONGEST)
zone_stacking.different_casters.antimobility	任一 H033 生效即禁止位移施放 (OR 语义)

E.36 第二轮关联扩散审计修订裁决记录（108—170）

编号	级别	裁决	落点
108	P0	被动 44/装备 48/怪物技能/召唤物建强类型权威表，运行时禁止解析分号字符串	E.31 / E.32 / E.33 / E.34

109	P0	唯一单向投影生成器+source_hash；手工维护禁止	第 0 章约定表 / G_PROJECTION_DRIFT
110	P0	拆分 BASE/DAMAGE/HEAL/SUMMON 族；owner_credit 仅伤害治疗必填	E.30
111	P0	57 类型逐类型闭合字段矩阵	E.30
112	P0	公式 DSL 定义并全部编译 AST	E.29 / 各载荷 ast 字段
113	P0	统一 INT64/Q4 定点+强制单位	E.28 / 全部载荷
114	P0	LIST 废止，改强类型 ARRAY/SET	E.28 / 全部载荷
115	P0	H033/H035 开放 STRING 改闭合枚举	E.26 载荷 / E.23 枚举注册
116	P0	回放/快照/根种子域升级 V3_0+旧域兼容映射	15.x 域行 / domain_compat
117	P0	B19 闪现：COMBAT_PROC 流单抽八方向取最远合法点，失败不重抽	第 7 章 B19 / E.31
118	P0	H009 施加瞬间快照；已生成实例不回退	E.35
119	P0	H009 白名单字段逐类饱和上下限	E.35
120	P0	周期刷新只延长结束时间不重置 tick 相位	E.35
121	P0	同施法者刷新不叠；异施法者伤害独立减速取最强	E.35
122	P0	H002/H004 改 damage_events[] 逐事件声明	E.26 载荷
123	P0	B11 闭合候选池 12 项+落空规则	E.31 B11 池表
124	P0	阶段切换保当前生命百分比；死亡实体按重生阶段	E.13 行
125	P0	技能固伤×攻击阶段倍率；CD 倍率首领 1/1/0.95/0.90，龙王核心 1/1/0.90/0.85	E.13 / E.33
126	P0	龙王普攻 0.45+0.25s，核心 0.60+0.35s	11.2 表 / E.33
127	P0	经济门槛扩为守恒+分位带+极端构筑+售罄断言	17.1 gates[10]
128	P0	全文档空单元格清零并设 G_NO_EMPTY_CELLS	全表 / 17.1

129	P0	38 概念图入库清单：源文件名/SHA-256/版权状态/派生资产	16.1.1 清单表
130	P0	MVS_019-022 改 LEADER 骨架 12000/6000/2000 面 50 骨	16.3 台账
131	P0	MonsterSkillShape 闭合枚举+逐形状必填几何	E.33 / E.23
132	P0	DISTRICT_PATH_0-5 共享边界路径表+闭合覆盖零重叠校验	9.1a 路径表
133	P1	D13 强制 A≠B；重复选择无效继续等待	第 6 章 D13 / E.26
134	P1	D20 陷阱 250 血可破坏无奖励不爆炸不挡路	第 6 章 D20 / E.26
135	P1	D15 斩杀=施法者正常野怪击杀；触发链一次；不触发普攻链	第 6 章 D15 / E.26
136	P1	H010 无限等待合法普攻；真死/替换/换英雄清除	第 6 章 H010 / E.26
137	P1	B28 猎杀目标 Nav 距离+实体 ID 唯一化，0.25s 重选	第 7 章 B28 / E.31
138	P1	B29 仅正伤害触发并重置	第 7 章 B29 / E.31
139	P1	B37 剔除 WORLD_DAMAGE 触发	第 7 章 B37 / E.10 / E.31
140	P1	B39 石像禁堵唯一通路；0.5s 让路；失败临时无碰撞	第 7 章 B39 / E.31 / E.34
141	P1	B06 旁路余数：总暴伤先算，HP 子事件 FLOOR，余数入护盾子事件	第 7 章 B06 / E.31
142	P1	B24 更名化伤；文案明确不能防止致死	第 7 章 / 附录 B.1
143	P1	G4 六米内无合法落点：交互失败保持飞行不扣费	第 8 章 G4 / E.32
144	P1	隐身来源独立；敌对动作清隐并伏石显形 1.5s 后重隐	E.7
145	P1	D12 显形覆盖隐身伏石但不穿墙锁定	第 6 章 D12 / E.7
146	P1	D20 短按引爆最近/长按选择器/无陷阱才放	第 6 章 D20 / E.6

		置	
147	P1	CORE_V6 本体伤害规格删除，移入镜像召唤载荷	11.3 / E.34
148	P1	五行龙王逐元素装饰/阵纹/龙息/尾击/冲锋/命中资产 ID	16.2 / 16.3 新行
149	P1	动态墙困住实体：下 tick 重寻路，1 秒无路回退最后合法点	E.8
150	P1	全实体 NavMesh 代理规格表	E.8
151	P1	地面点=相机射线首个合法面；方向技能用瞄准高度；弹道 3D 扫掠	E.6
152	P1	小地图与锁定的上下层箭头及高度差图标	E.20
153	P1	锁定回看 $\leq 250\text{ms}$ ；投射物不回溯；移动墙体不倒带	15.1
154	P1	预测误差 $\leq 1.5\text{m}$ 平滑 100ms，超过立即校正	15.1
155	P1	input_seq+估算 tick；5s 校时；服务端唯一权威	15.1
156	P1	重复丢弃/移动不回溯/普攻 250ms 验证/交易到达即结算	15.1
157	P1	快照分块+哈希+ACK+BOT 托管+下 tick 交还+按序补事件	13.3
158	P1	版本不符强制更新；席位存活可重连；不算逃跑	13.3
159	P1	客户端 100ms 表现插值；权威判定只读 20Hz	15.1
160	P1	第 4 设备登录返回会话列表用户明选；撤销即断线	15.2
161	P1	全部写端点错误/HTTP/retryable/幂等冲突矩阵	15.2 矩阵表
162	P1	证据包 180 天+人工复核状态机+撤销原子恢复	15.6

163	P1	交互优先级闭合表：安全提示>英雄>商店>空投>宝箱>遗宝堆>拾取	E.16
164	P2	全部载荷行带 schema_version+payload_sha256	E.26/E.31-E.34
165	P2	遗宝堆默认焦点/翻页保序/消失后继任规则	E.16
166	P2	强闪预警必须有低闪三合一替代资产	E.24 / G_LOW_FLASH_ALTERNATES
167	P2	首发锁简体中文 CN；其余闭合逐项开启	14.3
168	P2	接受 5 秒观战延迟残余风险，不加限制	15.6
169	P2	门槛绑定 evidence_id/checker_version/result_hash， 构建器生成状态	17.1
170	P2	封面状态改为审计闭合态，复核通过后终审	封面

E.37 第三轮收敛审计修订记录（171—179）

编号	级别	发现的漏洞	修复裁决与落点
171	P0	单位系统缺陷：*_per_s 速率字段误标单位 s；伤害/治疗量纲未定	单位推导规则修订为速率优先；全部载荷重生成（E.28）
172	P0	14 类被动产生的独立伤害/反弹事件缺形态/命中/五行/归属五元组	E.31 逐被动补 damage_events 语义声明（magnitude_field 指针引用，不复制数值）
173	P0	B13 火灵'点燃 3 秒'无伤害数值，无法实现	定案 burn_damage_per_s=6/8/10/12/15， 写入第 7 章与 E.31
174	P0	11.3 核心签名技能 V1-V6 未强类型化	全部入 E.33 载荷表；V5 补 shape=SINGLE_TARGET_CHAIN，V2 补车道字段，V6 为 shape=NONE 召唤类
175	P0	召唤物数值在第 7 章 /E.26/11.3 与 E.34 双重存在，无唯一权威	E.34 声明唯一权威；E.10 增字节级一致构建断言（六组对照）

176	P1	E.30 矩阵'可选'字段与'闭合 字段集全部必填'表述矛盾	改为必填/允许缺省/矩阵外禁止三段 语义
177	P1	P10 两处 § 章节引用存为开 放 ENUM	新增 REF 闭合类型（运行时不求 值），注册 E.23/E.28
178	P1	EquipmentQuality/StatKind/ 扩展 DamageEventId 未注 册闭合枚举	全部注册 E.23
179	P2	公式变量大小写规则未声明	E.29 增行：大小写敏感闭合集